FRM Part II 2026

Risk Management and Investment Management

Gold-source question bank

617 questions, sorted by chapter and learning objective

Scope and LO wording: SchweserNotes 2026 (final authority). Every question solved and independently verified.

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Reading 83. Factor Theory

Andrew Ang, Asset Management: A Systematic Approach to Factor Investing (New York, NY: Oxford University Press, 2014). Chapter 6.

LO	Learning objective	Qs
83.a	describe factors that impact asset prices and explain the theory of factor risk premiums.	7
83.b	discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM.	11
83.c	explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM.	14
83.d	describe multifactor models and compare and contrast multifactor models to the CAPM.	3
83.e	explain how stochastic discount factors are created and apply them in the valuation of assets.	6
83.f	describe efficient market theory and explain how markets can be inefficient.	3

LO 83.a

Describe factors that impact asset prices and explain the theory of factor risk premiums.

Q1 AnalystPrep Q.1646

According to the capital asset pricing model, assets whose returns are positively correlated with aggregate consumption or wealth will earn:

A. a return equal to the growth rate of GDP.
B. a risk premium.
C. the risk-free rate of return.
D. a return equal to the rate of inflation.

Answer: B

Step by step

1. Assets whose returns are positively correlated with aggregate wealth or consumption pay off well in good times and poorly in bad times.
2. Factor theory says exposure to bad times is what earns compensation, so investors demand an expected return above the risk-free rate.
3. That excess expected return is a (positive) risk premium.

Why B is correct: Positive co-movement with the economy means the asset loses value in bad times, which investors must be paid to bear, so it earns a positive risk premium.

Why the others are wrong

- **A:** GDP growth is a macroeconomic outcome, not the required return on any asset; confuses an economic factor with its price.
- **C:** The risk-free rate is earned only by assets with no exposure to bad times; this asset has such exposure.
- **D:** Inflation compensates for purchasing-power loss, not for factor risk; mixes nominal and risk-premium components.

Q2 AnalystPrep Q.1647

REBUILT — Options 'zero' and 'neutral' risk premium were near-synonyms (two defensible labels); replaced 'neutral' with a distinct plausible error and tightened the stem.

An investor adds an asset whose returns are negatively correlated with the economy; it pays off in bad times and acts as a hedge for the rest of the portfolio. According to factor theory, the asset is expected to have:

A. a zero risk premium, because its risk is fully diversifiable.
B. a positive risk premium, because all risky assets must be compensated for volatility.
C. a negative risk premium, because investors accept a lower return for the insurance it provides.
D. a risk premium equal to the market risk premium, because all assets are priced relative to the market.

Answer: C

Step by step

1. Factor risk premiums compensate investors for losses in bad times.

2. This asset gains in bad times, so it reduces the investor's exposure to those bad times, like insurance.
3. Investors pay for insurance, so they accept an expected return below the risk-free rate: a negative risk premium.

Why C is correct: An asset that pays off when the economy does badly hedges factor risk, so investors bid up its price and accept a negative expected excess return.

Why the others are wrong

- **A:** Treats the asset as having no priced risk; but the issue is systematic co-movement, not diversifiability, and the hedge has value.
- **B:** Assumes total volatility is rewarded; only exposure to bad times (systematic risk) is priced, and here the exposure is reversed.
- **D:** Applies the market premium to every asset; the premium depends on each asset's factor exposure (beta), which here is negative.

Q3

BT answers P2.T8.700.1 · BT notes P2.T9.700.1

Andrew Ang develops an analogy, writing "factors are to assets what nutrients are to food." His theory of factor risk premiums includes each of the following three ideas EXCEPT which idea?

A. Assets are bundles of factors (just as most foods are combinations of nutrients).
B. Factors do matter but asset classes do not (just as healthy eating is about the nutrients not the labels).
C. Different investors prefer and/or need different factors (just as different people have different nutritional needs).
D. Because factors represent different good times, most investors should seek exposure to most investable factors (just as most people should seek a balanced diet of most nutrients).

Answer: D

Step by step

1. Principle 1: factors, not assets, matter (B).
2. Principle 2: assets are bundles of factors (A).
3. Principle 3: investors have different optimal factor exposures (C).
4. D fails because factors represent different sets of BAD times, not good times, and the theory does not say everyone should hold most factors.

Why D is correct: Factor risk premiums reward exposure to bad times, and optimal exposures differ across investors, so D is contrary to the theory.

Why the others are wrong

- **A:** Is principle 2 of the theory, so it is part of it.
- **B:** Is principle 1 (look through assets to factors), so it is part of it.
- **C:** Is principle 3 (different optimal exposures), so it is part of it.

Q4

BT answers P2.T9.20.1.2 · BT notes P2.T9.20.1.2

Andrew Ang says that "the factor risk premium has an economic story." Which of the following is the BEST summary of the factor risk premium?

A. The factor risk premium is compensation for suffering losses during a specific set of bad times.
B. The factor risk premium is a reward for selecting a set of equities that outperforms over a given period.
C. The factor risk premium is an orthogonal linear combination of variables that maximizes the total variance.
D. The factor risk premium is the excess return due to the selection of a set of style factors within a single asset class; aka, smart beta.

Answer: A

Step by step

1. Factor theory defines risk as exposure to bad times; each factor represents a different set of bad times.
2. The premium is the compensation investors require in normal times for bearing those losses.
3. Only A states this economic story.

Why A is correct: A factor risk premium is the reward for bearing losses in a particular set of bad times.

Why the others are wrong

- **B:** Describes ex post outperformance (luck or skill in selection), not a risk-based compensation.
- **C:** Describes the first principal component in PCA, a statistical construct with no economic rationale.
- **D:** Describes smart-beta style tilts within one asset class; too narrow and not the economic definition of a premium.

Q5 GARP PE1 Q79

A group of junior analysts is attending a training session on how risk factors are used in investment management. During the training, several macroeconomic factors as well as investment style factors are described and their effects on investors in different situations are evaluated. Which of the following is correct according to the theory of factor risk?

A. Investors in an asset that is likely to have a high payoff during adverse market conditions usually earn high risk premiums on that asset.
B. Investors who invest in the same assets are exposed to the same set of risk factors and have the same level of optimal exposure.
C. Investors are affected negatively when there is an unexpected decline in economic growth, regardless of the type of their investments.
D. Investors in an asset with a large exposure to volatility risk can hedge against this risk by purchasing out-of-the-money put options.

Answer: D

Step by step

1. A: premiums reward losses in bad times; an asset paying off in bad times is a hedge and earns a low (even negative) premium.
2. B: the same assets carry the same factor exposures, but optimal exposures differ across investors (principle 3).
3. C: some holdings (e.g., government bonds) gain when growth falls, so effects are not uniform.
4. D: volatility rises in bad times, increasing put values, so buying OTM puts offsets volatility-risk losses.

Why D is correct: Put option values rise when volatility spikes, so puts hedge volatility-factor exposure; the other statements contradict factor theory.

Why the others are wrong

- **A:** Reverses the logic: high payoff in bad times means a hedge, which earns a low premium, not a high one.
- **B:** Confuses identical factor exposures with identical optimal exposures; investors differ in risk aversion and needs.
- **C:** Assumes all investors lose in a growth decline; defensive assets such as government bonds may gain.

Q6 Schweser Module Quiz 83.1 Q1

Which of the following concepts would least likely meet the definition of a factor?

A. Market.
B. Volatility.
C. Hedge funds.
D. Momentum investing style.

Answer: C

Step by step

1. Factors include the market, interest rates, volatility, macro variables and investing styles.
2. Hedge funds are assets that bundle many factors (equity, interest rate, volatility, default risk).
3. So hedge funds are least likely to be a factor.

Why C is correct: Hedge funds are bundles of underlying factor exposures rather than a factor themselves.

Why the others are wrong

- **A:** The market is a tradeable investment factor.
- **B:** Volatility is a key factor whose exposure is rewarded with a volatility premium.
- **D:** Momentum is a style factor.

Q7 BT Mock Q80 · BT answers P2.T9.20.1.1 · BT notes P2.T9.20.1.1

Janice is preparing to conduct factor regressions to discover the factor loadings in her firm's equity portfolio. She will begin by regressing portfolio returns on three explanatory variables: the market factor (MKT) plus two additional factors. Each of the following pairs of factors is a good candidate for the two additional factors EXCEPT which pair is NOT a good candidate?

A. Quality and momentum (style factors)
B. Strategy and reputation (style factors)
C. Economic growth and inflation (macro factors)

D. Real interest rates and liquidity (macro factors)

Answer: B

Step by step

1. Factor theory treats assets as bundles of systematic factor risks that earn risk premiums for exposure to 'bad times'; a candidate regressor must be a systematic, measurable, investable risk factor.
2. Style factors include value/growth, quality, low volatility, size and momentum; these are systematic and can be implemented with long-short portfolios. Pair A qualifies.
3. Macro factors include economic growth, inflation, real interest rates and liquidity. Pairs C and D qualify.
4. Strategy and reputation are firm-specific characteristics: they cannot be measured as a return series on a tradeable portfolio and do not represent a systematic bad time shared across investors. Pair B does not qualify, so B is the EXCEPT answer.

Why B is correct: Factors must be systematic, measurable and investable sources of risk that earn a premium for bad-times exposure; strategy and reputation are idiosyncratic, non-investable firm attributes, not style factors.

Why the others are wrong

- **A:** Quality and momentum are recognised style factors with long-short implementations; rejecting them confuses style factors with firm-specific traits.
- **C:** Growth and inflation are the classic fundamental macroeconomic factors; rejecting them wrongly assumes only market-traded factors qualify.
- **D:** Real rates and liquidity are macro factors whose bad times (rate spikes, liquidity crises) are rewarded with premiums; rejecting them misreads 'macro' as non-priced.

LO 83.b

Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM.

Q8

AnalystPrep Q.2375

REBUILT — Original mixed CAPM assumptions with results (equilibrium, diversification) so 'all correct' was loosely defensible; replaced with a clean assumption-identification item.

Which of the following is an assumption of the CAPM?

A. Investors have heterogeneous expectations about asset returns
B. Investors face taxes and transaction costs that differ across securities
C. Large investors are price setters whose trades move the market
D. Investors have mean-variance utility

Answer: D

Step by step

1. The CAPM assumes investors care only about the mean and variance of returns (mean-variance utility).
2. It also assumes homogeneous expectations, frictionless markets (no taxes or transaction costs), price-taking investors, a single-period horizon, and free information available to all.
3. A, B and C each state the opposite of a CAPM assumption, so only D is an assumption.

Why D is correct: Mean-variance utility is one of the simplifying CAPM assumptions; the other options describe real-world departures from the model.

Why the others are wrong

- **A:** The CAPM assumes homogeneous (identical) expectations; heterogeneous expectations are a real-world departure.
- **B:** The CAPM assumes frictionless markets with no taxes or transaction costs.
- **C:** The CAPM assumes all investors are price takers; price-setting is a real-world departure.

Q9

BT notes P2.T9.700.3

Andrew Ang catalogs both the failures and successes of the CAPM, where a success is an idea the CAPM gets right. Each of the following is an assumption or implication of the CAPM, but only one is true in practice and therefore useful. Which one is a genuine success?

A. Information is costless and available to all investors

B. Risk is factor exposure: an asset's risk is measured by its exposure to the factor
C. Investors have mean-variance utility
D. Investors have homogeneous expectations

Answer: B
Step by step
1. Test each against the real world: information is costly and unevenly available (A fails); investors dislike losses more than they like gains, so utility is not mean-variance (C fails); expectations are heterogeneous (D fails).
2. Risk being driven by factor exposure (beta) rather than by an asset in isolation holds in practice and underlies factor investing.
3. Only B is a genuine success.

Why B is correct: Measuring an asset's risk by its factor exposure is the CAPM idea that carries over to real markets (and to multifactor models).
Why the others are wrong
- **A:** Information is often costly and unavailable to some investors.
- **C:** Investors view risk asymmetrically (downside risk), so mean-variance utility fails.
- **D:** Investors hold heterogeneous expectations in practice.

Q10 AnalystPrep Q.3023

REBUILT — Two distractors were unexplained numbers; replaced with options that each reflect a specific formula error.
The expected return of the market portfolio is 6.2% and the risk-free rate is 3.9%. An asset has a beta of 1.3. Using the CAPM, the asset's expected return is closest to:
A. 8.06%
B. 2.99%
C. 5.20%
D. 6.89%

Answer: D
Step by step
1. $E(R_i) = R_F + \beta_i[E(R_M) - R_F]$
2. Market risk premium = 6.2% - 3.9% = 2.3%.
3. $E(R_i) = 3.9\% + 1.3 \times 2.3\% = 3.9\% + 2.99\% = 6.89\%$.

Why D is correct: The CAPM expected return is the risk-free rate plus beta times the market risk premium.
Why the others are wrong
- **A:** 1.3 x 6.2% = 8.06%: multiplied beta by the market return rather than the market risk premium and omitted rf.
- **B:** 1.3 x 2.3% = 2.99%: computed the asset's risk premium but forgot to add back the risk-free rate.
- **C:** 3.9% + 1.3 = 5.20%: added beta (as a percentage) to the risk-free rate instead of multiplying by the premium.

Q11 AnalystPrep Q.4586

Which of the following is LEAST likely an assumption of the CAPM?
A. Investors have a long-period (multi-period) investment horizon
B. Investors have identical expectations
C. Investors trade without transaction costs or taxes
D. Investors can access all information at the same time

Answer: A
Step by step
1. The CAPM is a single-period model: investors hold the market portfolio and no rebalancing is required.
2. Homogeneous expectations (B), frictionless markets (C) and free, universally available information (D) are CAPM assumptions.
3. A contradicts the single-period assumption, so it is the least likely assumption.

Why A is correct: The CAPM assumes a single-period horizon, not a multi-period one.
Why the others are wrong
- **B:** Homogeneous expectations is a CAPM assumption.

- **C:** Frictionless markets (no taxes or transaction costs) is a CAPM assumption.
- **D:** Free information available to everyone is a CAPM assumption.

Q12 AnalystPrep Q.4591

REBUILT — Original options conflated risk premium with factor exposure and were loosely worded; replaced with a numeric item showing risk is beta exposure, not total volatility.

Under the CAPM the risk-free rate is 3% and the market risk premium is 6%. Asset X has a market beta of 0.5 and a total volatility of 40%. Asset Y has a market beta of 1.5 and a total volatility of 20%. Which statement is correct?

A. X has the higher expected return because its total volatility is higher.
B. Y has the higher expected return (12.0% vs. 6.0%) because it has greater exposure to the market factor.
C. X and Y have the same expected return because both earn the same market risk premium.
D. X has the higher expected return because idiosyncratic risk is rewarded.

Answer: B

Step by step

1. $E(R_X) = 3\% + 0.5 \times 6\% = 6.0\%$.
2. $E(R_Y) = 3\% + 1.5 \times 6\% = 12.0\%$.
3. Y has higher expected return because the CAPM rewards exposure to the market factor (beta), not total volatility.

Why B is correct: In the CAPM, risk is measured by factor (market) exposure; higher beta means higher expected return.

Why the others are wrong

- **A:** Uses total volatility as the risk measure; the CAPM rewards only beta.
- **C:** The premium per unit of beta is the same, but the exposure (beta) differs, so expected returns differ.
- **D:** Idiosyncratic risk is diversifiable and carries no risk premium.

Q13 AnalystPrep Q.4592

In both the CAPM and multifactor models, risk is measured by factor betas. Which of the following is most likely true about factor betas and risk premiums?

A. Low betas imply high risk premiums
B. High betas imply low risk premiums
C. High betas imply high risk premiums
D. None of the above

Answer: C

Step by step

1. Beta measures sensitivity to the factor; high-beta assets lose more in bad times.
2. Investors require compensation for bearing this bad-times risk, so the premium rises with beta ($\text{RP}_i = \beta_i \times$ factor premium).
3. Hence high betas imply high risk premiums.

Why C is correct: Risk premium is proportional to factor beta, so more factor exposure is rewarded with a higher premium.

Why the others are wrong

- **A:** Reverses the relation; low-beta assets pay off relatively well in bad times and need low premiums.
- **B:** Reverses the relation; high beta means more bad-times risk and a higher premium.
- **D:** Option C is a correct statement.

Q14 AnalystPrep Q.3018

REBUILT — Original stem and answer kept; option B was reworded to state the amplified-magnitude feature of a high beta so it is unambiguous.

Which of the following best describes assets with a high positive beta?

A. Assets with lower volatility than the market, which tend to move slowly in response to market changes.
B. Assets that tend to go up when the market goes up and go down when the market goes down, by more than the market does.
C. Assets that tend to go down when the market goes up and up when the market goes down.

D. Assets whose prices are less volatile than the market portfolio.

Answer: B

Step by step

1. Beta is $\beta_i = \text{cov}(R_i, R_M)/\text{var}(R_M)$, the asset's sensitivity to the market factor.
2. A high positive beta ($\beta > 1$) means the asset co-moves in the same direction as the market with larger magnitude.
3. Only B describes same-direction, amplified movement.

Why B is correct: Under the CAPM, beta measures sensitivity to the market factor; a high positive beta means large same-direction co-movement, hence a high risk premium.

Why the others are wrong

- **A:** Describes a low-beta asset (beta below 1); confuses high beta with low market sensitivity.
- **C:** Describes a negative-beta asset, which moves opposite to the market.
- **D:** Also describes a low-beta asset; high beta means more, not less, market-related volatility.

Q15 AnalystPrep Q.4595

Security K has a standard deviation of returns of 4%. The standard deviation of the market return is 5%, the correlation between K's return and the market return is 0.75, the risk-free rate is 5%, and the expected market return is 10%. What is the beta of Security K?

A. 0.30
B. 0.375
C. 0.60
D. 0.75

Answer: C

Step by step

1. $\beta_K = \text{cov}(K, M)/\sigma_M^2 = \rho\,\sigma_K\sigma_M/\sigma_M^2 = \rho\,\sigma_K/\sigma_M$.
2. $\beta_K = 0.75 \times 4\%/5\% = 0.60$.
3. The risk-free rate and expected market return are not needed for beta.

Why C is correct: Beta equals correlation times the ratio of the asset's volatility to the market's volatility: $0.75 \times 0.8 = 0.60$.

Why the others are wrong

- **A:** Divided by the expected market return (10%) instead of the market volatility: 0.75 Ã— 4/10 = 0.30.
- **B:** Used 0.75 x 5%/10% = 0.375, dividing the market volatility by the expected market return instead of using σ_K/σ_M.
- **D:** Reported the correlation itself as beta, ignoring the volatility ratio.

Q16 AnalystPrep Q.4634

A portfolio manager wants to replicate the expected return of a stock with a CAPM beta of 1.6 using only the risk-free asset and the market portfolio. Which combination replicates the stock's expected return?

A. 0.6 Rf + 1.6 E(Rm)
B. -0.6 Rf + 1.6 E(Rm)
C. 1.6 E(Rm)
D. It cannot be determined with the information given.

Answer: B

Step by step

1. CAPM: $E(R_s) = R_f + \beta[E(R_m) - R_f]$.
2. $E(R_s) = R_f + 1.6E(R_m) - 1.6R_f = -0.6R_f + 1.6E(R_m)$.
3. Weights: 160% in the market and -60% in the risk-free asset (borrowing); they sum to 100%.

Why B is correct: Expanding the SML gives a weight of β in the market and $1 - \beta$ in the risk-free asset; with beta 1.6 the risk-free weight is negative (leverage).

Why the others are wrong

- **A:** Gets the sign on the risk-free weight wrong; weights would sum to 220% rather than 100%.
- **C:** Omits the risk-free component, forgetting that $1 - \beta$ must be invested at Rf so weights sum to one.

- **D:** The CAPM provides everything needed: beta alone fixes the market and risk-free weights.

Q17 AnalystPrep Q.2403

REBUILT — Original item (beta in a financial crisis, with garbled regression wording and no theory-determined answer) was replaced with a self-contained CAPM expected-return calculation using the regression beta.

The CAPM risk-free rate is 2% and the expected return on the market portfolio is 8%. A regression of a stock's excess returns on the market's excess returns gives a slope of 1.2. Using the CAPM, what is the stock's expected return?

A. 7.2%
B. 9.2%
C. 9.6%
D. 11.6%

Answer: B

Step by step

1. Beta is the regression slope: $\beta = 1.2$.
2. Market risk premium $= 8\% - 2\% = 6\%$.
3. $E(R) = R_f + \beta(E(R_m) - R_f) = 2\% + 1.2 \times 6\% = 9.2\%$.

Why B is correct: CAPM expected return is the risk-free rate plus beta times the market risk premium.

Why the others are wrong

- **A:** Computed only beta Ã— premium (7.2%) and forgot to add the risk-free rate.
- **C:** Multiplied beta by the full market return (1.2 Ã— 8% = 9.6%) instead of the premium, and omitted Rf.
- **D:** Applied beta to the market return rather than the premium, then added Rf: 2% + 1.2 Ã— 8% = 11.6%.

Q18 AP Mock 2 Q27

REBUILT — Original asked which accept/reject rule is best with an unusual coefficient-of-variation twist and distractors that were verdict-only; rebuilt as a single numeric question on the CAPM required return and the resulting maximum acceptable risk.

Jim Korner is evaluating an investment in an Indian textile company whose systematic risk is twice that of the market ($\beta = 2$). The expected market return is 12% and the risk-free rate is 4%. Assume CAPM holds. The company's coefficient of variation is 0.10, meaning that for every 1% of return required the project carries 0.10% of overall risk. Jim will accept the investment only if its overall risk does not exceed the level implied by the CAPM required return. What is the maximum overall risk at which the project should be accepted?

A. 1.2%
B. 2.0%
C. 2.4%
D. 2.8%

Answer: B

Step by step

1. CAPM required return: $E(R) = R_f + \beta(E(R_m) - R_f) = 4\% + 2(12\% - 4\%) = 20\%$.
2. Risk allowed per unit of return: 0.10% of risk per 1% of return, i.e. 0.10.
3. Maximum acceptable overall risk $= 0.10 \times 20\% = 2.0\%$.
4. Accept if the project's risk is 2.0% or less; reject above that.

Why B is correct: The CAPM prices systematic risk through beta times the market risk premium (8%), giving a 20% required return; scaling by the 0.10 coefficient of variation converts it into a risk ceiling of 2.0%.

Why the others are wrong

- **A:** Ignores beta and uses the market return: $0.10 \times 12\% = 1.2\%$.
- **C:** Uses beta times the market return and omits the risk-free rate: $2 \times 12\% = 24\%$, so $0.10 \times 24\% = 2.4\%$.
- **D:** Uses beta times the full market return plus the risk-free rate ($4\% + 2 \times 12\% = 28\%$) rather than the premium (12% - 4% = 8%), giving $0.10 \times 28\% = 2.8\%$.

LO 83.c

Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM.

Q19

AnalystPrep Q.2406

REBUILT — Original relied on an unstated assumption about age and risk aversion and had a nonsensical option (premium = 1); made the data explicit, made it numeric using the CAPM market-risk-premium relation, and moved to LO 83.c.

Under the CAPM, the market risk premium equals the average investor's risk aversion times the variance of the market portfolio. Two countries have the same market volatility of 18%. Investors in Country A are young, with average risk aversion of 2.0; investors in Country B are older and more risk averse, with average risk aversion of 3.5. What is the market risk premium in Country B?

A. 11.34%
B. 63.00%
C. 6.48%
D. 5.67%

Answer: A

Step by step

1. Risk premium: $E(R_M) - R_F = \gamma\sigma_M^2$.
2. $\sigma_M^2 = 0.18^2 = 0.0324$.
3. Country B: $3.5 \times 0.0324 = 0.1134 = 11.34\%$.
4. Country A would be $2.0 \times 0.0324 = 6.48\%$, lower, consistent with lower risk aversion.

Why A is correct: The premium is proportional to risk aversion and market variance, so more risk-averse Country B has a higher premium: $3.5 \times 0.18^2 = 11.34\%$.

Why the others are wrong

- **B:** Used volatility instead of variance: $3.5 \times 0.18 = 63\%$.
- **C:** Used Country A's risk aversion (2.0) instead of Country B's, giving 6.48%.
- **D:** Applied a mean-variance utility factor of one-half: $0.5 \times 3.5 \times 0.0324 = 5.67\%$; the CAPM relation has no one-half.

Q20

AnalystPrep Q.2385

REBUILT — Exhibit (SML graph) was missing; replaced with numeric data so the stock's position relative to the SML can be computed.

The risk-free rate is 3.0% and the expected return on the market portfolio is 8.0%. Stock A has a beta of 1.2 and an analyst-forecast expected return of 10.5%. According to the CAPM, stock A is:

A. Undervalued
B. Overvalued
C. Appropriately valued
D. Impossible to classify without the stock's total volatility

Answer: A

Step by step

1. Required return from the SML: $E(R_A) = R_F + \beta_A[E(R_M) - R_F] = 3.0\% + 1.2 \times 5.0\% = 9.0\%$.
2. Forecast return 10.5% exceeds the required 9.0%: $\alpha = +1.5\%$.
3. The stock plots above the SML, so it offers more return than its beta risk requires: it is undervalued (price too low).

Why A is correct: A stock whose expected return lies above the SML offers a positive alpha for its systematic risk, so it is undervalued.

Why the others are wrong

- **B:** Would apply if the forecast return were below the SML (below 9.0%); the sign of alpha was reversed.
- **C:** Appropriately valued requires forecast return equal to 9.0%; it is 1.5% higher.
- **D:** The CAPM prices risk by beta only, so total volatility is irrelevant to the SML comparison.

Q21

AnalystPrep Q.2386

REBUILT — Exhibit (SML graph) was missing; replaced with numeric data so the stock's position relative to the SML can be computed.

The risk-free rate is 3.0% and the expected return on the market portfolio is 9.0%. Stock C has a beta of 0.8 and an analyst-forecast expected return of 6.5%. According to the CAPM, stock C is:

A. Undervalued
B. Appropriately valued
C. Overvalued
D. Impossible to classify without the stock's total volatility

Answer: C

Step by step

1. Required return: $E(R_C) = 3.0\% + 0.8 \times (9.0\% - 3.0\%) = 3.0\% + 4.8\% = 7.8\%$.
2. Forecast 6.5% is below required 7.8%: $\alpha = -1.3\%$.
3. The stock plots below the SML, so its price is too high for the return it offers: overvalued.

Why C is correct: A forecast return below the CAPM-required return means negative alpha: the stock plots below the SML and is overvalued.

Why the others are wrong

- **A:** Reverses the sign of alpha; undervalued stocks plot above the SML.
- **B:** Would require forecast return of 7.8%; the stock falls 1.3% short.
- **D:** Total volatility is not a CAPM pricing input; beta alone determines the required return.

Q22

AnalystPrep Q.2388

Which statement is NOT true regarding the capital market line (CML)?

A. The risk measure for the CML is standard deviation.
B. The CML is the line from the risk-free rate through the market portfolio.
C. The CML is the best attainable capital allocation line.
D. The CML is also called the security market line.

Answer: D

Step by step

1. In equilibrium all investors hold the same mean-variance efficient (market) portfolio, so the capital allocation line through it is the CML.
2. The CML plots expected return against total risk (standard deviation); A, B and C are true.
3. The SML is a different line: it plots expected return against beta for individual assets. D is false.

Why D is correct: The CML (return vs. standard deviation, efficient portfolios) and the SML (return vs. beta, individual assets) are distinct lines.

Why the others are wrong

- **A:** True statement; the CML's horizontal axis is standard deviation.
- **B:** True; it starts at the risk-free rate and passes through the market (MVE) portfolio.
- **C:** True; it has the maximum Sharpe ratio, so it is the best attainable CAL.

Q23

AnalystPrep Q.4554

John Mathews, a portfolio risk manager, wants to re-assess the average investor's risk aversion using CAPM theory. His forecasts are:

Variable	Forecast
Risk-free rate	3%
Market risk premium	7%
Standard deviation of market returns	18%

The average investor's risk aversion coefficient is closest to:

A. 2.16
B. 0.39
C. 0.22
D. 1.23

Answer: A

Step by step

1. CAPM equilibrium: $E(R_M) - R_F = \bar{\gamma}\,\sigma_M^2$.
2. $\sigma_M^2 = 0.18^2 = 0.0324$.
3. $\bar{\gamma} = 0.07/0.0324 = 2.16$.

Why A is correct: The market risk premium equals average risk aversion times market variance, so risk aversion is the premium divided by variance.

Why the others are wrong

- **B:** 0.07/0.18 = 0.39: divided by the standard deviation instead of the variance.
- **C:** (7% - 3%)/0.18 = 0.22: subtracted rf from a premium that is already net of rf and divided by volatility.
- **D:** (7% - 3%)/0.0324 = 1.23: used the variance correctly but subtracted rf from the premium again.

Q24 AnalystPrep Q.2377

Country A, a developing economy, has recently witnessed government's renewed interest and effort in reforming the financial sector. A recent report published by a non-governmental organization (NGO) indicates the presence of rampant insider trading. The report also points towards the lack of depth in the financial market which poses significant problems for investors and market participants. However, the presence of natural resources in the country makes it an attractive destination for foreign investors. A fund manager investing in country A utilizes the CAPM to compute the expected return. Which of the following options is (are) correct?

I. Deviations from the CAPM will be observed due to insider trading
II. Deviations from the CAPM will be observed due to the developing economy
III. Deviations from the CAPM will be observed due to a lack of market depth
IV. Deviations from the CAPM will be observed due to the country's underdeveloped corporate sector

A. I only
B. I and II
C. I and III
D. III and IV

Answer: C

Step by step

1. Map each statement to a CAPM assumption.
2. I: insider trading means information is not free and equally available to all investors, which violates the CAPM information assumption, so deviations arise.
3. III: a shallow market means large trades move prices, so investors are price setters rather than price takers, which violates a CAPM assumption.
4. II and IV: being a developing economy or having an underdeveloped corporate sector is not itself a CAPM assumption; only the specific violations above matter.
5. Only I and III are correct.

Why C is correct: Deviations from the CAPM arise when its simplifying assumptions (free, equal information; price-taking investors) are violated, which insider trading and thin market depth do.

Why the others are wrong

- **A:** Omits III; thin markets make trades move prices, violating the price-taker assumption.
- **B:** Includes II, which is a label rather than a violated CAPM assumption, and omits III.
- **D:** Includes IV (not a CAPM assumption) and omits I, the information violation.

Q25 AnalystPrep Q.2382

REBUILT — Original asked about a constant-correlation assumption that the Schweser reading does not cover and gave obviously false distractors; rebuilt around the mean-variance utility shortcoming in the reading.

A fund manager in a developed market builds well-diversified portfolios using ten years of historical data and mean-variance utility, consistent with the CAPM framework. Which of the following best describes a shortcoming of the CAPM's assumption of mean-variance utility?

A. It ignores covariances between assets, so the benefit of diversification cannot be measured.
B. It assumes investors are price setters whose trades move asset prices.
C. It treats gains and losses symmetrically, whereas investors dislike losses more than they like equal-sized gains.
D. It assumes investors have a multi-period horizon and rebalance continuously.

Answer: C

Step by step

1. Mean-variance utility measures risk with variance, which treats upside and downside deviations alike.
2. In reality investors have an asymmetric view of risk and dislike losses more than they like gains.
3. Stocks with greater downside risk should therefore offer higher returns than a variance-only (beta) measure implies; this is a recognized CAPM shortcoming.

Why C is correct: Variance penalizes positive and negative deviations equally, while real investors are loss averse, so mean-variance utility understates the reward required for downside risk.

Why the others are wrong

- **A:** Mean-variance analysis uses covariances explicitly; diversification benefits are measured through them.
- **B:** Price-taking is a separate CAPM assumption, and mean-variance utility does not imply price setting.
- **D:** The CAPM assumes a single-period horizon; a multi-period horizon is the realistic alternative, not the assumption.

Q26

AnalystPrep Q.2387

REBUILT — Original relied on a missing SML graph and a murky claim about observed returns; rebuilt as a self-contained calculation of the required pre-tax return when markets are not frictionless.

Under the CAPM, the expected return on stock X, given its beta, is 10.0%. The CAPM assumes no taxes. Suppose instead that investors in stock X pay a 25% tax on capital gains and all of the stock's return is a capital gain. If investors require an after-tax return of 10.0%, the required pre-tax expected return on X and its position relative to the CAPM security market line (SML) is closest to:

A. 7.50%, below the SML
B. 10.00%, on the SML
C. 12.50%, above the SML
D. 13.33%, above the SML

Answer: D

Step by step

1. After-tax return = pre-tax return × (1 − 0.25).
2. Required pre-tax return = 10.0% / 0.75 = 13.33%.
3. 13.33% exceeds the 10.0% CAPM return, so stock X must offer a return above the SML to compensate for the tax, a deviation caused by the no-taxes (frictionless markets) assumption.

Why D is correct: Taxes drive a wedge between pre- and after-tax returns, so the taxed asset must offer a higher pre-tax return than the frictionless CAPM predicts.

Why the others are wrong

- **A:** Multiplied 10.0% by 0.75, deducting tax from the required return instead of grossing up.
- **B:** Assumes taxes have no effect, ignoring that the CAPM assumes frictionless markets.
- **C:** Added 25% of 10.0% (10.0% × 1.25) rather than dividing by (1 − 0.25).

Q27

AnalystPrep Q.2389

REBUILT — Original relied on a missing graph and a debatable answer; rebuilt as a calculation using the CAPM market risk premium formula.

In a CAPM world the market risk premium equals the average investor's risk aversion times the market variance. In normal times, risk aversion is 2.5 and market volatility is 16%. During a recession, risk aversion rises to 3.0 and market volatility rises to 20%. The slope of the SML (the market risk premium) during the recession is closest to:

A. 10.00%
B. 12.00%
C. 6.40%
D. 7.68%

Answer: B

Step by step

1. Market risk premium = $\gamma \times \sigma_M^2$.
2. Normal times: $2.5 \times 0.16^2 = 6.40\%$.
3. Recession: $3.0 \times 0.20^2 = 3.0 \times 0.04 = 12.00\%$.
4. The SML steepens because both risk aversion and market variance increase.

Why B is correct: The market risk premium is proportional to market variance and risk aversion, so the SML slope

rises when both increase.

Why the others are wrong

- **A:** Changed only the volatility (2.5×0.20^2) and kept risk aversion at 2.5.
- **C:** Normal-times premium; ignores that the SML slope changes with risk aversion and variance.
- **D:** Changed only risk aversion (3.0×0.16^2) and ignored the higher volatility.

Q28

AnalystPrep Q.2467

A portfolio manager holds a portfolio which consists only of IT stocks. Borrowing some knowledge of the CAPM, he argues that since β measures the contribution of each security to total portfolio risk, he has to include securities with low β in his portfolio in order to reduce his portfolio risk. Is the manager's assertion correct?

A. Yes because β measures the contribution of one security to total portfolio risk.

B. No because the portfolio held by the manager is not well diversified, β is not an appropriate risk parameter.

C. No because β measures non-systematic risk.

D. Yes because β measures systematic risk.

Answer: B

Step by step

1. Under the CAPM, beta measures an asset's covariance with the market, i.e., systematic risk, which is the relevant risk for a well-diversified investor.
2. The manager holds only IT stocks, so idiosyncratic and sector risk remain in the portfolio and total risk is not captured by beta.
3. Beta is therefore not the appropriate risk measure for this portfolio; diversifying across sectors would reduce risk more.

Why B is correct: Beta is the right risk measure only for investors who have diversified away idiosyncratic risk, which a single-sector portfolio has not.

Why the others are wrong

- **A:** Beta measures contribution to market (systematic) risk of a diversified portfolio, not total risk of a concentrated one.
- **C:** Beta measures systematic, not non-systematic, risk; the conclusion is right but the reason is wrong.
- **D:** Beta does measure systematic risk, but that does not make the manager's conclusion valid for an undiversified portfolio.

Q29

AnalystPrep Q.3017

REBUILT — Option D was a giveaway ('cannot be determined') and B and C were incoherent; rewrote distractors as plausible misconceptions.

Which of the following best explains, under the CAPM, why asset owners should hold the market portfolio rather than individual stocks?

A. The market portfolio earns a larger risk premium than any individual stock for the same exposure to the market factor.

B. Individual stocks carry idiosyncratic risk as well as market factor risk; the market portfolio diversifies away the uncompensated idiosyncratic risk while retaining the market risk premium.

C. Individual stocks have no exposure to the market factor, so they earn no risk premium.

D. Under the CAPM only a few investors hold the market portfolio, and the rest hold the stocks with the highest expected returns.

Answer: B

Step by step

1. The CAPM's first lesson is to hold the factor (market), not the individual asset.
2. Each stock's return = market factor exposure + idiosyncratic risk.
3. Only the market factor is compensated with a risk premium; idiosyncratic risk is diversifiable and unrewarded.
4. Holding the market portfolio removes idiosyncratic risk and keeps the premium, improving the Sharpe ratio.

Why B is correct: Diversification eliminates unrewarded idiosyncratic risk while keeping the market risk premium, so the market portfolio dominates holding stocks in isolation.

Why the others are wrong

- **A:** Risk premium depends on factor exposure (beta); for the same exposure the premium is the same.
- **C:** Stocks do have market exposure (beta) and earn premiums for it; the problem is the extra idiosyncratic risk.

- **D:** In the CAPM all investors hold the same risky mean-variance efficient portfolio, the market portfolio, in different proportions.

Q30 BT answers P2.T8.700.2 · BT notes P2.T9.700.2

REBUILT — Source asked what Ang 'else says' is true, relying on text not in the Schweser reading, with one-sided distractors; rebuilt to test CAPM equilibrium and optimal holdings.

Which of the following statements about the CAPM in equilibrium is correct?

A. The relevant risk of an asset is its own volatility, since risk is evaluated in isolation.

B. Investors with different risk aversion hold different risky portfolios, because they have different expected returns.

C. Arbitrageurs eventually eliminate the market factor risk premium, since it can be hedged away.

D. All investors hold the same risky mean-variance efficient portfolio, the market portfolio, and differ only in how much they hold relative to the risk-free asset.

Answer: D

Key correction: Source key (A) picked an Ang statement ('spectacular failure') not covered by the reading; the rebuilt item tests equilibrium and optimal holdings, and the correct answer is D.

Step by step

1. In the CAPM all investors share identical expectations, so they all identify the same mean-variance efficient risky portfolio (the market).
2. Differences in risk aversion show up only as different mixes of the risk-free asset and the market portfolio along the capital allocation line.
3. Systematic risk cannot be arbitraged away, so the market risk premium persists; relevant risk is beta, not own volatility.

Why D is correct: With homogeneous expectations the market is the only risky portfolio held; risk aversion only determines the position along the CAL.

Why the others are wrong

- **A:** The CAPM says own volatility is irrelevant; the relevant measure is covariance with the market (beta).
- **B:** Expectations are homogeneous, so the risky portfolio is the same; only the proportions differ.
- **C:** Systematic market risk cannot be diversified or arbitraged away, so its premium does not disappear.

Q31 BT answers P2.T8.700.3

In regard to the capital asset pricing model (CAPM), Andrew Ang is able to catalog both its failures and successes, where success refers to "ideas the CAPM gets right." Each of the following is an assumption of the CAPM, but only one is TRUE IN PRACTICE and therefore useful. Which of the following assumptions (or implications) of the CAPM is a genuine success such that it is both true in practice and useful to us?

A. Information is costless and available to all investors: technology has reduced information friction to roughly zero

B. Risk is factor exposure: The risk of an individual asset is measured in terms of the factor exposure of that asset

C. Investors have mean-variance utility: asset owners care only about means (which they like) and variances (which they dislike)

D. Investors have homogeneous expectations: investors have identical expectations with respect to the necessary inputs into the portfolio decision

Answer: B

Step by step

1. Shortcomings of the CAPM include costly or unequal information, non-mean-variance (asymmetric) utility, and heterogeneous expectations.
2. Lesson 5 of the CAPM, that risk is measured by exposure to a factor (beta), remains valid and useful in practice.
3. Therefore B is the success; A, C and D are unrealistic assumptions.

Why B is correct: Measuring an asset's risk by its factor exposure is the CAPM's enduring insight, whereas its information, utility and expectations assumptions do not hold in the real world.

Why the others are wrong

- **A:** Information is often costly and unavailable to some investors, so this assumption fails.
- **C:** Investors view risk asymmetrically (dislike losses more than gains), so variance alone is inadequate.

- **D:** Investors have heterogeneous expectations in reality.

Q32 BT answers P2.T9.20.1.3 · BT notes P2.T9.20.1.3

REBUILT — Statement V mixed CAPM beta with covariance to the bad-times index (an SDF concept) and was ambiguous, and the all-true answer made the item a giveaway; V was rewritten in CAPM terms as a false statement and the options reworked so exactly one is correct.

A study group is reviewing the lessons of the capital asset pricing model (CAPM), which Ang describes as a failure in practice but whose basic intuition still holds. Each member states a lesson:

I. Allison: allocating to asset class labels such as hedge funds is not the optimal way to build a portfolio, because what is rewarded is exposure to underlying factors, not the asset label.

II. Betty: each investor has their own optimal exposure to the factor risk, just as in the CAPM each investor holds a different mix of the risk-free asset and the market portfolio.

III. Chris: in equilibrium the market factor's risk premium cannot be arbitraged away, because systematic risk cannot be diversified, so the premium persists.

IV. Donald: an asset's risk is measured by its factor beta (its exposure to the factor), not by its stand-alone volatility.

V. Eric: assets with high CAPM betas are valuable because they pay off well in bad times, so they have lower risk premiums and lower expected returns.

Which of the statements are TRUE?

A. All five statements are true.

B. Allison and Betty are incorrect; the other statements are true.

C. Eric is incorrect; the other statements are true.

D. Chris and Donald are incorrect; the other statements are true.

Answer: C

Key correction: Source key (A) said all five true; that relied on an ambiguous statement V mixing CAPM beta with the covariance to the bad-times index m. Restated in CAPM terms, V is false, so the answer is C.

Step by step

1. I is true (Lesson 1): hold the factor, not the individual asset or label; assets are bundles of factor risks.
2. II is true (Lesson 2): every investor holds the same risky MVE portfolio but in different proportions along the CAL.
3. III is true: in equilibrium the market factor has a risk premium that persists because investors cannot arbitrage away systematic risk.
4. IV is true (Lesson 5): risk is measured by beta exposure to the factor.
5. V is false (Lessons 5 and 6): a high beta ($\beta_i = \text{cov}(R_i, R_M)/\text{var}(R_M)$) means the asset loses when the market is low, so it is risky and earns a HIGHER risk premium; it is low-beta assets that are valuable and have low premiums.
6. Only Eric is incorrect, so the answer is C.

Why C is correct: In the CAPM, risk premiums rise with beta because high-beta assets perform badly in bad times (low market returns); valuable bad-times hedges are low-beta and carry low premiums, so only statement V is wrong.

Why the others are wrong

- **A:** Treats V as true by confusing 'valuable assets have low premiums' with high beta; it is low beta assets that are valuable.
- **B:** Wrongly rejects Lessons 1 and 2 (factor exposure over asset labels; investor-specific optimal exposures), both of which are true, and also misses that V is false.
- **D:** Wrongly rejects III (systematic risk cannot be arbitraged away) and IV (risk is measured by beta, not total volatility), and misses that V is false.

LO 83.d

Describe multifactor models and compare and contrast multifactor models to the CAPM.

Q33 AnalystPrep Q.2376

REBUILT — Original 'all of the above' item had a vague statement on idiosyncratic risk; reworded so each statement is precise and one option is clearly correct.

Which of the following statements about multifactor models, relative to the CAPM, is CORRECT?

A. They treat the market portfolio as the only source of bad times, so an asset has a single beta.
B. An asset's risk is measured by its exposures (betas) to several factors, each of which carries a risk premium.
C. They abandon diversification, because idiosyncratic risk is priced when several factors are present.
D. They imply that the average investor does not hold the market portfolio.

Answer: B

Step by step

1. The CAPM is a single-factor model: bad times are low market returns and risk is one beta.
2. Multifactor models (e.g., APT) add other factors such as low GDP growth or low consumption; expected return is a linear function of factor exposures.
3. Each factor with non-diversifiable risk earns its own risk premium, so an asset has multiple betas. This is statement B.

Why B is correct: Multifactor models generalise the CAPM by measuring risk with several factor betas, each rewarded by a risk premium.

Why the others are wrong

- **A:** Describes the CAPM, not multifactor models; confuses single-factor with multifactor.
- **C:** Diversification still holds; in multifactor models the tradeable factor portfolios remove idiosyncratic risk, which is not priced.
- **D:** The average investor holds the market portfolio under both the CAPM and multifactor models.

Q34 AnalystPrep Q.4589

Which of the following lessons from multifactor models are similar to the lessons from the CAPM?

A. The average investor holds the market portfolio
B. Exposure to factor risk must be rewarded
C. Diversification of idiosyncratic risks works
D. All of the above

Answer: D

Step by step

1. Schweser lists six lessons shared by the CAPM and multifactor models.
2. Shared lessons include: diversification is beneficial, the average investor holds the market portfolio, and exposure to factor risk must be rewarded.
3. All of A, B and C appear on that list, so D is correct.

Why D is correct: Each statement is one of the six lessons common to both frameworks, so all three apply.

Why the others are wrong

- **A:** True, but incomplete: B and C are also shared lessons.
- **B:** True, but incomplete: A and C are also shared lessons (market factor priced in CAPM, each factor priced in multifactor).
- **C:** True, but incomplete: A and B are also shared lessons (the tradeable factor removes idiosyncratic risk).

Q35 BT answers P2.T8.701.3 · BT notes P2.T9.701.3

REBUILT — Source tested an outside-reading claim about the financial crisis and efficiency (not supported by the Schweser text); rebuilt as a CAPM vs multifactor lesson question under 83.d.

Which of the following is a lesson shared by the CAPM and multifactor models?

A. Bad times are defined only as periods of low market returns.
B. Valuable assets, which pay off in bad times, have low risk premiums.
C. An asset's risk is measured by a single market beta.
D. Idiosyncratic risk is rewarded with a risk premium.

Answer: B

Key correction: Source key (B) referred to the financial crisis being consistent with the multifactor model, a claim not in the Schweser text; the item was rebuilt and its answer is B for a different reason (the low-risk-premium lesson).

Step by step

1. Shared lessons: diversification is beneficial, optimal exposures, average investor holds the market, factor risk is rewarded, risk is measured by betas, valuable assets have low risk premiums.
2. Assets paying off in bad times are attractive and so carry low risk premiums in both models (B).
3. A and C are CAPM-specific (single factor/market only); multifactor models use several factors and betas.
4. D is false in both models: idiosyncratic risk is diversified away and unpriced.

Why B is correct: Assets with positive payoffs in bad times are valuable and therefore have low risk premiums, whichever factors define bad times.

Why the others are wrong

- **A:** True only in the CAPM; multifactor models also define bad times by low GDP growth, consumption, etc.
- **C:** True only in the CAPM; multifactor models use multiple factor betas.
- **D:** Idiosyncratic risk is diversified away and not rewarded in either model.

LO 83.e

Explain how stochastic discount factors are created and apply them in the valuation of assets.

Q36

AnalystPrep Q.2381

REBUILT — Original keyed 'pricing kernel' against distractors outside the reading (Fama-French, Gaussian copula) with a debatable key; rewrote the stem and options so only the SDF model clearly fits the LO.

Fund managers in Country A estimate asset correlations from the last 10 years of calm-period data and use mean-variance utility. Their well-diversified portfolios fell sharply together in the most recent crisis. Which approach is best suited to modeling how assets behave in bad times?

A. The CAPM, with the market return as the only factor and a constant beta for each asset
B. A single-index model estimated on the last 10 years of data
C. A pricing kernel (SDF) model, in which m indexes bad times across several factors and may depend nonlinearly on them
D. A traditional discount-rate model that discounts expected cash flows at a fixed rate

Answer: C

Step by step

1. Bad times are periods when an extra dollar of income is very valuable (high marginal utility): job loss, low growth, high inflation, market stress.
2. The SDF m is a single variable that is an index of these bad times across many factors and states.
3. The CAPM is the special case in which m is linear in the market return; this linearity is its shortcoming. The pricing kernel allows nonlinearity and several factors.
4. Hence the SDF/pricing kernel model is the best choice for crisis behavior.

Why C is correct: The pricing kernel summarizes bad times over multiple factors and permits nonlinear payoffs, so it can capture how assets co-move in a crisis, whereas calm-period linear models cannot.

Why the others are wrong

- **A:** The CAPM is the special case with m linear in the market return and only one factor; it cannot capture non-linear, multi-factor bad times.
- **B:** A single-index model is just a one-factor linear model estimated on calm-period data; it shares the CAPM's limitations and the data-window problem.
- **D:** A fixed discount-rate model ignores the state-dependent value of payoffs; the SDF discounts payoffs by m, which is high in bad times.

Q37

AnalystPrep Q.4590

Multifactor models use the stochastic discount factor (SDF) to define bad times over multiple factors. Which of the following statements about the SDF is most likely true?

A. The SDF represents an index of bad times
B. The CAPM is a special case of this SDF model
C. The SDF is also called a pricing kernel
D. All of the above

Answer: D

Step by step

1. A: m is an index of bad times indexed by many factors and states. True.
2. B: the CAPM is the special case in which m is linear in the market return. True.
3. C: the SDF is also called a pricing kernel. True.
4. All three statements are true, so D.

Why D is correct: Each of the three statements is a stated property of the SDF, so 'all of the above' is correct.

Why the others are wrong

- **A:** True but incomplete; B and C are also true.
- **B:** True but incomplete; A and C are also true.
- **C:** True but incomplete; A and B are also true.

Q38

BT Mock Q20 · BT answers P2.T9.20.2.2 · BT notes P2.T9.20.2.2

REBUILT — Answer was correct but the distractors (-5%, +4%, +8%) did not correspond to realistic errors; replaced with options produced by sign, unit and component mistakes.

In factor theory, an asset's risk premium is $E(r_i) - R_f = \beta_{i,m}\lambda_m$, where $\beta_{i,m} = \text{cov}(r_i, m)/\text{var}(m)$ is the asset's beta with respect to the SDF (pricing kernel) m, an index of bad times, and $\lambda_m = -\text{var}(m)/E(m)$ is the price of bad-times risk. Assume:

Parameter	Value
Risk-free rate	3.0%, so $E(m) = 1/1.03 = 0.9709$
Variance of m	$30.0\%^2 = 0.090$
Covariance(r_i, m)	−0.10680

Which is closest to the asset's risk premium, $E(r_i) - R_f$?

A. −11.0%

B. +3.3%

C. +9.3%

D. +11.0%

Answer: D

Step by step

1. $\beta_{i,m} = -0.10680/0.090 = -1.1867$
2. $\lambda_m = -0.090/0.9709 = -0.09270$
3. $E(r_i) - R_f = (-1.1867)(-0.09270) = 0.1100 = +11.0\%$
4. Check: $-\text{cov}(r_i, m)/E(m) = 0.10680/0.9709 = 11.0\%$.

Why D is correct: Both beta and the price of bad-times risk are negative, so their product is a positive premium: an asset that does badly when m is high (bad times) must pay a higher expected return.

Why the others are wrong

- **A:** Right magnitude but wrong sign; forgetting the negative sign in λ(m) = −var(m)/E(m) gives −11.0%. A negative-covariance asset earns a positive premium.
- **B:** Used the standard deviation 0.30 instead of the variance 0.090 when computing beta (−0.356 × −0.0927 = 3.3%).
- **C:** Reported only the price of bad-times risk |λ(m)| = 0.0927 and ignored the asset's beta.

Q39

BT answers P2.T8.701.1 · BT notes P2.T9.701.1

Ang writes the risk premium of an asset using the stochastic discount factor (SDF) m, an index of bad times: $E(r_i) - R_f = -\frac{\text{cov}(r_i,m)}{\text{var}(m)} \times \frac{\text{var}(m)}{E(m)} = \beta_{i,m}\lambda_m$. Note that β(i,m) is a generalized beta with respect to the SDF, not the CAPM market beta. Each of the following statements is true EXCEPT which is false?

A. The expected return for the asset can be negative if β(i,m) is high enough

B. The m denotes a pricing kernel or stochastic discount factor (SDF) and is an "index of bad times"

C. A higher cov(r_i, m) corresponds to a higher β(i,m), which implies a higher risk premium and a higher expected return for the asset

D. The SDF can depend on a vector of factors, F = [f1, f2, ..., fk], where each of the k factors defines different bad times, e.g., high inflation or low economic growth

Answer: C

Step by step

1. $\lambda_m = -\text{var}(m)/E(m) < 0$, so premium $= \beta_{i,m}\lambda_m$ falls as $\beta_{i,m}$ rises.
2. A: if cov(r_i, m) is positive (asset pays off in bad times, a hedge), β is positive and the premium is negative. True.
3. B and D restate the definition of the SDF as an index of bad times that can depend on several factors. True.
4. C: higher cov means higher β, which means a LOWER risk premium, not a higher one. False.

Why C is correct: Because the price of bad-times risk is negative, a higher covariance with m (a better hedge in bad times) lowers, not raises, the required return.

Why the others are wrong

- **A:** True: a positive β (positive covariance with m) times the negative λ gives a negative premium; the asset is valuable as insurance.
- **B:** True: this is the definition of the SDF.
- **D:** True: the SDF can be a function of multiple factors, each defining different bad times.

Q40 BT answers P2.T8.701.2 · BT notes P2.T9.701.2

Ang says, "The $64,000 question with multifactor pricing kernel models is: how do you define bad times?" With respect to the multifactor model, which of the following is the best definition of "bad times"?

A. Low wealth level
B. Low happiness level
C. High marginal utility of the representative agent
D. Low marginal utility among heterogeneous agents

Answer: C

Step by step

1. Bad times are periods when an additional $1 of income is very valuable (e.g., job loss, low GDP growth, low consumption).
2. That is, the SDF is interpreted as the marginal utility of the representative agent: high marginal utility = bad times.
3. Hence C.

Why C is correct: The SDF is interpreted as marginal utility; bad times are when an extra dollar is most valuable, i.e., marginal utility is high.

Why the others are wrong

- **A:** Low wealth often accompanies bad times but is not the definition; the definition is based on marginal utility.
- **B:** Happiness is not the measure; bad times are defined through the value of an extra dollar of income.
- **D:** Reverses the relationship (low rather than high marginal utility) and refers to heterogeneous agents rather than the representative agent.

Q41 BT notes P2.T9.20.2.1

In the CAPM, the expected excess return of an asset is $E(r_i) - R_f = \beta_{i,M}[E(r_M) - R_f]$. In factor theory, $E(r_i) - R_f = \beta_{i,m}\lambda_m$, where $\beta_{i,m} = \text{cov}(r_i, m)/\text{var}(m)$ is the beta of the asset with respect to the SDF m. For typical assets, $\beta_{i,m}$ and λ_m are both negative. Each of the following statements is true EXCEPT which is false?

A. λ(m) = −var(m)/E(m), and it is the price of bad-times risk
B. An increase in β(i,m), for example from −0.95 to −0.33, implies a decrease in the asset's expected return; i.e., higher beta implies lower risk premium
C. Because the value of the bad-times index m is negative, an asset with a non-zero beta with respect to the SDF has a negative expected excess return
D. An increase in the asset's covariance with the bad-times index, for example from −0.0980 to −0.0210, implies a decrease in the asset's expected return

Answer: C

Step by step

1. A: λ(m) = −var(m)/E(m) is the price of bad-times risk. True.
2. B: with λ = −var/E(m) < 0, premium = βλ; moving β from −0.95 to −0.33 shrinks the product (e.g., with λ = −0.10: 9.5% to 3.3%). True.
3. D: cov increases means β increases (var(m) > 0), so the premium falls. True.

4. C: m is a discount factor, which is positive (about 0.97 here), not negative. The premium is negative × negative = positive for typical assets. False.

Why C is correct: The SDF is a positive discount factor; it is β(i,m) that is typically negative, and a negative β times a negative λ gives a positive risk premium.

Why the others are wrong

- **A:** True: this is the stated price of bad-times risk.
- **B:** True: a less negative β means a smaller product with the negative λ, so a lower premium.
- **D:** True: a higher covariance raises β and lowers the expected return.

LO 83.f

Describe efficient market theory and explain how markets can be inefficient.

Q42 BT answers P2.T9.20.2.3 · BT notes P2.T9.20.2.3

Richard Thaler notes that the efficient market hypothesis (EMH) asserts two things: "the price is right" (prices equal intrinsic value) and "there is no free lunch" (active management cannot add sustainable value, because higher expected returns only come from bearing more risk). According to Ang's factor theory, which of the following statements about market efficiency is TRUE?

A. Factor theory says that markets are efficient; that is, there are no deviations from efficiency and no enduring risk premiums.

B. The rational explanation for market inefficiency is that high returns compensate investors for exposure to idiosyncratic risk.

C. The behavioral explanation for market inefficiency is that high returns result from inefficient updating of beliefs and from over- or under-reaction to news.

D. Risk premiums should not persist, because rational inefficiencies will be arbitraged away and behavioral inefficiencies cannot, by definition, persist beyond a few days.

Answer: C

Step by step

1. Recall the two explanations for deviations from EMH: rational (losses in bad times are compensated by high returns) and behavioral (agents under- or over-react to news, or update beliefs inefficiently).
2. Statement C matches the behavioral explanation exactly.
3. Test A, B and D against the reading: factor theory accepts enduring risk premiums; the rational story is about systematic bad-times risk; behavioral biases can persist because of barriers to capital.

Why C is correct: The behavioral explanation attributes high returns to mistakes in processing news (under- or overreaction, poor belief updating), which can create mispricing that is not purely compensation for risk.

Why the others are wrong

- **A:** Factor theory does not claim perfect efficiency; it allows enduring risk premiums for bad-times exposure and frictions/biases. Mistake: equating EMH with factor theory.
- **B:** Rational explanation concerns compensation for systematic (bad-times, factor) risk, not idiosyncratic risk, which is diversifiable and unrewarded.
- **D:** Behavioral inefficiencies can persist for long periods because of structural and regulatory barriers to entry of capital; also rational premiums are compensation and are not arbitraged away.

Q43 Schweser Module Quiz 83.2 Q1

Market efficiency can be described with the efficient market hypothesis (EMH). Regarding the definition of EMH and the rational and behavioral explanations for this approach, the EMH suggests that:

A. speculative trading is costless.

B. active managers cannot generally beat the market.

C. under the behavioral explanation, losses during bad times are compensated for by high returns.

D. under the rational explanation, it is agents' under- or overreactions to news that generates high returns.

Answer: B

Step by step

1. The EMH implies speculative trading is costly and the average investor holding the market portfolio can beat active managers by saving transaction costs.
2. Hence active managers cannot generally beat the market: B.
3. Check C and D: the explanations are swapped. Rational = bad-times compensation; behavioral = reaction to news.

Why B is correct: Under EMH prices already reflect available information, so active management cannot generally add value after costs.

Why the others are wrong

- **A:** EMH implies speculative trading is costly (transaction costs and fees erode returns), not costless. Mistake: assuming trading is free.
- **C:** Bad-times compensation is the rational explanation, not the behavioral one (labels swapped).
- **D:** Under- or overreaction to news is the behavioral explanation, not the rational one (labels swapped).

Q44

AnalystPrep Q.2384

REBUILT — Original relied on a missing price chart; replaced the exhibit with a verbal description of an immediate, complete price adjustment and rebuilt options so exactly one is correct.

Country A has a well-developed financial market in which information is freely available to every participant. An oil exploration company announces the discovery of huge reserves. On the announcement day its stock price jumps immediately to a new level and then fluctuates randomly around that level, with no further drift in the following weeks. An investment management firm in Country A promises clients a return in excess of the market return. Which statement is most appropriate?

A. Consistent with market efficiency, the firm is unlikely to deliver sustainable risk-adjusted excess returns net of costs; trading on public news like this would not help.

B. The firm can earn excess returns by buying the stock after the announcement, since good news continues to be incorporated gradually.

C. Passive management should earn excess returns over the market because the market portfolio is mispriced after such announcements.

D. The price jump shows that information was not freely available, so active managers can systematically exploit the information gap.

Answer: A

Key correction: Source key (B) depended on an unavailable chart assumed to show a delayed price reaction; with an immediate full adjustment the efficient-market answer is A.

Step by step

1. The price adjusts fully and immediately to public news, then shows no drift.
2. This is the behaviour predicted by EMH: no predictable profit from trading on public information after the announcement.
3. So active managers cannot generally beat the market net of costs; a promise of excess return is unlikely to be sustainable without taking extra risk.

Why A is correct: Immediate, complete incorporation of freely available news leaves no exploitable mispricing, so sustained excess returns from active management are not expected.

Why the others are wrong

- **B:** Assumes a delayed (drifting) reaction, which contradicts the described instantaneous adjustment.
- **C:** Passive holders of the market do not earn excess returns relative to the market by definition; no mispricing exists here.
- **D:** A jump on release shows information was quickly impounded, not that it was unavailable; the stem says information is free.

Reading 84. Factors

Andrew Ang, Asset Management: A Systematic Approach to Factor Investing (New York, NY: Oxford University Press, 2014). Chapter 7.

LO	Learning objective	Qs
84.a	describe the process of value investing and explain why a value premium may exist.	3
84.b	explain how different macroeconomic risk factors, including economic growth, inflation, and volatility, affect asset returns and risk premiums.	13
84.c	assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk.	4
84.d	explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama- French model as an example.	13
84.e	compare value and momentum investment strategies, including their return and risk profiles.	6

LO 84.a

Describe the process of value investing and explain why a value premium may exist.

Q45 AnalystPrep Q.2396

All of the following are explanations for the so-called value premium, EXCEPT:

A. Value firms are less flexible and have more unproductive capital during bad times.
B. Investors have psychological biases such as mental accounting and loss aversion.
C. Investors underestimate the growth prospects of value stocks.
D. Value stocks generate higher returns during bad economic times.

Answer: D

Step by step

1. Classify each statement as a rational (risk-based) or behavioral explanation of the value premium.
2. A: value firms hold fixed capital that cannot be redeployed (high, asymmetric adjustment costs), so they are riskier in bad times - rational explanation.
3. B: loss aversion and mental accounting (Barberis and Huang) make investors view beaten-down value stocks as riskier and demand a higher return - behavioral explanation.
4. C: overextrapolation leads investors to overestimate growth stocks and underestimate value stocks' prospects, so value stocks are cheap - behavioral explanation.
5. D: value stocks perform poorly, not well, in bad times; the premium compensates for those losses. D is not an explanation, so it is the EXCEPT answer.

Why D is correct: The rational story says the value premium is compensation for losing money in bad times; a claim that value stocks do better in bad times contradicts it.

Why the others are wrong

- **A:** This is the rational explanation (inflexible capital, high adjustment costs); it is a valid reason, so it is not the exception.
- **B:** Loss aversion and mental accounting are a standard behavioral explanation; choosing it overlooks that the question asks for the invalid statement.
- **C:** Underestimating value firms' growth prospects is the overextrapolation/overreaction behavioral explanation; it is valid.

Q46 AnalystPrep Q.2397 · Schweser Module Quiz 84.1 Q1

A high book-to-market ratio is indicative of a:

A. small-cap stock.
B. large-cap stock.
C. value stock.
D. growth stock.

Answer: C

Step by step

1. Book-to-market = book value per share / market price per share.
2. A high ratio means the market price is low relative to book value, i.e. the stock is cheap relative to fundamentals.
3. Such stocks are classified as value stocks; growth stocks have low book-to-market ratios.
4. Size (small vs large cap) is based on market capitalization, a separate classification.

Why C is correct: Value stocks have high book-to-market ratios because their prices are low relative to book value.

Why the others are wrong

- **A:** Confuses book-to-market (valuation) with market capitalization (size); the ratio does not identify size.
- **B:** Same confusion as A; large-cap status depends on market capitalization, not on book-to-market.
- **D:** Reverses the relationship; growth stocks have high prices relative to book value and hence low book-to-market ratios.

Q47 BT answers P2.T9.20.3.1

According to Andrew Ang, is it reasonable for a value strategy investor to expect excess returns?

A. No, because value is not a factor.
B. No, because the historical value premium is not statistically significant.
C. Yes, but the value factor is only explained by behavioral biases.
D. Yes, and the value factor might be explained by either a rational story and/or behavioral biases.

Answer: D

Step by step

1. Historically, value stocks have outperformed growth stocks (a value-growth strategy turned $1 into over $6 between 1965 and about 2012), so value investing appears to work.
2. Value is a systematic risk factor, ruling out A and B.
3. Two explanations exist: rational (premium compensates for poor performance in bad times) and behavioral (overextrapolation, loss aversion, mental accounting).
4. Neither is exclusive, so C ('only behavioral') is too strong; D is correct.

Why D is correct: The value premium is historically robust and can be explained by rational risk compensation and/or behavioral biases.

Why the others are wrong

- **A:** Value is a recognized risk factor (HML in Fama-French); denying this contradicts the reading.
- **B:** The historical premium is large and persistent, despite episodes of poor performance such as the late 1990s and 2007-2009.
- **C:** Ignores the rational explanation (compensation for bad-times risk); 'only' makes the statement wrong.

LO 84.b

Explain how different macroeconomic risk factors, including economic growth, inflation, and volatility, affect asset returns and risk premiums.

Q48 AnalystPrep Q.2391

REBUILT — Original said investors 'must invest in' an asset class with an ambiguous country table and a zero-coupon-bond distractor; rewrote as a clear return-maximizing question with standard asset-class options.

Country	Economic growth	Inflation	Volatility
A	High	Low	Low
B	Low	High	High

Based on the historical relationship between macroeconomic factors and asset returns, an investor in Country A who wants to maximize expected return and can tolerate equity risk would be best served by an allocation to:

A. Government bonds
B. Equities
C. Investment-grade corporate bonds
D. High-yield bonds

Answer: B

Step by step

1. Growth: equities outperform bonds in expansions/high-growth periods (e.g. large stocks 12.4%, small stocks 16.8%); high-yield bond returns are about the same in recessions and expansions (7.4% vs 7.7%).
2. Inflation: all asset classes do better when inflation is low; stocks return 14.7% (large) and 17.6% (small) in low inflation versus bond yields of roughly 8.6%-9.2%.
3. Volatility: low volatility is favorable for risky assets (stock returns fall when the VIX rises), so Country A's regime supports risk-taking.
4. High growth, low inflation and low volatility all favor equities.

Why B is correct: High growth, low inflation and low volatility are the macro conditions in which risky equities historically earn their highest returns.

Why the others are wrong

- **A:** Government bonds do best in low-growth recessions (flight to quality); in a high-growth regime they lag equities. This choice applies the Country B logic.
- **C:** Investment-grade bonds also shine in recessions, not expansions, and earn about 8.8% in low inflation, well below equities.
- **D:** High-yield bonds are largely insensitive to growth (7.4% vs 7.7%) and return about 9.2% in low inflation, so they do not capture the equity upside.

Q49

AnalystPrep Q.2392

During a recession, which of the following categories of investments will most likely outperform the other categories listed?

A. Large-cap stocks
B. High-yield bonds
C. Small-cap stocks
D. Government bonds

Answer: D

Step by step

1. Slow growth hurts risky assets: equities earn less in recessions than in expansions (large 12.4% and small 16.8% in expansions).
2. High-yield returns are nearly unchanged across the cycle (7.4% in recessions vs 7.7% in expansions), so they do not benefit from the downturn.
3. Government bonds earn about 12.3% in recessions as investors flee to safety and rates fall, and are the best performer among the listed choices.

Why D is correct: In recessions there is a flight to quality and falling rates, which lifts government bond prices while risky equity and junk-bond returns lag.

Why the others are wrong

- **A:** Large-cap stocks are cyclical and perform poorly in low-growth periods; they earn far less than government bonds in recessions.
- **B:** High-yield bonds carry credit/equity-like risk and earn only about 7.4% in recessions; the mistake is assuming bonds as a whole behave alike.
- **C:** Small caps are the most growth-sensitive asset (best in expansions at 16.8%), so they are the worst choice during a recession.

Q50

AnalystPrep Q.2408

REBUILT — Original 'least concerned' item rested on an unsupported investment-horizon argument; rebuilt to test the Schweser point that productivity shocks drive firm output and stock returns (correlation about 50%).

Productivity shocks affect firm output, and the correlation between productivity shocks and stock returns is relatively high (about 50%). Stock prices fell during the productivity slowdown of the 1960s and 1970s. A portfolio manager expects a negative productivity shock. Which of the following holdings would be most exposed to this macroeconomic risk?

A. A diversified portfolio of equities
B. A portfolio of 3-month Treasury bills
C. Cash held in insured bank deposits
D. A portfolio of overnight government repurchase agreements

Answer: A

Step by step

1. Productivity shocks change firm output and profitability, so they flow directly into equity cash flows and prices.
2. Equity returns have a positive correlation of about 50% with productivity shocks; falling productivity (1960s-1970s) was accompanied by falling stock prices.
3. Short-dated, fixed-payoff instruments (T-bills, deposits, overnight repo) are not claims on firm output, so their value is largely insulated.
4. The diversified equity portfolio has the greatest exposure.

Why A is correct: Equities are claims on firms' productive output, so a negative productivity shock reduces expected cash flows and prices; short-term fixed-payoff instruments are not claims on output.

Why the others are wrong

- **B:** T-bills have fixed payoffs and a short maturity, so they have little sensitivity to firm productivity.
- **C:** Insured deposits pay a contractual amount and are unaffected by firm productivity.
- **D:** Overnight repo is collateralized, very short-term, fixed-return lending; it has essentially no equity-like productivity exposure.

Q51

BT Mock Q15 · BT answers P2.T9.20.3.3 · BT notes P2.T9.20.3.3

Volatility is an important risk factor according to Andrew Ang. The strong negative relationship between VIX changes and stock returns is explained by (at least) two dynamics. The first is the leverage effect. The second channel is a time-varying risk premium story, which is the one the basic CAPM advocates.

Assume the risk-free rate is 3.0% and the equity risk premium is 6.0%, so the market's expected return is 9.0%. According to the CAPM, and consistent with Ang's transmission channel between volatility and stock returns, which of the following should cause a DECLINE in a stock's price?

A. The VIX Index declines from 21.00 to 19.00

B. The market's volatility decreases, and the stock's volatility decreases along with it

C. The stock's volatility decreases, which corresponds to an increase in its leverage via the leverage effect

D. The stock's volatility increases (along with an increase in its correlation to the market), while the market's volatility is unchanged

Answer: D

Step by step

1. CAPM required return: $E(r_i) = r_f + \beta_i \times 6.0\%$, with $\beta_i = \rho_{i,M}\,\sigma_i/\sigma_M$.
2. Illustration for D: $\rho = 0.8, \sigma_i = 20\%, \sigma_M = 15\%$ gives $\beta = 1.07$ and required return 9.4%. If ρ rises to 0.9 and σ_i to 25% with σ_M fixed, $\beta = 1.50$ and required return 12.0%.
3. A higher required return means future cash flows are discounted more heavily, so the price falls today.
4. A: a lower VIX lowers the market risk premium ($\gamma\sigma_M^2$ falls), lowering required returns and raising prices. B: if σ_i and σ_M fall proportionally, β is unchanged (e.g. 1.07 either way) while the market premium falls, so the price rises. C: lower stock volatility with σ_M fixed lowers β (e.g. 0.85 and required return 8.1% for $\sigma_i = 16\%$, $\rho = 0.8$), so the price rises.

Why D is correct: With market volatility fixed, higher stock volatility and correlation raise β and the required return, forcing the price down today so that higher future returns compensate for the extra risk.

Why the others are wrong

- **A:** A falling VIX signals lower market risk and a lower risk premium, so discount rates fall and prices rise (the opposite sign).
- **B:** Beta depends on the ratio σ_i/σ_M, which is unchanged when both fall together; the lower market variance reduces the risk premium, so the price rises.
- **C:** Lower stock volatility with unchanged market volatility lowers beta and the required return, raising the price; the leverage-effect wording is also backwards, since higher leverage raises, not lowers, volatility.

Q52

BT answers P2.T8.702.1 · BT notes P2.T9.702.1 · BT Mock Q25

REBUILT — Mock Q25 inverted the BT 'EXCEPT' item to 'TRUE' with several true options; restored the single-answer EXCEPT format and replaced the 'all five asset classes' volatility claim with the Schweser-supported statement.

Andrew Ang identifies growth, inflation and volatility as the three most important macro factors. His historical evidence covers 1952:Q1 to 2011:Q4 for large and small stocks and government, investment-grade and high-yield bonds. In regard to this analysis, each of the following statements is true EXCEPT which is false?

A. Government and investment-grade bonds performed better during economic recessions than during expan-

sions.

B. Both large and small stocks performed significantly better during economic expansions than during recessions.

C. During periods of high inflation, all of the asset classes performed significantly better than during periods of low inflation.

D. Stock returns and government bond returns were more volatile during recessions than during expansions.

Answer: C

Key correction: BT Mock Q25 (TRUE version) said A; but B and D (as originally worded) are also true, so that version had multiple correct options. In the EXCEPT version the only false statement is C.

Step by step

1. A true: recession returns of 12.3% (government) and 12.6% (investment grade) exceed their expansion returns and equity returns in recessions.
2. B true: in expansions large stocks return 12.4% and small stocks 16.8%, well above their recession returns.
3. C false: high inflation hurts everything. Large/small stocks earn 14.7%/17.6% in low inflation versus 8.0%/13.0% in high inflation, and bonds earn about 3 percentage points less.
4. D true: large-stock volatility is 23.7% in recessions versus 14.0% in expansions; government bond volatility is 15.5% versus 9.3%.

Why C is correct: Inflation is a negative factor for every asset class studied; returns are higher in low-inflation than in high-inflation periods, so the claim of better performance in high inflation is false.

Why the others are wrong

- **A:** This is a true statement (flight to quality in recessions); choosing it means misreading the EXCEPT format.
- **B:** True: equities are the growth-sensitive assets; choosing it reflects thinking bonds also beat stocks in expansions.
- **D:** True: volatility rises for both stocks and bonds in downturns; choosing it reflects the belief that safe assets are not more volatile in recessions.

Q53

BT answers P2.T8.702.2 · BT notes P2.T9.702.2

REBUILT — Option B relied on claims outside the Schweser reading (price of risk across all asset classes) and D was garbled; rebuilt options to match the Schweser treatment of volatility while keeping the EXCEPT format.

Volatility is one of the three most important macroeconomic factors. Each of the following statements about volatility as a macro factor is true EXCEPT which is false?

A. The correlation between the VIX and stock returns has historically been negative.

B. When volatility rises, companies' financial leverage rises because debt stays roughly constant while the market value of equity falls.

C. An increase in volatility is associated with higher contemporaneous stock returns because investors are rewarded immediately for bearing more risk.

D. Higher volatility raises required rates of return on equities, which pushes current stock prices down.

Answer: C

Step by step

1. Recall the reading: the VIX-stock return correlation is negative (about -0.39, 1986-2011), so stock returns tend to fall when volatility rises.
2. Two channels explain this: (1) the leverage effect, in which a fall in equity value raises financial leverage and makes equity riskier; (2) higher volatility raises discount rates, lowering prices today (and implying higher future returns).
3. A, B and D restate these facts. C claims a positive contemporaneous relation, which contradicts both channels, so C is false.

Why C is correct: Rising volatility coincides with falling stock prices (leverage effect and higher discount rates), so any claim of higher contemporaneous returns is false.

Why the others are wrong

- **A:** True: the VIX-return correlation is negative (-0.39); choosing it means misreading the sign.
- **B:** True: debt is roughly unchanged while equity value drops, so leverage increases; this is the leverage effect.
- **D:** True: the second channel is that higher volatility raises required returns and so depresses current prices.

Q54 BT answers P2.T8.702.3 · BT notes P2.T9.702.3

REBUILT — Option D (inflation neutral in real business cycle models) is not in the Schweser reading and made the item unfair; replaced with the reading's productivity-shock statement and reworded 'NOT necessarily true' to 'FALSE'.

In addition to economic growth, inflation and volatility, other macroeconomic factors include productivity risk, demographic risk and political risk. Which of the following statements about these factors is FALSE?

A. As the population ages, retirees buy more financial assets, and this extra demand pushes asset prices up.

B. Risk aversion increases with age, so as the average age of the population rises, the equity risk premium should rise and equity prices fall.

C. Political risk, once thought important only in emerging markets, was shown by the 2007-2009 crisis to raise risk premiums in developed countries too.

D. Falling productivity has coincided with falling stock prices, while improving productivity has coincided with rising stock prices.

Answer: A

Step by step

1. Demographic risk: in overlapping-generations models, retirees withdraw savings and liquidate assets to fund consumption.
2. More liquidation with relatively fewer young and middle-aged buyers puts downward, not upward, pressure on prices, so A is false.
3. B (risk aversion rises with age, raising the equity risk premium), C (political risk matters in developed markets after 2007-2009) and D (productivity shocks and stock prices move together) match the reading.

Why A is correct: Retirees are net sellers of financial assets, so aging tends to depress, not raise, asset prices.

Why the others are wrong

- **B:** True: higher average age implies higher risk aversion and a higher equity risk premium.
- **C:** True: the crisis showed political (sovereign) risk is not confined to emerging markets.
- **D:** True: productivity shocks are positively related to stock prices (weak 1960s-70s, strong 1980s-90s).

Q55 BT answers P2.T9.20.3.2 · BT notes P2.T9.20.3.2

For five asset classes (large stocks, small stocks, government bonds, investment grade corporate bonds and high-yield corporate bonds), Ang reported mean returns and volatilities conditioned on macroeconomic factors. According to this historical analysis (1952-2011), which of the following findings is TRUE?

A. High inflation is good for bonds.

B. Because asset prices respond to factors with a delay of two or three quarters, the level of the factor is what matters most.

C. Compared with large stocks, small stocks had higher returns and greater volatility over the sample period.

D. Low economic growth (recession) is bad for all asset classes, including government and corporate bonds.

Answer: C

Step by step

1. Inflation: all asset classes do better in low-inflation than high-inflation periods, so A is false.
2. Macro factor effects come from shocks (unanticipated changes), not levels, so B is false.
3. Recessions: government and investment grade bonds outperform (about 12.3% and 12.6%) while equities and high-yield lag, so D is false.
4. Small stocks earned higher mean returns than large stocks (e.g., 16.8% vs 12.4% in expansions, 17.6% vs 14.7% in low inflation) and are riskier, so C is true.

Why C is correct: Small stocks carry a size premium and are riskier: higher mean return and higher volatility than large stocks.

Why the others are wrong

- **A:** Reverses the inflation result: high inflation lowers real bond returns and bonds do worse.
- **B:** Confuses levels with shocks; it is unanticipated changes that move prices.
- **D:** Over-generalizes: government and investment grade bonds perform best in recessions.

Q56

BT answers P2.T9.20.4.3

REBUILT — Original options were ambiguous (volatility and expected returns 'increase or decrease', with the time-varying option not supported by the Schweser text); rebuilt around the leverage-effect and VIX evidence.

Which of the following statements about equity volatility as a macroeconomic risk factor is correct?

A. The VIX and stock returns are positively correlated, so stock prices tend to rise when volatility rises.

B. Higher volatility lowers stock returns through the leverage effect and through higher required rates of return that push down current prices.

C. Volatility is only a property of an individual asset and is not a risk factor affecting asset returns.

D. The leverage effect arises because debt falls sharply when volatility rises, which makes equities safer.

Answer: B

Key correction: Source key (D) rested on a time-varying risk-aversion claim not supported in the Schweser reading; the rebuilt item has answer B.

Step by step

1. The VIX-stock return correlation is negative (-0.39), so A has the wrong sign.
2. Path 1 (leverage effect): a rise in volatility is linked to falling equity value, hence higher financial leverage and riskier equity.
3. Path 2: higher volatility raises discount rates, so current prices fall (and future returns can be higher). Both paths give lower current returns, so B is correct.
4. Volatility is an important risk factor (C false), and leverage rises because equity falls while debt is about constant (D false).

Why B is correct: Both the leverage effect and higher required returns produce lower stock prices when volatility rises, giving a negative relation.

Why the others are wrong

- **A:** Gets the correlation sign wrong; it is negative.
- **C:** Ignores that volatility is one of the three key macro factors.
- **D:** Reverses the mechanism: debt stays about constant while equity value falls, so leverage increases and equity becomes riskier.

Q57

GARP PE2 Q31 · Mock 4 Q32

A group of newly hired analysts at a large pension fund are discussing how changes in different macroeconomic risk factors are likely to affect the fund's fixed-income and equity positions. The analysts consider economic growth, inflation and volatility. Which of the following observations is correct?

A. Government bonds tend to generate lower returns during recessionary periods than during expansionary periods.

B. Corporate bonds tend to be less volatile during periods of high real GDP growth than during periods of low real GDP growth.

C. Large-capitalization stocks tend to generate higher returns during periods of high inflation than during periods of low inflation.

D. Small-capitalization stocks tend to be more volatile during periods of high consumption growth than during periods of low consumption growth.

Answer: B

Step by step

1. Both stocks and bonds are more volatile in downturns and low-growth periods, so corporate bond volatility is lower when real GDP growth is high: B is true.
2. Government bonds perform best in recessions (flight to quality), so A is false.
3. High inflation is bad for stocks (large stocks 14.7% in low inflation vs 8.0% in high inflation), so C is false.
4. Equity volatility is higher in low-growth periods, so D is reversed.

Why B is correct: Volatility rises in downturns and low-growth periods for both stocks and bonds, so it is lower when growth is high.

Why the others are wrong

- **A:** Reverses the result: government bonds do better in recessions.
- **C:** Reverses the inflation effect: high inflation lowers stock returns.
- **D:** Reverses the growth effect: volatility is higher when consumption growth is low.

Q58 Schweser Module Quiz 84.1 Q2

Which of the following asset classes has approximately the same returns in high economic growth periods and low economic growth periods?

A. Small-cap stocks.
B. Large-cap stocks.
C. Government bonds.
D. High-yield bonds.

Answer: D

Step by step

1. Compare returns in recessions versus expansions (NBER data 1952-2011).
2. Equities do much better in expansions (large stocks 12.4%, small stocks 16.8% in expansions) and government bonds do better in recessions (12.3%).
3. High-yield bonds earned 7.4% in recessions and 7.7% in expansions, essentially unchanged.
4. Therefore high-yield bonds are the asset class indifferent to growth.

Why D is correct: High-yield bonds combine credit (equity-like) risk with bond cash flows, so their returns show almost no sensitivity to the growth cycle.

Why the others are wrong

- **A:** Small caps are the most growth-sensitive (strongly outperform in expansions); chosen by treating all risky assets alike.
- **B:** Large-cap stocks also do far better in expansions than recessions; confuses lower sensitivity than small caps with no sensitivity.
- **C:** Government bonds outperform in recessions (flight to quality) and lag in expansions; chosen by assuming safe assets are insensitive to growth.

Q59 AnalystPrep Q.2390

REBUILT — Original keyed a 150bp cut as the only shock although 95bp and 105bp are also (small) surprises, giving ambiguous options; rebuilt so the options differ clearly in size of surprise and level is a plausible trap.

Market participants expect the central bank to cut its policy rate by 100 basis points at its next meeting. Which of the following actual announcements would represent the largest shock to this macroeconomic factor?

A. A cut of 100 basis points.
B. A cut of 60 basis points.
C. A cut of 130 basis points.
D. A cut of 125 basis points.

Answer: B

Key correction: Source key (AnalystPrep) said a 150bp cut is the shock; that option is replaced because the 95bp and 105bp options were also nonzero surprises, so the original was not cleanly single-answer.

Step by step

1. A shock is the unanticipated change: actual minus expected, not the level of the change.
2. Surprise = |actual cut - 100bp|: A = 0; B = |60-100| = 40; C = |130-100| = 30; D = |125-100| = 25.
3. The largest surprise is 40bp, option B (a smaller-than-expected cut).

Why B is correct: Factor shocks, not factor levels, drive asset price moves; B deviates most from the expected 100bp cut.

Why the others are wrong

- **A:** Exactly as expected, so the shock is zero (already priced in); confuses a large level of cut with a shock.
- **C:** Surprise is 30bp, not the largest; chosen by picking the largest cut (level) rather than the largest deviation from expectation.
- **D:** Surprise is only 25bp; again a level-based choice that ignores expectations.

Q60 AnalystPrep Q.2405

REBUILT — Original asked for 'major risk factors for emerging market equities', which Reading 84 does not cover and had a vague all-of-the-above key; rebuilt to test the reading's treatment of productivity, demographic and political risk.

Which of the following statements about macroeconomic risk factors other than growth, inflation and volatility is most accurate?

A. Positive productivity shocks, such as the computer revolution of the 1980s and 1990s, were generally accompanied by rising stock prices.
B. Political (sovereign) risk lowers risk premiums because it is diversifiable across countries.
C. Political risk matters only in emerging markets; the 2007-2009 crisis showed developed markets are unaffected.
D. In overlapping generations models, demographic shocks affect asset returns only through inflation, not through labor output.

Answer: A
Step by step
1. Productivity shocks affect firm output; falling productivity (1960s-70s) coincided with falling stock prices and improving productivity (1980s-90s) with rising prices; correlation with stock returns is about 50%. A is correct.
2. Political risk increases risk premiums and, as 2007-2009 showed, affects developed as well as emerging markets, so B and C are false.
3. Demographic risk is a shock to labor output and firm production (OLG models), so D is false.

Why A is correct: Productivity is a shock to firm output, so positive shocks raise cash flows and stock prices.
Why the others are wrong
- **B:** Political risk increases (not lowers) risk premiums; confuses it with diversifiable risk.
- **C:** Reading states political risk, once thought important only in emerging markets, now affects developed markets too.
- **D:** Demographic risk is interpreted as a shock to labor output and to the composition of savers and dissavers, not an inflation-only effect.

LO 84.c

Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk.

Q61 BT notes P2.T9.703.1

REBUILT — Original relied on Ang's rebalancing-as-short-volatility point, which is outside the Schweser reading, and had a garbled formula; rebuilt around the volatility premium and the risk-premium/volatility relationship covered in LO 84.c.

Ang relates the expected market risk premium to market variance as $E(r_M) - r_f = \gamma\sigma_M^2$, where γ is the risk aversion of the average investor. Regarding the volatility risk premium and strategies that sell volatility, which of the following statements is TRUE?

A. In theory γ is negative, but empirical estimates are always positive.
B. Selling out-of-the-money puts tends to earn steady premiums in normal periods but exposes the seller to large losses in a crash, so only investors who can withstand such losses should sell volatility.
C. Realized volatility is on average higher than VIX-implied volatility, so options are cheap and buying volatility protection earns a positive premium.
D. Selling volatility protection is ultimately a low-risk strategy because volatility mean-reverts over the long run.

Answer: B
Step by step
1. Theory: γ is positive (risk-averse investors demand a higher premium for more variance); empirical estimates are positive, negative or zero. A is therefore wrong on both halves.
2. Realized volatility is on average about 2%-3% below VIX implied volatility, so options are expensive on average and the seller of protection collects the volatility premium. C reverses this.
3. Selling volatility (e.g. out-of-the-money puts) pays steady premiums in normal times, but in a crash such as the GFC sellers lost nearly 70% (Sep-Nov 2008) and return skewness fell from -0.37 to -8.26 once the crisis is included.
4. Hence selling volatility resembles selling insurance: high-probability small gains, low-probability huge losses. B is the only true statement.

Why B is correct: The volatility premium is a payment for bearing crash (tail) risk; sellers earn it in calm periods but suffer large negative returns in crises, so the strategy is far from low-risk.
Why the others are wrong

- **A:** Reverses theory and data: γ is positive in theory, and estimates in the data are mixed (positive, negative or zero), not always positive.
- **C:** Reverses the relationship: realized volatility is on average lower than implied, so options are expensive and it is the seller who collects the premium (the premium on volatility is negative for the buyer).
- **D:** Ignores negative skewness and crash risk; mean reversion does not protect the seller from the large losses incurred during a crisis.

Q62

GARP PE2 Q77 · Mock 4 Q78

A risk analyst at a bank is asked to prepare a report that tracks the relationship between volatility and asset performance. The analyst assesses the performance of various asset classes using empirical evidence over the last three decades and compares returns on those asset classes with changes in market volatility. Which of the following would be a correct statement for the analyst to include in the report?

A. Currency strategies such as currency carry trades tend to perform poorly during periods of high volatility.
B. When volatility is rising, all assets are either positively or negatively affected, with the exception of risk-free bonds.
C. Whether the relationship between stock returns and volatility is positive or negative depends on the phase of the business cycle.
D. When volatility is rising, stock returns tend to increase but bond returns tend to decrease.

Answer: A

Step by step

1. Recall the empirical evidence on volatility as a risk factor: equity returns are negatively correlated with the VIX (about -0.39 over 1986-2011), explained by the leverage effect and higher discount rates.
2. Currency strategies, including carry (long high-yield, short low-yield currencies), have large exposure to volatility risk and perform poorly when volatility is high.
3. Test each option against this evidence: only A is consistent with it.

Why A is correct: Carry-type currency strategies earn steady returns in calm markets but suffer sharp losses when volatility spikes, so they are effectively short volatility.

Why the others are wrong

- **B:** An unsupported universal claim; the evidence does not say every asset is affected (and says nothing special about risk-free bonds), so it is not a documented finding.
- **C:** The stock-volatility relationship is documented as negative (leverage effect, higher required returns), not sign-switching with the business cycle; confuses volatility with growth effects.
- **D:** Reverses the stock result: rising volatility coincides with falling stock prices (correlation about -0.39); bonds are less affected, not systematically hurt.

Q63

BT answers P2.T8.703.1

REBUILT — Original statement set relied on a garbled formula and on claims (rebalancing as short volatility, derivatives volatility trading being non-directional) not supported by the reading; rewritten around the reading's volatility-premium material with one defensible answer.

In the relationship $E(r_m) - r_f = \gamma \times \sigma_m^2$, γ is the risk aversion of the average investor and σ_m^2 is the variance of the market return. Which of the following statements about volatility risk and premiums is most accurate?

A. The coefficient γ must be negative in theory because investors dislike volatility, so a positive estimate in the data indicates a mis-specified model.
B. Because realized volatility is on average 2%-3% below VIX-implied volatility, selling volatility protection earns a premium, but the seller is exposed to large losses in a crash.
C. Selling out-of-the-money put options is a low-risk strategy because the payoffs are stable over long samples that include the 2007-2009 crisis.
D. Volatility carries a positive premium, so an investor earns it simply by holding a long position in equities.

Answer: B

Key correction: Source key (BT) said C (rebalancing is a short-volatility strategy earning the volatility risk premium); that statement is not supported by the Schweser reading, so the item was rebuilt around the reading's content.

Step by step

1. Theory: γ is positive in theory (higher risk aversion means a higher required premium per unit of variance); empirical estimates have been positive, negative or zero.

2. Volatility has a negative premium: options are expensive, since realized volatility is about 2%-3% lower than VIX-implied volatility.
3. So to collect the volatility premium one must sell volatility protection (e.g., sell OTM puts).
4. But the payoff is like selling insurance: steady gains in normal times and large losses in a crash (losses of nearly 70% in Sep-Nov 2008; skewness -8.26 with the GFC vs -0.37 without).
5. Only B states both the premium and the crash risk correctly.

Why B is correct: Implied volatility exceeds realized volatility on average, so volatility sellers earn a premium, but this is compensation for severe negative skewness and crash losses.

Why the others are wrong

- **A:** γ is positive in theory; the empirical estimates (not the theory) vary in sign. This reverses theory and data.
- **C:** Ignores the crisis: sellers suffered losses near 70% in 2008 and negative skewness of -8.26 over the full sample; stable payoffs appear only if the crisis is excluded.
- **D:** Confuses asset premiums with the volatility premium; assets have positive premiums, but volatility has a negative premium, so it is collected by shorting volatility, not by being long equities.

Q64

Schweser Module Quiz 84.2 Q1

Which of the following investment options provides a means of mitigating volatility risk?

A. Buying put options.
B. Selling put options.
C. Buying equities.
D. Buying call options.

Answer: A

Step by step

1. Volatility risk can be mitigated by holding less volatile assets (e.g., bonds) or by buying volatility protection in derivatives markets.
2. Out-of-the-money puts pay off in periods of high volatility, so buying them hedges volatility risk.
3. Hence buying puts is the correct choice.

Why A is correct: Long put options gain when markets fall and volatility spikes, offsetting portfolio losses and providing volatility protection.

Why the others are wrong

- **B:** Selling puts is short volatility: it collects the volatility premium but increases exposure to crashes.
- **C:** Equities have large exposure to volatility risk; buying them adds, not reduces, it.
- **D:** Calls benefit from rising prices and are not the hedge for volatility-driven declines; the reading names puts as the protection.

LO 84.d

Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama- French model as an example.

Q65

BT answers P2.T8.703.2 · BT notes P2.T9.703.2

REBUILT — Original options relied on outside claims about the size effect disappearing and had an internally contradictory option; rebuilt to test the HML/SMB beta interpretation in the Schweser reading.

The Fama-French three-factor model is $E(r_i) = r_f + \beta_{i,MKT}E(r_m - r_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML)$. Which of the following statements about the model is TRUE?

A. A value investor has a positive HML beta, so the expected return is adjusted upward relative to the CAPM expected return.
B. Growth stocks have positive HML betas, so their expected returns are adjusted upward relative to the CAPM.
C. The average stock has an SMB beta of 1 and an HML beta of 1, as the average stock has a market beta of 1.
D. Value stocks are expected to outperform growth stocks especially during bad times for the economy, which is why they earn a premium.

Answer: A

Step by step

1. HML = returns on high book-to-market (value) stocks minus low book-to-market (growth) stocks; E(HML) > 0 historically.
2. A value investor is long value, so $\beta_{HML} > 0$ and the expected return rises by $\beta_{HML}E(HML)$ above CAPM.
3. Growth stocks have negative HML betas, so their expected returns are adjusted down (B false).
4. SMB and HML betas are centered on zero for the average stock (only the market beta averages 1), so C is false.
5. Value stocks perform poorly in bad times; the premium compensates for that risk (D false).

Why A is correct: A positive HML beta for value stocks adds $\beta_{HML}E(HML)$ to the CAPM expected return, raising the risk premium.

Why the others are wrong

- **B:** Reverses the sign: growth stocks have negative HML betas, lowering their expected returns.
- **C:** Applies the CAPM average-beta-of-one result to SMB and HML; those betas are centered on zero, so the average investor has no size or value tilt.
- **D:** Reverses the rational explanation; value stocks underperform in bad times, and the premium is compensation for bearing that risk.

Q66

AnalystPrep Q.4551

REBUILT — Original left the carry-return definition implicit and key mixed rounded figures; stem now defines the FX carry factor return and the beta relationship, and distractors are tied to specific mistakes.

A US portfolio manager considers an FX carry trade modeled as a one-factor dynamic model: the expected carry return is $\beta_{i,FX} \times$ (factor return), where the factor return is the USD return on the carry trade minus the US interest rate. She converts USD 100 million into the emerging-market currency (SER) at the start, invests for one year at the SER rate of 7%, and converts back at the end-of-year spot rate. The US interest rate is 2%.

Variable	Value
Exchange rate at T0	SER 2.050/USD
Exchange rate at T1	SER 2.071/USD
Î²(i,FX)	1.45

What is the carry return attributable to the FX factor for the US manager?

A. 3.92%

B. 5.92%

C. 5.68%

D. 8.58%

Answer: C

Step by step

1. Convert: $100 \times 2.050 =$ SER 205.0 million.
2. Interest at 7%: $205.0 \times 1.07 =$ SER 219.35 million.
3. Convert back: $219.35/2.071 = 105.915$ USD million, so USD return = 5.915%.
4. Factor return = $5.915\% - 2\% = 3.915\%$.
5. Carry return = $1.45 \times 3.915\% = 5.68\%$.

Why C is correct: The factor return is the excess of the USD return on the trade over the US rate (3.915%), and exposure to the factor scales it by the beta of 1.45.

Why the others are wrong

- **A:** Reports the factor return (3.92%) without multiplying by the beta of 1.45.
- **B:** Reports the gross USD return of the trade (5.92%); neither subtracts the 2% US rate nor applies the beta.
- **D:** Applies the beta to the gross USD return (1.45 x 5.915%) and forgets to subtract the 2% US rate.

Q67

AnalystPrep Q.2395

REBUILT — Option B said 'small-beta' (garbled); reworded as low-beta and made the stem self-contained.

All of the following are factors in the Fama-French three-factor model, EXCEPT:

A. The market factor.

B. The difference in returns of high-beta stocks minus low-beta stocks.

C. The difference in returns of a portfolio of high book-to-market stocks minus a portfolio of low book-to-market stocks.

D. The difference in returns of small-capitalization stocks minus large-capitalization stocks.

Answer: B

Step by step

1. Fama-French: $E(r_i) = r_f + \beta_{i,MKT}E(r_m - r_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML)$.
2. Factors are MKT (market), SMB (small minus big) and HML (high minus low book-to-market).
3. A high-beta minus low-beta spread is not one of them.

Why B is correct: The model has exactly three factors: market, size (SMB) and value (HML); a beta-sorted spread is not among them.

Why the others are wrong

- **A:** Market excess return is the first FF factor, so it is included.
- **C:** This is HML, the value factor, so it is included.
- **D:** This is SMB, the size factor, so it is included.

Q68

AnalystPrep Q.2424

A fund's factsheet states that it invests in large value stocks. A prospective investor regresses the fund's excess return on the Fama-French factors and obtains:

$r_{fund} - r_f = 0.82\% + 0.74(r_{mkt} - r_f) + 0.32\,SMB + 0.54\,HML + \varepsilon$

Which statement about the fund's style exposure is correct?

A. The fund has more exposure towards value stocks.
B. The fund has more exposure towards growth stocks.
C. The fund has equal exposure towards value and growth stocks.
D. The fund is concentrated in growth stocks.

Answer: A

Step by step

1. HML = returns of high book-to-market (value) minus low book-to-market (growth) stocks.
2. The loading on HML is +0.54, so fund returns move with value minus growth: a value tilt.
3. Equal exposure would require an HML loading near zero; a growth tilt would require a negative loading.

Why A is correct: A positive HML beta means returns co-move with value stocks relative to growth stocks, i.e., a value tilt.

Why the others are wrong

- **B:** Reverses the sign interpretation of HML; growth stocks have negative HML betas.
- **C:** Would require an HML beta near zero, but it is 0.54.
- **D:** Also a growth conclusion; a growth tilt needs a negative HML loading.

Q69

AnalystPrep Q.2423

A fund's factsheet states that it invests in large value stocks. A prospective investor regresses the fund's excess return on the Fama-French factors and obtains:

$r_{fund} - r_f = 0.82\% + 0.74(r_{mkt} - r_f) + 0.32\,SMB + 0.54\,HML + \varepsilon$

Which statement about the fund's size exposure is correct?

A. The fund has more exposure towards large stocks.
B. The fund has more exposure towards small stocks.
C. The fund has equal exposure towards small and large stocks.
D. The fund has 100% exposure towards large stocks.

Answer: B

Step by step

1. SMB = returns on small stocks minus returns on big stocks.
2. The SMB loading is +0.32, so the fund moves with small minus big: a small-cap tilt.
3. This contradicts the factsheet claim of investing in large stocks.

Why B is correct: A positive SMB beta indicates a tilt toward small-capitalization stocks.

Why the others are wrong

- **A:** Takes the factsheet claim at face value or reads SMB as big minus small; a large-cap tilt needs a negative SMB beta.
- **C:** Equal exposure requires an SMB beta near zero, not 0.32.
- **D:** An SMB beta cannot imply 100% exposure, and its positive sign points to small, not large.

Q70 AnalystPrep Q.3019

In the Fama-French three-factor model, in addition to the market factor and the HML factor, what is the third factor and what does it capture?

A. The momentum factor, WML, which captures the outperformance of past winners relative to past losers.
B. The SMB factor, which captures the outperformance of high book-to-market stocks relative to low book-to-market stocks.
C. The SMB factor, which captures the outperformance of small firms relative to larger firms.
D. The momentum factor, WML, which captures the outperformance of high-growth stocks relative to low-growth stocks.

Answer: C

Step by step

1. The three factors are MKT, SMB (small minus big) and HML (high minus low book-to-market).
2. So the third factor is SMB, the return of small-cap stocks over large-cap stocks.

Why C is correct: SMB (size) is the third Fama-French factor and measures small-cap minus large-cap returns.

Why the others are wrong

- **A:** WML (momentum) is a real factor but is not one of the three Fama-French factors.
- **B:** Correct name, wrong definition: high-minus-low book-to-market describes HML, already in the model.
- **D:** Wrong factor and wrong definition; WML is winners minus losers, not growth versus low-growth.

Q71 AnalystPrep Q.4547

REBUILT — Original ('NOT an investment-style factor') had an arguable option because the market factor is also tradeable; reframed as macroeconomic vs. dynamic factor with one clear answer.

Which of the following is a macroeconomic (fundamental-based) risk factor rather than a tradeable dynamic (style) factor?

A. Size (SMB)
B. Economic growth
C. Value (HML)
D. Momentum (WML)

Answer: B

Step by step

1. Dynamic (style) factors are tradeable long-short portfolios: size (SMB), value (HML), momentum (WML).
2. Economic growth, inflation and similar macro variables are economy-wide factors that are not directly traded (volatility is the exception).
3. So economic growth is the macro factor.

Why B is correct: Economic growth is an economy-wide fundamental factor, while SMB, HML and WML are tradeable factor-mimicking portfolios.

Why the others are wrong

- **A:** SMB is a long small / short large tradeable portfolio, a dynamic factor.
- **C:** HML is a long value / short growth tradeable portfolio, a dynamic factor.
- **D:** WML is a long winners / short losers tradeable portfolio, a dynamic factor.

Q72 GARP PE1 Q9 · Mock 3 Q71

A risk analyst regresses thirty years of weekly portfolio returns on the three Fama-French factors and obtains:

Coefficient	Value
Alpha	0.10
Market coefficient	0.52
SMB coefficient	0.18
HML coefficient	-0.70

Assuming all coefficients are statistically significant, which statement is correct?

A. There is a positive correlation between portfolio return and the size factor, which indicates that the portfolio moves together with large-cap stocks.
B. There is a positive correlation between portfolio return and the value factor, which indicates that the portfolio moves together with growth stocks.

C. There is a negative correlation between portfolio return and the size factor, which indicates that the portfolio moves together with large-cap stocks.
D. There is a negative correlation between portfolio return and the value factor, which indicates that the portfolio moves together with growth stocks.

Answer: D
Step by step
1. SMB = small minus big; the loading is +0.18, so the portfolio moves with small-caps (positive relation to the size factor).
2. HML = high minus low book-to-market (value minus growth); the loading is -0.70, so the portfolio is negatively related to the value factor and moves with growth stocks.
3. Only D gives a correct sign and interpretation.

Why D is correct: A negative HML beta means the portfolio moves opposite to value-minus-growth, i.e., with growth stocks.
Why the others are wrong
- **A:** Sign is right (+0.18) but a positive SMB beta means small-cap, not large-cap, co-movement.
- **B:** A positive value-factor correlation is wrong; the HML beta is negative.
- **C:** The size correlation is positive (+0.18), not negative, and positive SMB means small-cap.

Q73 Schweser Module Quiz 84.2 Q2

Which of the following is not a factor in the Fama-French three-factor model?
A. The capital asset pricing model market risk factor.
B. The small capitalization minus big capitalization risk factor.
C. The winners minus losers risk factor.
D. The high book-to-market value minus low book-to-market value risk factor.

Answer: C
Step by step
1. The Fama-French three-factor model has MKT (market), SMB (size) and HML (value/growth).
2. WML (winners minus losers) is the momentum factor, which is added as a fourth factor in an extension of the model.
3. So the WML factor is the one that is not part of the original three-factor model.

Why C is correct: WML momentum was identified by Jegadeesh and Titman and is only an add-on to the Fama-French model, not one of its three factors.
Why the others are wrong
- **A:** MKT is the first of the three factors, so it is in the model.
- **B:** SMB captures the size effect and is the second factor.
- **D:** HML captures the value/growth effect and is the third factor.

Q74 AP Mock 3 Q29

REBUILT — Stem clarified (four-factor model, book-to-market wording) and options expressed as percentages; calculation unchanged.
Daniel Kevin is estimating returns on stock i using a four-factor model (market, size, value, momentum). The risk-free rate is 2.5%.

Factor	Loading	Portfolio returns	Value
MKT	1.10	Expected return on the market	7.5%
SMB	1.30	Small-cap stocks / large-cap stocks	11.0% / 4.4%
HML	0.90	High book-to-market stocks / low book-to-market stocks	13.0% / 5.5%
WML	1.20	Winner stocks / loser stocks	15.0% / 7.0%

Calculate the expected return of the stock considering the momentum effect.
A. 13.73%
B. 30.43%
C. 32.93%
D. 35.68%

Answer: C

Step by step

1. Factor premiums: MKT = 7.5% − 2.5% = 5.0%; SMB = 11.0% − 4.4% = 6.6%; HML = 13.0% − 5.5% = 7.5%; WML = 15.0% − 7.0% = 8.0%.
2. $E(R_i) = R_f + \beta_{MKT}(5.0\%) + \beta_{SMB}(6.6\%) + \beta_{HML}(7.5\%) + \beta_{WML}(8.0\%)$
3. = 2.5% + 1.10(5.0%) + 1.30(6.6%) + 0.90(7.5%) + 1.20(8.0%)
4. = 2.5 + 5.50 + 8.58 + 6.75 + 9.60 = 32.93%

Why C is correct: Each factor premium is the long-minus-short return, multiplied by its loading and added to the risk-free rate.

Why the others are wrong

- **A:** Subtracted the momentum term instead of adding it (2.5 + 5.5 + 8.58 + 6.75 − 9.6 = 13.73%).
- **B:** Forgot to add the risk-free rate (32.93 − 2.5 = 30.43%).
- **D:** Used the market return of 7.5% instead of the market premium of 5.0% for MKT (1.10 × 7.5 = 8.25, giving 35.68%).

Q75

AnalystPrep Q.4553

REBUILT — Original options C and D were implausible; replaced with distractors reflecting realistic errors (omitted rf, used market return as premium). Answer unchanged.

John Edwin, a financial risk manager, is forecasting stock returns using the Fama-French three-factor model. He gathers the following for stock i:

Variable	Value	Factor beta	Value
Expected return on the market	9%	β(i,MKT)	1.20
Value (HML) premium	3%	β(i,HML)	1.25
Size (SMB) premium	7%	β(i,SMB)	1.10

If the risk-free rate is 3%, what is the expected return of stock i?

A. 21.65%

B. 22.25%

C. 18.65%

D. 25.25%

Answer: A

Step by step

1. Market premium = 9% − 3% = 6%.
2. $E(R_i) = R_f + \beta_{MKT}(R_m - R_f) + \beta_{SMB}\,\text{SMB} + \beta_{HML}\,\text{HML}$
3. = 3% + 1.20(6%) + 1.10(7%) + 1.25(3%)
4. = 3 + 7.2 + 7.7 + 3.75 = 21.65%

Why A is correct: The market beta multiplies the market excess return, while size and value betas multiply their own premiums.

Why the others are wrong

- **B:** Swapped the premiums (SMB 3%, HML 7%): 3 + 7.2 + 3.3 + 8.75 = 22.25%.
- **C:** Omitted the risk-free rate: 7.2 + 7.7 + 3.75 = 18.65%.
- **D:** Used the full 9% market return instead of the 6% premium: 3 + 10.8 + 7.7 + 3.75 = 25.25%.

Q76

AnalystPrep Q.4555

REBUILT — Original option D ('cannot be determined') was a giveaway and other options were unexplained; replaced with distractors from realistic errors. Answer unchanged.

Mark Truman, FRM, is estimating the expected return of portfolio M using the Fama-French three-factor model. The risk-free rate is 4.5%, and other data are:

Factor	Premium	Factor beta
Size (SMB)	−3.30%	−1.10
Value (HML)	3.50%	1.77
Market risk premium	6.00%	1.30

Calculate the expected return of portfolio M.

A. 17.63%

B. 22.13%

C. 14.87%
D. 8.67%

Answer: B
Step by step
1. Market term: 1.30 × 6.00% = 7.80%.
2. Size term: (−1.10)(−3.30%) = +3.63%.
3. Value term: 1.77 × 3.50% = 6.195%.
4. E(R) = 4.5 + 7.80 + 3.63 + 6.195 = 22.125% ≈ 22.13%.

Why B is correct: A negative beta on a negative premium contributes a positive amount; all three factor terms are added to the risk-free rate.
Why the others are wrong
- **A:** Omitted the risk-free rate (7.80 + 3.63 + 6.195 = 17.625%).
- **C:** Mishandled the sign of the size term (−3.63 instead of +3.63): 4.5 + 7.8 − 3.63 + 6.195 = 14.87%.
- **D:** Dropped the value factor and mishandled the size sign: 4.5 + 7.8 − 3.63 = 8.67%.

Q77 AnalystPrep Q.3020

The size effect was documented by Banz in 1981. What does it imply?
A. After adjusting for beta, large stocks tended to outperform small stocks.
B. Although value outperforms over the long run, large stocks outperform small stocks during certain periods.
C. After adjusting for beta, small stocks tended to outperform large stocks.
D. All of the above are correct.

Answer: C
Step by step
1. Banz (1981) found that small stocks earn higher returns than large stocks even after adjusting for market beta.
2. This is the reverse of A; B is not the size effect; hence D is false.

Why C is correct: The size effect is the beta-adjusted outperformance of small stocks over large stocks (which later disappeared after the early 1980s).
Why the others are wrong
- **A:** Reverses the direction of the size effect.
- **B:** Mixes up the value premium with the size effect and does not describe Banz's finding.
- **D:** Only C is correct, so 'all of the above' fails.

LO 84.e

Compare value and momentum investment strategies, including their return and risk profiles.

Q78 AP Mock 3 Q13

A portfolio manager focused on buying stocks that have gone up in recent months and shorting stocks with the lowest returns during the same period is most likely pursuing which of these investment strategies?
A. Value investing.
B. Growth investing.
C. Momentum investing.
D. Income investing.

Answer: C
Step by step
1. The selection rule is based on past returns over a recent window, not on fundamentals or valuation.
2. Buying past winners and shorting past losers is the momentum (WML, winners minus losers) strategy.

Why C is correct: Momentum (trend) investing buys stocks that have risen over a recent period (about six months) and shorts those that have fallen over the same period.
Why the others are wrong
- **A:** Value investing sorts on valuation (high book-to-market), not on past price performance.

- **B:** Growth investing favors low book-to-market stocks with high expected growth; it is not defined by recent price trends.
- **D:** Income investing targets dividend or coupon yield, not past returns.

Q79 AnalystPrep Q.2399

The Fama-French model with a momentum factor ($\beta_{i,WML}$) added is:
$E(r_i) = r_f + \beta_{i,MKT} E(r_m - r_f) + \beta_{i,SMB} E(SMB) + \beta_{i,HML} E(HML) + \beta_{i,WML} E(WML)$
A fund manager buys stocks with increasing prices and sells stocks with decreasing prices. For stocks with increasing prices, $\beta_{i,WML}$ will be:

A. Positive
B. Negative
C. Zero
D. Equal to 1

Answer: A

Step by step

1. WML = return on past winners minus return on past losers.
2. A stock with rising prices is a past winner, so its returns move in the same direction as WML.
3. Positive co-movement with the factor means a positive loading.

Why A is correct: Past winners are the long leg of WML, so their returns are positively related to the factor and $\beta_{i,WML} > 0$; past losers would load negatively.

Why the others are wrong

- **B:** A negative loading describes past losers (the short leg of WML), not rising-price stocks.
- **C:** Zero would mean no relation to the momentum factor, which contradicts winners being the long leg.
- **D:** A beta is a regression sensitivity, not a fixed value of 1; nothing implies the WML loading equals 1.

Q80 AnalystPrep Q.4548

Which of the following statements is most likely correct about momentum investing?

A. It involves going long on stocks that have low prices and shorting stocks that have high prices.
B. It involves going long on stocks that have upward-trending returns over a specific period and short on stocks with downward-trending returns over the same period.
C. It involves going long on small-cap stocks and shorting large-cap stocks.
D. None of the above.

Answer: B

Step by step

1. Momentum sorts on past returns over a lookback window (e.g., six months).
2. Long past winners, short past losers (WML).
3. Choice A is a price-level tilt (value-like) and C describes the size factor (SMB), so only B is correct.

Why B is correct: Momentum strategies buy stocks whose prices have trended up and short those that have trended down over the same period.

Why the others are wrong

- **A:** Confuses momentum with a value (buy cheap, sell expensive) approach; low price level is not the same as poor past return.
- **C:** Describes the size factor SMB (small minus big), not momentum.
- **D:** Incorrect because B is a correct statement.

Q81 BT Mock Q70 · BT answers P2.T8.703.3 · BT notes P2.T9.703.3

Adding a momentum factor, WML (winners minus losers) or UMD (up minus down), to the Fama-French model produces a four-factor model. Over the period January 1965 to December 2011, which of the following is TRUE about the momentum factor?

A. Momentum is a negative feedback strategy which is inherently stabilizing.
B. The momentum factor is observed in equities but is NOT observed in bonds, commodities, and real estate.
C. Momentum investing by definition is an anti-value strategy; correlations between HML and WML are strongly

negative.

D. The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on either size or value.

Answer: D

Step by step

1. Check each statement against the reading.
2. A: momentum is positive feedback and destabilizing, so false.
3. B: a momentum premium appears in bonds, international equities, commodities and real estate, so false.
4. C: the HML-WML correlation was only about −0.16, so false.
5. D: one dollar in WML reached above 60 dollars, far more than SMB or HML, so true.

Why D is correct: Momentum returns exceed size and value premiums by a wide margin over 1965-2011 (the WML dollar peaked above 60 dollars), though with crash risk.

Why the others are wrong

- **A:** Reverses the mechanism: value is negative-feedback and stabilizing; momentum is positive-feedback and destabilizing.
- **B:** Momentum is observed across asset classes (fixed income, international equities, commodities, real estate), not just equities.
- **C:** The HML-WML correlation was only about −0.16, weak, so value returns are not simply the opposite of momentum returns.

Q82

BT answers P2.T9.20.4.2 · BT notes P2.T9.20.4.2

Ang analyzed the historical performance of several macroeconomic (fundamental) factors and style (investment) factors. Style factors include value, momentum, and size. According to Ang's analysis, which of the following is TRUE about style factors?

A. Style factors do not persist because eventually, investors discover the factor.

B. Investing in momentum is the most profitable strategy among all style factors but momentum strategies exhibit negative skew.

C. The opposite of a momentum strategy is portfolio rebalancing; i.e., an investor who rebalances cannot invest in the momentum factor by definition.

D. The opposite of a momentum strategy is a value strategy; i.e., a factor model with both value and momentum contradicts itself or at least suffers from perfect multicollinearity.

Answer: B

Step by step

1. Momentum returns exceed size and value by a wide margin, but the strategy is prone to crashes (negative skew).
2. A: only the size premium disappeared after publication; value has persisted for over 50 years.
3. C: a momentum investor should still rebalance.
4. D: value and momentum are opposite only in feedback behavior; correlation was about −0.16, so no perfect multicollinearity.

Why B is correct: Momentum has the highest premium of the style factors but is exposed to large crashes, giving negative skewness.

Why the others are wrong

- **A:** Overgeneralizes the disappearing size premium; the value premium has persisted for more than 50 years.
- **C:** Rebalancing is compatible with momentum; the reading says a momentum investor should still rebalance.
- **D:** Value and momentum are opposite only in the sense of stabilizing versus destabilizing; their correlation is only about −0.16, not −1.

Q83

Schweser Module Quiz 84.3 Q1

Which of the following investment strategies stabilizes asset prices?

A. A value investment strategy.

B. A momentum investment strategy.

C. A size investment strategy.

D. Value, momentum, and size strategies all stabilize asset prices.

Answer: A

Step by step

1. Value is a negative feedback strategy: fallen stocks become cheap enough to buy, pushing prices back up.
2. Momentum is positive feedback: rising prices attract buyers who push prices up further, which is destabilizing.
3. Size has no stabilizing feedback mechanism, so only value stabilizes.

Why A is correct: Value investing buys what has fallen, pushing prices back toward fundamentals (negative feedback), so it is inherently stabilizing.

Why the others are wrong

- **B:** Momentum buys what has risen and sells what has fallen, amplifying price moves (positive feedback, destabilizing).
- **C:** The size strategy has no feedback mechanism that stabilizes prices; the size premium has also largely disappeared.
- **D:** Wrong because momentum is destabilizing, so not all three stabilize.

Reading 85. Alpha (and the Low-Risk Anomaly)

Andrew Ang, Asset Management: A Systematic Approach to Factor Investing (New York, NY: Oxford University Press, 2014). Chapter 10.

LO	Learning objective	Qs
85.a	describe and evaluate the low-risk anomaly of asset returns.	4
85.b	define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio.	12
85.c	explain the impact of benchmark choice on alpha and describe characteristics of an effective benchmark to measure alpha.	9
85.d	describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law.	13
85.e	apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark.	12
85.f	explain how to use style analysis to handle time-varying factor exposures.	2
85.g	describe issues that arise when measuring alphas for nonlinear strategies.	2
85.h	compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly.	4
85.i	describe potential explanations for the risk anomaly.	3

LO 85.a

Describe and evaluate the low-risk anomaly of asset returns.

Q84 AP Mock 1 Q51

A pension fund portfolio manager is running a training session on low volatility investment strategies, using past performance data and explaining the benchmarks applied. Which assertion about low volatility strategies would be accurate for the manager to put forward?

A. These strategies typically demonstrate significant alphas when compared to conventional market capitalization benchmarks.

B. When evaluated against dynamic factors such as value or momentum, these strategies usually yield negative alphas.

C. Over the risk-free rate, these strategies tend to yield high alphas, but when compared to any other benchmark, the alphas are insignificant.

D. These strategies usually generate low alphas if the benchmark is risk-adjusted and high alphas otherwise.

Answer: A

Step by step

1. The low-risk anomaly: lower-beta, lower-volatility stocks have earned returns equal to or higher than the market, contradicting the CAPM's positive risk-return relationship.
2. Measured against a cap-weighted market benchmark (beta of 1), this produces a positive, significant alpha.
3. Against other factor benchmarks (value, momentum, etc.) the alpha shrinks toward insignificance, but it does not turn systematically negative.
4. Only A is consistent with this evidence.

Why A is correct: Low-volatility portfolios have delivered market-like or better returns with lower risk, so against a cap-weighted benchmark their alpha is positive and significant.

Why the others are wrong

- **B:** Alpha against value/momentum benchmarks tends to be insignificant, not negative; confuses 'reduced' with 'reversed'.
- **C:** Alpha over the risk-free rate is just excess return, and the claim that all other benchmarks give insignificant alpha ignores the significant alpha versus the market benchmark.
- **D:** Reverses the logic: the anomaly shows up as alpha even against the market benchmark and shrinks as benchmarks add factors; it is not 'high alpha' when risk adjustment is ignored.

Q85 AnalystPrep Q.4594

REBUILT — Original options were weak (A is not a general property of low-beta assets, C is a giveaway, D 'none of the above'); rewrote options so exactly one is correct and tied to the CAPM-contradiction evidence.

Which of the following is the most likely reason why low-beta stocks are attractive to investors?

A. They have historically delivered higher risk-adjusted returns (Sharpe ratios) than high-beta stocks, contrary to the CAPM's prediction that return rises with beta.

B. The CAPM predicts that low-beta stocks earn a higher expected return than the market to compensate for their lower diversification.

C. They are guaranteed to outperform the market in every period, including strongly rising markets.

D. They earn a larger risk premium because a low beta signals greater systematic risk.

Answer: A

Step by step

1. CAPM: expected return is increasing in beta, so low-beta stocks should earn less than the market.
2. Empirically (low-risk anomaly) low-beta/low-volatility stocks earned equal or higher returns with lower risk, so their Sharpe ratios are higher (e.g., Sharpe fell from 0.9 for the lowest-beta quintile to 0.4 for the highest).
3. This contradicts CAPM and is why low-beta stocks are attractive: better risk-adjusted performance.
4. Hence A.

Why A is correct: The anomaly shows that lower beta has been paired with higher risk-adjusted returns, the opposite of the CAPM relationship.

Why the others are wrong

- **B:** Reverses the CAPM: it predicts a lower, not higher, expected return for low beta.
- **C:** Overstates: low-beta stocks tend to lag in strong up markets; nothing is guaranteed in every period.
- **D:** Confuses direction: low beta means lower systematic risk, hence a lower, not higher, CAPM risk premium.

Q86 BT answers P2.T9.20.5.1 · BT notes P2.T9.20.5.1

Consider the following four portfolios:
I. Albert's portfolio has a high beta and generates high (strong) returns
II. Betty's portfolio exhibits low volatility with high (strong) returns
III. Chuck's portfolio exhibits low beta (systematic risk) but generates a high volatility (total risk)
IV. Donald's tracks the market's minimum variance portfolio with low tracking error and generates low (weak) returns
Which of the portfolios best illustrates and confirms the low-risk anomaly?

A. Albert's portfolio

B. Betty's portfolio

C. Chuck's portfolio

D. Donald's portfolio

Answer: B

Step by step

1. The low-risk anomaly: lower risk (volatility or beta) is associated with higher, not lower, returns, the opposite of the CAPM.
2. Albert: high risk, high return - consistent with CAPM, not an anomaly.
3. Betty: low volatility and high return - exactly the anomaly.
4. Chuck compares two different risk measures and says nothing about return. Donald: low risk, low return - consistent with CAPM.

Why B is correct: Only Betty combines low risk with high returns, contradicting the positive risk-return trade-off.

Why the others are wrong

- **A:** High risk with high return is what CAPM predicts, so it does not illustrate the anomaly.
- **C:** Low beta with high volatility mixes two risk measures but gives no return information.
- **D:** Low risk with low return is again the conventional risk-return relationship.

Q87 Schweser Module Quiz 85.1 Q1

Which of the following statements is correct concerning the relationship between the low-risk anomaly and the capital asset pricing model (CAPM)?

A. The low-risk anomaly provides support for the CAPM.
B. The notion that the low-risk anomaly violates the CAPM has not been proven empirically.
C. The low-risk anomaly violates the CAPM and suggests that low-beta stocks will outperform high-beta stocks.
D. Both CAPM and the low-risk anomaly point to a positive relationship between risk and reward.

Answer: C

Step by step

1. CAPM: higher beta implies higher expected return (positive risk-reward).
2. Low-risk anomaly: lower beta and lower volatility have been associated with higher returns, an inverse relationship.
3. So the anomaly contradicts CAPM and implies low-beta stocks outperform high-beta stocks.

Why C is correct: The anomaly is an inverse risk-return relationship, directly opposite to CAPM's positive one.

Why the others are wrong

- **A:** The anomaly contradicts, not supports, CAPM.
- **B:** Empirical studies (e.g., Sharpe ratio falling with beta and volatility) do document the violation.
- **D:** Only CAPM implies a positive relationship; the anomaly implies a negative one.

LO 85.b

Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio.

Q88 AnalystPrep Q.2411

The performance parameters of two fund managers (measured against the same benchmark) in an investment firm are shown below:

Parameter	Fund Manager 1	Fund Manager 2
Return	12.35%	13.00%
α	2.35%	3.00%
Tracking error	25%	34%

Which of the following statements is correct?

A. Fund manager 1 deviated more from the benchmark index.
B. Fund manager 2 deviated more from the benchmark index.
C. Fund manager 1 generated a higher excess return.
D. Fund manager 2 has a higher information ratio.

Answer: B

Step by step

1. Tracking error is the standard deviation of excess returns, so it measures dispersion around the benchmark: 34% (manager 2) > 25% (manager 1).
2. Excess return (alpha): manager 1 = 2.35%, manager 2 = 3.00%, so manager 2 generated the higher excess return.
3. $IR_1 = 2.35/25 = 0.0940$; $IR_2 = 3.00/34 = 0.0882$, so manager 1 has the higher IR.

Why B is correct: Tracking error is the standard deviation of active returns, so the manager with the larger tracking error (manager 2) deviated more from the benchmark.

Why the others are wrong

- **A:** Reverses the comparison; manager 1 has the smaller tracking error (25% vs 34%).
- **C:** Reverses the comparison; manager 2's alpha (3.00%) exceeds manager 1's (2.35%).
- **D:** Manager 2's higher alpha is offset by a much higher tracking error: $3/34 = 0.0882 < 2.35/25 = 0.0940$. Judging by alpha alone ignores risk.

Q89

AnalystPrep Q.2419

REBUILT — Original conflated beta with a percentage of total market risk and left the annualization convention unstated; restated as a regression of excess returns with explicit annualization and a clean beta interpretation.

A fund manager regresses the monthly excess returns of ABC Corporation on the monthly excess returns of the market over the past 5 years (60 months). The regression output is:

Parameter	Coefficient
Alpha (intercept)	0.82% per month
Beta	0.69
Assets under management	$10 million

Annualize alpha by multiplying the monthly alpha by 12. Which of the following is correct?

A. Annualized alpha is 0.82%; ABC's excess return moves about 0.69% for each 1% move in the market excess return.

B. Annualized alpha is 9.84%; ABC's excess return moves about 0.69% for each 1% move in the market excess return.

C. Annualized alpha is 9.84%; ABC's excess return moves about 1.45% for each 1% move in the market excess return.

D. Annualized alpha is 2.84%; ABC's excess return moves about 0.69% for each 1% move in the market excess return.

Answer: B

Step by step

1. Alpha is the regression intercept: 0.82% per month of return beyond what the market exposure explains.
2. Annualize by the stated convention: $0.82\% \times 12 = 9.84\%$.
3. Beta = 0.69 is the slope: a 1% change in the market excess return is associated with about a 0.69% change in ABC's excess return.
4. AUM ($10 million) is irrelevant to alpha.

Why B is correct: Alpha is the monthly intercept scaled by 12 (9.84%), and beta is the slope of the stock's excess return on the market's.

Why the others are wrong

- **A:** Quotes the monthly alpha without annualizing.
- **C:** Alpha is right, but the beta is inverted ($1/0.69 = 1.45$).
- **D:** Scales alpha by $\sqrt{12}$ ($0.82 \times 3.46 = 2.84\%$), the rule for volatility, not for an average return.

Q90

AnalystPrep Q.2753

You have been given the following information for a portfolio:

Portfolio return	10%
Beta	1.2
Standard deviation of the portfolio	5%
Benchmark return	8%
Tracking error	10%
Risk-free rate	3%

Calculate the Sharpe and the information ratios for the portfolio.

A. Sharpe ratio: 1.4; Information ratio: 0.2

B. Sharpe ratio: 1.4; Information ratio: 0.7

C. Sharpe ratio: 0.4; Information ratio: 0.2

D. Sharpe ratio: 0.4; Information ratio: 0.7

Answer: A

Step by step

1. Sharpe ratio uses the risk-free rate as benchmark and total volatility: $(10\% - 3\%)/5\% = 1.4$.
2. Information ratio uses the benchmark return and tracking error: $(10\% - 8\%)/10\% = 0.2$.
3. Beta (1.2) is not needed for either ratio.

Why A is correct: Sharpe divides excess return over the risk-free rate by portfolio standard deviation; IR divides excess return over the benchmark by tracking error.

Why the others are wrong

- **B:** Correct Sharpe, but IR used the risk-free rate as benchmark: $7/10 = 0.7$.

- **C:** Sharpe used the benchmark instead of the risk-free rate ($2/5 = 0.4$); IR correct.
- **D:** Both ratios use the wrong benchmark: Sharpe uses the benchmark return (0.4) and IR uses the risk-free rate (0.7).

Q91

AnalystPrep Q.2755

REBUILT — Original distractors (0.6, 0.83, 1.0) did not correspond to realistic errors; replaced with options from specific mistakes.

A portfolio manager invested $1 million in a portfolio of equities. At that time the benchmark index against which the portfolio is tracked stood at 2000 points. One year later the portfolio was worth $1.12 million and the benchmark index stood at 2180 points. Calculate the information ratio for the portfolio if the tracking error is 4%.

A. 0.94
B. 1.33
C. 0.75
D. 3.00

Answer: C

Step by step

1. Portfolio return = $1.12/1.00 - 1 = 12\%$.
2. Benchmark return = $2180/2000 - 1 = 9\%$.
3. Alpha (active return) = $12\% - 9\% = 3\%$.
4. IR = $3\%/4\% = 0.75$.

Why C is correct: IR is active return over the benchmark divided by tracking error: 3%/4% = 0.75.

Why the others are wrong

- **A:** Computed benchmark return as $180/2180 = 8.26\%$ (divided by the ending level), giving alpha 3.74% and $3.74/4 = 0.94$.
- **B:** Inverted the ratio: $4/3 = 1.33$.
- **D:** Ignored the benchmark and divided the raw portfolio return by tracking error: $12/4 = 3.0$.

Q92

BT answers P2.T8.409.2

REBUILT — Source relied on an outside text (Litterman) with statements not verifiable in the Schweser reading; rewritten as a self-contained EXCEPT item on tracking error and IR.

Which of the following statements about tracking error (TE) and the information ratio (IR) is INCORRECT?

A. Tracking error is the standard deviation of a portfolio's excess returns relative to its benchmark.
B. The IR divides alpha by tracking error, so it measures active return per unit of active risk.
C. A larger tracking error allowance gives a manager more freedom in decision making.
D. Because TE and IR are computed from achieved returns, only a few observations are needed to conclude reliably that a manager is skilled, and the results are unaffected by the market environment.

Answer: D

Step by step

1. A is the definition of tracking error in the reading.
2. B is the definition of IR: α/TE.
3. C is true: mandates cap TE, so a higher cap permits larger active bets.
4. D is false: TE and IR are sample statistics built on standard deviations, need many observations to be statistically meaningful, and achieved values depend heavily on the market environment during the sample.

Why D is correct: TE and IR are noisy estimates that need long samples to be informative and are affected by the sample period's environment.

Why the others are wrong

- **A:** True statement (definition of tracking error), so not the exception.
- **B:** True statement (definition of IR), so not the exception.
- **C:** True statement; larger permitted TE means more active freedom, so not the exception.

Q93 BT answers P2.T8.704.1 · BT notes P2.T9.704.2

REBUILT — Source had OCR-damaged/inconsistent numbers (benchmark volatility 7.99 vs 1.99, active return 2.37 vs 2.31) and missing options; data reconstructed consistently (check: 3.29 - 1.5237x0.98 = 1.80%) and four options built from common errors.

A portfolio's excess returns (return above the risk-free rate) are regressed against the benchmark's excess returns over 60 observations. Key output:
- Portfolio: average excess return 3.29%, volatility 3.67%
- Benchmark: average excess return 0.98%, volatility 1.99%
- Average difference between portfolio and benchmark returns (active return): 2.31%
- Regression intercept 1.80%; slope 1.5237
- Tracking error (standard error of the regression): 2.10%

Which of the following is nearest to the information ratio if IR is measured as residual return (alpha) per unit of residual risk?

A. 0.490
B. 0.857
C. 0.896
D. 1.100

Answer: B

Step by step

1. Residual return (alpha) is the regression intercept: 1.80% (check: $3.29 - 1.5237 \times 0.98 = 1.80$).
2. Residual risk is the standard error of the regression: 2.10%.
3. IR $= 1.80/2.10 = 0.857$.

Why B is correct: Beta-adjusted IR divides the regression alpha by the regression residual volatility: 1.80%/2.10% = 0.857.

Why the others are wrong

- **A:** Divided alpha by the portfolio's total volatility: $1.80/3.67 = 0.490$.
- **C:** This is the Sharpe ratio of the portfolio: $3.29/3.67 = 0.896$.
- **D:** Used the unadjusted active return (2.31%, beta assumed 1) instead of the regression alpha: $2.31/2.10 = 1.100$.

Q94 BT answers P2.T9.20.5.2 · BT notes P2.T9.20.5.2

REBUILT — Two identical sources merged; removed the confusing rhetorical '(why?)' clause from the stem.

Regarding the relationship between alpha, tracking error, the information ratio (IR) and the Sharpe ratio, each of the following is true EXCEPT which is false?

A. The Sharpe ratio is a special case of the information ratio in which the benchmark is the risk-free asset.
B. The information ratio is the portfolio's average active return (alpha) divided by its tracking error (the standard deviation of that active return).
C. If two funds use the same benchmark, the fund with the higher information ratio must have the lower tracking error.
D. If the benchmark is risk-adjusted (beta estimated by regressing the fund's excess returns on the benchmark's excess returns), tracking error can be interpreted as idiosyncratic volatility.

Answer: C

Step by step

1. A: with benchmark = risk-free rate, active return is excess return and TE is the portfolio's own volatility, which is the Sharpe ratio. True.
2. B: this is the definition of IR. True.
3. C: IR $= \alpha/\text{TE}$ can rise with a higher alpha even if TE is also higher. Example: $2\%/4\% = 0.50$ vs $6\%/8\% = 0.75$; the higher-IR fund has the larger TE. False.
4. D: with a regression benchmark, the residual standard deviation is idiosyncratic risk. True.

Why C is correct: IR depends on both alpha and tracking error, so a higher IR does not imply a lower tracking error.

Why the others are wrong

- **A:** True: Sharpe ratio is the IR with the risk-free rate as the benchmark.
- **B:** True: standard definition of the IR.
- **D:** True: the regression residual volatility is the fund's idiosyncratic volatility.

Q95 BT notes P2.T9.704.2 (704.3)

REBUILT — Source gave no answer and had a debatable option D (fees); restated as a numerical alpha calculation with distractors from specific errors.

An analyst estimates the alpha of a low-volatility fund against the Russell 1000 index using monthly data. Over the sample, the regression slope (beta) is 0.40, the fund's average excess return is 3.15% per month, and the Russell index's average excess return is 0.65% per month (excess returns are above the risk-free rate). What is the fund's monthly alpha?

A. +250 basis points
B. +275 basis points
C. +289 basis points
D. +315 basis points

Answer: C

Step by step

1. Regression alpha = fund excess return - beta x benchmark excess return.
2. $\alpha = 3.15\% - 0.40 \times 0.65\% = 3.15\% - 0.26\% = 2.89\%$.
3. 2.89% = 289 basis points per month.

Why C is correct: Alpha is measured against a beta-adjusted benchmark, so the benchmark's contribution is 0.40 x 0.65% = 0.26%, not the full 0.65%.

Why the others are wrong

- **A:** Assumed beta = 1: $3.15 - 0.65 = 2.50\%$, ignoring the estimated beta of 0.40.
- **B:** Subtracted beta alone from the return ($3.15 - 0.40 = 2.75\%$) rather than beta times the benchmark excess return.
- **D:** Reported the fund's raw excess return, with no benchmark adjustment at all.

Q96 AnalystPrep Q.2542

The excess returns of a fund are regressed against the excess returns of the benchmark. The intercept represents:

A. Alpha
B. Beta
C. The residual error
D. The correlation coefficient

Answer: A

Step by step

1. Regression: $R_t - R_F = \alpha + \beta(R_{B,t} - R_F) + \varepsilon_t$.
2. The intercept is α: the expected excess return when the benchmark's excess return is zero.
3. The slope is β; ε_t is the error term; correlation is a separate statistic.

Why A is correct: The intercept of the excess-return regression is the part of the fund's return not explained by benchmark exposure, which is alpha.

Why the others are wrong

- **B:** Beta is the slope coefficient, not the intercept.
- **C:** The residual error ε_t is the unexplained deviation in each period, not a regression coefficient.
- **D:** Correlation is a standardized co-movement measure and is not a parameter of the fitted line.

Q97 AnalystPrep Q.2558

REBUILT — Original distractors were not traceable to realistic errors; replaced them with options generated by common calculation mistakes (correct answer unchanged).

Consider the following data for a fund for 2017:

Item	Portfolio	Market
Average return	32%	25%
Beta	1.15	1.00
Standard deviation	40%	30%
Tracking error	16%	0%
T-bill rate	6%	6%

Calculate the alpha generated by the portfolio and by the market, respectively.

A. 7.00% and 0%

B. 4.15% and 0%
C. 3.25% and 0%
D. -2.75% and 0%

Answer: B
Step by step
1. Benchmark-adjusted expected return: $6\% + 1.15 \times (25\% - 6\%) = 27.85\%$.
2. Portfolio alpha: $32\% - 27.85\% = 4.15\%$.
3. Market: $25\% - [6\% + 1.00 \times 19\%] = 0\%$.

Why B is correct: Alpha is the return in excess of the beta-adjusted benchmark return (risk-free plus beta times market excess return); the market has beta 1 so its alpha is zero by construction.
Why the others are wrong
- **A:** 7.00% is the simple difference 32% - 25%, ignoring that the portfolio's beta is 1.15 rather than 1.
- **C:** 3.25% = 32% - 1.15 Ã— 25%: the risk-free rate was not subtracted from the market return before applying beta.
- **D:** -2.75% = 32% - (6% + 1.15 Ã— 25%): the risk-free rate was added to beta times the total market return, double counting it.

Q98

AnalystPrep Q.5384

REBUILT — Original was a vague conceptual item with a loosely reasoned key; rebuilt as a calculation of the information ratio under a beta-adjusted versus a naive benchmark (alpha, tracking error and IR).

A fund earned an average return of 11% per year. The risk-free rate is 3% and the S&P 500 averaged 9%. Regressing the fund's excess returns on the S&P 500's excess returns gives a beta of 0.80, and the tracking error (standard deviation of the regression residuals) is 5%. What is the fund's information ratio measured against the beta-adjusted benchmark?

A. 0.40
B. 0.64
C. 0.76
D. 1.60

Answer: B
Step by step
1. Beta-adjusted benchmark return: $3\% + 0.80 \times (9\% - 3\%) = 7.8\%$.
2. Alpha: $11\% - 7.8\% = 3.2\%$.
3. IR $= \alpha$/tracking error $= 3.2\%/5\% = 0.64$.

Why B is correct: The information ratio is alpha divided by tracking error, and alpha must be measured against a benchmark that reflects the fund's actual beta (here 80% S&P 500, 20% risk-free).
Why the others are wrong
- **A:** 0.40 = (11% - 9%)/5%: benchmarked to the S&P 500 alone, which wrongly assumes beta = 1 and understates alpha and IR.
- **C:** 0.76 = (11% - 0.80 Ã— 9%)/5%: the risk-free component was omitted from the beta-adjusted benchmark.
- **D:** 1.60 = (11% - 3%)/5%: the excess return over the risk-free rate divided by tracking error, ignoring the S&P 500 exposure.

Q99

GARP PE1 Q8 · Mock 3 Q72

REBUILT — Merged two identical items; replaced the unsupported distractors with ones from realistic errors (portfolio-return volatility, mean alone, beta-adjusted returns).

A portfolio manager at a hedge fund manages an equity portfolio benchmarked to an index. Performance over the last 5 years:

Year	Portfolio return	Benchmark return	Portfolio beta vs. benchmark
1	0.072	0.070	0.92
2	0.052	0.054	0.88
3	0.052	0.047	0.90
4	0.060	0.060	0.84
5	0.048	0.033	0.89

Using the sample standard deviation, what is the approximate information ratio of the manager measured against the benchmark?

A. 0.40
B. 0.60
C. 0.42
D. 1.90

Answer: B

Step by step

1. Excess returns (%): 0.2, -0.2, 0.5, 0.0, 1.5; mean (alpha) = 0.40%.
2. Squared deviations from 0.40: 0.04, 0.36, 0.01, 0.16, 1.21; sum = 1.78.
3. Tracking error (sample, n-1): $\sqrt{1.78/4} = 0.667\%$.
4. IR $= 0.40/0.667 = 0.60$. The betas are not needed.

Why B is correct: IR is average excess return over the benchmark divided by the standard deviation of those excess returns (tracking error).

Why the others are wrong

- **A:** 0.40 is just the average excess return in percent; it was not divided by tracking error.
- **C:** 0.42 divides alpha by the standard deviation of the portfolio's own returns (about 0.95%) rather than of excess returns.
- **D:** 1.90 uses betas to compute $R_p - \beta R_b$ (mean 1.00%, sd 0.527%), omitting the risk-free rate, which gives a meaningless alpha; the question asks for excess return over the benchmark itself.

LO 85.c

Explain the impact of benchmark choice on alpha and describe characteristics of an effective benchmark to measure alpha.

Q100

AnalystPrep Q.2412 · Schweser Module Quiz 85.1 Q2

Which of the following is NOT a characteristic of an appropriate benchmark for measuring alpha?

A. Tradeable
B. Replicable
C. Well-defined
D. Applied equally to all risky assets irrespective of their risk exposure

Answer: D

Step by step

1. An effective benchmark is well-defined (independent provider, verifiable, unambiguous), tradeable, replicable and adjusted for risk.
2. Options A, B and C are three of these four characteristics.
3. Option D is the opposite of risk adjustment: a benchmark must match the risk of the investment being evaluated (e.g., beta-adjusted), not be applied uniformly.

Why D is correct: A benchmark must be on the same risk scale as the assets under review; a one-size-fits-all benchmark makes the comparison unfair.

Why the others are wrong

- **A:** Tradeability is required so the benchmark could be held as a real alternative; it is a characteristic, so not the answer.
- **B:** Replicability is required (absolute return benchmarks fail it); it is a characteristic.
- **C:** Being well-defined (independent, verifiable) is a characteristic.

Q101

AnalystPrep Q.2414

A large investment management firm plans to launch a new fund for retail investors and is choosing a benchmark. Fund manager 1 proposes the highest-returning benchmark among the firm's existing funds. Fund manager 2 proposes the market return because a similar fund has beaten it for several periods. Fund manager 3 says: "Many absolute return funds have consistently beaten the market. We must take one of those funds as the benchmark."

Which of the following is correct with respect to the statement made by Fund manager 3?

A. Absolute return funds generate positive returns irrespective of market conditions and, therefore, it would be in order to use one of them as the benchmark.

B. Absolute return funds take excessive risks, a characteristic which effectively nullifies them from being used as benchmarks.

C. Absolute return funds employ different strategies which are usually not replicable, and hence one of them must not be used as the benchmark.

D. Absolute return benchmarks are usually not diversified, and hence one of them must not be used as the benchmark.

Answer: C

Step by step

1. Recall the four benchmark criteria: well-defined, tradeable, replicable, risk-adjusted.
2. Absolute return funds follow varied, often proprietary strategies that an investor cannot feasibly replicate.
3. If a benchmark cannot be replicated, tracking error will be very high and the comparison is not meaningful, so they should not be used.

Why C is correct: Absolute return benchmarks fail the replicability criterion, so they are unsuitable.

Why the others are wrong

- **A:** Positive returns in all conditions does not make a benchmark appropriate; it ignores replicability and risk adjustment.
- **B:** Excess risk is not the stated reason; the problem is non-replicability, and a risk-adjusted benchmark could in principle handle risk.
- **D:** Diversification is not one of the benchmark criteria; lack of replicability is the actual defect.

Q102 AnalystPrep Q.3022

Tradability is one of the properties of a sound benchmark. Which of the following best explains the link between tradability and a sound benchmark?

A. If alphas are not evaluated relative to a tradable benchmark, the implementable returns on investment strategies are not a representation of the calculated alphas.

B. Both the asset owner and the fund manager are supposed to display the ability to trade the benchmark.

C. Produced by an independent asset provider, a tradable benchmark should be verifiable and not ambiguous about its contents.

D. Alphas must not be measured in relation to a tradable benchmark because if so, the computed alphas will not give the returns on investment strategies that can be implemented.

Answer: A

Step by step

1. Tradeable means the benchmark is a basket of securities that could be invested in directly as an alternative.
2. Alpha measured against a tradeable benchmark is therefore the return gained over an investable alternative, i.e., an implementable return.
3. Without tradeability, calculated alphas do not represent returns an investor could actually capture.

Why A is correct: Tradeability makes the benchmark an investable alternative, so alpha reflects implementable excess return.

Why the others are wrong

- **B:** No requirement exists that both parties demonstrate the ability to trade the benchmark; the criterion concerns the benchmark's investability.
- **C:** Describes the 'well-defined' criterion (independent provider, verifiable, unambiguous), not tradeability.
- **D:** Reverses the logic: measuring against a tradeable benchmark is what makes alphas implementable.

Q103 AnalystPrep Q.4632

REBUILT — Original distractors (0.85, 1.17, 0.87) were not traceable to realistic errors; rebuilt options around benchmark-choice and ratio mistakes.

A portfolio manager evaluates a fund whose mandate is large-cap stocks. Data:

Variable	Value
Benchmark LC (90% large-cap, 10% cash) annual return	8%
Benchmark Hybrid (50% large-cap, 45% small-cap, 5% cash) annual return	10%
Fund annual return	12%
Tracking error	6%

Using the appropriate benchmark, what is the excess return per unit of tracking error?

A. 0.33
B. 0.67
C. 1.33
D. 2.00

Answer: B

Step by step

1. The mandate is large-cap, so the LC benchmark (mostly large-cap) is the appropriate, risk-matched benchmark.
2. Active return = 12% - 8% = 4%.
3. Information ratio = 4% / 6% = 0.667, about 0.67.

Why B is correct: Excess return over the mandate-appropriate benchmark divided by tracking error is the information ratio: 4%/6% = 0.67.

Why the others are wrong

- **A:** Used the Hybrid benchmark: (12%-10%)/6% = 0.33, an inappropriate benchmark for a large-cap mandate.
- **C:** Divided the benchmark return by tracking error: 8%/6% = 1.33.
- **D:** Divided the fund's total return by tracking error: 12%/6% = 2.00, ignoring the benchmark.

Q104 BT answers P2.T8.704.3

An analyst estimates the alpha of a new low-volatility fund using the Russell 1000 as a naive benchmark. Monthly (ex ante) statistics: regression slope coefficient beta = 0.40; the portfolio's average excess return = 3.15% per month; the Russell index's average excess return = 0.65% per month. Excess returns are above the risk-free rate. Which of the following is TRUE?

A. The portfolio's alpha is about +289 basis points
B. He should assume a beta of 1.0 such that the portfolio's alpha is about +250 basis points; this is a lower alpha due to the implicit risk-adjustment
C. The Russell is NOT an appropriate benchmark because the low beta implies that the Russell 1000 cannot be combined with another asset in order to generate a market-adjusted portfolio
D. The Russell is NOT an appropriate benchmark because it represents a tradeable, low-cost alternative but the firm's low-volatility fund is active and charges high fees; an ideal benchmark charges comparable fees

Answer: A

Step by step

1. The regression line passes through the sample averages: alpha = 3.15% - 0.40 x 0.65% = 3.15% - 0.26% = 2.89% per month.
2. That is about 289 basis points.
3. A risk-adjusted benchmark here is 40% Russell 1000 and 60% risk-free asset, so the Russell remains usable.

Why A is correct: Alpha is the average excess return minus beta times the benchmark's excess return, using the fund's true beta of 0.40.

Why the others are wrong

- **B:** Assuming beta = 1 gives 3.15% - 0.65% = 2.50%, but that ignores risk adjustment; it understates alpha because the benchmark is riskier than the fund.
- **C:** False: the Russell can be blended with the risk-free asset (40%/60%) to form a beta-matched benchmark.
- **D:** Benchmark criteria are well-defined, tradeable, replicable and risk-adjusted; being a low-cost tradeable index is a virtue, not a flaw, and comparable fees are not a criterion.

Q105 BT answers P2.T9.20.5.3 · BT notes P2.T9.20.5.3

William evaluates a low-volatility fund against its naive benchmark, the Russell 1000. Over three years the fund's average return exceeded the Russell 1000 by +0.28% per year, which he tentatively calls the fund's "alpha". His colleague Emily argues the correct benchmark is a risk-adjusted portfolio of 65.0% Russell 1000 and 35.0% risk-free

asset (the risk-free return is significantly less than the Russell 1000 return). If William re-evaluates performance over the same period using this risk-adjusted benchmark, which is the most likely outcome compared with his initial analysis?

A. The fund's alpha will increase
B. The fund's beta will increase
C. The fund's active return will decrease
D. The fund's information ratio will decrease because its tracking error will increase

Answer: A

Step by step

1. Using the Russell 1000 alone implicitly assumes beta = 1; a low-volatility fund likely has beta below 1 (Emily's benchmark implies a beta of 0.65).
2. The 65/35 benchmark has a lower return than the Russell 1000 because the risk-free asset earns much less. Illustration: Russell 8%, risk-free 2% gives 0.65 x 8% + 0.35 x 2% = 5.9% vs 8%.
3. If the fund returns 8.28%, the excess return rises from 0.28% to 2.38%, so alpha increases.

Why A is correct: A lower-risk benchmark has a lower return hurdle, so measured alpha (and generally the information ratio) rises.

Why the others are wrong

- **B:** Beta is a property of the fund's returns; changing the benchmark does not raise it, it only reveals that beta is below 1.
- **C:** Active return versus a lower-returning benchmark increases, not decreases.
- **D:** The higher alpha raises the information ratio; the fund is not made riskier relative to the benchmark, so the claim that tracking error rises and the ratio falls is backwards.

Q106

BT answers P2.T9.21.2.2 · BT notes P2.T9.21.2.2

REBUILT — Original relied on Ang's joint-hypothesis/Hansen-Jagannathan discussion not in the Schweser reading; rebuilt around the Schweser benchmark-selection criteria (benchmark chosen ex ante, independent of results).

Betty, a CIO at a fund of funds, evaluates a discretionary multi-factor active strategy. Her analyst runs about 100 regressions of the strategy's five-year returns on different sets of factors and then reports the one factor set for which the strategy shows zero alpha. Which of the following is the key weakness of this analysis?

A. There is no problem; selecting the factor set with the best fit gives the most accurate benchmark
B. The benchmark was chosen after the fact by searching until alpha disappeared, so it is not an independent, well-defined benchmark and the zero alpha is of little value for judging the strategy going forward
C. Factor benchmarks can never be used to measure alpha because alpha is only defined relative to a single market index
D. A zero alpha proves that markets are perfectly efficient, so the strategy should not be evaluated further

Answer: B

Step by step

1. An effective benchmark is well-defined and independent, and the choice of benchmark determines the measured alpha.
2. Trying many factor sets until alpha is zero lets the benchmark be fitted to the outcome (data-mining), so the benchmark is not specified independently of the result.
3. A benchmark found ex post this way says little about the strategy's future alpha or an investable alternative.

Why B is correct: Benchmarks should be set independently of, and before seeing, results; searching for a benchmark that removes alpha defeats that.

Why the others are wrong

- **A:** Fit-maximizing across many trials is data-mining, not accuracy.
- **C:** Multifactor (e.g., Fama-French) benchmarks are explicitly valid for measuring alpha.
- **D:** Zero alpha against one fitted benchmark does not imply market efficiency.

Q107

AP Mock 3 Q6

REBUILT — Original asked for the 'most reliable' qualitative conclusion with vague, overstated options and a weakly justified key; rebuilt as a self-contained benchmark-choice calculation on the same LO.

A technology fund earned an average return of 14% over the sample period. Over the same period the risk-free

rate was 3% and the S&P 500 returned 10%. A regression of the fund's excess returns on the S&P 500's excess returns gives a beta of 1.30, and the fund's tracking error is 5%. The manager reports alpha and the information ratio (IR) by simply comparing the fund with the S&P 500 (implicitly assuming a beta of 1.0). Using a benchmark that reflects the fund's true beta, what are the fund's alpha and IR, and how does the reported figure compare?

A. Alpha of 1.9% and IR of 0.38; the reported alpha (4.0%) and IR (0.80) are overstated because the S&P 500 is less risky than the fund.
B. Alpha of 4.0% and IR of 0.80; the S&P 500 is an appropriate benchmark.
C. Alpha of 1.0% and IR of 0.20; the reported figures are overstated.
D. Alpha of 4.9% and IR of 0.98; the reported figures are understated.

Answer: A

Step by step

1. Beta-adjusted benchmark return: $R_F + \beta(R_M - R_F) = 3\% + 1.30(10\% - 3\%) = 12.1\%$ (i.e. -30% in the risk-free asset, 130% in the S&P 500).
2. Alpha vs. adjusted benchmark: $14\% - 12.1\% = 1.9\%$.
3. IR: $1.9\%/5\% = 0.38$.
4. Naive benchmark (beta = 1): alpha $= 14\% - 10\% = 4.0\%$, IR $= 4.0/5 = 0.80$.
5. The fund is riskier than the S&P 500 (beta 1.30), so the naive comparison credits the manager with return that is merely compensation for extra market risk: reported alpha and IR are too high.

Why A is correct: An effective benchmark must be risk-adjusted. When the benchmark is less risky than the investment, alpha and IR are overstated; correcting for beta cuts alpha from 4.0% to 1.9%.

Why the others are wrong

- **B:** Accepts the naive S&P 500 comparison, which wrongly assumes beta = 1.0.
- **C:** Computed $14\% - 1.30 \times 10\% = 1.0\%$, applying beta to the total market return and omitting the risk-free component of the benchmark (should be β times the excess return plus R_F).
- **D:** Computed $14\% - 1.30(10\% - 3\%) = 4.9\%$, forgetting to add back the risk-free rate in the benchmark; this also wrongly implies understatement.

Q108

AnalystPrep Q.2415

REBUILT — Original options left D (benchmark for alternative investments) defensible and tested no clear criterion; rewrote around the investability criterion with one best answer.

A consultant proposes using the index ABC as the benchmark to measure a manager's alpha. The composition of ABC is:

Assets	Proportion
Exotic Stones	10%
Real Estate	75%
Art	15%

Which of the following is the best assessment of ABC as a benchmark?

A. Suitable, because it is diversified across three different asset classes.
B. Unsuitable, because it is not investable: its constituents trade rarely and in small amounts, so replicating it passively would be impractical and costly.
C. Unsuitable, because an effective benchmark must consist only of liquid securities that trade daily.
D. Suitable, because illiquid assets earn an illiquidity premium, which makes the benchmark more demanding.

Answer: B

Step by step

1. An effective benchmark must be investable: the investor should be able to hold the benchmark's constituents passively at reasonable cost.
2. All three components of ABC (exotic stones, real estate, art) are illiquid: they trade infrequently, in small amounts, with low turnover (Schweser: art trades once every 40-70 years, real estate turnover is about 5-7% per year).
3. Replicating the 10/75/15 mix would involve large transaction costs, search frictions and price impact, so ABC fails the investability test.
4. Therefore ABC is not an appropriate benchmark for measuring alpha (answer B).

Why B is correct: A benchmark is a passive alternative the manager could have held instead; if its constituents cannot be bought and sold at reasonable cost, it is not a realistic opportunity set and cannot fairly measure alpha.

Why the others are wrong

- **A:** Diversification across three asset types does not make an index investable; this confuses breadth with replicability.
- **C:** Overstates the rule: benchmarks for illiquid asset classes exist; the problem is the cost and difficulty of replication, not that every constituent must trade daily.
- **D:** A higher expected return is not a benchmark criterion; a benchmark must be unambiguous, investable and appropriate, not 'demanding'.

LO 85.d

Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law.

Q109

AnalystPrep Q.2416

An investment management firm allows each fund manager a maximum of 100 transactions per year to control transaction costs. One manager wants an information ratio of 0.75. Assuming the fundamental law applies, her information coefficient must be equal to:

A. 0.75
B. 0.0075
C. 0.075
D. 7.5

Answer: C

Step by step

1. Fundamental law: $\text{IR} = \text{IC} \times \sqrt{\text{BR}}$.
2. Breadth BR = 100, so $\sqrt{100} = 10$.
3. $\text{IC} = 0.75/10 = 0.075$.

Why C is correct: With breadth capped at 100 independent bets, IC must equal IR divided by the square root of breadth.

Why the others are wrong

- **A:** Ignores breadth (sets IC = IR).
- **B:** Divides by BR (100) instead of its square root.
- **D:** Multiplies IR by $\sqrt{\text{BR}}$ instead of dividing.

Q110

AnalystPrep Q.2417

An investment management firm allows each fund manager a maximum of 100 transactions per year. A manager assumes IC = 7.5% and uses $\text{IR} = \text{IC} \times \sqrt{\text{BR}}$ with BR = 100, which implies IR = 0.75. When she instead computes IR from the observed excess return and tracking error, the result is below 0.75. This is most likely due to the:

A. Correlation between the different bets of the fund manager.
B. Cap on the number of transactions per year.
C. Differences in portfolio weights.
D. Differences in the number of transactions from one manager to another.

Answer: A

Step by step

1. Predicted IR: $0.075 \times \sqrt{100} = 0.75$, valid only if the 100 bets are independent.
2. The fundamental law assumes forecasts are independent of one another.
3. If bets are correlated, effective breadth is below 100, so realized IR is below 0.75.

Why A is correct: Correlated bets reduce the effective number of independent bets (breadth), so the realized IR falls short of the law's prediction.

Why the others are wrong

- **B:** The cap is already built into BR = 100; it does not explain a shortfall against that calculation.
- **C:** Portfolio weights are not an input to the law and do not address the independence assumption.
- **D:** Differences across managers are irrelevant to one manager's own shortfall.

Q111 AnalystPrep Q.2418

An investment management firm allows each fund manager a maximum of 100 transactions per year. The assets under management of one manager's fund increase from $1 million to $20 million over 5 years. Other things equal, how does this most likely affect the manager's information ratio?

A. It increases
B. It remains unchanged
C. It decreases
D. It moves towards a value of 1

Answer: C

Step by step

1. $IR = IC \times \sqrt{BR}$; breadth is fixed at 100.
2. As AUM grows, the ability to generate high-quality forecasts (IC) tends to decline.
3. Lower IC with unchanged BR means a lower IR.

Why C is correct: IC tends to fall as assets under management rise, so IR falls when breadth is unchanged.

Why the others are wrong

- **A:** Assumes scale improves skill; the reading says IC declines with AUM.
- **B:** Treats IR as depending on breadth only; ignores the deterioration in IC.
- **D:** No basis in the law; IR does not converge to 1.

Q112 AnalystPrep Q.2761

Which of the following managers has the highest information ratio based on Grinold's fundamental law of active management, $IR = IC \times \sqrt{BR}$?

A. A manager with an information coefficient of 0.5 and a breadth of 20.
B. A manager with an information coefficient of 0.05 and a breadth of 200.
C. A manager with an information coefficient of 0.1 and a breadth of 100.
D. A manager with an information coefficient of 0.8 and a breadth of 10.

Answer: D

Step by step

1. A: $0.5\sqrt{20} = 2.236$.
2. B: $0.05\sqrt{200} = 0.707$.
3. C: $0.1\sqrt{100} = 1.000$.
4. D: $0.8\sqrt{10} = 2.530$, the highest.

Why D is correct: IR rises linearly with IC but only with the square root of breadth, so the high-IC manager wins here.

Why the others are wrong

- **A:** 2.24 is second highest; its IC is lower than D's and breadth of 20 does not compensate.
- **B:** 0.71; assumes many bets can compensate for a tiny IC, but $\sqrt{200}$ is only 14.1.
- **C:** 1.00; a moderate IC with a large breadth still falls well short of D's 2.53.

Q113 AnalystPrep Q.4633

REBUILT — Stem used the vague term 'forecast precision' and option D ('cannot be calculated') was a giveaway; restated as information coefficient and replaced D with a computational error.

A chief investment officer aims to achieve an annual information ratio of 0.83 on a firm-wide basis. The firm places one bet every month (12 independent bets per year). Using the fundamental law, what information coefficient (forecast accuracy) is required?

A. 0.24
B. 2.88
C. 4.17
D. 0.07

Answer: A

Step by step

1. $BR = 12$, $\sqrt{12} = 3.4641$.

2. $IC = IR/\sqrt{BR} = 0.83/3.4641 = 0.2396 \approx 0.24$.

Why A is correct: Rearranging IR = IC×√BR gives IC = IR/√BR.

Why the others are wrong

- **B:** Multiplied IR by √BR (0.83×3.464 = 2.88) instead of dividing.
- **C:** Inverted the correct result (1/0.2396 = 4.17).
- **D:** Divided by BR = 12 rather than √BR (0.83/12 = 0.069).

Q114 AnalystPrep Q.4706

A company has three active managers, each with an expected alpha of 5% and a tracking error of 8.4%. The information ratio is the same for each manager, all strategies have equal weights, and the strategies are independent. What is the total (combined) information ratio?

A. 1.031
B. 0.595
C. 1.68
D. 0.007

Answer: A

Step by step

1. Single-manager IR = 0.05/0.084 = 0.5952.
2. Independent, equally weighted strategies with equal IRs: $IR_{total} = IR\sqrt{N}$.
3. $0.5952 \times \sqrt{3} = 1.031$.

Why A is correct: Combining independent strategies of equal IR multiplies the IR by √N, as breadth does in the fundamental law.

Why the others are wrong

- **B:** The single-manager IR; no benefit from combining strategies counted.
- **C:** Inverted ratio (8.4/5 = 1.68).
- **D:** Used the tracking error variance (0.084^2 = 0.007).

Q115 BT answers P2.T8.705.3 · BT notes P2.T9.705.3

REBUILT — Option A tied IC decline to breadth, which is not the reading's stated limitation; reworded to the AUM relationship from the reading, keeping B as the false statement.

Grinold's fundamental law of active management states that the maximum attainable information ratio is $IR = IC \times \sqrt{BR}$. Each of the following statements about the law is true EXCEPT:

A. As assets under management grow, the information coefficient tends to decline.
B. A key advantage of the law is that it incorporates downside risk and higher-moment risk, adjusting for skewness and excess kurtosis.
C. An IR of 0.50 can be achieved by a market timer with IC = 0.25 making four bets a year, or by a manager with IC = 0.025 making 400 bets a year.
D. A crucial assumption is that forecasts are independent; overweighting 1,000 value stocks against underweighting 1,000 growth stocks is not 1,000 independent bets.

Answer: B

Step by step

1. A: reading states IC tends to decline as AUM rises: true.
2. C: $0.25\sqrt{4} = 0.50$ and $0.025\sqrt{400} = 0.50$: true.
3. D: correlated factor bets reduce independent breadth: true.
4. B: the framework ignores downside risk; it does not adjust for skew or kurtosis: false.

Why B is correct: The law ignores downside and higher-moment risk and assumes independent forecasts, so B is the false statement.

Why the others are wrong

- **A:** True: skill diminishes as AUM grows.
- **C:** True: both combinations give IR = 0.50.
- **D:** True: the independence assumption is violated by correlated factor bets.

Q116 BT answers P2.T9.21.1.1 · BT notes P2.T9.21.1.1

A portfolio manager has an information coefficient (IC) of 0.150 and breadth (BR) of 12.0. If she triples her breadth to 36, what happens to the information ratio (IR)?

A. The IR drops by 1/3rd to 0.346
B. It has no effect on the IR
C. The IR increases by 0.380 to 0.900
D. The IR increases by 1.039 to 1.559

Answer: C

Step by step

1. Initial: $0.150\sqrt{12} = 0.5196$.
2. New: $0.150\sqrt{36} = 0.150 \times 6 = 0.900$.
3. Change: $0.900 - 0.5196 = 0.380$.

Why C is correct: IR scales with √BR, so tripling breadth raises IR by √3 = 1.73, not by 3.

Why the others are wrong

- **A:** Reduces IR by a third (0.5196×2/3 = 0.346); the direction is wrong.
- **B:** Ignores that breadth enters the law.
- **D:** Triples the IR (0.5196×3 = 1.559) by treating IR as linear in BR instead of √BR.

Q117 BT answers P2.T9.21.3.1 · BT notes P2.T9.21.3.1

Grinold's fundamental law of active management states that the expected information ratio of an active strategy is $IR = IC \times \sqrt{BR}$, where IC is the information coefficient and BR is breadth; the extended law adds the transfer coefficient, $IR = IC \times \sqrt{BR} \times TC$, where TC measures how efficiently constraints and transaction costs allow the actual portfolio to reflect the optimal one.

Which of the following is the BEST measure of breadth, BR, in the fundamental law?

A. The number of independent bets per year
B. The ratio of active return to active risk
C. The number of positions (holdings) in the portfolio
D. The number of securities in the relevant index, e.g., S&P 1500, Russell 3000, or Wilshire 5000

Answer: A

Step by step

1. Breadth in the fundamental law is the number of independent investment decisions (bets) made per year.
2. It is not the number of holdings or the index size, because a position can be held for years and many positions may reflect the same underlying bet.
3. Active return divided by active risk is the information ratio itself, the output of the law.

Why A is correct: IR rises with the square root of the number of independent forecasts per year; breadth counts independent bets over a year, not securities held.

Why the others are wrong

- **B:** This is the definition of the information ratio (IR), not breadth.
- **C:** Confuses holdings with bets: positions may be correlated or unchanged, so they are not necessarily independent annual decisions.
- **D:** Confuses the size of the opportunity set with the number of independent bets actually made.

Q118 Mock 1 Q25

Using the fundamental law of active management (FLOAM), an investor requires an information ratio of 0.5 and is considering a stock selection strategy based upon momentum plays. Assuming the manager places 200 independent bets per year, what is the required information coefficient?

A. 0.035
B. 0.350
C. 0.050
D. 0.500

Answer: A

Step by step

1. $IR = IC \times \sqrt{BR}$, so $IC = IR/\sqrt{BR}$.
2. $\sqrt{200} = 14.142$.
3. $IC = 0.5/14.142 = 0.0354 \approx 0.035$.

Why A is correct: Rearranging the fundamental law gives $IC = IR/\sqrt{BR}$, which is small because many bets are made.

Why the others are wrong

- **B:** Divided by $\sqrt{2}$ ($0.5/1.414 = 0.354$), a slip in the square root of breadth.
- **C:** Divided by 10 ($\sqrt{100}$) rather than $\sqrt{200}$, i.e., $0.5/10 = 0.05$.
- **D:** Set IC equal to IR, ignoring breadth entirely.

Q119

Mock 2 Q54

REBUILT — Original last blank ('which theory is more conservative') was subjective and ambiguous; replaced with a clean computation of the two required ICs.

A mutual fund targets an information ratio (IR) of at least 0.60. Analyst 1 argues the fund should place at least 250 independent bets per year; Analyst 2 argues that only 25 independent bets per year suffice. Using the fundamental law (with $TC = 1$), what information coefficient is required to just reach the target under Analyst 1's breadth and under Analyst 2's breadth, respectively?

A. 0.0379 and 0.120
B. 0.120 and 0.0379
C. 0.0024 and 0.024
D. 9.49 and 3.00

Answer: A

Step by step

1. $IC = IR/\sqrt{BR}$.
2. Analyst 1: $IC = 0.60/\sqrt{250} = 0.60/15.811 = 0.0379$.
3. Analyst 2: $IC = 0.60/\sqrt{25} = 0.60/5 = 0.120$.
4. More bets allow a lower skill level to reach the same IR; fewer bets require higher skill.

Why A is correct: $IC = IR/\sqrt{BR}$: 250 bets need only 0.0379, while 25 bets need 0.120.

Why the others are wrong

- **B:** Reverses the pairing: assumes fewer bets need less skill, contrary to $IR = IC \times \sqrt{BR}$.
- **C:** Divided IR by BR instead of $\sqrt{BR}$ ($0.60/250$, $0.60/25$).
- **D:** Multiplied IR by $\sqrt{BR}$ (0.60×15.81, 0.60×5) instead of dividing.

Q120

Mock 2 Q78

Quantitative Equity Design (QED) is a quantitative equity fund manager. QED forecasts monthly alphas for stocks in its 3,000 stock universe. The number of independent forecasts made each month is 25. If the model employed by QED has an information coefficient of 0.15, the annual information ratio for QED is closest to:

A. 0.75
B. 2.60
C. 28.46
D. 45.00

Answer: B

Step by step

1. Breadth must be annual: $BR = 25 \times 12 = 300$ independent forecasts per year.
2. $IR = IC \times \sqrt{BR} = 0.15 \times \sqrt{300} = 0.15 \times 17.321 = 2.598 \approx 2.60$.
3. The 3,000-stock universe is irrelevant; only independent forecasts count.

Why B is correct: The law uses annual independent bets, 300 here, multiplied by IC after taking the square root.

Why the others are wrong

- **A:** Used the monthly breadth of 25: $0.15 \times \sqrt{25} = 0.75$, forgetting to annualize.
- **C:** Used $3{,}000 \times 12 = 36{,}000$ bets: $0.15 \times \sqrt{36{,}000} = 28.46$, treating every stock as an independent bet.
- **D:** Used breadth without the square root: $0.15 \times 300 = 45$.

Q121 Schweser Module Quiz 85.1 Q3

Grinold's fundamental law of active management suggests that:

A. investors should focus on increasing only their predictive ability relative to stock price movements.

B. sector allocation is the most important factor in active management.

C. a small number of investment bets decreases the chances of making a mistake and, therefore, increases the expected investment performance.

D. to maximize the information ratio, active investors need to either have high-quality predictions or place a large number of investment bets in a given year.

Answer: D

Step by step

1. The law is $IR = IC \times \sqrt{BR}$.
2. IR rises with skill (IC) and with the number of independent bets (BR); either can compensate for the other.
3. Only option D states this trade-off.

Why D is correct: Since $IR = IC \times \sqrt{BR}$, a high IR comes from high-quality forecasts, many independent bets, or both.

Why the others are wrong

- **A:** Ignores breadth; the law says breadth matters as much as skill.
- **B:** Sector allocation is not a feature of the law.
- **C:** Reverses the logic: fewer bets lower IR since IR increases with $\sqrt{BR}$.

LO 85.e

Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark.

Q122 Mock 4 Q41 · GARP PE2 Q40

An individual investor decides to create a USD 1 million portfolio that mimics the strategy of Fund CRN, which has consistently generated high alphas. The investor regresses Fund CRN's monthly excess returns over the risk-free rate against the three Fama-French factors plus a momentum factor and obtains the following statistically significant estimates:

Factor	Coefficient	T-statistic
Alpha	0.08	2.10
MKT loading	0.55	6.72
SMB loading	-0.63	4.35
HML loading	0.36	3.20
UMD loading	-0.07	2.77
Adjusted R2	0.56	-

Which of the following positions is a component of the mimicking portfolio?

A. A USD 7,000 long position in stocks showing positive momentum

B. A USD 360,000 long position in value stocks

C. A USD 450,000 short position in T-bills

D. A USD 630,000 short position in growth stocks

Answer: B

Step by step

1. Each loading times the USD 1,000,000 portfolio gives the dollar position in that factor.
2. MKT 0.55: $0.55 \times 1{,}000{,}000 = 550{,}000$ long the market; the remainder $(1 - 0.55) \times 1{,}000{,}000 = 450{,}000$ is LONG T-bills (weights sum to 1).
3. SMB -0.63: $-630{,}000$, i.e. short small caps / long large caps.
4. HML 0.36: $+360{,}000$, i.e. long value (high book-to-market) stocks / short growth stocks.
5. UMD -0.07: $-70{,}000$, i.e. short up-trending (winner) stocks / long down-trending stocks. Alpha is a return, not a position.
6. Only B matches a computed position.

Why B is correct: A factor loading scales a long-short factor portfolio, so a positive HML loading of 0.36 means a

long USD 360,000 position in value stocks (funded by a short in growth stocks).

Why the others are wrong

- **A:** Decimal slip (0.07 gives 70,000, not 7,000) and sign error: the UMD loading is negative, so the momentum position is short, not long.
- **C:** Sign error on T-bills: with a market loading of 0.55 (below 1) the balance of 450,000 is invested long in T-bills, not borrowed.
- **D:** Wrong side/label: the negative SMB loading implies short small-cap / long large-cap stocks; the size factor has nothing to do with growth stocks (that is HML).

Q123

AnalystPrep Q.2440

REBUILT — Original had a defensible-alternative option (daily returns add observations) and irrelevant clutter; stem condensed and distractors replaced so only the multifactor option is correct.

An analyst estimates the alpha and beta of the real estate sector by regressing sector excess returns on market excess returns, using returns from a consultancy report covering only a handful of transactions. The regression gives a beta of 0.65 and an alpha of 3.45%, and the analyst concludes that the sector is less risky than the market and offers an easy excess return of 3.45%. Which of the following would make the alpha estimate a more reliable measure of excess return?

A. Using fewer observations so that only the most recent transactions are included
B. Using a multifactor model (e.g. adding size and value factors) as the benchmark instead of the market alone
C. Regressing raw returns rather than returns in excess of the risk-free rate
D. Annualizing the estimated alpha before drawing conclusions

Answer: B

Step by step

1. A single-factor alpha is the intercept relative to the market only, so any return earned from other priced risk exposures (size, value, momentum) shows up as 'alpha'.
2. Adding further factors to the regression benchmark removes the part of the return explained by those exposures, leaving a cleaner alpha and a better-specified beta.
3. Fewer observations widen standard errors; using raw returns drops the risk-free benchmark; annualizing only rescales the same estimate.

Why B is correct: A factor regression with multiple factors constructs a benchmark that captures more systematic return sources, so the remaining intercept is a purer alpha.

Why the others are wrong

- **A:** Reducing the sample lowers statistical precision (larger standard errors); it does not remove omitted-factor bias.
- **C:** Alpha must be measured on excess returns over the risk-free rate; using raw returns misstates both alpha and beta.
- **D:** Annualizing multiplies the same flawed estimate by a constant; it cannot fix the omitted factors.

Q124

AnalystPrep Q.2541

The excess returns of a fund are regressed against the excess returns of the benchmark. The slope coefficient of this regression is:

A. Alpha
B. Beta
C. The residual error
D. The correlation coefficient

Answer: B

Step by step

1. Regression: $R_p - R_f = \alpha + \beta(R_B - R_f) + \varepsilon$.
2. The intercept is alpha, the slope on the benchmark's excess return is beta (the fund's sensitivity to the benchmark), and the leftover is the residual ε.
3. The slope equals $\rho\,\sigma_p/\sigma_B$, which is beta, not the correlation ρ itself.

Why B is correct: The slope measures how many units the fund's excess return moves per unit of benchmark excess return, which is beta.

Why the others are wrong

- **A:** Alpha is the intercept (return unexplained by the benchmark), not the slope.
- **C:** The residual error is the regression noise term, not a coefficient.
- **D:** Correlation is bounded by [-1, 1]; the slope is ρ scaled by σ_p/σ_B, so the two differ unless volatilities are equal.

Q125 AnalystPrep Q.2544

A manager's excess returns are regressed against the excess returns of the manager's peer group. The regression output is used to determine:

A. Whether the manager demonstrates skill over and above the peer group
B. Whether the manager's above-average returns result from unauthorized trades
C. The absolute level of leverage employed by the manager compared with the peer group
D. The absolute level of return generated by the manager

Answer: A

Step by step

1. The regression is $R_p - R_f = \alpha + \beta(R_{peer} - R_f) + \varepsilon$, with the peer group acting as the benchmark.
2. The intercept (alpha), if statistically significant, is the return in excess of what the peer-group exposure explains, i.e. evidence of skill relative to peers.
3. The regression says nothing about trade authorization, and beta is only a relative sensitivity (not an absolute leverage level); raw returns are inputs, not an output.

Why A is correct: Alpha from regressing on the peer group measures manager skill beyond what peer-group exposure explains.

Why the others are wrong

- **B:** A regression of returns cannot detect compliance breaches or unauthorized trades.
- **C:** Beta shows relative sensitivity to the peer group, not the absolute amount of leverage.
- **D:** Absolute return is an input observed directly; the regression measures relative (excess) performance.

Q126 BT Mock Q5 · BT answers P2.T9.21.1.2 · BT notes P2.T9.21.1.2

Richard ran a factor regression for his firm's Flagship Portfolio using the three-factor Fama-French specification (MKT, SMB = small minus big, HML = high minus low, i.e. value minus growth). Results are based on monthly excess returns, R(p) - Rf:

Variable	Coefficient	Standard Error
Alpha	0.300%	0.033%
MKT loading	1.250	0.096
SMB loading	0.640	0.043
HML loading	-0.920	0.066
Adjusted R2	0.290	-

Each of the following statements about these results is TRUE EXCEPT which one is false?

A. The portfolio's style appears to be growth-oriented, small-cap.
B. The portfolio manager appears to be skillful; that is, the alpha is significant.
C. The low adjusted R2 and low standard errors imply these factors are not appropriate to this portfolio.
D. The implied CAPM-type benchmark is leveraged: it borrows cash (T-bills) to invest more than 100% in the market portfolio.

Answer: C

Step by step

1. Alpha t-stat = 0.300/0.033 = 9.09, far above 2: alpha is significant (B true).
2. SMB = +0.64 with t = 0.64/0.043 = 14.9: small-cap tilt. HML = −0.92 with t = 0.92/0.066 = 13.9: growth tilt (A true).
3. MKT loading 1.25 (t = 13.0): benchmark holds 1.25 in the market financed by a -0.25 T-bill position, i.e. leverage (D true).
4. All loadings are highly significant (small standard errors) and an adjusted R2 of 0.29 is respectable for a single portfolio, so the factors ARE relevant. C is false.

Why C is correct: Low standard errors mean the loadings are precisely estimated and significant, which supports, not undermines, the factors' relevance.

Why the others are wrong

- **A:** True statement: positive SMB = small-cap, negative HML = growth.
- **B:** True statement: t-stat of about 9 means the alpha is significant.
- **D:** True statement: a market loading of 1.25 requires borrowing 0.25 at the risk-free rate to hold 1.25 in the market.

Q127

BT answers P2.T8.705.1 · BT notes P2.T9.705.1

REBUILT — Original option C was borderline (FF alpha t-stat 1.958 is just under 1.96, so 'insignificant alphas' was arguable); reworded C to a clearly false statement about R2 and fixed the OCR 0.570 vs 0.510 inconsistency.

Andrew Ang regressed Berkshire Hathaway's monthly excess returns (1990 to 2012) first on a CAPM benchmark and then on the Fama-French three-factor benchmark:

CAPM benchmark	Coefficient	Standard Error
Alpha	0.720%	0.356%
Beta	0.510	0.0783
Adjusted R2	0.140	-
Fama-French benchmark	Coefficient	Standard Error
Alpha	0.650%	0.332%
MKT loading	0.670	0.075
SMB loading	-0.500	0.102
HML loading	0.380	0.108
Adjusted R2	0.270	-

All of the following statements about these results are true EXCEPT which one is false?

A. The SMB loading of -0.50 and HML loading of +0.38 imply Berkshire is a large-cap, value investor.

B. Berkshire's Fama-French alpha is significant at the 90% confidence level, and adding SMB and HML improves the fit of the model.

C. The rise in adjusted R2 from 0.14 to 0.27 shows that SMB and HML add no explanatory power, so the CAPM benchmark is adequate.

D. The Fama-French benchmark implied by the estimates is USD 0.33 in T-bills + USD 0.67 in the market + (long USD 0.50 large caps, short USD 0.50 small caps) + (long USD 0.38 value, short USD 0.38 growth).

Answer: C

Key correction: Source key (BT) said C, which is also correct here; the source's option C wording (statistically insignificant alphas) was borderline, so it was rewritten.

Step by step

1. Signs: SMB < 0 = large-cap bias; HML > 0 = value focus, so A is true.
2. Alpha t-stat: $0.650/0.332 = 1.96$ (CAPM: $0.720/0.356 = 2.02$); 1.96 exceeds the 90% critical value of 1.645. Adjusted R2 rises from 0.14 to 0.27, so the fit improves: B true.
3. Benchmark weights: market $= 0.67$, T-bills $= 1 - 0.67 = 0.33$, plus long-short SMB (-0.50: long large / short small) and HML (+0.38: long value / short growth) overlays: D true.
4. Adjusted R2 nearly doubles, so SMB and HML clearly add explanatory power; C says the opposite and is false.

Why C is correct: Adjusted R2 rising from 0.14 to 0.27 shows that SMB and HML add explanatory power, so the multifactor benchmark fits better than CAPM.

Why the others are wrong

- **A:** True: negative SMB means co-movement with large caps; positive HML means value.
- **B:** True: alpha t-stat of about 1.96 exceeds 1.645, and adjusted R2 improves.
- **D:** True: market loading 0.67 funds the rest (0.33) in T-bills so weights sum to 1, with long-short size and value overlays.

Q128

BT answers P2.T8.705.2 · BT notes P2.T9.705.2

Below are Andrew Ang's regression results for the annual gross returns of the CalPERS pension fund against a passive portfolio of index funds in stocks and bonds. The returns are gross returns, not excess returns (they are NOT net of the risk-free rate).

CalPERS gross annual returns, R(p), versus benchmark portfolio	Coefficient	Standard Error (SE)
Alpha, α	-1.110%	0.957%
Bond Loading, β(B)	0.320	0.0229
Stock Loading, β(S)	0.680	0.0487
Adjusted R^2	0.900	

Which of the following statements about these regression results is TRUE?

A. The high adjusted R^2 validates a hypothesis that the CalPERS active fund managers do add value relative to the benchmark.

B. Because the factor loadings are not statistically significant, different factors should be repeatedly tested until a set is found that is significant.

C. This regression should be re-run with at least one additional factor because robust benchmark portfolios must include at least one risk-free asset.

D. It is acceptable to exclude risk-free assets and regress gross returns against the benchmark portfolio subject to the constraint that the factor loadings sum to one; here $\beta_B + \beta_S = 1.0$.

Answer: D

Step by step

1. The dependent variable is the gross return, so no risk-free asset is needed as long as the benchmark weights sum to 100%: the benchmark is a fully invested blend of stocks and bonds.
2. Check the constraint: $\beta_B + \beta_S = 0.320 + 0.680 = 1.000$, so the loadings are valid portfolio weights (32% bonds, 68% stocks).
3. Alpha is measured against this benchmark: $t = -1.110/0.957 = -1.16$, which is not significant at 95% (|t| < 2) and is in any case negative.
4. The high adjusted R^2 (0.90) only says the benchmark explains most of the fund's return variation; it says nothing about value added.

Why D is correct: With gross returns, the benchmark can omit the risk-free asset provided the loadings are constrained to sum to one. Here 0.32 + 0.68 = 1.0, so the regression is a valid benchmark construction.

Why the others are wrong

- **A:** High R^2 means the fund behaves like its passive benchmark; value added is measured by alpha, which is negative (-1.11%) and statistically insignificant (t = -1.16). Confuses fit with skill.
- **B:** The loadings are highly significant (t = 0.320/0.0229 = 13.97 and 0.680/0.0487 = 13.96), and repeatedly searching for factors that give significance is data mining, not valid practice.
- **C:** A risk-free asset is not required when gross returns are used and the loadings sum to one; it is needed only when using excess returns with unconstrained loadings.

Q129 Schweser Module Quiz 85.2 Q1

Why would an investor include multiple factors in a regression study?
I. To attempt to improve the adjusted R^2 measure.
II. To reduce the t-statistic values on the respective regression coefficients.

A. I only.

B. II only.

C. Both I and II.

D. Neither I nor II.

Answer: A

Step by step

1. Additional relevant factors (e.g., SMB and HML added to the market factor) can raise explanatory power, so adjusted R^2 may rise (Berkshire: 0.14 to 0.27). Statement I is true.
2. The aim is also a benchmark more representative of the portfolio's style, with potentially more significant coefficients; reducing t-statistics is never a goal. Statement II is false.
3. Only I is true: answer A.

Why A is correct: Multifactor regressions are used to improve explanatory power (adjusted R^2) and find a more representative benchmark; lowering t-statistics is not an objective.

Why the others are wrong

- **B:** Reversed: lower t-statistics would weaken, not support, the case for a factor.
- **C:** Statement II is false, since a lower t-statistic means less significance, which no investor seeks.
- **D:** Statement I is true; adding factors can raise adjusted R^2, as in the Berkshire Fama-French example.

Q130 AP Mock 2 Q31

James is a portfolio manager of the ABC Fund. He is constructing a blended benchmark from the S&P 500 and a risk-free portfolio of T-bills. The CAPM regression of the fund's excess returns yields:

Coefficient	Estimate	t-statistic
Alpha	0.65%	2.45
MKT loading	0.79	6.70
Adjusted R^2	0.17	

Which of the following best describes the benchmark?

A. 0.65 + 0.79 RTbill + 0.79 RSP500
B. 0.65 + 0.21 RTbill + 0.79 RSP500
C. 0.21 RTbill + 0.79 RSP500
D. 0.79 RTbill + 0.79 RSP500

Answer: C

Step by step

1. The market loading is the weight on the S&P 500: 0.79.
2. Weights must sum to one, so the T-bill weight is $1 - 0.79 = 0.21$.
3. Alpha (0.65%) is the return earned above the benchmark; it is not part of the benchmark.
4. Benchmark: $0.21\,R_{Tbill} + 0.79\,R_{S\&P500}$.

Why C is correct: A single-factor CAPM benchmark is a blend of the market with weight β and the risk-free asset with weight 1 - β; alpha is the excess over this blend and is excluded.

Why the others are wrong

- **A:** Includes alpha in the benchmark and puts 0.79 on T-bills, so the weights total 1.58 rather than 1.
- **B:** Includes alpha in the benchmark; alpha is the manager's out-performance relative to the benchmark, not a component of it.
- **D:** Sets the T-bill weight equal to β instead of 1 - β; the weights sum to 1.58.

Q131 AnalystPrep Q.2421

A fund manager regresses the monthly excess return of ABC Corporation, an asset management company, against the excess market return for the past 5 years. The output is:

Parameter	Coefficient
Alpha	0.82%
Beta	0.69

An investor wants to replicate ABC Corporation's market-related returns without holding its stock. Which portfolio mimics the stock of ABC Corp?

A. 25% T-bills and 75% market portfolio
B. 35% T-bills and 65% market portfolio
C. 31% T-bills and 69% market portfolio
D. 33% T-bills and 67% market portfolio

Answer: C

Step by step

1. Market weight = beta = 0.69, i.e., 69%.
2. T-bill weight = $1 - 0.69 = 0.31$, i.e., 31%.
3. Alpha of 0.82% is the excess above this replicating portfolio, so it is not replicated.

Why C is correct: The replicating (benchmark) portfolio holds β in the market and 1 - β in the risk-free asset.

Why the others are wrong

- **A:** 75/25 corresponds to a beta of 0.75, not 0.69.
- **B:** 65/35 corresponds to a beta of 0.65, not 0.69.
- **D:** 67/33 corresponds to a beta of 0.67, not 0.69.

Q132 BT notes P2.T9.18.1

REBUILT — Original distractors were unexplained numbers; replaced with options produced by realistic mistakes while keeping the same data and correct answer.

The single-index model describes the excess returns of two securities: R(A) = 1.0% + 0.80 R(M) + e(A) and R(B) = 2.0% + 1.10 R(M) + e(B). R(M) is the excess return of the market index, with volatility of 10.0%. The residual volatilities are 15.0% for e(A) and 30.0% for e(B), and residuals are uncorrelated with each other and with the market. What is the correlation between securities A and B? (Hint: Cov(A,B) = β(A)β(B)Var(M).)

A. 0.1956
B. 0.1620
C. 0.3442
D. 0.4706

Answer: B

Step by step

1. $\sigma_A = \sqrt{0.8^2(0.10)^2 + 0.15^2} = \sqrt{0.0064 + 0.0225} = 17.00\%$.
2. $\sigma_B = \sqrt{1.1^2(0.10)^2 + 0.30^2} = \sqrt{0.0121 + 0.09} = 31.95\%$.
3. $\text{Cov}(A, B) = 0.8 \times 1.1 \times 0.10^2 = 0.0088$.
4. $\rho = 0.0088/(0.1700 \times 0.3195) = 0.1620$.

Why B is correct: Only the common market factor links the two securities, so covariance is the product of betas times market variance, divided by the product of total volatilities.

Why the others are wrong

- **A:** Divided by the residual volatilities only (0.15 × 0.30 = 0.045) instead of total volatilities: 0.0088/0.045 = 0.1956.
- **C:** Used A's systematic volatility (0.08) instead of its total volatility: 0.0088/(0.08×0.3195) = 0.3442.
- **D:** Used B's systematic volatility (0.11) instead of its total volatility: 0.0088/(0.17×0.11) = 0.4706.

Q133 BT notes P2.T9.18.2

REBUILT — Original distractors were unexplained numbers; replaced with options from realistic mistakes (R-squared reported as correlation, wrong total variance, beta as correlation).

Over a measurement period, a portfolio had a monthly alpha of 50 basis points and a beta of 0.630 relative to a single-factor market index. The monthly standard deviation of the portfolio's residual (non-systematic) return was 5.0% and the monthly standard deviation of the market index was 9.0%. What was the correlation between the portfolio and the market index? (Hint: R^2 = ESS/TSS.)

A. 0.551
B. 0.563
C. 0.750
D. 0.630

Answer: C

Step by step

1. Explained variance: $\beta^2\sigma_M^2 = 0.63^2 \times 0.09^2 = 0.003215$.
2. Residual variance: $0.05^2 = 0.0025$; total variance = 0.005715.
3. $R^2 = 0.003215/0.005715 = 0.5625$.
4. Correlation = $\sqrt{0.5625} = 0.750$ (positive since beta > 0).

Why C is correct: In a single-factor regression, R^2 is the squared correlation, so the correlation is the square root of explained over total variance.

Why the others are wrong

- **A:** Used $\sigma_M^2 + \sigma_e^2$ as total variance (market variance without beta): 0.63×0.09/√(0.0081+0.0025) = 0.551.
- **B:** Reported $R^2 = 0.5625$ as the correlation without taking the square root.
- **D:** Took the beta (0.63) as the correlation; beta equals $\rho\,\sigma_p/\sigma_M$, not the correlation.

LO 85.f

Explain how to use style analysis to handle time-varying factor exposures.

Q134 BT answers P2.T8.706.1 · BT notes P2.T9.706.1

REBUILT — Equation was garbled in both OCR copies; rewrote it cleanly with the sum-to-one restriction stated, and kept the EXCEPT question.

A risk committee evaluates fund performance with a static three-factor Fama-French regression. A proposal would replace it with style analysis, illustrated by the following benchmark for next-period return, estimated each month using information up to time t:

$R_{t+1} = \alpha_t + \beta_{SPY,t}\, R_{t+1}^{SPY} + \beta_{SPYV,t}\, R_{t+1}^{SPYV} + \beta_{SPYG,t}\, R_{t+1}^{SPYG} + \varepsilon_{t+1}$, with $\beta_{SPY,t} + \beta_{SPYV,t} + \beta_{SPYG,t} = 1$

where SPY, SPYV and SPYG are the SPDR S&P 500, S&P 500 Value and S&P 500 Growth ETFs. Each of the following statements about style analysis versus a static benchmark is true EXCEPT which is false?

A. A drawback of style analysis is that it cannot introduce short positions into the benchmark.

B. A drawback of style analysis is that statistical significance is harder to detect because of larger standard errors.

C. An advantage of style analysis is that factor loadings can vary over time, as shown by $\beta_{SPY,t}$ rather than β_{SPY}.

D. An advantage of style analysis is that actual tradeable funds, such as SPDR ETFs, can be used as the factor benchmark.

Answer: A

Step by step

1. Style analysis uses tradeable funds (SPY, SPYV, SPYG) as factors, so D is true.
2. The loadings carry a time subscript t and are re-estimated each month with information up to t, so C is true.
3. Re-estimating loadings over changing windows uses less data per estimate, giving larger standard errors and weaker significance, so B is true.
4. The only restriction is that the loadings sum to one; nothing prevents a negative (short) loading, so A is false.

Why A is correct: Style analysis imposes only a sum-to-one constraint on loadings, so short positions are not ruled out; the statement that they cannot be included is the false one.

Why the others are wrong

- **B:** True statement: time-varying estimation relies on less data per estimate, so standard errors are larger. Choosing it means misreading the EXCEPT format.
- **C:** True statement: time-varying loadings are the defining feature of style analysis. Choosing it means misreading the EXCEPT format.
- **D:** True statement: unlike the untradeable SMB and HML indices, ETFs can be bought. Choosing it means misreading the EXCEPT format.

Q135 BT answers P2.T9.21.1.3 · BT notes P2.T9.21.1.3

In contrast to factor regressions that assume constant factor weights, which of the following is the primary advantage of style analysis for determining the factor benchmark?

A. The factor benchmark uses actual tradeable funds.

B. The factor benchmark can include a momentum factor.

C. The factor benchmark naturally overcomes the beta anomaly.

D. Statistical inference is easier when the factor loadings vary over time.

Answer: A

Step by step

1. Fama-French SMB and HML are untradeable indices; style analysis instead uses tradeable assets such as the SPDR ETFs SPY, SPYV and SPYG.
2. Its loadings are re-estimated each period, so it also handles time variation.
3. Check each option: only A describes a defining advantage of style analysis over static regressions.

Why A is correct: Style analysis replaces untradeable factor indices with actual tradeable funds and lets loadings change over time.

Why the others are wrong

- **B:** A momentum factor (UMD) can be added to a static multifactor regression too, so it is not an advantage specific to style analysis.
- **C:** Style analysis does not address the beta (low-risk) anomaly; that is an unrelated empirical finding.
- **D:** Time-varying loadings are estimated from less data, so inference is harder, not easier.

LO 85.g

Describe issues that arise when measuring alphas for nonlinear strategies.

Q136

BT answers P2.T8.706.2 · BT notes P2.T9.706.2

REBUILT — Original options were loosely worded (a vague 'joint hypothesis test' option and an obviously false option D); rewrote the distractors around realistic misconceptions while keeping the same Ang-benchmark scenario.

An investment committee evaluates external managers partly on alpha. A member worries that alpha is unreliable for managers whose strategies have nonlinear payoffs, for example managers effectively short volatility. The committee requires benchmarks that meet Ang's criteria: well defined, tradeable, replicable and risk-adjusted. Which approach to measuring alpha for these strategies is the most viable?

A. Add tradeable nonlinear factors, such as option-based return series, to the benchmark regression

B. Add quadratic terms such as r_t^2 or $\max(r_t, 0)$ of the market return to the regression and treat the intercept as the tradeable alpha

C. Keep the single-factor linear regression, because alpha and the Sharpe ratio are valid for return distributions of any shape

D. Use only the residual standard error of the linear regression to adjust the alpha for the nonlinearity

Answer: A

Step by step

1. Alpha is the intercept of a linear regression, so option-like or quadratic payoffs can show up as a spurious positive alpha (and alpha ignores negative skewness).
2. Fix: model the nonlinearity explicitly using factors that capture it.
3. Ang's criteria require the benchmark to be tradeable and replicable, so the added factors must be investable (for example option strategies or tradeable nonlinear index returns).
4. Mathematical terms like r_t^2 or $\max(r_t, 0)$ are not investable positions, so an alpha measured against them is not a tradeable alpha. Only A satisfies the criteria.

Why A is correct: Including tradeable nonlinear factors removes the false alpha caused by option-like payoffs while keeping the benchmark tradeable and replicable.

Why the others are wrong

- **B:** Nonlinear terms can capture the shape, but they are not tradeable, so the benchmark fails Ang's tradeable and replicable criteria.
- **C:** Ignores the problem: alpha is estimated in a linear framework and is distorted by nonlinear payoffs and skewness.
- **D:** The residual standard error measures unexplained variation, not nonlinearity; it does not remove the false alpha in the intercept.

Q137

BT answers P2.T9.21.2.1 · BT notes P2.T9.21.2.1

Robert runs a portfolio with a focused strategy of selling put options. He regresses the portfolio's excess returns on the market's excess returns using the most recent 48 months and obtains $R_p = 0.0520 + 0.67\,R_m + \varepsilon$. He declares that the portfolio generates an alpha of +5.20%. His colleague Mary criticizes this conclusion. Which of the following is probably the BEST criticism?

A. It is impossible to evaluate nonlinear payoffs

B. Alpha is the residual term, not the regression intercept

C. The strategy's return distribution is very negatively skewed

D. Returns are distorted by the short put strategy that collects up-front cash premiums

Answer: C

Step by step

1. Alpha comes from a linear regression, but a short put has an option-like, nonlinear payoff.
2. Short put strategies earn steady premiums most of the time and suffer large losses rarely, giving a negatively skewed, non-normal return distribution.
3. Over a short 48-month sample such a strategy can show a large positive intercept that is really compensation for tail risk, not skill; alpha ignores skewness.
4. So the best criticism is the negative skewness/nonlinearity that the linear alpha cannot capture.

Why C is correct: Nonlinear short-option payoffs produce negatively skewed returns that a linear regression in-

tercept misreads as alpha.

Why the others are wrong

- **A:** Too absolute: nonlinear payoffs can be evaluated, for example by adding tradeable nonlinear factors.
- **B:** Wrong definition: alpha is the regression intercept; the residual is the error term ε.
- **D:** Collecting premiums is the source of the return, not a distortion by itself; the issue is the skewed risk behind it.

LO 85.h

Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly.

Q138

BT notes P2.T9.704.2

The low-risk anomaly is a combination of three true effects. Which of the following statements is FALSE (and not technically included in the low-risk anomaly)?

A. Both contemporaneous and lagged volatility are inversely (negatively) related to returns
B. Contemporaneous beta is inversely (negatively) related to raw returns
C. Lagged beta is inversely (negatively) related to risk-adjusted returns
D. Minimum variance portfolios do better than the market

Answer: B

Step by step

1. Volatility anomaly: higher volatility (contemporaneous or lagged) goes with lower returns and Sharpe ratios - A is true.
2. Beta anomaly: high-beta stocks do not have low raw returns; they have lower risk-adjusted returns (Sharpe fell from 0.9 to 0.4) - C is true.
3. CAPM itself predicts a positive contemporaneous beta-return relationship, and this holds; it is not part of the anomaly - B is false.
4. Minimum variance portfolios have beaten the market on a risk-adjusted basis - D is part of the anomaly.

Why B is correct: Contemporaneous beta is positively related to raw returns (as CAPM implies); the anomaly concerns risk-adjusted returns, so B is the false statement.

Why the others are wrong

- **A:** True: volatility is negatively related to returns, an element of the anomaly.
- **C:** True: lagged beta is inversely related to risk-adjusted returns (Sharpe).
- **D:** True: minimum variance portfolios have outperformed the market on a risk-adjusted basis.

Q139

AnalystPrep Q.4630

REBUILT — Original option C was arguably true and option D 'none' was a weak distractor; rewrote options around the Ang et al. evidence (returns and Sharpe ratios by volatility quintile).

Ang, Hodrick, Xing, and Zhang (2006) sorted U.S. stocks into quintiles by volatility (standard deviation). Which of the following statements best describes the volatility anomaly they documented?

A. Average returns and Sharpe ratios rise as volatility rises, with the highest-volatility quintile performing best.
B. Average returns and Sharpe ratios fall as volatility rises, with the lowest-volatility stocks performing best.
C. Average returns fall as volatility rises, but Sharpe ratios rise because volatility is in the numerator.
D. Volatility is related to returns only contemporaneously, and lagged volatility has no relation to risk-adjusted returns.

Answer: B

Step by step

1. The anomaly is an inverse relation between risk (standard deviation) and performance across volatility quintiles.
2. Average return was above 10% for the three lowest-volatility quintiles, fell to 6.8% for quintile 4 and to 0.1% for the highest-volatility quintile.
3. Sharpe ratios fell from 0.8 (lowest volatility) to 0.0 (highest volatility).
4. Least volatile stocks did best, most volatile did worst: B.

Why B is correct: Under CAPM higher risk should earn higher return, but the data show both average returns and

Sharpe ratios decline as volatility increases.

Why the others are wrong

- **A:** States the CAPM-style positive risk-return relation, the reverse of the evidence.
- **C:** Volatility is the denominator of the Sharpe ratio, not the numerator; higher volatility with lower return makes the Sharpe ratio fall further.
- **D:** Invents a contemporaneous-only relation; the study found the inverse relation in risk-adjusted performance, not a lack of one.

Q140

BT answers P2.T8.704.2

REBUILT — Original EXCEPT item rested on contemporaneous vs lagged volatility/beta nuances not supported by the Schweser text; rebuilt with options based on the reading's volatility and beta anomaly evidence.

Regarding the volatility anomaly and the beta anomaly, each of the following statements is consistent with the empirical evidence EXCEPT:

A. Across volatility quintiles, both average returns and Sharpe ratios decreased as standard deviation increased.

B. Stocks with higher betas earned lower raw returns than stocks with lower betas, which is the essence of the beta anomaly.

C. Across beta quintiles, the Sharpe ratio fell from about 0.9 for the lowest-beta stocks to about 0.4 for the highest-beta stocks.

D. Higher-beta stocks tend to have higher standard deviations, which is the denominator of the Sharpe ratio.

Answer: B

Step by step

1. The beta anomaly is about risk-adjusted performance, not raw returns: high-beta stocks do not have low returns.
2. Their lower Sharpe ratios arise because higher betas come with higher standard deviation (the denominator).
3. A (volatility anomaly), C (0.9 to 0.4) and D match the reading.
4. B misstates the beta anomaly, so B is the exception.

Why B is correct: The beta anomaly means higher-beta stocks have lower Sharpe ratios, not lower raw returns.

Why the others are wrong

- **A:** True statement of the volatility anomaly; not the exception.
- **C:** True: the Sharpe ratio dropped from 0.9 to 0.4 from lowest to highest beta quintile.
- **D:** True: the higher volatility paired with high beta drives the lower Sharpe ratio.

Q141

BT answers P2.T9.21.2.3 · BT notes P2.T9.21.2.3

REBUILT — Original relied on a missing plot and asked about 'best returns' ambiguously (leveraged D could be argued); replaced with a self-contained table and asked for the highest expected Sharpe ratio.

An analyst compares four equity portfolios (risk-free rate 6.0%):

Portfolio	Standard deviation	Beta
A	8%	0.6
B	11%	1.3
C	15%	1.0
D	24%	1.6

Assuming the volatility anomaly and the beta anomaly hold, which portfolio is expected to have the highest Sharpe ratio?

A. Portfolio A

B. Portfolio B

C. Portfolio C

D. Portfolio D

Answer: A

Step by step

1. Volatility anomaly: Sharpe ratios decline as standard deviation rises. A has the lowest standard deviation (8%).
2. Beta anomaly: Sharpe ratios decline as beta rises. A has the lowest beta (0.6).
3. A ranks best on both criteria; B, C and D have higher volatility and beta, so lower expected Sharpe ratios.

Why A is correct: Both anomalies say higher risk (standard deviation or beta) goes with lower risk-adjusted re-

turns, so the lowest-volatility, lowest-beta portfolio has the highest Sharpe ratio.

Why the others are wrong

- **B:** Takes the second-lowest volatility but ignores its high beta (1.3), which the beta anomaly penalizes.
- **C:** Applies CAPM or market-portfolio reasoning (beta 1.0 as the efficient choice), ignoring that anomalies favor lower risk.
- **D:** Assumes more risk earns more reward; highest volatility and beta give the lowest expected Sharpe ratio.

LO 85.i

Describe potential explanations for the risk anomaly.

Q142 AnalystPrep Q.4631

Which of the following is a potential reason for the low-risk anomaly?

A. Leverage constraints
B. Low alpha
C. Inappropriate benchmark
D. None of the above

Answer: A

Step by step

1. Recall Ang's explanations for the low-risk anomaly: data mining, leverage constraints, institutional constraints (short-selling and tracking-error limits), and investor preferences.
2. Leverage-constrained investors cannot borrow, so they buy high-beta stocks that carry built-in leverage.
3. This extra demand bids up high-beta prices, lowering their subsequent risk-adjusted returns and leaving low-beta stocks relatively cheap.
4. Only option A matches an accepted explanation.

Why A is correct: Leverage-constrained investors overpay for high-beta stocks, depressing their future returns relative to low-beta stocks, which produces the anomaly.

Why the others are wrong

- **B:** Low alpha is an outcome of overpriced high-risk stocks, not a cause; it is not an explanation for the anomaly itself.
- **C:** An inappropriate benchmark affects how alpha is measured, but it is not one of the reading's explanations for the risk anomaly, and the anomaly is found across many benchmarks and asset classes.
- **D:** A is a recognised explanation, so 'none of the above' is wrong; picking it reflects not knowing the leverage-constraint argument.

Q143 BT answers P2.T8.706.3 · BT notes P2.T9.706.3

Which of the following statements about the risk anomaly is TRUE?

A. The primary argument against the risk anomaly, with respect to either low volatility or low beta, is that economists have generally demonstrated that markets are efficient; i.e., efficient market hypothesis (EMH) is true
B. Due to time-varying reality, it is theoretically intractable (i.e., not possible) to create a reproducible benchmark for either the low beta or low volatility risk anomalies such that empirical tests of the risk anomalies are not robust and the discussion remains "largely theoretical"
C. The risk anomaly might be explained by investors who are leverage constrained (i.e., who cannot borrow so instead bid up high beta stocks) and/or have an "agency problem" created by a need to minimize tracking error with the benchmark (e.g., they cannot short low volatility or short low beta)
D. The presence of the low-risk anomaly (aka, low-risk effect) in several different contexts–including U.S. equities, international stock markets, Treasury bonds, corporate bonds (across credit rating classes), commodity, option and foreign exchange markets–is compelling evidence that "data mining" almost surely creates an illusion of a relationship between idiosyncratic return volatility (IVOL) and future returns because diversification generally eliminates the impact of IVOL

Answer: C

Step by step

1. Candidate explanations for the risk anomaly: data mining, investor leverage constraints, institutional constraints (no shorting, tracking-error limits), and preferences.
2. Option C states leverage constraints and institutional agency/tracking-error constraints, both accepted explanations.
3. Option A is false: the anomaly contradicts CAPM and is not an argument that markets are efficient.
4. Option B is false: benchmarks can be built and the anomaly is documented empirically.
5. Option D is false: the anomaly appears across many markets, which weakens (not supports) the data-mining argument.

Why C is correct: Leverage-constrained investors bid up high-beta stocks, and institutional managers limited by tracking error or short-sale bans cannot arbitrage the mispricing away.

Why the others are wrong

- **A:** Reverses the logic: the anomaly is a puzzle relative to CAPM and no source shows markets are demonstrably efficient; EMH is not the 'primary argument'.
- **B:** Claims the anomaly cannot be tested; in fact it has been documented empirically with tradable low-volatility/low-beta portfolios.
- **D:** Evidence across many asset classes and regimes argues against data mining; the statement also wrongly says diversification eliminates idiosyncratic-volatility effects.

Q144 Schweser Module Quiz 85.3 Q1

Which of the following characteristics is a potential explanation for the risk anomaly?

A. Investor preferences.
B. The presence of highly leveraged retail investors.
C. Lack of short selling constraints for institutional investors.
D. Lack of tracking error constraints for institutional investors.

Answer: A

Step by step

1. Potential explanations: data mining, leverage constraints on retail investors, institutional constraints, and investor preferences.
2. Investor preferences (e.g., a taste for high-volatility stocks, disagreement) bid up high-risk assets and lower their future returns.
3. The other options misstate the constraints: the relevant ones are leverage constraints and the presence of short-selling and tracking-error constraints.

Why A is correct: A preference for high-beta, high-volatility stocks pushes their prices up and future returns down, producing the anomaly.

Why the others are wrong

- **B:** The explanation is that retail investors are leverage constrained (cannot borrow), not that they are highly leveraged.
- **C:** Reverses the argument: it is the presence of short-selling constraints that stops institutions from exploiting mispricing.
- **D:** Reverses the argument: it is the presence of tracking-error constraints that prevents institutions from arbitraging the anomaly.

Reading 86. Portfolio Construction

Richard Grinold and Ronald Kahn, Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk, 2nd Edition (New York, NY: McGraw-Hill, 2000). Chapter 14.

LO	Learning objective	Qs
86.a	describe the inputs to the portfolio construction process and explain challenges faced when using these inputs.	2
86.b	evaluate the motivation for and the methods used for refining alphas in the implementation process.	7
86.c	describe neutralization and the different approaches used for refining alphas to be neutral.	6
86.d	explain the implications of transaction costs on portfolio construction.	2
86.e	describe practical issues in portfolio construction, including the determination of an appropriate risk aversion, aversions to specific risks, and proper alpha coverage.	3
86.f	describe portfolio revisions and rebalancing, and analyze the tradeoffs between alpha, risk, transaction costs, and time horizon.	3
86.g	determine the optimal no-trade region for rebalancing with transaction costs.	2
86.h	evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming.	5
86.i	describe dispersion, explain its causes, and describe methods for controlling forms of dispersion.	4

LO 86.a

Describe the inputs to the portfolio construction process and explain challenges faced when using these inputs.

Q145

Schweser Module Quiz 86.1 Q1

Which of the following inputs to the portfolio construction process can be measured with the most certainty?

A. Position alphas.
B. Transaction costs.
C. The current portfolio.
D. Active risk aversion.

Answer: C

Step by step

1. Inputs to portfolio construction: current portfolio, alphas, covariances, transaction costs, and active risk aversion.
2. Alphas, covariances, and transaction costs are all forecasts or estimates and are subject to estimation error or bias.
3. Active risk aversion is a preference parameter that must be inferred and matched to the manager's target active risk.
4. The current portfolio (assets and weights) is directly observable, so it is measured with the most certainty.

Why C is correct: The current portfolio is a known holding, whereas every other input must be estimated or judged and is therefore subject to error.

Why the others are wrong

- **A:** Alphas are forecasts of expected excess returns and carry forecast error and bias, so they cannot be measured with certainty.
- **B:** Transaction costs are estimated (and rise with trading frequency), so they are subject to estimation error.
- **D:** Active risk aversion is a subjective preference that is hard to calibrate to the target level of active risk, not an observable quantity.

Q146

AnalystPrep Q.2441

REBUILT — Distractor C (managers face no constraints) was a giveaway and D was weak; rewrote options so each wrong answer reflects a realistic misconception, and remapped from 86.e to 86.a (constraints as a portfolio-construction challenge).

A fund manager makes the following comment: "Active management should be easy with the right alphas." Which assessment is most appropriate?

A. Incorrect: real portfolios are built under constraints (e.g., no short sales, cash and position limits), which make the constrained portfolio less efficient than the unconstrained optimum even when the alphas are right.
B. Correct: constraints change the weights but do not make the portfolio any less efficient, so the right alphas guarantee the optimal result.

C. Correct: refining the alphas for each constraint before optimizing fully eliminates the cost of the constraints, so the constrained portfolio is as efficient as the unconstrained one.
D. Correct: because most active managers do not short and hold little cash, their portfolios are more efficient than unconstrained portfolios.

Answer: A
Step by step
1. Unconstrained, a manager with the right alphas, risks, and costs could simply optimize and obtain the most efficient portfolio.
2. In practice, client and manager constraints (no shorting, cash limits, position/sector limits, tracking error limits) restrict the feasible set.
3. Any restriction on the feasible set cannot raise, and generally lowers, the efficiency (risk-adjusted alpha) of the portfolio.
4. Hence the statement is incorrect: even with right alphas, active management is not easy once constraints bind.

Why A is correct: Constraints shrink the set of feasible portfolios, so the constrained optimum is less efficient than the unconstrained one; correct alphas alone do not make active management easy.
Why the others are wrong
- **B:** Wrong: claims constraints cost nothing; a restricted feasible set cannot be more efficient than an unrestricted one.
- **C:** Wrong: refining alphas is a way to impose a constraint within an unconstrained optimizer, not a way to remove its cost; the efficiency loss remains.
- **D:** Wrong: reverses the effect of constraints; no-short and cash limits are constraints that reduce, not increase, efficiency.

LO 86.b

Evaluate the motivation for and the methods used for refining alphas in the implementation process.

Q147 Mock 2 Q62

An analyst regresses the returns of 500 stocks against the S&P 500. The resulting pool of 500 alphas has a residual risk of 18% and an information coefficient of 10%. Assuming the alphas are normally distributed with a mean of 0%, roughly how many stocks have an alpha greater than 3.6% or less than -3.6%?
A. 5
B. 250
C. 475
D. 25

Answer: D
Step by step
1. Scale of alphas = volatility × IC = $18\% \times 0.10 = 1.8\%$.
2. Threshold: $3.6\%/1.8\% = 2$ standard deviations.
3. Two-tailed tail probability: $2 \times (1 - N(2)) = 2 \times 0.02275 = 4.55\%$.
4. Count: $500 \times 4.55\% = 22.75$, which is closest to 25.

Why D is correct: Alphas have mean 0 and standard deviation equal to residual risk × IC, so ±3.6% is ±2 standard deviations, outside which about 4.6% of a normal distribution lies.
Why the others are wrong
- **A:** Too few: 5 stocks is only 1% of the pool, whereas about 4.6% lie beyond 2 standard deviations (5 would be nearer a 3-standard-deviation cutoff, $500 \times 0.27\% \approx 1.35$).
- **B:** Half of the stocks: the count on one side of the mean, not beyond 2 standard deviations.
- **C:** The complement: the stocks within ±2 sd ($500 \times 95.45\% \approx 477$) rather than outside.

Q148 AnalystPrep Q.2442

Which of the following statements is (are) correct?
I. Any complex portfolio construction process can be replaced by a process which first refines the alphas and uses a simple unconstrained mean/variance optimization to determine the active positions.
II. Simple models are always better than complicated implementation schemes.

A. Both I and II
B. None of the above
C. Only I
D. Only II

Answer: C

Step by step

1. Statement I: refining alphas (to embed investor and manager constraints) followed by a simple unconstrained mean-variance optimization can replace even the most sophisticated construction process. True.
2. Statement II: 'always' is unsupported; simple models are not universally better than complex implementation schemes. False.
3. Only I is correct.

Why C is correct: Refining the alphas lets constraints be reflected in the inputs, so a simple unconstrained optimizer can replace complex constrained procedures; the blanket claim in II is false.

Why the others are wrong

- **A:** Accepts II, an over-generalization ('always').
- **B:** Rejects I, though it is the stated result in the reading.
- **D:** Accepts II and rejects I, the reverse of the correct evaluation.

Q149 AnalystPrep Q.2444 · AnalystPrep Q.2744

Which of the following is NOT a procedure used to refine alphas?

A. Trimming
B. Interpolation
C. Scaling
D. Neutralization

Answer: B

Step by step

1. Alpha refinement procedures: neutralization (remove unintended biases/bets), scaling (give alphas the appropriate scale, volatility × IC) and trimming (reduce or remove extreme alphas).
2. Interpolation is not an alpha-refinement procedure.

Why B is correct: Refinement consists of neutralizing, scaling and trimming alphas; interpolation is a numerical technique not used for this purpose.

Why the others are wrong

- **A:** Trimming is a standard refinement: large alphas are examined and zeroed or capped.
- **C:** Scaling is a standard refinement: alphas are rescaled to volatility × IC.
- **D:** Neutralization is a standard refinement: alphas are adjusted to remove unwanted bets.

Q150 AnalystPrep Q.2457

REBUILT — Original options were loosely worded (option A 'standardized measure' and vague distractors); rewrote options so one is clearly the Grinold-Kahn definition and distractors are the other refinement techniques.

In refining alphas for portfolio construction, what is the primary purpose of scaling the alphas?

A. To give the alphas the appropriate scale, with a standard deviation approximately equal to residual volatility × information coefficient
B. To reduce or set to zero extremely large alphas that may stem from questionable data
C. To remove unintended bets, such as a beta or industry tilt, from the alphas
D. To adjust each alpha for the transaction costs of establishing the position

Answer: A

Step by step

1. Alpha = volatility × IC × score, with score ~ mean 0, sd 1.
2. So alphas should have mean 0 and scale ≈ volatility × IC (e.g., 0.10 × 30% = 3%).
3. If constraints or adjustments change the scale, alphas are rescaled to restore it; this is scaling.

Why A is correct: Scaling restores alphas to the scale implied by the manager's skill (IC) and the asset's residual risk, so the optimizer sees consistent magnitudes.

Why the others are wrong

- **B:** Describes trimming, not scaling.
- **C:** Describes neutralization, not scaling.
- **D:** Transaction costs are handled in the optimization/rebalancing decision, not by scaling alphas.

Q151

BT answers P2.T9.21.3.2 · BT notes P2.T9.21.3.2

REBUILT — Options rebuilt so each distractor stems from a specific error (the original 2.40% and 60.0% had no clear derivation).

According to Grinold, information coefficients (IC) are correlations but in practice they are small. An Average Manager has an IC of zero, a Good Manager has an IC of 0.060 and a Great Manager has an IC of 0.120. A certain stock has a residual volatility of 30.0%. A Great Manager has a stock recommendation about which she is highly positive, quantified by a z-score (score) of 1.50. What is the implied alpha of this recommendation?

A. 2.70%
B. 3.60%
C. 5.40%
D. 45.0%

Answer: C

Step by step

1. Alpha = volatility × IC × score.
2. 30% × 0.120 × 1.50 = 5.40%.

Why C is correct: Alpha is residual volatility times IC (the scale) times the score.

Why the others are wrong

- **A:** Used the Good Manager's IC of 0.060: 30% × 0.06 × 1.5 = 2.70%.
- **B:** Omitted the score: 30% × 0.12 = 3.60%.
- **D:** Omitted the IC: 30% × 1.5 = 45%.

Q152

BT answers P2.T9.21.3.3 · BT notes P2.T9.21.3.3

Kelly is constructing a portfolio where her initial unconstrained optimization generates two outcomes that violate her fund's style policy. The optimization implies several short positions and over-weights in three positions that would exceed her self-imposed concentration limit (positions cannot exceed benchmark holdings by more than 10.0%). Kelly refines the alphas by scaling them and then trimming the extreme alphas. What is the most likely reason for Kelly to refine the alphas?

A. Performance: the refined alphas generate higher information ratios (IR)
B. Precision: the active risk aversion coefficient is more accurately calibrated
C. Sophistication: the refined alphas are more conducive to quadratic programming
D. Simplification: to be able to size the active positions with a simple unconstrained mean-variance optimization

Answer: D

Step by step

1. Constraints (no shorts, concentration limits) normally require a complicated constrained optimization.
2. Refining alphas (scaling, trimming) builds the constraints into the inputs, so a simple unconstrained mean-variance optimization can be used.
3. Hence the motive is simplification.

Why D is correct: Refined alphas embed constraints so that a simple unconstrained optimizer yields acceptable positions.

Why the others are wrong

- **A:** Refining for constraints typically lowers the IR (alphas are altered from the manager's raw forecasts); it does not aim to raise it.
- **B:** Refinement does not calibrate risk aversion.

- **C:** The aim is to avoid complex constrained (quadratic) programming, not to feed it.

Q153 Schweser Module Quiz 86.1 Q2

Which of the following is correct with respect to adjusting the optimal portfolio for portfolio constraints?

A. No reliable method exists.

B. By refining the alphas and then optimizing, it is possible to include constraints of both the investor and the manager.

C. By refining the alphas and then optimizing, it is possible to include constraints of the investor, but not the manager.

D. By optimizing and then refining the alphas, it is possible to include constraints of both the investor and the manager.

Answer: B

Step by step

1. Constraints can be embedded by adjusting the alphas first (e.g., alphas that produce non-negative weights for a no-short-sales client).
2. The adjusted alphas are then run through an unconstrained optimization.
3. This handles both investor and manager constraints.

Why B is correct: Refining alphas before optimizing can accommodate both investor and manager constraints, replacing even sophisticated constrained processes.

Why the others are wrong

- **A:** A reliable method exists: refine the alphas.
- **C:** The method covers manager constraints (e.g., position limits) as well.
- **D:** Wrong order: alphas must be refined before the optimization.

LO 86.c

Describe neutralization and the different approaches used for refining alphas to be neutral.

Q154 AnalystPrep Q.2443

REBUILT — Original was a purely verbal item whose options A and B (cap-weighted vs simple average) were not clearly distinguishable under Schweser wording; replaced with a numeric industry-neutralization item that tests the same method.

An industry contains three stocks with benchmark (capitalization) weights within the industry of 50%, 30% and 20% and raw alphas of 4%, 2% and -1%, respectively. The manager wants alphas that are neutral to the industry risk factor, so that the active portfolio has no active industry bet relative to the benchmark. After neutralization, the alpha of the second stock (raw alpha 2%) is closest to:

A. -0.40%

B. 0.33%

C. 0.40%

D. 4.40%

Answer: A

Step by step

1. Industry neutralization subtracts the industry's capitalization-weighted average alpha from each alpha in that industry.
2. Cap-weighted average alpha: $0.50(4\%) + 0.30(2\%) + 0.20(-1\%) = 2.0\% + 0.6\% - 0.2\% = 2.4\%$.
3. Neutralized alpha of stock 2: $2\% - 2.4\% = -0.40\%$.
4. Check: the cap-weighted average of the neutralized alphas is 0, so the industry bet is removed.

Why A is correct: Subtracting the cap-weighted industry average alpha makes the benchmark-weighted alpha of the industry zero, so the active portfolio carries no industry bet.

Why the others are wrong

- **B:** Subtracted the simple (equal-weighted) average of 1.67% instead of the cap-weighted average: $2\% - 1.67\% = 0.33\%$. This leaves a residual industry bet.
- **C:** Sign error: computed $2.4\% - 2\% = +0.40\%$ (average minus alpha).

- **D:** Added the industry average instead of subtracting it: $2\% + 2.4\% = 4.4\%$, which doubles the industry bet.

Q155 AnalystPrep Q.2445

A portfolio manager is evaluating an active position in a stock with the following characteristics: active position alpha of 0.6%, active position beta of 1.3, benchmark alpha of 0.020%, and benchmark beta of 1.0. The manager wants to neutralize the active position with respect to the benchmark beta. After the neutralization, what is the modified benchmark-neutral alpha of the active position?

A. 0.54%
B. 0.57%
C. 0.62%
D. 0.66%

Answer: B

Step by step

1. Modified alpha = position alpha - (benchmark alpha × position beta).
2. Adjustment: $0.020\% \times 1.3 = 0.026\%$.
3. Modified alpha: $0.6\% - 0.026\% = 0.574\% \approx 0.57\%$.

Why B is correct: Benchmark neutralization removes the portion of alpha attributable to the benchmark's own alpha times the position's beta.

Why the others are wrong

- **A:** Subtracts an overstated adjustment (about 0.06% instead of 0.026%); it does not follow from the formula.
- **C:** Added the benchmark alpha without multiplying by beta (0.6% + 0.02%): wrong sign and omits the beta.
- **D:** Adds an overstated adjustment instead of subtracting the 0.026%; sign error.

Q156 AnalystPrep Q.3180

James Simpson, a balanced fund portfolio manager, wants to make his portfolio alphas benchmark-neutral. One position, Tara Industries, has an alpha of 1.9%, an information coefficient of 0.23, and a beta of 1.53. The benchmark has an alpha of 0.9%. The modified alpha that makes the position benchmark-neutral is closest to:

A. 0.523%
B. 0.46%
C. 3.277%
D. 2.754%

Answer: A

Step by step

1. Modified alpha = position alpha - (benchmark alpha × position beta).
2. Adjustment: $0.9\% \times 1.53 = 1.377\%$.
3. Modified alpha: $1.9\% - 1.377\% = 0.523\%$.
4. The information coefficient is irrelevant to this calculation.

Why A is correct: The neutralized alpha strips out the beta-scaled benchmark alpha from the raw alpha.

Why the others are wrong

- **B:** Does not follow from the formula; it results from applying a mis-sized adjustment (about 1.44%) rather than beta × benchmark alpha = 1.377%.
- **C:** Added instead of subtracted: $1.9\% + 1.377\% = 3.277\%$.
- **D:** Doubled the benchmark alpha in the adjustment: $1.53 \times 1.8\% = 2.754\%$, which is a mis-scaled adjustment term, not the modified alpha.

Q157 BT answers P2.T9.21.4.2 · BT notes P2.T9.21.4.2

Robert is a value manager who picks among stocks with a relatively low EV/EBIT. However, he observes that these ratios are biased by industry; for example, homebuilders have much lower ratios than technology companies. How can he neutralize his alphas so they are not biased by industry; that is, how can he achieve alphas that are risk-factor-neutral?

A. If we assume industry is a risk factor, he should forecast the risk factors
B. He can subtract each industry's capitalization-weighted average alpha from his initial alphas

C. He can conduct a three-step screen: rank stocks by alpha, choose the first X stocks, and then equal-weight those stocks
D. His best approach is quadratic programming (QP) with the maximum number of inputs

Answer: B
Step by step
1. The bias is an unintended industry bet in the alphas, so the alphas must be neutralized with respect to industry (a risk factor).
2. Group stocks by industry, compute each industry's capitalization-weighted average alpha, and subtract it from every alpha in that industry.
3. Result: the benchmark-weighted alpha of each industry is zero, so no active industry bets arise.

Why B is correct: Subtracting the cap-weighted industry average alpha removes the industry component, leaving only stock-specific information.
Why the others are wrong
- **A:** The manager does not forecast risk factors (no ability to add value there); the aim is to remove factor bets, not forecast them.
- **C:** A screen is a portfolio construction technique, and equal-weighting does nothing to remove the industry bias in alphas.
- **D:** QP is an optimization method; more inputs do not neutralize alphas and may amplify the bias.

Q158
GARP PE2 Q63

A portfolio manager is constructing an equity portfolio that will include several equities from multiple industries. The manager aims to achieve mean/variance optimization and asks a group of analysts to perform alpha analysis and also to estimate other portfolio inputs. As part of the alpha analysis, the analysts adjust alphas based on the portfolio's constraints and exclude very large positive and negative alphas from their calculations. Additionally, the analysts plan to remove biases or undesirable bets from the alphas. Which of the following procedures should the analysts apply to execute this plan?
A. Stratification
B. Neutralization
C. Scaling
D. Trimming

Answer: B
Step by step
1. Excluding very large positive and negative alphas is trimming, already performed in the stem.
2. The remaining plan, removing biases and undesirable bets from alphas, is the definition of neutralization (benchmark, cash, or risk-factor).
3. Answer: neutralization.

Why B is correct: Neutralization is the refinement that removes biases and unintended bets (e.g., beta, industry or cash) from alphas.
Why the others are wrong
- **A:** Stratification is a portfolio construction technique (sorting stocks into cells), not an alpha refinement.
- **C:** Scaling adjusts the magnitude (standard deviation) of alphas to match volatility × IC; it does not remove biases.
- **D:** Trimming handles extreme values and has already been applied in the stem.

Q159
Mock 4 Q64

A quantitative team produces raw alpha forecasts for a universe of stocks. Before the forecasts are passed to the optimizer, the team applies two adjustments. First, it removes any unintended systematic exposures, for example a persistent tilt toward small capitalization stocks, that were not part of the intended strategy. Second, it caps the most extreme positive and negative alpha values at a defined number of standard deviations from the mean. Which procedures has the team applied, in that order?
A. Trimming, followed by scaling.
B. Scaling, followed by neutralization.
C. Neutralization, followed by trimming.
D. Trimming, followed by neutralization.

Answer: C
Step by step
1. Removing unintended factor exposures (e.g., a small-cap tilt) is risk-factor neutralization.
2. Capping extreme alphas at a multiple of the alpha scale is trimming.
3. In the stated order: neutralization, then trimming.

Why C is correct: Eliminating unintended factor bets is neutralization; limiting extreme alphas is trimming.
Why the others are wrong
- **A:** Neither step is scaling, and the order is wrong; the first step removes factor exposures, not outliers.
- **B:** Scaling rescales alpha volatility and is not described; the capping step is trimming, not neutralization.
- **D:** Right procedures but reversed order: the stem removes exposures first, then caps extremes.

LO 86.d

Explain the implications of transaction costs on portfolio construction.

Q160 AP Mock 1 Q61 · AnalystPrep Q.2446

A portfolio manager makes the following comments about transaction costs in portfolio construction:
I. The correct way to compare transaction costs with the annual rate of gain from alpha and the annual rate of loss from active risk is to amortize the transaction costs, with the rate of amortization depending on the anticipated holding period.
II. The annualized transaction cost is given by the round-trip cost divided by the holding period in years.
Which of the comments is/are correct?

A. Only I is correct.
B. Only II is correct.
C. Both I and II are correct.
D. Neither I nor II is correct.

Answer: C
Step by step
1. Alpha and active risk are annual quantities, while transaction costs are incurred once, when the trade is made.
2. To compare like with like, the one-time cost must be spread (amortized) over the anticipated holding period, so I is correct.
3. Annualized cost = round-trip cost / holding period in years, so II is correct. Example: a 1% round trip held 6 months (0.5 years) costs 2% per year; held 2 years it costs 0.5% per year.
4. Both statements are correct.

Why C is correct: Costs are a one-off, benefits (alpha) and risk are annual, so costs are amortized over the holding period using round-trip cost divided by holding period in years.
Why the others are wrong
- **A:** Accepts the amortization principle but wrongly rejects the formula; the annualized cost is exactly round-trip cost divided by holding period in years.
- **B:** Accepts the formula but rejects the principle; the formula is only needed precisely because costs must be amortized to be comparable with annual alpha and risk.
- **D:** Would hold only if costs could be compared with annual alpha without adjustment, ignoring the time-mismatch between a one-time cost and annual benefits.

Q161 Schweser Module Quiz 86.1 Q3

Which of the following statements most correctly describes a consideration that complicates the incorporation of transaction costs into the portfolio construction process?

A. The transaction costs and the benefits always occur in two distinct time periods.
B. The transaction costs are uncertain while the benefits are relatively certain.
C. There are no complicating factors from the introduction of transaction costs.
D. The transaction costs must be amortized over the horizon of the benefit from the trade.

Answer: D
Step by step

1. Transaction costs arise at a specific time, when the portfolio is adjusted.
2. The benefits (e.g., higher alpha) accrue over time, with an uncertain holding period.
3. The challenge is therefore to assign the cost to the projected future benefits, i.e., amortize it over the horizon of the benefit from the trade (annual cost = round-trip cost / holding period in years).

Why D is correct: A one-time cost must be matched against benefits earned over time, so it must be amortized over the benefit horizon.

Why the others are wrong

- **A:** The cost and benefits are not 'always' in distinct periods, and timing alone is not the issue; the complication is the mismatch of a lump-sum cost against a stream of benefits.
- **B:** Reverses the reality: alpha forecasts are the noisy input, while transaction costs are comparatively estimable.
- **C:** Wrong because costs introduce a genuine complication: the need to assign them to future benefits over an uncertain holding period.

LO 86.e

Describe practical issues in portfolio construction, including the determination of an appropriate risk aversion, aversions to specific risks, and proper alpha coverage.

Q162

AnalystPrep Q.2447

REBUILT — Original left the units of active risk unspecified, so both 0.05 and 5 were defensible; stem now states active risk is in percentage points (Schweser convention) and options adjusted.

A fund manager desires an active risk of 5% and the portfolio has an information ratio of 0.5. Using the convention that desired active risk is entered in percentage points (5, not 0.05), the implied active risk aversion parameter is closest to:

A. 0.50
B. 5.00
C. 0.05
D. 0.005

Answer: C

Step by step

1. Optimal active risk: $\omega^* = \text{IR}/(2\lambda)$, so $\lambda = \text{IR}/(2\omega^*)$.
2. Substitute with $\omega^* = 5$ (percentage points): $\lambda = 0.5/(2 \times 5)$.
3. $\lambda = 0.5/10 = 0.05$.

Why C is correct: Risk aversion equals the information ratio divided by twice the desired active risk, with active risk in percentage points: 0.5/10 = 0.05.

Why the others are wrong

- **A:** Used IR itself (0.5) and ignored the division by 2ω.
- **B:** Entered active risk as a decimal (0.05): 0.5/(2×0.05)=5, violating the percentage-point convention.
- **D:** Divided by an extra factor of 10 (e.g., 0.5/(2×50)); a scaling slip.

Q163

BT Mock Q10 · BT answers P2.T9.21.4.1 · BT notes P2.T9.21.4.1

According to Grinold, typical or illustrative information ratios (IR) include: 0.500 is good, 0.750 is very good and 1.00 is exceptional. In regard to risk aversion, λ, 0.150 is restrained, 0.100 is moderate, and 0.050 is aggressive. Patricia is a manager with a "very good" information ratio of 0.750 and a "moderate" risk aversion of 0.10. If she shifts from this moderate risk aversion to an aggressive risk aversion of 0.050, what is the impact on the optimal level of active risk?

A. No effect
B. Optimal active risk reduces by one-half
C. Optimal active risk increases by about +41%
D. Optimal active risk doubles

Answer: D

Key correction: BT explanation printed the second optimal active risk as 3.750; the correct value is 7.50 (answer D unchanged).

Step by step
1. $\omega^* = \text{IR}/(2\lambda)$.
2. Moderate: $0.750/(2 \times 0.10) = 3.75\%$.
3. Aggressive: $0.750/(2 \times 0.05) = 7.50\%$.
4. Ratio $7.50/3.75 = 2$: optimal active risk doubles.

Why D is correct: Optimal active risk is inversely proportional to risk aversion, so halving λ doubles ω*.

Why the others are wrong
- **A:** Ignores that ω* depends on λ; IR is unchanged but λ is not.
- **B:** Inverted the relationship: lower risk aversion raises, not lowers, desired active risk.
- **C:** Applied a square-root relationship ($\sqrt{2} \approx 1.41$); ω* is linear in $1/\lambda$, not $1/\sqrt{\lambda}$.

Q164 Schweser Module Quiz 86.2 Q2

A manager has forecasts of Stocks A, B, and C, but not of Stocks D and E. Stocks A, B, and D are in the benchmark portfolio. Stocks C and E are not in the benchmark portfolio. Which of the following is correct concerning specific weights the manager should assign in tracking the benchmark portfolio?

A. wC = 0.
B. wD = 0.
C. wC = (wA + wB) / 2.
D. wC = wD = wE.

Answer: A

Step by step
1. Stocks outside the benchmark (C, E) get a benchmark (tracking-portfolio) weight of zero; active weights on C can still be taken to exploit its alpha.
2. Stock D is in the benchmark but has no forecast; it needs a nonzero tracking weight, inferred as a function of the alphas of the stocks with forecasts.
3. Check: wC = 0 is the only statement that is always true.

Why A is correct: A stock not in the benchmark has zero benchmark weight, regardless of whether the manager has a forecast for it.

Why the others are wrong
- **B:** D is in the benchmark with no forecast, so its weight is inferred from other alphas, not set to zero.
- **C:** C is outside the benchmark; its weight is not an average of other weights.
- **D:** C and E are outside the benchmark (zero) while D is inside (nonzero), so the weights are not equal.

LO 86.f

Describe portfolio revisions and rebalancing, and analyze the tradeoffs between alpha, risk, transaction costs, and time horizon.

Q165 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO; written from the SchweserNotes 2026 reading in GARP practice-exam style.

An active equity manager is comparing two trade ideas that use the same capital. Trade X is expected to earn an alpha of 1.5% over 3 months and costs 0.8% round trip; the manager expects to find a similar replacement opportunity every quarter. Trade Y is expected to earn an alpha of 3.6% over 12 months and costs 1.2% round trip. The manager amortizes transaction costs over each trade's holding period. Which of the following is correct?

A. Both trades have the same annualized net alpha because transaction costs should always be amortized over one year regardless of the holding period.
B. Trade X has an annualized net alpha of 0.7% versus 2.4% for Trade Y, so Trade Y clearly dominates and horizon uncertainty is irrelevant to the decision.
C. Trade X has an annualized net alpha of 5.2% versus 2.4% for Trade Y, and because shorter-horizon alpha forecasts are more reliable, Trade X should be preferred without further analysis.
D. Trade X has an annualized net alpha of 2.8% versus 2.4% for Trade Y, but because shorter-horizon alpha forecasts are less certain, the manager should require greater confidence in Trade X's alpha before acting on it.

Answer: D

Step by step

1. Annualize Trade X's alpha: four quarterly trades a year give $1.5\% \times 4 = 6.0\%$.
2. Amortize Trade X's cost over its 0.25-year holding period: $0.8\%/0.25 = 3.2\%$ a year.
3. Trade X net: $6.0\% - 3.2\% = 2.8\%$. Trade Y net (1-year holding period): $3.6\% - 1.2\% = 2.4\%$.
4. Apply the horizon tradeoff: the shorter the horizon, the more uncertain the alpha, so X's higher net figure only justifies trading if the manager is confident enough in the short-horizon alpha.

Why D is correct: Annual transaction cost is the round-trip cost divided by the holding period in years. On that basis X earns more net of cost, but the reading stresses that short-horizon alphas are less certain, so the manager must weigh the extra net alpha against that uncertainty.

Why the others are wrong

- **A:** Costs are amortized over each trade's actual holding period; forcing a one-year amortization ignores how often X's cost is incurred.
- **B:** Compares X's single-quarter figure (1.5% − 0.8%) with Y's annual figure, mixing horizons, and wrongly says horizon does not matter.
- **C:** Subtracts X's cost only once a year (6.0% − 0.8%) instead of every quarter, and reverses the horizon effect: shorter horizons make alpha less certain, not more.

Q166 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO; written from the SchweserNotes 2026 reading in GARP practice-exam style.

The risk committee of an asset manager is reviewing the rebalancing policy of a quantitative equity portfolio. The portfolio manager revises holdings using updated alpha forecasts, estimated active risk, and estimated transaction costs, all of which are subject to estimation error. Which of the following statements about the manager's revision and rebalancing decisions is correct?

A. If transaction costs were zero, the manager should revise the portfolio every time new information about alphas arrives.

B. Underestimating transaction costs will cause the manager to trade too infrequently, so the portfolio will drift away from its optimal holdings.

C. Shortening the horizon over which alphas are forecast increases the certainty of those alphas, which justifies more frequent rebalancing.

D. Because alpha, risk, and transaction cost estimates are uncertain, the manager should trade more aggressively to capture alpha before the forecasts decay.

Answer: A

Step by step

1. With no trading frictions there is no cost to acting on new information, so every alpha update should be implemented.
2. With transaction costs, each trade must be justified by expected active return relative to active risk and cost.
3. Because all three inputs are estimated with error, the reading recommends a conservative approach, and it notes that short horizons make alpha estimates more uncertain.

Why A is correct: The reading states directly that with zero transaction costs a manager should revise the portfolio every time new information arrives; costs are what create a reason not to trade.

Why the others are wrong

- **B:** Reverses the effect: underestimating costs makes trades look cheaper than they are, so the manager trades too frequently, not too infrequently.
- **C:** Reverses the horizon effect: shorter horizons make alpha estimates more uncertain, which argues against frequent trading.
- **D:** Uncertainty in the inputs calls for a more conservative approach, not more aggressive trading.

Q167 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A portfolio manager receives a revised alpha forecast for a stock held in the portfolio. Rebalancing the position to reflect the new forecast is expected to add 0.9% per year of active return on that position. The round-trip transaction cost of the rebalancing trade is 0.6%, and the manager expects to hold the revised position for 6 months before the forecast is updated again. Ignoring the effect on active risk, which of the following is the best decision?

A. Rebalance, because the 0.9% expected annual benefit exceeds the 0.6% transaction cost.

B. Do not rebalance, because the annualized transaction cost of 1.2% exceeds the 0.9% expected annual benefit.
C. Rebalance, because the annualized transaction cost of 0.3% is well below the 0.9% expected annual benefit.
D. Do not rebalance, because any trade that incurs transaction costs reduces the portfolio's active return.

Answer: B
Step by step
1. Amortize the round-trip cost over the 0.5-year holding period: $0.6\%/0.5 = 1.2\%$ per year.
2. Compare with the expected annual benefit: $0.9\% < 1.2\%$.
3. The benefit does not cover the annualized cost, so the trade destroys value over the expected holding period.

Why B is correct: Transaction costs are incurred at once while the benefit accrues over time, so the cost must be amortized over the holding period before it is compared with the annual benefit.
Why the others are wrong
- **A:** Compares an annual benefit with an unamortized one-off cost; with a 6-month holding period the cost is effectively paid twice a year.
- **C:** Multiplies the cost by the holding period (0.6% × 0.5) instead of dividing by it.
- **D:** Overgeneralizes: a trade is worthwhile whenever its expected benefit exceeds its amortized cost, so costs alone do not rule out trading.

LO 86.g

Determine the optimal no-trade region for rebalancing with transaction costs.

Q168

GARP PE2 Q55

REBUILT — Source units were ambiguous (MCVA quoted as -0.02% but arithmetic gives about -0.00024); stated the MCVA formula and units (percent) so the no-trade comparison is unambiguous.

A portfolio manager holds stock QYZ and is deciding whether to buy more, sell, or hold after new information. All quantities are in percent per year. Inputs: alpha $\alpha = 0.25$; marginal contribution to active risk MCAR $= 0.17$; active risk $\psi = 2.3$; active risk aversion $\lambda_A = 0.32$; purchase cost $PC = 0.30$; sales cost $SC = 0.40$. The manager uses MCVA $= \alpha - 2\lambda_A\psi \cdot$ MCAR. Which is the best action and rationale?

A. Purchase more of QYZ because its MCVA is greater than the purchase cost.
B. Purchase more of QYZ because its MCVA is greater than its MCAR.
C. Sell the position in QYZ because its MCVA is negative.
D. Maintain the position in QYZ because its MCVA is greater than the negative of the sales cost (and below the purchase cost).

Answer: D
Step by step
1. $2\lambda_A\psi \cdot \text{MCAR} = 2(0.32)(2.3)(0.17) = 0.25024$.
2. MCVA $= 0.25 - 0.25024 = -0.00024$, essentially zero.
3. No-trade region: $-SC \le$ MCVA $\le PC$, i.e. $-0.40 \le$ MCVA ≤ 0.30.
4. -0.00024 lies inside the band, so the benefit of trading is below the cost: hold.

Why D is correct: A trade is justified only if MCVA exceeds the purchase cost (buy) or falls below the negative of the sales cost (sell); an MCVA near zero sits inside the no-trade region.
Why the others are wrong
- **A:** MCVA (about -0.00024) is not above the 0.30 purchase cost; this would need MCVA > 0.30.
- **B:** Compares MCVA with MCAR, which is not the buy criterion; the hurdle is the purchase cost.
- **C:** Negative MCVA alone does not justify selling; it must be below -0.40 (the sales cost) to overcome costs.

Q169

Schweser Module Quiz 86.2 Q1

REBUILT — Source asked which factors 'increase the no-trade region' (key C), which is ambiguous because risk aversion and MCAR shift the band rather than widen it; rebuilt as a numeric question on the band's bounds.

For a stock, the purchase cost is $PC = 0.30\%$ and the sales cost is $SC = 0.40\%$. Trading is not worthwhile when $-SC \le \alpha - 2\lambda_A\psi \cdot \text{MCAR} \le PC$. Initially $2\lambda_A\psi \cdot \text{MCAR} = 0.20\%$. If the manager's active risk aversion λ_A doubles, holding all else equal, the no-trade region for the stock's alpha becomes:

A. -0.40% to 0.30%

B. -0.20% to 0.50%
C. 0.00% to 0.70%
D. 0.10% to 0.80%

Answer: C
Key correction: Source key (C: both risk aversion and MCAR widen the region) conflates widening with shifting: the width equals PC + SC, so risk aversion and MCAR move the band rather than enlarge it.
Step by step
1. Rearrange: $2\lambda_A\psi$ MCAR $- SC \leq \alpha \leq 2\lambda_A\psi$ MCAR $+ PC$.
2. Doubling λ_A doubles the term: 0.20% → 0.40%.
3. Lower bound = 0.40 − 0.40 = 0.00%; upper bound = 0.40 + 0.30 = 0.70%.
4. Width = 0.70% = $PC + SC$, unchanged from before (-0.20% to 0.50%); the band just shifted up.

Why C is correct: Risk aversion (and MCAR) set the centre of the band at 2Î»ÏˆÂ·MCAR, while transaction costs set its width, so doubling Î» shifts the band to 0.00%â€“0.70%.
Why the others are wrong
- **A:** Ignores the risk term entirely (centre at zero), giving just -SC to PC.
- **B:** The original band (term 0.20%); fails to update for the doubled risk aversion.
- **D:** Applies the costs to the wrong sides (adds SC on the upper bound and subtracts PC on the lower), giving 0.40-0.30 to 0.40+0.40.

LO 86.h

Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming.

Q170 AP Mock 1 Q65

Which of the following statements regarding portfolio construction techniques is correct?
A. Although screening ignores all information in alphas except the rankings, it limits transaction costs through judicious choice of the buy, sell, and hold lists.
B. The stratification technique provides superior risk control by overweighting the categories with lower risks and underweighting the categories with higher risks.
C. The linear programming approach characterizes stocks along dimensions of risk, and requires that these dimensions distinctly and exclusively partition the stocks.
D. Quadratic programming requires many more inputs than other portfolio construction techniques and as a result minimizes both noise and the potential for poor calibration.

Answer: A
Step by step
1. Screening ranks stocks by alpha and uses only the ranking, discarding the alpha magnitudes.
2. Turnover (and hence transaction costs) is controlled by sizing the buy, hold and sell categories.
3. Check B, C and D against the technique each describes: B misdescribes stratification, C describes stratification rather than LP, D reverses the effect of QP's many inputs.

Why A is correct: Screens throw away alpha magnitudes but allow turnover to be managed through the size of the buy/hold/sell lists.
Why the others are wrong
- **B:** Stratification controls risk by matching the benchmark's category weights, not by over/underweighting categories by risk level.
- **C:** Mutually exclusive, exhaustive categories are required by stratification; LP drops that requirement and allows overlapping risk characteristics.
- **D:** QP's many inputs (volatilities and covariances) add estimation noise and the risk of poor calibration; they do not minimize it.

Q171 AnalystPrep Q.2452

All the following are shortcomings of screen technique, EXCEPT:

A. It ignores all information in the alphas apart from rankings.
B. It does not protect against biases in the alphas.
C. It may result in more risky portfolios.
D. It is hard to code.

Answer: D

Step by step

1. Recall screen weaknesses: only rankings used, no protection against alpha bias, and no explicit risk control so portfolios may be riskier.
2. Screens are simple, intuitive and easy to code and explain; that is a strength.
3. So D is the only statement that is not a shortcoming.

Why D is correct: Simplicity and ease of implementation are strengths of screens, not weaknesses.

Why the others are wrong

- **A:** This is a genuine weakness: alpha magnitudes are discarded; choosing it confuses a weakness with a strength.
- **B:** Genuine weakness: screens do not neutralize bias in alphas (unlike stratification across categories).
- **C:** Genuine weakness: with no explicit risk model the resulting portfolio can carry higher active risk.

Q172 AnalystPrep Q.2455

Quadratic programming is a type of mathematical optimization that deals with certain constraints. Which of the following options does NOT accurately describe the characteristics or considerations of quadratic programming?

A. It includes the linear program as a special case.
B. It considers only two elements - risk and transactions costs.
C. It requires a large number of inputs.
D. The use of a large number of inputs increase the noise in the portfolio construction process.

Answer: B

Step by step

1. QP can include alphas, risk and transaction costs together, plus any number of constraints.
2. So saying it considers only risk and transaction costs omits alpha and is false.
3. The other statements are true: it generalizes LP, needs many inputs (e.g., 400 volatilities and 79,800 covariances for 400 stocks), and estimation error adds noise.

Why B is correct: QP explicitly optimizes alpha against risk and transaction costs, so 'only two elements' is wrong.

Why the others are wrong

- **A:** True: LP is a special case of QP; choosing it overlooks that the question asks for the inaccurate statement.
- **C:** True: covariance matrix for large universes requires many estimates.
- **D:** True: more inputs bring more estimation error and noise.

Q173 AnalystPrep Q.2745

A portfolio manager has a specific approach to building a portfolio. The manager first ranks stocks based on their alpha, then selects the top ten stocks from this ranking. These selected stocks are then used to create a portfolio where each stock is given equal weight. Which portfolio construction technique is the manager most likely using?

A. Stratification
B. Linear programming
C. Quadratic programming
D. Screens

Answer: D

Step by step

1. Ranking by alpha and keeping the top group is the definition of a screen.
2. Equal-weighting the survivors is a standard way to build the portfolio from a screen.
3. No categories, benchmark-matching or risk/cost optimization is used, ruling out the other techniques.

Why D is correct: Rank by alpha, keep the best, equal-weight: this is a screen.

Why the others are wrong

- **A:** Stratification would first split stocks into mutually exclusive categories and weight them as in the benchmark.
- **B:** LP matches the benchmark's risk characteristics; nothing here does that.
- **C:** QP optimizes alpha, risk and transaction costs jointly; no risk or cost model is used here.

Q174 GARP PE1 Q23 · Mock 3 Q57

The risk committee of an equity mutual fund is reviewing a portfolio construction technique proposed by a new portfolio manager who has recently been allocated capital to manage. The fund typically grants its portfolio managers flexibility in selecting and implementing appropriate portfolio construction procedures but requires that any methodology adopted fulfill key risk control objectives set by the firm. Which of the following is correct regarding a portfolio construction technique and its approach to risk control?

A. The quadratic programming technique takes into account additional risk parameters compared to other major portfolio construction techniques but also requires more inputs, which leads to more noise.

B. The screening technique ranks stocks by risk-adjusted alpha but it does not apply any additional risk control measures such as weighting the selected stocks by their relative capitalization.

C. The stratification technique splits the list of stocks into categories and maintains risk control by overweighting the categories with lower risks and underweighting the categories with higher risks.

D. The linear programming technique focuses on the pair-wise correlations of stocks rather than characterizing each stock along multiple dimensions of risk such as size, industry, volatility, or beta.

Answer: A

Step by step

1. QP uses alphas, the full covariance matrix and transaction costs, so it models more risk parameters than other techniques.
2. That requires many estimates (e.g., 400 volatilities and 79,800 covariances for 400 stocks), adding estimation noise.
3. Check the others: screens can be equal- or value-weighted; stratification matches benchmark category weights; LP uses risk characteristics, not pair-wise correlations.

Why A is correct: QP captures risk most fully through the covariance matrix, at the cost of many noisy inputs.

Why the others are wrong

- **B:** Screens rank by alpha and may be equal- or value-weighted (cap-weighted), so the claim that they cannot apply such weighting is wrong.
- **C:** Stratification sets category weights equal to benchmark weights; it does not tilt toward low-risk categories.
- **D:** Pair-wise correlations belong to QP's covariance matrix; LP characterizes stocks along risk dimensions such as size, industry, volatility and beta.

LO 86.i

Describe dispersion, explain its causes, and describe methods for controlling forms of dispersion.

Q175 AnalystPrep Q.2458

REBUILT — Original EXCEPT item had a questionable option D (number of portfolios does affect optimal dispersion); rewrote the options so exactly one is clearly not a cause.

A manager runs separately managed accounts for several clients and measures dispersion as the difference between the maximum and minimum account return over a period. Which of the following is NOT a cause of dispersion across the accounts?

A. Differences in portfolio betas across accounts.

B. Differences in the histories and cash flows of the clients.

C. Client-specific constraints, such as a ban on derivatives in some accounts.

D. Identical holdings in every account.

Answer: D

Step by step

1. Dispersion arises when accounts are not identical.
2. Differing betas, client histories/cash flows and different constraints all make holdings differ and therefore re-

turns differ.
3. Identical holdings in every account produce identical returns, so dispersion would be zero.

Why D is correct: Dispersion exists only because account holdings differ; identical holdings cannot generate any difference between maximum and minimum returns.

Why the others are wrong

- **A:** A real cause: different betas give different market exposure and returns; not the exception.
- **B:** A real cause: Schweser names different histories and cash flows as the basic cause; picking it misreads the EXCEPT wording.
- **C:** A real cause: unavoidable constraints make accounts differ and increase dispersion; picking it ignores that constraints are outside manager control but still cause dispersion.

Q176

AnalystPrep Q.2459

REBUILT — Source key (higher average returns) was not the reasoning in Schweser/Grinold-Kahn; reframed so the correct answer is the transaction-cost tradeoff.

Why might a manager of separately managed accounts rationally accept a nonzero level of dispersion across client accounts?

A. Dispersion by itself raises the average return of the accounts.
B. Eliminating dispersion requires trading, and the transaction costs of that trading can exceed the benefit.
C. Dispersion stays constant over time, so it cannot be reduced.
D. Dispersion has no effect on any client's return.

Answer: B

Key correction: Source key (AnalystPrep) said dispersion is optimal because it raises average returns; the correct reasoning is the transaction-cost tradeoff.

Step by step

1. Eliminating dispersion means matching accounts, either by giving new clients the old portfolio (sacrificing new information) or by rebalancing existing accounts (incurring transaction costs).
2. With nonzero transaction costs the cost of full convergence can exceed its benefit.
3. Hence there is an optimal, nonzero level of dispersion that balances transaction costs against gains from rebalancing.

Why B is correct: Nonzero transaction costs make full convergence costly, so the optimal dispersion is positive.

Why the others are wrong

- **A:** Dispersion is not itself a source of return; confuses the benefit of using new information with dispersion.
- **C:** False: with changing alphas and risk, dispersion tends to decrease over time and converge.
- **D:** False: each client's return differs from other accounts; dispersion is exactly differing client outcomes.

Q177

BT answers P2.T9.21.4.3 · BT notes P2.T9.21.4.3

In the portfolio construction context, where the manager maintains separate accounts for multiple clients, dispersion is the difference between the maximum return and minimum return for these separate account portfolios. In regard to this type of dispersion, each of the following statements is true EXCEPT which is false?

A. For a given tracking error, more portfolios imply greater dispersion
B. In the presence of nonzero transaction costs, some dispersion is optimal
C. Dispersion is a theoretical measure but not a practical problem because separate accounts are segregated
D. Dispersion is proportional to tracking error where the constant of proportionality is a function of the number of portfolios

Answer: C

Step by step

1. A: a greater number of portfolios raises optimal dispersion for given active risk: true.
2. B: transaction costs imply an optimal nonzero dispersion: true.
3. D: for a given number of portfolios, dispersion is proportional to active risk: true.
4. C: dispersion is a practical concern, since clients compare their returns and differences can be large; segregation does not remove it: false.

Why C is correct: Segregated accounts still hold different portfolios and clients compare outcomes, so dispersion is a real practical issue.

Why the others are wrong

- **A:** True statement; selecting it misses that the question asks for the false one.
- **B:** True statement; transaction costs make full convergence uneconomic.
- **D:** True statement; dispersion scales with active risk with a factor depending on number of portfolios.

Q178 GARP PE1 Q68 · Mock 3 Q13

A portfolio manager at an investment firm manages a number of accounts for multiple clients. The manager is analyzing the dispersion that occurs among these accounts, with dispersion defined as the difference between the maximum and minimum return for the accounts. The manager explores the various drivers of dispersion and deliberates over how dispersion can be minimized. Which of the following conclusions is correct for the manager to reach?

A. Dual-benchmark optimization can reduce dispersion and help achieve higher average returns.

B. A portfolio manager's tracking error and dispersion tend to be proportional to each other over time.

C. Dispersion is always client-driven since it refers to the variance in the performances of client portfolios managed by the same manager.

D. Portfolio managers can control dispersion and should aim to reduce any existing dispersion to zero.

Answer: B

Step by step

1. For a given number of portfolios, dispersion is proportional to active risk (tracking error).
2. Dispersion has client-driven causes (constraints, histories, cash flows) and manager-driven causes (holdings and beta differences), so C is false.
3. Transaction costs make some dispersion optimal, so driving it to zero is not desirable (D false).
4. Dual-benchmark optimization reduces dispersion only at the cost of expected return, not with higher returns (A false).

Why B is correct: Dispersion scales with tracking error for a given number of accounts.

Why the others are wrong

- **A:** Reducing dispersion through such constraints costs return; it does not raise average returns.
- **C:** Ignores manager-controlled sources such as differing holdings and betas.
- **D:** Transaction costs make full elimination uneconomic; optimal dispersion is positive.

Reading 87. Portfolio Risk: Analytical Methods

Philippe Jorion, Value-at-Risk: The New Benchmark for Managing Financial Risk, 3rd Edition (New York, NY: McGraw-Hill, 2007). Chapter 7.

LO	Learning objective	Qs
87.a	define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR.	55
87.b	explain the impact of correlation on portfolio risk.	21
87.c	apply the concept of marginal VaR in making portfolio management decisions.	4
87.d	explain and calculate the risk-minimizing position and the position that maximizes the ratio of expected return to risk.	7
87.e	explain the difference between risk management and portfolio management and describe how to use marginal VaR in portfolio management.	2

LO 87.a

Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR.

Q179

AnalystPrep Q.3151

REBUILT — Stem said 'daily expected loss' (made explicit as daily VaR) and the options were OCR-damaged/rounded; options rebuilt as realistic error results.

A fund holds Goodman Fertilizers and HiEdge Autos, whose returns are uncorrelated (ρ = 0). The annual 95% VaR of the whole portfolio is USD 7.2 million, and Goodman's annual 95% VaR is USD 5 million. Assume a 250-day year and that VaR scales with the square root of time. What is the daily 95% VaR of HiEdge's stock (the loss exceeded 5% of the time), closest to:

A. USD 139,140
B. USD 327,660
C. USD 5,180,000
D. USD 107,360

Answer: B

Step by step

1. With $\rho = 0$: $\text{VaR}_P^2 = \text{VaR}_G^2 + \text{VaR}_H^2$.
2. $\text{VaR}_H^2 = 7.2^2 - 5^2 = 51.84 - 25 = 26.84$, so annual $\text{VaR}_H = \sqrt{26.84} = 5.1807$ million.
3. Daily VaR = annual VaR $/\sqrt{250} = 5{,}180{,}734/15.811 \approx 327{,}658$.

Why B is correct: With zero correlation, individual VaRs combine in quadrature, so HiEdge's annual VaR is backed out of the portfolio VaR and then converted to daily by dividing by $\sqrt{250}$.

Why the others are wrong

- **A:** Subtracted the VaRs linearly (7.2 - 5 = 2.2 million) instead of in quadrature, then divided by $\sqrt{250}$: 139,140.
- **C:** Stopped at the annual figure (5.18 million) and never converted to a daily VaR.
- **D:** Divided the squared annual VaR (26.84, a variance) by 250 rather than taking the square root and dividing by $\sqrt{250}$: 107,360.

Q180

BT notes P2.T9.3.3 · BT answers P2.T8.3.3

REBUILT — Two duplicate qualitative method-choice items (not clearly tied to LO 87.a) were merged and rebuilt as a numeric incremental VaR (full revaluation) question for a large position.

A USD 100 million portfolio P has an annual volatility of 10%. A manager considers adding a large USD 40 million position A with annual volatility of 20%. The correlation between A's returns and P's returns is 0.5. Use a 95% confidence level (z = 1.65) and normally distributed returns. Using full revaluation of the portfolio before and after the addition, the incremental VaR of A is closest to:

A. USD 6.60 million
B. USD 9.27 million
C. USD 13.20 million
D. USD 25.77 million

Answer: B

Step by step

1. Dollar volatilities: $\sigma_P = 10$ million, $\sigma_A = 0.20 \times 40 = 8$ million. Old VaR = $1.65 \times 10 = 16.50$ million.
2. New variance = $10^2 + 8^2 + 2(0.5)(10)(8) = 100 + 64 + 80 = 244$; new $\sigma = 15.62$ million.
3. New VaR = $1.65 \times 15.62 = 25.77$ million.
4. Incremental VaR = $25.77 - 16.50 = 9.27$ million.

Why B is correct: Incremental VaR is the difference between the fully revalued portfolio VaR after and before the addition; for a large position it captures the nonlinearity that the marginal VaR shortcut ignores.

Why the others are wrong

- **A:** Marginal VaR approximation: $\beta = 0.5 \times 0.20/0.10 = 1.0$, MVaR = $1.0 \times 0.165 = 0.165$ per dollar, times 40 = 6.60. It is a linear approximation and understates the change for a large position.
- **C:** Stand-alone (individual) VaR of A, $1.65 \times 8 = 13.20$; ignores the diversification with the existing portfolio.
- **D:** The new total portfolio VaR; forgot to subtract the original VaR of 16.50.

Q181

AnalystPrep Mock 1 Q5

REBUILT — Stem asked for the 'effect' without direction and options had no sign; made direction explicit and rebuilt distractors as realistic errors.

A portfolio holds two assets:

Asset	Value	99% 1-day VaR
A	USD 5 million	USD 0.58 million
B	USD 10 million	USD 1.86 million

The correlation between the two assets is 0.7. The manager rebalances to equal weights (USD 7.5 million in each) with no change in volatilities. The change in portfolio VaR is closest to:

A. A decrease of USD 0.30 million
B. A decrease of USD 0.17 million
C. An increase of USD 0.20 million
D. A decrease of USD 0.20 million

Answer: D

Step by step

1. Initial: $\text{VaR}_P = \sqrt{0.58^2 + 1.86^2 + 2(0.7)(0.58)(1.86)} = 2.3035$ million.
2. VaR is proportional to position size: new $\text{VaR}_A = 0.58 \times 1.5 = 0.87$, new $\text{VaR}_B = 1.86 \times 0.75 = 1.395$.
3. New $\text{VaR}_P = \sqrt{0.87^2 + 1.395^2 + 2(0.7)(0.87)(1.395)} = 2.0981$ million.
4. Change = $2.0981 - 2.3035 = -0.2054$, a decrease of about 0.20 million.

Why D is correct: Shifting weight toward the lower-volatility asset reduces diversified VaR; the full correlation formula must be applied to both the old and new individual VaRs.

Why the others are wrong

- **A:** Set the correlation to zero (adding VaRs in quadrature): 1.948 to 1.644, a change of 0.30.
- **B:** Compared undiversified VaRs (2.44 to 2.265), which ignore correlation: 0.175.
- **C:** Right size but wrong sign; moving to the lower-volatility asset reduces VaR.

Q182

AnalystPrep Mock 2 Q13

REBUILT — Given betas (1.4583, 0.6250) were inconsistent with the stated volatilities (weighted betas did not sum to 1); rebuilt with the correlation stated and consistent figures.

A portfolio manager holds two assets:

Asset	Position (GBP m)	Standard deviation (annual)
X	240	18%
Y	260	10%

The correlation between X and Y is 0.6, and returns are normally distributed. At the 95% level (z = 1.645), which option gives the marginal VaR of X (per GBP of position), Y's percentage contribution to portfolio VaR, and the diversification benefit?

A. Marginal VaR of X 0.2961; Y contributes 52.0%; benefit GBP 113.83 million
B. Marginal VaR of X 0.2791; Y contributes 37.6%; benefit GBP 11.23 million
C. Marginal VaR of X 0.2791; Y contributes 34.7%; benefit GBP 11.23 million
D. Marginal VaR of X 0.2791; Y contributes 34.7%; benefit GBP 102.60 million

Answer: C

Step by step

1. Weights: $w_X = 0.48$, $w_Y = 0.52$. $\sigma_P^2 = (0.48)^2(0.18)^2 + (0.52)^2(0.10)^2 + 2(0.48)(0.52)(0.6)(0.18)(0.10) = 0.015559$, so $\sigma_P = 12.474\%$.
2. $\beta_X = \text{Cov}(R_X, R_P)/\sigma_P^2 = (0.48 \times 0.0324 + 0.52 \times 0.6 \times 0.018)/0.015559 = 1.3604$; $\beta_Y = 0.6673$ (check: $0.48 \times 1.3604 + 0.52 \times 0.6673 = 1$).
3. Marginal VaR of X $= z\,\beta_X\,\sigma_P = 1.645 \times 1.3604 \times 0.12474 = 0.2791$.
4. Y's contribution $= w_Y\beta_Y = 0.52 \times 0.6673 = 34.7\%$.
5. Diversified VaR $= 1.645 \times 0.12474 \times 500 = 102.60$ million; undiversified $= 1.645(240 \times 0.18 + 260 \times 0.10) = 113.83$ million; benefit $= 11.23$ million.

Why C is correct: Marginal VaR is beta times the portfolio's per-unit VaR, component contribution is weight times beta, and the diversification benefit is undiversified minus diversified VaR.

Why the others are wrong

- **A:** Used X's stand-alone VaR per unit (1.645 x 0.18) as marginal VaR, Y's weight (52%) as its contribution, and the undiversified VaR as the benefit.
- **B:** Right marginal VaR and benefit, but measured Y's share by stand-alone VaRs (0.052/0.1384 = 37.6%), which ignores correlation (component VaRs use weight x beta).
- **D:** Right marginal VaR and contribution, but reported the diversified VaR (102.60) as the benefit instead of undiversified minus diversified.

Q183

AnalystPrep Mock 2 Q32

REBUILT — Stem did not state the z-value (2.33 vs 2.326 matters for the options); added z = 2.33 and replaced an arbitrary distractor.

A portfolio manager runs a USD 200 million portfolio that includes USD 18 million of Alphabet stock. The correlation between Alphabet's return and the portfolio return is 0.54. Annual standard deviations are 22% for Alphabet and 27% for the portfolio. Assuming normally distributed returns, use a 1-year 99% VaR (z = 2.33). The component VaR of the Alphabet position is closest to:

A. USD 3.518 million
B. USD 4.982 million
C. USD 9.227 million
D. USD 6.115 million

Answer: B

Step by step

1. $\beta = \rho\,\sigma_A/\sigma_P = 0.54 \times 0.22/0.27 = 0.44$.
2. Marginal VaR $= z\,\sigma_P\,\beta = 2.33 \times 0.27 \times 0.44 = 0.2768$ per USD.
3. Component VaR $= 0.2768 \times 18$ million $= 4.982$ million.

Why B is correct: Component VaR = position size × marginal VaR = position × β × portfolio per-unit VaR, which reflects the position's covariance with the portfolio rather than its stand-alone risk.

Why the others are wrong

- **A:** Used the 95% multiplier (1.645) instead of 2.33 for a 99% VaR: 1.645 x 0.27 x 0.44 x 18 = 3.518.
- **C:** This is Alphabet's stand-alone (individual) VaR, 2.33 x 0.22 x 18; it ignores diversification within the portfolio.
- **D:** Used the correlation (0.54) in place of beta: 2.33 x 0.27 x 0.54 x 18 = 6.115.

Q184

AnalystPrep Mock 2 Q35

REBUILT — Original mixed a rebalance with a change of VaR measure (1-day 95% to 10-day 99%), so 'change in VaR' compared different measures; removed the measure switch and rebuilt options.

A bank holds CAD 72 million of Canadian equities and CAD 58 million of emerging market equities. Each position has a 1-day 95% VaR of CAD 2.5 million, and the correlation between the two is 0.47. The bank sells CAD 20 million of Canadian equities and buys CAD 20 million of emerging market equities. The change in the portfolio's 1-day 95% VaR is closest to:

A. An increase of CAD 0.22 million
B. An increase of CAD 0.17 million
C. An increase of CAD 0.28 million
D. A decrease of CAD 0.22 million

Answer: A

Step by step

1. Before: $\sqrt{2.5^2 + 2.5^2 + 2(0.47)(2.5)(2.5)} = \sqrt{18.375} = 4.2866$ million.
2. After (positions 52 and 78): individual VaRs $2.5 \times 52/72 = 1.8056$ and $2.5 \times 78/58 = 3.3621$.
3. New VaR $= \sqrt{1.8056^2 + 3.3621^2 + 2(0.47)(1.8056)(3.3621)} = 4.5022$ million.
4. Change $= 4.5022 - 4.2866 = +0.2156$, about +0.22 million.

Why A is correct: The trade tilts the portfolio toward the position that is far riskier per dollar (EM equities, 2.5/58 vs 2.5/72), so individual VaRs scale with position size and diversified VaR rises.

Why the others are wrong

- **B:** Compared undiversified VaRs (1.806 + 3.362 = 5.168 vs 5.000): 0.17; ignores correlation.
- **C:** Set the correlation to zero in both VaRs (3.817 vs 3.536): 0.28.
- **D:** Right size, wrong sign; shifting money to the higher-VaR-per-dollar asset raises VaR.

Q185

AnalystPrep Mock 3 Q18

REBUILT — Betas in the table were inconsistent with the volatilities and unnecessary for the answer; removed them and rebuilt distractors as realistic errors, stating z = 1.645.

A portfolio holds two airline stocks:

Asset	Position (USD m)	Standard deviation (annual)
Venus	450	16%
Mars	250	11%
Portfolio	700	14%

At the 95% confidence level (z = 1.645), the portfolio's diversification benefit is closest to:

A. USD 163.7 million
B. USD 161.2 million
C. USD 2.5 million
D. USD 3.5 million

Answer: C

Step by step

1. Diversified VaR $= 1.645 \times 0.14 \times 700 = 161.21$ million.
2. Undiversified VaR $= 1.645(450 \times 0.16 + 250 \times 0.11) = 1.645 \times 99.5 = 163.68$ million.
3. Benefit $= 163.68 - 161.21 = 2.47 \approx 2.5$ million.

Why C is correct: Diversification benefit is the sum of individual VaRs (undiversified) minus the portfolio VaR computed with the portfolio's own volatility.

Why the others are wrong

- **A:** Reported the undiversified VaR itself.
- **B:** Reported the diversified VaR itself.
- **D:** Used z = 2.326 (99%) instead of 1.645: 2.326 x (99.5 - 98) = 3.49.

Q186

AnalystPrep Mock 3 Q30

REBUILT — Option C summed to 100.2% (defective); rebuilt all distractors as realistic mistakes.

A new portfolio holds USD 2.5 million in asset A (volatility 6%, beta to the portfolio 0.97) and USD 1.0 million in asset B (volatility 10%, beta 1.075). At the 95% confidence level, the percentage contributions of A and B to portfolio VaR (component VaR / portfolio VaR) are closest to:

A. A 60.0% / B 40.0%
B. A 47.4% / B 52.6%
C. A 57.5% / B 42.5%
D. A 69.3% / B 30.7%

Answer: D

Step by step

1. Weights: $w_A = 2.5/3.5 = 0.7143$, $w_B = 0.2857$.
2. Percentage contribution $= w_i\beta_i$: A $= 0.7143 \times 0.97 = 69.3\%$; B $= 0.2857 \times 1.075 = 30.7\%$.
3. These sum to 100%, as component VaRs must; volatilities and the z-value cancel.

Why D is correct: Component VaR $= w_i\beta_i\,\mathrm{VaR}_P$, so the percentage contribution is $w_i\beta_i$ and the contributions sum

to 100%.

Why the others are wrong

- **A:** Used stand-alone VaR shares (2.5 x 6 = 15 vs 1.0 x 10 = 10), which ignores correlation with the portfolio.
- **B:** Used betas alone (0.97/2.045) and ignored position sizes.
- **C:** Used $w\,\beta\,\sigma$ shares (0.0416 vs 0.0307), double-counting volatility.

Q187

AnalystPrep Q.2469

REBUILT — Stem never said how large the new position is, so options referring to 'this case' were ambiguous; stated the position is large and rewrote options.

A portfolio manager plans to add a position that is large relative to the existing portfolio. He argues that marginal VaR and incremental VaR will give the same result, since both measure the change in portfolio VaR from the new position. Which statement is accurate?

A. He is correct; the two measures are identical for any size of position.

B. He should use marginal VaR, because for large additions VaR changes linearly with position size.

C. He should use incremental VaR, because marginal VaR is only a linear approximation valid for small changes, while full revaluation captures nonlinearity.

D. He should use marginal VaR, because it is based on full revaluation and is exact for large positions.

Answer: C

Step by step

1. Marginal VaR is the derivative of portfolio VaR with respect to a position, so it gives a linear approximation that holds only for small changes.
2. Incremental VaR is the difference between VaR after and before adding the entire new position (full revaluation) and so captures nonlinearity and the change in risk of existing positions.
3. For a large position the two diverge, so incremental VaR should be used.

Why C is correct: Marginal VaR assumes a linear, small change; incremental VaR is computed by full revaluation and is accurate for large additions (though costlier).

Why the others are wrong

- **A:** The two are only approximately equal for small additions; incremental VaR is generally larger and nonlinear.
- **B:** VaR does not change linearly for large changes; that linearity is exactly the marginal VaR assumption that fails.
- **D:** Reverses the measures: full revaluation defines incremental VaR, not marginal VaR.

Q188

AnalystPrep Q.2470

A portfolio manager intends to decompose the portfolio risk in order to compute the contribution of each asset to the overall portfolio VaR. The manager argues that taking individual VaRs, adding them up, and computing their individual percentages will provide the component VaR of each asset.

Which of the following statements is accurate?

A. The manager is correct.

B. The manager is incorrect as the method adopted completely ignores the diversification effects.

C. The manager is incorrect as the method adopted assumes a non-linear VaR.

D. The manager is incorrect as the method adopted assumes a linear VaR.

Answer: B

Step by step

1. Individual (stand-alone) VaRs are computed as if each position were held alone.
2. Summing them gives the undiversified VaR, which equals portfolio VaR only if all correlations are +1.
3. Component VaR = marginal VaR × position size = $\text{VaR}_P \times \beta_i \times w_i$; it depends on each position's beta (correlation with the portfolio), so it captures diversification.
4. Because individual VaRs ignore correlations, their percentage shares are not the component VaR shares.

Why B is correct: Component VaRs sum to the diversified portfolio VaR and reflect each asset's correlation with the portfolio; stand-alone VaRs do not, so their shares are wrong whenever diversification exists.

Why the others are wrong

- **A:** Treats the sum of stand-alone VaRs as a risk decomposition, ignoring that this is the undiversified VaR.
- **C:** The flaw is not non-linearity; component VaR is defined with a linear (Euler/beta) approximation, and the method fails because it omits correlation.
- **D:** Linearity is an assumption of the marginal/component approach itself, not the defect of this method.

Q189

AnalystPrep Q.3148

REBUILT — Betas were inconsistent (exposure-weighted average beta is 1.13, not 1) so the data could not describe a real portfolio; Lira beta set to 1.13, and distractors redesigned from specific errors. Correct answer unchanged.

Graham Bell is a currency strategist at the currency trading desk of Himilton Commercial Bank. His portfolio has a current exposure of $26 million in British pounds and $40 million in Turkish lira. The betas relative to the portfolio are 0.80 for the pound and 1.13 for the lira. The diversified portfolio VaR is $7 million. What are the marginal VaR and the component VaR of the British pound position?

A. MVaR(GBP): 0.085; CVaR(GBP): $2.2 million
B. MVaR(GBP): 0.106; CVaR(GBP): $2.76 million
C. MVaR(GBP): 0.215; CVaR(GBP): $5.6 million
D. MVaR(GBP): 0.120; CVaR(GBP): $4.79 million

Answer: A

Step by step

1. Portfolio value $W = 26 + 40 = 66$ million; weight of GBP = 26/66.
2. Check: $(26 \times 0.80 + 40 \times 1.13)/66 = 1.00$, so the betas are consistent.
3. $\text{MVaR}_{GBP} = (7/66) \times 0.80 = 0.0848 \approx 0.085$
4. $\text{CVaR}_{GBP} = \text{MVaR} \times 26 = 0.0848 \times 26 \approx 2.2$ million (= $7 \times 0.80 \times 26/66$)

Why A is correct: Marginal VaR is VaR per dollar of portfolio times beta; component VaR multiplies it by the dollar position.

Why the others are wrong

- **B:** 0.106 and $2.76M = 7/66 and 7 × 26/66: omitted the beta of 0.80.
- **C:** 0.215 and $5.6M = (7/26) × 0.80 and 7 × 0.80: divided by the GBP position instead of the whole portfolio and omitted the weight in the component VaR.
- **D:** 0.120 and $4.79M = (7/66) × 1.13 and 7 × 1.13 × 40/66: used the lira's beta and position.

Q190

AnalystPrep Q.2477

REBUILT — Distractors were not tied to identifiable mistakes (two were arbitrary); replaced with options from realistic errors, correct answer unchanged.

Portfolio X has a VaR of $100,000. The portfolio is made up of 5 assets, A, B, C, D, and E. These assets are equally weighted. If Asset C has a beta of 1.5, then the component VaR of Asset C is equal to:

A. $30,000
B. $20,000
C. $150,000
D. $13,333

Answer: A

Step by step

1. Weight of C = 1/5 = 0.20.
2. $\text{CVaR}_C = \text{VaR}_P \times \beta_C \times w_C$
3. $= 100{,}000 \times 1.5 \times 0.20 = 30{,}000$

Why A is correct: Component VaR is the portfolio VaR scaled by the asset's beta and its portfolio weight.

Why the others are wrong

- **B:** $20,000 = 100,000 × 0.20: omitted the beta.
- **C:** $150,000 = 100,000 × 1.5: omitted the weight.
- **D:** $13,333 = 20,000 / 1.5: divided by beta instead of multiplying.

Q191

AnalystPrep Q.2478

REBUILT — Two distractors (0.02, 0.025) did not follow from any realistic error; replaced with options from specific mistakes.

Portfolio X has a VaR of $100,000. The portfolio is made up of 5 assets, A, B, C, D, and E, each valued at $2,000,000. These assets are equally weighted. If Asset C has a beta of 1.5, then the marginal VaR of Asset C is equal to:

A. 0.075
B. 0.015
C. 0.010
D. 0.0225

Answer: B

Step by step

1. Portfolio value $W = 5 \times 2{,}000{,}000 = 10{,}000{,}000$.
2. $\text{VaR}_P/W = 100{,}000/10{,}000{,}000 = 0.01$.
3. $\text{MVaR}_C = (\text{VaR}_P/W) \times \beta_C = 0.01 \times 1.5 = 0.015$

Why B is correct: Marginal VaR is the change in portfolio VaR per dollar added to the position: portfolio VaR per dollar of portfolio value times beta.

Why the others are wrong

- **A:** 0.075 = (100,000 / 2,000,000) × 1.5: divided by one asset's value instead of the total portfolio value.
- **C:** 0.010: omitted the beta.
- **D:** 0.0225 = 0.015 × 1.5: applied beta twice.

Q192

AnalystPrep Q.2649

REBUILT — Source data garbled (values 3 and 7 against a $1 million portfolio) and the keyed 0.92 million was off by a factor of 10; restated as $300,000/$700,000 positions with options recomputed.

A portfolio of $1 million consists of two assets, A and B. Asset A has a value of $300,000 and a daily standard deviation of 3%. Asset B has a value of $700,000 and a daily standard deviation of 5%. The correlation between A and B is 0.4. Using the variance-covariance approach with a z-value of 2.33, the 1-day 99% VaR of the portfolio is closest to:

A. $92,000
B. $84,000
C. $103,000
D. $186,000

Answer: A

Key correction: Source key (A, '0.92 million') has the right digits but the wrong scale for a $1 million portfolio; the VaR is about $0.092 million.

Step by step

1. Dollar volatilities: A = 300,000 × 0.03 = 9,000; B = 700,000 × 0.05 = 35,000.
2. $\sigma_P^2 = 9{,}000^2 + 35{,}000^2 + 2(0.4)(9{,}000)(35{,}000) = 81\text{M} + 1{,}225\text{M} + 252\text{M} = 1{,}558\text{M}$
3. $\sigma_P = \sqrt{1{,}558{,}000{,}000} \approx 39{,}472$
4. $\text{VaR}_{99\%} = 2.33 \times 39{,}472 \approx 91{,}969 \approx 92{,}000$

Why A is correct: Diversified VaR uses the portfolio standard deviation including the covariance term, then scales by z.

Why the others are wrong

- **B:** $84,000: set the correlation to zero (dropped the 252M covariance term).
- **C:** $103,000: assumed perfect correlation (summed the individual VaRs, 2.33 × 44,000) giving the undiversified VaR.
- **D:** $186,000: applied z to the sum of the percentage volatilities (3% + 5%) times $1 million, ignoring the weights and correlation.

Q193

AnalystPrep Q.2748

REBUILT — Two distractors (1.47, 0.93) were not explained by any realistic error; replaced with options from specific mistakes.

A portfolio of four equally weighted assets has a total value of $2 million. The VaR of the portfolio is $450,000. If one of the assets in the portfolio has a beta of 1.2, calculate the marginal VaR of this asset.

A. 0.19
B. 0.27
C. 1.08
D. 0.225

Answer: B

Step by step

1. $\text{VaR}_P/W = 450{,}000/2{,}000{,}000 = 0.225$.
2. $\text{MVaR} = 0.225 \times 1.2 = 0.27$

Why B is correct: Marginal VaR = (portfolio VaR / portfolio value) × beta.

Why the others are wrong

- **A:** 0.19 = 0.225 / 1.2: divided by beta instead of multiplying.
- **C:** 1.08 = (450,000 / 500,000) × 1.2: divided by one position's value ($500,000) rather than the portfolio value.
- **D:** 0.225: forgot to multiply by beta.

Q194

AnalystPrep Q.2749

REBUILT — Distractors were clustered arbitrary values; replaced with options from specific mistakes.

A portfolio consists of five equally weighted assets each having a value of $200,000. The VaR of the portfolio is $350,000. If the beta of one of the assets is 0.75, calculate the component VaR for this asset.

A. $52,500
B. $70,000
C. $262,500
D. $93,333

Answer: A

Step by step

1. Portfolio value = 5 × 200,000 = 1,000,000; weight = 0.20.
2. CVaR = 350,000 × 0.75 × 0.20 = 52,500

Why A is correct: Component VaR = portfolio VaR × beta × weight.

Why the others are wrong

- **B:** $70,000 = 350,000 × 0.20: omitted beta.
- **C:** $262,500 = 350,000 × 0.75: omitted the weight.
- **D:** $93,333 = 70,000 / 0.75: divided by beta instead of multiplying.

Q195

AnalystPrep Q.3026

REBUILT — Distractors were arbitrary values; replaced with options from specific mistakes (correct answer unchanged).

A portfolio holds two foreign currencies, the Australian dollar (AUD) and the sterling pound (GBP), which are uncorrelated. Their standard deviations against the dollar are 6.5% and 10%, respectively. The portfolio has USD 4 million invested in AUD and USD 3 million in GBP. Compute the portfolio VaR at the 95% confidence level, using α = 1.65.

A. USD 924,000
B. USD 396,989
C. USD 655,032
D. USD 260,040

Answer: C

Step by step

1. Dollar volatilities: AUD = 4,000,000 × 0.065 = 260,000; GBP = 3,000,000 × 0.10 = 300,000.
2. Uncorrelated: $\sigma_P = \sqrt{260{,}000^2 + 300{,}000^2} = \sqrt{0.1576 \times 10^{12}} \approx 396{,}989$
3. VaR = 1.65 × 396,989 ≈ 655,032

Why C is correct: With zero correlation, risk adds in quadrature; the diversified VaR is z times the portfolio dollar volatility.

Why the others are wrong

- **A:** USD 924,000 = 1.65 × (260,000 + 300,000): summed the volatilities, i.e. the undiversified VaR.
- **B:** USD 396,989: forgot to multiply by 1.65.
- **D:** USD 260,040 = 1.65 × 0.1576 × 10^6: multiplied z by the variance instead of the standard deviation.

Q196

AnalystPrep Q.3144

REBUILT — One distractor (0.14) did not follow from a realistic error; replaced with 0.20 (beta omitted).

Jack Wills is a portfolio manager at a sovereign wealth fund. His portfolio has a VaR of $8,000,000. The portfolio is made up of investments in four companies: Sahara Petroleum, Burki Refinery, Dumsim Drillers, and RedSea Exploration. The four company assets are equally weighted within the portfolio with each position valued at $10,000,000:
Sahara Petroleum has a beta of 1.2
Burki Refinery has a beta of 1.4
Dumsim Drillers has a beta of 0.8

RedSea Exploration has a beta of 0.6
Based on the above information, what is the marginal VaR of Burki Refinery?

A. 0.8
B. 0.20
C. 1.12
D. 0.28

Answer: D

Step by step

1. Portfolio value = 4 × 10,000,000 = 40,000,000.
2. $\text{VaR}_P/W = 8{,}000{,}000/40{,}000{,}000 = 0.20$.
3. $\text{MVaR}_{Burki} = 0.20 \times 1.4 = 0.28$

Why D is correct: Marginal VaR is VaR per dollar of portfolio value times the asset's beta.

Why the others are wrong

- **A:** 0.8 = 8M / 10M: divided by one position instead of the portfolio value, and omitted beta.
- **B:** 0.20: omitted the beta.
- **C:** 1.12 = (8M / 10M) × 1.4: divided by one position's value rather than total portfolio value.

Q197

AnalystPrep Q.3145

REBUILT — Distractors were arbitrary values; replaced with options from specific mistakes.

Jack Wills is a portfolio manager at a sovereign wealth fund. His portfolio has a VaR of $8,000,000. The portfolio is made up of investments in four companies: Sahara Petroleum, Burki Refinery, Dumsim Drillers, and RedSea Exploration. The four company assets are equally weighted within the portfolio with each position valued at $10,000,000:
Sahara Petroleum has a beta of 1.2
Burki Refinery has a beta of 1.4
Dumsim Drillers has a beta of 0.8
RedSea Exploration has a beta of 0.6
Based on the above information, the component VaR for Sahara Petroleum is closest to:

A. $2,400,000
B. $1,920,000
C. $9,600,000
D. $2,000,000

Answer: A

Step by step

1. Weight of Sahara = 10M / 40M = 0.25.
2. CVaR = 8,000,000 × 1.2 × 0.25 = 2,400,000
3. Check: the four component VaRs are 2.4M, 2.8M, 1.6M and 1.2M, summing to the $8M portfolio VaR.

Why A is correct: Component VaR = portfolio VaR × beta × weight, and the components sum to portfolio VaR.

Why the others are wrong

- **B:** $1.92M = 8M × 1.2 × 0.20: assumed a 20% weight (five positions) instead of 25%.
- **C:** $9.6M = 8M × 1.2: omitted the weight.
- **D:** $2.0M = 8M × 0.25: omitted the beta.

Q198

AnalystPrep Q.3150

REBUILT — Distractor 62.09 had no identifiable error; replaced with the VaR computed using portfolio volatility, and stated the z-score of 1.65.

Sasha Lamb is a portfolio manager at Lizard Financial. Her portfolio's net asset value (NAV) is $950 million and contains $98 million invested in the common stock of Bambino Entertainment. The annual standard deviation of returns is 24% for Bambino and 19% for the overall portfolio, and the correlation between Bambino's returns and the portfolio's returns is 0.6. Using a 1-year 95% VaR with a z-score of 1.65 and assuming normally distributed returns, the component VaR of Bambino is closest to:

A. $30.72 million
B. $23.28 million
C. $38.81 million
D. $15.52 million

Answer: B

Step by step

1. Individual VaR = position $\times z \times \sigma_i = 98 \times 1.65 \times 0.24 = 38.81$ million.
2. Beta of Bambino vs. portfolio: $\beta_i = \rho\sigma_i/\sigma_P = 0.6 \times 0.24/0.19 = 0.7579$.
3. Diversified VaR per dollar of NAV = $1.65 \times 0.19 = 0.3135$; marginal VaR = $\beta_i \times 0.3135 = 0.7579 \times 0.3135 = 0.2376$.
4. Component VaR = marginal VaR × position = $0.2376 \times 98 = 23.28$ million.
5. Shortcut: component VaR = individual VaR × ρ = $38.81 \times 0.6 = 23.28$ million.

Why B is correct: Component VaR equals $w_i\beta_i \text{VaR}_P = \text{VaR}_i \times \rho_{i,P}$, so the stand-alone VaR is scaled down by the correlation with the portfolio.

Why the others are wrong

- **A:** Used the portfolio volatility (19%) instead of the stock's 24% in the individual-VaR formula: $98 \times 1.65 \times 0.19 = 30.72$; also no correlation adjustment.
- **C:** Stopped at the individual (stand-alone) VaR and never multiplied by the correlation of 0.6, so diversification was ignored.
- **D:** Multiplied individual VaR by $1 - \rho = 0.4$ instead of $\rho = 0.6$: $38.81 \times 0.4 = 15.52$.

Q199

AnalystPrep Q.3162

REBUILT — Original distractors (0.80, 32,000, 99,999.9, 48,000) were not tied to realistic errors; rebuilt options around identifiable mistakes (dollar-scaled marginal VaR, omitted beta, wrong asset's weight and beta).

Consider a two-asset portfolio worth USD 1 million. Asset A has weight 0.6 and beta 0.8; Asset B has weight 0.4 and beta 1.3 (betas measured against the portfolio). The standard deviation of the portfolio's decimal return is 0.060606. Using a 95% confidence level (z = 1.645), the marginal VaR of Asset A (per dollar invested) and the component VaR of Asset B are closest to:

A. USD 79,757 and USD 51,842
B. 0.08 and USD 39,879
C. 0.08 and USD 51,842
D. 0.08 and USD 47,854

Answer: C

Step by step

1. Portfolio VaR = $1.645 \times 0.060606 \times 1{,}000{,}000 = 99{,}696.87$.
2. Marginal VaR of A = $(\text{VaR}_P/V_P) \times \beta_A = 0.099697 \times 0.8 = 0.0798 \approx 0.08$ per dollar.
3. Component VaR of B = $\text{VaR}_P \times w_B \times \beta_B = 99{,}696.87 \times 0.4 \times 1.3 = 51{,}842.37$.
4. Check: $0.6(0.8) + 0.4(1.3) = 1.0$, so the components sum to the portfolio VaR.

Why C is correct: Marginal VaR is VaR per dollar times beta; component VaR scales marginal VaR by the dollars held, i.e. $\text{VaR}_P w_i \beta_i$.

Why the others are wrong

- **A:** Quoted marginal VaR in dollars (VaR × β = 79,757) instead of per dollar (not dividing by portfolio value); the second figure is right but the first is wrong.
- **B:** Omitted beta in the component VaR: $99{,}696.87 \times 0.4 = 39{,}879$.
- **D:** Used Asset A's weight and beta (0.6×0.8) for the component VaR: $99{,}696.87 \times 0.48 = 47{,}854$.

Q200

AnalystPrep Q.4710

Which of the following correctly describes individual VaR?

A. It explains VaR in terms of the change in the size of a position in the portfolio.
B. Each position is evaluated separately, as if it were held in isolation.
C. It incorporates the diversification benefits between positions.
D. It is the additional risk that adding a new security brings to an existing portfolio.

Answer: B

Step by step

1. Individual VaR is the VaR of one position on its own: $|w_i| \times z \times \sigma_i \times V$.
2. It ignores correlations with other positions, so diversification is not reflected.
3. Match definitions: change per unit of position = marginal VaR; new-position addition = incremental VaR.

Why B is correct: Individual VaR treats every position separately, ignoring the other holdings and their correla-

tions.

Why the others are wrong

- **A:** This describes marginal VaR (change in portfolio VaR per unit change in a position).
- **C:** Individual VaR ignores correlations; diversification benefits appear in diversified, marginal and component VaR.
- **D:** This describes incremental VaR, the change in portfolio VaR from adding a new position.

Q201

AnalystPrep Q.4711

REBUILT — Replaced two unexplained distractors (0.4750, 0.009) with the perfect-correlation sum and the total-exposure error.

An investment company holds two uncorrelated currency positions: US$2.5 million in the Japanese yen (volatility 6%) and US$1 million in the euro (volatility 10%). What is the dollar volatility of the portfolio?

A. $0.0325 million
B. $0.2500 million
C. $0.1803 million
D. $0.4082 million

Answer: C

Step by step

1. Dollar volatilities: JPY $2.5 \times 0.06 = 0.15$; EUR $1 \times 0.10 = 0.10$ (millions).
2. Variance with $\rho = 0$: $0.15^2 + 0.10^2 = 0.0225 + 0.0100 = 0.0325$.
3. Dollar volatility = $\sqrt{0.0325} = 0.1803$ million.

Why C is correct: With zero correlation there is no covariance term, so portfolio dollar volatility is the square root of the sum of squared position dollar volatilities.

Why the others are wrong

- **A:** Reported the dollar variance (0.0325) without taking the square root.
- **B:** Added the dollar volatilities (0.15 + 0.10), correct only for perfect correlation ($\rho = 1$).
- **D:** Applied $\sqrt{0.06^2 + 0.10^2}$ to the total $3.5 million exposure instead of weighting each volatility by its own position.

Q202

AnalystPrep Q.4715

REBUILT — Stated the 1.65 z-score so the answer is exact, and replaced unexplainable distractors with errors from swapped volatilities, omitted z, and wrong z.

A portfolio manager holds two assets that are perfectly correlated. Asset A has volatility 6% and $2.5 million invested; Asset B has volatility 10% and $1 million invested. Using a z-score of 1.65 (95% confidence), what are the individual VaRs of the two assets?

A. Asset A: $247,500; Asset B: $165,000
B. Asset A: $412,500; Asset B: $99,000
C. Asset A: $150,000; Asset B: $100,000
D. Asset A: $348,900; Asset B: $232,600

Answer: A

Step by step

1. Individual VaR = $V \times z \times \sigma$.
2. Asset A: $2{,}500{,}000 \times 1.65 \times 0.06 = 247{,}500$.
3. Asset B: $1{,}000{,}000 \times 1.65 \times 0.10 = 165{,}000$.
4. Correlation does not enter individual VaR (it matters only for the portfolio VaR).

Why A is correct: Individual VaR is each position's stand-alone value times z times its own volatility.

Why the others are wrong

- **B:** Swapped the volatilities between the assets (A with 10%, B with 6%).
- **C:** Omitted the z-score, giving one-standard-deviation dollar losses.
- **D:** Used the 99% z-score of 2.326 instead of 1.65.

Q203

AnalystPrep Q.5289

REBUILT — Replaced distractors with unidentifiable derivations by ones from realistic errors (component VaR, portfolio volatility, wrong z).

An investment banker is evaluating a CAD 150 million bond portfolio containing CAD 20 million in bond X. The annualized standard deviations of returns are 12% for the portfolio and 9% for bond X, and the correlation between bond X and the portfolio is 0.60. Using a 1-year 99% VaR (z = 2.326) and normally distributed returns, what is the individual VaR of bond X?

A. CAD 2,512,080
B. CAD 4,186,800
C. CAD 5,582,400
D. CAD 4,636,800

Answer: B

Step by step

1. Individual VaR uses only the position's own value and volatility.
2. $\text{VaR}_X = 20{,}000{,}000 \times 2.326 \times 0.09 = 4{,}186{,}800$.
3. Portfolio volatility, portfolio value and correlation are not needed.

Why B is correct: Stand-alone VaR is position value × z × position volatility, independent of the rest of the portfolio.

Why the others are wrong

- **A:** Multiplied by the correlation (0.60), which gives bond X's component VaR, not its individual VaR: 4,186,800 × 0.6.
- **C:** Used the portfolio volatility (12%) instead of the bond's 9%: 20,000,000 × 2.326 × 0.12.
- **D:** Used the 99.5% z-score of 2.576 rather than 2.326: 20,000,000 × 2.576 × 0.09.

Q204

BT answers P2.T5.24.1.3

REBUILT — Original stem was OCR-damaged (Asset 1 SD) and used an ambiguous 'which statement is correct' format; rebuilt as a single numeric question with z = 2.326 and error-based distractors.

A risk manager assesses a two-asset portfolio. Asset 1: $60 million invested, daily return standard deviation 1%. Asset 2: $7 million invested, daily return standard deviation 4%. The correlation between the assets is 0.6. Assuming normally distributed returns and using z = 2.326 for 99% confidence, the 5-day diversified VaR of the portfolio is closest to:

A. USD 4.16 million
B. USD 4.58 million
C. USD 3.44 million
D. USD 1.86 million

Answer: A

Step by step

1. Dollar volatilities: $60 \times 0.01 = 0.60$; $7 \times 0.04 = 0.28$ (millions).
2. Daily variance = $0.60^2 + 0.28^2 + 2(0.6)(0.60)(0.28) = 0.36 + 0.0784 + 0.2016 = 0.64$; daily $\sigma = 0.80$ million.
3. 1-day 99% VaR = $2.326 \times 0.80 = 1.8608$ million.
4. 5-day VaR = $1.8608 \times \sqrt{5} = 4.1609$ million.

Why A is correct: Diversified VaR uses the full covariance term, and the horizon is scaled by the square root of time.

Why the others are wrong

- **B:** Undiversified VaR (assumed $\rho = 1$): $2.326 \times (0.60 + 0.28) \times \sqrt{5} = 4.58$.
- **C:** Ignored the correlation term (treated assets as uncorrelated): $2.326 \times \sqrt{0.36 + 0.0784} \times \sqrt{5} = 3.44$.
- **D:** Stopped at the 1-day VaR and did not scale to 5 days.

Q205

BT answers P2.T8.1.1 · BT notes P2.T9.1.1

REBUILT — Merged the two identical sources; replaced distractors that had no identifiable error.

Portfolio returns are normally distributed. A two-asset portfolio has a diversified VaR of $76 million. The first asset has an individual VaR of $70 million, and the two assets have zero correlation. What is the individual VaR of the second asset?

A. $6.0 million
B. $2.4 million
C. $103.3 million

D. $29.6 million

Answer: D

Step by step

1. With $\rho = 0$: $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$.
2. $\text{VaR}_2^2 = 76^2 - 70^2 = 5776 - 4900 = 876$.
3. $\text{VaR}_2 = \sqrt{876} = 29.6$ million.

Why D is correct: With zero correlation the diversified VaR is the root-sum-of-squares of the individual VaRs, which can be inverted for the missing one.

Why the others are wrong

- **A:** Subtracted linearly (76 - 70), which assumes perfect correlation ($\text{VaR}_P = \text{VaR}_1 + \text{VaR}_2$).
- **B:** Took the square root of the linear difference, $\sqrt{76 - 70}$, instead of the difference of squares.
- **C:** Added the squares, $\sqrt{76^2 + 70^2}$; this is larger than the diversified VaR, so it cannot be an individual VaR here.

Q206

BT answers P2.T8.1.2 · BT notes P2.T9.1.2

A two-asset portfolio with a value of $20 million contains two equally-weighted assets (each asset has a value of $10 million). The volatility of the first asset is 10%, and the volatility of the second asset is 20% (asset returns are normally distributed). What is, respectively, the 95% diversified portfolio value at risk (VaR) if (i) the assets are uncorrelated, (ii) the assets have a correlation (rho) of 0.5, and (iii) the assets are perfectly correlated?

A. $3.0, 3.9 and 4.5 million
B. $3.7, 4.4 and 4.9 million
C. $3.9, 5.2 and 5.8 million
D. $4.1, 5.6 and 6.3 million

Answer: B

Step by step

1. Individual VaRs: $\text{VaR}_1 = 10 \times 0.10 \times 1.645 = 1.645$ million; $\text{VaR}_2 = 10 \times 0.20 \times 1.645 = 3.290$ million.
2. Diversified VaR: $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$.
3. (i) $\rho = 0$: $\sqrt{1.645^2 + 3.290^2} = 3.678$ million.
4. (ii) $\rho = 0.5$: $\sqrt{2.706 + 10.824 + 5.412} = 4.352$ million.
5. (iii) $\rho = 1$: $1.645 + 3.290 = 4.935$ million (equals undiversified VaR).
6. Rounded: 3.7, 4.4 and 4.9 million.

Why B is correct: Portfolio VaR is the correlation-adjusted root-sum-of-squares of the individual VaRs, rising from 3.68 at $\rho = 0$ to the simple sum 4.94 at $\rho = 1$.

Why the others are wrong

- **A:** All three figures are too low (e.g. from a z-value well below 1.645). With perfect correlation VaR must equal the sum of the individual VaRs, 4.935, not 4.5.
- **C:** All three are too high. Perfect-correlation VaR cannot exceed the sum of individual VaRs (4.935), so 5.8 is impossible; looks like a larger z-value was used.
- **D:** Too high in every case; 6.3 exceeds the undiversified VaR of 4.935, the upper bound for any correlation.

Q207

BT answers P2.T8.1.3 · BT notes P2.T9.1.3

REBUILT — Option B (6.73) was flagged as OCR-doubtful and matched no realistic error; replaced with 6.98 (half the diversified VaR) so each wrong option stems from a specific mistake.

A two-asset portfolio with a value of $40 million contains two equally-weighted assets (each asset has a value of $20 million). The volatility of both assets is 30%. The assets are uncorrelated (i.e., their correlation is zero). What is the incremental value at risk (VaR), assuming 95% confident VaR, if we subtract one asset from the portfolio, leaving only the remaining asset in the portfolio?

A. $4.09 million
B. $6.98 million
C. $9.87 million
D. $13.96 million

Answer: A

Step by step

1. Individual VaR of each asset: $20 \times 0.30 \times 1.645 = 9.870$ million.
2. Diversified VaR of the two-asset portfolio ($\rho = 0$): $\sqrt{9.870^2 + 9.870^2} = 13.958$ million.
3. Incremental VaR of the removed asset = VaR with it minus VaR without it: $13.958 - 9.870 = 4.088$ million.

Why A is correct: Incremental VaR is the change in portfolio VaR from the position: the portfolio VaR with both assets less the VaR of the remaining asset alone.

Why the others are wrong

- **B:** 6.98 is half of the diversified VaR (13.958 / 2); the incremental VaR is a difference of portfolio VaRs, not an average share.
- **C:** 9.87 is the stand-alone (individual) VaR of the remaining asset, i.e. the portfolio VaR after removal, not the change.
- **D:** 13.96 is the full diversified VaR of the two-asset portfolio; the subtraction of the remaining asset's VaR was omitted.

Q208

BT answers P2.T8.1.4 · BT notes P2.T9.1.4

If computed for a portfolio where correlations are imperfect (less than 1), which of the following value at risk (VaR) measures will be greatest?

A. Undiversified VaR
B. Diversified VaR
C. Individual VaR
D. Incremental VaR

Answer: A

Step by step

1. Undiversified VaR is the sum of the individual VaRs and assumes $\rho = 1$ between all positions.
2. Diversified VaR $= \sqrt{\sum\sum w_i w_j \rho_{ij} \sigma_i \sigma_j} \times z$ is below the sum whenever some $\rho < 1$.
3. Any single individual VaR is only one term of the undiversified sum, so it is smaller than the sum; incremental VaR is a change in portfolio VaR and cannot exceed the position's own individual VaR.

Why A is correct: With imperfect correlation, diversification lowers the portfolio VaR below the sum of the stand-alone VaRs, so the undiversified (sum) VaR is the largest.

Why the others are wrong

- **B:** Diversified VaR is strictly less than undiversified VaR when correlations are below 1; it equals it only if all correlations are 1.
- **C:** Individual VaR covers one position only, so it is a component of the undiversified sum and smaller than it.
- **D:** Incremental VaR is the change in portfolio VaR from adding a position; it is at most that position's individual VaR and is smaller than the total.

Q209

BT answers P2.T8.1.5 · BT notes P2.T9.1.5

REBUILT — Original stem ('which approach') was vague; reworded to state that a measure is approximated with marginal (local, delta-normal) VaR, asking which needs full revaluation for accuracy.

A risk manager can approximate some VaR measures using local (delta-normal) valuation based on marginal VaRs. Which of the following VaR measures requires a full revaluation of the portfolio to be computed accurately, so that a local-valuation approach is most likely to be insufficient for large position changes?

A. Undiversified VaR
B. Diversified VaR
C. Individual VaR
D. Incremental VaR

Answer: D

Step by step

1. Incremental VaR is the difference between the portfolio VaR after and before adding (or removing) a position.
2. Accuracy requires revaluing the whole portfolio, including the changed risk of existing positions, because the marginal-VaR approximation is linear and assumes away non-linear effects.
3. Undiversified, diversified and individual VaR are computed directly from the current positions' volatilities and correlations (no portfolio revaluation for a position change).

Why D is correct: Marginal VaR gives a linear (local) approximation; the exact incremental VaR for a sizeable position change needs a full revaluation, which is costly for large portfolios.

Why the others are wrong

- **A:** Undiversified VaR is just the sum of individual VaRs; no revaluation of a changed portfolio is needed.
- **B:** Diversified VaR of the current portfolio is calculated from its covariance matrix; it is not a change measure requiring a second valuation.
- **C:** Individual VaR values one position alone; a local approximation is not at issue.

Q210

BT answers P2.T8.1.6 · BT notes P2.T9.1.6

Which of the following is equal to the sum of the component VaRs of a portfolio?

A. Undiversified VaR
B. Diversified VaR
C. Sum of individual VaRs
D. Sum of incremental VaRs

Answer: B

Step by step

1. Component VaR of position i (dollar position x_i): $\text{CVaR}_i = x_i \times \text{MVaR}_i = x_i \beta_i \, z\sigma_P$.
2. Dollar-weighted betas satisfy $\sum_i x_i \beta_i = V$ (portfolio value), since the portfolio's own beta is 1.
3. Hence $\sum_i \text{CVaR}_i = z\sigma_P V = \text{VaR}_P$, the diversified portfolio VaR.

Why B is correct: Component VaRs are a decomposition of diversified portfolio VaR across positions (Euler allocation), so they add up to it exactly.

Why the others are wrong

- **A:** Undiversified VaR is the sum of individual VaRs (perfect correlation); component VaRs reflect actual correlations and sum to less (or equal).
- **C:** The sum of individual VaRs ignores diversification; this equals undiversified VaR, not the sum of components.
- **D:** Incremental VaR measures the change from adding or removing a whole position with revaluation; incremental VaRs generally do not sum to portfolio VaR.

Q211

BT answers P2.T8.1.7 · BT notes P2.T9.1.7

Which of the following is equal to the sum of the individual VaRs of the positions in a portfolio?

A. Undiversified VaR
B. Diversified VaR
C. Sum of component VaRs
D. Sum of incremental VaRs

Answer: A

Step by step

1. Individual VaR of position i: $\text{VaR}_i = z\,\sigma_i |x_i|$, ignoring other positions.
2. Portfolio VaR with all correlations equal to 1 is $z \sum_i \sigma_i |x_i| = \sum_i \text{VaR}_i$.
3. This no-diversification figure is called undiversified VaR.

Why A is correct: Adding stand-alone VaRs implicitly assumes perfect correlation, which is the definition of undiversified VaR.

Why the others are wrong

- **B:** Diversified VaR uses actual correlations and is less than or equal to the sum of individual VaRs.
- **C:** Component VaRs sum to diversified VaR, which is smaller than the sum of individual VaRs unless correlations are all 1.
- **D:** Incremental VaRs are changes in portfolio VaR from adding a position and do not sum to the undiversified VaR.

Q212

BT answers P2.T8.2.1 · BT notes P2.T9.2.1

REBUILT — Source options had OCR damage and several distractors had no identifiable error; kept the data and answer but rebuilt options so each wrong answer reflects a specific mistake.

A $20 million portfolio is equally invested in two currencies: $10 million in US dollars (USD) and $10 million in euros (EUR). The volatility of EUR is 20% and the volatility of USD is 30%. The correlation between the two currencies

is 0.60. What is the beta of the USD position with respect to the two-asset portfolio?

A. 0.89
B. 1.00
C. 1.25
D. 1.78

Answer: C

Step by step

1. Weights are 0.5 each. Portfolio variance $= 0.25(0.30)^2 + 0.25(0.20)^2 + 2(0.5)(0.5)(0.6)(0.30)(0.20) = 0.0225 + 0.0100 + 0.0180 = 0.0505$.
2. $\text{Cov}(USD, P) = 0.5(0.30)^2 + 0.5(0.6)(0.30)(0.20) = 0.045 + 0.018 = 0.063$.
3. $\beta = 0.063/0.0505 = 1.2475 \approx 1.25$.

Why C is correct: Beta is the position's covariance with the portfolio divided by the portfolio variance; USD is the riskier asset, so its beta exceeds 1.

Why the others are wrong

- **A:** Omitted the covariance term: $0.5(0.09)/0.0505 = 0.89$.
- **B:** Assumed a beta of 1 (the beta of the whole portfolio, or the weighted-average beta), ignoring that USD is more volatile than the portfolio.
- **D:** Used USD's own variance over portfolio variance: $0.09/0.0505 = 1.78$, forgetting the weight and covariance with EUR.

Q213

BT answers P2.T8.2.2 · BT notes P2.T9.2.2

REBUILT — Source options B-D were OCR-damaged and inconsistent across the two sources (D 1.724 vs 1.124); kept the stem and answer and rebuilt distractors from realistic errors.

A $10 million portfolio is equally invested in two currencies: $5 million in Swiss francs (CHF) and $5 million in Japanese yen (JPY). The volatility of CHF is 10% and the volatility of JPY is 20%. The correlation is 0.30. Using a 95% delta-normal VaR (z = 1.645), what is the marginal VaR of the CHF position with respect to the two-asset portfolio?

A. 0.106
B. 0.064
C. 0.205
D. 0.849

Answer: A

Step by step

1. Portfolio variance $= 0.25(0.10)^2 + 0.25(0.20)^2 + 2(0.5)(0.5)(0.3)(0.10)(0.20) = 0.0155$; $\sigma_P = 12.45\%$.
2. $\text{Cov}(CHF, P) = 0.5(0.10)^2 + 0.5(0.3)(0.10)(0.20) = 0.0080$.
3. $\beta_{CHF} = 0.0080/0.0155 = 0.516$.
4. MVaR $= z\,\beta\,\sigma_P = 1.645 \times 0.516 \times 0.1245 = 0.1057 \approx 0.106$ (per dollar invested in CHF).

Why A is correct: Marginal VaR is the VaR change per extra dollar in the position: $z\,\text{Cov}(R_i, R_P)/\sigma_P = z\beta_i\sigma_P$.

Why the others are wrong

- **B:** Left out the z-score: $\beta\sigma_P = 0.516 \times 0.1245 = 0.064$.
- **C:** Computed portfolio VaR per dollar, $z\sigma_P = 1.645 \times 0.1245 = 0.205$, without multiplying by beta.
- **D:** Multiplied z by beta only: $1.645 \times 0.516 = 0.849$, omitting portfolio volatility.

Q214

BT answers P2.T8.2.3 · BT notes P2.T9.2.3

REBUILT — Source answer 0.55 sits against a computed 0.559 (0.56) and other options had no identifiable error; rebuilt options around the exact result.

A $10 million portfolio with a volatility of 30% is invested in two positions, Y and Z. The beta of position Y with respect to the portfolio is 0.80. Using z = 2.33 for 99% confidence, what is the marginal VaR of position Y?

A. 0.24
B. 0.40
C. 0.56
D. 0.70

Answer: C

Key correction: Source key (B, 0.55) is a rounding of 0.559; the exact value is 0.56, so options were rebuilt with 0.56 as the key.

Step by step

1. $\text{MVaR}_Y = z\,\beta_Y\,\sigma_P$.
2. $= 2.33 \times 0.80 \times 0.30 = 0.5592 \approx 0.56$ per dollar of Y.

Why C is correct: Marginal VaR equals the portfolio VaR rate $z\sigma_P$ scaled by the position's beta.

Why the others are wrong

- **A:** Omitted the z-score: $0.80 \times 0.30 = 0.24$.
- **B:** Used the 95% deviate 1.645: $1.645 \times 0.8 \times 0.3 = 0.395 \approx 0.40$.
- **D:** Omitted beta: $2.33 \times 0.30 = 0.70$.

Q215

BT answers P2.T8.2.4 · BT notes P2.T9.2.4

REBUILT — Original data were internally inconsistent (marginal VaR of 0.30 with 30% portfolio volatility cannot hold for two identical positions); restated with a portfolio VaR of $6.0 million so the component VaRs reconcile.

A $20 million portfolio has a 95% VaR of $6.0 million. It is invested equally in two identical positions, A and B ($10 million in each). The 95% marginal VaR of Position A with respect to the portfolio is 0.30. What is the 95% component VaR of Position A?

A. $900,000
B. $3.0 million
C. $6.0 million
D. $1.5 million

Answer: B

Step by step

1. Component VaR = marginal VaR × dollar position (i.e., MVaR × $w_i P$).
2. Position A: 0.30 × $10 million = $3.0 million.
3. Check: the two identical positions each contribute \$3.0 million, summing to the portfolio VaR of \$6.0 million.

Why B is correct: Component VaR is the marginal VaR (VaR change per dollar) scaled by the dollars held; component VaRs add up to the diversified portfolio VaR.

Why the others are wrong

- **A:** Multiplied marginal VaR by 30% again ($0.30 \times 0.30 \times \$10$ million) instead of by the dollar position.
- **C:** Applied marginal VaR to the whole portfolio ($0.30 \times \$20$ million); this equals total portfolio VaR, not A's share.
- **D:** Applied the 50% weight on top of the dollar position ($0.30 \times 0.5 \times \$10$ million), double-counting the weight.

Q216

BT answers P2.T8.2.5 · BT notes P2.T9.2.5

A $20.0 million portfolio is invested in two positions, Y and Z: $14.0 million in Y and $6.0 million in Z. The volatility of each position is 10% and the positions are uncorrelated. The beta of Y with respect to the portfolio is 1.207 and the beta of Z is 0.517. What is the 95% component VaR of Position Y?

A. $389,000
B. $1.7 million
C. $2.1 million
D. $2.3 million

Answer: C

Step by step

1. Portfolio volatility $= \sqrt{0.7^2(0.10)^2 + 0.3^2(0.10)^2} = 7.616\%$.
2. Portfolio VaR $= 1.645 \times 0.07616 \times \$20$ million $= \$2.506$ million.
3. Component VaR (Y) $= \text{VaR}_P \times w_Y \times \beta_Y = 2.506 \times 0.70 \times 1.207 = \$2.117$ million.
4. Check: component (Z) $= 2.506 \times 0.30 \times 0.517 = \$0.389$ million; the two sum to \$2.506 million.

Why C is correct: Component VaR allocates portfolio VaR by weight times beta, so the components sum to the diversified VaR.

Why the others are wrong

- **A:** This is the component VaR of Position Z ($2.506 \times 0.30 \times 0.517$), the wrong position.
- **B:** Omitted beta: $2.506 \times 0.70 = \$1.75$ million, a weight share of VaR that ignores Y's contribution relative to portfolio risk.
- **D:** Individual (stand-alone) VaR of Y: $1.645 \times 0.10 \times \$14$ million $= \$2.303$ million, which ignores diversification.

Q217 BT answers P2.T8.2.6 · BT notes P2.T9.2.6

A $10 million portfolio is invested in two currencies: $6.0 million in Canadian dollars (CAD) and $4.0 million in US dollars (USD). The volatility of CAD is 10% and of USD is 20%. Assume the currencies are uncorrelated. The beta of the CAD position with respect to the portfolio is 0.60. At the 95% confidence level, what are, respectively, the component VaR and the incremental VaR of the CAD position?

A. $329,000 (component VaR) and $329,000 (incremental VaR)
B. $329,000 (component VaR) and $592,000 (incremental VaR)
C. $592,000 (component VaR) and $329,000 (incremental VaR)
D. $592,000 (component VaR) and $592,000 (incremental VaR)

Answer: C

Step by step

1. Portfolio volatility $= \sqrt{0.6^2(0.10)^2 + 0.4^2(0.20)^2} = 10\%$; portfolio VaR $= 1.645 \times 0.10 \times \$10$ million $= \$1.645$ million.
2. Component VaR (CAD) $= 1.645 \times 0.6 \times 0.60 = \$592,200$.
3. With uncorrelated positions, removing CAD leaves the USD position alone: stand-alone VaR $= 1.645 \times 0.20 \times \$4$ million $= \$1.316$ million.
4. Incremental VaR (CAD) $= 1.645 - 1.316 = \$329,000$.

Why C is correct: Component VaR is weight times beta times portfolio VaR (a linear allocation); incremental VaR is the full change in portfolio VaR from removing the position.

Why the others are wrong

- **A:** Set component VaR equal to incremental VaR; the two differ because component VaR is a linear approximation and the position is large.
- **B:** Swapped the two measures: $592,000 is the component VaR and $329,000 is the incremental VaR.
- **D:** Used the component VaR for both; the incremental VaR requires the before-and-after difference $1.645 - 1.316$.

Q218 BT answers P2.T8.2.7 · BT notes P2.T9.2.7

Each of the following is TRUE about component VaR EXCEPT:

A. By construction, component VaRs sum to the (diversified) portfolio VaR
B. Components with a negative sign act as hedges against the remainder of the portfolio
C. Component VaR = marginal VaR x portfolio value ($) x position weight (%)
D. If correlations are imperfect, component VaRs must be greater than individual VaRs

Answer: D

Step by step

1. Component VaR $= \text{MVaR}_i \times w_i P = \text{VaR}_i \times \rho_{i,P}$, where VaR_i is individual VaR.
2. With imperfect correlation $\rho_{i,P} < 1$, so component VaR is below individual VaR (components sum to the diversified VaR, below the undiversified sum of individual VaRs).
3. So D is false; A, B (a negative component reduces portfolio VaR) and C are true.

Why D is correct: Individual VaRs sum to undiversified VaR while component VaRs sum to the smaller diversified VaR, so components are smaller, not greater.

Why the others are wrong

- **A:** True statement (the defining additive property), so not the exception.
- **B:** True: a negative component VaR means the position reduces portfolio VaR, i.e., it is a hedge.
- **C:** True: this is the component VaR formula.

Q219 BT answers P2.T8.2.8 · BT notes P2.T9.2.8

Each of the following is TRUE about incremental VaR EXCEPT:

A. Incremental VaR is not equal to component VaR
B. Incremental VaR = marginal VaR x portfolio value ($) x position weight (%)
C. Incremental VaR is the change in portfolio VaR if a position is deleted
D. Incremental VaR probably requires a "before and after" full revaluation of the portfolio, if accuracy is desired

Answer: B

Step by step

1. Incremental VaR is the change in portfolio VaR from adding or removing a position; accuracy requires full revaluation before and after.
2. The expression MVaR × position is the linear approximation, i.e., the component VaR (valid only for small changes).
3. So B describes component VaR, not incremental VaR, and is the false statement.

Why B is correct: Incremental VaR is a discrete before/after difference; the marginal-VaR product is only a linear approximation (component VaR).

Why the others are wrong

- **A:** True: the two coincide only approximately for small positions.
- **C:** True: this is the definition (removal is the mirror of addition).
- **D:** True: an exact figure needs full portfolio revaluation.

Q220

BT answers P2.T8.3.1 · BT notes P2.T9.3.1

A bank is proposing to add Position (A) to Portfolio (P). (A) is small relative to the large but simple portfolio (P), which is accurately summarized by risk factors that are jointly normally distributed. The bank's Risk Manager seeks a fast and efficient method. Which is the best method given the situation and the risk manager's preference?

A. Parametric VaR with marginal VaR
B. Full revaluation with incremental VaR
C. Historical simulation
D. Monte Carlo simulation

Answer: A

Step by step

1. The added position is small relative to the portfolio, so the change in portfolio VaR is well approximated by a linear (local) measure: $\Delta\text{VaR} \approx \text{marginal VaR} \times \text{size of addition}$.
2. Risk factors are jointly normal, so delta-normal (parametric) VaR applies and marginal VaR comes analytically from the covariance matrix.
3. No full revaluation of the portfolio is needed, so this is the fastest and most efficient choice.

Why A is correct: For a small addition to a normally distributed portfolio, marginal VaR under the parametric approach approximates incremental VaR accurately and avoids re-running the whole portfolio.

Why the others are wrong

- **B:** Full revaluation gives the exact incremental VaR, but it is slow; the small size of A makes the extra accuracy unnecessary.
- **C:** Historical simulation is a full-revaluation method and is slow; it is not needed when returns are well described by a normal distribution.
- **D:** Monte Carlo is the most computationally intensive full-revaluation method and conflicts with the preference for speed.

Q221

BT answers P2.T9.21.5.1 · BT notes P2.T9.21.5.1

REBUILT — Source tables had OCR-garbled cells (individual VaR $471,200/$411,200 and placement of the portfolio VaR); restored the consistent table (1.645 x 25% x $1m = $411,250) and replaced two options that matched no realistic error.

Patricia manages a two-asset $2.0 million portfolio equally weighted between a Technology and a Real Estate position. Returns are assumed jointly normal. The exhibit shows each position's size, volatility, individual VaR and marginal VaR at 95.0% confidence, and the portfolio's volatility and diversified VaR.

Asset	Position	Volatility	Individual VaR	Marginal VaR
Technology	$1,000,000	25.00%	$411,200	0.31530
Real Estate	$1,000,000	36.00%	$592,100	0.53000
Portfolio	$2,000,000	25.70%	$845,301 (diversified)	

What is nearest to the 95.0% component VaR of the Technology position?

A. $315,300
B. $411,200
C. $530,000
D. $422,700

Answer: A

Step by step

1. Component VaR = marginal VaR × position value.
2. Technology: 0.31530 × 1,000,000 = 315,300.
3. Check: Real Estate component = 0.53000 × 1,000,000 = 530,000; the sum 315,300 + 530,000 = 845,300 equals the diversified portfolio VaR.

Why A is correct: Component VaRs are marginal VaR times position size and add up to the diversified portfolio VaR.

Why the others are wrong

- **B:** $411,200 is Technology's stand-alone individual VaR (1.645 × 25% × 1m); it ignores diversification.
- **C:** $530,000 is the Real Estate component VaR, taken from the wrong row.
- **D:** $422,700 is half of the diversified portfolio VaR, wrongly allocating risk by equal weights instead of by marginal VaR.

Q222

BT answers P2.T9.21.5.2 · BT notes P2.T9.21.5.2

REBUILT — The portfolio VaR cell was garbled/misplaced in the source tables; labelled it explicitly as the diversified VaR and replaced the unexplained option $433,753 with the undiversified VaR.

Doug manages a two-asset portfolio worth $1.30 million: $500,000 in a Financial fund and $800,000 in an Energy fund. Returns are assumed jointly normal. The exhibit shows 95.0% VaR information.

Asset	Position	Volatility	Individual VaR	Marginal VaR
Financial	$500,000	40.0%	$329,000	0.58970
Energy	$800,000	20.0%	$263,200	0.27380
Portfolio	$1,300,000	24.0%	$513,900 (diversified)	

What is nearest to the incremental VaR of the Financial position?

A. $184,900
B. $194,011
C. $250,700
D. $592,200

Answer: C

Step by step

1. Incremental VaR = VaR of portfolio − VaR of portfolio without the position.
2. Without Financial, only Energy remains: VaR = $263,200.
3. Incremental VaR = 513,900 − 263,200 = 250,700.

Why C is correct: Incremental VaR is the full change in portfolio VaR from adding the Financial position to the Energy holding.

Why the others are wrong

- **A:** 513,900−329,000 = 184,900 is the incremental VaR of Energy (subtracting Financial's individual VaR instead of Energy's).
- **B:** 0.58970×329,000 = 194,011 multiplies marginal VaR by individual VaR; component VaR would use position size (0.5897 × 500,000 = 294,850).
- **D:** 329,000 + 263,200 = 592,200 is the undiversified VaR, not an incremental amount.

Q223

BT answers P2.T9.21.5.3 · BT notes P2.T9.21.5.3

REBUILT — Replaced option $319,118 (no identifiable mistake behind it) with a realistic formula error; Consumer marginal VaR digit fixed to 0.42890.

Mary manages a two-asset portfolio worth $400,000: $100,000 in the Materials sector and $300,000 in the Consumer sector. Returns are jointly normal. The exhibit shows 95.0% VaR information.

Asset	Position	Volatility	Individual VaR	Marginal VaR
Materials	$100,000	50.0%	$82,200	0.53330
Consumer	$300,000	29.0%	$143,100	0.42890
Portfolio	$400,000	27.7%		

If the correlation between the positions is 0.250, what is nearest to the 95.0% diversified portfolio VaR?

A. $165,000
B. $182,000
C. $225,300

D. $173,700

Answer: B

Step by step

1. $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$.
2. $= \sqrt{82{,}200^2 + 143{,}100^2 + 2(0.25)(82{,}200)(143{,}100)} = 181{,}978$.
3. Cross-check: $1.645 \times 27.7\% \times 400{,}000 \approx 182{,}266$; component VaRs $0.5333 \times 100{,}000 + 0.4289 \times 300{,}000 = 182{,}000$.

Why B is correct: Individual VaRs combine like volatilities, including the correlation term, so the diversified VaR lies between the zero-correlation and perfect-correlation values.

Why the others are wrong

- **A:** $\sqrt{82{,}200^2 + 143{,}100^2} = 165{,}029$ sets the correlation to zero, ignoring the $\rho = 0.25$ term.
- **C:** $82{,}200 + 143{,}100 = 225{,}300$ is the undiversified VaR (perfect correlation).
- **D:** Dropping the factor 2 in the cross term ($\rho\,\text{VaR}_1\text{VaR}_2$ instead of $2\rho\,\text{VaR}_1\text{VaR}_2$) understates the covariance contribution and gives about $173,700.

Q224

BT answers P2.T9.21.6.3 · BT notes P2.T9.21.6.3

Consider a $30.0 million portfolio with two positions at 95% confidence (z = 1.645):
- $20.0 million in a Consumer ETF: expected excess return 4.0%, volatility 12.0%, marginal VaR 0.122
- $10.0 million in a Materials ETF: expected excess return 8.0%, volatility 38.0%, marginal VaR 0.543
- Correlation between the two positions: +0.150

Each of the following is true EXCEPT which is false?

A. The incremental VaR of the Materials ETF position is larger than the incremental VaR of the Consumer position

B. Each position's component VaR is less than its incremental VaR

C. If the correlation doubles to +0.30, each position's component VaR will increase

D. To decrease portfolio VaR or increase the Sharpe ratio, reallocate from the Materials ETF to the Consumer ETF

Answer: B

Step by step

1. Individual VaR: Consumer $1.645 \times 0.12 \times 20 = 3.948$ million; Materials $1.645 \times 0.38 \times 10 = 6.251$ million.
2. Component VaR = marginal × position: Consumer $0.122 \times 20 = 2.44$ million; Materials $0.543 \times 10 = 5.43$ million; diversified VaR = 7.87 million.
3. Incremental VaR = portfolio VaR − VaR of the other position alone: Consumer $7.87 - 6.251 = 1.62$ million; Materials $7.87 - 3.948 = 3.92$ million.
4. A: 3.92 > 1.62, true. B: 2.44 > 1.62 and 5.43 > 3.92, so component exceeds incremental; B is false.
5. C: recomputing at ρ = 0.30 gives components 2.76 and 5.58 million (up from 2.45 and 5.43), true.
6. D: Materials has the higher marginal VaR and the lower excess-return-to-marginal-VaR ratio ($0.08/0.543 = 0.147$ vs $0.04/0.122 = 0.328$), so shifting to Consumer lowers VaR and raises the ratio; true.

Why B is correct: Because the portfolio VaR is convex in position size, the linear component VaR exceeds the exact incremental VaR for both positions, so the statement that component is less than incremental is false.

Why the others are wrong

- **A:** True statement (3.92m vs 1.62m); mixing up which position is removed when computing incremental VaR would make it look false.
- **C:** True: higher correlation raises each position's covariance with the portfolio, so both components rise (2.45 to 2.76 and 5.43 to 5.58).
- **D:** True: Materials' ratio of excess return to marginal VaR (0.147) is below Consumer's (0.328), so moving money to Consumer is the right direction.

Q225

GARP PE1 Q18 · Mock 3 Q62

A risk manager is evaluating the risks of a portfolio of stocks. Currently, the portfolio is valued at CAD 248 million and contains CAD 15 million in stock T. The annualized standard deviations of returns of the overall portfolio and of stock T are 16% and 13%, respectively. The correlation of returns between the portfolio and stock T is 0.45. Assuming the risk analyst uses a 1-year 95% VaR and the returns are normally distributed, what is the component VaR of stock T?

A. CAD 0.096 million

B. CAD 1.444 million
C. CAD 2.041 million
D. CAD 3.948 million

Answer: B
Step by step
1. Portfolio VaR per dollar: $1.645 \times 0.16 = 0.2632$.
2. Beta of T to the portfolio: $\beta_T = \rho\sigma_T/\sigma_P = 0.45 \times 0.13/0.16 = 0.36563$.
3. Marginal VaR: $0.2632 \times 0.36563 = 0.09623$.
4. Component VaR = marginal VaR × dollar position: $0.09623 \times 15 =$ CAD 1.4435 million.

Why B is correct: Component VaR is the marginal VaR (per-dollar contribution, $\text{VaR}_P/P \times \beta$) scaled by the CAD position, so it reflects T's correlation with the portfolio, not its stand-alone risk.

Why the others are wrong
- **A:** This is the marginal VaR (0.0962 per CAD) reported as if it were a dollar amount; it was not multiplied by the CAD 15 million position.
- **C:** Used the 99% z-score 2.326 instead of 1.645: $15 \times 0.13 \times 0.45 \times 2.326 = 2.041$.
- **D:** Used the portfolio volatility instead of T's: $15 \times 0.16 \times 1.645 = 3.948$, which also ignores the correlation.

Q226

Mock 1 Q21 · Mock 2 Q25

REBUILT — Source betas (0.65, 1.35 with weights 0.35/0.65) did not weight-average to 1, which is impossible for betas measured against the portfolio; changed to weights 0.40/0.60 and betas 0.70/1.20 and rebuilt the options.

Consider a two-asset portfolio worth USD 3.4 million. The portfolio weight of Asset X is 0.40 and of Asset Y is 0.60. The standard deviation of portfolio returns is 9.8%. The betas of X and Y with respect to the portfolio are 0.70 and 1.20, respectively. Assuming normally distributed returns, what is the marginal VaR of Asset X and the component VaR of Asset Y at a 99% confidence level (z = 2.33)?

A. Marginal VaR (X): 0.2283; Component VaR (Y): USD 558,976
B. Marginal VaR (X): 0.1598; Component VaR (Y): USD 372,651
C. Marginal VaR (X): 0.1598; Component VaR (Y): USD 558,976
D. Marginal VaR (X): 0.1598; Component VaR (Y): USD 465,814

Answer: C
Step by step
1. Portfolio VaR: $2.33 \times 0.098 \times 3.4\text{m} =$ USD 776,356, so $\text{VaR}_P/P = 0.22834$.
2. Check: $0.40 \times 0.70 + 0.60 \times 1.20 = 1.00$, as required for portfolio betas.
3. Marginal VaR of X: $0.22834 \times 0.70 = 0.1598$.
4. Marginal VaR of Y: $0.22834 \times 1.20 = 0.27401$; Y's position is $0.60 \times 3.4\text{m} = 2.04\text{m}$.
5. Component VaR of Y: $0.27401 \times 2{,}040{,}000 =$ USD 558,976.

Why C is correct: Marginal VaR is $(\text{VaR}_P/P) \times \beta_i$ per dollar; component VaR scales it by the dollar position in the asset.

Why the others are wrong
- **A:** Reports the portfolio VaR per dollar (0.2283) as X's marginal VaR, ignoring beta (0.70).
- **B:** Right marginal VaR, but component VaR of Y computed with X's position (0.40 x 3.4m = 1.36m): 0.27401 x 1.36m = 372,651.
- **D:** Computes Y's stand-alone (individual) VaR, $2.33 \times 0.098 \times 2.04\text{m} = 465{,}814$, ignoring Y's beta and diversification.

Q227

Schweser Module Quiz 5.2 Q2

If portfolio assets are perfectly correlated, portfolio VaR will equal:

A. marginal VaR.
B. component VaR.
C. undiversified VaR.
D. diversified VaR.

Answer: C
Step by step

1. With $\rho = 1$: $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\,\text{VaR}_1\text{VaR}_2} = \text{VaR}_1 + \text{VaR}_2$.
2. The sum of individual VaRs (long positions) is the undiversified VaR, so there is no diversification benefit.

Why C is correct: Perfect correlation removes all diversification, so portfolio VaR collapses to the sum of the individual VaRs, i.e. undiversified VaR.

Why the others are wrong

- **A:** Marginal VaR is a per-dollar sensitivity of portfolio VaR to a position, not a portfolio total.
- **B:** Component VaR is one position's contribution; only the sum of components equals portfolio VaR, and this holds for any correlation.
- **D:** Diversified VaR accounts for correlation below 1 and is lower than undiversified VaR; the question describes the case where no diversification benefit exists, which is the definition of undiversified VaR.

Q228

Schweser Module Quiz 87.1 Q1

REBUILT — Original distractors were invented terms (vector, position, incidental VaR); replaced with a question comparing diversified and undiversified VaR with plausible distractors.

Which of the following statements about diversified VaR and undiversified VaR for a portfolio of long positions is correct?

A. Diversified VaR uses the portfolio's covariance structure and is less than or equal to undiversified VaR.
B. Undiversified VaR uses the covariance structure and is less than diversified VaR.
C. Diversified VaR is the sum of the individual VaRs of the positions.
D. Diversified VaR can exceed undiversified VaR when correlations are positive.

Answer: A

Step by step

1. Diversified (portfolio) VaR = $z\sigma_P P$, where σ_P reflects all volatilities and correlations.
2. Undiversified VaR = sum of individual VaRs = diversified VaR with $\rho = 1$ for all pairs.
3. Since $\rho \leq 1$, σ_P cannot exceed its value at $\rho = 1$, so diversified VaR $\leq$ undiversified VaR.

Why A is correct: Correlations of at most 1 mean portfolio volatility never exceeds the sum of weighted volatilities, so diversified VaR is bounded above by undiversified VaR.

Why the others are wrong

- **B:** Reverses the definitions: undiversified VaR ignores correlation (sums the individual VaRs) and is the upper bound.
- **C:** Describes undiversified VaR, not diversified VaR.
- **D:** Even with positive correlation (below 1) diversified VaR stays below the sum of individual VaRs; it equals it only at $\rho = 1$.

Q229

Schweser Module Quiz 87.1 Q2

REBUILT — Original options were vague and D was also defensible; replaced with a numerical short-position individual VaR question testing the absolute-value convention.

A portfolio contains a short position of USD 5 million in an asset with annualized return volatility of 20%. Assuming normally distributed returns, what is the individual 1-year VaR of this position at the 99% confidence level (z = 2.33)?

A. USD 2.33 million
B. -USD 2.33 million
C. USD 1.65 million
D. USD 0.466 million

Answer: A

Step by step

1. Individual VaR = $z\,\sigma\,|P_i|$: the absolute value is used because long and short positions both carry risk.
2. $2.33 \times 0.20 \times |-5| =$ USD 2.33 million.

Why A is correct: Individual VaR uses the absolute value of the position, so a short position has positive VaR given by the same formula as a long position.

Why the others are wrong

- **B:** Carries the negative sign of the short position through; risk is measured with the absolute value and cannot be negative.

- **C:** Used the 95% z-score 1.645: $1.645 \times 0.20 \times 5 = 1.65$.
- **D:** Omitted the position size: $2.33 \times 0.20 = 0.466$ (a percentage of position, not dollars).

Q230 Schweser Module Quiz 87.1 Q3

Which of the following is true with respect to computing incremental VaR? Compared with using marginal VaRs, computing with full revaluation is:

A. more costly, but less accurate.
B. less costly, but more accurate.
C. less costly, but also less accurate.
D. more costly, but also more accurate.

Answer: D

Step by step

1. Full revaluation: re-compute VaR of the whole portfolio with the new position; incremental VaR = new VaR minus old VaR.
2. This captures nonlinearities and changes in risk of existing positions, so it is accurate but expensive for large portfolios.
3. The marginal VaR shortcut uses already-available sensitivities and is linear, so it is cheap but approximate (good for small additions).

Why D is correct: Full revaluation captures nonlinear effects and the shift in risk of existing positions, at greater computational cost; marginal VaR is a cheap linear approximation.

Why the others are wrong

- **A:** Claims revaluation is less accurate; it is the accurate benchmark.
- **B:** Reverses cost: revaluation is the expensive approach, not the cheap one.
- **C:** Reverses both attributes; the cheap and less accurate method is the marginal VaR approximation.

Q231 AnalystPrep Q.3147

A commodity trader computes the VaR of a gold futures position at USD 67 million and the VaR of a palladium position at USD 32 million. If the correlation between changes in the two commodity prices is 0.4, what is the combined (diversified) portfolio VaR?

A. USD 99.00 million
B. USD 85.02 million
C. USD 74.25 million
D. USD 35.00 million

Answer: B

Step by step

1. $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$.
2. $67^2 + 32^2 = 4489 + 1024 = 5513$; $2 \times 0.4 \times 67 \times 32 = 1715.2$.
3. $\sqrt{5513 + 1715.2} = \sqrt{7228.2} = 85.02$ million.

Why B is correct: With both positions long and correlation of 0.4, VaRs combine as a square root of the quadratic form, giving a value between the uncorrelated (74.25) and fully correlated (99) bounds.

Why the others are wrong

- **A:** Simple sum $67 + 32$: assumes $\rho = 1$ (undiversified VaR).
- **C:** $\sqrt{67^2 + 32^2} = 74.25$: assumes zero correlation, dropping the cross term.
- **D:** Difference 67 - 32: nonsensical for long positions with positive correlation.

Q232 AnalystPrep Q.3157

A portfolio manager holds USD 700 million of Grinold Bank shares (return volatility 12%) and considers adding USD 300 million of Fujarah Bank shares (return volatility 9%, correlation with Grinold of 0.8). Assuming normally distributed returns, what would be the 99% VaR (z = 2.33) of the portfolio after investing USD 300 million in Fujarah Bank?

A. USD 248.92 million
B. USD 195.72 million

C. USD 444.64 million
D. USD 53.2 million

Answer: A
Step by step
1. Weights on USD 1,000 million: 0.7 and 0.3.
2. $\sigma_P^2 = 0.7^2(0.12)^2 + 0.3^2(0.09)^2 + 2(0.7)(0.3)(0.8)(0.12)(0.09) = 0.007056 + 0.000729 + 0.0036288 = 0.0114138$.
3. $\sigma_P = 10.68\%$.
4. VaR $= 2.33 \times 0.1068 \times 1000 =$ USD 248.92 million.

Why A is correct: Diversified VaR of the combined USD 1 billion portfolio uses the portfolio volatility including the 0.8 correlation.
Why the others are wrong
- **B:** Existing Grinold-only VaR $2.33 \times 0.12 \times 700 = 195.72$; ignores the new position.
- **C:** Adds the existing Grinold-only VaR (195.72) to a portfolio VaR (248.92) that already includes Grinold, double counting the existing position: 195.72 + 248.92 = 444.64.
- **D:** This is the incremental VaR (248.92 - 195.72), the change in VaR, not the portfolio VaR.

Q233

Mock 1 Q42

REBUILT — Added that returns are normal, volatilities are annual and the new position is long, so the portfolio-VaR calculation is unambiguous.

A manager holds a portfolio consisting of one position: a USD 50 million long investment in Asset P, whose annual return volatility is 7%. She is considering adding a USD 50 million long position in either Asset R or Asset S. She wants to limit portfolio VaR to USD 13.5 million at the 95% confidence level ($z = 1.645$), assuming normally distributed returns. R has an annual volatility of 11% and a correlation with P of 0.6. S has an annual volatility of 14% and a correlation with P of 0.2. Which of the two proposed additions will keep the manager within her risk budget?

A. Adding R only.
B. Adding S only.
C. Adding either R or S.
D. Adding neither R nor S.

Answer: A
Step by step
1. Each combined portfolio is USD 100 million with weights $w_P = w_X = 0.5$.
2. P + R: $\sigma_p^2 = 0.25(0.07)^2 + 0.25(0.11)^2 + 2(0.5)(0.5)(0.6)(0.07)(0.11) = 0.006560$, so $\sigma_p = 8.099\%$.
3. P + R: VaR $= 1.645 \times 0.08099 \times 100 =$ USD 13.32 million $\leq$ 13.5 million, within budget.
4. P + S: $\sigma_p^2 = 0.25(0.07)^2 + 0.25(0.14)^2 + 2(0.25)(0.2)(0.07)(0.14) = 0.007105$, so $\sigma_p = 8.429\%$.
5. P + S: VaR $= 1.645 \times 0.08429 \times 100 =$ USD 13.87 million $>$ 13.5 million, breach.
6. Only R satisfies the budget, despite its higher volatility and correlation, because S is a larger stand-alone risk (individual VaR USD 11.52m vs 9.05m for R).

Why A is correct: The portfolio VaR depends on the combined variance, which includes correlation; R's lower stand-alone risk outweighs its higher correlation with P, so P+R has VaR of 13.32m while P+S has 13.87m.
Why the others are wrong
- **B:** Reverses the result: assumes the lower correlation of S (0.2) automatically gives the lower portfolio VaR, ignoring that S's 14% volatility is much larger.
- **C:** Treats the positions as uncorrelated ($\rho = 0$): VaR is then about 10.7m (R) and 12.9m (S), both within budget, but the stated correlations raise the true VaR.
- **D:** Uses undiversified VaR (sum of individual VaRs): 5.76 + 9.05 = 14.8m and 5.76 + 11.52 = 17.3m, both over 13.5m, ignoring diversification benefits.

LO 87.b

Explain the impact of correlation on portfolio risk.

Q234 AnalystPrep Q.2464

A fund manager is studying the relationship between covariance and diversification. Which of the following statements would be correct for the manager to make in this context?

A. A higher covariance implies a stronger relationship between the assets' returns, indicating more diversification benefits.
B. We can still achieve diversification with positive covariance.
C. Diversification is not possible with positive covariance.
D. Covariance is not relevant to diversification.

Answer: B

Step by step

1. Portfolio variance for two assets: $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\,\mathrm{Cov}_{12}$.
2. A diversification benefit exists whenever $\rho < 1$, because then $\sigma_p < w_1\sigma_1 + w_2\sigma_2$.
3. Positive covariance only means $\rho > 0$; as long as $\rho < 1$ some diversification remains, though less than with low or negative covariance.

Why B is correct: Diversification benefits vanish only when correlation equals +1; any positive covariance with $\rho < 1$ still lowers portfolio risk below the sum of stand-alone risks.

Why the others are wrong

- **A:** Reverses the relationship: higher covariance means less diversification benefit, since the cross term increases portfolio variance.
- **C:** Over-states the point: benefits are eliminated only at perfect correlation ($\rho = +1$), not for every positive covariance.
- **D:** Covariance (correlation) is the key driver of portfolio variance and hence of diversification.

Q235 AnalystPrep Q.2746

REBUILT — Added 'long-only positions' to the stem (with short positions the sign of the effective correlation flips) and tidied option D.

For a portfolio of long-only positions, the portfolio VaR is maximized when the pairwise correlations between the returns of the underlying assets are:

A. Perfectly positive
B. Perfectly negative
C. Zero
D. Strictly between -1 and +1 (neither perfectly positive nor perfectly negative)

Answer: A

Step by step

1. For long positions, $\mathrm{VaR}_P = \sqrt{\sum_i \sum_j \rho_{ij}\mathrm{VaR}_i\mathrm{VaR}_j}$ is increasing in each ρ_{ij}.
2. The upper bound is at $\rho_{ij} = +1$: $\mathrm{VaR}_P = \sum_i \mathrm{VaR}_i$ (undiversified VaR).

Why A is correct: Portfolio VaR rises with correlation and is capped at the sum of individual VaRs, reached only when all correlations are +1.

Why the others are wrong

- **B:** Perfect negative correlation gives the minimum VaR (potentially zero for a suitably weighted pair).
- **C:** Zero correlation drops the cross terms and gives VaR below the sum.
- **D:** Any correlation strictly below +1 yields some diversification, so VaR is below the maximum.

Q236 AnalystPrep Mock 1 Q20

REBUILT — Replaced an unexplained distractor (USD 63.45 million) with the rho = -1 result so every wrong option reflects a realistic error.

Simon Sarkar is an analyst at Indo-Sino Investments. He is evaluating the risk of a portfolio and computes the VaR for the two positions: VaR1 = USD 7.3 million and VaR2 = USD 5.4 million. If the returns of the two assets are perfectly correlated ($\rho = +1$), the VaR of the portfolio is closest to:

A. USD 1.90 million.
B. USD 12.70 million.
C. USD 82.45 million.
D. USD 9.08 million.

Answer: B

Step by step

1. $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$.
2. With $\rho = 1$: $\sqrt{53.29 + 29.16 + 2(7.3)(5.4)} = \sqrt{161.29}$.
3. This equals $7.3 + 5.4 = 12.70$ million: no diversification, so VaR is the simple sum.

Why B is correct: With perfect positive correlation the cross term completes a perfect square, so portfolio VaR equals the sum of the individual VaRs (the undiversified VaR).

Why the others are wrong

- **A:** This is the $\rho = -1$ result, $|7.3 - 5.4|$; the question specifies perfect positive correlation.
- **C:** Forgot the square root: $7.3^2 + 5.4^2 = 82.45$ is a squared quantity, not a VaR.
- **D:** This is the zero-correlation result, $\sqrt{7.3^2 + 5.4^2} = 9.08$, which omits the cross term.

Q237

AnalystPrep Mock 3 Q17

REBUILT — Original asked which addition 'keeps the risk budget optimal', which is ambiguous; reframed as the addition giving the lower portfolio VaR (z = 2.33 stated) and replaced distractors with realistic errors.

Anthony James has a portfolio with a single position of USD 700 million in shares of Grinold Bank. He is considering adding a USD 300 million position in either Fujarah Bank or Yamama Bank.

Share	Volatility	Correlation with Grinold
Grinold Bank (existing, USD 700m)	12%	-
Fujarah Bank	9%	0.8
Yamama Bank	12%	0.0

Using a 99% confidence level (z = 2.33), which addition results in the lower portfolio VaR, and what is that VaR?

A. Fujarah; VaRp = USD 195.72 million.

B. Yamama; VaRp = USD 212.94 million.

C. Fujarah; VaRp = USD 248.93 million.

D. Yamama; VaRp = USD 279.60 million.

Answer: B

Step by step

1. Weights: Grinold 0.7, new bank 0.3 (total USD 1,000m).
2. Fujarah: $\sigma_p^2 = 0.7^2(0.12)^2 + 0.3^2(0.09)^2 + 2(0.7)(0.3)(0.12)(0.09)(0.8) = 0.011414$, $\sigma_p = 10.68\%$, VaR $= 2.33 \times 0.1068 \times 1000 = 248.93$ million.
3. Yamama: $\sigma_p^2 = 0.007056 + 0.001296 + 0 = 0.008352$, $\sigma_p = 9.14\%$, VaR $= 2.33 \times 0.0914 \times 1000 = 212.94$ million.
4. Yamama gives the lower portfolio VaR (USD 212.94m versus USD 248.93m).

Why B is correct: Yamama's zero correlation removes the cross term, so despite its higher stand-alone volatility the diversification benefit makes portfolio risk lower than with the 0.8-correlated Fujarah.

Why the others are wrong

- **A:** USD 195.72m is just the stand-alone VaR of the existing USD 700m position ($2.33 \times 0.12 \times 700$); it ignores the added position entirely.
- **C:** Correct arithmetic for Fujarah, but it is the higher-VaR portfolio: the 0.8 correlation leaves little diversification. Choosing it reflects judging by stand-alone volatility (9% < 12%).
- **D:** Yamama with correlation wrongly treated as +1 (undiversified sum $195.72 + 83.88 = 279.60$); the correlation is 0, so the cross term should be dropped.

Q238

AnalystPrep Q.2462

REBUILT — Original statement II ('risk will be amplified') was vague and the source rationale was flawed; added individual VaRs and restated the statements precisely, focusing on the long-short sign effect.

A hedge fund is USD 1 billion long corporate bonds and USD 1 billion short Treasury bonds. The stand-alone VaR of the long position is USD 30 million and that of the short position is USD 20 million. Which of the following statements is (are) correct?

I. If the returns of the corporate and Treasury bonds are uncorrelated, the portfolio VaR is zero.

II. If the returns of the corporate and Treasury bonds have a correlation of -1, the portfolio VaR equals the sum of the two individual VaRs (USD 50 million).

A. Only I

B. Only II
C. Both I and II
D. Neither I nor II

Answer: B

Key correction: Source rationale for II was flawed (it cited an undiversified VaR 'only if correlation is unity'); the key B stands, but the correct reason is that the short position flips the sign of the P&L correlation.

Step by step

1. Because the Treasury position is short, the P&L correlation of the two positions is $-\rho$, where ρ is the correlation of bond returns.
2. I: with $\rho = 0$, P&L correlation is 0, so $\text{VaR}_P = \sqrt{30^2 + 20^2} = 36.06$ million, not zero. False.
3. II: with $\rho = -1$, P&L correlation is +1, so $\text{VaR}_P = 30 + 20 = 50$ million: no diversification, risk is at its maximum. True.
4. (Only a return correlation of +1 would hedge the book: VaR $= \sqrt{900 + 400 - 1200} = 10$ million.)

Why B is correct: For a long-short book the effective correlation between position P&Ls is the negative of the return correlation, so a return correlation of -1 means perfectly correlated P&Ls and undiversified (additive) VaR.

Why the others are wrong

- **A:** Statement I is false and II is true: zero correlation only removes the cross term, giving USD 36.06m, not zero.
- **C:** Statement I is false (zero correlation gives USD 36.06m); the error is confusing 'uncorrelated' with 'perfectly hedged'.
- **D:** Statement II is true; this answer comes from ignoring the short position's sign flip and assuming a correlation of -1 must diversify.

Q239

AnalystPrep Q.2463

REBUILT — Options C and D were OCR-garbled; reconstructed with plausible distractors based on common counting errors.

A portfolio manager wants to compute the variance of a portfolio consisting of n different securities. How many distinct covariance terms (pairs of different securities) must be estimated to compute the portfolio variance?

A. $n(n-1)/2$
B. $n(n+1)/2$
C. n^2
D. $n(n-1)$

Answer: A

Step by step

1. The covariance matrix is $n \times n$, with n variances on the diagonal and $n^2 - n = n(n-1)$ off-diagonal entries.
2. $\text{Cov}(i,j) = \text{Cov}(j,i)$, so the off-diagonal entries are symmetric and each distinct pair counts once.
3. Distinct covariances $= n(n-1)/2$ (for example, $n = 10$ gives 45).

Why A is correct: The number of distinct pairs from n securities is n choose 2, which equals $n(n-1)/2$.

Why the others are wrong

- **B:** $n(n+1)/2$ is the number of distinct elements in the whole symmetric matrix, including the n variances on the diagonal.
- **C:** n^2 counts every cell of the covariance matrix, including variances and the duplicate lower triangle.
- **D:** $n(n-1)$ counts both $\text{Cov}(i,j)$ and $\text{Cov}(j,i)$, double-counting each pair.

Q240

AnalystPrep Q.2465

REBUILT — Original two-statement 'both reduce risk' question was a giveaway and ignored the diversification floor; rebuilt as a calculation using the equal-weight, equal-correlation formula.

An equally weighted portfolio holds N positions, each with a return volatility of 30%. The correlation between each pair of returns is 0.2. As N becomes very large, the portfolio volatility approaches which of the following?

A. 0%
B. 6.00%
C. 13.42%
D. 15.87%

Answer: C

Step by step

1. For an equally weighted portfolio with equal volatilities and correlations: $\sigma_p = \sigma\sqrt{\frac{1}{N} + \rho\left(1 - \frac{1}{N}\right)}$.
2. As $N \to \infty$, $\sigma_p \to \sigma\sqrt{\rho} = 0.30\sqrt{0.2} = 13.42\%$.
3. Check: N = 10 gives 15.87%, N = 100 gives 13.68%, converging to 13.42%.

Why C is correct: Adding assets removes only the idiosyncratic (1/N) part of variance; the correlated component $\rho\sigma^2$ cannot be diversified away, so volatility has a floor at $\sigma\sqrt{\rho}$.

Why the others are wrong

- **A:** Assumes enough securities eliminate all risk; this is true only when correlations are zero.
- **B:** Forgot the square root, using $\sigma\rho = 0.30 \times 0.2$ instead of $\sigma\sqrt{\rho}$.
- **D:** This is the volatility for N = 10 ($30\%\sqrt{0.1 + 0.2 \times 0.9}$), not the limit as N grows.

Q241

AnalystPrep Q.2466

REBUILT — Original options 'All of the above' / 'None of the above' were giveaways and its rationale wrongly claimed rho = -1 also gives the sum; rebuilt with numbers and statements about specific correlations.

A portfolio consists of two long positions with stand-alone VaRs of USD 8 million and USD 6 million. Which of the following statements about the portfolio VaR is correct?

A. At a correlation of -1, portfolio VaR equals USD 14 million.

B. At a correlation of 0, portfolio VaR equals USD 14 million.

C. At a correlation of +1, portfolio VaR equals USD 14 million, and for any correlation below +1 it is less than USD 14 million.

D. At a correlation of +1, portfolio VaR equals USD 10 million.

Answer: C

Key correction: Source explanation wrongly implied rho = -1 also gives the sum of VaRs; only rho = +1 does.

Step by step

1. $\text{VaR}_P = \sqrt{8^2 + 6^2 + 2\rho(8)(6)}$.
2. $\rho = +1$: $\sqrt{64 + 36 + 96} = 14$. $\rho = 0$: $\sqrt{100} = 10$. $\rho = -1$: $\sqrt{4} = 2$.
3. Any $\rho < 1$ shrinks the cross term, so $\text{VaR}_P < \text{VaR}_1 + \text{VaR}_2$.

Why C is correct: Portfolio VaR equals the sum of the individual VaRs (undiversified VaR) only when correlation is +1; otherwise it is strictly smaller.

Why the others are wrong

- **A:** At $\rho = -1$ VaR is $|8 - 6| = 2$ million, the minimum, not the sum.
- **B:** At $\rho = 0$ the cross term disappears and VaR is $\sqrt{64 + 36} = 10$ million; adding the VaRs ignores the square-root combination.
- **D:** USD 10 million is the zero-correlation value, mistakenly attached to perfect correlation.

Q242

AnalystPrep Q.2646

In risk management, the diversification status of a portfolio directly impacts its Value at Risk (VaR). The VaR of a long-only portfolio is said to be undiversified, that is, equal to the sum of the individual VaRs, when the assets in the portfolio are:

A. Positively correlated

B. Perfectly correlated

C. Negatively correlated

D. Not correlated

Answer: B

Step by step

1. $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$.
2. Only at $\rho = +1$ does this reduce to $\text{VaR}_1 + \text{VaR}_2$, the undiversified VaR.

Why B is correct: Undiversified VaR arises only when correlation is exactly +1; any $\rho < 1$ produces a diversification benefit.

Why the others are wrong

- **A:** Positive but imperfect correlation ($0 < \rho < 1$) still yields VaR below the sum.
- **C:** Negative correlation gives the largest diversification benefit, the opposite of undiversified.
- **D:** Zero correlation drops the cross term, giving a VaR smaller than the sum (square root of the sum of squares).

Q243

AnalystPrep Q.2648

REBUILT — Rounding depended on the z-value (2.33 vs 2.326) and distractors were weak; fixed z = 2.326 in the stem and replaced distractors with realistic mistakes.

USD 10 million is invested in assets X and Y in the ratio 6:4. The return volatilities of X and Y are 5% and 10%, respectively. Using z = 2.326 for a 99% confidence level, what is the portfolio VaR if the assets are (i) perfectly correlated and (ii) uncorrelated? (Ignore the time horizon and expected returns.)

A. (i) USD 1.16 million; (ii) USD 1.63 million
B. (i) USD 1.63 million; (ii) USD 1.16 million
C. (i) USD 1.74 million; (ii) USD 1.30 million
D. (i) USD 1.15 million; (ii) USD 0.82 million

Answer: B

Step by step

1. $\text{VaR}_X = 2.326 \times 0.05 \times 6 = 0.698$ million; $\text{VaR}_Y = 2.326 \times 0.10 \times 4 = 0.930$ million.
2. (i) $\rho = 1$: $0.698 + 0.930 = 1.63$ million.
3. (ii) $\rho = 0$: $\sqrt{0.698^2 + 0.930^2} = 1.16$ million.

Why B is correct: Perfect correlation gives the simple sum of the individual VaRs, while zero correlation gives the square root of the sum of squares, which must be smaller.

Why the others are wrong

- **A:** Swaps the two cases: the uncorrelated VaR cannot exceed the perfectly correlated (undiversified) VaR.
- **C:** Splits the money 5:5 instead of 6:4 (giving 1.74 and 1.30).
- **D:** Uses z = 1.645 (95% confidence) instead of the 99% value 2.326.

Q244

AnalystPrep Q.2747

REBUILT — Source figure ($177,989) came from rounding volatility to 10.82%; unrounded result is $177,934, so the stem now asks for 'closest to' and the options were rebuilt with error-based distractors.

A portfolio consists of five equally weighted positions, each worth $200,000. Each position has a return standard deviation of 15%, and the correlation between every pair of positions is 0.4. Using a z-score of 1.645 (95% confidence), the portfolio VaR is closest to:

A. $110,350
B. $177,934
C. $191,132
D. $251,595

Answer: B

Step by step

1. Equal weights, equal volatility, common correlation: $\sigma_P = \sigma\sqrt{\frac{1}{N} + \left(1 - \frac{1}{N}\right)\rho}$.
2. $\sigma_P = 0.15\sqrt{0.20 + 0.80 \times 0.4} = 0.15\sqrt{0.52} = 10.817\%$.
3. Portfolio value = 5 × $200,000 = $1,000,000.
4. VaR $= 1.645 \times 0.10817 \times 1{,}000{,}000 = \$177{,}934$.

Why B is correct: Pairwise correlation of 0.4 raises portfolio volatility above the zero-correlation case but well below the no-diversification case, and the closed-form equal-weight formula captures this.

Why the others are wrong

- **A:** Set $\rho = 0$ ($\sigma_P = \sigma/\sqrt{N}$), ignoring the positive correlation, which understates risk.
- **C:** Used $\sqrt{1/N + \rho}$ and forgot to multiply ρ by (1 – 1/N), overstating risk.
- **D:** Used the 99% z-score of 2.326 instead of 1.645.

Q245 Mock 4 Q54

A risk analyst estimates the VaR of an equally weighted, two-asset portfolio to demonstrate the impact of correlation. Each asset has a daily return volatility of 2%, the portfolio value is USD 20 million, and the 10-day 99% VaR is USD 2.33 million. If the correlation between the two assets doubles, which of the following is closest to the new 10-day 99% portfolio VaR?

A. USD 1.16 million.
B. USD 2.55 million.
C. USD 4.66 million.
D. USD 5.43 million.

Answer: B

Step by step

1. Equal weights, equal vols: $\sigma_P = \sigma\sqrt{(1+\rho)/2}$.
2. Implied daily $\sigma_P = 2.33/(20 \times 2.326 \times \sqrt{10}) = 1.584\%$, so $(1+\rho)/2 = (1.584/2)^2 = 0.627$ and $\rho \approx 0.25$.
3. Doubled $\rho \approx 0.50$: $\sigma_P = 2\%\sqrt{0.75} = 1.732\%$.
4. New VaR = $20 \times 0.01732 \times 2.326 \times \sqrt{10} \approx \$2.55$ million.

Why B is correct: VaR is proportional to portfolio volatility, which grows only with the square root of $(1+\rho)/2$; doubling ρ raises VaR by roughly 10%, not 100%.

Why the others are wrong

- **A:** Halved the VaR, getting the direction of the correlation effect wrong.
- **C:** Doubled the VaR, treating VaR as linear in correlation.
- **D:** Multiplied 2.33 by itself; not a valid calculation.

Q246 AnalystPrep Q.3041

Understanding the correlation between portfolio assets is crucial in risk management, especially when calculating Value at Risk (VaR). The degree of correlation significantly impacts the overall risk profile of the portfolio. If portfolio assets are perfectly correlated, portfolio VaR will equal:

A. Component VaR
B. Marginal VaR
C. Diversified VaR
D. Undiversified VaR

Answer: D

Step by step

1. For two positions, $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$.
2. With $\rho = 1$, this becomes $\sqrt{(\text{VaR}_1 + \text{VaR}_2)^2} = \text{VaR}_1 + \text{VaR}_2$.
3. The sum of the individual VaRs (for long positions) is the undiversified VaR, the upper bound on portfolio VaR.

Why D is correct: With perfect correlation there is no diversification benefit, so portfolio VaR equals the sum of the stand-alone VaRs, which is undiversified VaR.

Why the others are wrong

- **A:** Component VaR is each position's contribution to diversified VaR; the components sum to diversified VaR, not the undiversified sum.
- **B:** Marginal VaR is the change in portfolio VaR per unit added to a position; it is not a portfolio total.
- **C:** Diversified VaR uses the actual correlations and is below the sum of stand-alone VaRs unless every pair has $\rho = 1$; the sum of stand-alone VaRs is by definition called undiversified VaR.

Q247 AnalystPrep Q.3141

REBUILT — Distractor options replaced so each reflects a specific realistic error.

Bob William is an equity strategist who computes the VaR of the two positions in his portfolio as VaR1 = $3.6 million and VaR2 = $1.2 million. If the returns of the two assets are uncorrelated, the portfolio VaR is closest to:

A. $3.794 million
B. $14.40 million
C. $4.80 million
D. $3.394 million

Answer: A

Step by step

1. Uncorrelated: $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$.
2. $3.6^2 + 1.2^2 = 12.96 + 1.44 = 14.40$.
3. $\sqrt{14.40} = \$3.794$ million.

Why A is correct: With zero correlation the cross term vanishes, so VaR combines in quadrature: $\sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$, below the simple sum.

Why the others are wrong

- **B:** Reported the sum of squares (14.40) and forgot the square root.
- **C:** Added the VaRs (3.6 + 1.2), which is the perfectly correlated ($\rho = 1$) case.
- **D:** Subtracted instead of added the squares: $\sqrt{3.6^2 - 1.2^2}$.

Q248 AnalystPrep Q.3146

REBUILT — Original distractors ($100m, $300m, $20m) were not tied to any realistic error; replaced.

A portfolio consists of two positions with VaRs of $10 million and $40 million. If the returns of the two positions are not correlated, the portfolio VaR is closest to:

A. $50.00 million
B. $1,700 million
C. $38.73 million
D. $41.23 million

Answer: D

Step by step

1. Uncorrelated: $\text{VaR}_P = \sqrt{10^2 + 40^2}$.
2. $100 + 1600 = 1700$.
3. $\sqrt{1700} = \$41.23$ million.

Why D is correct: With zero correlation the cross term vanishes, so VaR combines in quadrature: $\sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$, below the simple sum.

Why the others are wrong

- **A:** Summed the VaRs (10 + 40), the perfectly correlated case.
- **B:** Used the sum of squares (1,700) and forgot the square root.
- **C:** Subtracted the squares: $\sqrt{40^2 - 10^2}$.

Q249 Schweser Module Quiz 87.1 Q4

A portfolio consists of two positions. The VaRs of the two positions are $10 million and $20 million. If the returns of the two positions are not correlated, the VaR of the portfolio would be closest to:

A. $5.48 million.
B. $15.00 million.
C. $22.36 million.
D. $25.00 million.

Answer: C

Step by step

1. Uncorrelated: $\text{VaR}_P = \sqrt{10^2 + 20^2}$.
2. $100 + 400 = 500$.
3. $\sqrt{500} = \$22.36$ million.

Why C is correct: With zero correlation the cross term vanishes, so VaR combines in quadrature: $\sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$, below the simple sum.

Why the others are wrong

- **A:** Took the square root of the sum of the VaRs: $\sqrt{10 + 20}$.
- **B:** Averaged the two VaRs instead of combining them.
- **D:** Not produced by a valid method; it lies between the uncorrelated value and the $30 million perfect-correlation

bound and would require an arbitrary positive correlation.

Q250

AnalystPrep Q.3159

REBUILT — Original distractors ($6.3m, $10.52m) were not tied to realistic errors; replaced.

Donald Clayton constructs a portfolio of two stocks, A and B, with VaR_A = $7.9 million and VaR_B = $5.7 million. Assuming the two stocks are uncorrelated, the portfolio VaR is closest to:

A. $5.47 million
B. $94.90 million
C. $13.60 million
D. $9.74 million

Answer: D

Step by step

1. Uncorrelated: $\text{VaR}_P = \sqrt{7.9^2 + 5.7^2}$.
2. $62.41 + 32.49 = 94.90$.
3. $\sqrt{94.90} = \$9.74$ million.

Why D is correct: With zero correlation the cross term vanishes, so VaR combines in quadrature: $\sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$, below the simple sum.

Why the others are wrong

- **A:** Subtracted the squares: $\sqrt{7.9^2 - 5.7^2}$.
- **B:** Stopped at the sum of squares (94.90) and forgot the square root.
- **C:** Added the VaRs (7.9 + 5.7), the perfectly correlated case.

Q251

AnalystPrep Q.4712

REBUILT — Original distractors ($1.198m, $2.504m) were not tied to realistic errors; replaced.

A portfolio is made up of two uncorrelated positions. The VaR of position one is $3 million and the VaR of position two is $2 million. The portfolio VaR is closest to:

A. $3.606 million
B. $5.000 million
C. $2.236 million
D. $13.000 million

Answer: A

Step by step

1. Uncorrelated: $\text{VaR}_P = \sqrt{3^2 + 2^2}$.
2. $9 + 4 = 13$.
3. $\sqrt{13} = \$3.606$ million.

Why A is correct: With zero correlation the cross term vanishes, so VaR combines in quadrature: $\sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$, below the simple sum.

Why the others are wrong

- **B:** Added the VaRs (3 + 2), the perfectly correlated case.
- **C:** Subtracted the squares: $\sqrt{3^2 - 2^2}$.
- **D:** Used the sum of squares (13) and forgot the square root.

Q252

AnalystPrep Q.3165

REBUILT — Source option said 'increase by $60,000' (rounded); exact change is about $58,000, and the options were restated with error-based distractors.

Abraham Bell manages $40 million in JetAero and $25 million in Indigo. The 95% 1-day VaRs are $2.3 million for JetAero and $1.7 million for Indigo, and the correlation between the two stocks' returns is 0.6. The committee sells $7 million of JetAero and uses the proceeds to buy Indigo. Assume normally distributed returns and that the stocks' volatilities are unchanged. Which of the following best describes the effect on the 95% 1-day portfolio VaR?

A. It remains unchanged
B. It increases by approximately $58,000
C. It decreases by approximately $58,000

D. It decreases by approximately $127,000

Answer: B

Key correction: Source key (B, increase by $60,000) was right in direction but rounded; the exact change is about $58,000.

Step by step

1. Initial: $\sqrt{2.3^2 + 1.7^2 + 2(0.6)(2.3)(1.7)} = \sqrt{5.29 + 2.89 + 4.692} = \$3.588$ million.
2. VaR scales with position size: JetAero $2.3 \times 33/40 = 1.8975$; Indigo $1.7 \times 32/25 = 2.176$.
3. New: $\sqrt{1.8975^2 + 2.176^2 + 2(0.6)(1.8975)(2.176)} = \$3.646$ million.
4. Change = 3.646 - 3.588 ≈ +$0.058 million, an increase of about $58,000.

Why B is correct: Indigo carries a higher VaR per dollar (6.8% vs 5.75%), and with a correlation of 0.6 moving money into it raises portfolio VaR.

Why the others are wrong

- **A:** Assumed shifting money between the two assets cannot change risk; the assets have different volatilities.
- **C:** Sign error: the shift is into the more volatile stock, so VaR rises.
- **D:** Looked at the change in the undiversified sum (4.2 to 4.0735 = -$0.1265m), which falls, instead of the diversified VaR.

Q253

AnalystPrep Q.3166

REBUILT — Dangling 'this' in the stem was fixed and the implausible 'decrease by 3.4 million' option replaced with an error-based distractor.

A portfolio manager holds $60 million in BeiXin Shipping and $40 million in TongChi Logistics, with 95% 1-day VaRs of $6.3 million and $4.1 million respectively. Assume normal returns and that the individual VaRs do not change. The correlation between the two stocks' returns moves from zero in 2017 to perfect positive correlation (+1) in 2018. The effect on the 95% 1-day portfolio VaR is:

A. No change
B. An increase of $2.88 million
C. A decrease of $2.88 million
D. An increase of $7.52 million

Answer: B

Step by step

1. 2017 ($\rho = 0$): $\sqrt{6.3^2 + 4.1^2} = \sqrt{39.69 + 16.81} = \$7.517$ million.
2. 2018 ($\rho = 1$): $6.3 + 4.1 = \$10.40$ million.
3. Change = 10.40 - 7.517 = +$2.88 million.

Why B is correct: Higher correlation removes diversification benefit, pushing portfolio VaR from the quadrature value up to the simple sum.

Why the others are wrong

- **A:** Ignored that the correlation changed; constant individual VaRs do not fix portfolio VaR.
- **C:** Wrong direction: rising correlation increases portfolio VaR.
- **D:** Reported the 2017 VaR ($7.52m) as if it were the change.

Q254

AnalystPrep Q.2471

REBUILT — Original asked which positions 'hedge' two negatively correlated securities, with a wrong key (long both is not a hedge in a risk-minimizing sense) and a vague stem; rebuilt as a numeric test of how correlation drives portfolio VaR.

A portfolio holds long positions in two assets, A and B. The standalone 95% VaR is $4 million for A and $3 million for B. The returns on A and B have a correlation of -0.5. Assuming normally distributed returns, what is the diversified portfolio VaR?

A. $7.00 million
B. $6.08 million
C. $5.00 million
D. $3.61 million

Answer: D

Key correction: Source key (C, 'go long on both') was not a defensible hedging answer; with negative correlation, holding both long already lowers risk relative to the sum of individual VaRs, which is what the rebuilt question quantifies.

Step by step

1. Portfolio VaR for two positions: $\text{VaR}_p = \sqrt{\text{VaR}_A^2 + \text{VaR}_B^2 + 2\rho\,\text{VaR}_A\text{VaR}_B}$.
2. Substitute: $4^2 + 3^2 + 2(-0.5)(4)(3) = 16 + 9 - 12 = 13$.
3. $\text{VaR}_p = \sqrt{13} = 3.61$ million, below the undiversified VaR of 7 and even below the larger individual VaR of 4.

Why D is correct: Negative correlation makes the cross term negative, so the portfolio VaR falls below the sum of individual VaRs and, here, below either position's own VaR.

Why the others are wrong

- **A:** Simple sum $4 + 3$; valid only for $\rho = +1$, so it ignores diversification.
- **B:** Sign error: used $+2\rho\,\text{VaR}_A\text{VaR}_B$ with $|\rho|$, i.e. $\sqrt{16 + 9 + 12}$, treating the correlation as positive.
- **C:** Set the cross term to zero, $\sqrt{16 + 9}$; that is the zero-correlation answer and ignores the negative correlation.

LO 87.c

Apply the concept of marginal VaR in making portfolio management decisions.

Q255

AnalystPrep Q.2468

REBUILT — Original options were vague and partly nonsensical (B, D); rebuilt with numeric marginal VaRs and realistic distractors keyed to the same concept.

An investor holds three positions and wants to lower portfolio VaR by reducing exactly one position by \$100,000 (the proceeds are held in cash, which has no risk). Position data:

Position	Weight	Standalone volatility	Marginal VaR (per \$1 of position)
X	50%	30%	0.040
Y	30%	20%	0.065
Z	20%	12%	0.018

Which position should be reduced?

A. Position X, because it has the largest weight and the highest standalone volatility

B. Position Y, because it has the largest marginal VaR

C. Position Z, because it has the smallest marginal VaR

D. Any position, because the reduction is the same \$100,000 in each case

Answer: B

Step by step

1. Reducing a position by Δw lowers portfolio VaR by approximately $\text{MVaR}_i \times \Delta w$.
2. X: $0.040 \times 100{,}000 = \$4{,}000$.
3. Y: $0.065 \times 100{,}000 = \$6{,}500$.
4. Z: $0.018 \times 100{,}000 = \$1{,}800$.
5. Y gives the largest VaR reduction, so Y is the position to cut.

Why B is correct: Portfolio VaR changes by roughly marginal VaR times the dollar change, so a fixed-size cut is most effective in the position with the highest marginal VaR.

Why the others are wrong

- **A:** Ranks on size or standalone risk; marginal VaR also reflects correlation with the rest of the portfolio, and X's reduction is only about \$4,000.
- **C:** Reverses the rule: cutting the lowest-marginal-VaR position gives the smallest reduction (about \$1,800); that is the position to add to.
- **D:** Equal dollar size does not mean equal risk impact; each position's contribution to portfolio VaR per dollar differs.

Q256

AnalystPrep Q.2472

REBUILT — Clarified that total capital is held fixed (cut one, add the other) so the answer is unambiguous; same data and concept.

A fund manager holds a two-currency portfolio and must keep total invested capital fixed:

Asset	Weight	Marginal VaR
CAD	66.67%	0.0528
EUR	33.33%	0.1521

To reduce portfolio risk, the manager should:

A. Reduce the CAD position and increase the EUR position

B. Reduce the EUR position and increase the CAD position
C. Increase both the CAD and EUR positions
D. Make no change, because the portfolio is already at the minimum-risk allocation

Answer: B
Step by step
1. Risk is minimized when marginal VaRs are equal across positions.
2. Here $0.1521 > 0.0528$, so the portfolio is not at the minimum (D is ruled out).
3. Shift capital out of the higher-MVaR asset (EUR) and into the lower-MVaR asset (CAD).
4. Capital is fixed, so both cannot be increased (C is infeasible).

Why B is correct: Moving funds from the high-marginal-VaR asset to the low-marginal-VaR asset lowers portfolio VaR until marginal VaRs equalize.
Why the others are wrong
- **A:** Direction reversed: it adds to the asset with the higher marginal VaR, raising portfolio VaR.
- **C:** Increasing both requires extra capital, which violates the fixed-capital condition and would raise VaR.
- **D:** Unequal marginal VaRs (0.0528 vs 0.1521) show the allocation is not risk-minimizing.

Q257
BT answers P2.T8.14.1 · BT notes P2.T9.14.1

A portfolio manager has identified two stocks, a utility stock and a technology stock, each with an expected return of +20% over the next year. If she can only add one position, which of the following criteria is best?
A. Stock with lower marginal value at risk (VaR)
B. Stock with higher marginal VaR
C. Stock with lower volatility
D. Stock with higher incremental VaR

Answer: A
Step by step
1. Both stocks offer the same expected return, so return does not discriminate.
2. Adding a position raises portfolio VaR by about MVaR times the amount added.
3. Choose the stock with the lowest marginal VaR, which adds the least risk to the portfolio.
4. Standalone volatility ignores correlation with the existing portfolio, so it is not the right criterion.

Why A is correct: With equal expected returns, the best addition is the one that increases portfolio risk least, i.e. the lowest marginal VaR.
Why the others are wrong
- **B:** Higher marginal VaR adds more portfolio risk for the same return.
- **C:** Standalone volatility ignores correlation with the portfolio; a low-volatility stock can still have a higher marginal VaR.
- **D:** Higher incremental VaR also means a larger increase in portfolio VaR, the opposite of the goal.

Q258
GARP PE1 Q65 · Mock 3 Q16

A money manager who has recently received a small amount of new capital is planning to invest it in an existing fund, which is benchmarked to an index. Rather than investing in a new asset, the manager plans to increase the holding of one of the fund's four assets. Information about these assets and their performance during the most recent evaluation period is given below:

Asset	Portfolio weight	Return	Volatility of return	Beta to the portfolio
BDE	0.35	14%	19%	1.20
JKL	0.30	13%	18%	0.90
MNO	0.25	13%	16%	1.00
STU	0.10	10%	10%	0.80

The portfolio manager wants to select the asset that has the lowest marginal VaR as long as its Jensen's alpha is greater than or equal to the market risk premium. Assuming the risk-free interest rate is 3% and the market return is 8%, which asset should the portfolio manager select?
A. Asset BDE
B. Asset JKL
C. Asset MNO

D. Asset STU

Answer: B

Step by step

1. Market risk premium = 8% − 3% = 5%.
2. Jensen's alpha = $R_i - [R_f + \beta_i(R_m - R_f)]$.
3. BDE: $14 - (3 + 1.20 \times 5) = 5.0\%$; JKL: $13 - (3 + 0.90 \times 5) = 5.5\%$; MNO: $13 - (3 + 1.00 \times 5) = 5.0\%$; STU: $10 - (3 + 0.80 \times 5) = 3.0\%$.
4. STU fails the screen (3.0% < 5%); the other three pass.
5. Marginal VaR is proportional to beta to the portfolio, so the lowest MVaR among the passers is JKL ($\beta = 0.90$).

Why B is correct: Marginal VaR is proportional to beta, so after screening on alpha the lowest-beta qualifying asset, JKL, has the lowest marginal VaR.

Why the others are wrong

- **A:** BDE passes the alpha screen but has the highest beta (1.20), hence the highest marginal VaR.
- **C:** MNO passes the screen but its beta of 1.00 is higher than JKL's 0.90.
- **D:** STU has the lowest beta, but its alpha is 3.0%, below the 5% threshold; ignoring the screen leads here.

LO 87.d

Explain and calculate the risk-minimizing position and the position that maximizes the ratio of expected return to risk.

Q259

AnalystPrep Q.2473

REBUILT — Original table was OCR-garbled and only asked a trivial weighted average; rebuilt with a clean table, a risk-free rate and a question on expected excess return plus the 87.d rebalancing rule (excess return to beta).

A manager holds a currency portfolio with the risk-free rate at 2.0%. Each beta is the position's beta relative to the portfolio.

Position	Weight	Expected return	Beta
CAD	66.67%	8.00%	0.615
EUR	33.33%	5.00%	1.770

Which of the following correctly gives the portfolio's expected excess return and the adjustment that moves the portfolio toward the position maximizing the ratio of expected excess return to risk?

A. Expected excess return 7.0%; increase EUR and decrease CAD

B. Expected excess return 5.0%; increase CAD and decrease EUR

C. Expected excess return 5.0%; increase EUR and decrease CAD

D. Expected excess return 7.0%; increase CAD and decrease EUR

Answer: B

Key correction: Source key (B) gave a 7% expected return, which is the plain weighted average; the garbled item never addressed excess return or rebalancing, so the rebuilt item's answer is B for different reasons.

Step by step

1. Expected return: $0.6667(8\%) + 0.3333(5\%) = 7.0\%$.
2. Expected excess return: $7.0\% - 2.0\% = 5.0\%$ (equivalently $0.6667(6\%) + 0.3333(3\%) = 5.0\%$).
3. Excess return to beta: CAD $6\%/0.615 = 9.76\%$; EUR $3\%/1.770 = 1.69\%$.
4. Optimum requires $(E_i - R_F)/\beta_i$ equal across positions; add to the high-ratio position (CAD), trim the low-ratio one (EUR).

Why B is correct: At the optimal (maximum return-to-risk) portfolio the ratio of excess return to beta is equal across positions, so capital moves toward the position with the higher ratio, CAD.

Why the others are wrong

- **A:** Reports total expected return (7.0%) instead of excess return and moves the wrong way (toward the lower ratio).
- **C:** Excess return is correct, but the direction is reversed: EUR has the lower excess-return-to-beta ratio, so it should be trimmed.
- **D:** Direction is right, but 7.0% is the expected return before subtracting the 2.0% risk-free rate.

Q260

BT Mock Q30

Emily manages a USD 20.0 million portfolio allocated between a Technology ETF and a REIT ETF:

Item	Value
Tech ETF volatility	25%
REIT ETF volatility	18%
Correlation	0.30

She compares four allocations, with the Tech ETF weight at 35.0%, 50.0%, 65.0% or 80.0% (the remainder in the REIT ETF). If her goal is to minimise the portfolio's diversified VaR, which allocation is best?

A. Mix #1 with 35.0% allocated to the Tech ETF
B. Mix #2 with 50.0% allocated to the Tech ETF
C. Mix #3 with 65.0% allocated to the Tech ETF
D. Mix #4 with 80.0% allocated to the Tech ETF

Answer: A

Step by step

1. Diversified VaR $= z\,V\,\sigma_p$ with z and V identical across mixes, so minimise σ_p.
2. $\sigma_p = \sqrt{w^2(0.25)^2 + (1-w)^2(0.18)^2 + 2w(1-w)(0.30)(0.25)(0.18)}$: $w = 35\% \to 16.58\%$; $50\% \to 17.46\%$; $65\% \to 19.11\%$; $80\% \to 21.36\%$.
3. Check: minimum-variance Tech weight $w^* = \frac{0.18^2 - 0.30(0.25)(0.18)}{0.25^2 + 0.18^2 - 2(0.30)(0.25)(0.18)} = \frac{0.0189}{0.0679} = 27.8\%$, below all options, and σ_p rises as w moves away from it.
4. The lowest Tech weight (35%) is therefore closest to the minimum and best.

Why A is correct: Portfolio risk is lowest at the minimum-variance weight (27.8% Tech); every offered mix lies above it, so the one nearest to it (35%) has the lowest VaR.

Why the others are wrong

- **B:** Equal weighting is not the minimum-variance mix; its volatility is 17.46%, above 16.58%.
- **C:** Volatility rises to 19.11% because it overweights the higher-volatility Tech ETF.
- **D:** The heaviest Tech weight has the highest volatility (21.36%); it is the worst choice, reflecting the mistake of ignoring that Tech is the riskier asset.

Q261

BT answers P2.T9.21.6.1 · BT notes P2.T9.21.6.1

REBUILT — Source tables had OCR-corrupted and mutually inconsistent figures (marginal VaRs, undiversified VaR, and a diversified-VaR explanation that did not match the table); rebuilt from stated volatilities and correlation so the answer is computed directly.

Emily manages a $20.0 million portfolio allocated between a Technology ETF and a REIT ETF. Annual volatilities are 20% (Tech) and 30% (REIT); the correlation is -0.20. Use a 95% confidence level (z = 1.645). She compares four allocations:

Mix	#1	#2	#3	#4
Weight in Tech ETF	35.0%	50.0%	65.0%	80.0%
Weight in REIT ETF	65.0%	50.0%	35.0%	20.0%

If her goal is to minimise the portfolio's diversified VaR, which allocation is best?

A. Mix #1 with 35.0% allocated to the Tech ETF
B. Mix #2 with 50.0% allocated to the Tech ETF
C. Mix #3 with 65.0% allocated to the Tech ETF
D. Mix #4 with 80.0% allocated to the Tech ETF

Answer: C

Key correction: Source key (C) is retained. Its quoted diversified VaRs ($6,131 / $5,043 / $4,621 / $5,037 thousand) are inconsistent with its own table; recomputed with the stated parameters they are $6.37m / $5.36m / $4.93m / $5.24m, with Mix #3 still lowest.

Step by step

1. Diversified VaR $= 1.645\sqrt{(20w \cdot 0.20)^2 + (20(1-w) \cdot 0.30)^2 + 2(-0.20)(20w \cdot 0.20)(20(1-w) \cdot 0.30)}$ in $ millions.
2. Mix #1: $6.368m; Mix #2: $5.356m; Mix #3: $4.931m; Mix #4: $5.239m.
3. Cross-check with marginal VaR: at Mix #3 the marginal VaRs are 0.239 (Tech) and 0.260 (REIT), nearly equal; at Mix #1 they are 0.053 vs 0.461.

4. Minimum-variance Tech weight $w^* = \frac{0.30^2-(-0.2)(0.2)(0.3)}{0.2^2+0.3^2-2(-0.2)(0.2)(0.3)} = 66.2\%$, closest to 65%.

Why C is correct: Risk is minimised where marginal VaRs are equal; this occurs at about 66% Tech, so the 65% mix has the lowest diversified VaR ($4.93m).

Why the others are wrong

- **A:** Puts 65% in the higher-volatility REIT ETF; VaR is highest ($6.37m) and marginal VaRs are far apart (0.053 vs 0.461).
- **B:** Equal weighting is a naive guess; VaR is $5.36m, above Mix #3, because marginal VaRs (0.141 vs 0.394) are still unequal.
- **D:** Over-tilts to the lower-volatility asset, ignoring the diversification benefit from the negative correlation; VaR is $5.24m and Tech's marginal VaR (0.306) now exceeds REIT's (0.087).

Q262

BT answers P2.T8.3.4 · BT notes P2.T9.3.4

If the goal is to shift a portfolio to its global risk minimum (global minimum-risk), per the perspective of a risk manager, which of the following is most effective?

A. Add (trim) positions with low (high) Sharpe ratios until Sharpe ratios are equal
B. Add (trim) positions with low (high) marginal VaR until marginal VaRs are equal
C. Add (trim) positions with low (high) ratios of excess return to beta until ratios (excess return/beta) are equal
D. Add (trim) positions with low (high) ratios of individual VaR to portfolio VaR until ratios (individual VaR/portfolio VaR) are equal

Answer: B

Step by step

1. Portfolio VaR is at its global minimum when marginal VaR is equal across all positions.
2. If marginal VaRs differ, moving capital from the highest-marginal-VaR position to the lowest-marginal-VaR position lowers portfolio VaR.
3. So add to low-MVaR positions and trim high-MVaR positions until the marginal VaRs are equal.

Why B is correct: Marginal VaR is the change in portfolio VaR per unit of added position; equalising it removes any VaR-reducing reallocation, defining the minimum-risk portfolio.

Why the others are wrong

- **A:** Equalising Sharpe ratios involves expected returns; it is a reward/risk criterion, not a pure risk-minimisation one.
- **C:** Equalising excess return/beta is the optimality condition for maximising the return-to-risk ratio (portfolio manager), not for minimising risk, and the direction stated is also reversed.
- **D:** Individual VaR to portfolio VaR is not the marginal contribution to risk (it ignores correlations) and equalising it does not minimise VaR.

Q263

BT answers P2.T8.3.5 · BT notes P2.T9.3.5

If the goal is to shift a portfolio to the optimal portfolio, per the reward/risk perspective of a portfolio manager, which of the following is most effective?

A. Add (trim) positions with low (high) Sharpe ratios until Sharpe ratios are equal
B. Add (trim) positions with low (high) marginal VaR until marginal VaRs are equal
C. Add (trim) positions with high (low) ratios of excess return to beta until ratios (excess return/beta) are equal
D. Add (trim) positions with low (high) ratios of individual VaR to portfolio VaR until ratios (individual VaR/portfolio VaR) are equal

Answer: C

Step by step

1. The optimal portfolio maximises the ratio of excess return to risk, which requires $(E_i - R_F)/\text{MVaR}_i$ (equivalently $(E_i - R_F)/\beta_i$) to be equal for all positions.
2. A position with a higher ratio earns more excess return per unit of risk added, so increase it; one with a lower ratio is trimmed.
3. Continue until the ratios are equal across positions.

Why C is correct: Return per unit of risk is maximised when each position's excess return divided by its marginal risk (beta or MVaR) is the same, so capital flows to high-ratio positions.

Why the others are wrong

- **A:** Direction is backwards (adds low-ratio positions) and position-level Sharpe ratios are not the marginal condition.
- **B:** Equal marginal VaRs gives the minimum-risk portfolio, which ignores returns and is generally not optimal.
- **D:** Individual VaR to portfolio VaR ignores expected returns and correlation effects, and its direction is also reversed.

Q264

BT answers P2.T9.21.6.2 · BT notes P2.T9.21.6.2

REBUILT — OCR-damaged table (a mis-read Healthcare marginal VaR and an inconsistent undiversified-VaR row); repaired to consistent values and removed the irrelevant undiversified VaR row.

Peter manages a $10.0 million portfolio allocated between an Energy ETF and a Healthcare ETF. He compares four allocations. The Energy ETF offers an expected excess return (over the risk-free rate) of 6.0%, and the Healthcare ETF 12.0%. Betas are each ETF's beta relative to the portfolio.

Mix	#1	#2	#3	#4
Weight in Energy ETF	45.0%	60.0%	75.0%	90.0%
Marginal VaR (Energy)	0.051	0.085	0.126	0.158
Beta (Energy, portfolio)	0.215	0.441	0.787	1.032
Weight in Healthcare ETF	55.0%	40.0%	25.0%	10.0%
Marginal VaR (Healthcare)	0.391	0.353	0.263	0.110
Beta (Healthcare, portfolio)	1.642	1.838	1.639	0.716

Peter's goal is not risk minimisation but to maximise the ratio of expected excess return to risk. Which allocation is best?

A. Mix #1 with 45.0% allocated to the Energy ETF
B. Mix #2 with 60.0% allocated to the Energy ETF
C. Mix #3 with 75.0% allocated to the Energy ETF
D. Mix #4 with 90.0% allocated to the Energy ETF

Answer: C

Step by step

1. Optimum: $(E_E - R_F)/\beta_E = (E_H - R_F)/\beta_H$ (equivalently equal excess return/MVaR).
2. Excess return/beta (Energy, Healthcare): #1 0.279, 0.073; #2 0.136, 0.065; #3 0.076, 0.073; #4 0.058, 0.168.
3. Excess return/MVaR (Energy, Healthcare): #1 1.176, 0.307; #2 0.706, 0.340; #3 0.476, 0.456; #4 0.380, 1.091.
4. Mix #3 has the two ratios nearly equal on both measures, so it is the most efficient allocation.

Why C is correct: Return per unit of risk is maximised where excess return/beta (or /MVaR) is equal across positions; Mix #3 is the closest to equality.

Why the others are wrong

- **A:** Energy's ratio (0.279) is almost four times Healthcare's (0.073), so the portfolio is far from balanced: too much in Healthcare.
- **B:** Ratios are still far apart (0.136 vs 0.065; 0.706 vs 0.340); an in-between weight is not the equal-ratio point.
- **D:** Healthcare's ratio (0.168) far exceeds Energy's (0.058): Energy is overweight. Choosing it reflects minimum-risk logic (lowest Healthcare marginal VaR) rather than equal ratios.

Q265

GARP PE2 Q16 · Mock 4 Q17

A portfolio manager wants to reach the optimal portfolio on the portfolio's efficient frontier by replacing one stock with a new stock not currently in the portfolio, keeping portfolio value constant. The manager considers the following actions:

Action 1: Sell the stock with the highest marginal VaR and buy an equivalent value of a new stock that would have the lowest marginal VaR in the portfolio.

Action 2: Sell a stock and buy an equivalent value of a new stock, which would cause the ratio of expected excess return to portfolio beta to be equal for all stocks in the portfolio.

Action 3: Sell a stock and buy an equivalent value of a new stock, which would cause the portfolio betas of all stocks to be equal.

Action 4: Sell a stock and buy an equivalent value of a new stock, which would significantly decrease the portfolio standard deviation without changing the average excess portfolio return.

Which action would create the optimal portfolio?

A. Action 1.
B. Action 2.

C. Action 3.
D. Action 4.

Answer: B
Step by step
1. The optimal (tangency) portfolio maximizes the Sharpe ratio, which requires equal excess-return-to-marginal-VaR ratios across holdings: $(R_i - R_F)/\text{MVaR}_i$ is the same for all i.
2. Marginal VaR is proportional to the position's beta relative to the portfolio, $\text{MVaR}_i \propto \beta_i$, so the condition is equivalent to equal $(R_i - R_F)/\beta_i$ for all holdings.
3. If the ratios differ, shifting money from the low-ratio asset to the high-ratio asset raises return per unit of risk, so the portfolio is not yet optimal.
4. Action 2 imposes exactly this equality, so it describes the optimal portfolio.

Why B is correct: Optimality requires that no reallocation can improve the return-to-risk trade-off, which holds only when every position offers the same excess return per unit of marginal risk (beta or marginal VaR).

Why the others are wrong
- **A:** Minimizes risk only: it ignores expected returns, and the minimum-risk portfolio is not the tangency portfolio.
- **C:** Equal betas is not the condition; optimality needs equal ratios of excess return to beta, and assets with higher beta should have proportionally higher excess return.
- **D:** Lowering standard deviation at constant return is an improvement, but it does not guarantee that excess-return-to-beta ratios are equal, so the portfolio need not be at the optimum.

LO 87.e

Explain the difference between risk management and portfolio management and describe how to use marginal VaR in portfolio management.

Q266

Schweser Module Quiz 87.2 Q1

A portfolio has an equal amount invested in two positions, X and Y. The expected excess return of X is 9% and that of Y is 12%. Their marginal VaRs are 0.06 and 0.075, respectively. To move toward the optimal portfolio, the manager will probably:

A. increase the allocation in Y and/or lower that in X.
B. increase the allocation in X and/or lower that in Y.
C. do nothing because the information is insufficient.
D. not change the portfolio because it is already optimal.

Answer: A
Step by step
1. At the optimum, $\frac{\mu_i - r_f}{\text{MVaR}_i}$ is equal across all positions.
2. X: $0.09/0.06 = 1.500$.
3. Y: $0.12/0.075 = 1.600$.
4. The ratios differ, so the portfolio is not optimal; Y gives more excess return per unit of marginal VaR.
5. Shift funds from X to Y until the ratios equalize.

Why A is correct: The optimum (maximum excess return to VaR) requires equal excess return/marginal VaR ratios; Y's ratio (1.6) exceeds X's (1.5), so increasing Y and reducing X improves the portfolio.

Why the others are wrong
- **B:** Reverses the comparison; it adds to the position with the lower return per unit of marginal VaR (X).
- **C:** Information is sufficient: excess returns and marginal VaRs are all the ratio test needs.
- **D:** Would be true only if the ratios were equal; 1.5 is not 1.6. (Equal marginal VaRs would instead give the minimum-VaR portfolio, not the optimum.)

Q267

BT answers P2.T8.409.1 · BT notes P2.T9.409.1

According to Rosengarten and Zangari (Litterman), many organizations and asset managers have formed independent risk management units (RMUs) that oversee the risk exposures of portfolios and ensure that such exposures are authorized and in line with risk budgets. Which of the following is the LEAST LIKELY to be an objective of a risk

management unit (RMU)?

A. To manage risk, especially by exercising the authority to replace or reduce the decision methods of portfolio managers

B. To be a catalyst for a comprehensive discussion of risk-related matters, including matters that do not easily lend themselves to measurement

C. To develop an inventory of risk data for use in evaluating portfolio managers and market environments

D. To develop performance attribution analytical tools and to assess the quality of models used to measure risk, for example by back-testing and proactive research into model risk

Answer: A

Step by step

1. Risk management (measuring, monitoring, reporting risk) is distinct from portfolio management (choosing positions by weighing risk against return).
2. An independent RMU measures and monitors risk for those with a vested interest; portfolio managers retain responsibility for managing risk and for decisions.
3. B, C and D are communication, data, attribution and model-validation roles typical of an RMU.
4. A gives the RMU authority over portfolio managers' decision methods, which is portfolio management, so A is the least likely objective.

Why A is correct: The RMU's role is independent measurement, monitoring and communication of risk; overriding portfolio managers' decisions is portfolio management, not an RMU objective.

Why the others are wrong

- **B:** A genuine objective: the RMU promotes broad risk discussion, including qualitative risks hard to measure.
- **C:** A genuine objective: a risk-data inventory supports evaluating managers and market conditions.
- **D:** A genuine objective: attribution tools and model validation (back-testing, model-risk research) are core RMU tasks.

Reading 88. VaR and Risk Budgeting in Investment Management

Philippe Jorion, Value-at-Risk: The New Benchmark for Managing Financial Risk, 3rd Edition (New York, NY: McGraw-Hill, 2007). Chapter 17.

LO	Learning objective	Qs
88.a	define risk budgeting.	4
88.b	describe the impact of horizon, turnover, and leverage on the risk management process in the investment management industry.	5
88.c	describe the investment process of large investors such as pension funds.	3
88.d	describe the risk management challenges associated with investments in hedge funds.	2
88.e	describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk.	30
88.f	explain the use of VaR to monitor risk.	10
88.g	explain how VaR can be used in the development of investment guidelines and for improving the investment process.	2
88.h	describe the risk budgeting process and calculate risk budgets across asset classes and active managers.	17

LO 88.a

Define risk budgeting.

Q268

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO; written from the SchweserNotes 2026 reading in GARP practice-exam style.

The chief investment officer of a public pension fund tells the board that the fund will "move from allocating capital to allocating risk" by adopting risk budgeting. A trustee asks the risk manager to explain what this means. Which of the following descriptions of risk budgeting is correct?

A. It is the allocation of market value to asset classes in proportion to their expected returns, subject to a limit on the fund's total VaR.

B. It is a bottom-up process in which each manager's VaR is measured and the sum of those VaRs becomes the fund's risk budget for the following year.

C. It is a top-down process in which the fund first sets an acceptable level of total risk and then allocates that risk to asset classes, managers, and positions.

D. It is the practice of holding a capital reserve equal to the fund's total VaR so that losses up to that amount can be absorbed without changing the allocation.

Answer: C

Step by step

1. Risk budgeting starts from the top: the fund decides how much total risk it is willing to take.
2. That total risk budget is then allocated to the components of the portfolio (asset classes, managers, positions).
3. What is allocated is risk, which distinguishes risk budgeting from a traditional market value (capital) allocation.

Why C is correct: The reading defines risk budgeting as a top-down process of choosing and managing risk exposures: a risk budget is set for the whole portfolio and then allocated to individual positions based on a predetermined fund risk level.

Why the others are wrong

- **A:** Describes a market value allocation; risk budgeting allocates risk rather than capital.
- **B:** Describes a bottom-up aggregation; risk budgeting sets the total first and allocates downward, and summing VaRs ignores diversification.
- **D:** Describes holding economic capital as a loss buffer, which is not the same as budgeting and allocating risk exposures.

Q269

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO; written from the SchweserNotes 2026 reading in GARP practice-exam style.

An endowment allocates USD 600 million to an equity manager and USD 400 million to a global macro manager. After adopting risk budgeting, the investment committee reviews several statements prepared by a junior analyst. Which of the following statements is correct?

A. Because the equity manager receives the larger market value allocation, the equity manager must also receive

the larger share of the endowment's risk budget.

B. Risk budgeting allocates target returns to each manager, and risk is managed afterwards by adjusting market value weights.

C. Once risk budgets are assigned, positions no longer need to be monitored because each manager is constrained by its budget.

D. Under risk budgeting the endowment first sets its total acceptable risk and then allocates it, so the macro manager could be given a larger share of the risk budget despite managing less capital.

Answer: D

Step by step

1. Market value allocation splits capital (60%/40% here); risk budgeting splits risk.
2. The endowment sets its total risk first (top-down) and then decides how much of that risk each manager may use.
3. A manager with less capital but a higher-risk mandate, such as global macro, can legitimately be given a larger share of the risk budget.

Why D is correct: Risk budgeting differs from market value allocation because it allocates risk, so the risk shares need not match the capital shares.

Why the others are wrong

- **A:** Treats risk shares as identical to capital shares, which is exactly the market value allocation view that risk budgeting replaces.
- **B:** Reverses the logic: risk budgeting allocates risk directly rather than allocating returns and then adjusting capital weights.
- **C:** Budgets must be monitored; the point of measuring risk is to check whether each manager stays within its budget and why any excess occurs.

Q270

BT answers P2.T8.408.3

REBUILT — Original relied on an outside reading (Rosengarten/Zangari risk plan, RORC) not in the 2026 text and used an EXCEPT format with arguable options; rebuilt to test the Schweser/Jorion definition of risk budgeting.

Which of the following statements best describes risk budgeting as applied by a large investment fund?

A. A top-down process in which total acceptable portfolio VaR is set first and then allocated to asset classes and managers, taking diversification effects into account

B. A bottom-up process in which stand-alone VaRs are computed for each position and simply added to obtain the fund's risk level

C. A process equivalent to market value allocation, in which capital is divided among asset classes in proportion to their weights in the benchmark

D. A top-down process in which the total VaR budget is divided so that the stand-alone VaRs of the asset classes sum exactly to the fund's total VaR limit

Answer: A

Step by step

1. Risk budgeting begins with the total amount of risk (VaR) the fund is willing to accept.
2. That risk is then allocated across asset classes and managers, which differs from allocating market value.
3. Unless assets are perfectly correlated, the sum of stand-alone VaRs exceeds portfolio VaR, so the allocation must recognise diversification.
4. Only A states all of this.

Why A is correct: Risk budgeting is top-down: fix the portfolio-level risk first, then allocate risk (not just dollars) to components while accounting for diversification.

Why the others are wrong

- **B:** Describes a bottom-up aggregation and ignores diversification; risk budgeting is top-down.
- **C:** Confuses risk allocation with market value allocation; the text stresses that risk budgeting differs from the latter.
- **D:** Top-down is right, but forcing stand-alone VaRs to sum to the limit ignores diversification (the sum of stand-alone VaRs exceeds portfolio VaR), making the budget too conservative.

Q271 BT notes P2.T9.21.9.1

REBUILT — Original asked about elements of a 'risk plan' (a framework not covered in the Schweser reading) with a vague 'EXCEPT' stem; rebuilt to test the core LO 88.a definition of risk budgeting, keeping the risk-plan/budget/monitoring context.

Acme Industrial Corporation has just finished establishing financial accounting controls (planning, budgeting, and variance monitoring). It now wants a parallel risk management framework built around a risk budget for the whole firm. Which of the following statements best defines risk budgeting?

A. A top-down process in which a risk level is set for the entire portfolio and then allocated to individual positions or managers.

B. A bottom-up process in which each manager sets his own risk level and the total portfolio risk is the sum of these amounts.

C. An allocation of the portfolio's market value across asset classes so that each class holds an equal dollar amount.

D. A limit on the notional size of each position that is independent of the volatility of the position.

Answer: A

Step by step

1. Risk budgeting is a top-down process of choosing and managing exposures to risk.
2. The risk manager first sets a risk budget (for example a VaR level) for the entire portfolio.
3. That total risk is then allocated to asset classes, managers, or positions based on the predetermined fund-level risk.
4. Only (A) describes top-down allocation of risk rather than of market value or notional.

Why A is correct: Risk budgeting starts from a total fund risk level and allocates risk, not dollars, down to individual positions or managers.

Why the others are wrong

- **B:** Reverses the process: risk budgeting is top-down, not a sum of independently chosen bottom-up risk levels.
- **C:** Describes market-value (capital) allocation; the reading stresses that risk budgeting allocates risk, which differs from allocating market value.
- **D:** Describes a traditional notional/position limit, which ignores risk; risk budgets are expressed in risk terms such as VaR.

LO 88.b

Describe the impact of horizon, turnover, and leverage on the risk management process in the investment management industry.

Q272 AnalystPrep Q.2479

The daily application of VaR measures has become a requirement of bank trading portfolios due to all the following factors, EXCEPT:

A. Short horizon

B. Rapid turnover

C. Historical returns

D. High leverage

Answer: C

Step by step

1. Sell-side bank trading portfolios have a short horizon, rapid turnover and high leverage.
2. Because positions change quickly, yesterday's risk may bear no relation to today's positions, so risk must be measured daily with forward-looking VaR.
3. Historical returns are a backward-looking measure that cannot keep up with changing positions; they are not a driver of the need for VaR.
4. The EXCEPT item is therefore C.

Why C is correct: Short horizon, rapid turnover and high leverage make historical measures stale and downside risk critical, which forces daily VaR; historical returns are the inadequate alternative, not a cause.

Why the others are wrong

- **A:** Short horizon is a defining sell-side trait that drives daily VaR; choosing it misreads the EXCEPT format.
- **B:** Rapid turnover makes past risk irrelevant to current positions, so it is a driver.

- **D:** High leverage raises the need to control downside risk, so it is a driver.

Q273 AnalystPrep Q.4704

Which one of the following is true about the ”buy side” of the investment management industry? Investors:

A. have a shorter horizon.
B. have a slow turnover.
C. have higher leverage.
D. control risks by utilizing position limits and stop-loss rules.

Answer: B

Step by step

1. Compared with the sell side, the buy side has a longer horizon, slower turnover and lower leverage.
2. A (shorter horizon) and C (higher leverage) reverse these relations.
3. Position limits and stop-loss rules are sell-side (trading desk) controls; the buy side relies on diversification, benchmarks and guidelines.
4. Only B is true.

Why B is correct: Buy-side investors hold positions for years, so turnover is slow.

Why the others are wrong

- **A:** Reversed: the buy-side horizon is longer, not shorter.
- **C:** Reversed: buy-side leverage is lower because of strict constraints.
- **D:** Attributes sell-side trading controls to the buy side.

Q274 BT answers P2.T8.11.2 · BT notes P2.T9.11.2

According to Jorion, which of the following risk controls and/or risk measures is LEAST LIKELY to be effective in managing the risk of an institutional trading portfolio when the horizon is short, turnover is rapid, and leverage is high (i.e., an environment typified by traditional ”sell-side” banks)?

A. Investment guidelines and asset allocation informed by long-term historical studies of the equity risk premium
B. Value at risk (VaR) and VaR limits
C. Stress testing
D. Stop-loss rules

Answer: A

Step by step

1. The environment described is the sell side: short horizon, rapid turnover, high leverage.
2. Such desks need dynamic, forward-looking controls: VaR and VaR limits, stress tests, stop-loss rules.
3. Long-term asset allocation based on historical risk-premium studies suits long-horizon, slow-turnover, low-leverage buy-side investors, not a fast-changing trading book.
4. So A is least effective.

Why A is correct: Long-horizon, historically based allocation tools do not track positions that turn over rapidly with high leverage.

Why the others are wrong

- **B:** VaR and VaR limits are the core sell-side tools for rapidly changing, levered books.
- **C:** Stress testing complements VaR in capturing losses from extreme moves in levered positions.
- **D:** Stop-loss rules are standard trading-desk controls for short horizons.

Q275 BT answers P2.T9.21.7.2 · BT notes P2.T9.21.7.2

Jane is evaluating job opportunities on the buy side and the sell side. She considers a buy side firm where the investment style (or characteristics) include a long-term time horizon, slow turnover, and low leverage. Unsurprisingly, this buy side firm also emphasizes diversification as a risk control. Additionally, each of the following is a valid risk management technique, tactic, or approach EXCEPT which of the following is she LEAST LIKELY to observe at this buy-side firm?

A. The buy side firm benchmarks against the S&P 500 and similar established equity indices
B. The firm minimizes tracking error risk (aka, tracking error volatility) by increasing its cash allocation to at least 50%

C. The buy side firm recently centralizes with a global custodian and reduced the number of disparate risk systems
D. The buy side firm determines a risk appetite and allocate a risk budget to its major asset classes that cascades to each of its active managers

Answer: B
Step by step
1. Buy-side investors benchmark to passive indices, centralize risk systems and set risk budgets cascading to managers (A, C, D are all typical).
2. Holding 50% or more in cash to cut tracking error would abandon the equity mandate and create large tracking error versus an equity benchmark.
3. Hence B is least likely to be observed.

Why B is correct: Benchmarking, centralization and risk budgeting are standard buy-side practice; shifting half the portfolio to cash is inconsistent with an equity benchmark and mandate, and would itself produce large tracking error.

Why the others are wrong
- **A:** Benchmarking to indices such as the S&P 500 is a defining buy-side practice.
- **C:** Centralizing custody and risk systems is a recognized buy-side response to complexity.
- **D:** A risk appetite cascaded as risk budgets to asset classes and managers is the risk-budgeting approach.

Q276
Schweser Module Quiz 88.1 Q1

With respect to the buy side and sell side of the investment industry, the:
I. buy side uses more leverage.
II. sell side has relied more on VaR measures.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: B
Step by step
1. Statement I: buy-side investors face stronger leverage constraints, so they use less leverage than banks; false.
2. Statement II: banks, with short horizons, rapid turnover and high leverage, developed and have long used VaR; true.
3. Only II is correct.

Why B is correct: The sell side, with high leverage and rapid turnover, pioneered forward-looking VaR; the buy side uses less leverage.

Why the others are wrong
- **A:** Treats statement I as true, but buy-side leverage is lower.
- **C:** Accepts the false statement I.
- **D:** Rejects II, ignoring that banks have long relied on VaR.

LO 88.c

Describe the investment process of large investors such as pension funds.

Q277
New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

The investment committee of a large public pension fund is documenting its investment process for a new board member. The committee asks the chief risk officer to describe the first step in the process. Which of the following best describes this first step?

A. Determining long-term strategic asset allocations across asset classes, typically by balancing return and risk with a method such as mean-variance optimization.
B. Selecting external managers and setting guidelines on beta and duration for each of them.
C. Evaluating each manager's tracking error relative to its benchmark on a periodic basis.

D. Replacing historical risk measures with a VaR system covering all managers' positions.

Answer: A

Step by step

1. The investment process of a large investor has two main steps.
2. Step 1 sets the long-term strategic asset allocation (domestic and foreign stocks and bonds, real estate, venture capital, hedge funds), usually balancing risk and return with mean-variance optimization and relying on passive indices as benchmarks.
3. Step 2 selects and monitors managers, which comes after the allocation is set.

Why A is correct: The first step is the strategic asset allocation decision, commonly made with mean-variance optimization and passive benchmark indices.

Why the others are wrong

- **B:** This is the second step; manager selection and guidelines follow the allocation decision.
- **C:** Tracking error is used to evaluate managers after they have been selected, so it is part of the second step.
- **D:** Adopting VaR is a response to globalization and complexity, not a step in the investment process.

Q278 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager at a large pension fund reviews the fund's second stage of investment, which concerns the managers that run the strategic allocation. Which of the following is most consistent with how this stage is typically conducted?

A. Managers must either track a passive benchmark or be given unrestricted discretion, because guidelines would prevent them from outperforming.

B. Managers may be passive or active, are reviewed periodically, must follow guidelines on investment types and risk exposures such as beta and duration, and are often evaluated using tracking error.

C. Managers are chosen first, and the strategic asset allocation is then derived from the aggregate of their holdings.

D. Manager performance is evaluated mainly by absolute volatility, since benchmarks are unavailable for most asset classes.

Answer: B

Step by step

1. Step 2 is manager selection: passive managers track benchmarks, active managers try to beat them.
2. Managers are reviewed periodically and must conform to guidelines including investment types and risk restrictions such as beta and duration.
3. Performance is evaluated by analyzing tracking error relative to the benchmark.

Why B is correct: The reading describes periodic review, guidelines with risk restrictions like beta and duration, and tracking error as the performance measure.

Why the others are wrong

- **A:** Guidelines apply to managers generally; they are not removed for active managers.
- **C:** The sequence is reversed; the strategic allocation is set first and benchmarks from passive indices make the allocation feasible.
- **D:** Passive indices make benchmarks available and managers are evaluated relative to them by tracking error, not by absolute volatility.

Q279 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

The CIO of a large pension fund employs many external managers, each allowed to change positions within its own constraints. The CIO is concerned that historical risk measures no longer capture total fund risk. Which of the following is the most appropriate reason this concern has increased demand for VaR systems?

A. Historical measures cannot be computed for passive managers who simply track benchmarks.

B. VaR eliminates the need to set strategic asset allocations and guidelines because it constrains managers automatically.

C. Individual managers' changes within constraints may be modest, but their collective effect is hard to gauge with historical measures, and globalization, complexity, and dynamic management make these measures inadequate.

D. Regulation requires VaR to replace tracking error as the only measure of manager performance.

Answer: C

Step by step

1. The investment industry has become more global, complex and dynamic.
2. With many managers, each changing positions within constraints, the combined changes are hard to assess with historical measures.
3. This is why VaR risk systems are becoming more important.

Why C is correct: The reading attributes rising VaR use to globalization, complexity, and dynamic management, where aggregate changes by many managers cannot be gauged with historical measures.

Why the others are wrong

- **A:** Historical measures can be computed for passive managers; this is not the reason given.
- **B:** VaR complements, not replaces, allocation and guidelines; the reading does not claim it removes them.
- **D:** The reading cites no regulation, and tracking error remains a performance evaluation tool.

LO 88.d

Describe the risk management challenges associated with investments in hedge funds.

Q280

AnalystPrep Q.2482

A hedge fund manager makes the following comment: "Hedge funds are transparent and reveal all information about their positions. However, due to group heterogeneity, risk measurement problems arise." Is the manager's comment correct?

A. Yes.

B. No, because hedge funds are usually homogenous.

C. No, because hedge funds lack transparency and often refuse to reveal information about their positions.

D. No, because the law requires hedge funds to be discreet and conceal investment-related information in line with the principle of client confidentiality.

Answer: C

Step by step

1. Hedge funds are a very heterogeneous group of strategies, so the second half of the comment (heterogeneity) is true.
2. Hedge funds have low transparency, which makes both the size and type of risk hard to measure; the first half of the comment is false.
3. Hence the comment is incorrect, and the reason is the lack of transparency (C).

Why C is correct: Low transparency is a core hedge fund risk-measurement challenge, so the claim that they reveal all position information is false.

Why the others are wrong

- **A:** Accepts the claim of transparency; the heterogeneity part is true but the transparency part is false.
- **B:** Hedge funds are a heterogeneous, not homogeneous, class of assets covering many strategies.
- **D:** No law requires concealment; the opacity is a practice chosen by managers, not a legal duty.

Q281

BT answers P2.T8.11.3 · BT notes P2.T9.11.3

REBUILT — Original keyed on an unstated Jorion benchmark point not in the Schweser reading and had a debatable 'least likely'; rewrote the answer set so exactly one statement is clearly false per Schweser.

Which of the following statements about the risk measurement challenges of hedge funds is LEAST accurate?

A. Hedge funds typically use little leverage and trade infrequently, so their risk profile resembles that of the buy side.

B. Low liquidity tends to reduce the volatility of historical prices and the correlations of positions, leading to underestimation of traditional risk measures.

C. Low transparency makes it difficult to measure both the size and the type of risk.

D. Hedge funds are a very heterogeneous class of assets that include a variety of trading strategies.

Answer: A

Step by step

1. Schweser: hedge funds often use leverage and trade a great deal, so their risks resemble the sell side, not the buy side. A is therefore false.
2. B is correct: illiquid positions have smoothed prices, biasing volatility and correlation downward.
3. C is correct: low transparency hinders measurement of size and type of risk.
4. D is correct: hedge funds span many strategies.

Why A is correct: Hedge funds use high leverage and high turnover, making them more like the sell side; A states the opposite.

Why the others are wrong

- **B:** True statement (liquidity risk causes downward bias in volatility and correlation); picking it overlooks the smoothing effect.
- **C:** True statement; transparency is a key measurement problem.
- **D:** True statement; heterogeneity is a defining feature of hedge funds.

LO 88.e

Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk.

Q282

AnalystPrep Q.2488

REBUILT — Original options were ambiguous (equities vs. fund equity) and the keyed answer was internally inconsistent; rebuilt with explicit durations and balances to test funding risk (LO 88.e) with one defensible answer.

A pension fund has assets of $1,000 million with a modified duration of 5 and liabilities of $900 million, consisting mainly of nominal payments, with a modified duration of 12. The manager expects interest rates to fall by 100 bps across the curve. If rates do fall by 100 bps, what is the approximate impact on the fund's surplus?

A. Surplus increases by about $58 million, because both assets and liabilities gain in value.
B. Surplus decreases by about $58 million, because liabilities rise by more than assets.
C. Surplus decreases by about $108 million, because liabilities rise and assets are unaffected.
D. Surplus increases by about $50 million, because assets rise and nominal liabilities are fixed.

Answer: B

Key correction: Source key (C) concluded that liabilities rise by more than equity/assets, hurting the surplus, but its option wording ('value of equity will increase' while liabilities increase more) is ambiguous and the explanation inconsistent; the conclusion that the surplus falls is retained and quantified.

Step by step

1. Asset change: $\Delta A \approx D_A \times A \times \Delta y = 5 \times 1{,}000 \times 0.01 = +\$50$ million (rates fall, so value rises).
2. Liability change: $\Delta L \approx 12 \times 900 \times 0.01 = +\$108$ million; nominal liabilities behave like a short position in a long-duration bond.
3. Change in surplus: $\Delta S = \Delta A - \Delta L = 50 - 108 = -\$58$ million.
4. Check: initial surplus $1{,}000 - 900 = 100$; new surplus $1{,}050 - 1{,}008 = 42$, a fall of 58.

Why B is correct: Funding risk comes from the mismatch between asset and liability interest-rate sensitivity. Long-duration nominal liabilities gain more in value than the assets when rates fall, so the surplus shrinks.

Why the others are wrong

- **A:** Gets the sign wrong: it notes both sides rise but ignores that the liability increase (108) exceeds the asset increase (50), so surplus falls.
- **C:** Ignores the 50 gain on assets and counts only the liability increase of 108.
- **D:** Treats nominal liabilities as insensitive to rates and counts only the asset gain of 50; liabilities are present values of fixed payments and rise when rates fall.

Q283

AP Mock 1 Q66

REBUILT — Source asked for an upper 95% bound on the surplus (answer 187), which is not a risk measure; rebuilt to ask for the lower-tail worst-case surplus, and removed unused durations.

An Indian pension fund reports the following (amounts in INR million):

Item	Assets	Liabilities
Amount	500	410
Expected growth per year	8%	9%
Annual volatility of growth	15%	8%

Annual asset returns and liability growth are jointly normal with a correlation of 0.7. Using a one-sided 95% confidence level (z = 1.645), the worst-case surplus value at the end of one year (surplus at risk) is closest to:

A. INR -0.8 million
B. INR 93.9 million
C. INR -18.8 million
D. INR -41.6 million

Answer: A

Key correction: Source key (D, INR 187m) is the upper bound 93.1 + 1.645(57.07); this is the best-case surplus, not surplus at risk.

Step by step

1. Expected surplus: $E[S] = 500(1.08) - 410(1.09) = 540 - 446.9 = 93.1$
2. Dollar volatilities: $\sigma_A = 500 \times 15\% = 75$; $\sigma_L = 410 \times 8\% = 32.8$
3. $\sigma_S = \sqrt{75^2 + 32.8^2 - 2(0.7)(75)(32.8)} = \sqrt{3256.8} = 57.07$
4. 95% one-sided worst case: $93.1 - 1.645 \times 57.07 = 93.1 - 93.88 = -0.78 \approx -0.8$

Why A is correct: Surplus at risk is the lower-tail surplus: expected surplus minus z times the volatility of the surplus, where surplus variance reflects asset and liability volatility and their correlation.

Why the others are wrong

- **B:** This is only $1.645\sigma_S = 93.9$, the loss relative to the mean; it omits the expected surplus of 93.1.
- **C:** Used the two-sided z of 1.96 instead of the one-sided 1.645: $93.1 - 1.96 \times 57.07 = -18.8$.
- **D:** Ignored the correlation (treated A and L as independent): $\sigma_S = \sqrt{75^2 + 32.8^2} = 81.9$, giving $93.1 - 1.645 \times 81.9 \approx -41.6$.

Q284

AnalystPrep Q.2495

REBUILT — Source option C was a vague 'may be less of a concern' judgement; rebuilt to test economic risk and the role of correlation with operating profits as defined in the reading.

Two firms sponsor identical defined benefit plans with the same surplus volatility. For Firm X, operating profits are strongly positively correlated with the plan's surplus (both fall in a downturn). For Firm Y, operating profits are uncorrelated with the surplus. Compared with Firm Y, Firm X's economic risk (variation in the sponsor's total economic earnings) is most likely:

A. higher, because the positive correlation compounds variation in total earnings
B. lower, because strong operating profits always offset pension surplus variation
C. the same, because economic risk depends only on surplus volatility
D. lower, because positive correlation reduces the variance of total earnings

Answer: A

Step by step

1. Economic risk is the variation in the sponsor's total economic earnings, considering how components relate (e.g., correlation of surplus and operating profits).
2. Variance of total earnings: $\sigma^2 = \sigma_{op}^2 + \sigma_S^2 + 2\rho\sigma_{op}\sigma_S$.
3. With the same σ_S, a positive ρ (X) gives a larger variance than $\rho = 0$ (Y).

Why A is correct: Economic risk depends on the correlation between surplus and operating profits; positive correlation adds to the variance of total earnings.

Why the others are wrong

- **B:** Assumes high profits eliminate pension risk; but when profits co-move with the surplus, bad outcomes coincide.
- **C:** Ignores that economic risk explicitly includes the correlation with operating profits.
- **D:** Sign error: positive correlation increases, not reduces, the combined variance (negative correlation would diversify).

Q285

AP Mock 2 Q34

A defined benefit scheme holds high-quality corporate bonds and gilts. The analyst reports:

Item	Assets	Liabilities
Amount (USD million)	396	308
Expected annual growth	8%	10%
Annual volatility of growth	30%	14%

Assume annual returns on assets and annual growth of liabilities are jointly normally distributed, and the correlation coefficient between them is 0.63. Determine the 95% confidence interval for the surplus value at the end of one year.

A. (USD -88.88m, USD 97.561m)
B. (USD -102.34m, USD 191.22m)
C. (USD -88.88m, USD 186.441m)
D. (USD -102.34m, USD 280.1m)

Answer: D

Step by step

1. $E[S] = 396(1.08) - 308(1.10) = 427.68 - 338.80 = 88.88$
2. $\sigma_A = 396 \times 30\% = 118.80$; $\sigma_L = 308 \times 14\% = 43.12$
3. $\sigma_S^2 = 118.8^2 + 43.12^2 - 2(0.63)(118.8)(43.12) = 9518.2$, so $\sigma_S = 97.561$
4. CI: $88.88 \pm 1.96(97.561) = 88.88 \pm 191.22 = (-102.34,\ 280.10)$

Why D is correct: A normal confidence interval is the mean surplus plus or minus z times the surplus volatility, which is built from asset and liability dollar volatilities and their correlation.

Why the others are wrong

- **A:** Used the (negated) expected surplus and σ_S themselves as the bounds, with no z multiplier.
- **B:** Correct lower bound, but the upper bound is just $1.96\sigma_S = 191.22$; the expected surplus of 88.88 was not added.
- **C:** Lower bound is the sign-flipped mean and upper bound is the mean plus one σ_S (88.88 + 97.561); no z = 1.96 and wrong sign on the lower bound.

Q286

AnalystPrep Q.2485

REBUILT — Source asked whether the manager can take more deviation without raising total VaR, with loose reasoning and giveaway size-comparison distractors; rebuilt as a computation of the implied correlation between policy-mix and active VaR using the same data.

A fund reports the following VaR decomposition (same confidence level and horizon, as % of assets):

Source of risk	VaR
Policy-mix VaR	19.6%
Active management VaR	1.6%
Total (asset) VaR	19.3%

Treating total VaR as the combination of the policy-mix VaR and the active management VaR, the implied correlation between the two components is closest to:

A. -0.23
B. -0.45
C. +0.23
D. -0.03

Answer: A

Key correction: Source key (B) reasoned that the components sum to more than asset VaR; that shows diversification but does not by itself justify larger benchmark deviations.

Step by step

1. Total VaR combines components: $\text{VaR}_T^2 = \text{VaR}_P^2 + \text{VaR}_A^2 + 2\rho\,\text{VaR}_P\text{VaR}_A$
2. $19.3^2 = 372.49$; $19.6^2 + 1.6^2 = 384.16 + 2.56 = 386.72$; difference = -14.23
3. $2 \times 19.6 \times 1.6 = 62.72$
4. $\rho = -14.23/62.72 = -0.227 \approx -0.23$
5. Negative correlation means active deviations diversify the policy mix: total VaR (19.3%) is below the simple sum (21.2%) and even below the policy-mix VaR alone (19.6%).

Why A is correct: Active management deviations can diversify the policy-mix risk, so total VaR is less than the sum

of the components; the implied correlation is backed out of the VaR-combination formula.

Why the others are wrong

- **B:** Omitted the factor 2 in the cross term (divided by 19.6 x 1.6 = 31.36), doubling the correlation.
- **C:** Sign error: total VaR below the root-sum-of-squares implies a negative, not positive, correlation.
- **D:** Used the linear shortfall $(19.3 - 19.6 - 1.6)/(2 \times 19.6 \times 1.6)$ instead of squared VaRs, which understates the correlation.

Q287

AnalystPrep Q.2487

A pension fund manager makes the following comments:
I. Volatility of assets is the appropriate risk parameter for a pension fund.
II. The value of pension fund assets may not be sufficient to cover the liabilities of the fund. Thus, an asset/liability management (ALM) framework is desirable.
III. Pension funds are exposed to funding risk, hence an ALM framework is desirable.
Which of the pension fund manager's comments is (are) accurate?

A. Only I
B. Both II and III
C. Both I and III
D. Only II

Answer: B

Step by step

1. Funding risk is the risk that assets will not cover liabilities, so the relevant quantity is the surplus (assets minus liabilities).
2. I is false: asset volatility alone ignores liabilities; surplus volatility is the appropriate risk parameter.
3. II is true: a possible asset shortfall relative to liabilities motivates ALM.
4. III is true: pension funds face funding risk, which ALM addresses. So II and III.

Why B is correct: For pension funds the risk of interest is surplus risk, so assets and liabilities must be managed jointly (ALM), not asset volatility alone.

Why the others are wrong

- **A:** Treats absolute asset risk as the pension fund's risk parameter, ignoring the liability side.
- **C:** Accepts I (asset volatility only) and drops II, which is also valid.
- **D:** Correct about II but misses that III (funding risk as the reason for ALM) is also accurate.

Q288

AnalystPrep Q.2490

A pension fund has $100,000 in assets and $90,000 in liabilities. Assume that the expected return on the surplus is 5%, and the annual VaR of the surplus is 22% at the 99% confidence level.
The initial surplus of the fund is equal to:

A. $10,000
B. $20,000
C. $5,000
D. $1,000

Answer: A

Step by step

1. Surplus = assets - liabilities = 100,000 - 90,000 = 10,000.

Why A is correct: Surplus is assets minus liabilities; the 5% and 22% figures apply to this surplus and are not needed here.

Why the others are wrong

- **B:** Doubled the surplus; not assets minus liabilities.
- **C:** Applied the 5% expected return to the 100,000 of assets (5,000) instead of computing the surplus.
- **D:** Not supported by the data; the surplus is assets minus liabilities, 100,000 - 90,000.

Q289 AnalystPrep Q.2491

A pension fund has $100,000 in assets and $90,000 in liabilities. Assume that the surplus is expected to grow at a rate of 5% per year, and the annual VaR of the surplus is 22% at the 99% confidence level.
The expected surplus after one year is equal to:

A. $10,000
B. $10,500
C. $5,000
D. $25,000

Answer: B
Step by step
1. Initial surplus = 100,000 - 90,000 = 10,000.
2. Expected surplus = 10,000 × 1.05 = 10,500.

Why B is correct: The growth rate applies to the surplus, not to the assets, so the expected surplus is the initial surplus times 1.05.
Why the others are wrong
- **A:** Forgot to apply the 5% growth; this is the initial surplus.
- **C:** Applied 5% to the 100,000 of assets (5,000) rather than to the surplus.
- **D:** Not supported by any correct step; it adds 15,000 to the initial surplus instead of growing it by 5%.

Q290 AnalystPrep Q.2492

REBUILT — Clarified that the 50% growth and 220% VaR are expressed relative to the initial surplus, which the source left ambiguous.
A pension fund has $100,000 in assets and $90,000 in liabilities. The surplus is expected to grow by 50% over the next year, and the annual VaR of the surplus is 220% of the initial surplus at the 99% confidence level. Which of the following statements is (are) accurate?
I. There is a 99% probability that, over the next year, the surplus will turn into a deficit of $7,000 or more.
II. There is a 1% probability that, over the next year, the surplus will turn into a deficit of $7,000 or more.
III. The surplus VaR is equal to $22,000.
IV. The surplus VaR is equal to $5,000.

A. Only I
B. Both II and III
C. Both II and IV
D. Both I and IV

Answer: B
Step by step
1. Initial surplus = 100,000 - 90,000 = 10,000.
2. Surplus VaR (SaR) = 220% × 10,000 = 22,000, so III is true and IV is false.
3. Expected surplus after one year = 10,000 × 1.5 = 15,000.
4. Worst-case 1% surplus = 15,000 − 22,000 = −7,000: a deficit of 7,000 with 1% probability, so II is true and I (99%) is false.

Why B is correct: VaR is measured from zero change in the surplus while the expected surplus grows, so the 1% tail outcome is expected surplus minus VaR, a deficit of 7,000 occurring with 1% (not 99%) probability.
Why the others are wrong
- **A:** Reads the 99% confidence level as the probability of the loss; it is the probability of NOT exceeding it.
- **C:** Correct on II, but IV (5,000) is the expected gain of the surplus (15,000 - 10,000), not the VaR.
- **D:** Combines the confidence/probability confusion (I) with the wrong VaR (IV).

Q291 AnalystPrep Q.2493

The ultimate responsibility for the pension fund rests with:

A. Employees
B. The fund manager
C. The plan sponsor
D. None of the above

Answer: C

Step by step

1. Plan sponsor risk concerns those who ultimately bear responsibility for the pension fund.
2. The sponsor funds the plan and must make additional contributions if the surplus turns negative.
3. So the answer is the plan sponsor.

Why C is correct: The plan sponsor stands behind the plan's obligations and bears the cost of any shortfall, hence sponsor risk.

Why the others are wrong

- **A:** Employees are beneficiaries of the plan, not the party that funds a shortfall.
- **B:** The manager implements the investment strategy but does not bear the plan's funding obligations.
- **D:** Incorrect since the sponsor is the responsible party.

Q292 AnalystPrep Q.2494

A pension fund manager comments: "Cash flow risk is the risk of year-to-year fluctuations in contributions to the pension fund. It restricts the plan sponsor from adopting a more volatile risk profile." Is the statement true?

A. Yes, the statement is true for all plan sponsors.

B. No, because a plan sponsor who can absorb greater variations in funding costs can adopt a more volatile risk profile.

C. No, because a plan sponsor who cannot absorb greater variations in funding costs can adopt a more volatile risk profile.

D. No, because cash flow risk can be mitigated by restricting the sponsor's funding options.

Answer: B

Step by step

1. Cash flow risk is the variation in contributions to the fund (correctly defined in the statement).
2. The ability to absorb cash-flow fluctuations allows a more volatile risk profile.
3. So the statement is not universally true: it restricts only sponsors that cannot absorb large variation in funding costs.

Why B is correct: The constraint depends on the sponsor's capacity to absorb contribution swings; strong sponsors can accept more volatility.

Why the others are wrong

- **A:** Generalizes to all sponsors, ignoring differences in capacity to absorb funding variation.
- **C:** Reverses the logic: a sponsor that cannot absorb variations is the one that must limit volatility.
- **D:** Restricting funding options does not mitigate cash flow risk and is not the reason the statement fails.

Q293 AnalystPrep Q.2496

A pension fund has heavily invested in stocks. The fund manager predicts a slowdown in the economy going forward. Which of the following is an accurate statement?

A. There will be no adverse effect on the pension fund.

B. The value of fund's assets will increase, thus increasing the surplus.

C. The value of fund's assets will decrease, and the economic downturn will severely restrict the sponsor's ability to make contributions.

D. The value of fund's assets will increase, but the economic downturn will severely restrict the sponsor's ability to make contributions.

Answer: C

Step by step

1. A slowdown depresses equity prices, so the value of a stock-heavy fund's assets falls and the surplus shrinks.
2. The same slowdown lowers the plan sponsor's operating earnings, so its capacity to make top-up contributions is reduced.
3. Asset losses and weaker sponsor support occur together (sponsor risk is positively linked to funding risk), so the fund suffers a double hit.

Why C is correct: Funding risk (falling asset values) and sponsor risk (weaker sponsor earnings and cash flow) are driven by the same economic factor, so they compound each other in a downturn.

Why the others are wrong

- **A:** Ignores both the fall in equity values and the sponsor's weaker ability to contribute.

- **B:** Assumes equities rise in a slowdown; the direction of the asset effect is reversed.
- **D:** Gets the sponsor effect right but wrongly assumes equity values rise in a slowdown.

Q294 AnalystPrep Q.2497

AlphaStop sponsors a defined benefits pension plan. The company's pension plan risk management system generates reports based on movements in the assets and the surplus. The pension management plan is not integrated with the firm's financial goals. The management has access to daily risk reports, and the board receives monthly risk reports. Which of the following is an accurate statement?

A. The company's pension plan management is satisfactory.
B. The reporting frequency to the board must be increased.
C. The company must integrate its pension plan management with its overall financial goals.
D. The company's pension plan must not generate reports based on movements in the assets.

Answer: C

Step by step

1. Plan sponsor risk is judged by its effect on the variation in the sponsor's total economic value (economic risk) and on contributions (cash flow risk).
2. Therefore asset and surplus risk must be integrated with the sponsor's overall financial goals, consistent with enterprise-wide risk management.
3. The plan reports on assets and surplus but is not integrated, so it is deficient; reporting frequency is not the problem.

Why C is correct: From the sponsor's viewpoint the focus should be the variation in the firm's economic value, so pension risk management must be integrated with the firm's overall financial goals.

Why the others are wrong

- **A:** Accepts a stand-alone system, overlooking that it is not tied to the sponsor's economic value.
- **B:** Treats the problem as reporting frequency; more frequent reports do not fix the missing integration.
- **D:** Asset and surplus movement reports are appropriate; the flaw is lack of integration, not the reports themselves.

Q295 AnalystPrep Q.3028

Which of the following correctly differentiates between absolute risk and relative risk?

A. Absolute risk is the risk of dollar loss owing to the policy mix selected by the fund; relative risk is the risk of dollar loss over the horizon and may also be called asset risk.
B. Absolute risk is the risk of a dollar loss over the horizon and is sometimes called asset risk; relative risk is the risk of a dollar loss in a fund relative to its benchmark.
C. Absolute risk is the risk of dollar loss in a fund relative to its benchmark; relative risk is the risk of dollar loss owing to the policy mix selected by the fund.
D. Absolute risk is the risk of dollar loss in a fund relative to its benchmark and sometimes is called asset risk; relative risk is the risk of a dollar loss over the horizon.

Answer: B

Step by step

1. Absolute (asset) risk = total possible dollar loss over the horizon, with no reference to a benchmark.
2. Relative risk = shortfall of the fund's return versus a benchmark (excess return), measured by tracking error.
3. Only B assigns each definition to the right term; the others swap them or substitute policy-mix risk.

Why B is correct: Absolute risk is the benchmark-free loss over the horizon, whereas relative risk is the benchmark-relative shortfall.

Why the others are wrong

- **A:** Defines absolute risk as policy-mix risk and gives the absolute-risk definition to relative risk.
- **C:** Reverses the terms and replaces relative risk with policy-mix risk.
- **D:** Swaps the two definitions: benchmark-relative loss is relative risk, and the horizon loss (asset risk) is absolute risk.

Q296 AnalystPrep Q.4702

Which of the following statements differentiates between absolute risk and relative risk?

A. Absolute risk is the possibility of loss caused by an insufficient cover to the liabilities of the fund, while relative risk is where the surplus ultimately applies to the risk of the plan owner.

B. Absolute risk arises from yearly changes in the contribution to the pension fund, while relative risk stems from a fluctuation of financial earnings of the plan sponsor.

C. Absolute risk is the possibility of a loss in base currency over the horizon, while relative risk is the base currency difference between the fund's return and the return on the benchmark.

D. Absolute risk is the base currency loss measured against the benchmark selected by the fund, while relative risk is the sum of profits or losses from active managers concerning the baseline.

Answer: C

Step by step

1. Absolute risk: possible loss in dollars (base currency) over the horizon, i.e., asset risk.
2. Relative risk: dollar shortfall of the fund return relative to the benchmark return, measured by tracking error.
3. C matches both; A describes funding risk and sponsor risk, B describes cash flow risk and economic risk of the sponsor, D mislabels absolute risk as benchmark-based.

Why C is correct: Absolute risk is a benchmark-free horizon loss, while relative risk is the difference between fund and benchmark returns.

Why the others are wrong

- **A:** Describes funding risk (assets insufficient to cover liabilities) and surplus/sponsor risk, not absolute and relative risk.
- **B:** Describes sponsor cash flow risk (variation in contributions) and economic risk (variation in sponsor earnings).
- **D:** Makes absolute risk benchmark-relative and describes relative risk as the active-manager total, which confuses relative risk with active management risk.

Q297 BT answers P2.T8.12.1 · BT notes P2.T9.12.1

REBUILT — Merged two identical sources and added the 99% z-value (2.33) to the stem so the answer is unique.

Public Employee Retirement Fund (PERF) has $800 million in assets and $720 million in liabilities, for a current surplus of $80 million. The expected annual return on the surplus, scaled by assets, is 5.0%. The volatility of assets is 18.0% and returns are normally distributed. Assume liabilities are unchanged over the year and use a 99% z-value of 2.33. At the end of one year, there is a 1.0% probability that the pension will have a deficit of what amount?

A. $117 million deficit

B. $215 million deficit

C. $335 million deficit

D. $455 million deficit

Answer: B

Step by step

1. Expected end-of-year surplus = 80 + 5.0% × 800 = $120 million.
2. Relative (1%) surplus-at-risk = 2.33 × 18.0% × 800 = $335.5 million (liabilities unchanged, so surplus volatility = asset volatility).
3. Worst-case end surplus = 120 − 335.5 = −$215.5 million, i.e., a deficit of about $215 million.

Why B is correct: The 1% worst-case loss is measured from the expected end surplus, so the deficit is the VaR less the expected surplus.

Why the others are wrong

- **A:** Used the 95% z-value of 1.645 (1.645 × 18% × 800 = 236.9, less 120 = 117).
- **C:** Reported the VaR itself without subtracting the $120 million expected surplus.
- **D:** Added the expected surplus to the VaR (335.5 + 120) instead of subtracting it.

Q298 BT answers P2.T8.12.2 · BT notes P2.T9.12.2

REBUILT — Merged two identical sources, removed the editorial note, and replaced an unexplained distractor ($85.6m) with the realistic sign error of adding the drift.

Public Employee Retirement Fund (PERF) has $600 million in assets and $600 million in liabilities, for a current

surplus of zero. The annual expected return on assets is 8.0% with 18.0% volatility; the annual expected growth of liabilities is 6.0% with 14.0% volatility. Both are normally distributed and the correlation between assets and liabilities is 0.60. Using a z-value of 1.645, what is the 95% absolute surplus at risk (the worst shortfall relative to the current surplus of zero)?

A. $12.0 million
B. $157.6 million
C. $133.6 million
D. $194.0 million

Answer: C

Step by step

1. Expected surplus growth = 8% × 600 − 6% × 600 = 48 − 36 = +$12.0 million.
2. Surplus variance = $600^2(0.18^2)+600^2(0.14^2)-2(600)(600)(0.18)(0.14)(0.60) = 11{,}664+7{,}056-10{,}886.4 = 7{,}833.6$ (in squared millions of dollars), so $\sigma = 88.51$ million dollars.
3. Relative SaR = 1.645 × 88.51 = $145.6 million.
4. Absolute SaR = 145.6 − 12.0 = $133.6 million.

Why C is correct: Absolute SaR is the 95% relative loss (z × surplus volatility) reduced by the expected positive surplus growth.

Why the others are wrong

- **A:** Reported only the expected surplus growth, not the risk.
- **B:** Added the $12 million expected growth to the relative SaR instead of subtracting it (145.6 + 12).
- **D:** Used the 99% z-value of 2.326 (2.326 × 88.51 − 12 ≈ 194).

Q299

BT answers P2.T8.13.1 · BT notes P2.T9.13.1

REBUILT — Merged two identical sources and replaced the unexplained -0.5% distractor with -1.5% (equity contribution only).

A pension fund invests in only two asset classes, equities and fixed income. Its policy mix is 50% equities and 50% fixed income. Over the last period the benchmark return for equities was 7.0% and for fixed income 5.0%. The fund did not deviate from its policy mix (50% / 50%). The equity managers returned +4.0% and the fixed income managers returned +10.0%. What is the fund's return attributable to active management?

A. −1.5%
B. 0.0%
C. +1.0%
D. +2.5%

Answer: C

Step by step

1. Actual fund return = 0.5 × 4.0% + 0.5 × 10.0% = 7.0%.
2. Policy-mix (benchmark) return = 0.5 × 7.0% + 0.5 × 5.0% = 6.0%.
3. Active management return = 7.0% − 6.0% = +1.0%, which equals 0.5(4 − 7) + 0.5(10 − 5) = −1.5% + 2.5%.

Why C is correct: With weights equal to the policy mix, the entire difference between fund return and policy-benchmark return comes from manager selection within each asset class.

Why the others are wrong

- **A:** Counted only the equity managers' contribution, 0.5 × (4% − 7%).
- **B:** Reasoned that sticking to the policy mix means there is no active risk; manager returns can still differ from the benchmarks.
- **D:** Counted only the fixed income managers' contribution, 0.5 × (10% − 5%).

Q300

BT answers P2.T8.13.2 · BT notes P2.T9.13.2

A pension fund invests in a small number of major asset classes. Following the Brinson research, the fund assumes that most of the variation in portfolio performance is attributable to its policy-mix choice of asset classes rather than to the selection of individual managers. Within each asset class the fund diversifies by allocating to many investment managers. Finally, the correlation between policy-mix VaR and active-management VaR is slightly negative. Which is the most plausible ordering of the sources of risk, from most risk to least?

A. Asset VaR (most), active-management VaR, policy-mix VaR (least)
B. Policy-mix VaR (most), asset VaR, active-management VaR (least)
C. Active-management VaR (most), policy-mix VaR, asset VaR (least)

D. Active-management VaR (most), asset VaR, policy-mix VaR (least)

Answer: B

Step by step

1. Asset (total) VaR combines the two sources: $\text{VaR}_A^2 = \text{VaR}_P^2 + \text{VaR}_{AM}^2 + 2\rho\,\text{VaR}_P\text{VaR}_{AM}$.
2. Policy-mix risk dominates (Brinson), so policy-mix VaR is large.
3. Active-management risk is small: managers are diversified within each asset class and deviations from the policy weights diversify across managers.
4. With a slightly negative correlation the cross term is negative and small, so asset VaR is pulled to roughly the policy-mix level (slightly below it for a small active component) and is far above active-management VaR.
5. Order: policy-mix VaR > asset VaR > active-management VaR.

Why B is correct: Total risk is the correlation-adjusted combination of a large policy-mix component and a small active component; a slightly negative correlation keeps total VaR close to (not above) the policy-mix VaR, yet well above the small active VaR.

Why the others are wrong

- **A:** Puts active-management VaR above policy-mix VaR, contradicting Brinson and the diversification across many managers; it also places asset VaR first, which ignores the negative correlation that keeps total VaR from exceeding the dominant policy-mix VaR by much.
- **C:** Ranks active-management risk highest, ignoring the Brinson finding and the diversification across many managers; asset VaR cannot be smallest because it is the combined risk.
- **D:** Again ranks active-management risk highest and policy-mix lowest, the reverse of the Brinson result.

Q301

BT answers P2.T8.13.3 · BT notes P2.T9.13.3

Each of the following definitions is true EXCEPT, which is false?

A. Cash-flow risk is the risk of year-to-year fluctuations in contributions to the pension fund

B. Economic risk is the risk of variation in total economic earnings of the plan sponsor

C. Funding risk is the risk that the value of assets will not be sufficient to cover the liabilities of the fund

D. Active-management risk is the risk of a dollar loss owing to the policy mix selected by the fund

Answer: D

Step by step

1. Cash-flow risk = variation in the sponsor's contributions to the fund (true).
2. Economic risk = variation in the sponsor's total economic earnings (true).
3. Funding risk = assets may be insufficient to cover liabilities (true).
4. Active-management risk arises from managers deviating from the policy weights; risk from the chosen policy weights is policy-mix risk. So D is false.

Why D is correct: Active-management risk comes from managers' deviations from the policy mix, not from the policy mix itself; D describes policy-mix risk.

Why the others are wrong

- **A:** A correct definition of cash-flow risk (a sponsor risk component).
- **B:** A correct definition of economic risk (a sponsor risk component).
- **C:** A correct definition of funding risk; choosing it confuses a true statement with the false one.

Q302

BT answers P2.T9.21.7.3 · BT notes P2.T9.21.7.3

Four colleagues at a state public employee retirement fund debate the investment process and its metrics. The fund is large and delegates to managers across traditional and alternative asset classes, including private equity funds and a successful hedge fund whose outperformance comes from skillful investing in mispriced, illiquid assets. You overhear:

- Albert says funding risk should be a top priority.
- Betty says the hedge fund VaR budget should be adjusted for artificially low correlations and volatility.
- Charles says most of the fund's downside risk is due to policy mix while only a small portion is due to active-management risk.
- Derek says the weakness of VaR is that it cannot accurately inform top-down risk budgets based on forward-looking risk, nor contribute useful insight into risk drivers.

Each of these people makes a valid point EXCEPT who is probably incorrect?

A. Albert

B. Betty
C. Charles
D. Derek

Answer: D
Step by step
1. Albert: a defined-benefit plan has the highest funding risk (assets may not cover liabilities), so a top priority is valid.
2. Betty: illiquid hedge fund holdings produce smoothed returns, biasing volatility and correlations downward, so the VaR budget should be adjusted; valid.
3. Charles: policy mix drives most of total risk and active risk is small (diversified managers); valid.
4. Derek: VaR is forward-looking, supports top-down risk budgeting, and decomposes risk into drivers (policy mix vs active, marginal/component VaR). His claim is wrong.

Why D is correct: VaR's strengths are exactly what Derek denies: a top-down, forward-looking budget and decomposition of risk drivers.

Why the others are wrong
- **A:** Albert is right; funding risk is central for pension funds.
- **B:** Betty is right; low liquidity biases volatility and correlation downward.
- **C:** Charles is right; policy-mix risk dominates, active risk is small and diversified.

Q303

BT answers P2.T9.21.8.1 · BT notes P2.T9.21.8.1

REBUILT — Source options were partly OCR-damaged (one shortfall figure unreadable); rewrote distractors as realistic computational mistakes with the same correct answer.

A pension fund has a surplus of $190.0 million:

Item	Assets	Liabilities
Amount (USD million)	700.0	510.0
Expected annual growth	9.0%	6.0%
Annual volatility	30.0%	15.0%
Correlation between asset and liability returns	0.60	0.60

Returns are normally distributed. Losses and shortfalls are expressed as positive numbers. The relative surplus at risk (rSaR) is the worst loss at 99% confidence relative to the expected end-of-year surplus; the absolute surplus at risk (aSaR) is the worst loss relative to the beginning-of-year surplus; the worst expected shortfall is the worst negative surplus (relative to zero) at the end of the year. For a one-year horizon at 99% confidence (z = 2.326), which of the following is nearest to rSaR, aSaR and the worst expected shortfall, respectively?

A. rSaR = $288 million, aSaR = $256 million, shortfall = $66 million
B. rSaR = $407 million, aSaR = $440 million, shortfall = $630 million
C. rSaR = $612 million, aSaR = $580 million, shortfall = $390 million
D. rSaR = $407 million, aSaR = $375 million, shortfall = $185 million

Answer: D
Step by step
1. Expected change in surplus = 700(9%) − 510(6%) = 63.0 − 30.6 = $32.4m; expected end surplus = 190 + 32.4 = $222.4m.
2. $\sigma_S = \sqrt{(700 \times 0.30)^2 + (510 \times 0.15)^2 - 2(0.60)(700 \times 0.30)(510 \times 0.15)} = \sqrt{44{,}100 + 5{,}852.25 - 19{,}278} = \$175.14\text{m}$.
3. rSaR = 2.326 × 175.14 = $407.4m.
4. aSaR = rSaR − expected change = 407.4 − 32.4 = $375.0m (loss versus the starting surplus of 190).
5. Worst shortfall = rSaR − expected end surplus = 407.4 − 222.4 = $185.0m (surplus of −$185m).

Why D is correct: Surplus risk is the volatility of A − L (the variance includes −2ρ cov terms), scaled by z; the loss is measured from the expected surplus (rSaR), the starting surplus (aSaR, after removing expected growth), or zero (shortfall, after removing the expected end surplus).

Why the others are wrong
- **A:** Used z = 1.645 (95%) instead of 2.326: 1.645 × 175.14 = 288, then 288 − 32 = 256 and 288 − 222 = 66.
- **B:** Right rSaR but adjusted in the wrong direction: added the expected growth (407 + 32 = 440) and the expected end surplus (407 + 222 = 630) instead of subtracting.
- **C:** Added the correlation term instead of subtracting it (σ = 263.1), giving 612, 612 − 32 = 580 and 612 − 222 =

390; the variance of A − L has −2·cov.

Q304

BT notes P2.T9.12.3 · BT answers P2.T8.12.3

REBUILT — Original options mixed maturity and duration and omitted units; rewrote options as long/short with durations of 16.0 vs 1.6 years.

A public employee retirement fund reports a projected benefit obligation (PBO) of $70.0 billion. If the discount rate decreases by 10 basis points, the PBO increases by $1.120 billion. The pension liabilities behave most nearly like which of the following?

A. A long position in a bond with modified duration of 16.0 years
B. A long position in a bond with modified duration of 1.6 years
C. A short position in a bond with modified duration of 1.6 years
D. A short position in a bond with modified duration of 16.0 years

Answer: D

Step by step

1. Modified duration $D = -\frac{1}{P}\frac{\Delta P}{\Delta y} = -\frac{1}{70} \times \frac{+1.120}{-0.0010} = 16.0$ years.
2. The liability value rises when rates fall, so the sign is that of a bond the fund has sold (an obligation, i.e., a negative asset).
3. So the liabilities behave like a short bond position with modified duration 16.0 years.

Why D is correct: A 1.6% rise in value for a 10 bp fall implies duration of 16 years, and because the PBO is an obligation it is equivalent to a short bond position.

Why the others are wrong

- **A:** Duration is right but treats the liability as an asset (long); a long bond would hedge the liability, not mimic it.
- **B:** Wrong magnitude (treated the 10 bp move as 1%, giving 1.6) and wrong direction.
- **C:** Direction (short) is right but the magnitude is wrong: treated the 10 bp move as 1%, giving 1.6 years instead of 16.

Q305

BT notes P2.T9.408.1 · BT answers P2.T8.408.1

Both value at risk (VaR) and tracking error (TE) are considered risk measures. Which of the following statements is TRUE about the relationship and/or contrast between VaR and tracking error?

A. Relative VaR, as opposed to absolute VaR, is analogous but not identical to tracking error (also called active risk), and either VaR or TE can set the risk constraints (limits) in a risk budget
B. Value at Risk (VaR) is mathematically identical to tracking error; the only difference is that VaR implies the user is a shareholder or owner of capital, while TE implies an asset manager
C. VaR is identical to tracking error only if asset returns are normally distributed. Otherwise VaR differs from TE because VaR does not require normal returns but TE requires normal returns
D. They are neither compatible nor analogous: tracking error is an absolute return metric which only has meaning in the context of an information ratio, not a risk budget; VaR is a relative risk metric which is necessarily expressed in different units

Answer: A

Step by step

1. Tracking error is the standard deviation of excess (portfolio minus benchmark) return; relative VaR is a quantile of the excess-return distribution, so the two are analogous (linked by a multiple of σ under normality) but not identical.
2. Either can be used to set limits in a risk budget (e.g., TE budgets by manager).
3. B is false: TE is one standard deviation, VaR a loss quantile at a confidence level.
4. C is false: even under normality VaR = zσ − μ differs from σ, and TE does not require normality.
5. D is false: TE is a relative measure and VaR can be absolute or relative.

Why A is correct: Relative VaR and TE both measure risk of excess return versus a benchmark; they differ in definition (quantile vs standard deviation) but both can serve as risk-budget limits.

Why the others are wrong

- **B:** Confuses two different statistics (a quantile vs a standard deviation) and invents a shareholder/manager distinction.
- **C:** Wrongly says normality makes them identical (VaR still scales σ by z) and that TE requires normality.
- **D:** Reverses the facts: TE is a relative measure used in risk budgeting, and VaR may be absolute or relative.

Q306 GARP PE2 Q72

A risk manager at a defined-benefit pension fund for a large automobile manufacturer is training newly hired analysts on the fund's risk measurement and management practices. The manager describes some of the types of risk that are most important for defined-benefit pension funds and provides examples of how each type of risk can be mitigated. Which of the following statements would be most appropriate for the manager to make?

A. Relative risk is mitigated when investment decisions made by portfolio managers decrease the fund's tracking error.
B. Cash-flow risk is minimized when the marginal VaR of all assets held by the fund is equal.
C. Funding risk is mitigated when the duration of the fund's liabilities is decreasing as interest rates are also decreasing.
D. Sponsor risk is eliminated when the policy mix selected by the fund generates the expected returns.

Answer: A

Step by step

1. Relative risk is the risk of deviating from a benchmark; it is measured by the excess return (portfolio minus benchmark) and its volatility, the tracking error.
2. Decisions that reduce tracking error therefore reduce relative risk, so A is correct.
3. Cash-flow risk is the variation in sponsor contributions; it is not controlled by equalizing marginal VaRs.
4. Funding risk is the risk that assets fall short of liabilities; falling rates raise liability value, so this is not mitigation.
5. Sponsor risk cannot be eliminated by hitting expected returns, since actual returns are uncertain.

Why A is correct: Relative risk is shortfall versus a benchmark, measured by tracking error, so lowering tracking error lowers relative risk.

Why the others are wrong

- **B:** Equal marginal VaRs is a portfolio-optimization condition, not a definition or mitigation of cash-flow risk (variation in contributions); confuses cash-flow risk with risk-budget optimality.
- **C:** Funding risk is the chance assets cannot cover liabilities. Falling rates raise liability PV, often by more than assets (liability duration is typically longer), so it worsens funding risk; decreasing duration is not a mitigant.
- **D:** Sponsor risk (economic and cash-flow risk to the sponsor) is never eliminated; returns are random and a policy mix only sets expected return. Confuses expected with realized outcomes.

Q307 Mock 1 Q46

REBUILT — Original distractors were mere sign/size variants (two of four were giveaways); replaced with options from realistic errors, same data and answer.

Xfinity Resorts, Inc., operates a defined benefit pension scheme with assets of USD 300 million and liabilities of USD 275 million. Xfinity's advisors, Fortune Capital Management, have estimated a 7% growth in assets and a 4% growth in liabilities over the next year. The volatility of the growth in assets and liabilities is estimated to be 13% and 5%, respectively. The correlation between asset and liability growth is estimated to be 0.40. What is the projected amended surplus at a 95% confidence level (use z = 1.65)?

A. -USD 24.05 million
B. -USD 34.06 million
C. -USD 41.31 million
D. USD 24.05 million

Answer: A

Step by step

1. Expected surplus: $300(1.07) - 275(1.04) = 321 - 286 = 35.00$ million.
2. Dollar volatilities: assets $0.13 \times 300 = 39$; liabilities $0.05 \times 275 = 13.75$.
3. $\sigma_S^2 = 39^2 + 13.75^2 - 2(0.40)(39)(13.75) = 1521 + 189.06 - 429 = 1281.06$, so $\sigma_S = 35.79$.
4. SaR $= 1.65 \times 35.79 = 59.05$ million.
5. Amended surplus $= 35.00 - 59.05 = -24.05$ million (a deficit).

Why A is correct: The worst-case surplus is the expected surplus minus z times the surplus volatility, where the variance of A - L subtracts twice the covariance.

Why the others are wrong

- **B:** Started from the current surplus of 25 instead of the expected surplus of 35: $25 - 59.05 = -34.05$ (approximately -34.06).
- **C:** Added the covariance term instead of subtracting it: $\sigma_S = 46.25$, SaR 76.31, $35 - 76.31 = -41.31$.
- **D:** Sign error: reported the size of the shortfall below zero as a positive surplus.

Q308

Mock 1 Q71

REBUILT — Option D (18.6) matched no coherent error; replaced with sign-error distractors, correct answer unchanged.

A pension fund has plan assets and liabilities currently valued at USD 150 million and USD 120 million, respectively. The assets are fully invested in equities, while the liabilities resemble fixed-income obligations and have an average duration of 14. Over the next year, the plan assets fall 30% and yields fall by 2 percentage points. Which of the following is the change in surplus?

A. -USD 48.6 million
B. -USD 78.6 million
C. -USD 11.4 million
D. USD 78.6 million

Answer: B

Step by step

1. Change in assets $= 150 \times (-0.30) = -45.0$ million.
2. Change in liabilities $= -D \times L \times \Delta y = -14 \times 120 \times (-0.02) = +33.6$ million (liabilities rise as yields fall).
3. Change in surplus $= \Delta A - \Delta L = -45.0 - 33.6 = -78.6$ million.
4. Check: surplus moves from 30 to 105 - 153.6 = -48.6.

Why B is correct: Assets lose value while falling yields increase the PV of the liabilities, so both effects reduce the surplus.

Why the others are wrong

- **A:** This is the new surplus level (-48.6 = 30 - 78.6), not the change in surplus.
- **C:** Treated the liability increase as a gain: -45 + 33.6 = -11.4; liabilities rising hurts the surplus.
- **D:** Right magnitude but wrong sign; the surplus falls.

Q309

Schweser Module Quiz 88.1 Q2

Compared to policy risk, which of the following is not a reason that management risk is not much of a problem?

A. There will be diversification effects across the deviations.
B. Managers tend to make the same style shifts at the same time.
C. For well-managed funds, it is usually fairly small for each of the individual funds.
D. There can be diversification with the policy mix VaR to actually lower the total portfolio VaR.

Answer: B

Step by step

1. Schweser gives three reasons active management risk is usually small relative to policy-mix risk: deviations are small for well-managed funds, deviations diversify across managers, and active risk can diversify against policy-mix VaR.
2. A, C and D restate these reasons.
3. B says managers make the same style shifts at the same time; this would make deviations correlated and increase management risk, so it is not a reason. Answer B.

Why B is correct: Correlated style shifts remove the diversification across managers' deviations that keeps management risk small.

Why the others are wrong

- **A:** A valid reason: deviations across managers diversify, so this is a reason management risk is small (the question asks for the non-reason).
- **C:** A valid reason: for well-managed funds individual deviations are small.
- **D:** A valid reason: active risk can diversify against policy-mix VaR and even lower total VaR.

Q310

AnalystPrep Q.4705

A pension fund allocates $4,550 million to different classes of assets. The volatility of the investment is 7%. Assuming normally distributed returns and a z-score of 1.65 for 95% confidence, what is the value-at-risk (VaR)?

A. $318.5 million
B. $742.11 million
C. $525.53 million
D. $382.20 million

Answer: C

Step by step

1. Absolute VaR = $\alpha\sigma W$, with $\alpha = 1.65$ at 95% confidence.
2. $\sigma W = 0.07 \times 4{,}550 = 318.5$ million.
3. VaR $= 1.65 \times 318.5 = 525.53$ million.

Why C is correct: VaR is the z-score for the confidence level times the dollar volatility of the portfolio.

Why the others are wrong

- **A:** One-standard-deviation loss ($0.07 \times 4{,}550$); the 1.65 multiplier was omitted.
- **B:** Used the 99% z-score of 2.33 instead of 1.65 ($2.33 \times 318.5 = 742.11$).
- **D:** Used an arbitrary multiplier of 1.2 on the dollar volatility; it matches neither the 95% nor the 99% z-score.

Q311 BT answers L2.T5.65.1

A fund with value of $10 million is benchmarked against an index. The fund's volatility is 10.0% and normally distributed. The benchmark volatility is 5.0% and normally distributed. The fund's active risk (tracking error) is 8.0% and normally distributed. What is the 95% tracking error value at risk (tracking error VaR)? Use z = 1.645.

A. $820,000
B. $1.32 million
C. $1.65 million
D. $1.96 million

Answer: B

Step by step

1. Relative VaR uses the volatility of the excess return, i.e. the 8.0% tracking error.
2. TEVaR $= 1.645 \times 0.08 \times 10{,}000{,}000$
3. $= \$1{,}316{,}000 \approx \$1.32$ million.

Why B is correct: Relative (tracking error) VaR measures the loss versus the benchmark, so only the volatility of the active return enters.

Why the others are wrong

- **A:** Used the benchmark volatility of 5% ($1.645 \times 0.05 \times 10 = 0.82$ million).
- **C:** Used the fund's total volatility of 10% ($1.645 \times 0.10 \times 10 = 1.645$ million), which is absolute VaR, not relative VaR.
- **D:** Used the fund volatility of 10% with the two-tailed z of 1.96 ($1.96 \times 0.10 \times 10$).

LO 88.f

Explain the use of VaR to monitor risk.

Q312 AnalystPrep Q.2480

A VaR risk management system is suited to the investment management industry due to the following reasons, EXCEPT:

A. The need for risk measures which take diversification into account.
B. Complex financial instruments which create a need for stronger, centralized risk management systems.
C. The dynamic nature of investment portfolios.
D. The presence of highly illiquid assets such as convertible bonds, traded infrequently.

Answer: D

Step by step

1. Schweser cites globalization, increased complexity of instruments and the dynamic nature of the industry as reasons VaR is needed in place of purely historical measures.
2. VaR also captures diversification across many managers and asset classes.
3. Illiquid, infrequently traded assets give little price data and make VaR harder to estimate; this is not a reason VaR suits the industry.
4. The EXCEPT item is D.

Why D is correct: VaR suits the industry because of complexity, dynamics and diversification; illiquid, thinly traded

assets are a limitation for VaR, not a justification.

Why the others are wrong

- **A:** VaR measures portfolio-level risk including diversification effects, so it is a valid reason.
- **B:** Complex instruments need a centralized, forward-looking system, a reason cited in the reading.
- **C:** Managers change positions within their constraints, so portfolios are dynamic and historical measures are inadequate; a valid reason.

Q313

AnalystPrep Q.2499

As a fund investor, you notice a sudden jump in the reported VaR of the fund. All of the following may provide an explanation for the sudden jump, EXCEPT:

A. The manager may be taking more risk.
B. Different managers may be taking similar bets.
C. Managers may be taking different bets.
D. The volatility of the markets may have increased.

Answer: C

Step by step

1. Schweser lists three explanations for a dramatic rise in portfolio risk: (1) a manager taking on more risk, (2) different managers taking similar bets, (3) more volatile markets.
2. A, B and D match these three explanations.
3. Managers taking different bets lowers correlation among their positions and diversifies the total portfolio, which reduces (or at least does not raise) VaR.
4. So C is the exception.

Why C is correct: Different (diversified) bets across managers reduce aggregate risk, whereas more risk, overlapping bets and higher volatility all raise VaR.

Why the others are wrong

- **A:** A valid cause: a manager exceeding the risk budget or drifting in weights raises VaR. Choosing it misses that it is a listed explanation.
- **B:** A valid cause: when managers independently converge on the same exposures, diversification disappears and aggregate VaR jumps.
- **D:** A valid cause: higher market volatility raises VaR even with unchanged positions.

Q314

AnalystPrep Q.2498

REBUILT — Original asked which risk VaR cannot measure, but the source's claim that VaR covers liquidity risk is not supported by the Schweser reading; replaced with the reading's three explanations for a rise in portfolio risk.

The total VaR of a large investment firm's portfolio has risen sharply. Top management wants to investigate the cause using the firm's VaR monitoring system. Which of the following is NOT one of the standard explanations for a dramatic increase in risk that management should examine?

A. A manager has exceeded her risk budget.
B. Different managers are taking similar style bets.
C. Markets have become more volatile.
D. Managers' realized alphas have been higher than forecast.

Answer: D

Step by step

1. The reading gives three explanations for a dramatic change in risk: (1) a manager taking on more risk (exceeding the risk budget, weight drift, unauthorized trades), (2) different managers taking similar bets, (3) more volatile markets.
2. Options A, B and C map to explanations 1, 2 and 3 respectively.
3. Realized alpha is a return outcome; it does not change the forward-looking risk (VaR) of the positions held, so D is not an explanation.

Why D is correct: VaR is a forward-looking risk measure driven by positions, correlations and volatilities, not by past realized excess returns; higher-than-forecast alpha does not raise VaR.

Why the others are wrong

- **A:** This is explanation (1): a manager exceeding her risk budget raises portfolio VaR, so it is a standard item to check.

- **B:** This is explanation (2): common style bets raise correlation across managers and thus total risk (e.g., all moving into long bonds).
- **C:** This is explanation (3): higher market volatility raises VaR even with unchanged positions.

Q315 AnalystPrep Q.2524

A risk manager at a large bank argues that the risk management unit (RMU) must monitor only the overall tracking error expectations. If the overall tracking error expectations are within the prescribed limits, there is no cause for concern. Which of the following statements is correct?

A. Monitoring the overall tracking error expectations is sufficient.
B. The overall tracking error analysis must be restricted to keep it within prescribed limits.
C. Managers must identify how such tracking error is decomposed into its constituent parts.
D. The tracking error expectations can exceed the prescribed limits.

Answer: C

Step by step

1. Total tracking error is a single aggregate number; offsetting over- and under-weights in different sources of risk can leave it unchanged.
2. Monitoring therefore requires decomposing forecast tracking error by source (e.g., sector, style, country, security selection) and comparing each part with its budget.
3. Only option C states this requirement.

Why C is correct: An aggregate within limits can hide risk concentrated in unintended sources, so risk must be monitored both in total and by component.

Why the others are wrong

- **A:** Treats the aggregate as sufficient; offsetting exposures can mask unintended bets.
- **B:** Confuses monitoring with restricting; the issue is where risk comes from, not limiting the analysis.
- **D:** Says limits may be exceeded; this does not answer the monitoring question and contradicts the idea of prescribed limits.

Q316 AnalystPrep Q.2538 · AP Mock 1 Q71

Pinnacle Capital has built its reputation on value investing. During quarterly risk monitoring, the risk unit discovers that despite maintaining an overall forecast tracking error consistent with target, most of the risk is placed in growth themes. Which statement best describes the situation?

A. Pinnacle Capital is successfully managing its overall tracking error and should continue with its current strategy.
B. Pinnacle Capital is experiencing style drift and may not be investing in accordance with its stated philosophy.
C. Pinnacle Capital is investing with greater risk due to the inconsistency in its risk themes.
D. Pinnacle Capital's risk forecasting model is not working properly.

Answer: B

Step by step

1. Tracking error measures the amount of active risk, not where it comes from.
2. Total tracking error is on target, but the decomposition shows the risk is concentrated in growth themes, inconsistent with a value mandate.
3. This is style drift: a departure from the stated philosophy, detected by decomposing risk.

Why B is correct: The decomposition, not the aggregate, reveals that active risk sits in growth themes, contradicting the value philosophy.

Why the others are wrong

- **A:** Relies only on the aggregate tracking error and ignores the decomposition.
- **C:** Total risk is on target, so risk is not necessarily greater; the problem is the mix of risk, not the amount.
- **D:** The model worked: it produced the decomposition that exposed the drift.

Q317 MK Mock 3 Q14 · GARP PE1 Q67

A risk manager at an investment firm is making a presentation about VaR to a group of newly hired junior analysts. The manager explains the use of VaR in controlling and monitoring risks and discusses the benefits and limitations of VaR. Which of the following statements would be correct for the manager to make?

A. VaR is a reliable and easily performed method to measure the riskiness of illiquid assets.
B. VaR systems can generate accurate market risk estimates but at the expense of making "rogue trading" easier.
C. VaR measures can help identify the different sources of an increase in total portfolio risk.
D. VaR systems can monitor the risk levels of investments at regular intervals but not in real time.

Answer: C

Step by step

1. Component and marginal VaR attribute portfolio VaR to positions and risk factors, so they show where an increase in total risk came from.
2. Illiquid/unique assets (real estate, hedge funds, venture capital) are hard to measure with VaR, so A is false.
3. Risk monitoring helps catch rogue traders, so B is false; nothing limits VaR to non-real-time use, so D is false.

Why C is correct: VaR decomposition tools (marginal, component) let management trace a rise in total risk to specific managers, positions or bets.

Why the others are wrong

- **A:** The reading names illiquid/unique assets as a problem area for risk measurement; stale prices make VaR unreliable.
- **B:** Forward-looking monitoring helps detect rogue traders rather than making them easier; this reverses the benefit.
- **D:** VaR systems can be run frequently, including near real time; the stated restriction is invented.

Q318 AP Mock 2 Q75

Allegro Investments' risk unit finds that a portfolio manager's forecast tracking error aligns perfectly with the mandate target. However, the same manager is significantly over budget in specific sector themes, with offsetting under-spending in other areas. As CRO, how should you assess this?

A. The situation is not problematic since the overall forecast tracking error is aligned with the target.
B. The overspending in specific sectors is a matter of the manager's discretion and should not be questioned.
C. The over-budget positions in specific sectors indicate potential deviations from the established investment strategy.
D. This situation suggests that the risk forecasting model is not functioning correctly.

Answer: C

Step by step

1. Over-budget sector risk is offset by under-budget risk elsewhere, so the aggregate tracking error equals target.
2. Risk budgets apply to each component as well as the total; the sector overruns breach the intended allocation.
3. This signals possible departure from the strategy and should be investigated.

Why C is correct: Component budgets bind independently of the aggregate; offsetting overruns and underruns mask strategy deviations.

Why the others are wrong

- **A:** Looks only at the aggregate; offsets hide the breach.
- **B:** Manager discretion operates within the risk budget; the RMU's job is to question overruns.
- **D:** The model correctly produced the decomposition that exposed the breach.

Q319 AnalystPrep Q.2502

REBUILT — Original exception option ('risk indicators based on historical data') was ambiguous versus the reading's forward-looking risk measures; rewrote options around the reading's custodian roles with a clear non-custodian role as the exception.

A global custodian can help an investment firm pursue centralized risk management. Which of the following is NOT a role the reading attributes to a global custodian?

A. Aggregating portfolios to give a consolidated picture of the fund's total exposures.
B. Combining reports on changes in positions with market data to produce forward-looking risk measures.
C. Setting each portfolio manager's risk budget and deciding the firm's investment philosophy.
D. Providing risk management products as part of a broader set of services.

Answer: C

Step by step

1. The reading says a global custodian aggregates portfolios (consolidated view of exposures), combines position changes with market data to produce forward-looking risk measures, and increasingly offers risk management products.
2. Options A, B and D restate these roles.
3. Setting risk budgets and the investment philosophy are decisions of the firm's management, not the custodian.

Why C is correct: The custodian supplies data, aggregation and risk reporting; risk budgets and philosophy are management decisions.

Why the others are wrong

- **A:** This is a stated custodian role: aggregation across portfolios.
- **B:** This is a stated custodian role: forward-looking risk measures from positions plus market data.
- **D:** Large custodian banks do provide risk management products.

Q320 AnalystPrep Q.2515 · AnalystPrep Q.2510

Transglobal Investments has built its reputation on value investing. Its Global Equity Fund, managed by Laura, meets its target tracking error, but the quarterly review shows a spike in allocation to technology companies, traditionally regarded as growth investments. To ensure Laura's activities are consistent with the firm's value philosophy, what would be the most effective step for the Risk Management Unit (RMU)?

A. Interview Laura to understand her rationale for the increased tech allocations.

B. Analyze the fund's performance over the last quarter to see if returns were affected by the tech allocations.

C. Investigate the risk decomposition to see whether the increased tech allocation is consistent with the firm's expectations.

D. Review client feedback about the fund's performance to understand their perception of the tech allocations.

Answer: C

Step by step

1. Meeting the tracking error target only confirms the amount of active risk.
2. To check consistency with the philosophy, compare the forecast tracking error with its budget and examine how it is allocated across subsections (sectors, styles).
3. This objective decomposition shows whether tech risk is within the expected allocation; interviews, performance and client views do not test this.

Why C is correct: Decomposing forecast tracking error against its budget directly tests whether risk is allocated as the philosophy expects.

Why the others are wrong

- **A:** Qualitative and after the fact; it does not measure where risk is actually allocated.
- **B:** Past performance does not show the risk allocation and can look good while style drift persists.
- **D:** Client perception is irrelevant to verifying risk allocation against the mandate.

Q321 Schweser Module Quiz 88.2 Q1

Using VaR to monitor risk is important for a large firm with many types of managers because:

A. it can help catch rogue traders, and it can detect changes in risk from changes in benchmark characteristics.

B. although it cannot help catch rogue traders, it can detect changes in risk from changes in benchmark characteristics.

C. although it cannot detect changes in risk from changes in benchmark characteristics, it can help detect rogue traders.

D. of no reason. VaR is not useful for monitoring risk in large firms.

Answer: A

Step by step

1. Rogue trading is more likely in large firms; deviations can be short-lived and missed by regular reports, but a forward-looking VaR system flags the risk jump.
2. Benchmark risk profiles change over time (e.g. the S&P 500 tech concentration in the late 1990s); forward-looking VaR picks this up even for passive portfolios.
3. Both are valid reasons, so A is correct.

Why A is correct: VaR-based monitoring both flags unauthorized or off-guideline positions and reveals drift in benchmark risk exposures.

Why the others are wrong

- **B:** Wrongly denies the rogue-trader benefit; unexpected jumps in VaR are a primary way such deviations are caught.
- **C:** Wrongly denies the benchmark benefit; the belief that passive/benchmarked portfolios need no monitoring is a common error.
- **D:** VaR is useful for monitoring in large firms, e.g. to check risk budgets and detect clustered style bets.

LO 88.g

Explain how VaR can be used in the development of investment guidelines and for improving the investment process.

Q322

BT answers P2.T8.14.2 · BT notes P2.T9.14.2

REBUILT — Removed the dependence on a verbatim Jorion quotation and reframed statement C so that it is judged against the Schweser reading's information-ratio budgeting logic.

Which of the following statements about using VaR in investment guidelines and the investment process is FALSE?

A. Guidelines based on VaR limits, such as a cap on tracking error, can replace limits on notional exposures because VaR reflects leverage, derivatives and correlations.

B. VaR can complement a top-down strategic asset allocation process that is based on mean-variance analysis.

C. Budgeting risk across active managers by using each manager's information ratio requires no assumption about the managers' expected excess returns.

D. A trader choosing between two positions with similar expected returns should prefer the one with the lower marginal VaR.

Answer: C

Step by step

1. Check each statement against the LO 88.g material and the IR budgeting discussion.
2. A: notional/sensitivity limits ignore variations in risk and correlations and fail with leverage and derivatives; VaR limits capture all of these. True.
3. B: strategic asset allocation normally uses mean-variance analysis, as do the basic VaR measures, so VaR complements it. True.
4. C: IR = expected excess return / tracking error. Allocating weights across managers by IR needs a forecast of each manager's expected excess return, so an assumption is required. False.
5. D: for similar returns, the lower marginal VaR adds less to portfolio risk. True.

Why C is correct: The information ratio has expected excess return in its numerator, so IR-based manager budgeting necessarily rests on forecasts of manager performance; the claim that no assumptions are needed is false.

Why the others are wrong

- **A:** True: VaR limits capture leverage, derivatives and correlation, which notional limits miss. Choosing it means rejecting a valid LO 88.g point.
- **B:** True: strategic allocation uses mean-variance analysis, as do basic VaR measures. Choosing it confuses 'complement' with 'conflict'.
- **D:** True: with similar returns, the lower marginal VaR is the better use of the risk budget. Choosing it confuses stand-alone with marginal risk logic.

Q323

Schweser Module Quiz 88.2 Q2

VaR can be used to compose better guidelines for investment companies by:
I. relying less on notionals.
II. focusing more on overall risk.

A. I only.

B. II only.

C. Both I and II.

D. Neither I nor II.

Answer: C

Step by step

1. Notional and sensitivity limits have been the traditional focus but are cumbersome, easily evaded, and ignore risk variations and correlations.
2. VaR limits incorporate volatility, correlation and leverage, so guidelines rely less on notionals (I true).
3. VaR also gives a measure of the possible loss over a period, so guidelines focus on overall portfolio risk (II true).
4. Both are true: C.

Why C is correct: VaR guidelines replace position-by-position notional limits with a single measure of total risk that reflects correlations and leverage.

Why the others are wrong

- **A:** Accepts I but overlooks that shifting the focus to total portfolio risk is the very purpose of VaR guidelines.
- **B:** Accepts II but wrongly assumes notionals remain the main control; VaR reduces reliance on them.
- **D:** Implies VaR adds nothing to guidelines, contradicting the reading that notional limits are ineffective.

LO 88.h

Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.

Q324

AnalystPrep Q.2762

REBUILT — Options A/B contained unexplained numbers (3.30/6.20/4.6); stem lacked the z-value; rewrote options so each wrong choice reflects a realistic error and stated z = 2.33.

A pension fund needs to allocate $20 million to three asset classes. Use z = 2.33 for the 99% confidence level.

Asset class	Volatility	Allocation
US stocks	12%	30%
US bonds	5%	40%
Non-US stocks	18%	30%

The total volatility target for the fund is 10%. Which of the following correctly gives the total VaR budget for the fund and the undiversified VaR at the 99% confidence level?

A. Total VaR: $4.66 million; undiversified VaR: $5.44 million
B. Total VaR: $2.00 million; undiversified VaR: $2.20 million
C. Total VaR: $4.66 million; undiversified VaR: $5.13 million
D. Total VaR: $3.29 million; undiversified VaR: $3.62 million

Answer: C

Step by step

1. Total VaR budget $= z\sigma_P V = 2.33 \times 0.10 \times 20 = 4.66$ million.
2. Stand-alone VaRs: US stocks $2.33 \times 0.12 \times 0.30 \times 20 = 1.677$; US bonds $2.33 \times 0.05 \times 0.40 \times 20 = 0.932$; non-US stocks $2.33 \times 0.18 \times 0.30 \times 20 = 2.516$.
3. Undiversified VaR is the sum of the stand-alone VaRs: $1.677 + 0.932 + 2.516 = 5.13$ million (equivalently $2.33 \times 0.11 \times 20$, where 11% is the weighted-average volatility).
4. The undiversified figure exceeds the total budget because the asset classes are not perfectly correlated.

Why C is correct: The fund-level budget uses the 10% target volatility, while undiversified VaR sums the weighted stand-alone VaRs (perfect correlation), so it is larger than the diversified budget.

Why the others are wrong

- **A:** Total is right, but undiversified VaR used equal weights (average volatility 11.67%) instead of the 30/40/30 allocation: $2.33 \times 0.1167 \times 20 = 5.44$.
- **B:** Omitted the z-value: $0.10 \times 20 = 2.00$ and $0.11 \times 20 = 2.20$.
- **D:** Used z = 1.645 (95% confidence) instead of 2.33: $1.645 \times 0.10 \times 20 = 3.29$ and $1.645 \times 0.11 \times 20 = 3.62$.

Q325

BT answers P2.T9.21.7.1 · BT notes P2.T9.21.7.1

REBUILT — OCR-garbled option D and table alignment; reworded option D and kept the data as reconstructed (values verified against the correlation matrix).

The State Public Employee Retirement Fund (SPERF) needs to allocate $1.0 billion to four asset classes. The only criterion is to minimize portfolio volatility. The allocation below provides a risk budget based on a 95% VaR, assuming multivariate normal returns (VaR figures in $ millions).

Asset	Expected return	Volatility	Corr 1	Corr 2	Corr 3	Corr 4	Allocation	Principal ($MM)	95% VaR b
Technology (1)	13.0%	25.0%	1.00	|	| 20.0%	$200.0	$82.2		
REITs (2)	7.0%	15.0%	0.30	1.00	|	33.0%	$330.0	$81.4	
Large Value (3)	9.0%	17.0%	-0.10	0.50	1.00	| 37.0%	$370.0	$103.5	
International (4)	11.0%	16.0%	0.20	0.60	0.70	1.00	10.0%	$100.0	$26.3
Portfolio	12.6%	|	|	| 100.0%	$1,000.0	$206.8			

The $200.0 million Technology principal will be divided equally between two active managers, each running a portfolio with a volatility of 35.0%, and the correlation between them is zero. Which of the following statements about this risk budget is TRUE?

A. The undiversified VaR is $206.8 million, but the fund's actual VaR budget is $293.4 million.

B. The 1.0 entries on the diagonal of the correlation matrix are suspicious and should be checked for errors.

C. SPERF can allocate a risk budget of approximately $58.0 million to each of the two Technology managers.

D. The approach is weak because expected returns drove the allocation and the benefits of diversification are not captured in the budget.

Answer: C

Step by step

1. Technology budget $= 1.645 \times 0.25 \times 200 = 82.2$ million; each manager's budget x must satisfy the two-asset VaR formula.
2. With $\rho = 0$: $\sqrt{x^2 + x^2} = \sqrt{2}\,x = 82.2$, so $x = 82.2/1.414 = 58.1$ million (about $58.0 million).
3. Check: each manager's stand-alone VaR $= 1.645 \times 0.35 \times 100 = 57.6$; combined $\sqrt{2} \times 57.6 = 81.4 \approx 82.2$.
4. Undiversified VaR $= 82.2 + 81.4 + 103.5 + 26.3 = 293.4$; the diversified portfolio VaR (the budget) is 206.8.

Why C is correct: Because the two managers are uncorrelated, their VaRs combine by the square root of the sum of squares, so each can take about $82.2/\sqrt{2} \approx 58$ million.

Why the others are wrong

- **A:** Reverses the two figures: the sum of the stand-alone VaRs (293.4) is the undiversified VaR, and the diversified portfolio VaR (206.8) is the budget.
- **B:** Every asset has a correlation of exactly 1.0 with itself, so the diagonal is correct.
- **D:** The allocation minimized volatility without using expected returns, and the portfolio VaR of 206.8 is below the 293.4 sum precisely because diversification is included.

Q326

AP Mock 1 Q62

The BC&C pension fund has 60% (about USD 21 billion) invested in equities. It measures absolute risk with a 95% one-year VaR of USD 3.4545 billion, assuming a normal distribution and 10% annual volatility. The fund wishes to allocate this risk to two equity managers, each with the same VaR budget. The correlation between them is 0.5. The VaR budget for each manager is closest to:

A. USD 1.9945 billion.

B. USD 3.4545 billion.

C. USD 2.4427 billion.

D. USD 1.0731 billion.

Answer: A

Step by step

1. Check: $1.645 \times 0.10 \times 21 = 3.4545$ billion total VaR.
2. With equal budgets x: $\text{VaR}_P^2 = x^2 + x^2 + 2(0.5)x^2 = 3x^2$.
3. $x = 3.4545/\sqrt{3} = 1.9945$ billion.

Why A is correct: With equal budgets and $\rho = 0.5$, the portfolio VaR equals $x\sqrt{3}$, so each budget is the total divided by $\sqrt{3}$.

Why the others are wrong

- **B:** This is the total fund VaR, not the individual budget.
- **C:** Assumes zero correlation: $3.4545/\sqrt{2} = 2.4427$.
- **D:** Lies below even the perfect-correlation budget ($3.4545/2 = 1.7273$); it would require a correlation above 1, so it cannot be right.

Q327 AP Mock 3 Q5

A pension fund wants to allocate USD 400 million across a pool of active managers so as to maximise the information ratio of the fund, subject to an overall tracking error volatility (TEV) of 4%. Any balance not allocated to active managers is placed in an index fund with zero tracking error.

Item	TEV	Information ratio
Manager 1	8.0%	0.48
Manager 2	6.0%	0.36
Index fund	0.0%	0.00

Assuming the managers' excess returns are independent of each other, the optimal allocation to the index fund is equal to:

A. USD 80 million.
B. USD 160 million.
C. USD 320 million.
D. USD 0.

Answer: A

Step by step

1. Portfolio IR $= \sqrt{0.48^2 + 0.36^2} = 0.60$.
2. Optimal weight $w_i = (\text{IR}_i/\text{IR}_P)(\text{TEV}_P/\text{TEV}_i)$: $w_1 = (0.48/0.60)(4/8) = 40\%$; $w_2 = (0.36/0.60)(4/6) = 40\%$.
3. Check: TEV $= \sqrt{(0.4 \times 8\%)^2 + (0.4 \times 6\%)^2} = 4\%$.
4. Index weight $= 1 - 0.8 = 20\%$, i.e. $0.20 \times 400 = 80$ million.

Why A is correct: The weights on the managers need not sum to 100%; the remainder (20%) goes to the zero-IR index fund.

Why the others are wrong

- **B:** This is the allocation to each manager (40% of 400), not to the index.
- **C:** This is the total allocated to the two managers (80% of 400), not the index remainder.
- **D:** Assumes the managers absorb all the capital; the weights sum to only 80%.

Q328 AnalystPrep Q.2503

REBUILT — Distractors were arbitrary (e.g. $240 million, $450 million); replaced with options produced by realistic formula errors.

A pension fund wants to allocate $300 million to a pool of active managers so as to maximize the information ratio of the fund subject to an overall tracking error volatility (TEV) of 4%.

Item	TEV	Information ratio
Manager 1	6.0%	0.60
Manager 2	8.0%	0.40
Index	0.0%	0.00
Portfolio	4.0%	0.72

Assuming the excess returns of the managers are independent of each other, the optimal allocation to Manager 1 is closest to:

A. $166.67 million
B. $250.00 million
C. $375.00 million
D. $83.33 million

Answer: A

Step by step

1. $w_1 = (\text{IR}_1/\text{IR}_P)(\text{TEV}_P/\text{TEV}_1) = (0.60/0.72)(4\%/6\%)$.
2. $= 0.8333 \times 0.6667 = 0.5556$.
3. Allocation $= 0.5556 \times 300 = 166.67$ million.

Why A is correct: The optimal weight scales the manager's IR relative to the portfolio IR and the portfolio TEV relative to the manager's TEV.

Why the others are wrong

- **B:** Omitted the TEV ratio: $0.8333 \times 300 = 250$.
- **C:** Inverted the TEV ratio (used 6%/4%): $0.8333 \times 1.5 \times 300 = 375$.
- **D:** This is Manager 2's allocation, not Manager 1's.

Q329

AnalystPrep Q.2504

REBUILT — Distractors were arbitrary (e.g. $80,000,000, $33,330,000); replaced with options produced by realistic formula errors.

A pension fund wants to allocate $300 million to a pool of active managers so as to maximize the information ratio of the fund subject to an overall tracking error volatility (TEV) of 4%.

Item	TEV	Information ratio
Manager 1	6.0%	0.60
Manager 2	8.0%	0.40
Index	0.0%	0.00
Portfolio	4.0%	0.72

Assuming the excess returns of the managers are independent of each other, the optimal allocation to Manager 2 is closest to:

A. $166.67 million
B. $83.33 million
C. $270.00 million
D. $111.11 million

Answer: B

Step by step

1. $w_2 = (\text{IR}_2/\text{IR}_P)(\text{TEV}_P/\text{TEV}_2) = (0.40/0.72)(4\%/8\%)$.
2. $= 0.5556 \times 0.5 = 0.2778$.
3. Allocation $= 0.2778 \times 300 = 83.33$ million.

Why B is correct: Manager 2 has a lower IR and a higher TEV than Manager 1, so it receives a smaller weight (27.8% versus 55.6%).

Why the others are wrong

- **A:** This is Manager 1's allocation (wrong manager).
- **C:** Inverted the IR ratio (0.72/0.40) times the TEV ratio: $1.8 \times 0.5 \times 300 = 270$.
- **D:** Used Manager 1's TEV (6%) in the TEV ratio: $0.5556 \times (4/6) \times 300 = 111.11$.

Q330

AnalystPrep Q.2763

REBUILT — Original options were a giveaway (only A did not sum to 100%) and omitted the benchmark; rebuilt to ask for all three weights.

Information about two portfolio managers and the overall portfolio:

Name	Tracking error volatility (TEV)	Information ratio
Manager 1	5%	0.5
Manager 2	4%	0.3
Portfolio	3%	0.6

Assuming the excess returns of the two managers are independent of each other, which of the following is the optimal allocation to Manager 1, Manager 2 and the benchmark (index), respectively?

A. 50.0%, 37.5%, 12.5%
B. 37.5%, 50.0%, 12.5%
C. 57.1%, 42.9%, 0.0%
D. 30.0%, 22.5%, 47.5%

Answer: A

Step by step

1. Use the stated portfolio IR of 0.6: $w_i = (\text{IR}_i/\text{IR}_P)(\text{TEV}_P/\text{TEV}_i)$.
2. $w_1 = (0.5/0.6)(3/5) = 0.50$; $w_2 = (0.3/0.6)(3/4) = 0.375$.
3. Benchmark $= 1 - 0.50 - 0.375 = 12.5\%$ (its IR is zero).

Why A is correct: The optimal manager weights come directly from the formula; they need not sum to 1 and the residual goes to the benchmark.

Why the others are wrong

- **B:** Swapped the managers' weights.
- **C:** Rescaled the two weights to sum to 100% (0.5/0.875, 0.375/0.875), leaving nothing for the benchmark.
- **D:** Dropped the division by IR_P: $0.5 \times 0.6 = 0.30$ and $0.3 \times 0.75 = 0.225$.

Q331 AnalystPrep Q.3029

REBUILT — Options were arbitrary numbers; rebuilt with distractors from realistic errors and stated the confidence-level z-value (2.33) explicitly in the stem.

A pension fund has allocated $650 million in U.S. stocks to two active managers, each receiving $325 million. The two portfolios each have a volatility of 15.8% and a correlation of 0.81 with each other. Using z = 2.33, the total VaR (risk budget) for the $650 million is closest to:

A. $239.29 million
B. $169.20 million
C. $227.64 million
D. $119.65 million

Answer: C

Step by step

1. Each manager's VaR $= 2.33 \times 0.158 \times 325 = 119.65$ million.
2. $\text{VaR}_P = \sqrt{119.65^2 + 119.65^2 + 2(0.81)(119.65)(119.65)} = 119.65\sqrt{3.62}$.
3. $= 119.65 \times 1.9026 = 227.64$ million.

Why C is correct: Because the correlation is below 1, diversification makes total VaR less than the sum of the individual VaRs.

Why the others are wrong

- **A:** Simply added the two VaRs (assumes correlation of 1): $2 \times 119.65 = 239.29$.
- **B:** Assumed zero correlation: $119.65\sqrt{2} = 169.20$.
- **D:** This is the VaR of only one manager's $325 million.

Q332 AnalystPrep Q.3167

REBUILT — Stem data were inconsistent (95% VaR of $1.4bn on $5.6bn NAV implies 15.2% volatility, not 12%); corrected volatility to 15.2% and replaced arbitrary distractors.

A large cap equity fund has a net asset value (NAV) of $5.6 billion and is fully invested in equities. Returns are normally distributed with annual volatility of 15.2%. The fund measures absolute risk with a 95%, one-year VaR of $1.4 billion (z = 1.645). The fund wants to allocate this risk to two equity managers with equal VaR budgets. If the correlation between the managers' returns is 1.0, the VaR budget for each manager should be:

A. $0.35 billion
B. $0.70 billion
C. $0.81 billion
D. $0.99 billion

Answer: B

Step by step

1. Check: $1.645 \times 0.152 \times 5.6 = 1.40$ billion.
2. With equal budgets x and $\rho = 1$: $\sqrt{x^2 + x^2 + 2x^2} = 2x = 1.4$.
3. $x = 0.70$ billion.

Why B is correct: With perfect correlation there is no diversification, so the individual budgets simply add up to the total VaR.

Why the others are wrong

- **A:** Divided by 4 (1.4/4), as if the variances rather than the VaRs added.
- **C:** Assumed correlation 0.5: $1.4/\sqrt{3} = 0.81$.
- **D:** Assumed zero correlation: $1.4/\sqrt{2} = 0.99$.

Q333 AnalystPrep Q.4707

REBUILT — Original stem was ambiguous about what the 10.1% refers to and had an unexplained distractor; restated as a fund-level VaR budget split equally and replaced distractor D.

A pension fund allocates $4,000 million equally to 4 active managers. The fund's risk budget is set from a return volatility target of 10.1% for the total portfolio. Assume normally distributed returns, a 95% confidence level (use z = 1.65), and ignore expected returns and diversification effects across managers. If the fund-level VaR budget is divided equally among the managers, what is the VaR risk budget for each manager?

A. $404.00 million
B. $166.65 million

C. $666.60 million
D. $234.93 million

Answer: B
Step by step
1. Fund-level VaR budget: 1.65 × 10.1% × $4,000 = $666.60 million.
2. Divide equally across 4 managers: 666.60/4 = $166.65 million.
3. Equivalently, each manager gets \$1,000 million, so 1.65 × 0.101 × 1,000 = 166.65.

Why B is correct: Risk budgeting is top-down: convert the volatility target into a VaR (z times sigma times assets) and then allocate that VaR across managers; with equal shares each receives one quarter.

Why the others are wrong
- **A:** $404 million is 10.1% × 4,000: it is the one-sigma loss, omitting the 1.65 multiplier and the division by 4.
- **C:** $666.60 million is the total fund VaR budget; it was not divided among the 4 managers.
- **D:** $234.93 million uses the 99% deviate 2.326 (2.326 × 0.101 × 1,000) instead of 1.65 for 95%.

Q334

AnalystPrep Q.4709 · AnalystPrep Q.4708

REBUILT — Merged the capital-allocation item (Q.4708, a trivial first step) into the risk-budget item; clarified that risk budget means stand-alone VaR and replaced unexplained distractors.

A company allocates $6,200 million to two asset classes, machinery and vehicles.

Asset	Weight	Volatility
Machinery	68%	14.2%
Vehicles	32%	18.8%

Using $\alpha = 1.65$ and ignoring diversification, what is the stand-alone VaR risk budget (in $ millions) for each asset class?

A. Machinery: $4,216; Vehicles: $1,984
B. Machinery: $988; Vehicles: $615
C. Machinery: $1,453; Vehicles: $1,923
D. Machinery: $465; Vehicles: $1,308

Answer: B
Step by step
1. Capital allocation: Machinery 0.68 × 6,200 = $4,216m; Vehicles 0.32 × 6,200 = $1,984m.
2. Machinery risk budget: 1.65 × 0.142 × 4,216 = $987.8m.
3. Vehicles risk budget: 1.65 × 0.188 × 1,984 = $615.4m.

Why B is correct: The risk budget for each asset class is $\alpha\sigma W$, using the dollar amount allocated to that class (weight times total), not the total fund value.

Why the others are wrong
- **A:** These are the capital allocations (weight times $6,200m); the VaR step ($\alpha\sigma$) was never applied.
- **C:** Applied $\alpha\sigma$ to the full $6,200m for each asset (1.65 x 14.2% x 6,200 = 1,453; 1.65 x 18.8% x 6,200 = 1,923), ignoring the weights.
- **D:** Swapped the weights between the assets (machinery on $1,984m, vehicles on $4,216m).

Q335

BT answers P2.T8.11.1 · BT notes P2.T9.11.1

REBUILT — Original distractors (3.7, 5.2) did not correspond to identifiable errors; replaced with options from realistic mistakes. Stem/answer unchanged.

A pension fund has $28.0 billion invested in an equity portfolio with a volatility of 14.0% and normally distributed returns. The fund measures absolute risk with a 95.0% one-year VaR (use z = 1.645). The fund wants to allocate this VaR budget between two managers, assigning each an identical VaR budget. If the correlation between the managers is 0.30, what is the VaR budget for each manager?

A. $3.2 billion
B. $4.0 billion
C. $4.6 billion
D. $6.4 billion

Answer: B

Step by step

1. Total VaR $= 28.0 \times 14.0\% \times 1.645 = \$6.448$ billion.
2. With equal VaRs V: $\text{VaR}_P^2 = V^2 + V^2 + 2(0.30)V^2 = 2.6V^2$.
3. $V = 6.448/\sqrt{2.6} = 6.448/1.6125 = \$3.999$ billion, about \$4.0 billion.

Why B is correct: Component VaRs combine like volatilities: portfolio VaR $= V\sqrt{2 + 2\rho}$, so each manager's budget is the total divided by $\sqrt{2.6}$, which exceeds half the total because of diversification.

Why the others are wrong

- **A:** \$3.2 billion is total/2, which assumes perfect correlation ($\rho = 1$) and so ignores diversification.
- **C:** \$4.6 billion is total/$\sqrt{2}$, treating the managers as uncorrelated ($\rho = 0$).
- **D:** \$6.4 billion is the fund's total VaR budget, not divided between the managers.

Q336

BT answers P2.T8.14.3 · BT notes P2.T9.14.3

REBUILT — Replaced the giveaway/unexplained distractors (zero, 44.2%, 56.9%) with options from specific formula errors; notation aligned to Schweser (TEV, IR).

When a fund maximizes its portfolio information ratio subject to a tracking error volatility (TEV) constraint, and manager excess returns are independent, the optimal weight of manager i is $x_i = \frac{IR_i}{IR_P} \times \frac{TEV_P}{TEV_i}$, so relative risk budgets ($x_i \times TEV_i$) are proportional to information ratios. The portfolio's maximum information ratio is 0.82 and its TEV constraint is 5.0%. What is the optimal allocation to a manager with an information ratio of 0.50 and a TEV of 8.0%?

A. 31.3%
B. 38.1%
C. 61.0%
D. 97.6%

Answer: B

Step by step

1. $x_i = (IR_i/IR_P) \times (TEV_P/TEV_i)$.
2. $IR_i/IR_P = 0.50/0.82 = 0.6098$.
3. $TEV_P/TEV_i = 5.0\%/8.0\% = 0.625$.
4. $x_i = 0.6098 \times 0.625 = 0.3811 = 38.1\%$.

Why B is correct: The manager's share of the risk budget equals IR_i/IR_P of the portfolio TEV; converting that risk share into a weight requires dividing by the manager's own TEV.

Why the others are wrong

- **A:** 31.3% is $0.50 \times 5/8$: omitted the division by the portfolio IR of 0.82.
- **C:** 61.0% is IR_i/IR_P alone; forgot to scale by TEV_P/TEV_i.
- **D:** 97.6% inverted the TEV ratio (8/5 instead of 5/8): 0.6098×1.6.

Q337

BT answers P2.T9.21.8.3 · BT notes P2.T9.21.8.3

REBUILT — Source exhibit was missing/garbled and option B contradicted the 3.0% TEV in the stem; rebuilt a clean self-contained exhibit and corrected the options.

A pension fund sub-allocates assets to two active managers whose excess returns are independent. It maximizes the portfolio information ratio subject to an overall TEV of 3.0%. Any remainder is invested in the benchmark index. The proposed allocation is:

Holding	TEV	IR	Weight	Relative risk budget (weight x TEV)
Manager 1	5.0%	0.63	36.0%	1.80%
Manager 2	7.0%	0.84	34.3%	2.40%
Index	0.0%	0.00	29.7%	0.00%
Portfolio	3.0%	1.05	100.0%	

Is this optimization correct (best answer)?

A. No, the index allocation should be zero, otherwise the information ratio is not maximized
B. No, the overall TEV of 3.0% cannot be less than each manager's TEV (5.0% and 7.0%)
C. Yes, because each weight satisfies the optimal condition so that relative risk budgets are proportional to the information ratios
D. Yes, because the index is allocated a relative risk budget of zero

Answer: C

Step by step

1. Portfolio IR for independent managers: $\sqrt{0.63^2 + 0.84^2} = 1.05$, as shown.
2. Optimal weights: $x_1 = (0.63/1.05)(3.0/5.0) = 36.0\%$; $x_2 = (0.84/1.05)(3.0/7.0) = 34.3\%$; index = $1 - 0.360 - 0.343 = 29.7\%$.
3. Relative risk budgets 1.80% and 2.40% have ratio 0.75 = 0.63/0.84, proportional to the IRs.
4. Check TEV: $\sqrt{1.80^2 + 2.40^2} = 3.0\%$, meeting the constraint.

Why C is correct: The optimal allocation requires x_i*TEV_i to be proportional to IR_i; the weights satisfy this, and the unallocated 29.7% sits in the benchmark, which has IR = 0.

Why the others are wrong

- **A:** Weights need not sum to one; the remainder goes to the benchmark (IR = 0), and holding it is exactly what scales risk down to the 3.0% TEV constraint.
- **B:** Because weights are below 100% and the managers are independent (diversification), portfolio TEV (3.0%) is below each manager's stand-alone TEV; this is not a flaw.
- **D:** The benchmark always has a zero relative risk budget, but that does not demonstrate optimality; the proportionality of risk budgets to IRs is the test.

Q338

GARP PE2 Q20 · Mock 4 Q21

REBUILT — Original asked for the 'sum of risk budgets' with ambiguous meaning (diversified vs stand-alone) and options built on a rounded z-value that did not match the data; rewritten to ask for the sum of the asset-class risk budgets that must equal the fund VaR, with a stand-alone-sum distractor.

An investment fund uses risk budgeting with delta-normal VaR at the 99% confidence level (z = 2.326). The fund has EUR 100 million of principal and maintains a total portfolio volatility of 5% (annual). The fund allocates this risk across four asset classes, with the risk budget of each class defined as the amount of fund VaR attributable to it (taking diversification into account). Information on the asset classes:

Asset class	Weight	Volatility
European stocks	40%	9.25%
Non-European stocks	12%	11.91%
European bonds	22%	5.76%
Non-European bonds	26%	11.94%

The pairwise correlations between the asset classes are all between -0.07 and +0.02. What is the sum of the risk budgets allocated to the four asset classes?

A. EUR 5.00 million
B. EUR 8.23 million
C. EUR 11.63 million
D. EUR 22.10 million

Answer: C

Key correction: Source key (GARP PE2 Q20 / Mock 4 Q21) said EUR 22.12 million (sum of stand-alone VaRs); correct is EUR 11.63 million because risk budgets are allocated top-down from the diversified fund VaR (Schweser: the sum of stand-alone VaRs exceeds portfolio VaR, so budgeting must account for diversification), and the sum of the budgets equals the fund VaR.

Step by step

1. Total risk is set top-down by the fund volatility target: $\sigma_P = 5\%$.
2. Fund VaR budget = $z\,\sigma_P\,V = 2.326 \times 0.05 \times 100 =$ EUR 11.63 million.
3. Risk budgets (component/diversified VaRs) are a decomposition of this total, so they sum to EUR 11.63 million.
4. Check: the stand-alone VaRs are $2.326 \times 100 \times w_i\sigma_i = 8.61, 3.32, 2.95, 7.22$, summing to EUR 22.10 million. This exceeds the fund VaR because it ignores diversification (the correlations are near zero, so the portfolio volatility is far below the weighted-average volatility of 9.5%); the 5% target volatility gives the diversified fund VaR of EUR 11.63 million.

Why C is correct: Risk budgeting starts from the fund-level VaR implied by the volatility target and allocates it across asset classes with diversification considered, so the budgets must sum to $z\sigma_P V =$ EUR 11.63 million.

Why the others are wrong

- **A:** Uses $\sigma_P V = 5$ and forgets to multiply by the 99% z-value of 2.326.
- **B:** Uses the 95% z-value of 1.645 instead of 2.326: $1.645 \times 0.05 \times 100 = 8.23$.
- **D:** Sums the stand-alone VaRs $z\,w_i\sigma_i V$ of each class, ignoring diversification; this exceeds the fund VaR and is not an allocation of it.

Q339 Mock 3 Q15

A pension fund allocates its active risk budget across three external managers, each running an independent strategy benchmarked to the same index. The fund's chief investment officer gathers the following data:

Manager	Allocation	Tracking error
Manager A	50%	4.0%
Manager B	30%	6.0%
Manager C	20%	3.0%

Assuming the managers' active returns are uncorrelated with one another, what is the tracking error of the fund as a whole?

A. 2.76%
B. 4.33%
C. 4.40%
D. 7.81%

Answer: A

Step by step

1. Weighted tracking errors: $0.50 \times 4.0\% = 2.00\%$; $0.30 \times 6.0\% = 1.80\%$; $0.20 \times 3.0\% = 0.60\%$.
2. Active returns are uncorrelated, so variances add: $\text{TE}_P^2 = 2.00^2 + 1.80^2 + 0.60^2 = 4.00 + 3.24 + 0.36 = 7.60$.
3. $\text{TE}_P = \sqrt{7.60} = 2.76\%$.

Why A is correct: With zero correlation the portfolio tracking error is the square root of the sum of squared weighted tracking errors, $\sqrt{\sum(w_i \text{TE}_i)^2}$.

Why the others are wrong

- **B:** Takes the simple average of the three tracking errors, (4+6+3)/3 = 4.33%, ignoring both the weights and the diversification from zero correlation.
- **C:** Adds the weighted tracking errors linearly (2.00+1.80+0.60 = 4.40%), which assumes perfect correlation among managers.
- **D:** Combines the unweighted tracking errors in quadrature, $\sqrt{16 + 36 + 9} = 7.81\%$, forgetting to apply the allocations.

Q340 Schweser Module Quiz 88.2 Q3

REBUILT — Option formulas were lost to OCR; reconstructed the four formulas (correct one is the Jorion/Schweser optimal weight) with distractors that invert or mis-pair the IR and TEV terms.

In allocating across independent active managers, let IR_i and TEV_i be the information ratio and tracking error volatility of manager i, and IR_P and TEV_P those of the total portfolio. Which of the following gives the optimal weight x_i to allocate to manager i?

A. $x_i = \dfrac{\text{IR}_P}{\text{TEV}_P} \times \dfrac{\text{IR}_i}{\text{TEV}_i}$

B. $x_i = \dfrac{\text{TEV}_P}{\text{IR}_P} \times \dfrac{\text{TEV}_i}{\text{IR}_i}$

C. $x_i = \dfrac{\text{TEV}_P}{\text{IR}_P} \times \dfrac{\text{IR}_i}{\text{TEV}_i}$

D. $x_i = \dfrac{\text{IR}_P}{\text{TEV}_P} \times \dfrac{\text{TEV}_i}{\text{IR}_i}$

Answer: C

Step by step

1. For independent managers the optimal allocation equates each manager's contribution to the portfolio information ratio: weight rises with the manager's IR_i and falls with its tracking error TEV_i, so $x_i \propto \text{IR}_i/\text{TEV}_i$.
2. The scale factor ties the weights to the portfolio's risk budget and quality: TEV_P/IR_P.
3. Hence $x_i = (\text{TEV}_P/\text{IR}_P)\,(\text{IR}_i/\text{TEV}_i)$.

Why C is correct: A manager with a higher IR deserves more weight and a manager with a larger tracking error uses up more of the TEV budget per unit of weight, so $x_i = (\text{TEV}_P/\text{IR}_P)(\text{IR}_i/\text{TEV}_i)$.

Why the others are wrong

- **A:** Inverts the scale factor to IR_P/TEV_P; the portfolio-level term must be TEV_P/IR_P.
- **B:** Inverts the manager term to TEV_i/IR_i, which would give more weight to high-risk, low-IR managers.
- **D:** Inverts both ratios, giving exactly the reciprocal of the correct weight.

Reading 89. Portfolio Performance Evaluation

Zvi Bodie, Alex Kane, and Alan J. Marcus, Investments, 12th Edition (New York, NY: McGraw-Hill, 2020). Chapter 24.

LO	Learning objective	Qs
89.a	differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses.	15
89.b	describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant.	28
89.c	describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures.	7
89.d	determine the statistical significance of a performance measure using standard error and the t-statistic.	3
89.e	describe style analysis.	6
89.f	explain the difficulties in measuring the performance of actively managed portfolios.	4
89.g	describe performance manipulation and the problems associated with using conventional performance measures.	3
89.h	describe techniques to measure the market timing ability of fund managers with a regression and with a call option model and calculate a manager's return due to market timing.	7
89.i	describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution.	8

LO 89.a

Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses.

Q341

BT notes P2.T9.15.1 · BT answers P2.T8.15.1

REBUILT — Distractors had no identifiable error source; replaced them with results of realistic mistakes (geometric return, subtracting instead of adding the variance term, adding the full variance).

Thirty months ago, USD 100 was invested and has grown to USD 175.00 today. Monthly returns were normally distributed with a monthly standard deviation of 10.0%. Without the return data, and using the geometric (time-weighted) return, which of the following is the best estimate of the arithmetic average monthly return?

A. 1.38%
B. 1.88%
C. 2.38%
D. 2.88%

Answer: C

Key correction: Source key (BT notes, earlier version) said B; the later corrected version and this solution give C because the question asks for the arithmetic mean, not the geometric return.

Step by step

1. Geometric monthly return: $(175/100)^{1/30} - 1 = 1.8829\%$.
2. Volatility drag: $R_G \approx R_A - \frac{1}{2}\sigma^2$, so $R_A \approx R_G + \frac{1}{2}\sigma^2$.
3. $\frac{1}{2}\sigma^2 = 0.5 \times (0.10)^2 = 0.50\%$.
4. $R_A \approx 1.8829\% + 0.50\% = 2.38\%$.

Why C is correct: The geometric (time-weighted) mean is below the arithmetic mean by about half the variance, so the arithmetic mean is the geometric mean plus $\sigma^2/2$.

Why the others are wrong

- **A:** Subtracted $\sigma^2/2$ from the geometric return (1.88% - 0.50%) instead of adding it.
- **B:** This is just the geometric return; it ignores the volatility adjustment needed to get the arithmetic mean.
- **D:** Added the full variance (1.00%) rather than half the variance.

Q342

AP Mock 2 Q18

REBUILT — Original distractors had no identifiable error source; replaced with a mis-specified cash-flow IRR, the time-weighted return and the first-year holding period return.

You buy one share of a stock at USD 40 now. The stock pays an annual dividend of USD 4 per share. At the end of year 1, after receiving the dividend, you buy another share at USD 42. At the end of year 2, you receive the dividend

on both shares and sell both shares at USD 43 each. The dollar-weighted rate of return on the investment is closest to:

A. 9.5%
B. 13.0%
C. 13.4%
D. 15.0%

Answer: B

Step by step

1. Cash flows: t0 = -40; t1 = +4 (dividend) - 42 (purchase) = -38; t2 = 2×4 (dividends) + 2×43 (sales) = 94.
2. Dollar-weighted return is the IRR: $-40 - \frac{38}{1+r} + \frac{94}{(1+r)^2} = 0$.
3. Solving numerically gives $r = 12.99\% \approx 13.0\%$.

Why B is correct: The dollar-weighted return is the IRR of all net cash flows, so the larger second-year investment (two shares) is weighted more heavily.

Why the others are wrong

- **A:** Left out the USD 4 dividend at t1 (used flows -40, -42, +94), which understates the IRR.
- **C:** Time-weighted return: HPR1 = (42-40+4)/40 = 15.0%, HPR2 = (43+4-42)/42 = 11.9%, geometric mean 13.4%. Not the dollar-weighted return.
- **D:** First-year holding period return only; ignores the second period and the added share.

Q343

AP Mock 3 Q19

REBUILT — Option A was printed as 29.94% while the exact answer is 29.96%; corrected the rounding so the key matches the computation.

David Parker purchased 500 shares of Yamacha Petroleum at USD 60 per share on 1 January 2016, and another 500 shares at USD 75 on 1 January 2017. The stock paid a dividend of USD 5 per share on 31 December 2016 and again on 31 December 2017. On 31 December 2017 he sold all his shares at USD 90 each. The time-weighted rate of return per year on the investment is closest to:

A. 26.67%
B. 29.96%
C. 33.33%
D. 68.89%

Answer: B

Step by step

1. Year 1 HPR: $(75 - 60 + 5)/60 = 33.33\%$.
2. Year 2 HPR: $(90 - 75 + 5)/75 = 26.67\%$.
3. Geometric mean: $\sqrt{1.3333 \times 1.2667} - 1 = \sqrt{1.6889} - 1 = 29.96\%$.

Why B is correct: Time-weighted return chain-links per-share sub-period returns and takes the geometric mean, so the number of shares held (cash-flow size) does not matter.

Why the others are wrong

- **A:** Second-year return only.
- **C:** First-year return only.
- **D:** Cumulative two-year return (1.6889 - 1), not annualized by taking the square root.

Q344

AnalystPrep Q.2559

REBUILT — Distractors had no identifiable error source (the 2.5% and 10% were arbitrary); replaced with the dollar-weighted IRR, the no-dividend return and the first-period return.

An investor buys one share at t = 0 for USD 300. At t = 1 the investor buys another share at USD 320. At the end of year 2, both shares are sold for USD 330 each. At the end of years 1 and 2 the stock paid a USD 5.00 per share dividend. The time-weighted rate of return is closest to:

A. 4.9%
B. 5.9%
C. 6.5%
D. 8.3%

Answer: C

Step by step

1. Year 1: $(320 + 5 - 300)/300 = 8.333\%$.
2. Year 2 (2 shares): start value 640, end value $660 + 10 = 670$; return $30/640 = 4.6875\%$.
3. Geometric mean: $\sqrt{1.08333 \times 1.046875} - 1 = 6.49\% \approx 6.5\%$.

Why C is correct: Time-weighted return computes each sub-period return on the capital actually invested in that period and links them geometrically.

Why the others are wrong

- **A:** Ignored dividends: $\sqrt{(1 + 20/300)(1 + 20/640)} - 1 = 4.9\%$.
- **B:** This is the dollar-weighted return (IRR of -300, -315, +670), not the time-weighted return.
- **D:** Year 1 return only (25/300).

Q345

AnalystPrep Q.3032

REBUILT — Source computed the year-2 return as 12% but (2×120 + 6 - 220)/220 = 11.82%; the correct 12.41% was not an option, so options were rebuilt.

A stock sells for USD 100 and pays an annual dividend of USD 3.00. An investor buys one share, collects the USD 3.00 dividend at the end of year 1 and, instead of selling the share (price USD 110), buys a second share at USD 110. Both shares are held until the end of year 2, when he receives a USD 3.00 dividend per share and sells each share for USD 120. The geometric (time-weighted) average annual return is closest to:

A. 11.03%
B. 12.22%
C. 12.41%
D. 12.50%

Answer: C

Key correction: Source key (AnalystPrep) said 12.49% (B); that used a year-2 return of 12% instead of 11.82%. Correct is 12.41%.

Step by step

1. Year 1: $(110 + 3 - 100)/100 = 13.00\%$.
2. Year 2 (2 shares): start value 220; end value $2 \times 120 + 2 \times 3 = 246$; return $26/220 = 11.818\%$.
3. Geometric mean: $\sqrt{1.13 \times 1.11818} - 1 = 12.41\%$.

Why C is correct: Link the two sub-period returns (13.00% and 11.82%) and take the square root for the annual geometric mean.

Why the others are wrong

- **A:** Omitted the year-2 dividends: year-2 return $20/220 = 9.09\%$, giving 11.03%.
- **B:** Dollar-weighted return (IRR of -100, -107, +246), not time-weighted.
- **D:** Used 12.00% for year 2 instead of 11.82% (the source's arithmetic slip), giving $\sqrt{1.13 \times 1.12} - 1 = 12.50\%$.

Q346

AnalystPrep Q.3137

REBUILT — Distractors had no identifiable error source; replaced with the time-weighted return and IRRs that omit one of the dividends.

You buy one share of AFC Bank at the beginning of year 1 for USD 36. At the end of year 1 you receive a USD 2 dividend and buy one more share for USD 30. At the end of year 2 you receive total dividends of USD 4 (USD 2 per share) and sell both shares for USD 36.45 each. The dollar-weighted return on the investment is closest to:

A. 6.74%
B. 8.63%
C. 10.31%
D. 12.35%

Answer: D

Step by step

1. Cash flows: t0 = -36; t1 = +2 - 30 = -28; t2 = 4 + 2×36.45 = 76.90.
2. Solve $-36 - \frac{28}{1+r} + \frac{76.9}{(1+r)^2} = 0$.
3. $r = 12.35\%$.

Why D is correct: The dollar-weighted return is the IRR of the actual net cash flows.

Why the others are wrong

- **A:** Time-weighted return: HPR1 = (30-36+2)/36 = -11.1%, HPR2 = (76.9-60)/60 = 28.2%, geometric mean 6.74%.

- **B:** Left out the year-2 dividends (flows -36, -28, +72.90).
- **C:** Left out the year-1 dividend (flows -36, -30, +76.90).

Q347

AnalystPrep Q.3172

REBUILT — Source answer treated the end-of-year contribution as investment gain (36.17%), and its 19.14% option was assigned to the wrong timing; stem clarified and options rebuilt.

A trust account was worth USD 4,700,000 at the start of the year, before a contribution of USD 800,000. At the end of the year the account was worth USD 6,400,000 (after the contribution). The manager did not record when the contribution was made. What is the rate of return if the contribution was received at the very start of the year versus at the very end of the year?

A. Start: 19.15%; End: 16.36%
B. Start: 16.36%; End: 19.15%
C. Start: 16.36%; End: 36.17%
D. Start: 36.17%; End: 19.15%

Answer: B

Key correction: Source key (AnalystPrep) said D (end-of-year case 36.17%); that counts the contribution as investment gain. Correct is B.

Step by step

1. Start of year: capital invested all year = 4.7 + 0.8 = 5.5m; return $(6.4 - 5.5)/5.5 = 16.36\%$.
2. End of year: the contribution earned nothing, so investment gain = $6.4 - 0.8 - 4.7 = 0.9$m on 4.7m; return $0.9/4.7 = 19.15\%$.
3. The later the contribution, the less capital worked all year, so the same gain is a higher return.

Why B is correct: Money contributed earlier is exposed for longer, so the same dollar gain divided by a larger capital base gives a lower return; a late contribution leaves the base at 4.7m.

Why the others are wrong

- **A:** Timing reversed: start-of-year and end-of-year returns swapped.
- **C:** For the end case it treated the 0.8m contribution as investment gain: (6.4 - 4.7)/4.7 = 36.17%.
- **D:** Both cases wrong: 36.17% for the start case ignores the contribution in the base, and the 19.15% belongs to the end case but is paired with the wrong start figure.

Q348

BT answers P2.T8.15.2 · BT notes P2.T9.15.2

REBUILT — The two sources differ in option D; options were rebuilt so every distractor reflects an identifiable error.

You make an investment over two years. At T0 you buy one share at USD 10.00. At the end of year 1 (T1) the share pays a USD 2.00 dividend and, with the price at USD 11.00, you buy a second share at USD 11.00. At the end of year 2 (T2) the price is USD 18.00; you collect a USD 2.00 dividend per share and sell both shares, for total proceeds (including dividends) of USD 40.00. What are, respectively, the time-weighted (geometric) and dollar-weighted (internal) rates of return?

A. 60.0% (time) and 53.7% (dollar)
B. 55.9% (time) and 60.0% (dollar)
C. 53.7% (time) and 60.0% (dollar)
D. 53.7% (time) and 52.4% (dollar)

Answer: C

Step by step

1. Year 1 return: $(11 + 2 - 10)/10 = 30.0\%$.
2. Year 2 return: $(18 + 2 - 11)/11 = 81.82\%$.
3. Time-weighted: $\sqrt{1.30 \times 1.8182} - 1 = 53.74\%$.
4. Dollar-weighted: flows -10, +2 - 11 = -9, +40. $-10 - 9/(1 + r) + 40/(1 + r)^2 = 0 \Rightarrow 10a^2 + 9a - 40 = 0$, $a = 1.6, r = 60.0\%$.

Why C is correct: Time-weighted return links per-share period returns, while the IRR weights the larger year-2 investment (two shares) and the very strong year-2 performance more, so it is higher here.

Why the others are wrong

- **A:** Labels swapped.
- **B:** Time-weighted figure is the arithmetic mean (30% + 81.82%)/2 = 55.9% rather than the geometric mean.

- **D:** Dollar-weighted figure omits the USD 2 year-1 dividend (flows -10, -11, +40 give 52.4%).

Q349

BT answers P2.T8.15.3 · BT notes P2.T9.15.3

REBUILT — Merged two identical copies; replaced distractors with ones traceable to specific errors (simple annualised gain, arithmetic mean, swapped measures).

Shares of XYZCorp pay a $2 dividend at the end of every year on December 31. An investor buys two shares on January 1 at $20 each, sells one share for $22 exactly one year later (the next January 1), and sells the second share one year after that for $19. What are, respectively, the dollar-weighted and time-weighted annual rates of return on the 2-year investment?

A. 8.8% (dollar) and 7.0% (time)
B. 11.9% (dollar) and 7.7% (time)
C. 7.0% (dollar) and 11.9% (time)
D. 11.9% (dollar) and 7.0% (time)

Answer: D

Key correction: Source key D agrees; some source solutions note that the textbook used an arithmetic average for the time-weighted return (about 7.7%), which is not the geometric time-weighted return.

Step by step

1. Cash flows: $t_0 = -40$; $t_1 = 2 \times 2 + 22 = +26$ (dividends on two shares plus sale of one share); $t_2 = 2 + 19 = +21$.
2. Dollar-weighted return is the IRR: $-40 + 26/(1+r) + 21/(1+r)^2 = 0 \Rightarrow r = 11.91\%$.
3. Year 1 HPR: $(22 - 20 + 2)/20 = 20.00\%$. Year 2 HPR: $(19 - 22 + 2)/22 = -4.55\%$.
4. Time-weighted return (geometric): $\sqrt{1.20 \times 0.9545} - 1 = 7.03\%$.

Why D is correct: The dollar-weighted return is the IRR of the actual cash flows (which gives more weight to the larger first-year balance, hence higher), while the time-weighted return geometrically links the sub-period HPRs.

Why the others are wrong

- **A:** Dollar return is wrong: 8.8% is total profit / initial investment $(47 - 40)/40 = 17.5\%$ divided by 2, ignoring the timing of cash flows (no discounting).
- **B:** Dollar return is right but time-weighted uses the arithmetic mean $(20\% - 4.55\%)/2 = 7.7\%$ instead of the geometric mean.
- **C:** The two measures are swapped.

Q350

BT answers P2.T9.21.10.1 · BT notes P2.T9.21.10.1

REBUILT — Merged two copies; supplied the missing cash-flow table from the stem and replaced unrecoverable (OCR-damaged) distractors.

Bertha buys one share at Time 0 for $100. At Time 1 the price has fallen to $80 and the share paid a $5.00 dividend; she buys a second share at $80. At Time 2 she receives $8.00 in dividends on two shares, the price is $92, and she buys a third share. At Time 3 the price is $106; she receives $13.80 in dividends on three shares and sells all three shares for $318.00.

Time	0	1	2	3
Share price	$100.00	$80.00	$92.00	$106.00
Dividend per share	| $5.00	$4.00	$4.60	
Net cash flow	-$100.00	-$75.00	-$84.00	+$331.80

What are, respectively, the time-weighted and dollar-weighted annual returns over the three-year period?

A. 7.0% (time) and 12.5% (dollar)
B. 8.4% (time) and 12.5% (dollar)
C. 7.0% (time) and 7.9% (dollar)
D. 12.5% (time) and 7.0% (dollar)

Answer: A

Step by step

1. HPR1 = $(80+5-100)/100 = -15.00\%$; HPR2 = $(92+4-80)/80 = 20.00\%$; HPR3 = $(106+4.60-92)/92 = 20.22\%$.
2. Time-weighted: $[(0.85)(1.20)(1.2022)]^{1/3} - 1 = 7.03\%$.
3. Net flows: $t_1 = 5 - 80 = -75$; $t_2 = 8 - 92 = -84$; $t_3 = 13.80 + 318 = 331.80$.
4. Dollar-weighted: solve $-100 - 75/(1+r) - 84/(1+r)^2 + 331.80/(1+r)^3 = 0 \Rightarrow r = 12.50\%$.

Why A is correct: Time-weighted return geometrically links HPRs and is unaffected by the size/timing of pur-

chases; the dollar-weighted return is the IRR of the net flows and here is higher because more money was invested after the price dip.

Why the others are wrong

- **B:** Time-weighted return uses the arithmetic mean of the HPRs $(-15 + 20 + 20.22)/3 = 8.4\%$ instead of the geometric mean.
- **C:** Dollar-weighted return ignores the dividends (IRR of -100, -80, -92, +318 is 7.9%).
- **D:** The two measures are swapped.

Q351

Mock 1 Q58

An investor purchases a security for USD 100. Six months later the security is priced at USD 108 and pays a dividend of USD 4, which the investor does not reinvest. At the end of twelve months the security is priced at USD 112. Assuming dividends are not reinvested, what is the time-weighted rate of return (TWRR) on this investment over the 12-month holding period?

A. 12.00%
B. 15.70%
C. 16.00%
D. 16.15%

Answer: D

Step by step

1. Split the year at the dividend date (six months).
2. HPR1 $= (108 + 4 - 100)/100 = 12.00\%$.
3. HPR2 $= (112 - 108)/108 = 3.70\%$.
4. TWRR $= 1.12 \times 1.037037 - 1 = 16.15\%$.

Why D is correct: TWRR compounds (geometrically links) the sub-period returns, so the 12% and 3.70% are multiplied, not added.

Why the others are wrong

- **A:** Only the first sub-period return; the second half is omitted.
- **B:** Added the HPRs (12.00% + 3.70%) instead of compounding.
- **C:** Treated the whole year as one period: $(112+4-100)/100 = 16.00\%$, ignoring that the dividend was received at mid-year and the sub-period base changed.

Q352

Mock 1 Q59

An investor purchases one share of a security for USD 100. At the end of Year 1 the share is worth USD 110, and the investor purchases a second share at that price. At the end of Year 2 each share is worth USD 99 and the investor sells both. There are no dividends. What is the dollar-weighted rate of return on this investment over the holding period?

A. -5.71%
B. -3.92%
C. -1.00%
D. 0.00%

Answer: B

Step by step

1. Cash flows: $t_0 = -100$; $t_1 = -110$; $t_2 = +2 \times 99 = +198$.
2. Solve $-100 - 110/(1 + r) + 198/(1 + r)^2 = 0$.
3. Result: $r = -3.92\%$ per year.

Why B is correct: The dollar-weighted return is the IRR of the actual cash flows; the larger investment was made just before the price fell, so it is worse than the time-weighted return.

Why the others are wrong

- **A:** Simple loss over total money invested: $(198 - 210)/210 = -5.71\%$ with no annualisation/IRR.
- **C:** This is the time-weighted return $(1.10)(0.90) - 1 = -1.00\%$.
- **D:** Arithmetic average of +10% and -10%.

Q353

Mock 2 Q12

REBUILT — Option A was ambiguously worded; reworded so that calendar month-end valuation is clearly the unnecessary step.

The time-weighted rate of return measures compound growth over a specified time horizon. Which of the following actions is NOT required to compute the annual time-weighted return for an investment?

A. Value the portfolio at every month-end, regardless of whether any cash flows occurred.
B. Compute the product of (1 + holding period return) across the sub-periods to obtain the total return for the entire measurement period.
C. Form sub-periods that correspond to the dates of deposits and withdrawals.
D. Compute the holding period return of the portfolio for each sub-period.

Answer: A

Step by step

1. Time-weighted return requires: (1) break the horizon at each external cash flow, (2) value the portfolio at those dates, (3) compute each sub-period HPR, (4) chain-link the HPRs.
2. Valuation dates are driven by external cash flows, not by the calendar; month-end valuations are not required.

Why A is correct: Sub-periods are determined by deposits and withdrawals, so valuing at every month-end irrespective of cash flows is unnecessary.

Why the others are wrong

- **B:** Required: geometric linking of HPRs is the final step.
- **C:** Required: sub-periods must start/end at cash flow dates.
- **D:** Required: each sub-period HPR is the building block.

Q354

Schweser Module Quiz 89.1 Q1

REBUILT — Exhibit text was missing from the item; restored the setup from the reading and replaced distractors with traceable errors.

Assume you purchase one share of stock for $50 at time t = 0 and another share at $65 at time t = 1. At the end of Year 1 and Year 2 the stock pays a $2.00 dividend per share. At the end of Year 2 you sell both shares for $70 each. What is the dollar-weighted rate of return on the investment?

A. 10.77%
B. 14.51%
C. 16.73%
D. 18.02%

Answer: D

Step by step

1. $t_0 = -50$.
2. $t_1 = +2$ (dividend) $- 65 = -63$.
3. $t_2 = 2 \times 2$ (dividends) $+ 2 \times 70 = +144$.
4. Solve $-50 - 63/(1 + r) + 144/(1 + r)^2 = 0 \Rightarrow r = 18.02\%$ (financial calculator IRR).

Why D is correct: The dollar-weighted return is the IRR of all actual cash flows including interim dividends and purchases.

Why the others are wrong

- **A:** This is the second-year HPR $(140 + 4)/130 - 1 = 10.77\%$, not an IRR.
- **B:** IRR with both dividends ignored (flows -50, -65, +140).
- **C:** IRR ignoring the Year 1 dividend (flows -50, -65, +144).

Q355

Schweser Module Quiz 89.1 Q2

REBUILT — Exhibit text was missing from the item; restored the setup from the reading and replaced distractors with traceable errors.

Assume you purchase one share of stock for $50 at time t = 0 and another share at $65 at time t = 1. At the end of Year 1 and Year 2 the stock pays a $2.00 dividend per share. At the end of Year 2 you sell both shares for $70 each. What is the time-weighted rate of return on the investment?

A. 18.02%
B. 18.32%
C. 21.83%
D. 22.39%

Answer: C

Step by step

1. HPR1 $= (65 + 2)/50 - 1 = 34.00\%$ (one share held in Year 1).
2. Value at $t = 1$ after buying second share: $2 \times 65 = 130$. HPR2 $= (140 + 4)/130 - 1 = 10.77\%$.
3. TWR $= [(1.34)(1.1077)]^{1/2} - 1 = 21.83\%$.

Why C is correct: The time-weighted return geometrically links the annual HPRs, with the Year 2 base of \$130 (two shares) after the second purchase.

Why the others are wrong

- **A:** This is the dollar-weighted return (IRR).
- **B:** Dividends ignored in both HPRs: $[(1.30)(140/130)]^{1/2} - 1 = 18.32\%$.
- **D:** Arithmetic mean of the HPRs $(34\% + 10.77\%)/2 = 22.39\%$ instead of the geometric mean.

LO 89.b

Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant.

Q356

AnalystPrep Q.3030

REBUILT — Original was a vague qualitative item with irrelevant firm/'green zone' context and a loosely worded key; replaced by a numeric Treynor-versus-Sharpe ranking item testing when each measure is relevant.

Over the same period the risk-free rate was 3% and the market return was 8%. Two funds are compared:

Fund	Return	Beta	Standard deviation
X	11%	0.80	20%
Y	10%	1.00	12%

Fund X ranks first on the Treynor measure but second on the Sharpe ratio. Which statement best explains the reversal?

A. Fund X is better diversified than Fund Y, which lowers its Sharpe ratio.

B. Fund Y has the higher Jensen's alpha, so the Treynor ranking is misleading.

C. Fund X carries a larger share of unsystematic risk relative to its systematic risk, which Treynor ignores but Sharpe penalizes.

D. The Sharpe ratio measures risk with beta and the Treynor measure uses standard deviation, so the two must disagree.

Answer: C

Step by step

1. Treynor: $X = (11 - 3)/0.80 = 10.0$; $Y = (10 - 3)/1.00 = 7.0$, so X ranks first.
2. Sharpe: $X = (11 - 3)/20 = 0.40$; $Y = (10 - 3)/12 = 0.583$, so Y ranks first.
3. Fund X has a low beta but a high total risk, so most of its risk is unsystematic (diversifiable).
4. Treynor and Jensen's alpha consider only beta; Sharpe uses total risk and so penalizes X's unsystematic risk.
5. Check alpha: $X = 11 - (3 + 0.8 \times 5) = 4.0\%$, $Y = 10 - (3 + 1.0 \times 5) = 2.0\%$; alpha agrees with Treynor.

Why C is correct: Treynor and Jensen's alpha measure only systematic risk while Sharpe measures total risk, so a fund with a large unsystematic component ranks better on Treynor/alpha than on Sharpe/M-squared.

Why the others are wrong

- **A:** Backwards: a better-diversified fund would rank better, not worse, on Sharpe relative to Treynor. X's low Sharpe reflects poor diversification.
- **B:** Fund X's alpha (4.0%) exceeds Y's (2.0%); the claim is wrong, and alpha ranks consistently with Treynor.
- **D:** The definitions are swapped: Sharpe uses standard deviation and Treynor uses beta.

Q357

AnalystPrep Q.2757

REBUILT — Stem clarified that beta is unchanged (otherwise Treynor is not strictly unaffected).

A portfolio manager increases the number of stocks in his portfolio from ten to twenty, spread across a range of industries. The added stocks have the same average beta as the existing holdings, so portfolio beta and excess return are unchanged. What is the most likely effect on the Sharpe and Treynor ratios?

A. Sharpe decreases; Treynor is unaffected.

B. Sharpe is unaffected; Treynor increases.
C. Sharpe increases; Treynor is unaffected.
D. Sharpe is unaffected; Treynor decreases.

Answer: C
Step by step
1. Treynor $= (R_p - R_f)/\beta_p$: excess return and beta unchanged, so Treynor is unchanged.
2. Adding more stocks across industries removes unsystematic risk, lowering σ_p while beta stays the same.
3. Sharpe $= (R_p - R_f)/\sigma_p$ rises because the denominator falls.

Why C is correct: Diversification reduces only unsystematic risk, which lowers total risk (raising Sharpe) but does not alter beta (leaving Treynor unchanged).
Why the others are wrong
- **A:** Reverses the effect: lower total risk raises, not lowers, Sharpe.
- **B:** Treynor depends on beta, not diversifiable risk, so it does not rise; Sharpe does change.
- **D:** Wrong on both: Sharpe changes via lower standard deviation and Treynor is unaffected.

Q358
AP Mock 1 Q47

REBUILT — Option D (0.29) could not be traced to any realistic error; replaced with the Sharpe-ratio-based difference (0.04). Stem and answer unchanged.

Bob Johnson gathered the following data on his equity fund and the EGX index over the same period:

Item	Equity fund	EGX index
Return	-12%	-16%
Standard deviation	15%	19%
Beta	1.18	1.00

Assuming a risk-free rate of 6%, the difference between the Treynor measure for the equity fund and the Treynor measure for the EGX index is closest to:

A. 0.15
B. 0.07
C. 0.37
D. 0.04

Answer: B
Step by step
1. Fund Treynor: $(-12\% - 6\%)/1.18 = -0.1525$.
2. Index Treynor: $(-16\% - 6\%)/1.00 = -0.2200$.
3. Difference: $-0.1525 - (-0.2200) = +0.0675 \approx 0.07$.
4. Standard deviations are irrelevant to Treynor, which divides by beta.

Why B is correct: Treynor is excess return per unit of beta; the fund lost less per unit of systematic risk than the index, so its Treynor measure is higher by about 0.07.
Why the others are wrong
- **A:** This is the magnitude of the fund's own Treynor (0.1525), not the difference between the two measures.
- **C:** Adds the magnitudes (0.1525 + 0.2200 = 0.3725) instead of subtracting the signed values.
- **D:** Uses Sharpe ratios (-1.20 vs -1.158, difference 0.04), i.e. divides by standard deviation instead of beta.

Q359
AP Mock 1 Q58

An investment firm has hired an active manager for its pension fund. Her benchmark is the Russell 2000 Growth index. Which statistics are most suitable for evaluating the manager's performance and risk?

A. VaR and information ratio.
B. Tracking error and Sharpe ratio.
C. Tracking error and information ratio.
D. VaR and Sharpe ratio.

Answer: C
Step by step
1. The mandate is benchmark-relative, so both risk and performance should be measured against the Russell 2000 Growth index.

2. Tracking error = standard deviation of (portfolio return - benchmark return): the active risk.
3. Information ratio = $(R_p - R_B)/\sigma(R_p - R_B)$: active return per unit of active risk.
4. The two form a matched benchmark-relative pair.

Why C is correct: For a manager measured against a benchmark, the information ratio (active return divided by tracking error) and tracking error use the same benchmark-relative basis.

Why the others are wrong

- **A:** VaR measures absolute loss risk, not deviation from the benchmark, so it does not pair with the information ratio's active-risk denominator.
- **B:** Sharpe uses the risk-free rate and total risk, ignoring the benchmark, so it does not match tracking error.
- **D:** Both are absolute measures that ignore the benchmark entirely.

Q360

AP Mock 2 Q26

REBUILT — Original did not state that the fund is a stand-alone absolute-return vehicle without a benchmark, leaving alpha/information ratio defensible; added that context.

A newly established hedge fund is a stand-alone investment that will represent the investor's entire risky portfolio. It has no market or index benchmark and pursues absolute returns. Which measure is most appropriate to evaluate its performance?

A. Sharpe ratio.
B. Treynor measure.
C. Information ratio.
D. Jensen's alpha.

Answer: A

Step by step

1. The fund is the investor's whole risky portfolio, so total risk (standard deviation) is the relevant risk: use $(R_p - R_f)/\sigma_p$.
2. Treynor and Jensen's alpha rely on beta against a market portfolio and assume the fund is one well-diversified piece of a larger portfolio.
3. The information ratio requires a benchmark, which this fund lacks.

Why A is correct: Sharpe measures excess return per unit of total risk and needs only the risk-free rate, so it suits a stand-alone, benchmark-free portfolio.

Why the others are wrong

- **B:** Uses only systematic risk (beta), which ignores the diversifiable risk that matters when the fund is held alone.
- **C:** Needs a benchmark to compute active return and tracking error; none exists.
- **D:** Measures return relative to CAPM using beta, again only systematic risk and requiring a market proxy.

Q361

AP Mock 3 Q51

REBUILT — Distractors were not traceable to realistic errors; options rebuilt around omitting the risk-free rate, using the information ratio, and mixing formulas. Stem and answer unchanged.

Consider the following data for a fund in 2017:

Item	Portfolio	Market
Average return	32%	25%
Beta	1.15	1.00
Standard deviation	40%	30%
Tracking error σ(e)	16%	0%
T-bill rate	6%	6%

Determine the Sharpe ratio of the portfolio and of the market, respectively.

A. 0.80 and 0.83
B. 0.65 and 0.63
C. 0.44 and 0.00
D. 0.65 and 0.83

Answer: B

Step by step

1. Portfolio: $(32\% - 6\%)/40\% = 0.650$.

2. Market: $(25\% - 6\%)/30\% = 0.633$.
3. Beta and tracking error are not needed for Sharpe.

Why B is correct: The Sharpe ratio is excess return over the risk-free rate divided by total risk (standard deviation).

Why the others are wrong

- **A:** Forgot to subtract the T-bill rate: 32/40 = 0.80 and 25/30 = 0.83.
- **C:** Used the information ratio for the portfolio: $(32 - 25)/16 = 0.44$; the market's tracking error is 0.
- **D:** Subtracted the risk-free rate for the portfolio but not for the market (25/30 = 0.83).

Q362

AnalystPrep Q.2474

REBUILT — OCR-garbled table with irrelevant beta/VaR columns replaced by a clean two-asset data set; options rebuilt around realistic errors.

A currency portfolio holds two positions:

Asset	Expected return	Weight
CAD	8.00%	66.67%
EUR	5.00%	33.33%

The portfolio standard deviation is 15.62% and the risk-free rate is 2%. Which of the following is closest to the portfolio's Sharpe ratio?

A. 0.38
B. 0.19
C. 0.45
D. 0.32

Answer: D

Step by step

1. Expected return: $0.6667 \times 8\% + 0.3333 \times 5\% = 7.0\%$.
2. Excess return: $7.0\% - 2\% = 5.0\%$.
3. Sharpe: $5.0/15.62 = 0.320$.

Why D is correct: Sharpe is portfolio excess return per unit of portfolio total risk, using the weighted-average expected return.

Why the others are wrong

- **A:** Used the CAD return only: $(8 - 2)/15.62 = 0.38$.
- **B:** Used the EUR return only: $(5 - 2)/15.62 = 0.19$.
- **C:** Forgot to subtract the risk-free rate: $7/15.62 = 0.45$.

Q363

AnalystPrep Q.2557

REBUILT — Distractors were arbitrary; options rebuilt around realistic errors (Jensen's alpha, omitting rf, multiplying by beta). Stem and answer unchanged.

Consider the following data for a particular fund for the year 2017:

Item	Portfolio	Market
Average return	32%	25%
Beta	1.15	1.00
Standard deviation	40%	30%
Tracking error σ(e)	16%	0%
T-bill rate	6%	6%

Determine the Treynor measure of the portfolio and of the market, respectively.

A. 0.28 and 0.25
B. 0.04 and 0.00
C. 0.23 and 0.19
D. 0.30 and 0.19

Answer: C

Step by step

1. Portfolio: $(32\% - 6\%)/1.15 = 0.226 \approx 0.23$.
2. Market: $(25\% - 6\%)/1.00 = 0.19$.
3. Standard deviation and tracking error are not used.

Why C is correct: Treynor divides excess return by beta (systematic risk); the market's beta is 1, so its Treynor equals its market risk premium.

Why the others are wrong

- **A:** Forgot to subtract the risk-free rate: 32/1.15 = 0.28 and 25/1.00 = 0.25.
- **B:** Computed Jensen's alpha: $26 - 1.15 \times 19 = 4.15\%$ for the portfolio and 0 for the market.
- **D:** Multiplied by beta instead of dividing: $0.26 \times 1.15 = 0.30$.

Q364

AnalystPrep Q.2752

REBUILT — 'Always the same rankings' is not strictly true mathematically; restated as the same performance conclusion relative to the market, matching the reading's statement that alpha and Treynor agree.

Jensen's alpha will lead to the same performance conclusion as which of the following measures? (Alpha is positive exactly when this measure exceeds the market's value.)

A. Sharpe ratio
B. M-squared
C. Information ratio
D. Treynor measure

Answer: D

Step by step

1. Jensen's alpha $= R_p - [R_f + \beta(R_m - R_f)] = \beta\,(T_p - T_m)$, where T is the Treynor measure.
2. So alpha is positive exactly when the portfolio's Treynor measure exceeds the market's.
3. Sharpe and M-squared use total risk, so they can disagree when a manager holds large unsystematic risk.

Why D is correct: Both alpha and Treynor consider only systematic (beta) risk, so they reach the same conclusion; Sharpe and M-squared agree with each other instead.

Why the others are wrong

- **A:** Uses total risk, so a large unsystematic component can lower Sharpe while leaving alpha unchanged.
- **B:** M-squared is a transform of the Sharpe ratio and shares its total-risk basis.
- **C:** Measures active return per unit of tracking error against a benchmark, not systematic risk.

Q365

AnalystPrep Q.2754

REBUILT — Options rebuilt so distractors correspond to realistic errors; original 3.2%, 1.7%, 4.2% were not traceable.

You have been provided the following information about a portfolio:

Return of the portfolio	10%
Beta of the portfolio	1.1
Standard deviation of portfolio	6%
Return on the market	7%
Risk-free rate	3%

Calculate the Jensen's alpha for the portfolio.

A. -0.7%
B. 5.6%
C. 3.0%
D. 2.6%

Answer: D

Step by step

1. CAPM required return: $3\% + 1.1 \times (7\% - 3\%) = 7.4\%$.
2. Alpha: $10\% - 7.4\% = 2.6\%$.
3. Standard deviation is not needed.

Why D is correct: Jensen's alpha is actual return minus the CAPM return required for the portfolio's beta.

Why the others are wrong

- **A:** Used βR_m instead of $\beta(R_m - R_f)$: $3 + 1.1 \times 7 = 10.7$, alpha $= -0.7\%$.
- **B:** Forgot to add back the risk-free rate: $10 - 1.1 \times 4 = 5.6\%$.
- **C:** Ignored beta: $10 - 7 = 3.0\%$ (return over the market).

Q366

AnalystPrep Q.3021

REBUILT — Original answer was correct but distractors were arbitrary; replaced with error-based distractors.

James Dunn is a retail investor who expects his stock portfolio to return 21.2% in the following year. If the return on risk-free Treasury notes is 11.6% and the portfolio has a standard deviation of 9% (0.09), which of the following is closest to the Sharpe ratio of Dunn's portfolio?

A. 1.29
B. 2.36
C. 1.07
D. 0.83

Answer: C

Step by step

1. Sharpe ratio $= (R_p - R_f)/\sigma_p$
2. Excess return $= 21.2\% - 11.6\% = 9.6\%$
3. $9.6\%/9\% = 1.0667 \approx 1.07$

Why C is correct: The Sharpe ratio is excess return per unit of total risk (standard deviation).

Why the others are wrong

- **A:** 1.29 = 11.6/9: divided the risk-free rate (not the excess return) by the standard deviation.
- **B:** 2.36 = 21.2/9: forgot to subtract the risk-free rate.
- **D:** 0.83 = 9.6/11.6: divided the excess return by the risk-free rate instead of the standard deviation.

Q367

AnalystPrep Q.4456

REBUILT — Original answer correct; replaced the arbitrary 0.5 distractor with an error-based one.

The expected return of an investor's portfolio is 31% with a standard deviation of 19%, while the expected return of the market is 22% with a standard deviation of 16%. The risk-free rate is 5% and the portfolio's beta is 0.9. What is the difference between the Sharpe ratio of the portfolio and the Sharpe ratio of the market?

A. 0.31
B. 0.47
C. 1.06
D. 0.12

Answer: A

Step by step

1. Portfolio Sharpe $= (31 - 5)/19 = 1.368$
2. Market Sharpe $= (22 - 5)/16 = 1.0625$
3. Difference $= 1.368 - 1.0625 = 0.306 \approx 0.31$

Why A is correct: Sharpe ratios use total risk, so beta is not needed; subtract the market's Sharpe ratio from the portfolio's.

Why the others are wrong

- **B:** 0.47 = (31 - 22)/19: used the return difference over the portfolio's standard deviation.
- **C:** 1.06 is the market's Sharpe ratio itself, not the difference.
- **D:** 0.12 = 0.119: the difference of Treynor measures, (31 - 5)/0.9 - (22 - 5)/1.0 = 28.9% - 17% = 11.9%, i.e. dividing by beta instead of standard deviation.

Q368

AnalystPrep Q.3173

REBUILT — Stated that the discount rate is the risk-free proxy and replaced arbitrary distractors with error-based ones.

Rohit Sharma, a treasury manager at Zannata Industries, is evaluating Kratnik Fund, a mutual fund with an average annual return of 12% over the last five years, a beta of 1.35, and a standard deviation of returns of 16.80%. Over the same period the average risk-free rate (proxied by the central bank discount rate) was 4.5% and the average annual return on the SENSEX 100 index was 18%. The Sharpe ratio of Kratnik Fund is closest to:

A. 0.80
B. 5.56
C. 0.71
D. 0.45

Answer: D

Step by step

1. Sharpe $= (R_p - R_f)/\sigma_p$
2. $(12\% - 4.5\%)/16.80\% = 7.5/16.8 = 0.446 \approx 0.45$

Why D is correct: Sharpe uses the fund's own excess return over the risk-free rate divided by its total standard deviation; beta and the market return are irrelevant.

Why the others are wrong

- **A:** 0.80 = (18 - 4.5)/16.8: used the market index return instead of the fund's return.
- **B:** 5.56 = (12 - 4.5)/1.35: this is the Treynor measure (divided by beta, in percent), not Sharpe.
- **C:** 0.71 = 12/16.8: forgot to subtract the risk-free rate.

Q369

AnalystPrep Q.3174

REBUILT — Original answer correct; replaced arbitrary distractors with plausible measure-confusion errors.

David Hale is an investment advisor evaluating two pension funds, Qintar Fund and Zombie Fund, for Gallant Motor Companies Pension Fund. Gallant's minimum acceptable return is 5.0%. The T-bill return over the last fiscal year was 4.5%. The S&P 500 (the market index) returned 10% with a standard deviation of 21% and a beta of 1.0. The fund data are:

Fund	Return	Standard deviation	Beta	Downside deviation
Qintar	16.5%	38.1%	0.8	14.9%
Zombie	15.9%	35.6%	1.25	14.0%

The Sharpe ratio for the Zombie Fund is closest to:

A. 0.78
B. 0.32
C. 0.26
D. 9.12

Answer: B

Step by step

1. Sharpe $= (R_p - R_f)/\sigma_p$, with $R_f = 4.5\%$ (T-bill)
2. $(15.9 - 4.5)/35.6 = 11.4/35.6 = 0.320$

Why B is correct: Sharpe divides excess return over the risk-free rate by total standard deviation.

Why the others are wrong

- **A:** 0.78 = (15.9 - 5.0)/14.0: used the minimum acceptable return and downside deviation (a Sortino-type ratio).
- **C:** 0.26 = (10 - 4.5)/21: computed the market's Sharpe ratio.
- **D:** 9.12 = (15.9 - 4.5)/1.25: divided by beta (Treynor measure, in percent).

Q370

AnalystPrep Q.3176

REBUILT — Fixed spelling (Quintar/Qintar) and replaced arbitrary distractors with error-based ones.

David Hale is an investment advisor evaluating two pension funds, Qintar Fund and Zombie Fund, for Gallant Motor Companies Pension Fund. Gallant's minimum acceptable return is 5.0%. The T-bill return over the last fiscal year was 4.5%. The S&P 500 (the market index) returned 10% with a standard deviation of 21% and a beta of 1.0. The fund data are:

Fund	Return	Standard deviation	Beta	Downside deviation
Qintar	16.5%	38.1%	0.8	14.9%
Zombie	15.9%	35.6%	1.25	14.0%

Jensen's alpha for the Qintar Fund is closest to:

A. 0.120
B. 0.065
C. 0.076
D. 0.089

Answer: C

Step by step

1. CAPM required return $= 4.5\% + 0.8(10\% - 4.5\%) = 4.5\% + 4.4\% = 8.9\%$
2. $\alpha = 16.5\% - 8.9\% = 7.6\% = 0.076$

Why C is correct: Jensen's alpha is the fund's return minus the return required by the CAPM given its beta.
Why the others are wrong
- **A:** 0.120 = 16.5 - 4.5: excess return over the risk-free rate, ignoring the beta-adjusted CAPM return.
- **B:** 0.065 = 16.5 - 10: subtracted the market return (ignored beta and the risk-free rate).
- **D:** 0.089 is the CAPM required return (8.9%), not the alpha.

Q371

AnalystPrep Q.3177

REBUILT — Replaced one arbitrary option (0.70 and 0.19) with an error-based one.

Jack Shiller is a portfolio manager at Quandra Investment. Thunder Growth Fund returned 14% with a standard deviation of 25% and a beta of 1.15. The index returned 12% with a standard deviation of 18% and a beta of 1.00. The risk-free rate is 4%. Which of the following represents the Sharpe ratio and the Treynor measure, respectively, for Thunder Growth Fund?

A. 0.56 and 0.12
B. 0.40 and 0.10
C. 0.08 and 0.02
D. 0.40 and 0.09

Answer: D
Step by step
1. Sharpe $= (14\% - 4\%)/25\% = 0.40$
2. Treynor $= (14\% - 4\%)/1.15 = 0.0870 \approx 0.09$

Why D is correct: Sharpe divides excess return by total risk (Ïƒ = 25%); Treynor divides excess return by systematic risk ($\hat{I}^2 = 1.15$).
Why the others are wrong
- **A:** 0.56 and 0.12: forgot to subtract the risk-free rate in both ratios (14/25 and 14/1.15).
- **B:** 0.40 and 0.10: Sharpe correct, but divided by the index beta of 1.00 instead of the fund's 1.15.
- **C:** 0.08 and 0.02: used the fund's excess return over the index (14 - 12 = 2%) instead of over the risk-free rate.

Q372

AnalystPrep Q.3178

REBUILT — Original answer correct; replaced arbitrary distractors with error-based ones.

Samuel Badree manages a hedge fund at Bumzee Investments. The fund returned 13.3% while its benchmark index returned 8.9%. The fund's volatility was 16.3% and the index's volatility 9.8%. The fund's tracking error was 1.34% and the risk-free rate is 5.7%. What is the fund's information ratio?

A. 0.47
B. 3.3
C. 5.67
D. 0.27

Answer: B
Step by step
1. $IR = (R_p - R_b)/\text{tracking error}$
2. Active return $= 13.3\% - 8.9\% = 4.4\%$
3. $4.4/1.34 = 3.28 \approx 3.3$

Why B is correct: The information ratio is active return over the benchmark per unit of tracking error (active risk).
Why the others are wrong
- **A:** 0.47 = (13.3 - 5.7)/16.3: that is the fund's Sharpe ratio.
- **C:** 5.67 = (13.3 - 5.7)/1.34: used excess return over the risk-free rate instead of over the benchmark.
- **D:** 0.27 = 4.4/16.3: divided active return by the fund's volatility instead of tracking error.

Q373

AnalystPrep Q.3181

REBUILT — Added 'Using the Treynor measure' to the stem so the criterion is explicit.

Graham Cook is an investment analyst at Gabriel Fund considering two stocks for long-term investment. Ketrick Fertilizer has an expected return of 16% and a beta of 1.6. Sanbros Chemical has an expected return of 13% and a beta of 1.2. The risk-free rate is 6.7%. Using the Treynor measure, which stock should be included in the portfolio?

A. Sanbros Chemical because it has a lower Treynor measure.

B. Sanbros Chemical because it has a higher Treynor measure.
C. Ketrick Fertilizer because it has a higher Treynor measure.
D. Ketrick Fertilizer because it has a lower Treynor measure.

Answer: C
Step by step
1. Treynor $= (R_p - R_f)/\beta$
2. Ketrick: $(16 - 6.7)/1.6 = 9.3/1.6 = 5.81\%$
3. Sanbros: $(13 - 6.7)/1.2 = 6.3/1.2 = 5.25\%$
4. Ketrick has the higher excess return per unit of systematic risk.

Why C is correct: A higher Treynor measure means greater compensation per unit of beta, so Ketrick is preferred.
Why the others are wrong
- **A:** Wrong stock and wrong logic: a lower Treynor measure is never a reason to select a stock.
- **B:** Picks Sanbros, but its Treynor measure (5.25%) is lower than Ketrick's (5.81%); the choice may reflect preferring its lower beta rather than comparing excess return per unit of beta.
- **D:** Right stock but reversed reasoning: Ketrick's Treynor measure is higher, not lower.

Q374

AnalystPrep Q.3190

Jack Marconi, an equity strategist at Gandhara Investment, ranks four large-cap equity portfolios using the Treynor measure and the Sharpe ratio:

Fund	Treynor Measure Rank	Sharpe Ratio Rank
Azgard	1	4
Lambda	2	3
Tricky	3	2
Jackpot	4	1

The difference in rankings for Azgard and Jackpot is most likely due to:
A. Different benchmarks used to evaluate each fund's performance.
B. A difference in risk premiums.
C. Poor diversification in Azgard Fund relative to Jackpot Fund.
D. None of the above.

Answer: C
Step by step
1. Both measures share the same numerator (excess return over the risk-free rate); neither uses a benchmark.
2. They differ only in the denominator: Sharpe uses total risk σ, Treynor uses systematic risk β.
3. Azgard ranks high on Treynor but low on Sharpe, so its beta is small relative to its total risk: it carries large unsystematic (diversifiable) risk.
4. Hence Azgard is poorly diversified relative to Jackpot.

Why C is correct: Rankings diverge only when total risk and systematic risk are not proportional, i.e., when diversification differs across funds.
Why the others are wrong
- **A:** Neither Sharpe nor Treynor depends on a benchmark, so benchmark choice cannot explain the rank gap.
- **B:** Both ratios use the same risk premium in the numerator, so premiums cannot make the two rankings differ.
- **D:** Incorrect because option C explains the divergence.

Q375

AnalystPrep Q.4455

REBUILT — Original answer correct; replaced arbitrary distractors with error-based ones.
The standard deviation of a portfolio is 15%. If the portfolio's return is 22% and the risk-free return is 6%, what is the Sharpe ratio of the portfolio?
A. 0.73
B. 1.07
C. 1.47
D. 0.40

Answer: B

Step by step

1. Sharpe $= (22\% - 6\%)/15\% = 16/15 = 1.067 \approx 1.07$

Why B is correct: Sharpe is excess return divided by total standard deviation.

Why the others are wrong

- **A:** 0.73 = 16/22: divided excess return by the portfolio return.
- **C:** 1.47 = 22/15: forgot to subtract the risk-free rate.
- **D:** 0.40 = 6/15: used the risk-free rate in the numerator.

Q376

AnalystPrep Q.4457

A 10-year study of three portfolios and the market reveals:

Portfolio	Average annual return	Standard deviation	Beta
1	14%	21%	1.15
2	16%	24%	1.00
3	20%	28%	1.25
S&P 500	12%	20%	-

The risk-free rate is 6%. Using the Sharpe measure, rank the portfolios from lowest to highest.

A. 1, 3, 2

B. 2, 3, 1

C. 2, 1, 3

D. 1, 2, 3

Answer: D

Step by step

1. Sharpe = $(R_p - R_f)/\sigma_p$.
2. Portfolio 1: $(14 - 6)/21 = 0.381$.
3. Portfolio 2: $(16 - 6)/24 = 0.417$.
4. Portfolio 3: $(20 - 6)/28 = 0.500$.
5. Lowest to highest: 1, 2, 3.

Why D is correct: The Sharpe ratio scales excess return by total risk (standard deviation), so the ranking follows 0.381 < 0.417 < 0.500.

Why the others are wrong

- **A:** Puts portfolio 3 (the highest Sharpe, 0.500) in the middle and portfolio 2 last; the order does not match the computed ratios.
- **B:** Places portfolio 1 (lowest Sharpe) last, i.e. closer to a highest-to-lowest reading; portfolio 3 should be last in a lowest-to-highest list.
- **C:** Puts portfolio 1 above portfolio 2; it ignores that 1 has the lowest ratio (8/21 = 0.381 versus 10/24 = 0.417), e.g. by ranking on beta or on a misread ratio.

Q377

BT answers P2.T8.16.1 · BT notes P2.T9.16.1

REBUILT — Source option D used alpha/tracking error (12%), inconsistent with Schweser's information ratio (surplus return over benchmark divided by tracking error); D is now 30.0% and A is now the correct-arithmetic but inappropriate Sharpe of 37.5%.

Consider the following performance data for a sample period:

Item	Portfolio (P)	Market (M)
Average return	15.0%	9.0%
Beta	1.60	1.00
Standard deviation	32.0%	24.0%
Tracking error	20%	0
Risk-free rate	3.0%	

Portfolio P is one sub-portfolio that will be combined with several other portfolios into a large investment fund. Which risk-adjusted performance measure is appropriate, and what is its value for P?

A. Sharpe of 37.5%

B. Treynor of 6.0%

C. Treynor of 7.5%

D. Information ratio of 30.0%

Answer: C

Step by step

1. A sub-portfolio inside a large diversified fund contributes only systematic risk, so use the Treynor measure.
2. $T_P = (15\% - 3\%)/1.60 = 7.5\%$.
3. Market Treynor for comparison: $(9\% - 3\%)/1.00 = 6.0\%$.

Why C is correct: When a portfolio is one component of a larger diversified fund, its diversifiable risk washes out, so excess return per unit of beta (Treynor) is the relevant measure.

Why the others are wrong

- **A:** Sharpe = 12%/32% = 37.5% is correct arithmetic but uses total risk, which suits a stand-alone portfolio, not a sub-portfolio.
- **B:** 6.0% is the market's Treynor measure (9% - 3% over beta of 1), not Portfolio P's.
- **D:** (15% - 9%)/20% = 30.0% is the benchmark-relative information ratio, suited to mixing an active and a passive portfolio, not to a component of a diversified fund.

Q378

BT answers P2.T8.16.2 · BT notes P2.T9.16.2

REBUILT — Source option D (0.2727) used alpha/tracking error, inconsistent with Schweser's information ratio; replaced with (11-9)/11 = 0.1818, and option A (which differed between the two sources) is now the market's Sharpe ratio 0.4167.

Consider the following performance data for a sample period:

Item	Portfolio (P)	Market (M)
Average return	11.0%	9.0%
Beta	0.80	1.00
Standard deviation	13.0%	12.0%
Tracking error	11%	0
Risk-free rate	4.0%	

Portfolio P represents the entire risky investment fund. Which risk-adjusted performance measure is appropriate, and what is its value for P?

A. Sharpe of 0.4167
B. Sharpe of 0.5385
C. Treynor of 8.750%
D. Information ratio of 0.1818

Answer: B

Step by step

1. P is the whole risky fund, so total risk matters: use Sharpe.
2. $S_P = (11\% - 4\%)/13\% = 0.5385$.
3. For comparison, $S_M = (9 - 4)/12 = 0.4167$, so P outperforms the market on this basis.

Why B is correct: For an entire investment fund, the investor bears all of its risk, not just systematic risk, so the Sharpe ratio using total standard deviation is appropriate.

Why the others are wrong

- **A:** 0.4167 is the market's Sharpe ratio (5%/12%), not Portfolio P's.
- **C:** Treynor = 7%/0.80 = 8.75% is correct arithmetic but uses beta, ignoring the unsystematic risk that matters for an entire fund.
- **D:** (11% - 9%)/11% = 0.1818 is the benchmark-relative information ratio, suited to active/passive mixing, not to evaluating an entire fund.

Q379

BT answers P2.T8.16.3 · BT notes P2.T9.16.3

REBUILT — Source keyed the information ratio as alpha/tracking error (0.380), which conflicts with Schweser's surplus-return definition; the stem now defines the ratio, giving 0.35, and Jensen's alpha is expressed as 7.60%.

Consider the following performance data for a sample period:

Item	Portfolio (P)	Market (M)
Average return	15.0%	8.0%
Beta	0.90	1.00
Standard deviation	27.0%	15.0%
Tracking error	20%	0
Risk-free rate	2.0%	

Portfolio P is the active portfolio that will be optimally mixed with the passive (market) portfolio. Using the informa-

tion ratio defined as (portfolio return - benchmark return) divided by tracking error, which measure is appropriate, and what is its value for P?

A. Sharpe of 0.4815
B. Jensen's alpha of 7.60%
C. Treynor of 14.44%
D. Information ratio of 0.35

Answer: D

Step by step

1. An active portfolio to be mixed with a passive benchmark is judged by the information ratio.
2. Surplus return = 15% - 8% = 7%.
3. $IR = 7\%/20\% = 0.35$.

Why D is correct: The information ratio measures benchmark-relative return per unit of benchmark-relative risk, which is what matters when an active bet is combined with a passive portfolio.

Why the others are wrong

- **A:** Sharpe = 13%/27% = 0.4815 uses total risk against the risk-free rate, not a benchmark-relative measure.
- **B:** Alpha = 15% - [2% + 0.9(6%)] = 7.6% is an unscaled return difference; it is not a risk-adjusted ratio and ignores the risk of the active bet.
- **C:** Treynor = 13%/0.90 = 14.44% is based on systematic risk and suits a sub-portfolio of a diversified fund.

Q380

BT answers P2.T9.21.10.2 · BT notes P2.T9.21.10.2

REBUILT — Source claimed the information ratio was also negative, but measured against the market it is positive; the information ratio was removed from the statements. Residual standard deviation corrected to 24.0% to be consistent with total variance (40^2 = 1.6^2 x 20^2 + 24^2).

The risk-free rate was 3.0%. The market's return was 11.0% with volatility of 20.0%. Donald's portfolio returned 15.0% with volatility of 40.0%, correlation with the market of 0.80, and residual standard deviation of 24.0%. Which statement is TRUE?

A. We cannot answer without the portfolio's beta.
B. Donald's Jensen's alpha was negative, and both his Sharpe ratio and Treynor measure were below the market's.
C. Donald's Jensen's alpha was positive, and both his Sharpe ratio and Treynor measure were above the market's.
D. Donald's Jensen's alpha was zero, and both his Sharpe ratio and Treynor measure matched the market's.

Answer: B

Step by step

1. Beta $= \rho\,\sigma_P/\sigma_M = 0.80 \times 40/20 = 1.60$.
2. Sharpe: P $= 12/40 = 0.300$; M $= 8/20 = 0.400$.
3. Treynor: P $= 12/1.6 = 7.5\%$; M $= 8/1 = 8.0\%$.
4. Alpha $= 15 - [3 + 1.6(8)] = -0.8\%$.

Why B is correct: Beta can be derived from correlation and volatilities; with beta of 1.6 the CAPM-required return is 15.8%, above the 15% earned, so alpha is negative and both ratios trail the market's.

Why the others are wrong

- **A:** Beta is derivable as correlation times the volatility ratio (0.80 x 40/20 = 1.6), so the question is answerable.
- **C:** Gets the sign wrong: the raw excess return (12% vs 8%) looks higher, but beta of 1.6 requires 12.8% excess return.
- **D:** Would need alpha of exactly zero; actual alpha is -0.8% and the ratios differ (0.30 vs 0.40; 7.5% vs 8.0%).

Q381

Mock 2 Q80

Money managers are routinely evaluated using risk-adjusted performance measures including Sharpe's measure, Treynor's measure, Jensen's alpha, and the information ratio. Which statement is most likely correct?

A. Jensen's alpha gives no indication of the diversification in the portfolio.
B. A lower information ratio indicates better performance.
C. When ranking portfolios, Sharpe will always give the same conclusion as Jensen's alpha.
D. The Treynor measure is very similar to the Sharpe ratio except that it uses standard deviation as the measure of risk.

Answer: A

Step by step

1. Jensen's alpha is based on beta, so it reflects only systematic risk.
2. Unsystematic risk (lack of diversification) does not enter it, so it says nothing about diversification.
3. Check the others against the reading: higher IR is better; Sharpe can disagree with alpha; Treynor uses beta.

Why A is correct: Like Treynor, Jensen's alpha considers only systematic risk, so it is blind to how diversified the portfolio is.

Why the others are wrong

- **B:** A higher information ratio (more surplus return per unit of tracking risk) indicates better performance.
- **C:** Sharpe (total risk) and alpha (systematic risk) can rank differently when portfolios hold different amounts of unsystematic risk.
- **D:** Reverses the facts: Treynor uses beta; it is Sharpe that uses standard deviation.

Q382

Schweser Module Quiz 89.2 Q1

REBUILT — The fund data exhibit was missing; reconstructed returns and betas consistent with the stated ranking and adjusted distractors to reflect realistic mistakes.

The average risk-free rate was 5%. The following information is available for Funds ABC, RST, JKL, and XYZ:

Fund	Average return	Beta
ABC	12%	0.90
RST	15%	1.20
JKL	14%	0.80
XYZ	10%	1.00

Rank the funds from best to worst according to their Treynor measure.

A. JKL, RST, ABC, XYZ.

B. JKL, ABC, RST, XYZ.

C. RST, JKL, ABC, XYZ.

D. XYZ, ABC, RST, JKL.

Answer: A

Step by step

1. Treynor measure $T_p = (R_p - R_f)/\beta_p$ with $R_f = 5\%$.
2. ABC: $(12 - 5)/0.90 = 7.78$
3. RST: $(15 - 5)/1.20 = 8.33$
4. JKL: $(14 - 5)/0.80 = 11.25$
5. XYZ: $(10 - 5)/1.00 = 5.00$
6. Ranking from highest to lowest: JKL, RST, ABC, XYZ.

Why A is correct: Treynor's measure is excess return per unit of systematic risk (beta); the higher the ratio, the better the risk-adjusted performance.

Why the others are wrong

- **B:** Forgot to subtract the risk-free rate: R_p/β gives 13.3, 12.5, 17.5, 10.0, which ranks ABC above RST.
- **C:** Ranked by excess return (RST 10, JKL 9, ABC 7, XYZ 5) and ignored beta, so the high-beta RST is wrongly placed first.
- **D:** Reversed the ranking (worst to best) instead of best to worst.

Q383

Schweser Module Quiz 89.2 Q2

REBUILT — The fund data exhibit (and market return) was missing; reconstructed consistent data and redesigned the distractors.

The following data has been collected to appraise Funds A, B, C, and D. The risk-free rate for the period was 4% and the average market return was 10%.

Fund	Average return	Beta
A	11%	0.80
B	14%	1.50
C	13%	1.10
D	9%	0.70

Calculate and rank the funds from best to worst using Jensen's alpha.

A. B, C, A, D.

B. A, C, D, B.
C. C, A, B, D.
D. D, B, A, C.

Answer: C
Step by step
1. Jensen's alpha $\alpha_p = R_p - [R_f + \beta_p(R_m - R_f)]$, with $R_m - R_f = 6\%$.
2. A: $11 - (4 + 0.80 \times 6) = 11 - 8.8 = 2.2\%$
3. B: $14 - (4 + 1.50 \times 6) = 14 - 13.0 = 1.0\%$
4. C: $13 - (4 + 1.10 \times 6) = 13 - 10.6 = 2.4\%$
5. D: $9 - (4 + 0.70 \times 6) = 9 - 8.2 = 0.8\%$
6. Ranking by alpha: C (2.4), A (2.2), B (1.0), D (0.8).

Why C is correct: Jensen's alpha is the return in excess of the CAPM-required return, so each fund's beta-adjusted benchmark must be subtracted before ranking.

Why the others are wrong
- **A:** Ranked by raw return (14, 13, 11, 9) without adjusting for beta.
- **B:** Ranked by Treynor measure (A 8.75, C 8.18, D 7.14, B 6.67) instead of Jensen's alpha.
- **D:** Reversed the ranking (worst to best).

LO 89.c

Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures.

Q384

AnalystPrep Q.3175

REBUILT — Original options (8.10%, 6.70%, 9.46%) were arbitrary giveaways and the stem had extraneous data; kept the data, replaced distractors with realistic errors.

An analyst evaluates the Qintar Fund against the market index. Over the last year the T-bill return was 4.5%. The market index returned 10.0% with a standard deviation of 21.0% and a beta of 1.0. The Qintar Fund returned 16.5% with a standard deviation of 38.1% and a beta of 0.8.

The M-squared (M^2) measure of the Qintar Fund is closest to:

A. 6.50%
B. 1.11%
C. 5.31%
D. 11.11%

Answer: B
Step by step
1. Weight in Qintar to match market risk: $21.0/38.1 = 0.5512$, i.e. 55.12% in Qintar and 44.88% in T-bills.
2. Return of the adjusted portfolio $P^* = 0.5512(16.5\%) + 0.4488(4.5\%) = 11.11\%$.
3. $M^2 = R_{P^*} - R_M = 11.11\% - 10.00\% = 1.11\%$.
4. Check: $M^2 = \sigma_M(S_P - S_M) = 21\%(0.3150 - 0.2619) = 1.11\%$.

Why B is correct: M^2 is the return of the fund after it is mixed with T-bills to have the market's standard deviation, minus the market return.

Why the others are wrong
- **A:** Raw return difference 16.5% - 10.0%; ignores the difference in risk.
- **C:** Difference in Sharpe ratios $(0.3150 - 0.2619) \times 100$; forgot to multiply by $\sigma_M = 21\%$.
- **D:** This is the return of the risk-matched portfolio P^*; forgot to subtract the market return of 10%.

Q385

BT answers P2.T8.17.1 · BT notes P2.T9.17.1

REBUILT — Distractors -2.5% and +0.5% could not be tied to realistic errors; replaced with plausible mistakes (Jensen alpha, raw excess return, inverted leverage).

The following data compares a Portfolio (P) to the Market (M):

Item	Portfolio (P)	Market (M)
Average return	15.0%	9.0%
Beta	1.60	1.00
Standard deviation	32.0%	24.0%
Tracking error	20%	0
Risk-free rate	3.0%	

What is the Modigliani-squared (M^2) measure of the portfolio?

A. 2.4%
B. 6.0%
C. 10.0%
D. 3.0%

Answer: D

Step by step

1. Weight in P to match market volatility: $24/32 = 0.75$; 25% in the risk-free asset.
2. $R_{P^*} = 0.75(15.0\%) + 0.25(3.0\%) = 12.0\%$.
3. $M^2 = 12.0\% - 9.0\% = +3.0\%$.

Why D is correct: Re-mixing P with cash to the market's 24% volatility gives 12.0%, which beats the market's 9.0% by 3.0%.

Why the others are wrong

- **A:** This is Jensen's alpha $15 - [3 + 1.6(6)] = 2.4\%$, a beta-based measure, not M^2.
- **B:** Raw return difference 15% - 9%; ignores that P is riskier than the market.
- **C:** Inverted the weight (32/24 = 1.333, borrowing at 3%): $1.333(15) - 0.333(3) - 9 = 10\%$; this levers P up instead of de-levering it.

Q386 BT answers P2.T8.17.2 · BT notes P2.T9.17.2

The following data compares two portfolios, Portfolio (A) and Portfolio (B), to the Market (M):

Item	Portfolio (A)	Portfolio (B)	Market (M)
Average return	15%	24%	12%
Beta	0.80	1.50	1.00
Standard deviation	30%	24%	16%
Tracking error	18%	15%	0
Risk-free rate	3.0%		

If the portfolios are ranked by the Modigliani-squared (M^2) measure and by the slope of the T-line (Treynor measure), how do they rank?

A. Portfolio (A) offers both a higher M^2 and a steeper T-line than Portfolio (B)
B. Portfolio (A) offers a higher M^2, but Portfolio (B) has a steeper T-line
C. Portfolio (B) offers a higher M^2, but Portfolio (A) has a steeper T-line
D. Portfolio (B) offers both a higher M^2 and a steeper T-line than Portfolio (A)

Answer: C

Step by step

1. A: weight in A = 16/30 = 0.5333; $R_{A^*} = 0.5333(15) + 0.4667(3) = 9.4\%$; $M_A^2 = 9.4 - 12 = -2.6\%$.
2. B: weight in B = 16/24 = 0.6667; $R_{B^*} = 0.6667(24) + 0.3333(3) = 17.0\%$; $M_B^2 = 17.0 - 12 = +5.0\%$. B has the higher M^2.
3. Treynor: A = $(15 - 3)/0.80 = 15.0\%$; B = $(24 - 3)/1.50 = 14.0\%$. A has the steeper T-line.
4. B wins on total risk (M^2); A wins on systematic risk (Treynor).

Why C is correct: M^2 is based on total risk (standard deviation) and Treynor on beta, so rankings can differ: B ranks first on M^2, A first on Treynor.

Why the others are wrong

- **A:** A's M^2 is -2.6%, below B's +5.0%, so A is not higher on M^2.
- **B:** Reverses both rankings: A is lower on M^2 and higher on Treynor.
- **D:** B's Treynor is 14.0%, below A's 15.0%; likely compared excess returns (21% vs 12%) without dividing by beta.

Q387 BT answers P2.T8.17.3 · BT notes P2.T9.17.3

REBUILT — Treynor-squared was not defined in the stem and distractors were unexplained numbers; added the definition and rebuilt distractors from realistic errors (Jensen alpha, M-squared, Treynor ratio).

The following data compares two portfolios, Portfolio (A) and Portfolio (B), to the Market (M):

Item	Portfolio (A)	Portfolio (B)	Market (M)
Average return	11%	22%	9.0%
Beta	0.70	1.20	1.00
Standard deviation	10%	20%	16%
Risk-free rate	4.0%		

The Treynor-squared (T^2) measure of a portfolio is the return of the portfolio after it is re-mixed with the risk-free asset to have a beta of 1.0, minus the market return. What are, respectively, the T^2 measures of Portfolio (A) and Portfolio (B)?

A. A = 5.0% and B = 10.0%
B. A = 6.2% and B = 9.4%
C. A = 3.5% and B = 12.0%
D. A = 10.0% and B = 15.0%

Answer: A

Step by step

1. Treynor ratios: $A = (11 - 4)/0.70 = 10.0\%$; $B = (22 - 4)/1.20 = 15.0\%$.
2. Market excess return $= 9 - 4 = 5.0\%$ (beta 1, Treynor ratio 5.0%).
3. $T_A^2 = 10.0 - 5.0 = 5.0\%$; $T_B^2 = 15.0 - 5.0 = 10.0\%$.
4. Check: re-mix A to beta 1: weight 1/0.70 = 1.4286 in A, return $4 + 1.4286(7) = 14.0\%$, minus 9% = 5.0%.

Why A is correct: Scaling excess return by beta puts each portfolio at beta 1, so T^2 is the Treynor ratio minus the market's excess return.

Why the others are wrong

- **B:** These are M^2 values (re-mixing to 16% standard deviation: A 6.2%, B 9.4%); uses total risk instead of beta.
- **C:** These are Jensen alphas $7 - 0.7(5) = 3.5\%$ and $18 - 1.2(5) = 12\%$; not rescaled to beta 1.
- **D:** These are the Treynor ratios; forgot to subtract the market's 5.0% excess return.

Q388 BT answers P2.T9.21.10.3 · BT notes P2.T9.21.10.3

Below is the performance of two portfolios, P and Q, in comparison to the market. The risk-free rate is 3.0%.

Item	Portfolio (P)	Portfolio (Q)	Market (M)
Beta	0.70	1.20	1.00
Excess return	10.0%	14.0%	5.0%
Volatility	12.0%	27.0%	16.0%

In regard to Portfolio P as it compares to Portfolio Q, which of the following statements is TRUE?

A. Portfolio P has a higher alpha (than Q) but is equal to Portfolio Q in regard to both M^2 and T^2
B. Portfolio P has a higher alpha (than Q) but is inferior (to Q) in regard to both M^2 and T^2
C. Portfolio P has a lower alpha (than Q) but is superior (to Q) in regard to both M^2 and T^2
D. Portfolio P has a lower alpha (than Q) and is also inferior (to Q) in regard to both M^2 and T^2

Answer: C

Step by step

1. Alpha = excess return $-\beta \times 5\%$: $P = 10 - 0.7(5) = 6.5\%$; $Q = 14 - 1.2(5) = 8.0\%$. P's alpha is lower.
2. M^2 = excess return $\times 16/\sigma - 5\%$: $P = 10(16/12) - 5 = 8.33\%$; $Q = 14(16/27) - 5 = 3.30\%$. P is higher.
3. T^2 = excess return/beta -5%: $P = 10/0.70 - 5 = 9.29\%$; $Q = 14/1.20 - 5 = 6.67\%$. P is higher.
4. P has the lower alpha but is superior on M^2 and T^2.

Why C is correct: Alpha is not scaled for the amount of beta taken, while T^2 (per unit of beta) and M^2 (per unit of total risk) are, so low-risk P beats Q on both despite a smaller alpha.

Why the others are wrong

- **A:** P's alpha (6.5%) is lower, not higher, and the M^2/T^2 values differ (8.33% vs 3.30%; 9.29% vs 6.67%).
- **B:** Gets the alpha direction wrong (P 6.5% < Q 8.0%) and the M^2/T^2 direction wrong.
- **D:** Right on alpha but P is not inferior on M^2/T^2; judging by raw excess return (10% vs 14%) rather than risk-adjusted performance gives this.

Q389

GARP PE2 Q8 · Mock 4 Q9

An investor is comparing the performances of two portfolio managers who have been allocated an equal amount of investment funds. The managers apply the same strategy with the same constraints, and their portfolios represent the investor's entire holdings (they are not diversified). The investor gathers the following data about the two managers and the market index:

Measure	Manager 1	Manager 2	Market index
Average return	32%	28%	22%
Beta with respect to market index	1.2	1.4	1.0
Standard deviation of returns	18%	14%	10%
Tracking error	8%	6%	0%

The risk-free rate of interest is 3%. Which of the following is an appropriate measure to use and the correct conclusion to reach when comparing the performances of the two managers?

A. The Modigliani-squared measure, which shows that Manager 1 outperforms Manager 2 by 2%
B. The Modigliani-squared measure, which shows that Manager 2 outperforms Manager 1 by 2%
C. Treynor's measure, which shows that Manager 1 outperforms Manager 2 by 6%
D. Treynor's measure, which shows that Manager 2 outperforms Manager 1 by 6%

Answer: B

Step by step

1. The portfolios are each the investor's entire holdings, so total risk (standard deviation) is the relevant risk: use M-squared (Sharpe-based), not Treynor.
2. Sharpe ratios: Manager 1 $(32-3)/18 = 1.611$; Manager 2 $(28-3)/14 = 1.786$.
3. Risk-adjust each to the market's 10% volatility: $R_{P'} = 3\% + \text{SR} \times 10\%$. Manager 1: 19.11%; Manager 2: 20.86%.
4. Manager 2 beats Manager 1 by $20.86 - 19.11 = 1.75\% \approx 2\%$ (market return is the same benchmark for both, so the M-squared difference equals this gap).
5. Treynor (for comparison only): Manager 1 $29/1.2 = 24.17$; Manager 2 $25/1.4 = 17.86$, a gap of about 6.3 points favoring Manager 1, but Treynor ignores unsystematic risk and is inappropriate here.

Why B is correct: Undiversified, stand-alone portfolios should be ranked on total risk, so M-squared (equivalently Sharpe) applies; Manager 2 has the higher risk-adjusted return by about 1.75%, i.e. roughly 2%.

Why the others are wrong

- **A:** Right measure but reversed ranking; Manager 1's higher raw return (32% vs 28%) ignores its much higher volatility.
- **C:** Uses Treynor, which only suits a portfolio that is one component of a diversified whole; it gives 24.17 vs 17.86, favoring Manager 1 by about 6, but the conclusion is built on the wrong measure.
- **D:** Uses Treynor (wrong for undiversified portfolios) and also misstates the direction: Treynor actually favors Manager 1 (24.17 vs 17.86).

Q390

Mock 1 Q61

REBUILT — Original options used M2 as the risk-adjusted return (10.04%) while Schweser defines M2 as its difference from the market return; reworded options to Schweser's definition so only one answer is correct.

A portfolio has a mean return of 11% and a volatility of 25%. The market portfolio has a mean return of 9.95% and a volatility of 21%. The risk-free rate is 5%. Using the Schweser definition of M-squared (the difference between the return of the risk-adjusted portfolio and the market return), which of the following is most accurate?

A. M^2 = 1.05%; the portfolio outperformed the market
B. M^2 = 2.19%; the portfolio outperformed the market
C. M^2 = 0.09%; the portfolio outperformed the market
D. M^2 = −0.09%; the portfolio underperformed the market

Answer: C

Step by step

1. Match the market's volatility by mixing the portfolio with the risk-free asset: weight in P = $21/25 = 0.84$.
2. Risk-adjusted return: $R_{P'} = 5\% + 0.84 \times (11\% - 5\%) = 5\% + 5.04\% = 10.04\%$.
3. $M^2 = R_{P'} - R_M = 10.04\% - 9.95\% = 0.09\%$.
4. Positive M-squared means the portfolio beat the market at equal risk: outperformed.

Why C is correct: M-squared is the return on the portfolio levered/de-levered to market volatility minus the market return; it is +0.09%, so the portfolio slightly outperformed.

Why the others are wrong
- **A:** Raw return difference $11 - 9.95$ with no volatility adjustment.
- **B:** Inverted the volatility ratio (used 25/21): $5 + 6 \times 1.19 = 12.14$, minus 9.95 = 2.19%; this would lever the portfolio the wrong way.
- **D:** Subtracted in the wrong order (market minus portfolio), flipping the sign and conclusion.

LO 89.d

Determine the statistical significance of a performance measure using standard error and the t-statistic.

Q391

BT answers P2.T8.18.3 · BT notes P2.T9.18.3

REBUILT — Merged two copies; option values were OCR-damaged (173/174, 576/516) and distractors were unexplained, so options were rebuilt around realistic mistakes.

Over a historical measurement period, hedge fund manager Joe Dart produced an alpha of 50 basis points per month (+6.0% per year). The monthly standard deviation of the residual (non-systematic) risk was 4.0%. At a two-tailed 90% significance level (critical value 1.645), approximately how many months (N) of data are required to reject the null hypothesis that his true alpha is zero?

A. 13 months
B. 105 months
C. 173 months
D. 246 months

Answer: C

Step by step
1. The t-statistic of alpha is $t = \alpha/(\sigma_e/\sqrt{N}) = \alpha\sqrt{N}/\sigma_e$.
2. Set $t = 1.645$ (two-tailed 90%) and solve: $N = (t\,\sigma_e/\alpha)^2$.
3. $N = (1.645 \times 4.0\%/0.50\%)^2 = (13.16)^2 = 173.2$.
4. Rounded, about 173 months (roughly 14.4 years).

Why C is correct: The standard error of alpha is $\sigma_e/\sqrt{N}$, so the sample size needed grows with the square of (critical t x residual volatility / alpha).

Why the others are wrong
- **A:** Forgot to square: 1.645 x 4.0/0.50 = 13.2 is the square root of N, not N.
- **B:** Used 1.282 (the one-tailed 90% / two-tailed 80% value) instead of 1.645: (1.282 x 8)^2 = 105.
- **D:** Used 1.96 (two-tailed 95%) instead of 1.645: (1.96 x 8)^2 = 246.

Q392

BT answers P2.T9.21.11.1 · BT notes P2.T9.21.11.1

REBUILT — Merged two copies (one with an OCR-damaged option); replaced weak distractors with ones produced by realistic errors.

Portfolio manager Jane Dart's portfolio produced an alpha of 190 basis points per month (+22.80% per year before compounding). Her beta is 1.30 and the monthly standard deviation of the residual is 4.0%; assume this equals the monthly standard deviation used in the alpha t-test. The null hypothesis is that her alpha is due to luck, tested at a two-sided 10.0% significance level (critical t = 1.645). Approximately how many years of this outperformance are required to reject the null hypothesis?

A. 0.3 years
B. 1.0 year
C. 1.4 years
D. 12.0 years

Answer: B

Step by step
1. $t = \alpha\sqrt{N}/\sigma_e \Rightarrow N = (t\,\sigma_e/\alpha)^2$, with N in months because alpha and sigma are monthly.
2. $N = (1.645 \times 4.0\%/1.90\%)^2 = (3.463)^2 = 11.99$ months.
3. 11.99 months is about 1.0 year.

Why B is correct: Her large monthly alpha relative to residual volatility means only about 12 monthly observations are needed to reach t = 1.645.

Why the others are wrong
- **A:** Forgot to square: 3.46 is the square root of N, giving about 3.5 months = 0.3 years.
- **C:** Used 1.96 (two-tailed 95%) instead of 1.645: (1.96 x 4.0/1.90)^2 = 17.0 months = 1.4 years.
- **D:** Read the 12 months result as 12 years (unit confusion); N is in months.

Q393 Mock 1 Q14

Fund managers A and B both claim superior performance due to active management. Manager A has an excess return (alpha) of 0.12% and a standard error of the alpha of 0.087%. Manager B has an alpha of 0.085% and a standard error of the alpha of 0.039%. Assuming a large sample size of monthly returns, which of the managers can statistically claim superior performance at the 5% significance level?

A. Manager A only.
B. Manager B only.
C. Both Manager A and Manager B.
D. Neither Manager A nor Manager B.

Answer: B
Step by step
1. Test $H_0 : \alpha = 0$ using $t = \alpha/SE(\alpha)$; critical value for large samples at 5% (two-tailed) is 1.96 (about 2).
2. Manager A: $t = 0.12/0.087 = 1.38 < 1.96$, fail to reject.
3. Manager B: $t = 0.085/0.039 = 2.18 > 1.96$, reject H0.
4. Only Manager B shows statistically significant alpha.

Why B is correct: Significance depends on alpha relative to its standard error, not alpha alone; B's smaller alpha is more precisely estimated.
Why the others are wrong
- **A:** Judged by the larger raw alpha (0.12% > 0.085%) and ignored standard errors.
- **C:** Assumed any positive alpha is superior performance; A's t of 1.38 is below the critical value.
- **D:** Over-conservative: B's t of 2.18 exceeds 1.96, so B is significant.

LO 89.e

Describe style analysis.

Q394 Mock 2 Q74

A pension fund manager wants to understand the underlying factors driving an actively managed equity portfolio's returns and compare them with the fund's stated strategy. Which approach would be most appropriate to decompose the portfolio's returns into style and factor exposures?

A. Conduct a returns-based style analysis using a multi-factor regression model.
B. Perform a holdings-based analysis using the individual security characteristics.
C. Apply a Sharpe ratio analysis to compare risk-adjusted returns across different time periods.
D. Implement a tracking error minimization technique to identify style drift.

Answer: A
Step by step
1. Sharpe's style analysis regresses the portfolio's returns on returns of asset class (style) indices.
2. The estimated slopes (nonnegative, summing to 100%) are the effective exposures to each style.
3. Comparing these effective exposures with the stated strategy reveals any mismatch.
4. Only option A describes a return-decomposition regression.

Why A is correct: Returns-based style analysis regresses fund returns on style index returns, and the slopes are the effective style exposures that can be compared with the stated strategy.
Why the others are wrong
- **B:** Holdings-based analysis describes composition, not a decomposition of returns into style exposures; it is not Sharpe's regression method.
- **C:** The Sharpe ratio is a single risk-adjusted performance number and says nothing about style exposures.
- **D:** Tracking error minimization is a portfolio-construction tool; it does not estimate the exposures behind real-

ized returns.

Q395

AnalystPrep Q.2566

REBUILT — Original answer was 'None of the above' with unexplained coefficients and an odd alpha; recast with weights summing to 100% and a direct 6% answer option.

A fund manager runs a Sharpe style analysis on his fund. He regresses the fund's returns on four style portfolios and obtains the following (slopes constrained to be nonnegative and to sum to 100%):

Style portfolio	Regression weight
T-Bills	23%
Small Cap	20%
High P/E	28%
Large Cap	29%
R^2	94%

Which of the following statements is correct?

A. 94% of the variability in the fund's returns is attributable to security selection and market timing.
B. 6% of the variability in the fund's returns is attributable to style (asset allocation) choice.
C. 6% of the variability in the fund's returns is attributable to security selection and market timing.
D. 23% of the variability in the fund's returns is attributable to security selection and market timing.

Answer: C

Key correction: Source (AnalystPrep) key said D (none of the above), which held only because the 6% figure was not offered; options were rebuilt so the 6% statement is listed.

Step by step

1. The weights 23% + 20% + 28% + 29% = 100%, consistent with Sharpe's constraints.
2. Attribution to style (asset allocation) = $R^2 = 94\%$.
3. Attribution to selection (security selection and timing) = $1 - R^2 = 1 - 0.94 = 6\%$.
4. Option C states this.

Why C is correct: R^2 measures the share of return variability explained by style; the residual $1 - R^2 = 6\%$ reflects security selection and timing.

Why the others are wrong

- **A:** Swaps the two components: 94% is the style share, not the selection share.
- **B:** Uses 1 - R^2 for style; the style share is R^2 = 94%.
- **D:** Takes a regression weight (T-Bills 23%) as the selection share; weights are asset allocation, not variability attribution.

Q396

AnalystPrep Q.3184

REBUILT — Source used a non-standard 'style return' (manager minus market index) and an ambiguous stem; recast as Sharpe style-analysis selection attribution against the style benchmark.

A performance analyst applies Sharpe's style analysis to a fund. The fund returned 7.6% over the year. The style benchmark built from the fund's regression-estimated style weights (applied to the asset class index returns) returned 6.2%, while the broad market index returned 8.8%. What return is attributable to the manager's selection (security selection and timing)?

A. +1.4%
B. -1.2%
C. -2.6%
D. +2.6%

Answer: A

Key correction: Source (AnalystPrep) defined style return as manager minus market and answered B (active management but not style); that definition is non-standard and the item was rebuilt.

Step by step

1. Style analysis measures performance against the passive style benchmark with the same effective weights.
2. Selection attribution = fund return - style benchmark return.
3. $7.6\% - 6.2\% = +1.4\%$.
4. The broad market index is not the relevant comparison.

Why A is correct: Selection (and timing) return is the excess of the fund's return over the passive style benchmark: 7.6% - 6.2% = 1.4%.

Why the others are wrong

- **B:** Compares the fund with the broad market (7.6% - 8.8%); the benchmark must match the fund's style.
- **C:** Computes style benchmark minus market (6.2% - 8.8%), a style effect rather than the manager's selection.
- **D:** Computes market minus style benchmark (8.8% - 6.2%), wrong comparison and sign.

Q397

BT answers P2.T8.20.2 · BT notes P2.T9.20.2

To conduct a style analysis on a mutual fund that cannot short securities, you regress the fund's returns against five regressors: T-bills, a small capitalization index, a large capitalization index, a growth index, and a value index. The regression returned is: R(t) = 0*TBILL - 0.2*SMALL + 1.4*LARGE + 0.3*GROWTH + 0.7*VALUE + 1.57. Your colleague Jane criticizes your style analysis with the following statements:

I. You should not have a zero coefficient (0*TBILL)
II. You should not have a non-zero intercept (1.57)
III. You should not have a negative coefficient (-0.2*SMALL)
IV. You should not have a coefficient that exceeds one (1.4*LARGE)
V. The sum of the coefficients should not be different than 1.0 (0 - 0.2 + 1.4 + 0.3 + 0.7 = 2.2, which is not 1.0)

Which of the criticisms is (are) correct?

A. None are correct; the regression is fine
B. I. and II. are correct only
C. III., IV. and V. are correct only
D. All are correct

Answer: C

Step by step

1. Sharpe's constraints: slopes nonnegative and summing to 100% (fully invested, no shorts).
2. I: a zero slope is allowed (no T-bill exposure), so I is wrong.
3. II: the intercept is the return not explained by style (selection and timing, i.e., alpha) and is allowed, so II is wrong.
4. III: -0.2 violates nonnegativity, so III is correct. IV: 1.4 > 1 is impossible with nonnegative weights summing to 1, so IV is correct.
5. V: $0 - 0.2 + 1.4 + 0.3 + 0.7 = 2.2 \neq 1$, so V is correct. Answer: III, IV and V only.

Why C is correct: Style weights must be nonnegative and sum to 100%, so negative, above-one and non-unit-sum coefficients are errors, while a zero slope and a non-zero intercept are acceptable.

Why the others are wrong

- **A:** Ignores that the weights break the nonnegativity and sum-to-one constraints.
- **B:** Wrongly treats a zero slope and the alpha intercept as errors; the intercept captures selection/timing return.
- **D:** Wrongly includes I and II, which are acceptable.

Q398

GARP PE1 Q64 · Mock 3 Q17

REBUILT — Original options B and C were both defensible (sources gave conflicting keys); option B replaced with a clearly false statement about unconstrained slopes.

A group of junior risk analysts at a wealth management firm is discussing whether style analysis would be an appropriate approach to assess a mutual fund's investment decisions. Which of the following statements about Sharpe's style analysis is correct?

A. Style analysis shows that the contribution of security selection to the variation in a fund's returns is usually much higher than the contribution of asset allocation.
B. The regression slopes are unconstrained and may be negative so that they capture short positions in the asset class indices.
C. Returns are regressed on relevant asset class benchmark indices, and the R-squared measures the percentage of return variability attributable to style choice.
D. Returns are regressed on non-tradable factors, and the intercept of the regression measures the average return attributable to style choice.

Answer: C

Key correction: The two sources gave conflicting keys (GARP PE1 said C; Mock 3 said B). C is the one consistent with Schweser; the 'constant exposures' option was replaced.

Step by step

1. Sharpe regresses fund returns on an exhaustive, mutually exclusive set of asset class indices.
2. Percentage of performance due to style (asset allocation) = R^2; due to selection = $1 - R^2$.
3. Sharpe's evidence (Magellan: 97.3% style, 2.7% selection) shows style dominates, so A is reversed.
4. Slopes are constrained nonnegative and sum to 100%, so B is false; regressors are tradable indices and the intercept reflects selection/timing, so D is false.

Why C is correct: Style analysis uses asset class indices as regressors and reads R^2 as the share of variability due to style choice.

Why the others are wrong

- **A:** Reversed: style/asset allocation explains most variation (97.3% for Magellan), selection little.
- **B:** Sharpe constrains slopes to be nonnegative and to sum to 100%; allowing negatives is not the method.
- **D:** Regressors are tradable asset class indices, and the intercept reflects selection and timing, not style.

Q399 Schweser Module Quiz 89.3 Q1

Sharpe's style analysis, used to evaluate an active portfolio manager's performance, measures performance relative to:

A. a passive benchmark of the same style.
B. broad-based market indices.
C. the performance of an equity index fund.
D. an average of similar actively managed investment funds.

Answer: A

Step by step

1. Style analysis regresses fund returns on asset class indices with constrained slopes.
2. The slopes define a passive portfolio with the same effective style (the style benchmark).
3. Performance is the fund's return relative to this custom passive benchmark.

Why A is correct: The estimated style weights form a passive benchmark of the same style, and the residual is the manager's selection return.

Why the others are wrong

- **B:** A broad market index ignores the fund's style tilts.
- **C:** An equity index fund is a single fixed benchmark and need not match the style.
- **D:** Peer-group averages are a different (universe) comparison and not what style analysis uses.

LO 89.f

Explain the difficulties in measuring the performance of actively managed portfolios.

Q400 BT answers P2.T9.21.9.3 · BT Mock Q50

Acme Investment Bank has built the third leg of its three-legged risk management stool, after the risk plan and the risk budget: a risk monitoring process. The bank shares the belief of Rosengarten and Zangari that "risk is the shadow cost behind returns." Each of the following is probably an element of the bank's risk monitoring process EXCEPT which is the least likely to be found there?

A. The bank's monitoring does not depend on any single tool or metric but relies on an ensemble of tools and metrics.
B. The bank measures alpha as a regression intercept; alpha can help evaluate manager skill, but its statistical significance often requires more data (a longer history) than is available.
C. The bank measures return on risk capital (RORC), using VaR as the measure of risk capital, and watches the variability of RORC as well as its level.
D. After a threshold level of data is collected, subjective articulations of each portfolio manager's philosophy become unnecessary and should be discarded in favor of quantitative performance evaluation.

Answer: D

Step by step

1. Evaluate each statement for realism in monitoring practice.
2. A, B and C are true: monitoring uses an ensemble of metrics; alpha significance needs long histories; RORC with VaR as risk capital is monitored in level and variability.
3. D is false: a manager's articulated philosophy remains a necessary complement to quantitative measures (e.g. to detect style drift), so it is never discarded.
4. D is the exception.

Why D is correct: Quantitative measures are noisy and slow to become significant, so qualitative understanding of a manager's philosophy is never made redundant.

Why the others are wrong

- **A:** True statement; relying on one metric would miss risks, so an ensemble is standard.
- **B:** True statement; alpha is a regression intercept and its t-statistic needs many observations.
- **C:** True statement; RORC with VaR as risk capital is a standard monitoring measure, and its variability matters too.

Q401

BT answers P2.T8.19.2 · BT notes P2.T9.19.2

REBUILT — Original options had two weak distractors (0.63, 0.72) and left the SD convention unstated; stated population SD convention and replaced distractors with realistic-mistake values.

In the first year, a hedge fund manager produces quarterly excess returns of -1.0%, +5.0%, -1.0%, +5.0%. In the second year, the manager shifts to a riskier strategy and produces quarterly excess returns of -12.0%, +36.0%, -12.0%, +36.0%. The manager's Sharpe ratio was 0.67 in the first year and 0.50 in the second year (using the population standard deviation of the four quarterly returns). What is the Sharpe ratio of the eight quarters measured as a single sequence, using the same convention?

A. 0.37
B. 0.39
C. 0.52
D. 0.59

Answer: B

Step by step

1. Year 1: mean = 2%, population SD = 3%, Sharpe = $2/3 = 0.67$; year 2: mean = 12%, SD = 24%, Sharpe = 0.50 (consistent with the stem).
2. Combined mean of the 8 quarters = $(8 + 48)/8 = 7.0\%$.
3. Squared deviations from 7%: $(-8)^2 = 64$, $2^2 = 4$, 64, 4, $(-19)^2 = 361$, $29^2 = 841$, 361, 841; sum = 2540; variance = $2540/8 = 317.5$; population SD = 17.82%.
4. Sharpe = $7.0/17.82 = 0.39$.

Why B is correct: The two years have different means, so pooling adds between-year dispersion to the within-year dispersion; the pooled SD (17.82%) is far above the average of the yearly SDs while the mean is only the average, so the pooled Sharpe ratio (0.39) is below both annual ratios. This is why the changing mean and risk of an active manager make Sharpe ratios hard to interpret.

Why the others are wrong

- **A:** Uses the sample (n-1) SD of 19.05% for the pooled series ($7/19.05 = 0.37$), inconsistent with the convention used for the annual figures.
- **C:** Divides the pooled mean by the simple average of the two yearly SDs, $(3 + 24)/2 = 13.5\%$ ($7/13.5 = 0.52$), ignoring the extra variance created by the shift in mean.
- **D:** Averages the two annual Sharpe ratios, $(0.67 + 0.50)/2 = 0.585$, wrongly treating the Sharpe ratio as additive across periods with different risk.

Q402

GARP PE2 Q68

A senior human resources manager at a large alternative asset management company is assessing the process the firm uses to evaluate and compensate portfolio managers based on their performance. The manager reviews several active portfolio management strategies and identifies difficulties in measuring portfolio performance with precision. Which of the following correctly describes a challenge in measuring the performance of actively managed portfolios?

A. Actively managed portfolios typically have long investment horizons, which can cover a variety of market conditions.

B. Portfolio managers typically utilize high leverage and are reluctant to reduce their leverage except when directed by their superiors or by the CRO.
C. Some performance measures can be manipulated both within a measurement period and across several measurement periods.
D. Since bull markets occur more frequently, measures of forecasting ability will tend to reward portfolio managers who always predict market conditions to be bullish.

Answer: C
Step by step
1. Evaluate each statement against the known difficulties: changing mean/risk of active portfolios, short track records, and manipulability of measures.
2. A reverses the problem: active portfolios usually have short, not long, histories, so too few observations cover varied conditions.
3. B and D are unsupported claims; leverage is adjusted by managers (which is itself a manipulation channel) and forecasting measures do not reward always-bullish calls.
4. C is correct: managers can alter leverage, risk or return timing within and across periods to inflate Sharpe-type measures.

Why C is correct: Because the return distribution of an active manager changes over time and the manager controls risk-taking and timing, conventional measures can be gamed both within and across periods.
Why the others are wrong
- **A:** Reverses the facts: the problem is short measurement histories, not long horizons spanning many market conditions.
- **B:** Invented behavior; managers often change leverage themselves (e.g., raising it after a poor start), so reluctance is not the challenge.
- **D:** Market-timing measures (e.g., Treynor-Mazuy/Henriksson-Merton) are built to give no credit for a constant bullish bias; this confuses bull-market frequency with timing skill.

Q403 BT answers P2.T8.19.1 · BT notes P2.T9.19.1

According to Bodie, Kane, and Marcus, EACH of the following makes it difficult to evaluate the performance of hedge funds EXCEPT for:
A. Liquidity risk: Hedge funds tend to hold more illiquid assets, such that compensation for illiquidity may mistakenly appear to be alpha
B. Tail risk: Some hedge fund strategies will earn consistent profits for a period, appearing to be high reward per unit of risk, but are exposed to tail risk; e.g., writing deep OTM puts
C. Incentive structure: Carried interest fee creates a circularity in the evaluation model that is difficult to overcome
D. Instability of risk attributes: As hedge funds have greater leeway to invest opportunistically, their factor loading and risk profile changes rapidly

Answer: C
Step by step
1. Recall the difficulties: nonlinear/tail risk, illiquid holdings (smoothed, serially correlated returns and illiquidity premium looking like alpha), unstable risk exposure over time (including higher correlation in crises).
2. A matches the illiquidity difficulty.
3. B matches nonlinear/tail risk (Sharpe ratio overstated by short-option strategies).
4. D matches the non-constant risk and factor loadings.
5. C (fee-induced circularity) is not a recognized difficulty, so it is the exception.

Why C is correct: The recognized hedge-fund evaluation problems are nonlinear tail risk, illiquidity/smoothed prices and time-varying risk; a circularity from the incentive fee is not among them.
Why the others are wrong
- **A:** A genuine difficulty (illiquidity premium misread as alpha), so it is not the exception; picking it ignores the illiquid-holdings problem.
- **B:** A genuine difficulty (short-option strategies look high-Sharpe until a tail event), so it is not the exception.
- **D:** A genuine difficulty (risk and factor loadings change rapidly, so historical estimates mislead), so it is not the exception.

LO 89.g

Describe performance manipulation and the problems associated with using conventional performance measures.

Q404

BT answers P2.T9.21.11.3 · BT notes P2.T9.21.11.3

REBUILT — Source tested a Bodie-text claim about MRAR that is not in the Schweser reading and used an 'impossible to manipulate' claim; rebuilt around the Schweser LO 89.g dynamic-risk Sharpe-ratio example with computed numbers.

A manager runs a low-risk strategy in Year 1 (average excess return 1% per year, volatility 3%) and, after lagging, switches to a high-risk, highly levered strategy in Year 2 (average excess return 5%, volatility 15%). The market index has a Sharpe ratio of 0.30. Over the combined two years, the portfolio's average excess return is 3% with volatility of 11%. Which statement is most accurate?

A. Each year's Sharpe ratio is 0.333, but the combined two-year Sharpe ratio is about 0.27, below the index; the change in risk level biases the full-period Sharpe ratio downward and can mislead the comparison.

B. The combined two-year Sharpe ratio is about 0.33, the same as each year, so the Sharpe ratio is robust to a mid-period change in risk level.

C. The combined two-year Sharpe ratio is about 0.27, which is the correct measure; the single-year ratios of 0.333 overstate performance and should be disregarded.

D. The combined two-year Sharpe ratio is about 0.20, so the leverage increase clearly destroyed value relative to the index.

Answer: A

Key correction: Source key (D) rested on a Bodie-text claim about MRAR that is not covered in the Schweser reading; the question was rebuilt around the Schweser dynamic-risk Sharpe example.

Step by step

1. Year 1 Sharpe = 1% / 3% = 0.3333 > 0.30 (index).
2. Year 2 Sharpe = 5% / 15% = 0.3333 > 0.30.
3. Combined Sharpe = 3% / 11% = 0.2727 < 0.30.
4. Each sub-period beats the index, yet the full period appears to underperform: the shift in volatility biases the Sharpe ratio downward.
5. Hence conventional measures like Sharpe are unreliable when risk levels change over time (and can be gamed by changing leverage).

Why A is correct: The Sharpe ratio assumes stable risk and return characteristics; when a manager changes risk mid-period, the pooled volatility rises and the full-period ratio no longer reflects either strategy, giving a misleading comparison to the benchmark.

Why the others are wrong

- **B:** Averaging the two yearly ratios (or assuming Sharpe is additive) gives 0.333, ignoring that pooled volatility (11%) is not the average of the yearly volatilities.
- **C:** The 0.27 figure is right, but treating it as the 'correct' measure ignores that it is distorted by the risk shift; it is the biased figure, not the truth.
- **D:** Dividing the 3% average excess return by the Year 2 volatility of 15% (3/15 = 0.20) uses the wrong denominator and overstates the damage.

Q405

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager at a pension fund reviews an active manager who ran a low-risk strategy in Year 1 (average excess return 2%, standard deviation 4%) and then switched to a high-risk strategy in Year 2 (average excess return 6%, standard deviation 12%). The market index has a Sharpe ratio of 0.45. Over the combined two years, the average excess return was 4% with a standard deviation of about 9.2%. Which of the following is correct?

A. The two-year Sharpe ratio is about 0.43, which is the most reliable figure because it uses the largest sample and therefore the sub-period ratios should be ignored.

B. Each year has a Sharpe ratio of 0.50, and the two-year Sharpe ratio is also 0.50 because the Sharpe ratio is unaffected by changes in risk level.

C. The Year 1 Sharpe ratio is 0.50 and the Year 2 Sharpe ratio is 1.50 because the higher excess return in Year 2 is rewarded more heavily.

D. Each year has a Sharpe ratio of 0.50, but the two-year Sharpe ratio is about 0.43, so the full-period measure wrongly suggests underperformance of the index.

Answer: D

Step by step

1. Year 1 Sharpe ratio = 2% / 4% = 0.50, above the index at 0.45.
2. Year 2 Sharpe ratio = 6% / 12% = 0.50, also above the index.
3. Two-year Sharpe ratio = 4% / 9.2% = about 0.43, below the index.
4. The shift in risk level raises the pooled volatility more than proportionally to the average excess return, so the full-period ratio is biased downward and gives a misleading ranking.

Why D is correct: Both sub-period Sharpe ratios (0.50) exceed the index (0.45), yet the pooled ratio (about 0.43) falls below it. Changing risk levels bias the Sharpe ratio downward over the combined period and can lead to incorrect ranking conclusions.

Why the others are wrong

- **A:** Wrong: treats the larger sample as more reliable. With dynamic risk the pooled ratio is the biased one, and the sub-period ratios carry the valid information.
- **B:** Wrong: assumes the Sharpe ratio is robust to dynamic risk. The pooled volatility (9.2%) is not the average of 4% and 12%, so the ratio falls to about 0.43.
- **C:** Wrong: divides Year 2 excess return by the Year 1 volatility (6%/4% = 1.5), mixing periods instead of using the 12% volatility of Year 2.

Q406 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

An investment committee is comparing an actively managed fund with a passive index fund using Sharpe ratios computed over a long history. The active manager is known to have materially changed the fund's leverage and asset mix several times during the sample. Which of the following is the most appropriate conclusion about the use of the Sharpe ratio?

A. The full-sample Sharpe ratio may be biased and may rank the active fund incorrectly, so the committee should account for changes in portfolio composition and risk level when interpreting it.

B. The Sharpe ratio is most reliable for the active fund because it measures total risk and therefore automatically adjusts for changes in leverage over time.

C. The full-sample Sharpe ratio will be biased upward for the active fund, because pooling periods of different risk reduces measured volatility relative to average excess return.

D. The Sharpe ratio is valid only for the passive fund; for the active fund the committee should replace it with the average excess return, since volatility is irrelevant when risk changes.

Answer: A

Step by step

1. The Sharpe ratio suits passive strategies whose risk and return characteristics are roughly constant over time.
2. For active strategies, changes in volatility bias the Sharpe ratio and can give incorrect conclusions versus a benchmark.
3. The committee should therefore consider changes in portfolio composition and risk when using the measure, for example by examining sub-periods.

Why A is correct: Dynamic risk levels bias the Sharpe ratio and can produce incorrect ranking conclusions, so changes in composition must be taken into account.

Why the others are wrong

- **B:** Wrong: total-risk measurement does not make the ratio robust to shifts in risk level; the ratio is best suited to constant-risk (passive) strategies.
- **C:** Wrong: reverses the direction of the bias. Pooling periods with different risk levels raises measured volatility, biasing the ratio downward in the reading example.
- **D:** Wrong: overcorrects. The Sharpe ratio is still informative, and dropping risk altogether ignores the risk adjustment; the problem is interpretation under changing risk, not that volatility is irrelevant.

LO 89.h

Describe techniques to measure the market timing ability of fund managers with a regression and with a call option model and calculate a manager's return due to market timing.

Q407

AnalystPrep Q.2563

REBUILT — Original never said whether the alpha and timing estimates were statistically significant and option B was only defensible by treating 0.05 as meaningful; added the significance assumption and made the timing conclusion explicit.

A portfolio analyst estimated the following market-timing regression results for two portfolios. Assume all estimates (including the timing coefficient) are statistically significant at the 5% level.

Measure	Portfolio A	Portfolio B
Alpha (α)	−1.0%	1.75%
Beta (β)	0.5	0.6
Timing coefficient	0.05	0.00

Which of the following is TRUE about the results?

A. Portfolio A had unsuccessful security selection but shows evidence of successful market timing.

B. Neither portfolio shows evidence of successful market timing.

C. Portfolio B shows evidence of successful market timing.

D. Portfolio B had unsuccessful security selection.

Answer: A

Step by step

1. Alpha measures security selection: Portfolio A's alpha is −1.0% (negative), so selection was unsuccessful; Portfolio B's alpha is +1.75%, so selection was successful.
2. The timing coefficient (M_P, the up-market minus down-market beta) is positive for a successful timer. Portfolio A: $M_P = 0.05 > 0$ and significant, so timing was successful.
3. Portfolio B: $M_P = 0$, so up- and down-market betas are equal and there is no timing evidence.
4. Only statement A is consistent with all of this.

Why A is correct: Alpha captures selection and a significantly positive timing coefficient captures timing; A combines a negative alpha with a positive significant M_P.

Why the others are wrong

- **B:** Wrong: Portfolio A's significant positive timing coefficient is evidence of timing; the error is treating any small coefficient as zero.
- **C:** Wrong: Portfolio B's timing coefficient is 0, so its beta does not change between up and down markets.
- **D:** Wrong: Portfolio B's alpha is positive (+1.75%), indicating successful selection; the sign of alpha was reversed.

Q408

AnalystPrep Q.2564

A fund manager tries to generate excess returns by timing the market and by shifting the fund between the market index and Treasury bills. The manager regresses the portfolio's excess return on the benchmark's excess return using the quadratic (Treynor-Mazuy) specification:

$rp - rf = ap + bp(rm - rf) + cp(rm - rf)^2 + ep$

Regression estimates:

Parameter	Estimate
ap	1.88
bp	0.65
cp	1.92
R^2	0.93

Assuming the estimates are statistically significant, a positive cp signifies that:

A. the fund manager has successfully timed the market.

B. the fund manager was not able to successfully time the market.

C. the beta of the portfolio is constant.

D. the fund manager has negative security-selection skill.

Answer: A

Step by step

1. Differentiate the fitted line with respect to the market excess return: effective beta $= b_p + 2c_p(r_m - r_f)$.
2. With $c_p = 1.92 > 0$, effective beta rises as the market excess return rises and falls as it declines, i.e. the manager holds more market exposure in up markets.
3. A beta that moves in the right direction with the market is the signature of successful timing, so the answer is A.

Why A is correct: A positive coefficient on the squared market excess return makes the portfolio's beta increase with the market return, which is what a successful timer does.

Why the others are wrong

- **B:** Wrong: this would follow from a zero or negative c_p; a positive coefficient is the evidence of timing.
- **C:** Wrong: a nonzero c_p means beta varies with the market return; a constant beta corresponds to $c_p = 0$.
- **D:** Wrong: security selection is measured by a_p (here +1.88), not c_p; the timing coefficient says nothing about selection.

Q409

BT Mock Q60 · BT answers P2.T9.21.12.1 · BT notes P2.T9.21.12.1

REBUILT — Merged three identical items; the source options (7.39%, 46.50%, 83.77%) were arbitrary or OCR-damaged, so replaced distractors with results of realistic formula mistakes.

Following Merton's analogy, perfect market timing is equivalent to owning a call option on the equity portfolio without paying the premium. When the strike is USD 1.00 × e^(rT), the Black-Scholes formula reduces to c = 2N[0.5 σe √T] − 1. Assuming annual forecasts (T = 1.0 year) and equity portfolio volatility σe = 36.0% per annum, which is nearest to the value of perfect market timing as a percentage of the portfolio's value?

A. 7.14%
B. 14.28%
C. 28.12%
D. 57.14%

Answer: B

Step by step

1. Argument: $0.5 \times 0.36 \times \sqrt{1} = 0.18$.
2. $N(0.18) = 0.57142$.
3. $c = 2(0.57142) - 1 = 0.14285$, i.e. about 14.28% of the portfolio value.

Why B is correct: Value of perfect timing = $2N(0.5\sigma_e\sqrt{T}) - 1$, which equals about 14.28% at $\sigma_e = 36\%$.

Why the others are wrong

- **A:** Computed $N(0.18) - 0.5 = 7.14\%$; forgot to multiply by 2.
- **C:** Omitted the 0.5 in the argument: $2N(0.36) - 1 = 28.12\%$.
- **D:** Stopped at $N(0.18) = 57.14\%$, forgetting to double and subtract 1.

Q410

BT answers P2.T8.19.3 · BT notes P2.T9.19.3

REBUILT — Merged two near-identical items; option B was internally contradictory (said research proves timing success yet profitability is not significant), so rewrote the options to a clean EXCEPT item aligned with the Schweser regression treatment.

Regarding techniques for measuring the market timing ability of fund managers, each of the following statements is correct EXCEPT:

A. A simple linear security characteristic line is inadequate for evaluating a market timer because the timer's beta changes with the market forecast.
B. Empirical regression studies of mutual funds show that most managers successfully time the market, with positive up-market minus down-market betas.
C. Adding a quadratic term, $c_p(r_m - r_f)^2$, to the index model is a viable way to measure both security selection and market timing.
D. A dummy-variable regression can be used so that the portfolio beta is β_P in down markets and $\beta_P + M_P$ in up markets.

Answer: B

Step by step

1. A: a timer shifts beta ahead of market moves, so a single constant beta (linear SCL) misspecifies the model: true.
2. C: the quadratic (Treynor-Mazuy) model gives a beta that rises with market return; true.
3. D: the dummy-variable model has D = 0 in down markets and 1 in up markets, so beta is β_P or $\beta_P + M_P$: true.
4. B: empirical evidence indicates M_P is negative for most funds, i.e. little or no timing ability: false, so B is the exception.

Why B is correct: Evidence shows most managers fail to time the market, so the claim that most succeed is the only false statement.

Why the others are wrong

- **A:** True statement, so not the exception; a constant beta cannot represent a timer.

- **C:** True statement describing the quadratic timing regression.
- **D:** True statement describing the dummy-variable regression with M_P as the beta difference.

Q411 BT answers P2.T8.20.1 · BT notes P2.T9.20.1

REBUILT — Merged duplicates; one source had no key and the original distractors (Zero, 0.20, 0.25) did not map to realistic errors, so replaced them.

A market timing score is defined as P = P1 + P2 − 1, where P1 is the proportion of correct forecasts of bull markets and P2 is the proportion of correct forecasts of bear markets. Over ten periods, the first five were bull markets and the last five were bear markets. Mr. Permabull forecast a bull market for the first eight periods and a bear market for the last two. What is his market timing score?

A. 0.70
B. 0.40
C. 1.40
D. 0.60

Answer: B

Step by step

1. Bull periods 1-5: all forecast bull, so $P_1 = 5/5 = 1.0$.
2. Bear periods 6-10: forecast bull in periods 6-8 (wrong) and bear in 9-10 (correct), so $P_2 = 2/5 = 0.4$.
3. $P = 1.0 + 0.4 - 1 = 0.40$.

Why B is correct: The score is P1 + P2 − 1 with P1 = 1.0 and P2 = 0.4, giving 0.40.

Why the others are wrong

- **A:** Used the overall hit rate, 7 correct out of 10 = 0.70, instead of the P1 + P2 − 1 formula.
- **C:** Forgot to subtract 1: $1.0 + 0.4 = 1.40$.
- **D:** Counted 3 of 5 bear forecasts as correct (mixing up the 3 wrong with the 2 right): $1.0 + 0.6 - 1 = 0.60$.

Q412 BT answers P2.T9.21.12.2 · BT notes P2.T9.21.12.2

Patty is an analyst who makes a bull-or-bear forecast each quarter. Over the last two years (eight quarters), the actual market was a bull (positive excess returns) for the first six quarters and a bear (negative excess returns) for the final two quarters. Patty forecast a bull market for the first five quarters and a bear market for the remaining three quarters; she was therefore one quarter early in switching from bull to bear. Which of the following is closest to her market timing score (the sum of the proportions of correct bull and bear forecasts, minus one)?

A. 0.375
B. 0.833
C. 1.000
D. 1.667

Answer: B

Step by step

1. Proportion of bull markets correctly predicted: $P_1 = 5/6 = 0.8333$.
2. Proportion of bear markets correctly predicted: $P_2 = 2/2 = 1.000$ (quarters 7 and 8 were both forecast bearish).
3. Timing score $= P_1 + P_2 - 1 = 0.8333 + 1.000 - 1 = 0.833$.

Why B is correct: A manager with no timing ability has $P_1 + P_2 = 1$ (score 0); the score measures how far the sum of the hit rates exceeds one. Here it is 0.833, against 1.000 for perfect timing.

Why the others are wrong

- **A:** 0.375 equals 3/8, the fraction of quarters in which she was bearish; it is not a timing statistic.
- **C:** 1.000 is the score for perfect timing (1 + 1 − 1); it ignores her one wrong call in quarter 6.
- **D:** 1.667 adds the two hit rates (0.833 + 1.000) but forgets to subtract 1.

Q413 BT answers P2.T9.21.12.3 · BT notes P2.T9.21.12.3

REBUILT — Original relied on a missing scatterplot, had OCR-damaged coefficients (beta 7.37 vs 1.37) and an unclear 'perhaps' answer; rebuilt as a self-contained market-timing regression test using the Schweser up/down-market dummy specification with decisive t-statistics.

A fund's monthly returns over 72 months are used to test for market timing with the regression $R_P - R_F = \alpha_P + \beta_P(R_M - R_F) + M_P D(R_M - R_F) + e_P$, where $D = 1$ in up markets ($R_M > R_F$) and $D = 0$ otherwise. Estimates (standard errors): $\alpha_P = 0.76$ (0.607), $\beta_P = 1.37$ (0.169), $M_P = 0.20$ (0.042). Use a two-sided 95% test with a

critical t-value of about 1.99. Which conclusion is best supported?

A. Beta and market timing are significant, but there is no significant evidence of stock selection skill (alpha)
B. Beta and alpha are significant, but there is no evidence of market timing ability
C. Alpha and market timing are significant, but beta is not significantly different from zero
D. Only beta is significant; there is no evidence of either stock selection skill or market timing ability

Answer: A

Step by step

1. Alpha: $t = 0.76/0.607 = 1.25 < 1.99$, not significant.
2. Beta: $t = 1.37/0.169 = 8.11 > 1.99$, significant.
3. Timing coefficient: $t = 0.20/0.042 = 4.76 > 1.99$, significant and positive, so up-market beta ($1.37 + 0.20 = 1.57$) exceeds down-market beta (1.37).
4. Conclusion: significant beta and timing ability, but no significant alpha.

Why A is correct: Each coefficient is tested as t = estimate / standard error against about 1.99; only alpha falls inside the acceptance region, while a significantly positive M_P indicates successful market timing.

Why the others are wrong

- **B:** Treats alpha (t = 1.25) as significant and overlooks that M_P has t = 4.76; likely from judging the 0.76 estimate as large rather than computing its t-statistic.
- **C:** Wrongly claims beta is insignificant (its t is 8.11) and alpha significant (t is only 1.25); inverts the t-statistics.
- **D:** Correctly finds beta significant and alpha insignificant, but misses that M_P (t = 4.76) is significant, perhaps from assuming market timing cannot be detected or confusing 0.20 with a small estimate.

LO 89.i

Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution.

Q414

AP Mock 3 Q7

REBUILT — Original options were unsigned magnitudes (0.80-1.00%) although the true excess return is negative; replaced with signed options built from realistic mistakes.

A portfolio manager manages a portfolio that tracks the XYZ index. The composition of the index and the portfolio performance are given below:

Sector	Index weight	Index return	Portfolio weight	Portfolio return
Infrastructure	0.20	10.00%	0.15	12.00%
Financial services	0.15	15.00%	0.10	15.00%
Utilities	0.20	20.00%	0.25	22.00%
Energy	0.10	8.00%	0.05	5.00%
IT	0.20	18.00%	0.25	10.00%
Fixed income	0.15	15.00%	0.20	12.00%

Which of the following is closest to the excess return (portfolio minus index) generated by the portfolio manager?

A. -0.95%
B. +0.95%
C. -1.95%
D. +1.00%

Answer: A

Step by step

1. Index return = $0.20(10) + 0.15(15) + 0.20(20) + 0.10(8) + 0.20(18) + 0.15(15) = 14.90\%$.
2. Portfolio return = $0.15(12) + 0.10(15) + 0.25(22) + 0.05(5) + 0.25(10) + 0.20(12) = 13.95\%$.
3. Excess return = $13.95\% - 14.90\% = -0.95\%$.
4. Check via attribution: allocation $\sum(w_p - w_b)R_b = +1.00\%$; selection $\sum w_p(R_p - R_b) = -1.95\%$; aggregate $= +1.00\% - 1.95\% = -0.95\%$.

Why A is correct: Aggregate contribution (asset allocation plus selection) equals portfolio return minus benchmark return, which is negative here because the large IT overweight earned 10% versus 18% for the index IT sector.

Why the others are wrong

- **B:** Right magnitude but wrong sign: the portfolio underperformed, it did not outperform.
- **C:** This is the selection effect alone, $\sum w_p(R_p - R_b)$, ignoring the +1.00% allocation effect.
- **D:** This is the allocation effect alone, $\sum(w_p - w_b)R_b$, ignoring the selection effect.

Q415 AnalystPrep Q.3185

Johny Nara is a research analyst at a large investment bank. He performed an attribution analysis for the Sethi Investment Portfolio and determined that the sector effect was 0.322%, the within-sector selection effect was -0.157%, and the allocation/selection interaction effect was 0.061%. If the benchmark return was 8.441%, how much value did the portfolio manager add, and what was the portfolio's return during the period?

A. 0.226%, 8.667%
B. 0.418%, 8.859%
C. 0.226%, 8.215%
D. 0.324%, 7.534%

Answer: A

Step by step

1. Value added (aggregate contribution) = sum of the effects = $0.322 - 0.157 + 0.061 = 0.226\%$.
2. Portfolio return = benchmark + value added = $8.441\% + 0.226\% = 8.667\%$.

Why A is correct: The attribution components sum to the portfolio's active return, which is added to the benchmark return.

Why the others are wrong

- **B:** Subtracted the negative selection effect and the interaction: $0.322 + 0.157 - 0.061 = 0.418\%$ (sign errors), then added to the benchmark.
- **C:** Correct value added but subtracted it from the benchmark instead of adding it.
- **D:** Components do not sum to 0.324%; the resulting return (7.534%) is not benchmark plus any combination of the effects.

Q416 AnalystPrep Q.3186

REBUILT — The original exhibit was garbled (unrecoverable ordering of ending values); rebuilt as a self-contained two-step attribution with the same 100,000 start, 7% passive and 16% actual returns.

A fund began the year with 100,000 and had no contributions or withdrawals. Its actual return was 16% (ending value 116,000). A performance attribution gives the following ending values:

Passive policy benchmark portfolio (benchmark weights, benchmark index returns)	107,000
Actual asset allocation held at benchmark index returns	110,000
Actual portfolio (actual allocation and actual security selection)	116,000

What was the contribution of the manager's asset allocation decision to the fund's return?

A. 3%
B. 6%
C. 9%
D. 10%

Answer: A

Step by step

1. Policy benchmark return = $107{,}000/100{,}000 - 1 = 7\%$.
2. Actual allocation at index returns = $110{,}000/100{,}000 - 1 = 10\%$.
3. Asset allocation contribution = $10\% - 7\% = 3\%$.
4. Selection contribution = $16\% - 10\% = 6\%$; total excess = 9%.

Why A is correct: Asset allocation is the return from deviating from benchmark weights, measured at benchmark index returns: actual-allocation return minus policy-benchmark return.

Why the others are wrong

- **B:** This is the selection contribution (16% - 10%).
- **C:** This is the total excess return over the passive benchmark (16% - 7%), allocation plus selection.
- **D:** This is the return of the actual allocation at index returns, not its excess over the benchmark.

Q417 AnalystPrep Q.3187

Lilly Johana is a bond portfolio manager decomposing the sources of return of her portfolio, which returned 10%. The actual treasury yield was 8%, which is 0.5% better than the expected yield of 7.5%. She has also determined that her portfolio benefited 0.50% from sector allocation and 0.25% from allocation/selection interaction. How much of the portfolio's overall return is attributable to within-sector selection?

A. 1.25%
B. 1.00%
C. 1.75%
D. 2.50%

Answer: A

Step by step

1. Total return = expected yield + unexpected yield + sector allocation + interaction + within-sector selection.
2. Known components: $7.50 + 0.50 + 0.50 + 0.25 = 8.75\%$.
3. Within-sector selection = $10.00 - 8.75 = 1.25\%$.

Why A is correct: The aggregate return is the sum of all attribution components, so selection is the residual after the other components are removed.

Why the others are wrong

- **B:** Treated the 8% actual yield as the base and also subtracted the 0.50% surprise: 10 - 8 - 0.5 - 0.5 = 1.0, double-counting the unexpected yield and omitting the interaction.
- **C:** Omitted the 0.50% sector allocation effect: 10 - 7.5 - 0.5 - 0.25 = 1.75.
- **D:** Subtracted only the expected yield: 10 - 7.5 = 2.5, attributing all excess return to selection.

Q418 BT answers P2.T8.20.3 · BT notes P2.T9.20.3

A portfolio manager's bogey is a benchmark portfolio invested 60.0% in the S&P 500, 30.0% in a Lehman bond index, and 10.0% in a money market fund. The manager's actual portfolio held 70.0% equities, 20.0% bonds and 10.0% cash. The managed portfolio returned 3.1% against 4.0% for the bogey.

Bogey portfolio:

Component	Benchmark index	Weight	Index return	Bogey contribution
Equity	S&P 500	60.0%	5.0%	3.0%
Bonds	Lehman	30.0%	3.0%	0.9%
Cash	Money market	10.0%	1.0%	0.1%
Return on bogey	|	| 4.0%		

Managed portfolio:

Component	Actual weight	Actual return	Contribution
Equity	70.0%	4.0%	2.8%
Bonds	20.0%	1.0%	0.2%
Cash	10.0%	1.0%	0.1%
Return on managed portfolio	|	3.1%	
Excess return of managed portfolio	|	-0.9%	

If the excess return of -0.9% is decomposed into asset allocation and security selection, what is the contribution to the excess return from security selection?

A. -1.10%
B. -0.70%
C. -0.20%
D. +0.20%

Answer: A

Step by step

1. Selection = $\sum w_p(R_p - R_b)$ (actual weights).
2. Equity: $0.70(4.0 - 5.0) = -0.70\%$; bonds: $0.20(1.0 - 3.0) = -0.40\%$; cash: $0.10(1.0 - 1.0) = 0$.
3. Selection = $-0.70 - 0.40 + 0 = -1.10\%$.
4. Check: allocation $\sum(w_p - w_b)R_b = 0.10(5) + (-0.10)(3) + 0 = +0.20\%$; $0.20 - 1.10 = -0.90\%$.

Why A is correct: Selection measures how much each asset class return deviated from its index, weighted by the weight actually held; it is the part of the -0.9% excess not explained by the +0.20% allocation effect.

Why the others are wrong

- **B:** Counted only the equity selection effect (-0.70%) and omitted bonds.
- **C:** Sign-flipped allocation effect; not a component of the decomposition.
- **D:** This is the asset allocation contribution, not selection.

Q419 BT answers P2.T9.21.11.2 · BT notes P2.T9.21.11.2

REBUILT — Two source copies were identical and one had a weak option set (0.75%, 4.0%); merged and replaced distractors with results of realistic errors.

Peter manages a portfolio measured against a bogey with 50.0% equity, 30.0% bonds and 20.0% cash.

Bogey portfolio:

Component	Benchmark index	Weight	Index return	Bogey contribution
Equity	S&P 500	50.0%	8.0%	4.0%
Bonds	Lehman	30.0%	4.0%	1.2%
Cash	Money market	20.0%	1.0%	0.2%
Return on bogey	|	| 5.4%		

Managed portfolio:

Component	Actual weight	Actual return	Contribution
Equity	80.0%	11.0%	8.8%
Bonds	10.0%	5.0%	0.5%
Cash	10.0%	1.0%	0.1%
Return on managed portfolio	|	9.4%	

Peter's portfolio outperformed the bogey by 4.0%. Of this excess return, how much is attributable to security selection?

A. 1.50%
B. 2.50%
C. 3.40%
D. 1.80%

Answer: B

Step by step

1. Selection = $\sum w_p(R_p - R_b)$.
2. Equity: $0.80(11 - 8) = 2.40\%$; bonds: $0.10(5 - 4) = 0.10\%$; cash: $0.10(1 - 1) = 0$.
3. Selection = $2.40 + 0.10 = 2.50\%$.
4. Check: allocation $0.30(8) + (-0.20)(4) + (-0.10)(1) = 1.50\%$; $1.50 + 2.50 = 4.00\%$.

Why B is correct: Security selection is the actual-weighted return difference versus the benchmark index in each asset class; it equals total excess return minus the allocation effect.

Why the others are wrong

- **A:** This is the asset allocation contribution, not selection.
- **C:** Added the unweighted bond return difference (1.0%) to the weighted equity effect: 2.40 + 1.0 = 3.40.
- **D:** Weighted return differences by bogey weights instead of actual weights: 0.50(3)+0.30(1)=1.80.

Q420 BT answers P2.T9.21.8.2 · BT notes P2.T9.21.8.2

A fund's benchmark (bogey) is a paper portfolio equally weighted between equities (S&P 500) and a bond index. After one year the benchmark's excess return was +3.50%.

Component	Benchmark weight	Benchmark excess return	Fund weight	Fund excess return
Equity (S&P 500)	50.0%	5.00%	80.0%	7.00%
Bonds	50.0%	2.00%	20.0%	1.00%
Total	| 3.50%	| 5.80%		

The fund's excess return was +5.80%, so it outperformed its benchmark by 2.30%. Asset allocation refers to the fund's decision to vary from the 50/50 mix; security selection refers to performance within an asset class. Which of the following statements about the fund's performance is TRUE?

A. Both asset allocation and security selection (active management) contributed positively to the fund's outperformance
B. Equities were a drag on the fund's performance in regard to both asset allocation and security selection, but bonds over-compensated
C. The fund's outperformance is due to a reduction in both policy-mix risk and active-management risk
D. Neither asset allocation nor security selection (active management) contributed positively to the fund's out-

performance

Answer: A

Step by step

1. Allocation $= \sum(w_p - w_b)R_b = (0.80 - 0.50)(5\%) + (0.20 - 0.50)(2\%) = 1.50 - 0.60 = +0.90\%$.
2. Selection $= \sum w_p(R_p - R_b) = 0.80(7 - 5) + 0.20(1 - 2) = 1.60 - 0.20 = +1.40\%$.
3. Total $0.90 + 1.40 = 2.30\%$, matching the outperformance; both components are positive.

Why A is correct: Both effects are positive: the overweight in the higher-returning equity class added 0.90% and net within-class selection added 1.40%.

Why the others are wrong

- **B:** Equities contributed positively to both effects (+1.50% allocation, +1.60% selection); the bond underweight/underperformance was the drag.
- **C:** Outperformance came from taking more risk (80% equity, a large departure from the policy mix), not less.
- **D:** Contradicted by the calculation: both effects (+0.90%, +1.40%) are positive.

Q421 GARP PE1 Q61 · Mock 3 Q20

A fund of funds (FOF) manager is conducting a performance attribution analysis for a portfolio consisting of equity and fixed-income securities to evaluate the effects of the portfolio manager's asset allocation and security selection decisions. The FOF manager uses the information presented below:

Asset class	Portfolio weight	Benchmark weight	Portfolio return	Benchmark return
Equity	58%	50%	8%	11%
Fixed income	42%	50%	6%	7%

What is the contribution of the portfolio manager's asset allocation decision to the portfolio's overall excess return?

A. -2.16%
B. -1.84%
C. -0.16%
D. 0.32%

Answer: D

Step by step

1. Asset allocation contribution = $\sum(w_P - w_B)\, r_B$ (portfolio weights versus benchmark weights, valued at benchmark returns).
2. Equity: $(0.58 - 0.50) \times 11\% = 0.08 \times 11\% = +0.88\%$.
3. Fixed income: $(0.42 - 0.50) \times 7\% = -0.08 \times 7\% = -0.56\%$.
4. Allocation = $0.88\% - 0.56\% = +0.32\%$. (Check: because the weight differences sum to zero, measuring against the total benchmark return of 9.0% gives the same answer: $0.08 \times 2\% + (-0.08)(-2\%) = 0.32\%$.)
5. Cross-check: portfolio return $= 0.58(8\%) + 0.42(6\%) = 7.16\%$, benchmark $= 9.0\%$, excess $= -1.84\%$; selection $\sum w_P(r_P - r_B) = -2.16\%$; allocation + selection $= 0.32\% - 2.16\% = -1.84\%$.

Why D is correct: Allocation isolates the effect of over- or underweighting asset classes relative to the benchmark, valued at benchmark returns. The manager overweighted equity (which returned 11%) and underweighted fixed income (7%), a favorable tilt worth +0.32%.

Why the others are wrong

- **A:** This is the selection effect $\sum w_P(r_P - r_B) = 0.58(-3\%) + 0.42(-1\%) = -2.16\%$, not the allocation effect.
- **B:** This is the total excess return (7.16% - 9.00%), which is allocation plus selection, not allocation alone.
- **C:** This is the interaction term $\sum(w_P - w_B)(r_P - r_B) = 0.08(-3\%) + (-0.08)(-1\%)$, i.e. weight differences multiplied by return differences, not benchmark returns.

Reading 90. Hedge Fund Investment Strategies

David P. Stowell and Paul Stowell, Investment Banks, Hedge Funds, and Private Equity, 4th Edition (Elsevier, 2023). Chapter 12.

LO	Learning objective	Qs
90.a	describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles.	8
90.b	compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies.	5
90.c	describe the mechanics and risks of event-driven strategies such as merger arbitrage.	6
90.d	explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy.	3

LO 90.a

Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles.

Q422 BT answers P2.T8.412.1 · BT notes P2.T9.412.1

You receive a general solicitation from a hedge fund, which is now possible under the 2012 JOBS Act. The fund advertises its style as a long/short equity manager. However, you are skeptical of each of the solicitation's claims. In fact, each of the following claims from the long/short equity manager is dubious or unlikely EXCEPT which is the MOST PLAUSIBLE or MOST LIKELY?

A. As a long/short equity manager, we are located in a rare (uncommon) style with few competitors who also employ a long/short strategy

B. As a long/short equity manager, we are non-directional with virtually no exposure to the stock market

C. As a long/short equity manager, you might find us long small-cap stocks and emerging markets but short large-cap stocks

D. As a long/short equity manager, we try to avoid the idiosyncratic risk that attaches to so-called "stock pickers" in favor of acknowledging that markets are efficient

Answer: C

Step by step

1. Long/short equity relies on fundamental stock selection: long positions in stocks expected to be underpriced and short positions in stocks expected to be overpriced, to earn alpha.
2. Its main goal is to reduce (not eliminate) market risk, so some directional exposure and other factor exposures (for example size or region tilts) typically remain.
3. Test each claim: A (rare style) is false because long/short equity is one of the most widely used hedge fund styles. B (virtually no market exposure) is false because the strategy keeps residual directional exposure. D (avoid stock-picking, markets are efficient) contradicts the strategy, whose alpha comes from exploiting mispricings.
4. C describes a net tilt toward small-cap and emerging-market longs against large-cap shorts, which is a realistic residual factor exposure for a stock picker, so it is the only plausible claim.

Why C is correct: Long/short equity managers pick stocks on both sides but do not fully hedge away market and factor exposures, so a long small-cap/emerging, short large-cap tilt is plausible while the other claims contradict the strategy's nature.

Why the others are wrong

- **A:** Wrong: long/short equity is among the most crowded and widely used hedge fund styles, not a rare niche; this answer confuses popularity with scarcity.
- **B:** Wrong: the strategy only reduces market risk; it keeps some directional exposure and factor exposures beyond beta, so claiming virtually zero market exposure confuses it with equity market neutral.
- **D:** Wrong: alpha comes from fundamental stock selection (idiosyncratic risk is the source of return); claiming market efficiency would remove the rationale for the strategy.

Q423 AP Mock 3 Q8

REBUILT — Original options included both 'managed futures' and 'trend follower' (essentially the same strategy, outside the Schweser reading); replaced with strategies from the reading so exactly one fits.

A fund manager has recently started his own hedge fund. The fund's investors demand that the fund take large, leveraged, discretionary positions across equity, fixed-income, currency and commodity markets, with a highly active asset allocation that bets on a diverse range of macroeconomic risk factors over time. Which of the following is the most suitable hedge fund style for this manager?

A. Merger arbitrage.
B. Global macro.
C. Dedicated short.
D. Nonhedged equity.

Answer: B

Step by step

1. The mandate is broad: multiple asset classes (equities, fixed income, FX, commodities) and discretionary, leveraged bets driven by macroeconomic views.
2. Global macro funds make large, leveraged directional bets across these markets, using futures, swaps and options, and have broad discretion on where to invest.
3. Merger arbitrage, dedicated short and nonhedged equity are each confined to a narrow opportunity set (deal completion, overvalued equities, long-only equities), so none matches the mandate.

Why B is correct: Only global macro allocates capital opportunistically across all major asset classes and macro risk factors with leverage and discretion.

Why the others are wrong

- **A:** Wrong: merger arbitrage is an event-driven strategy that bets on deal completion, a single narrow risk, not a rotation across risk factors.
- **C:** Wrong: dedicated short only sells overvalued securities short (mainly equities); it is not a diversified multi-asset allocator.
- **D:** Wrong: nonhedged equity is long-only stock picking in equities, so it cannot span currencies, rates and commodities.

Q424 BT Mock Q55 · BT answers P2.T9.21.13.1 · BT notes P2.T9.21.13.1

REBUILT — Chimera's 'classified as relative value' claim was a second arguable inconsistency not supported by the reading; removed it so only Daphne's description is inconsistent.

A pension fund wants to add one hedge fund strategy and is reviewing four brochures. The strategies are global macro, merger arbitrage, long/short equity and equity market neutral.

Acheilus - Global Macro: "We opportunistically make bets in different markets using a range of strategies; think of us as betting on factors such as commodities, interest rates and exchange rates."

Boreas - Merger Arbitrage: "We collect nickels in front of the proverbial steamroller! Like our sibling event-driven strategy (Distressed), we offer nonlinear returns but with tail exposure to extreme market conditions."

Chimera - Long/Short Equity: "Although we have some directional market exposure, we are stock pickers seeking alpha on both our long and short positions."

Daphne - Equity Market Neutral: "We bring you a basket of betas! As an event-driven strategy, our returns are highly linked to common market risk factors including equity indices, bond spreads and volatility factors."

Each description is consistent with its strategy EXCEPT one. Select the fund whose description is NOT consistent.

A. Acheilus.
B. Boreas.
C. Chimera.
D. Daphne.

Answer: D

Step by step

1. Acheilus: global macro takes leveraged, discretionary bets across commodities, rates and FX; consistent.
2. Boreas: merger arbitrage is event-driven and earns small, frequent gains with a large loss if a deal fails (short-put-like payoff, tail exposure); consistent.
3. Chimera: long/short equity reduces but does not eliminate market risk, so some directional exposure remains, and alpha comes from stock selection on long and short sides; consistent.
4. Daphne: equity market neutral is an equity-based long/short strategy designed to hold near-zero market ex-

posure, not an event-driven strategy, and it should not load heavily on common market factors. Both claims contradict the strategy, so Daphne is inconsistent.

Why D is correct: Equity market neutral aims to neutralize market (beta) exposure and is not event-driven, so a brochure claiming event-driven status and heavy common-factor exposure is inconsistent.

Why the others are wrong

- **A:** Wrong: the global macro description (leveraged bets on commodities, rates, FX) is accurate; picking it reflects not recognizing macro as multi-factor allocation.
- **B:** Wrong: merger arbitrage is event-driven with nonlinear, tail-exposed payoffs; the 'nickels in front of a steamroller' image is the standard description.
- **C:** Wrong: long/short equity retains some directional exposure and relies on stock-picking alpha, so the description is consistent; confusing it with market neutral leads to this choice.

Q425

BT answers P2.T8.411.2 · BT notes P2.T9.411.2

Fung and Hsieh describe one hedge fund strategy in the following way: "[These] fund managers are known to be highly dynamic traders, often taking highly leveraged bets on directional movements in exchange rates, interest rates, commodities, and stock indices in an opportunistic manner. We can think of [them] as a highly active asset allocator betting on a range of risk factors over time. Over a reasonably long time frame, opportunities in different risk factors will come and go." To which strategy does this description refer?

A. Merger arbitrage; aka, risk arb
B. Global Macro
C. Event-driven distressed
D. Convertible arbitrage

Answer: B

Step by step

1. Key phrases: leveraged directional bets on exchange rates, interest rates, commodities and stock indices; opportunistic; active asset allocator across risk factors.
2. These are the hallmarks of global macro, which makes large leveraged bets across asset classes based on macroeconomic views and has broad discretion.
3. Merger arbitrage, distressed and convertible arbitrage are each tied to a specific security-level opportunity, not a rotation across markets.

Why B is correct: Global macro managers act as active, leveraged asset allocators betting on macro risk factors across currencies, rates, commodities and equity indices.

Why the others are wrong

- **A:** Wrong: merger arbitrage profits from deal completion spreads, not from directional bets across asset classes.
- **C:** Wrong: distressed investing buys the securities of troubled firms and depends on restructuring outcomes, not macro factor rotation.
- **D:** Wrong: convertible arbitrage is long convertible bonds and short stock to isolate volatility and income, a relative-value rather than directional strategy.

Q426

BT notes P2.T9.410.1

Your wealthy Aunt Betty has capital to invest and is trying to choose between a hedge fund and a mutual fund. According to Fung and Hsieh, which of the following is the best argument in favor of a hedge fund investment rather than a mutual fund investment?

A. She wants to minimize the total fees charged, which reduce her net return
B. She prefers the investor protections associated with the greater regulations imposed on hedge funds
C. She seeks a manager with the skill set to market-time, fund leveraged positions, and/or carry short positions
D. She prefers a simple strategy that is conducive to a high degree of transparency

Answer: C

Step by step

1. Hedge funds differ from mutual funds mainly by flexibility: they can short sell, use leverage and derivatives, hold concentrated positions and invest across asset classes.
2. That flexibility requires a distinct skill set: timing markets, funding leveraged positions and carrying short positions for extended periods. This matches C.

3. A is backwards (hedge funds charge higher fees), B is backwards (hedge funds are less regulated), D is backwards (hedge fund strategies are complex and less transparent).

Why C is correct: The distinguishing advantage of a hedge fund is a manager able to short, lever and time markets, which long-only mutual funds generally do not do.

Why the others are wrong

- **A:** Wrong: hedge funds typically charge management plus performance fees, higher than mutual funds, so fees argue against hedge funds.
- **B:** Wrong: hedge funds face lighter regulation and fewer investor protections than mutual funds.
- **D:** Wrong: hedge fund strategies tend to be complex and opaque, so transparency favors mutual funds.

Q427 Schweser Module Quiz 90.1 Q1

Which of the following hedge fund strategies is most likely to earn an interest rebate as a source of income?

A. Global macro.
B. Equity long/short.
C. Emerging markets.
D. Nonhedged equity.

Answer: B

Step by step

1. An interest rebate is interest earned on the cash proceeds of short sales, net of broker and lender fees.
2. Only strategies that short securities generate such proceeds; equity long/short has short positions.
3. Global macro and emerging markets are not defined by short-sale proceeds; nonhedged equity is long only, so it has no short proceeds.

Why B is correct: The rebate arises from cash proceeds of short sales, which equity long/short (like dedicated short) routinely generates.

Why the others are wrong

- **A:** Global macro makes leveraged directional bets across markets; its returns do not rely on short-sale cash rebates.
- **C:** Emerging markets funds invest in developing-country securities and sovereign debt; a rebate is not a defining income source.
- **D:** Nonhedged equity is long only, so there are no short-sale proceeds to earn a rebate.

Q428 Schweser Module Quiz 90.1 Q2

Which of the following statements best describes the primary risk associated with a dedicated short strategy?

A. Lack of leverage prevents meaningful alpha generation.
B. Short sale proceeds generate insufficient interest rebates to offset costs.
C. Market prices tend to decline over long periods, limiting upside potential.
D. Rising share prices can trigger margin calls and forced covering of positions.

Answer: D

Step by step

1. A short position loses when the security's price rises, with potentially unlimited loss.
2. Rising prices trigger margin calls, short squeezes and forced covering driven by risk constraints, locking in losses.
3. Hence the primary risk is D; the other statements misstate the strategy's characteristics.

Why D is correct: Short positions lose as prices rise, which can force margin calls, short squeezes and forced covering.

Why the others are wrong

- **A:** Hedge funds can use leverage; leverage is not what limits dedicated short alpha, and this is not the primary risk.
- **B:** Interest rebates are a return source, not the primary risk; insufficient rebates is a cost issue, not the key danger.
- **C:** Reverses the market fact: prices tend to rise over time, which is the headwind for short sellers.

Q429 AnalystPrep Q.3033

REBUILT — Original asked for the composition of a specific vendor index with a giveaway 'All of the above'; rebuilt to test the emerging markets strategy's investment universe and risk profile per Schweser.

A hedge fund manager runs an emerging markets strategy. Which of the following statements best describes the strategy and its return and risk profile?

A. It invests in corporate and sovereign securities of developing countries, seeking higher returns but bearing added political, liquidity, and local economic risks.

B. It invests only in sovereign debt of developing countries, and its risk is limited because rapid economic growth removes liquidity and political risk.

C. It takes long and short positions in related developed-market equities to hedge market risk, so returns come almost entirely from security selection alpha.

D. It makes large leveraged directional bets across currencies, commodities, and fixed income based on macroeconomic forecasts, trading mainly very liquid instruments.

Answer: A

Step by step

1. Schweser: an emerging markets strategy invests in the securities of developing countries, both corporate securities and sovereign debt.
2. These markets feature rapid growth and industrialization, so the goal is higher returns.
3. The higher return potential comes with heightened risks unique to emerging markets: political conditions, liquidity constraints, and local economic factors.
4. Only option A states the full investment universe and the matching risk profile.

Why A is correct: Emerging markets funds cover both corporate and sovereign securities of developing countries and are compensated for political, liquidity, and local economic risks.

Why the others are wrong

- **B:** Wrongly restricts the universe to sovereign debt and claims growth eliminates risk; the strategy includes corporate securities and has heightened liquidity and political risk.
- **C:** Describes equity long/short, which is a developed-market style hedge of market exposure, not an emerging markets strategy.
- **D:** Describes global macro, which trades liquid instruments with leveraged directional bets across asset classes.

LO 90.b

Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies.

Q430 AnalystPrep Q.3034

REBUILT — Original swap-spread question had vague, overlapping options (including 'all of the above') and a muddled key; rebuilt as a fixed-income arbitrage convergence trade consistent with the Schweser example.

The yield on hedged Italian government bonds has temporarily risen above the yield on comparable U.S. Treasuries, with no change in fundamentals. A fixed-income arbitrageur expects the two yields to converge. Which position best exploits this view?

A. Buy the hedged Italian government bonds and short the U.S. Treasuries.

B. Short the hedged Italian government bonds and buy the U.S. Treasuries.

C. Buy both the Italian government bonds and the U.S. Treasuries.

D. Short both the Italian government bonds and the U.S. Treasuries.

Answer: A

Step by step

1. A higher yield means a lower price, so the hedged Italian bonds are cheap relative to Treasuries.
2. Convergence means the Italian yield falls (price rises) relative to the Treasury yield.
3. Profit is earned by buying the cheap Italian bonds and shorting the rich Treasuries.

Why A is correct: Fixed-income arbitrage is a long/short relative-value trade: buy the relatively cheap (higher-yield) security and short the rich one, profiting as the spread converges.

Why the others are wrong

- **B:** Reverses the direction: shorting the cheap bonds loses when the spread narrows.

- **C:** A long-only position in both is not a spread trade and does not profit from convergence.
- **D:** A short-only position in both is also not a spread trade; it is exposed to the general level of rates.

Q431 AnalystPrep Q.2576

REBUILT — Original asked only for a strategy label with giveaway distractors; rebuilt to test the actual on-the-run/off-the-run trade.

The on-the-run 30-year Treasury bond trades at a premium to the off-the-run 29½-year Treasury issued six months earlier. Which trade would a fixed-income arbitrageur most likely establish, and from what source does it profit?

A. Buy the on-the-run bond and short the off-the-run bond; profit as the premium widens.
B. Short the on-the-run bond and buy the off-the-run bond; profit as the on-the-run premium fades.
C. Buy both bonds; profit from the higher coupon income.
D. Short both bonds; profit from a rise in Treasury yields.

Answer: B

Step by step

1. The on-the-run bond is more expensive (more liquid); the off-the-run bond is cheaper.
2. The arbitrageur shorts the rich bond and buys the cheap bond.
3. As the premium fades over time, the short leg gains relative to the long leg, giving a small positive profit.

Why B is correct: The liquidity premium on the on-the-run issue fades as it ages, so shorting it against the cheaper off-the-run issue earns a small, hedged convergence profit.

Why the others are wrong

- **A:** Reverses the legs and bets on the premium widening, contrary to the expected fade.
- **C:** A long-only position is not an arbitrage and does not exploit the price difference.
- **D:** A short-only position is a directional bet on yields, not a relative-value trade.

Q432 AnalystPrep Q.2577

A company has issued bonds with an option to convert the bonds into equity after three years. Due to an error in the conversion factor, the convertible bonds appear undervalued when compared to the projected market price of the company's stock at the time of conversion. A hedge fund manager wants to exploit the trading opportunity arising from this underestimation. The most appropriate strategy is:

A. Going short on the convertible security and long on the common stock of the company.
B. Going short on the convertible security and short on the common stock of the company.
C. Going long on the convertible security and short on the common stock of the company.
D. Going long on common stock and long on the convertible security.

Answer: C

Step by step

1. The convertible is undervalued, so buy it.
2. The convertible carries equity (delta) risk; short the underlying stock to hedge it.
3. Result: long convertible, short stock (convertible arbitrage).

Why C is correct: Convertible arbitrage buys the cheap convertible and shorts the issuer's stock to neutralize equity exposure while capturing the mispricing.

Why the others are wrong

- **A:** Shorts the undervalued bond and buys the stock: the reverse of the hedge, so it bets against the mispricing and doubles equity exposure.
- **B:** Shorts the undervalued convertible, which loses when its price corrects, and the short stock does not hedge a short convertible.
- **D:** Buys both, so equity risk is doubled rather than hedged.

Q433 Schweser Module Quiz 90.2 Q1

Which of the following characteristics best defines a convertible arbitrage strategy?

A. Buying undervalued sovereign bonds and shorting corporate bonds.
B. Taking offsetting long and short positions across different yield curves.
C. Buying a convertible bond and shorting the underlying stock to hedge equity exposure.
D. Buying stocks with high dividend yields while shorting stocks with relatively low dividend yields.

Answer: C

Step by step

1. Convertible arbitrage = long the convertible, short the issuer's stock.
2. The short stock position offsets equity (delta) risk, with delta rebalanced as the stock moves.
3. Return comes from coupon income, short-sale interest, and monetizing volatility.

Why C is correct: The strategy hedges the embedded equity option's delta with short stock, leaving income and volatility as return sources.

Why the others are wrong

- **A:** Describes a sovereign-versus-corporate credit spread trade, not a convertible position.
- **B:** Describes a yield curve (fixed-income) trade.
- **D:** Describes a dividend-yield equity long/short, with no convertible involved.

Q434 Schweser Module Quiz 90.2 Q2

A pairs trading strategy is best classified as which type of arbitrage?

A. Merger arbitrage.
B. Convertible arbitrage.
C. Fixed-income arbitrage.
D. Relative value arbitrage.

Answer: D

Step by step

1. Pairs trading buys one stock and shorts a historically correlated peer after a temporary divergence.
2. It profits when the spread reverts, which is relative value arbitrage.

Why D is correct: Pairs trading exploits temporary pricing divergence between two related securities.

Why the others are wrong

- **A:** Merger arbitrage is event-driven and involves deal targets and acquirers.
- **B:** Convertible arbitrage involves a convertible bond and its underlying stock.
- **C:** Fixed-income arbitrage involves bonds, not peer equities.

LO 90.c

Describe the mechanics and risks of event-driven strategies such as merger arbitrage.

Q435 Mock 4 Q73

REBUILT — Original keyed on a lookback-straddle trend-following payoff (Fung-Hsieh), which is not in the 2026 RMIM LOs; rewritten as a statement question on equity, macro, event-driven and distressed strategies from Reading 90.

A research analyst is preparing a report on hedge fund strategies. Which of the following statements would be correct for the analyst to include?

A. Global macro funds are backward looking and make small, unleveraged bets on a single market.
B. In a merger arbitrage strategy, the arbitrageur buys the target's stock (and may short the acquirer's stock), profiting from the spread if the deal closes.
C. In a merger arbitrage strategy, the target's stock is sold short in the expectation that the deal fails.
D. An equity long/short strategy aims to maximize market (beta) exposure rather than to exploit security mispricings.

Answer: B

Key correction: Source key (Mock 4) said A (trend-following with perfect foresight resembles a lookback straddle); that statement is outside the 2026 RMIM LOs, so the item was rebuilt around a statement from Reading 90.

Step by step

1. Merger arbitrage profits from the spread between the acquirer's offer price and the target's post-announcement price.
2. The arbitrageur therefore buys the target and, in a stock deal, may short the acquirer; the position is unwound at the merger, ideally at a profit.
3. The risk is a failed deal: the target falls back toward its pre-announcement price, producing the loss.

4. Check the other statements against Reading 90 descriptions of global macro and long/short: A and D contradict them. Only B is correct.

Why B is correct: Merger arbitrage is long the target (often short the acquirer) to capture the deal spread, with deal failure as the main risk.

Why the others are wrong

- **A:** Global macro funds are forward looking and make large, leveraged bets across markets; this describes the opposite.
- **C:** Reverses the trade: shorting the target would profit from a failed deal, but the strategy is long the target to capture the spread.
- **D:** Equity long/short seeks alpha from mispricings while hedging market risk; maximizing beta is not its goal.

Q436

AnalystPrep Q.2574

REBUILT — Original 'asymmetric payoffs' wording was misleading for merger arbitrage and the distractors were not clearly excluded; restated the fund's goal as profiting from the spread on announced deals so only one strategy fits.

Quasar Hedge Fund (QHF) wants an event-driven strategy that profits from the price discrepancy between the current price of a target firm's shares and the final purchase price offered by the acquirer after a deal is announced. Management accepts the risk that a deal may be cancelled or modified, but does not want a strategy that is mainly a bet on interest rate movements or one that requires deep expertise in distressed securities. Which hedge fund strategy should QHF adopt?

A. Convertible arbitrage
B. Merger arbitrage
C. Fixed-income arbitrage
D. Distressed securities

Answer: B

Step by step

1. The target is to capture the spread between the target's post-announcement price and the offer price: this is merger (risk) arbitrage.
2. Its main risk is deal failure or modification, in which the target price falls back toward its pre-announcement level.
3. It does not depend primarily on interest rates, nor on analysis of bankrupt firms.
4. Therefore B.

Why B is correct: Merger arbitrage buys the target (and in share deals shorts the acquirer) to earn the arbitrage spread; deal-break risk is its principal risk.

Why the others are wrong

- **A:** Convertible arbitrage exploits mispricing between a convertible and its underlying stock; it is not tied to announced deals. Chosen by confusing any 'arbitrage' label with the deal spread.
- **C:** Fixed-income arbitrage trades relative value between bonds and is heavily rate/spread sensitive, which management excluded.
- **D:** Distressed securities is event-driven but targets firms in or near bankruptcy and needs deep legal/credit expertise, which management excluded.

Q437

AnalystPrep Q.2760

Which of the following hedge fund strategies can be described as event-driven?

A. Trend followers
B. Equity long/short
C. Global macro
D. Distressed securities

Answer: D

Step by step

1. Event-driven strategies profit from pricing inefficiencies around corporate events such as mergers, restructurings and bankruptcies.
2. Distressed securities (firms entering, in, or emerging from bankruptcy) is a core event-driven strategy, alongside merger arbitrage and activism.
3. Trend followers, equity long/short and global macro are driven by price trends, security selection and macro

views, not specific corporate events.

Why D is correct: Distressed investing exploits mispricing created by a specific corporate event: bankruptcy and restructuring.

Why the others are wrong

- **A:** Trend followers (managed futures) follow price momentum across markets, not corporate events.
- **B:** Equity long/short is an equity-based strategy built on security selection, not event catalysts.
- **C:** Global macro is a top-down bet on economic and policy developments, not corporate events.

Q438 BT answers P2.T8.411.3 · BT notes P2.T9.411.3

Fung and Hsieh include two major strategies in the event-driven category: risk (merger) arbitrage and distressed. About these event-driven strategies, each of the following is true EXCEPT which is false?

A. Risk arbitrage strategy returns (risk arbitrage index) are correlated to the S&P 500

B. Distressed strategy returns (Distress index) are correlated with high yield bonds

C. Both of these event-driven strategies–risk arb and distressed–exhibit nonlinear returns characteristics

D. Both of these event-driven strategies–risk arb and distressed–tend to benefit from extreme market moves in either direction

Answer: D

Step by step

1. Merger arbitrage earns small gains when deals close and suffers large losses when deals break, typically in sharp equity sell-offs: it behaves like a short put on the market, so returns are correlated with equities (A true) and nonlinear (C true).
2. Distressed strategies hold low-grade debt, so returns co-move with high-yield bonds (B true) and also have nonlinear, tail-risk exposure.
3. Because of this short-option-like payoff, extreme market moves hurt rather than help event-driven funds; only strategies like trend followers benefit from extreme moves in either direction.
4. So D is the false statement.

Why D is correct: Event-driven payoffs resemble selling insurance: steady small gains with tail losses in extreme markets, so they do not benefit from extreme moves.

Why the others are wrong

- **A:** True statement: risk arbitrage returns are correlated with the S&P 500, so it is not the exception.
- **B:** True statement: distressed debt returns track high-yield bonds, so it is not the exception.
- **C:** True statement: both strategies have option-like (nonlinear) payoffs, so it is not the exception.

Q439 Mock 1 Q33

A general partner of Quicksilver Capital Partners, an up-and-coming hedge fund based out of Geneva, identifies Diamond Technologies, Inc., as a strong candidate for a takeover by a larger player, Aztec Mining Corporation. Which of the following is incorrect in relation to implementing a merger arbitrage strategy?

I. Buy Aztec and sell Diamond Technologies.

II. The payoff is large profits if the merger goes ahead, and small losses if it doesn't.

III. The strategy is likened to selling insurance.

A. I and II.

B. I and III.

C. II and III.

D. I, II, and III.

Answer: A

Step by step

1. I: the arbitrageur buys the target (Diamond) and, in share-for-share deals, shorts the acquirer (Aztec). I reverses this: incorrect.
2. II: gains are the small spread if the deal closes; losses if the deal fails (target falls back) can far exceed the gains. II reverses the asymmetry: incorrect.
3. III: small steady gains with occasional large losses resemble writing insurance: correct.
4. Incorrect statements: I and II.

Why A is correct: Merger arbitrage is long the target, short the acquirer, with small likely gains and large unlikely losses, like selling insurance.

Why the others are wrong

- **B:** Treats III as incorrect while accepting II; but II is the incorrect one and the selling-insurance analogy in III is valid.
- **C:** Marks III incorrect and misses that I is wrong; the insurance analogy is valid.
- **D:** Includes III, but the selling-insurance analogy is correct.

Q440 Schweser Module Quiz 90.3 Q1

REBUILT — Answer A was correct; replaced two distractors (0.25%, 11.25%) with ones tied to realistic errors (not annualizing; ignoring the failure loss).

Novelty Healthcare has received a tender offer from a larger competitor. The offer terms specify a price of $30 per share. Before the announcement, Novelty's stock traded at $25, and it rose to $29 after the offer was made public. Analysts at Primus Fund estimate that the deal has a 75% probability of closing and expect completion within three months. What is the annualized expected return from a merger arbitrage position in Novelty's stock?

A. -3.45%
B. -3.33%
C. -0.86%
D. 10.34%

Answer: A

Step by step

1. Buy at P = $29. Gain if the deal closes: $G = 30 - 29 = \$1$. Loss if it fails: $L = 29 - 25 = \$4$.
2. Expected profit per share $= 0.75(1) - 0.25(4) = -\$0.25$.
3. 3-month return $= -0.25/29 = -0.862\%$.
4. Annualize: $Y = 3/12 = 0.25$, so $-0.862\%/0.25 = -3.45\%$.

Why A is correct: With a 25% chance of a $4 loss against a 75% chance of a $1 gain the expected value is negative; scaling by the $29 cost and annualizing by 4 gives -3.45%.

Why the others are wrong

- **B:** Divided the expected loss by the $30 offer price instead of the $29 purchase price (-0.25/30 x 4).
- **C:** Correct expected return but not annualized (3-month figure).
- **D:** Counted only the probability-weighted gain, ignoring the 25% failure loss: 0.75 x 3.45% x 4 = 10.34%.

LO 90.d

Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy.

Q441 AnalystPrep Q.2573

REBUILT — Original claimed distressed gives upside in both bull and bear markets and had loosely worded exclusions; rewrote the criteria around the Schweser features (illiquidity, default and timing risk accepted; no leverage reliance, no net-market hedging).

A newly formed hedge fund is choosing one strategy. It is willing to accept illiquid positions, default risk, and a restructuring process of uncertain length, and it expects to profit from securities of firms entering, in, or emerging from bankruptcy that are bought at deep discounts to risk-adjusted value. It does not want a strategy whose returns depend mainly on leverage applied to small mispricings, nor one that requires continuously hedging net market exposure. Which strategy best fits?

A. Equity long/short
B. Global macro
C. Distressed securities
D. Convertible arbitrage

Answer: C

Step by step

1. The stated target (securities of firms in or near bankruptcy, bought at discounts) is the definition of a distressed securities strategy.
2. The accepted risks (illiquidity, default risk, lengthy restructuring) are the typical distressed risks.
3. Distressed returns come mainly from price recovery and mispricing across the capital structure, not from lever-

age or from continuous market-risk hedging.
4. Equity long/short, global macro and convertible arbitrage do not target bankrupt-firm securities, so they do not fit.

Why C is correct: Distressed securities strategies buy securities of firms entering, in, or emerging from bankruptcy at discounts, accepting illiquidity, default and timing risk, without relying on leverage or net-market hedging.

Why the others are wrong

- **A:** Equity long/short targets mispricings across ordinary equities and aims to hedge market risk; it is not about bankrupt firms. Choosing it confuses 'hedged' with 'distressed'.
- **B:** Global macro takes directional bets on rates, currencies and economies, often with leverage; it does not involve bankruptcy securities.
- **D:** Convertible arbitrage buys convertibles and shorts the stock, relying on leverage and delta hedging of market exposure, the features the fund wants to avoid.

Q442

Mock 2 Q20

REBUILT — Original exception (liquidity buffer interest) is a long/short return source in Schweser, not a clearly wrong distressed statement; replaced option D with a clearly incorrect statement about pension-fund holdings and kept A-C.

Which of the following is NOT a recognized source of return opportunity in a distressed securities strategy?

A. Buying shares in a company and holding them until it emerges from bankruptcy.
B. Buying loans that banks sell to clean up their balance sheets.
C. Buying trade claims from original holders who lack the expertise or desire to stay involved through the bankruptcy process.
D. Buying non-investment-grade bonds that are cheap because pension funds and institutions are required to hold them.

Answer: D

Key correction: Mock key (D) is retained but the original option D (interest on a liquidity buffer) was not clearly wrong under Schweser; option D was rewritten so the exception is unambiguous.

Step by step

1. A, B and C are the examples listed in Schweser: hold equity through emergence, buy bank loans sold to clean balance sheets, buy trade claims at discounts.
2. Non-investment-grade bonds can be cheap because many institutions and pension funds are PROHIBITED from owning them, forcing sales.
3. Option D says they are required to hold them, which reverses the mechanism, so D is not a valid source of return.

Why D is correct: Distressed bonds become cheap because many institutions are prohibited from holding non-investment-grade debt (forced selling), not because they are required to hold it.

Why the others are wrong

- **A:** A valid return source: holding equity through emergence from bankruptcy is listed in the reading.
- **B:** A valid return source: banks sell distressed loans to clean up balance sheets, creating buying opportunities.
- **C:** A valid return source: original holders of trade claims often do not want to remain through bankruptcy, so claims trade at discounts.

Q443

Schweser Module Quiz 90.3 Q2

Which of the following scenarios best reflects the core risk faced by investors in distressed securities?

A. A restructuring process takes longer than expected.
B. Senior secured creditors receive the lowest recoveries.
C. Bankruptcy guarantees predictable outcomes for all creditors.
D. High trading volume in distressed debt limits position building.

Answer: A

Step by step

1. Schweser lists core risks: unreliable recovery estimates, illiquidity, legal and court uncertainty, and a restructuring or bankruptcy process that can take much longer than expected.
2. Option A matches this timing risk: a longer process delays and erodes returns.
3. B reverses seniority, C asserts certainty that does not exist, and D reverses the liquidity problem.

Why A is correct: Distressed securities carry significant timing and process risk because restructuring or bankruptcy can last much longer than expected, eroding returns.

Why the others are wrong

- **B:** Senior secured creditors typically recover the most, not the least; this reverses seniority.
- **C:** Bankruptcy outcomes are uncertain and depend on courts and negotiations; predictability for all creditors is the opposite of the risk.
- **D:** Distressed securities are typically illiquid with low trading volume, which makes exiting hard; high volume is not the risk.

Reading 91. Risk, Regulation and Organizational Structure

David P. Stowell and Paul Stowell, Investment Banks, Hedge Funds, and Private Equity, 4th Edition (Elsevier, 2023). Chapter 14.

LO	Learning objective	Qs
91.a	describe different risks faced by investors in hedge funds.	6
91.b	explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk.	3
91.c	summarize different approaches taken by regulators to mitigate risks related to hedge funds.	2
91.d	describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure.	3

LO 91.a

Describe different risks faced by investors in hedge funds.

Q444

AnalystPrep Q.3876

REBUILT — Original options were implausible and the stem was only loosely tied to the reading; rebuilt with realistic distractors. Reassigned to 91.a because the cause is hedge fund funding/leverage risk.

Convertible arbitrage hedge funds suffered large losses during the 2007-2009 crisis. Which of the following was the most likely cause?

A. The loss of financing from prime brokers forced leveraged funds to deleverage and sell convertibles at depressed prices.

B. A rise in issuers' stock prices caused losses on the short stock hedge that were not offset by the convertibles.

C. A surge in demand for convertibles from long-only investors pushed prices well above model values.

D. A collapse in equity volatility reduced the value of the embedded call options.

Answer: A

Step by step

1. Convertible arbitrage relies on leverage to magnify small mispricing profits.
2. In the crisis, financing was withdrawn, forcing deleveraging and fire sales.
3. Forced selling pushed convertible prices below model value, producing losses.

Why A is correct: Leverage-dependent strategies are exposed to funding liquidity risk; withdrawal of financing forced sales at distressed prices.

Why the others are wrong

- **B:** A delta-hedged position is largely protected from stock rises, since the convertible gains too; this was not the main driver.
- **C:** Excess demand would raise convertible prices and benefit longs; in the crisis demand vanished.
- **D:** Volatility rose sharply in the crisis, which increases option value; it did not collapse.

Q445

AnalystPrep Q.2585

REBUILT — Original asked about fund closure mitigation via multiple managers, which the Schweser reading does not cover and had a debatable key; rebuilt to test the compensation-driven 'risk tolerance' incremental risk.

Justin Boucher is considering an investment in a hedge fund whose managers receive 20% of the profits generated by the fund. He worries that this compensation structure will entice the managers to take large, risky bets because they share in the gains but not in the losses. Which of the five incremental risks of hedge fund investing does Boucher's concern most directly describe?

A. Leverage risk

B. Regulatory risk

C. Transparency risk

D. Risk tolerance

Answer: D

Step by step

1. Schweser lists five incremental hedge fund risks: leverage, regulation, short selling, transparency and risk tolerance.
2. The concern is that a profit-sharing fee rewards upside but not downside, so managers are incentivized to take

more risk.
3. Schweser states that hedge fund managers generally have higher risk tolerance than mutual fund managers and that compensation structures often incentivize additional risk.
4. This matches 'risk tolerance'.

Why D is correct: The incentive-fee structure motivates managers to take on additional risk, which Schweser classifies under the risk tolerance incremental risk.

Why the others are wrong

- **A:** Leverage amplifies gains and losses via borrowing or derivatives; the stem concerns incentives, not borrowing. Confuses a possible tool of risk-taking with its cause.
- **B:** Regulatory risk concerns limited oversight or rule changes; the stem is about fee incentives, not oversight.
- **C:** Transparency risk concerns secrecy of strategies that hinders monitoring; nothing in the stem addresses disclosure.

Q446

AnalystPrep Q.2588

REBUILT — Original options were vague (leverage and short selling also defensible); rebuilt around the named investment-level risk of correlation spikes.

A hedge fund manager runs several funds and uses different strategies across them to achieve diversification. During a severe market crisis, the funds suffer losses at the same time even though their strategies normally behave independently. Which investment-level risk best explains why the diversification failed?

A. Key person risk
B. Style drift
C. Correlation spikes during market stress
D. Unrelated business taxable income (UBTI)

Answer: C

Step by step

1. Schweser's investment-level risks: counterparty risk, conflicts of interest, key person risk, strategy failure, style drift, regulatory changes and correlation spikes.
2. Normally uncorrelated strategies moving together in a crisis is a correlation spike (as seen with LTCM and the GFC fire sales).
3. Diversification benefits that rely on low correlations therefore disappear when stress hits.
4. Answer: correlation spikes during market stress.

Why C is correct: In extreme stress, assets and strategies that usually move independently become highly correlated, so diversification across strategies fails exactly when it is needed.

Why the others are wrong

- **A:** Key person risk is dependence on a particular individual; it does not explain simultaneous losses across different strategies.
- **B:** Style drift is a manager deviating from the stated strategy; the stem says strategies were as designed but correlations rose.
- **D:** UBTI is a portfolio-level tax issue for tax-exempt investors with debt-financed income, unrelated to diversification failure.

Q447

AnalystPrep Q.2578

Richard Chang is a financial risk analyst at a global investment firm, Capital Dynamics. He is reviewing the performance of various hedge funds in which the company has significant investments. Richard is aware that principals (investors) and agents (hedge fund managers) have different risk-sharing preferences, leading to potential issues. One fund, Zeta Capital, has had a fantastic run in the past three years, consistently generating high returns. However, Richard notices that this performance was driven by high-risk strategies. He is concerned about the asymmetry in risk-sharing between his firm (as the principal) and Zeta Capital's managers (as the agents). Which of the following correctly describes the problem of risk-sharing asymmetry between principals and agents in the context of Richard's concerns?

A. Hedge fund managers might underperform if they are not given enough freedom to take risks, leading to lower returns for principals.
B. Hedge fund managers tend to avoid high-risk strategies due to the pressure from principals, leading to missed opportunities.
C. Hedge fund managers, being compensated for performance and not bearing the full downside, might take excessive risks, which might not align with the risk appetite of the principals.

D. Hedge fund managers always align their risk-taking decisions with the principals' expectations, leading to homogeneous investment outcomes.

Answer: C

Step by step

1. Principals (investors) bear the full downside of fund losses; agents (managers) earn performance-based fees and typically do not share in losses.
2. This payoff asymmetry is like a call option for the manager, who gains from upside but bears little downside.
3. So managers may take more risk than the principals' risk appetite warrants, which fits the concern about Zeta's high-risk strategies.

Why C is correct: Performance compensation without full participation in losses gives managers an incentive to take excessive risk relative to investors' preferences.

Why the others are wrong

- **A:** Reverses the problem: the concern is too much risk-taking incentive, not too little freedom.
- **B:** Reverses the incentive: managers paid on performance tend to seek risk, not avoid it.
- **D:** Assumes perfect alignment, which is exactly what the asymmetry undermines.

Q448

BT answers P2.T8.412.2 · BT notes P2.T9.412.2

REBUILT — Source is a Fung-Hsieh reading item with absolute-worded options not in the Schweser text; rewritten around Schweser's investment-level risks (correlation spikes) and leverage, with one defensible answer.

An investor holds a hedge fund portfolio spread across several strategies that show low historical correlations. Many of the funds use collateralized leverage. During a market-wide liquidity shock, the investor is most concerned that the diversification benefit will disappear. Which of the following best explains why?

A. Key person risk, because diversification depends on each fund's lead manager remaining in place

B. Correlation spikes, because leveraged funds facing margin calls and forced liquidation sell similar assets at the same time, pushing returns together

C. Unrelated business taxable income (UBTI), because debt-financed income becomes taxable for tax-exempt investors during stress

D. Survivorship bias, because reported historical correlations only include funds that survived

Answer: B

Key correction: Source key (D) endorsed a Fung-Hsieh statement about leverage and forced liquidation; that idea is retained in the rebuilt correct option B.

Step by step

1. Identify the problem: diversification across strategies is expected to hold, but it fails in a market-wide liquidity shock.
2. Schweser lists correlation spikes as an investment-level risk, and notes leverage raises a fund's sensitivity to liquidity shocks.
3. Leveraged funds hit by margin calls and collateral demands must deleverage, selling similar liquid assets at once.
4. This synchronized selling raises correlations across strategies, so diversification fails exactly when needed.
5. Therefore the best explanation is B.

Why B is correct: Leverage makes funds sensitive to liquidity shocks; forced, simultaneous deleveraging pushes strategy returns together (a correlation spike), eroding the diversification the investor relied on.

Why the others are wrong

- **A:** Key person risk is real but concerns loss of a specific manager at one fund; it does not explain a simultaneous, market-wide loss of diversification across strategies.
- **C:** UBTI is a portfolio-level tax issue for tax-exempt investors, unrelated to how fund returns co-move in a liquidity shock.
- **D:** Survivorship bias distorts measured historical performance and correlation data, but it is not a mechanism that makes correlations rise during a shock.

Q449 BT answers P2.T8.412.3

REBUILT — Source is a Fung-Hsieh principal-agent item whose co-investment mitigation is not in the Schweser text; rebuilt to test the Schweser incremental risk arising from manager compensation (risk tolerance).

A hedge fund manager's compensation is tied heavily to performance, which may encourage the manager to take more risk than a typical investor would prefer. Which of the five incremental risks that Schweser highlights as particularly pronounced in hedge funds does this best illustrate?

A. Leverage
B. Regulation
C. Transparency
D. Risk tolerance

Answer: D

Key correction: Source key (B) relied on Fung-Hsieh's co-investment mitigation, which is not in the Schweser text; the item was rebuilt around Schweser's risk tolerance risk.

Step by step

1. The five incremental risks are leverage, regulation, short selling, transparency, and risk tolerance.
2. The scenario describes an incentive structure that pushes the manager toward greater risk-taking than investors may want.
3. Schweser states hedge fund managers generally have higher risk tolerance than mutual fund managers and compensation structures incentivize additional risk.
4. This matches risk tolerance, so the answer is D.

Why D is correct: Performance-linked compensation incentivizes managers to take on additional risk, which Schweser classifies under the risk tolerance incremental risk.

Why the others are wrong

- **A:** Leverage concerns amplification of gains and losses through borrowing or derivatives, not the incentive effect of compensation.
- **B:** Regulation risk concerns lighter oversight and regulatory change, not manager incentives.
- **C:** Transparency risk concerns secrecy of strategies, which hinders monitoring, not compensation-driven risk appetite.

LO 91.b

Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk.

Q450 Schweser Module Quiz 91.1 Q2

Which of the following actions is least likely to reduce systemic risk associated with hedge funds?

A. Strict new regulations.
B. Lower leverage in hedge fund portfolios.
C. More conservative lending practices by banks.
D. Greater diversification in hedge fund portfolios.

Answer: A

Step by step

1. Schweser identifies three approaches to mitigating hedge fund systemic risk: (1) more conservative bank lending, (2) lower leverage and greater diversification in hedge fund portfolios, (3) reasonable regulation.
2. Options B, C and D match approaches (1) and (2) directly.
3. Option A calls for strict new regulation. Excessively strict regulation can reduce the liquidity hedge funds provide as investors of last resort (e.g., in distressed securities), which can have unintended negative effects on markets.
4. Therefore strict regulation is the action least likely to reduce systemic risk.

Why A is correct: Only reasonable, not excessively strict, regulation mitigates systemic risk; strict rules can drain market liquidity and unintentionally heighten systemic risk, while the other three actions directly reduce leverage-driven fire sales and lender losses.

Why the others are wrong

- **B:** Lower leverage reduces forced deleveraging and fire-sale contagion, so it does reduce systemic risk; choosing

it confuses leverage as a risk amplifier with a mitigant.
- **C:** Conservative lending (higher margins, tighter collateral) limits bank credit and counterparty losses from hedge fund failures, so it reduces systemic risk.
- **D:** Diversification lowers the chance that one strategy failure triggers a forced unwind, so it reduces systemic risk.

Q451 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager at a prime broker is reviewing a hedge fund client that holds $400 million of equities financed with a $300 million margin loan (leverage, measured as loan divided by equity, of 3.0x). The fund's policy is to restore leverage to 3.0x whenever it drifts. The equity position falls 10% and the loan balance is unchanged. Assuming the fund restores leverage to 3.0x solely by selling stock and repaying the loan with the proceeds, the forced sale is closest to:

A. $40 million.
B. $120 million.
C. $60 million.
D. $180 million.

Answer: B

Step by step

1. Initial equity = 400 − 300 = 100 ($ million); leverage = 300/100 = 3.0x.
2. After a 10% fall the position is worth $360 million; equity = 360 − 300 = 60 ($ million); leverage = 300/60 = 5.0x.
3. Target 3.0x with equity of $60 million (sales at market do not change equity) requires a loan of 3 × 60 = 180 ($ million).
4. Sale needed = 300 − 180 = 120 ($ million); remaining stock $240 million = $180 million loan + $60 million equity.

Why B is correct: Because equity has shrunk to $60 million while the loan is unchanged, the loan must fall to $180 million, requiring a $120 million sale. Such forced selling is unrelated to fundamentals and can feed contagion across markets.

Why the others are wrong

- **A:** Uses the $40 million price decline as the sale amount; this ignores that each dollar of lost equity requires repaying three dollars of debt at 3.0x leverage.
- **C:** Uses the post-loss equity of $60 million as the sale amount; equity is not the amount of debt to be repaid.
- **D:** Treats the required post-sale loan ($180 million) as the amount to sell, confusing the remaining loan with the amount repaid.

Q452 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A bank's chief risk officer is assessing the bank's prime brokerage business after the failure of a highly leveraged investment vehicle that used total return swaps. Which of the following statements best describes the credit risk the bank faces on margin loans to hedge funds?

A. Credit risk is negligible because margin loans are collateralized, so losses can arise only from derivatives transactions.
B. Credit risk arises mainly from the loss of fee income if several hedge fund clients fail simultaneously.
C. Credit risk arises because the value of the collateral can decline faster than anticipated, leaving the bank undersecured.
D. Credit risk is eliminated once the hedge fund reduces leverage, because only highly leveraged funds can default.

Answer: C

Step by step

1. Banks face revenue risk, credit risk, and counterparty risk from hedge fund exposure.
2. Margin loans are secured, but collateral values can fall quickly in stress, so the loan may exceed collateral value.
3. This undersecured exposure is the credit risk; fee loss is revenue risk; derivatives exposure is counterparty risk.

Why C is correct: The reading identifies credit risk on margin loans as the possibility that the collateral falls in

value more quickly than expected, leaving the bank undersecured.

Why the others are wrong

- **A:** Collateral reduces but does not remove credit risk, and derivatives exposure is a separate (counterparty) risk.
- **B:** Describes revenue risk (loss of trading, clearing, financing and other fees), not credit risk.
- **D:** Lower leverage reduces but does not eliminate default risk; the statement overstates the effect and ignores collateral value risk.

LO 91.c

Summarize different approaches taken by regulators to mitigate risks related to hedge funds.

Q453 Schweser Module Quiz 91.1 Q1

Hedge funds encounter the most significant restrictions on short selling in which of the following jurisdictions?

A. China
B. Singapore
C. Hong Kong
D. The European Union

Answer: A

Step by step

1. China's CSRC prohibits leverage and limits short selling to only 50 securities in the CSI 300 Index.
2. Brokers cannot use client shares to facilitate short sales, so shorting costs roughly 10% per year.
3. Singapore's rules (MAS registration above $250 million, reporting, license), Hong Kong's SFO (licensing, marketing limits) and the EU's AIFMD (registration, leverage limits, reporting) do not comparably restrict short selling.
4. China is the answer.

Why A is correct: China restricts shorting to 50 CSI 300 securities and bars brokers from lending client shares, making shorting extremely expensive.

Why the others are wrong

- **B:** Singapore's rules centre on registration, licensing and reporting, not short-selling limits.
- **C:** Hong Kong's SFO focuses on licensing, reporting and marketing limits rather than restricting shorting.
- **D:** The EU's AIFMD imposes registration, leverage limits and reporting; short-sale rules there are disclosure-oriented, not a near-ban.

Q454 Schweser Module Quiz 91.2 Q1

REBUILT — Option C (the Advisers Act) was also defensible as the statute governing registration; reworded the stem to ask which provision eliminated the exemption that let managers avoid SEC registration.

Before 2010, many U.S. hedge fund managers avoided SEC registration under the private adviser exemption (advising fewer than 15 funds). Which of the following eliminated that exemption and now requires U.S.-based hedge fund managers to register with the SEC?

A. Regulation D.
B. Title IV of the Dodd-Frank Act.
C. Investment Advisers Act of 1940.
D. Alternative Investment Fund Managers Directive.

Answer: B

Step by step

1. The private adviser exemption sat inside the Investment Advisers Act of 1940: managers advising fewer than 15 funds need not register.
2. Title IV of Dodd-Frank (Private Fund Investment Advisers Registration Act) eliminated that exemption.
3. Managers above the asset threshold must therefore register with the SEC and file reports such as Form PF.
4. Reg D is a private-placement rule and AIFMD is EU law, so neither applies.

Why B is correct: Title IV of Dodd-Frank removed the private adviser exemption from the Advisers Act, so U.S. hedge fund managers must now register with the SEC.

Why the others are wrong
- **A:** Regulation D governs private placements and limits marketing/investor eligibility; it does not address adviser registration.
- **C:** The 1940 Advisers Act is where the exemption lived; it is the law that was amended, not the provision that removed the exemption.
- **D:** AIFMD is the EU framework; it does not apply to U.S. managers' SEC registration.

LO 91.d

Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure.

Q455 Schweser Module Quiz 91.2 Q2

Which of the following is most likely an incentive for hedge funds to be structured as limited partnerships?
- **A.** Management fees are taxed at capital gains rates.
- **B.** Performance fees can be taxed at capital gains rates.
- **C.** All fees paid to the general partner are treated as capital gains.
- **D.** Management fees can be deferred indefinitely for tax purposes.

Answer: B

Step by step
1. Typical hedge fund structure: manager is the general partner (GP), investors are limited partners; fees follow a 2-and-20 model.
2. The 2% management fee is taxed as ordinary income.
3. The 20% performance fee (carried interest) may be taxed at the lower capital gains rate if the underlying assets were held at least one year.
4. This favorable treatment of carried interest is a key tax motivation for the limited partnership form, so B is the incentive.

Why B is correct: Carried interest (the performance fee) can receive capital gains treatment when the partnership holds assets for at least one year, which is a tax advantage of the limited partnership structure.

Why the others are wrong
- **A:** Management fees are ordinary income, not capital gains; confuses the management fee with the performance fee.
- **C:** Overgeneralizes: only the performance fee (carried interest) can get capital gains treatment, not all GP fees.
- **D:** The ability to defer management fees (Section 457) was eliminated in 2008, so indefinite deferral is no longer available.

Q456 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

The general partner of a U.S.-domiciled hedge fund is explaining the master-feeder structure to a prospective investor, a U.S. public pension plan. Which of the following statements best describes how the pension plan would most likely participate and why?
- **A.** It invests through the onshore feeder, because tax-exempt investors benefit from the onshore partnership's pass-through treatment.
- **B.** It invests directly in the master fund, because only the master fund holds the pooled capital and executes trades for all investors.
- **C.** It invests through the offshore feeder, because U.S. investors are required to use offshore vehicles to avoid registration with the SEC.
- **D.** It invests through the offshore feeder, which helps tax-exempt investors participate in a tax-efficient manner.

Answer: D

Step by step
1. The master fund pools capital and trades; investors access it through feeder funds.
2. A U.S.-domiciled fund typically has an onshore feeder for U.S. taxable investors and an offshore feeder for non-U.S. investors and U.S. tax-exempt investors.

3. A pension plan is a U.S. tax-exempt investor, so it uses the offshore feeder, which avoids multiple layers of taxation.

Why D is correct: The offshore feeder serves non-U.S. and U.S. tax-exempt investors such as pension funds, which allows them to participate tax-efficiently.

Why the others are wrong

- **A:** The onshore feeder is for U.S. taxable investors; tax-exempt investors are typically placed offshore.
- **B:** Investors normally do not invest directly in the master; they enter through feeders, each tailored to an investor group.
- **C:** The motive is tax efficiency, not avoiding SEC registration; registration is a manager-level requirement under the Investment Advisers Act.

Q457 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A hedge fund manager organized as a limited partnership earns a 2% management fee on $500 million of assets and a 20% performance fee on a $60 million gain from positions held at least one year. The manager's ordinary income tax rate is 37% and the capital gains rate is 20%. Assuming the tax treatment described for the typical partnership structure applies, the manager's total tax on these fees is closest to:

A. $6.1 million.
B. $4.4 million.
C. $6.4 million.
D. $8.1 million.

Answer: A

Step by step

1. Management fee = $2\% \times 500 = 10$ ($ million), taxed as ordinary income: $10 \times 0.37 = 3.7$.
2. Performance fee = $20\% \times 60 = 12$ ($ million), carried interest on gains held at least one year, taxed at the capital gains rate: $12 \times 0.20 = 2.4$.
3. Total tax = $3.7 + 2.4 = 6.1$ ($ million).

Why A is correct: Management fees are ordinary income while performance fees from gains held at least one year qualify for the lower capital gains rate, giving $6.1 million.

Why the others are wrong

- **B:** Taxes all $22 million of fees at 20% ($22 \times 0.20 = 4.4$); management fees do not get capital gains treatment.
- **C:** Reverses the treatment (management fee at 20%, performance fee at 37%: $2.0 + 4.44 = 6.44$).
- **D:** Taxes all $22 million of fees as ordinary income ($22 \times 0.37 = 8.14$), ignoring the carried interest treatment.

Reading 92. The Rise and Risks of Private Credit

"The Last Mile: Financial Vulnerabilities and Risks, Chapter 2: The Rise and Risks of Private Credit," IMF Global Financial Stability Report (April 2024).

LO	Learning objective	Qs
92.a	describe characteristics of private credit, including its typical investors and borrowers, and compare private credit to other types of loans and fixed-income instruments.	9
92.b	explain the return profile and growth profile of the private credit asset class, and compare the historical returns of private credit to those of other asset classes.	2
92.c	describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability.	11
92.d	assess potential policy recommendations that could help mitigate the risks associated with private credit.	3

LO 92.a

Describe characteristics of private credit, including its typical investors and borrowers, and compare private credit to other types of loans and fixed-income instruments.

Q458

AP Mock 3 Q44

REBUILT — Option D (PE in defensive cash-generating firms) was also defensible and option B was vague; replaced D with a clearly wrong option and sharpened the stem.

An institutional investor has a mandate to generate consistent income while limiting downside risk. Based on the characteristics of private credit and private equity, which allocation best fits this mandate?

A. Overweight private equity, because of its potential for higher capital appreciation.
B. Split the allocation equally between private credit and private equity to maximize diversification.
C. Overweight private credit, because of contractual interest payments, collateral and covenant protection.
D. Overweight venture capital, because it has outperformed private credit in recent years.

Answer: C

Step by step

1. Mandate = steady income plus downside protection.
2. Private credit: contractual (mostly floating-rate) interest, collateral and covenants protecting the lender, so cash flows are regular and losses are cushioned.
3. Private equity returns come mainly from lumpy exit gains as a residual claim, which does not match an income mandate.
4. Therefore overweight private credit.

Why C is correct: Private credit provides contractual income with collateral and covenants, matching income and downside objectives, whereas private equity returns are capital-gain driven and residual.

Why the others are wrong

- **A:** Chases capital appreciation, which is the PE objective, not income or downside protection.
- **B:** Diversification is not the stated goal; a 50% PE weight dilutes income and adds equity-like downside.
- **D:** Factual error: since 2020 private credit has significantly outperformed venture capital, and VC pays no income.

Q459

AnalystPrep Q.6380

Private credit often uses closed-end fund structures. Given the long-term nature of private credit deals, how does a closed-end structure primarily benefit fund managers and investors?

A. It provides flexibility to adjust investment focus quickly and ensures high liquidity.
B. It facilitates capital management over a set timeline without redemption pressure.
C. It allows unlimited capital expansion based on market demand.
D. It provides tax advantages through deferred income recognition.

Answer: B

Step by step

1. Private credit loans are illiquid and long-dated.

2. A closed-end fund raises fixed capital with capital calls and a defined life, with limited redemptions.
3. So managers can hold loans to maturity without forced sales; this matches assets to liabilities.

Why B is correct: Fixed capital and limited redemptions let managers hold illiquid loans over their full term without run risk.

Why the others are wrong

- **A:** Closed-end funds are less flexible and less liquid, the opposite of this statement.
- **C:** Closed-end funds raise a fixed amount of capital; unlimited expansion describes open-ended funds.
- **D:** Tax deferral is not the primary benefit of the structure; the key advantage is liquidity matching.

Q460 AnalystPrep Q.6393

An institutional investor compares private credit with high-yield bonds. Which of the following is a key difference between the two?

A. Private credit offers greater liquidity than high-yield bonds.
B. Private credit is less risky because of its direct lending structure.
C. Private credit offers greater flexibility in negotiating loan terms.
D. Private credit typically has lower yields.

Answer: C

Step by step

1. Private credit loans are negotiated bilaterally between lender and borrower.
2. This allows customized terms (deferred interest, repayment schedule, covenants, collateral).
3. High-yield bonds are standardized public securities.

Why C is correct: Direct bilateral negotiation allows tailored terms, unlike standardized public high-yield bonds.

Why the others are wrong

- **A:** Private loans are nontraded and illiquid; high-yield bonds trade publicly.
- **B:** Direct lending does not make the asset class less risky: borrowers typically have weaker earnings and higher leverage, and the loans are illiquid. Collateral and covenants only mitigate losses.
- **D:** Private credit costs and yields are generally higher, compensating for illiquidity and risk.

Q461 Schweser Module Quiz 92.1 Q1

To minimize liquidity risk, which of the following investment structures would most likely be used as an intermediary between borrowers and end investors in private credit?

A. Closed-end fund.
B. Open-ended fund.
C. Collateralized loan obligation (CLO).
D. Business development company (BDC).

Answer: A

Step by step

1. Liquidity risk arises when investors can redeem while assets are illiquid loans.
2. Closed-end funds (over 80% of the market) use capital calls and limit redemptions.
3. So the closed-end fund best minimizes liquidity risk.

Why A is correct: Limited redemptions in a closed-end fund align investor liquidity with illiquid loans.

Why the others are wrong

- **B:** Open-ended funds allow redemptions, creating run and liquidity risk.
- **C:** CLOs are a small share (about 5%) and are not the structure identified as minimizing liquidity risk.
- **D:** BDCs (about 14%) are less common and are not the structure identified as minimizing liquidity risk.

Q462 AnalystPrep Q.6378

Considering the target borrowers within the private credit landscape, which of the following best characterizes this market?

A. Large multinational corporations.
B. Middle-market firms.
C. Early-stage venture capital-backed startups.

D. Distressed companies undergoing restructuring.

Answer: B

Step by step

1. Private credit mainly serves mid-sized firms (revenues about $100 million to $1 billion).
2. They need more credit than a single bank provides but are too small for public offerings.
3. Answer: middle-market firms.

Why B is correct: Middle-market firms fall in the gap between bank lending capacity and public debt market access.

Why the others are wrong

- **A:** Large multinationals use public bonds or syndicated loans (private credit is expanding toward larger firms, but they are not the core).
- **C:** Startups rely on venture capital equity, not leveraged loans.
- **D:** Distressed lending is a niche, not the characteristic borrower base.

Q463 AnalystPrep Q.6379

Which feature of private credit loans protects lenders against rising interest rates and the erosion of returns?

A. Fixed interest rate terms.
B. Interest rates adjusted infrequently based on long-term economic forecasts.
C. Floating interest rates indexed to benchmarks.
D. Interest rates tied to the borrower's financial performance.

Answer: C

Step by step

1. Rate risk for a lender: rising market rates make a fixed coupon less valuable.
2. Floating-rate loans reset with a benchmark (e.g., SOFR) plus a spread.
3. So lender income rises when rates rise.

Why C is correct: Benchmark-indexed floating rates reset upward as market rates rise.

Why the others are wrong

- **A:** A fixed rate stays unchanged when rates rise, locking in lower real returns.
- **B:** Infrequent, forecast-based resets lag the market and offer little protection.
- **D:** Performance-based pricing is not the standard mechanism and does not track market rates.

Q464 AnalystPrep Q.6385

Summit Investments, a family office, is considering private credit but is concerned about the effect of rising interest rates on its returns. Which characteristic of private credit loans most directly mitigates this concern?

A. Enhanced covenants.
B. Floating interest rates indexed to benchmarks.
C. Closed-end fund structures.
D. Bespoke lending arrangements.

Answer: B

Step by step

1. The concern is interest rate risk on lender returns.
2. Floating rates reset with benchmarks, so income rises with market rates.
3. Covenants address credit risk, closed-end funds liquidity, bespoke terms flexibility.

Why B is correct: Floating-rate coupons adjust upward with benchmark rates, directly offsetting rate risk.

Why the others are wrong

- **A:** Covenants protect against credit deterioration, not rate moves.
- **C:** Closed-end structure addresses liquidity/redemption risk.
- **D:** Customization gives flexibility but does not hedge rates.

Q465 AnalystPrep Q.6386

Apex Corp, a middle-market manufacturer, seeks financing for a plant expansion but traditional banks are hesitant given its recent financial performance. What is the most likely reason Apex turns to private credit?

A. Access to lower interest rates than bank loans.
B. More standardized loan terms and faster approval.
C. Greater flexibility in loan structuring and covenant negotiation.
D. Greater transparency and regulatory oversight than bank lending.

Answer: C
Step by step
1. Borrowers with weak earnings or high leverage often fail bank underwriting.
2. Private lenders negotiate bilaterally and tailor terms (deferred interest, covenants, collateral).
3. Costs are usually higher than bank loans, but flexibility explains the appeal.

Why C is correct: Private credit's bilateral, customizable terms suit borrowers that banks will not serve on standard terms.
Why the others are wrong
- **A:** Private credit total costs are typically higher than bank loans.
- **B:** Terms are bespoke, not standardized.
- **D:** Private credit has little regulation and less transparency than bank or public debt.

Q466 AnalystPrep Q.6395

In the United States, a growing segment of retail investor participation in private credit is facilitated through which type of investment vehicle?

A. Hedge funds.
B. Business development companies (BDCs).
C. Sovereign wealth funds.
D. Private equity funds of funds.

Answer: B
Step by step
1. BDCs are one of the main private credit structures, along with closed-end funds and CLOs.
2. They are publicly traded, regulated vehicles lending to small and mid-sized firms, accessible to retail investors.
3. The other options serve institutions or are not private credit vehicles.

Why B is correct: BDCs are the vehicle through which retail investors access private credit loans.
Why the others are wrong
- **A:** Hedge funds are limited to accredited/institutional investors.
- **C:** Sovereign wealth funds are institutional investors, not retail vehicles.
- **D:** PE funds of funds are equity vehicles for sophisticated investors.

LO 92.b

Explain the return profile and growth profile of the private credit asset class, and compare the historical returns of private credit to those of other asset classes.

Q467 Schweser Module Quiz 92.1 Q2

In recent years, which of the following asset classes has outperformed private credit?

A. Real estate.
B. Private equity.
C. Venture capital.
D. Natural resources.

Answer: B
Step by step

1. Since 2020, private equity returns jumped dramatically ahead of private credit.
2. Private credit has performed similarly to natural resources and somewhat better than real estate.
3. Private credit has significantly outperformed the S&P 500 and venture capital.
4. Only private equity outperformed.

Why B is correct: Since about 2020 private equity has significantly outperformed private credit, reversing the post-crisis decade in which private credit led.

Why the others are wrong

- **A:** Real estate has performed somewhat worse than private credit.
- **C:** Venture capital has been significantly outperformed by private credit.
- **D:** Natural resources have had similar, not superior, performance to private credit.

Q468

AnalystPrep Q.6382

REBUILT — Original option A ('concentrated in North America') was also partly defensible since most private credit is managed there; rewrote options around the Schweser growth figures so exactly one is correct.

Which of the following statements best describes the growth profile of private credit in recent years?

A. Growth has been confined to the United States, with Europe and Asia showing little growth.

B. Annual growth has been about 20% in the United States and Asia and about 17% in Europe over five years, although it is still mainly managed in North America.

C. Growth in Europe has been much faster than in the United States and Asia.

D. Private credit now makes up more than one-fifth of total corporate credit in the United States.

Answer: B

Step by step

1. Schweser: US and Asia saw roughly 20% annual growth over the past five years; Europe about 17%.
2. Noticeable growth in Europe and Asia, yet private credit is mainly managed in North America.
3. As of June 2023, private credit was about 7% of US, 1.6% of European and 0.2% of Asian corporate credit.
4. Only B is consistent with these facts.

Why B is correct: Growth has been strong and broad (US and Asia about 20%, Europe about 17%) even though management remains concentrated in North America.

Why the others are wrong

- **A:** Ignores the strong European and Asian growth.
- **C:** Europe's 17% growth is slightly lower than the roughly 20% in the US and Asia.
- **D:** The US share is about 7% of corporate credit, not above 20%; 20% is the growth rate, confused with market share.

LO 92.c

Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability.

Q469

AnalystPrep Q.6392

REBUILT — Original had two defensible answers (A and B) and tested underwriting risk rather than return profile; reframed around the Schweser underwriting-standards risk with one clear answer and re-mapped to 92.c.

A large private credit fund has recently received a surge of investor inflows and now holds substantial undeployed capital (dry powder). Given the resulting competition for borrowers, which of the following is the most likely consequence for the private credit market?

A. Tighter covenants and higher risk premiums, as lenders become more selective.

B. Diminished underwriting standards, with loans priced too low for their risk and less strict covenants.

C. Lower floating-rate exposure, because lenders shift toward fixed-rate loans.

D. Smaller average borrower size, as lenders return to the small-company segment.

Answer: B

Key correction: Source key (B, 'downward pressure on future returns') was one defensible reading, but the Schweser text ties competition for deals directly to weaker underwriting; the stem and options were rebuilt around that.

Step by step
1. More capital chasing a limited pool of borrowers increases competition among lenders.
2. Schweser: the rapid rise of private credit and increased supply lead to reduced risk premiums and less onerous covenants.
3. Result: diminished underwriting standards (loans priced too low or not commensurate with risk) and looser covenants.
4. Only B matches this; A is the opposite effect.

Why B is correct: Excess supply of lending capital forces lenders to compete on price and terms, so risk premiums compress and covenants loosen, which increases expected losses.

Why the others are wrong
- **A:** Reverses the effect: competition lowers, not raises, risk premiums and covenant strictness.
- **C:** Private credit loans remain mainly floating-rate; nothing in competition shifts them to fixed-rate.
- **D:** Nothing in the Schweser text suggests competition for deals shrinks borrower size; the effect is weaker pricing and covenants, not a return to smaller companies.

Q470 AP Mock 2 Q47

Vanguard Credit, a private credit fund investing primarily in loans to technology startups, is experiencing rapid growth in assets under management. Which potential risk is most likely to increase for the fund in this scenario?

A. Stale valuations due to a lack of market prices.
B. Deteriorating underwriting standards due to increased competition.
C. Increased exposure to interest rate risk due to floating rate loans.
D. Liquidity risk due to the use of closed-end fund structures.

Answer: B

Step by step
1. Rapid inflows mean more capital chasing a limited pool of borrowers.
2. Competition for deals pushes lenders to cut spreads, loosen covenants and relax credit standards (underwriting risk).
3. Valuation, floating-rate and closed-end features are structural and are not changed by growth.

Why B is correct: Growth increases competition to deploy capital, which erodes underwriting standards and covenants, raising expected losses.

Why the others are wrong
- **A:** Stale valuations are a permanent feature of illiquid, unquoted loans; they do not intensify because of growth.
- **C:** Floating-rate loans reduce the lender's interest rate risk (the risk shifts to borrower credit risk), and growth does not change this.
- **D:** Closed-end structures do not permit withdrawals, so they mitigate liquidity risk rather than increase it.

Q471 AnalystPrep Q.6394

Regulators are concerned about the growing participation of retail investors in private credit, particularly through Business Development Companies (BDCs). Which of the following is most likely a key risk associated with this trend?

A. Retail investors may not fully understand private credit risks.
B. Lower returns for institutional investors.
C. Their participation will increase the demand for more standardized loan terms.
D. A lower amount of dry powder in the market

Answer: A

Step by step
1. Retail participation brings less sophisticated investors.
2. Credit, liquidity and redemption restrictions may not be well explained: conduct risk.
3. Other options are not identified risks.

Why A is correct: Unsophisticated investors may be inadequately informed of risks and redemption limits, creating conduct risk.

Why the others are wrong
- **B:** No direct link between retail inflows and institutional returns.

- **C:** Standardized terms would be a benefit, not a risk.
- **D:** Dry powder is not a stated conduct or stability risk.

Q472 AnalystPrep Q.6384

REBUILT — Original stem was vague (borrower leverage could be read as pointing to defaults or systemic risk); restated around the Schweser credit-risk point (small, highly leveraged, floating-rate borrowers) so only one option fits.

Private credit borrowers tend to be small and highly leveraged, and most private credit loans are floating rate. Which risk is most directly raised by this borrower profile when interest rates rise?

A. Credit risk: higher debt-service burdens could trigger a large number of borrower defaults.
B. Liquidity risk: investors may rush to withdraw from semiliquid funds.
C. Valuation risk: the lack of price discovery for unquoted loans.
D. Conduct risk: unsophisticated investors may not understand fund restrictions.

Answer: A

Step by step

1. Borrowers are small and highly leveraged, so they have thin cushions.
2. Floating-rate loans pass higher rates to borrowers, raising interest burdens.
3. Higher burdens lead to defaults, i.e., credit risk, reducing lender profitability.

Why A is correct: Leverage plus floating-rate payments make borrowers sensitive to rate rises, so defaults (credit risk) are the direct consequence.

Why the others are wrong

- **B:** Liquidity risk stems from fund redemption structures, not borrower leverage.
- **C:** Valuation risk stems from lack of price discovery and transparency, not borrower leverage.
- **D:** Conduct risk stems from investor suitability and disclosure, unrelated to borrower leverage.

Q473 AnalystPrep Q.6387

REBUILT — Stale valuations could also plausibly relate to the solvency concern; reworded to ask which vulnerability is the direct source of the credit losses.

Global Insurance, a large insurer, has increased its allocation to private credit. It fears that credit losses on these assets in an economic downturn could threaten its solvency. Which vulnerability of private credit is the direct source of those credit losses?

A. Stale valuations.
B. Fragile borrowers.
C. Growing share of semiliquid investment vehicles.
D. Multiple layers of leverage.

Answer: B

Step by step

1. Credit losses arise when borrowers cannot service their debt.
2. Private credit borrowers are small, highly leveraged and exposed to floating-rate increases, so they are fragile and default more in a downturn.
3. Those defaults generate the losses that hit insurers' capital.

Why B is correct: Fragile (small, highly leveraged) borrowers default in downturns, directly generating the losses insurers bear.

Why the others are wrong

- **A:** Stale valuations delay the recognition of losses but do not cause them.
- **C:** Semiliquid vehicles create redemption/liquidity risk, not credit losses.
- **D:** Layered leverage drives interconnectedness and contagion rather than being the origin of borrower losses.

Q474 AnalystPrep Q.6389

REBUILT — Options A and B were both defensible; stem now asks for the credit risk, and distractors B and C were replaced with realistic misreadings of PIK.

Frontier Investments, a private credit fund, is considering lending to a highly leveraged private-equity-backed company on a payment-in-kind (PIK) basis, so unpaid interest is added to the loan principal. Which statement best describes the main credit risk from this PIK structure?

A. The outstanding debt grows, and the lender faces default if cash flow stays insufficient to pay interest and

reduce the debt.
B. Cash interest is received earlier, which accelerates the lender's recovery of capital.
C. Deferral of interest lowers expected loss because the borrower gets temporary relief.
D. The borrower's leverage declines over the life of the loan.

Answer: A
Step by step
1. PIK adds unpaid interest to the outstanding debt.
2. The balance (and the borrower's leverage) rises over time.
3. If cash flow remains insufficient, the lender faces default on a larger claim.

Why A is correct: PIK gives the borrower temporary relief but compounds the debt, so the lender is exposed to default risk if cash flow does not recover.
Why the others are wrong
- **B:** Reverses the mechanism: PIK defers cash interest, so cash is received later.
- **C:** Relief for the borrower does not reduce lender expected loss; the larger balance raises it.
- **D:** Accrued interest increases debt, so leverage rises, not falls.

Q475 AnalystPrep Q.6390

REBUILT — Option C (floating-rate loans) is itself a Schweser-listed vulnerability, so two options were defensible; replaced C and tightened the stem to loss recognition.

A regulator worries that a sudden downturn will hit institutional investors with large private credit allocations. Which of the following is a key vulnerability that could amplify the downturn's effects by delaying the recognition of losses?
A. Infrequent, model-based valuations that lag the true value of the loans.
B. Diversification of private credit portfolios across sectors.
C. Closed-end fund structures that do not permit withdrawals.
D. Strong covenants that give lenders early warning.

Answer: A
Step by step
1. Private loans lack price discovery and regulatory monitoring.
2. Managers can defer unrealized losses, and stale values overstate assets.
3. When losses are finally recognized they can occur all at once, potentially triggering a wave of defaults.

Why A is correct: Stale valuations understate losses and postpone recognition, so stress appears suddenly and amplifies the downturn.
Why the others are wrong
- **B:** Diversification is a mitigant, reducing concentration of losses.
- **C:** Closed-end structures restrict redemptions and mitigate liquidity risk.
- **D:** Strong covenants are protective features, not vulnerabilities.

Q476 AnalystPrep Q.6391

Horizon Life, an insurance company, is evaluating the risk profile of its private credit portfolio. They are particularly focused on the potential for "contagion" within the financial system. Which aspect of the private credit market most directly contributes to this contagion risk?
A. The use of floating rate loans.
B. The multiple layers of leverage within the private credit ecosystem.
C. The focus on lending to middle-market firms.
D. The use of closed-end fund structure.

Answer: B
Step by step
1. Contagion corresponds to interconnectedness risk.
2. Borrowers, funds and investors each use leverage, so a default or capital call in one place can cascade to other participants.
3. Only option B describes this chain.

Why B is correct: Layered leverage links participants so a shortfall at one spreads (domino effect) and funding dries up across the system.

Why the others are wrong

- **A:** Floating rates mainly raise borrower credit risk; they do not link participants.
- **C:** Borrower size is a credit-risk characteristic, not a transmission channel.
- **D:** Closed-end structures restrict withdrawals and reduce, not spread, liquidity stress.

Q477 Schweser Module Quiz 92.2 Q1

Historically speaking, which of the following asset classes has experienced the highest level of losses?

A. Leveraged loans.
B. High-yield bonds.
C. Private credit loans.
D. Broadly syndicated loans.

Answer: B

Step by step

1. Private credit losses have been lower than high-yield bonds and similar to leveraged loans (PE backing and collateral help).
2. Broadly syndicated loans have deep secondary markets and are likely the lowest.
3. So high-yield bonds are highest.

Why B is correct: Historical private credit and leveraged loan losses sit below those of high-yield bonds.

Why the others are wrong

- **A:** Leveraged loans have losses similar to private credit, below high-yield.
- **C:** Private credit losses are lower than high-yield thanks to collateral and PE sponsor support.
- **D:** Syndicated loans, being liquid and often investment grade, likely have the lowest losses.

Q478 Schweser Module Quiz 92.2 Q2

REBUILT — Original asked which risk is 'currently of greatest concern', which conflicts with the reading (liquidity and interconnectedness risks currently appear low); rebuilt to ask what drives interconnectedness risk.

Which of the following best describes the main source of interconnectedness risk in corporate private credit?

A. Multiple layers of leverage among borrowers, funds and investors, combined with close ties between private equity and private credit.
B. The predominance of floating-rate loans in fund portfolios.
C. The use of gates and suspensions on semiliquid fund redemptions.
D. Infrequent valuation of unquoted loans.

Answer: A

Key correction: Source key (Schweser Module Quiz 92.2 Q2) said D (interconnectedness risk) as the greatest current concern, but the reading states liquidity and interconnectedness risks currently appear low; the item was rebuilt to remove the ambiguity.

Step by step

1. Interconnectedness arises from a labyrinth of leverage with many participants.
2. Many private credit managers are PE managers and many borrowers are PE-backed, so weakness in one spills to the other.
3. A default or capital call shortfall can have a domino effect.

Why A is correct: Layered leverage and PE-credit linkages let stress transmit between participants and sectors.

Why the others are wrong

- **B:** Floating rates drive borrower credit risk, not linkages.
- **C:** Gates and suspensions are liquidity-management tools.
- **D:** Infrequent valuation is valuation risk.

Q479 AP Mock 1 Q74

A private credit fund is evaluating a potential loan to a company in a cyclical industry. Which loan feature is particularly important to consider in order to mitigate potential downside risk?

A. Floating interest rates.

B. Closed-end fund structure.
C. Enhanced covenants.
D. Short loan maturity.

Answer: C
Step by step
1. Cyclical borrowers face earnings drops in downturns.
2. Enhanced covenants give early warning and control rights, limiting leverage and distributions before losses build.
3. Other options are either not loan-level protections or add risk.

Why C is correct: Covenants are the loan-level protection against deteriorating credit quality.
Why the others are wrong
- **A:** Floating rates raise the borrower's interest burden in stress, increasing credit risk.
- **B:** A closed-end structure is a fund feature (mitigating liquidity risk), not a loan feature.
- **D:** Short maturity creates refinancing risk for a cyclical borrower.

LO 92.d

Assess potential policy recommendations that could help mitigate the risks associated with private credit.

Q480

Schweser Module Quiz 92.2 Q3

REBUILT — Original option D (mandating external valuations or stronger internal governance) was also defensible under the reading's valuation recommendations; replaced with a clearly incorrect valuation statement so exactly one option is correct.

Which of the following statements regarding policy recommendations on private credit risks is most accurate?

A. Allowing more frequent redemptions in private credit funds can create conduct risk, because managers may be tempted to manipulate valuations to discourage redemptions.
B. Given that private credit risks are largely mitigated, authorities should avoid an overly regulatory approach that would limit the growth of this emerging asset class.
C. Regulatory arbitrage across borders and sectors has helped mitigate excessive concentration of private credit risk in certain jurisdictions and sectors.
D. Because current regulations already account for asset valuation, no further regulatory oversight of the valuation methods used by private credit investors is needed.

Answer: A
Step by step
1. Conduct risk arises from conflicts of interest and unsophisticated investors.
2. More frequent redemptions give managers an incentive to manipulate valuations to discourage redemptions, or to make redemptions conditional on reinvestment in manager-controlled investments.
3. Hence A is consistent with the reading; B, C and D each contradict it (see distractors).

Why A is correct: More frequent redemptions create a conflict of interest for the manager, who may manipulate valuations or steer redeemed money into affiliated investments, which is a conduct risk.
Why the others are wrong
- **B:** The reading argues risks are significant and call for stronger oversight and cross-sectoral, cross-border supervision, not that they are largely mitigated.
- **C:** Regulatory arbitrage concentrates risk in jurisdictions and sectors with lax rules; harmonizing risk assessment is recommended to reduce it.
- **D:** Regulations currently do not consider asset valuation, leaving it to investors; the reading recommends greater regulatory oversight of valuation methods, more frequent external valuations and audits.

Q481

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A financial stability regulator is reviewing how pension funds and insurers value their private credit holdings. Each institution currently selects its own valuation method, and the regulator has found that the same loan is carried at materially different values across institutions. Several loans are large and have valuations that are prone to

estimation error. Which of the following policy recommendations is most consistent with mitigating the valuation risk in this situation?

A. Leave valuation to each institution because it has the best information about its own loans, and rely on external credit rating agencies to confirm the carrying values.

B. Require more frequent external valuations for the significant, error-prone positions and audits of valuation methods and internal controls, with write-downs recognized in the appropriate period.

C. Prohibit write-downs on semiliquid loans until the loans are realized, so that temporary valuation noise does not trigger investor redemptions.

D. Require the fund managers to produce internal valuations at a higher frequency, because the same managers can then be relied on without any external review or audit.

Answer: B

Step by step

1. Regulations currently do not address valuation; the onus is on each institution, which produces large differences in value for the same asset depending on method.
2. The reading's remedy is greater regulatory oversight of valuation methods; where valuations are significant or error-prone, more frequent external valuations may be imposed.
3. Audits of valuations should also be performed to find deficiencies in methods and internal controls, and write-downs of illiquid and semiliquid assets must be recognized in the appropriate period.
4. Option B combines these elements.

Why B is correct: Greater regulatory oversight of valuation methods, more frequent external valuations for significant or error-prone positions, and audits of valuation methods and controls directly address inconsistent valuations and the risk of unrecognized write-downs.

Why the others are wrong

- **A:** Wrong. This is the status quo that creates the problem (self-determined valuations with no oversight), and credit rating agencies are described as providing little confidence. The error is treating the current reliance on institutions as a remedy.
- **C:** Wrong. The key valuation risk is failure to recognize write-downs of illiquid and semiliquid assets; losses must be accurately determined and recognized in the appropriate period. Deferring write-downs worsens stale valuations. The error is confusing smoothing of redemptions with risk mitigation.
- **D:** Wrong. Higher-frequency internal valuations still lack independent verification; the policy calls for external valuations where errors are likely and audits of methods and internal controls. The error is assuming frequency alone, produced by the interested party, fixes valuation risk.

Q482 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A macroprudential supervisor observes that insurers and pension funds in jurisdictions with lenient rules hold substantially larger private credit positions than their peers in stricter jurisdictions. The supervisor also lacks consistent data on leverage and interconnections between funds, banks, and institutional investors. Which of the following is the most appropriate policy response?

A. Have each national regulator separately tighten its own rules and reporting, since cross-border coordination would add little given that private credit is bilateral and nonbank.

B. Allow regulatory competition to continue, because moving risk to lenient jurisdictions disperses it and reduces the concentration of private credit risk in any one jurisdiction.

C. Harmonize risk assessment processes across sectors and borders to reduce regulatory arbitrage, and strengthen cooperation and reporting requirements among regulators to close the data gaps.

D. Rely on the leverage reported by each fund to set leverage caps, since fund-level reporting already captures leverage at SPVs, feeder and master funds, investors, and borrowers.

Answer: C

Step by step

1. Differences in strictness across countries and sectors invite regulatory arbitrage, with more risk taken where rules are lax; this may be contributing to large insurer and pension holdings.
2. Concentrated risks may require harmonizing risk assessment processes across sectors and borders.
3. Data deficiencies on leverage and interconnectedness call for more reporting requirements within and outside borders and enhanced regulator cooperation.
4. Only option C matches both elements.

Why C is correct: Harmonization reduces or eliminates regulatory arbitrage, while cross-border and cross-sector cooperation plus expanded reporting fills the data gaps on leverage and interconnectedness and reduces contagion risk.

Why the others are wrong

- **A:** Wrong. Purely national action leaves arbitrage and cross-border data gaps in place; the reading says the most effective approach is stronger cross-sectoral and cross-border regulation and supervision. The error is ignoring the cross-border nature of the problem.
- **B:** Wrong. Regulatory arbitrage concentrates risk in lax jurisdictions and sectors rather than mitigating concentration; it is a warning sign. The error is reading arbitrage as diversification.
- **D:** Wrong. Fund-level reports may omit or obscure leverage from SPVs, feeder/master funds, investors and borrowers, so caps based on them miss total leverage. The reading calls for more reporting requirements and regulator cooperation to overcome these data deficiencies. The error is assuming current reporting is complete.

Reading 93. Private Markets Investing

Cyril Demaria, Maurice Pedergnana, Remy He, Roger Rissi, and Sarah Debrand, Asset Allocation and Private Markets: A Guide to Investing with Private Equity, Private Debt, and Private Real Assets (Wiley, 2021). Chapter 3.

LO	Learning objective	Qs
93.a	explain the motivations for investors to invest in private markets.	3
93.b	describe different investment structures that can be used to invest in private assets and compare their characteristics.	3
93.c	apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages.	2
93.d	explain how different risks can arise for investors in private markets and assess how these risks can be measured.	3
93.e	describe the practice of co-investing in private markets and assess its advantages and disadvantages.	3
93.f	explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund.	2

LO 93.a

Explain the motivations for investors to invest in private markets.

Q483

Schweser Module Quiz 93.1 Q1

Which of the following statements best explains why investors may prefer private investment vehicles over public markets? Private investment vehicles:

A. require less expertise from investors.
B. harness the power of home country bias.
C. guarantee higher returns than in public markets.
D. offer investors greater control over the value creation process.

Answer: D

Step by step

1. Recall the Schweser motivations for private markets: performance and diversification, control and value creation, access to specialized structures, and related benefits.
2. Control and value creation: private investors can monitor, guide and influence the company and contribute capital, connections and operating expertise, which is generally unavailable in public markets without a very large stake.
3. Test each option: A and C contradict the reading; B is not a stated motivation; D matches the control and value creation motivation.

Why D is correct: Private investors can monitor, guide and influence portfolio companies and add operational expertise, a degree of control generally not available in public markets.

Why the others are wrong

- **A:** Private investing typically demands more expertise (due diligence, monitoring, illiquidity management), not less.
- **B:** Private markets help reduce home country bias by widening access to global opportunities; they do not harness it, and it is not a stated motivation.
- **C:** Private markets offer the potential for higher returns, but nothing is guaranteed; confusing potential with a guarantee.

Q484

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

The chief investment officer of a pension fund notes that the fund's public equity holdings are concentrated in its domestic market, which has only a small number of listed firms. She is considering private market allocations in several emerging economies that have underdeveloped or nonexistent public markets. Which of the following motivations for investing in private markets is most consistent with her objective?

A. Potential for absolute returns, because returns in emerging economies are guaranteed to be uncorrelated.
B. Control and value creation, because private investors can influence management in foreign firms.
C. Specialized investment structures, because funds of funds are available only in emerging economies.

D. Wider geographic reach, which expands diversification opportunities and helps reduce home country bias.

Answer: D

Step by step

1. Identify the objective: reduce domestic concentration and reach regions with limited public markets.
2. Map it to the six motivations: this is wider geographic reach.
3. Check the others: they describe different motivations and add claims the reading does not support.

Why D is correct: Private investments offer exposure to regions where public markets are underdeveloped or nonexistent, widening diversification and reducing home country bias.

Why the others are wrong

- **A:** Absolute-return potential rests on low correlation with public indices, which is a tendency, not a guarantee; it also does not address the geographic need.
- **B:** Control and value creation is a real motivation, but it concerns influence over company performance, not the access to regions without public markets that the CIO seeks.
- **C:** Specialized structures such as funds of funds and co-investments provide unusual risk exposures; they are not restricted to emerging economies, so the premise is false.

Q485 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk analyst at an endowment is preparing a note on why the endowment should consider private markets. Which of the following statements in the note is most accurate?

A. Private firms are far more numerous than listed firms, roughly 300 million versus 52,000 worldwide, giving access to more industries, business models, and lifecycle stages.

B. Because private assets show low correlation with public indices, they are not exposed to macroeconomic risks such as inflation or interest rate changes.

C. Private investments in public equities (PIPEs) are investments in firms with no listed securities and no ability to ever trade publicly.

D. Private investors typically obtain the same level of influence over company performance as a small minority shareholder in a listed firm.

Answer: A

Step by step

1. Recall the broader investment universe motivation: more than 300 million private companies versus about 52,000 listed.
2. This gives unique diversification across industries, models, and lifecycle stages.
3. Eliminate statements that overstate diversification, understate control, or misdefine PIPEs.

Why A is correct: The sheer number of private firms and their presence at all lifecycle stages creates a more diverse investment universe than public markets alone.

Why the others are wrong

- **B:** Low correlation with public indices does not remove exposure to macro risks; private assets remain exposed to downturns, inflation, and interest rate policy.
- **C:** PIPEs are private transactions through which publicly traded firms with limited trading volume raise capital, so the issuer is listed.
- **D:** Private investors can monitor, guide, and influence companies; this control is generally unavailable in public markets unless the stake is very large.

LO 93.b

Describe different investment structures that can be used to invest in private assets and compare their characteristics.

Q486 Schweser Module Quiz 93.1 Q2

Which of the following private market investment structures requires the highest level of investor expertise and involvement, offers high control, but results in concentrated risk and low diversification unless many positions are held?

A. Private funds.
B. Direct investment.
C. Managed accounts.
D. Private funds of funds.

Answer: B

Step by step

1. Match the profile: highest involvement/expertise, high control, concentrated risk, low diversification.
2. Direct investment (owning a private company or asset outright) fits every attribute; diversification requires many separate positions.
3. Compare the alternatives: private funds pool capital under a GP (low to moderate involvement, limited control); managed accounts give moderate involvement and control with high diversification; FoFs have very low involvement and very high diversification.

Why B is correct: In a direct investment the investor owns the asset and does the sourcing, diligence and oversight itself, giving maximum control but one-asset concentration unless many positions are built.

Why the others are wrong

- **A:** Private funds pool capital and delegate decisions to the GP, so involvement is low to moderate and diversification moderate to high; the mistake is treating fund investing as hands-on.
- **C:** Managed accounts offer customization and moderate control with high diversification; confusing negotiated control with the full ownership control of direct investing.
- **D:** FoFs are closest to passive investing with very low involvement and very high diversification, the opposite of the profile described.

Q487 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A family office with limited experience in private markets wants an entry point that places a high priority on diversification and comes closest to a passive approach. The office accepts that it will not control individual investment decisions. Which of the following structures is most appropriate, and what is its main added cost?

A. Special purpose vehicle; it concentrates exposure in one asset or deal.
B. Private fund of funds; it adds an extra layer of management fees.
C. Managed account; it usually requires a large capital commitment and negotiation.
D. Direct investment; it requires substantial local networks to source deals.

Answer: B

Step by step

1. Match the profile: limited experience, diversification priority, near-passive approach.
2. Funds of funds invest in a portfolio of private funds, diversifying beyond a single fund.
3. The trade-off is a second layer of fees.

Why B is correct: A private fund of funds offers the closest approximation to a passive approach among private market options and enhances diversification versus a single fund, at the cost of additional management fees.

Why the others are wrong

- **A:** An SPV holds a specific asset or group of assets for targeted exposure, suited to sophisticated investors, not a diversified entry point.
- **C:** Managed accounts suit large institutions seeking tailored mandates; they need big commitments and are not a passive diversified entry point.
- **D:** Direct investment requires strong expertise, networks, and active involvement and gives low diversification; it is the opposite of passive.

Q488 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A large pension plan negotiates a separate managed account with a private equity manager under an advisory mandate. Which of the following is most likely true of this arrangement?

A. The manager makes all investment decisions, and the arrangement typically involves carried interest.
B. The plan's capital is pooled with other investors in a closed-end limited partnership managed by the general partner.
C. The plan retains veto authority over investments, and the arrangement often does not involve carried interest.
D. The arrangement is mainly suited to retail investors because it requires only a small capital commitment.

Answer: C

Step by step

1. Managed accounts are separate accounts tailored to the investor's mandate.
2. Advisory mandates leave the investor with veto authority; discretionary mandates give the manager full authority.
3. Discretionary mandates typically carry carried interest; advisory ones often do not.

Why C is correct: In an advisory mandate the investor retains veto authority, and performance-based fees (carried interest) are typically associated with discretionary mandates rather than advisory ones.

Why the others are wrong

- **A:** This describes a discretionary mandate, where the manager makes all decisions and carried interest is typical.
- **B:** Pooling in a closed-end limited partnership describes a private fund, not a separate managed account.
- **D:** Managed accounts usually need a substantial capital commitment and negotiation, so they suit large institutions, not retail investors.

LO 93.c

Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages.

Q489 Schweser Module Quiz 93.2 Q1

Which of the following performance metrics for private equity best incorporates the timing of cash flows but can be artificially inflated by early distributions or the use of credit lines?

A. Internal rate of return (IRR).
B. Residual value to paid-in (RVPI).
C. Public market equivalent (PME).
D. Multiple of invested capital (MOIC).

Answer: A

Step by step

1. The metric must reflect when cash is called and distributed, which rules out the multiples (MOIC and its sub-metrics), since they ignore holding period.
2. Among time-sensitive metrics, only IRR is sensitive to early distributions: returning cash early shortens the effective holding period and raises IRR.
3. Subscription credit lines delay capital calls, which shortens the measured investment period and also lifts IRR.
4. Therefore the answer is IRR.

Why A is correct: IRR is the time-weighted, cash-flow-based rate of return, so accelerating distributions or delaying capital calls (via credit lines) mechanically raises it without improving true economic performance.

Why the others are wrong

- **B:** RVPI is residual value divided by paid-in capital, a multiple with no timing component; it is vulnerable to overstated valuations, not to early distributions.
- **C:** PME does account for timing by comparing cash flows with a public index, but it is not the metric described as inflated by early distributions or credit lines; that weakness is specific to IRR.
- **D:** MOIC is a pure multiple of capital and ignores holding period, so it does not incorporate timing at all.

Q490 Schweser Module Quiz 93.2 Q2

REBUILT — Original option A (DPI) is itself a MOIC submetric that also ignores holding period, making two options defensible; replaced it with time-zero IRR (modified IRR).

Which of the following private market performance metrics is most useful for assessing absolute performance (total value relative to capital invested) without regard to holding periods?

A. Time-zero IRR (modified IRR).
B. Internal rate of return (IRR).
C. Public market equivalent (PME).
D. Multiple of invested capital (MOIC).

Answer: D

Step by step

1. The metric must be a plain multiple of value to capital, with no time dimension.
2. MOIC = (distributions + residual value) / capital invested; a 2.0x MOIC means value doubled, whether in 3 years or 10.
3. IRR, time-zero IRR and PME all depend on the timing of cash flows.

Why D is correct: MOIC measures total value generated per dollar invested as an absolute multiple and does not adjust for how long the capital was held.

Why the others are wrong

- **A:** Time-zero IRR still is a rate of return driven by the timing of cash flows; it only changes the starting point to reduce the effect of delayed capital calls.
- **B:** IRR is an annualized rate that is explicitly sensitive to holding period and cash flow timing.
- **C:** PME benchmarks timed cash flows against a public index, so it is a relative, timing-dependent measure rather than an absolute multiple.

LO 93.d

Explain how different risks can arise for investors in private markets and assess how these risks can be measured.

Q491 Schweser Module Quiz 93.2 Q3

Which of the following metrics best helps investors monitor the impact of macro risks and selection risk in private market performance?

A. Value at risk (VaR).
B. Default and loss rates.
C. Performance dispersion.
D. Strategic asset allocation (SAA) weights.

Answer: C

Step by step

1. Identify the risks named: macro (systematic) risk and selection risk (choosing the wrong manager or deal).
2. Match each risk to its Schweser measurement tool: macro risk is assessed by the spread between the highest and lowest pooled average TVPI; the same dispersion, and a fund's TVPI versus its category's pooled average, monitors selection risk.
3. Only performance dispersion addresses both risks, so the answer is C.

Why C is correct: Wide dispersion of TVPI across managers shows both how much macro conditions move returns and how much manager or deal choice matters, so one metric covers both risks.

Why the others are wrong

- **A:** VaR and stress testing model worst-case outcomes for strategy-specific risks and macro shocks, but they do not capture manager selection risk.
- **B:** Historical default and loss rates benchmark recovery and operational risk in private debt, not macro or selection risk.
- **D:** Comparing portfolio weights to a benchmark measures strategic asset allocation risk, not macro or selection risk.

Q492 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager reviews a private equity fund with a DPI of 0.55x and a TVPI of 1.65x. The fund's net asset value is determined by the manager. Which of the following conclusions is most appropriate?

A. RVPI is 0.55x, so the gap between DPI and TVPI indicates selection risk relative to the category pooled average.

B. RVPI is 2.20x, so the fund has distributed most of its value and reporting risk is low.

C. RVPI is 1.10x, so the fund is largely realized and DPI is more exposed to valuation manipulation than TVPI.

D. RVPI is 1.10x, so about two-thirds of reported value rests on manager-determined NAV, which signals potential reporting risk.

Answer: D

Step by step

1. TVPI = DPI + RVPI, so RVPI = 1.65 − 0.55 = 1.10x.
2. Share of unrealized value: 1.10/1.65 = 66.7%.
3. A substantial gap between DPI (realized cash) and TVPI (NAV-dependent) may indicate reporting risk.

Why D is correct: Comparing DPI with TVPI assesses reporting risk; the large unrealized portion (RVPI 1.10x, two-thirds of TVPI) depends on subjective NAV estimates.

Why the others are wrong

- **A:** Mistakes RVPI for DPI; selection risk is assessed by comparing a fund's TVPI with the pooled category average, not by the DPI-TVPI gap.
- **B:** Adds DPI and TVPI instead of subtracting (1.65 + 0.55 = 2.20) and misreads the result; the large gap actually signals reporting risk.
- **C:** RVPI is correct, but DPI is based on realized cash flows and is less exposed to manipulation than TVPI or RVPI; the fund is also mostly unrealized.

Q493 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

After a sharp decline in listed equity prices, a CIO sees that the portfolio's private equity allocation now appears well above its strategic target. Private fund valuations were last updated before the decline. The CIO proposes selling private assets immediately to restore the target. Which of the following is the most appropriate assessment?

A. The apparent overweight is partly artificial because stale private valuations are compared with current liquid public prices (denominator effect), so rebalancing may be unnecessary.

B. The overweight reflects settlement risk, which is higher in private markets than in public transactions.

C. The overweight is a true tactical asset allocation error caused by market timing and should be corrected immediately.

D. The overweight proves that private valuations are more accurate than public prices because they did not fall.

Answer: A

Step by step

1. Public prices fell quickly; private valuations adjust slowly.
2. Comparing an outdated private value with a current public value distorts weights (denominator effect, a valuation risk).
3. Differences in weighting may therefore be artificial, causing unnecessary rebalancing; valuation risk is assessed via third-party appraisals, DCF, comparables, and IPEV guidelines.

Why A is correct: Private asset values adjust slowly relative to public prices, so part of the weight difference is an artifact of stale pricing rather than a genuine deviation.

Why the others are wrong

- **B:** Settlement risk arises with counterparties and is typically lowered in private markets by extensive due diligence; it does not explain the weight shift.
- **C:** TAA risk concerns market timing forecasts, not a weight distortion produced by stale valuations.
- **D:** Slow-moving private valuations reflect stale pricing and subjectivity, not superior accuracy.

LO 93.e

Describe the practice of co-investing in private markets and assess its advantages and disadvantages.

Q494 Schweser Module Quiz 93.3 Q1

Which of the following is a key advantage of co-investing compared with investing as a limited partner in the primary fund?

A. Increased legal protections.
B. Reduced management fees.
C. Reduced need for due diligence.
D. Increased portfolio diversification.

Answer: B

Step by step

1. Recall the advantages of co-investing: lower (or no) management fee and carried interest, greater control over deal selection and exposure, higher potential net returns, and learning/access.
2. Recall the disadvantages: adverse selection, speed and complexity, resource needs, concentration, and weaker legal protections.
3. Compare each option: only reduced management fees is an advantage; the others are either disadvantages or the opposite of reality.

Why B is correct: Co-investors typically pay reduced or no management fees and carry on the specific deal, which raises net returns relative to the primary fund (while still sharing transaction and due diligence costs).

Why the others are wrong

- **A:** Co-investors are usually passive and accept fewer legal protections than LPs in the primary fund; this reverses the actual disadvantage.
- **C:** Co-investments often arise on short notice and require accelerated due diligence by the investor's own team; due diligence is not reduced, only compressed.
- **D:** Co-investments are larger, more concentrated single-deal positions, so diversification falls, not rises.

Q495 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A fund manager invites a large limited partner to co-invest in a leveraged buyout that exceeds the fund's usual investment limit. Compared with investing only through the primary fund, which of the following is most likely a disadvantage for the co-investor?

A. The co-investor bears none of the transaction and due diligence costs, reducing its incentive to monitor.
B. The co-investment is typically a larger, more concentrated position, leaving the portfolio less diversified and more exposed to deal-specific risk.
C. The co-investor has less ability to shape its exposures and portfolio weights.
D. The co-investor pays higher management fees and carried interest than primary fund investors.

Answer: B

Step by step

1. Compare co-investment characteristics with primary fund exposure.
2. Co-investments are larger, concentrated positions, often in leveraged buyouts or infrastructure.
3. The other options contradict listed advantages (lower fees, greater control, shared costs).

Why B is correct: Concentration and diversification risk is a listed disadvantage: co-investments are typically larger and more concentrated than standard fund exposures.

Why the others are wrong

- **A:** Co-investors still share in transaction and due diligence costs.
- **C:** Co-investing gives greater control by allowing investors to target specific deals and shape weights.
- **D:** Co-investors typically pay reduced or no management fees and carry, which is an advantage.

Q496 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

An investor with a small internal team is offered several co-investment opportunities by its general partners, each requiring a decision within two weeks. A consultant warns the investor about adverse selection. Which of the following best describes this concern?

A. Co-investors are given more legal protections than fund investors, which raises the cost of each deal.
B. Co-investment access is open to all investors on equal terms, so the best deals are quickly competed away.

C. Offered deals may be less attractive or outside the manager's core expertise, and deal flow tends to be procyclical, appearing when valuations are already high.
D. Co-investors always receive better deals than the fund, which leads to crowding out of the fund's limited partners.

Answer: C
Step by step
1. Adverse selection means the opportunities offered may be worse than average.
2. The reading lists less attractive deals, deals outside core expertise, and procyclical deal flow.
3. Short deadlines compound the problem by forcing accelerated due diligence.

Why C is correct: Adverse selection arises because co-investors may be offered less attractive deals or ones outside the manager's expertise, and deal flow is procyclical.
Why the others are wrong
- **A:** Co-investors usually accept fewer legal protections than primary fund limited partners, not more.
- **B:** Access is usually restricted to large, established investors with strong manager relationships, not open to all.
- **D:** This is the opposite of adverse selection; there is no support for co-investors receiving systematically better deals.

LO 93.f

Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund.

Q497

Schweser Module Quiz 93.3 Q2

Which of the following is a key benefit of investing in private markets through intermediated vehicles such as private equity or private debt funds?
A. Access to expertise.
B. More access to liquidity.
C. Elimination of management fees.
D. Increased transparency and control.

Answer: A
Step by step
1. Benefits of intermediation (Schweser Reading 93): manager expertise, diversification/risk reduction, deal flow access, cost and resource sharing.
2. Fund managers bring skills in sourcing, due diligence, structuring, monitoring and exiting, which reduces operational risk for investors lacking those skills.
3. Only option A matches a listed benefit; B, C and D each describe the opposite of a known challenge.

Why A is correct: Fund managers provide specialized skills in sourcing, due diligence, structuring, monitoring and exit, which is a core benefit of intermediation.
Why the others are wrong
- **B:** Private funds are illiquid: capital is locked up long-term with no redemption rights and only a limited secondary market (often at NAV discounts). Confuses pooling with liquidity.
- **C:** Intermediation adds fees: management fees plus carried interest. Fees are a challenge, not eliminated.
- **D:** Intermediation adds a layer between investors and assets, reducing transparency and control (information asymmetry, blind pools).

Q498

Schweser Module Quiz 93.3 Q3

Which of the following is a key challenge of investing in private markets through intermediated vehicles such as private equity or private debt funds?
A. Reinvestment risk.
B. Low information asymmetry.
C. Increased concentration risk.
D. Performance hurdle from fees.

Answer: D

Step by step

1. Challenges of intermediation (Schweser Reading 93): high fees, principal-agent risk, lack of transparency/information asymmetry, blind pool risk, illiquidity.
2. Investors pay management fees plus carried interest, so the fund must earn enough just to cover fees: a natural performance hurdle.
3. Check A-C against the list: none is a stated challenge of intermediation, and B and C contradict it.

Why D is correct: Management fees and carried interest are paid on top of underlying returns, creating a performance hurdle the manager must clear for net returns to be attractive.

Why the others are wrong

- **A:** Reinvestment risk relates to the IRR's assumed reinvestment rate (a performance-measurement issue), not a challenge specific to intermediation.
- **B:** Intermediation increases information asymmetry through limited disclosure and infrequent valuations; 'low' is the opposite.
- **C:** Pooling capital across many deals, sectors and geographies reduces concentration risk; concentration is more a feature of direct or co-investment.

Reading 94. Performing Due Diligence on Specific Managers and Funds

Kevin R. Mirabile, Hedge Fund Investing: A Practical Approach to Understanding Investor Motivation, Manager Profits, and Fund Performance, 2nd Edition (Hoboken, NJ: Wiley Finance, 2016). Chapter 12.

LO	Learning objective	Qs
94.a	identify reasons for the failures of hedge funds in the past.	3
94.b	explain elements of the due diligence process used to assess investment managers.	6
94.c	identify themes and questions investors can consider when evaluating a hedge fund manager.	6
94.d	describe criteria that can be evaluated in assessing a hedge fund's risk management process.	6
94.e	explain how due diligence can be performed on a hedge fund's operational environment.	2
94.f	explain how a hedge fund's business model risk and its fraud risk can be assessed.	8
94.g	describe elements that can be included as part of a due diligence questionnaire.	3

LO 94.a

Identify reasons for the failures of hedge funds in the past.

Q499

Schweser Module Quiz 94.1 Q1

Based on historical evidence, which of the following factors is least likely to result in the eventual failure of a hedge fund?

A. Excessive internal controls.
B. Taking on more systematic risk.
C. Making decisions in a committee setting.
D. Materially misstated financial statements.

Answer: B

Step by step

1. Recall the Schweser list of past failure causes: poor investment decisions, fraud (accounting, valuation, theft), extreme events, excess leverage, lack of liquidity, poor controls, insufficient questioning, insufficient attention to returns.
2. Excessive controls: listed under insufficient attention to returns (too many expenses, too little return), so a cause.
3. Committee decisions: listed under insufficient questioning (dominant members, fear of voicing concerns), so a cause.
4. Misstated financial statements: accounting fraud, so a cause.
5. Taking on more systematic risk is not on the list; by itself it raises exposure to market moves but is not a historical failure driver, so it is the least likely.

Why B is correct: Historical failures stem from fraud, leverage, illiquidity, weak or excessive controls and groupthink, not from systematic (market) risk exposure as such.

Why the others are wrong

- **A:** Wrong: over-engineering controls to cut operational risk inflates expenses and depresses returns, a documented failure cause.
- **C:** Wrong: committee processes can be dominated by one member or inhibit dissent (insufficient questioning), a documented failure cause.
- **D:** Wrong: misstated financials are accounting fraud, a documented failure cause.

Q500

AnalystPrep Q.2592

REBUILT — Option A was defensible in general; reworded it as an absolute claim so only B (dominant head / silent members) is correct.

A large investment firm manages multiple funds. The investment committee decides on the investment strategy and takes the final decision on the investment opportunities available in the market. While making the investment decision, the committee relies heavily on the past performance of a similar strategy. The Chief Investment Officer (CIO) heads the investment committee with fund managers as members. The firm carries out a performance appraisal of its fund managers quarterly, and the CIO heads the appraisal committee.

Which of the following is the most appropriate statement with regards to the committee's role in deciding the investment strategy?

A. Since the committee consists of multiple members, its decisions are necessarily more robust than those of an individual, whatever the committee's dynamics.
B. Members of the committee could sometimes not speak out during controversial scenarios, and the committee head may be able to direct the decisions for the whole team.
C. Since the committee consists of multiple members, reaching a consensus may be difficult.
D. Since the committee consists of multiple members, diverse opinions may cause confusion.

Answer: B
Step by step
1. Insufficient questioning is a cited cause of fund failures: in committees a dominant member (here the CIO) can sway decisions.
2. Members may be afraid to voice concerns, so dissent is suppressed.
3. B describes this; A is an unconditional claim, C and D are not inherent.

Why B is correct: A dominant committee head and silent members defeat the benefit of group decision-making (insufficient questioning).
Why the others are wrong
- **A:** Assumes group size guarantees robustness, ignoring dominance and silence.
- **C:** Difficulty reaching consensus is not the documented risk; the risk is too little dissent.
- **D:** Diverse opinions are desirable; the problem is that they are not voiced.

Q501 AnalystPrep Q.2593

A large investment firm manages multiple funds. The investment committee decides on the investment strategy and takes the final decision on the investment opportunities available in the market. While making the investment decision, the committee relies heavily on the past performance of a similar strategy. The Chief Investment Officer (CIO) heads the investment committee with fund managers as members. The firm carries out a performance appraisal of its fund managers quarterly, and the CIO heads the appraisal committee.
With regards to the CIO heading the appraisal committee, which of the following is the most appropriate statement?
A. The arrangement ensures that the interests of the fund managers, head of the committee, and the firm are aligned.
B. The arrangement guarantees robust investment decisions by the committee.
C. The arrangement may result in suboptimal investment decisions.
D. The arrangement may result in better discussions on investment opportunities.

Answer: C
Step by step
1. The CIO heads both the investment committee and the managers' appraisal.
2. Managers whose pay and careers depend on the CIO are reluctant to challenge him or her.
3. Dissent is suppressed, so decisions may be suboptimal (insufficient questioning).

Why C is correct: The appraisal power gives the CIO undue influence over members, discouraging questioning and risking suboptimal decisions.
Why the others are wrong
- **A:** Appraisal control creates dependence, not alignment of interests.
- **B:** No structure guarantees robust decisions, and this one weakens them.
- **D:** Better discussion is the opposite of the likely effect: members hold back.

LO 94.b

Explain elements of the due diligence process used to assess investment managers.

Q502 AnalystPrep Q.2595

REBUILT — Original was stem-driven and weak (answer depended on invented investor criteria, past returns ignored); rebuilt to test the attribution-analysis element of due diligence.
An investor is performing due diligence on an investment manager whose fund has posted strong returns. The investor has already read the fund's prospectus and confirmed that an independent auditor verified the reported

returns. The investor now wants to determine whether the returns resulted from manager skill and control, from luck, or from factors beyond the manager's control, and whether the return was in line with the risk taken. Which element of the due diligence process is most appropriate for this purpose?

A. Review of the fund's offering memorandum
B. Reference checks on the manager's former employers
C. Return attribution analysis
D. Review of the redemption terms and blackout periods

Answer: C

Step by step

1. The question is about WHY the returns occurred (skill vs. luck vs. outside factors) and whether return matched risk.
2. Per the reading, an attribution analysis is the element used to determine how returns were generated and whether they were in line with the risk taken.
3. The prospectus and third-party verification are already done and address strategy description and accuracy of reported numbers, not the source of returns.
4. Therefore, perform a return attribution analysis.

Why C is correct: Attribution analysis decomposes performance into its sources, separating manager skill from luck and market factors and relating return to risk taken.

Why the others are wrong

- **A:** The offering memorandum describes the strategy and terms; it is the starting point but does not explain what generated realized returns.
- **B:** Reference checks assess the manager's character and track as a person (leader/follower, integrity), not the source of the fund's returns.
- **D:** Redemption terms and blackout periods are an operations/business-model item; they say nothing about the origin of returns.

Q503

BT Mock Q65

REBUILT — Original options included an invented SEC filter and a Mirabile-specific answer not in the Schweser text; rebuilt to test that traditional criteria alone are insufficient.

An investor is performing due diligence on a start-up hedge fund manager with an impressive educational and employment background, a strong reputation, and a short but strong track record. Which approach is most appropriate?

A. Rely mainly on the manager's background and reputation, since the fund has too little history to assess further.
B. Focus exclusively on past performance, because it is the best predictor of future returns.
C. Skip an assessment of operations, because a small start-up has few controls to evaluate.
D. Supplement background, reputation and performance with an assessment of the investment process, risk controls, operations and business model.

Answer: D

Step by step

1. The reading states that background, reputation (education, employment) and past performance have traditionally been key but are no longer sufficient on their own.
2. Additional elements are the investment process and risk controls (starting with the prospectus/offering memorandum) and the fund's operations and business model (internal controls, segregation of duties, valuation, fees, redemption limits).
3. A start-up especially needs this since it has little track record; only D reflects the full process.

Why D is correct: Modern due diligence combines traditional criteria with assessment of investment process, risk controls, operations and business model.

Why the others are wrong

- **A:** Relies only on traditional criteria, which the reading says are insufficient; the short history makes this more dangerous, not less.
- **B:** Overweights past performance, from which investors should not get carried away; it is not an assurance of future returns.
- **C:** Operations and controls (segregation of duties, valuation, fraud prevention) are core elements, and are especially relevant in young firms.

Q504 BT answers P2.T8.414.3 · BT notes P2.T9.414.3

According to Mirabile, which of the following is TRUE about hedge fund due diligence?

A. An investor must use a comprehensive checklist to ensure that nothing is left out or omitted yet remain free to ask open-ended questions that provide insights into a firm's philosophy or culture.

B. Due diligence is an activity with diminishing marginal returns which consumes the time of the fund's investment professionals; therefore a potential investor is wise to limit the inquiry to a quantitative analysis which tends to be objective and fact-based.

C. Hedge funds are still largely unregulated and, unfortunately, anyone who feels that they have been defrauded by a hedge fund cannot realistically report to a government agency as no single agency has jurisdiction.

D. Because due diligence is focused on the firm's risk management process, fund operating environment, business model and fraud risk, the investment strategy (e.g., equities, fixed-income, distressed) is largely irrelevant.

Answer: A

Step by step

1. B is false: the process includes qualitative work (manager interviews, reference and background checks, ownership, business model), not only quantitative analysis.
2. C is false: the reading refers to filings with the SEC and state securities authorities (Form ADV) and to checking SEC databases for litigation, so regulators are relevant.
3. D is false: due diligence includes strategy questions (style, derivatives, shorting, leverage, liquidity) tailored to the strategy.
4. A is the only true statement: a comprehensive (questionnaire/checklist) approach is combined with open-ended qualitative questioning.

Why A is correct: Due diligence is comprehensive and structured (checklist/questionnaire) yet also qualitative and exploratory.

Why the others are wrong

- **B:** Reduces due diligence to quantitative analysis; the reading stresses interviews, references and background checks.
- **C:** Incorrect claim about regulation; regulators and public databases are part of the process.
- **D:** Strategy-specific questions are a core part of manager evaluation, so strategy is not irrelevant.

Q505 BT notes P2.T9.413.1

REBUILT — Original keyed on Mirabile's 'dinosaur' CEO/CIO point not in the Schweser reading; rebuilt as an EXCEPT item on Schweser's manager-evaluation elements.

In performing due diligence on an investment manager, each of the following is an appropriate element EXCEPT:

A. Reference checks with former employers, colleagues, clients and investors.

B. Reviewing the manager's Form ADV and public databases for litigation or criminal history.

C. Asking whether members of the investment team have ownership interests in the firm.

D. Relying on the manager's reputation and past performance in place of assessing the investment process and risk controls.

Answer: D

Step by step

1. A: reference checks are listed under investment management evaluation.
2. B: background checks include Form ADV review and examining public databases and the SEC website.
3. C: ownership interests help align the team with investors and are a stated inquiry.
4. D: reputation and past performance are no longer sufficient on their own; process and risk controls must also be assessed. D is the exception.

Why D is correct: Reputation and track record are inputs, not substitutes, for assessing process, risk controls, operations and business model.

Why the others are wrong

- **A:** Appropriate element (reference checks), so not the exception.
- **B:** Appropriate element (background check), so not the exception.
- **C:** Appropriate inquiry (ownership aligns interests and retains staff), so not the exception.

Q506 Schweser Module Quiz 94.1 Q2

In performing due diligence on a potential investment manager, which of the following factors is the least important for the investor to consider?

A. Risk controls.
B. Business model.
C. Past performance.
D. Investment process.

Answer: C

Step by step

1. Elements of due diligence: investment background/reputation, past performance, investment process, risk controls, operations and business model.
2. Past performance has traditionally been key but is no longer sufficient on its own, and investors should not get caught up in past successes.
3. Risk controls, business model and investment process are fundamental to the assessment.
4. Therefore past performance is the least important of the four.

Why C is correct: Past performance is not indicative of future results, whereas process, controls and business model determine whether results are repeatable and safe.

Why the others are wrong

- **A:** Risk controls are fundamental (fraud prevention, limits, segregation of duties).
- **B:** Business model (fees, valuation, redemption terms) is a core element.
- **D:** Investment process is a core element, reviewed starting with the prospectus.

Q507 AnalystPrep Q.2591

A large investment firm manages multiple funds. The investment committee decides on the investment strategy and takes the final decision on the investment opportunities available in the market. While making the investment decision, the committee relies heavily on the past performance of a similar strategy. The Chief Investment Officer (CIO) heads the investment committee with fund managers as members. The firm carries out a performance appraisal of its fund managers quarterly, and the CIO heads the appraisal committee.

Which of the following statements is most accurate?

A. Past performances often predict future performances; hence, the committee must rely on past performances as its best quantitative tool.
B. The committee must not rely entirely only on past performances to decide on the strategy.
C. The committee must examine at least the past 10 years of performances to select a strategy.
D. The committee must examine performance data since the firm's inception to select a strategy.

Answer: B

Step by step

1. Due diligence should not rely on past performance alone; investors should not get caught up in a manager's past successes.
2. Process, risk controls, operations and business model must be assessed too.
3. Hence the committee should not rely entirely on past performance.

Why B is correct: Past performance is no longer sufficient on its own; it must be combined with assessment of process, controls and business model.

Why the others are wrong

- **A:** Treats past performance as a reliable predictor and 'best' tool, the very over-reliance to avoid.
- **C:** No fixed 10-year minimum is prescribed; the issue is reliance on history alone.
- **D:** Inception-to-date data is not required and still only covers performance.

LO 94.c

Identify themes and questions investors can consider when evaluating a hedge fund manager.

Q508 AnalystPrep Q.2596

REBUILT — Original relied on a Mirabile-specific 'mixed seniority' point not in the Schweser reading and was ambiguous; rebuilt around the reference/background-check and verification elements of manager evaluation.

Gong Fang attended an investor meeting at which a large investment management firm launched a new fund and presented the past performance of its earlier funds. Fang's questions were answered by investor relations staff and the CEO, and he is about to invest. He has not contacted anyone outside the firm or verified any of the performance figures. Which advice is most consistent with a proper manager due diligence process?

A. Conduct independent reference checks and background checks on the manager and verify the presented track record with a third party before investing.

B. Invest, because attending the meeting and having his questions answered constitutes due diligence.

C. Invest, because a strong presentation of past performance is the most important part of due diligence.

D. Decline, because a firm's marketing presentation is never an acceptable source of information.

Answer: A

Step by step

1. Everything Fang has seen came from the firm itself (a sales-oriented presentation).
2. The reading's manager evaluation calls for reference checks (former employers, colleagues, clients, investors), background checks (Form ADV, public databases, credit reports) and asking whether past performance was audited or verified by a third party.
3. Only A adds independent verification; B and C rely solely on firm-provided information, and D overreaches (the presentation may be used, just not relied on alone).

Why A is correct: Due diligence requires independent verification (references, background checks, third-party verified performance), not reliance on information supplied by the manager's own marketing.

Why the others are wrong

- **B:** Treats attending a firm-run meeting as due diligence; it provides no independent verification.
- **C:** Past performance is a traditional consideration but is not sufficient on its own and here is unverified.
- **D:** Overreaction: the presentation is a legitimate input but must be supplemented with independent checks; the reading does not say to reject it.

Q509 AnalystPrep Q.2580

John McAdams, a pension fund manager, is considering diversifying the fund's portfolio by adding investments in hedge funds. However, John has serious concerns about the possible asymmetry of risk-sharing that typically comes with hedge funds. What strategy should John adopt to mitigate the potential risk asymmetry inherent in hedge fund investments?

A. Distribute the funds across various hedge fund strategies to ensure diversification of risk.

B. Choose a hedge fund manager who has an excellent reputation and manages portfolios for other large pension funds.

C. Verify that the chosen hedge fund managers have a substantial portion of their own assets invested in their respective funds.

D. Mandate daily position reports from the hedge fund to monitor closely for any possible risk asymmetry.

Answer: C

Step by step

1. The risk asymmetry arises because managers share in gains but not in losses.
2. The remedy must make managers bear losses alongside investors, i.e. a large personal investment in the fund (ownership alignment).
3. Diversification, reputation and reporting do not change the manager's payoff.

Why C is correct: Manager co-investment ('skin in the game') aligns manager and investor interests, which is the ownership theme of manager evaluation.

Why the others are wrong

- **A:** Diversifying across strategies reduces portfolio risk but leaves each manager's asymmetric payoff unchanged.
- **B:** Reputation and big clients do not make the manager bear losses; they are not a proxy for alignment.
- **D:** Daily reports improve monitoring but do not remove the incentive misalignment, and are rarely feasible.

Q510 AnalystPrep Q.2597

A fund manager states that: "A firm's ownership structure impacts performance." Which of the following is the most accurate statement?

A. A firm's ownership structure helps in aligning the interests of the fund managers with those of investors.
B. A firm's ownership does not affect its performance.
C. The presence of large personal investments on the part of fund managers increases the firm's returns.
D. The presence of significant personal investments on the part of fund managers decreases the firm's returns.

Answer: A

Step by step

1. Ownership interests help align the interests of the investment team and investors.
2. They also help attract and retain quality staff.
3. They do not mechanically raise or lower returns, so C and D overstate, and B denies any effect.

Why A is correct: Ownership by the investment team aligns incentives with investors, which is why ownership is a manager evaluation theme.

Why the others are wrong

- **B:** Ownership structure does matter through alignment and staff retention.
- **C:** Overstates: alignment may help, but higher returns are not guaranteed.
- **D:** No basis; co-investment raises, not lowers, alignment.

Q511 BT answers P2.T8.413.1

REBUILT — Original referenced an outside text with garbled wording; restated as a self-contained item with the reason D is undesirable made explicit.

In performing due diligence on a hedge fund manager, each of the following is a desirable step or characteristic EXCEPT:

A. Performing reference checks and background checks on the firm's principals.
B. Obtaining specific information on the firm's ownership structure.
C. Having access to the actual decision makers and risk takers of the fund.
D. The founder/CEO also serving as CIO with sole responsibility for investment, risk and business decisions, because this concentrates accountability.

Answer: D

Step by step

1. A: reference and background checks are core elements of manager evaluation.
2. B: ownership interests indicate alignment and retention, so investors inquire into them.
3. C: access to decision makers lets investors assess them directly.
4. D: concentrating investment, risk and business roles in one person removes independent checks and creates key-person risk, so it is not desirable.

Why D is correct: One person controlling investment, risk and business functions removes oversight and creates key-person risk.

Why the others are wrong

- **A:** Desirable step, so not the exception.
- **B:** Desirable; ownership information is a required inquiry.
- **C:** Desirable; direct access to decision makers is expected.

Q512 Mock 2 Q14

REBUILT — Original had two defensible answers (peer comparison is a track record question in Schweser, risk management is a separate theme); rewritten to ask which question belongs to a different theme.

An investor evaluates a hedge fund manager using four themes: strategy, ownership, track record and investment management. Which of the following questions is LEAST likely to be asked when examining the manager's strategy?

A. Are most of the investment team members who generated the past results still employed by the firm?
B. Are derivatives used for hedging or for speculative purposes?
C. What is the fund's level of turnover and liquidity?
D. To what extent are quantitative models used in the investment process?

Answer: A
Step by step
1. Strategy questions cover style, turnover and liquidity, quantitative modeling, short sales and derivatives use.
2. B, C and D are all in this list.
3. A (continuity of the team behind past results) belongs to the track record theme.

Why A is correct: Team continuity relates to whether the track record is repeatable, so it is a track record question.
Why the others are wrong
- **B:** Derivatives use is a strategy question.
- **C:** Turnover and liquidity is a strategy question.
- **D:** Quantitative modeling is a strategy question.

Q513 Schweser Module Quiz 94.1 Q4

Lisa Tahara, FRM, is considering an institutional investment in a hedge fund that has experienced volatile and generally positive returns in the past. Which of the following considerations about the fund's track record is least relevant for consideration in her investment decision?

A. Size of investment assets.
B. Absolute level of past returns.
C. Verification of returns by a third party.
D. Employment continuity of the investment team.

Answer: B
Step by step
1. Track record questions: peer comparison, third-party verification, enough history, returns relative to asset size, and continuity of the investment team.
2. A, C and D are all on this list.
3. The absolute level of returns, with no context (risk, peers, asset size), is not on it and is least informative.

Why B is correct: Returns matter relative to peers, risk and assets, not in absolute terms.
Why the others are wrong
- **A:** Returns relative to asset size is a track record question.
- **C:** Third-party verification gives assurance that returns are real.
- **D:** Team continuity shows whether past results are repeatable.

LO 94.d

Describe criteria that can be evaluated in assessing a hedge fund's risk management process.

Q514 AnalystPrep Q.2536

Global Finance's Asia-Pacific Equity Fund has an overall risk level within its expected limits. However, the risk team finds that the fund is over-exposed to certain U.S. equity themes, which deviates from the fund's stated objective of focusing on Asia-Pacific equities. Which of the following actions should Global Finance take?

A. Re-evaluate the risk monitoring model because it appears to be producing inaccurate results.
B. Modify the fund's stated objective to include more U.S. equities.
C. Investigate the fund's investment activities for potential style drift and realign the portfolio if needed.
D. Ignore the deviation because the total risk level is within acceptable limits.

Answer: C
Step by step
1. Total risk within limits does not mean the risk is of the intended type: the exposure is to U.S. themes, outside the stated Asia-Pacific mandate.
2. A mismatch between actual holdings and the stated strategy is style drift.
3. Due diligence on risk management requires fund terms and risk to be consistent with the investment strategy, so the manager should investigate the drift and realign the portfolio.

Why C is correct: Style drift changes the fund's risk-return profile away from what investors signed up for, so it must be investigated and corrected even when aggregate risk is within limits.

Why the others are wrong

- **A:** The model correctly flagged the exposure; the problem is the portfolio, not the model. Blaming the model confuses detection with the underlying drift.
- **B:** Rewriting the objective to fit the drift would mislead investors who committed capital for Asia-Pacific exposure.
- **D:** Looks only at total risk and ignores the composition of risk; the drift alters the risk-return characteristics even if the total is acceptable.

Q515

AnalystPrep Q.2599

REBUILT — Original distractors were near-duplicate vague statements and the stem contained irrelevant details; reworded the stem around the independence of the risk function and rewrote the options so exactly one is correct.

A hedge fund hires a risk manager through its employee referral programme, with no background check. The risk manager joins the global equity portfolio team and reports to the head of that portfolio. The fund has no risk committee, although the CIO receives daily risk reports, and its exotic derivatives valuation model was last audited two years ago. Which of the following is the most appropriate conclusion about the fund's risk management structure?

A. Having the risk manager report to the portfolio head improves risk management because the two are closely aligned.

B. The two-year-old model audit is the only significant weakness because the reporting line is standard practice.

C. The risk manager should report independently of the portfolio managers, for example to the CIO, so that risk measurement is separate from risk taking.

D. The daily reports to the CIO make a risk committee and an independent reporting line unnecessary.

Answer: C

Step by step

1. Risk measurement should be separated from risk taking, so the risk manager should not report to the person whose positions are being monitored (conflict of interest).
2. Reporting to the portfolio head compromises independence; reporting to the CIO (or a risk committee) restores it.
3. A and D therefore describe weaker structures, and B wrongly treats the reporting line as acceptable.

Why C is correct: An effective risk function is independent of the portfolio managers whose risk it measures; reporting to the CIO provides that independence.

Why the others are wrong

- **A:** Alignment is the problem, not a benefit: a risk manager reporting to the portfolio head faces a conflict of interest and may not escalate risk concerns.
- **B:** The reporting line is itself a red flag; the model audit is a concern, but it is not the only one.
- **D:** Daily reports are useful but do not substitute for independence of the risk function or for oversight by a risk committee.

Q516

BT answers P2.T8.413.2 · BT notes P2.T9.413.2

You are performing due diligence on a hedge fund and specifically on its risk management process. According to Mirabile, which of the following is MOST LIKELY to be a red flag?

A. The fund's risk manager reports to the portfolio managers, with dotted-line reporting to the traders.

B. The fund employs an independent risk service provider, but the provider is not co-located at the firm's headquarters.

C. The fund claims that its process for originating and controlling risk is "complex".

D. The fund employs different strategies, but each tends to incur a different set of risks rather than a common set of risks.

Answer: A

Step by step

1. Check each item against the independence principle: risk measurement must be separate from risk taking.
2. A has the risk manager reporting to the people who take the risk (portfolio managers and traders): no independence, a red flag.
3. B, C and D describe acceptable features: location does not matter, complex strategies often need complex risk processes, and different strategies naturally carry different risks.

Why A is correct: Lack of independence between the risk function and risk takers undermines the integrity of risk reporting.

Why the others are wrong

- **B:** Physical co-location is not required; what matters is independence and quality of the risk measurement.
- **C:** Complexity is often unavoidable for sophisticated strategies and is not itself a warning sign.
- **D:** Different strategies carrying different risks is normal; it is not a concern.

Q517 BT answers P2.T8.413.3 · BT notes P2.T9.413.3

A due diligence team is evaluating the risk management process of a relatively new long/short equity hedge fund. According to Mirabile, none of the following facts is by itself (without further context) necessarily alarming, HOWEVER which one is a red flag or unacceptable?

A. The firm has both a high-water mark and a one-year lockup.
B. The firm does not share equity ownership among its portfolio managers or traders.
C. The firm is new and small and consequently does not yet provide periodic reports to investors.
D. The firm is new and small but uses only one prime broker.

Answer: C

Step by step

1. Separate acceptable features from unacceptable ones.
2. Fee terms (high-water mark with lockup), the ownership structure and a single prime broker at a small new fund can each be reasonable.
3. Regular risk and performance reports to investors are required regardless of fund age or size, so C is unacceptable.

Why C is correct: Periodic reporting to investors is a basic requirement that does not depend on the size or age of the fund.

Why the others are wrong

- **A:** A high-water mark and a lockup are common, investor-friendly or neutral terms and are consistent with the strategy.
- **B:** Equity sharing is not required; incentives can be aligned through other compensation.
- **D:** A single prime broker is common for a small new fund and not by itself a problem.

Q518 GARP PE1 Q62

REBUILT — Original option A was arguably defensible, creating a second plausible answer; replaced it with a clearly incorrect statement about internal versus third-party risk reporting.

At a conference on hedge funds, investors discuss how to evaluate the risk management systems of large multi-strategy hedge funds. Risk management questions should be part of the evaluation of both the investment process and the operational environment. Which of the following is correct?

A. Investors should prefer that risk reports be prepared by the fund's own risk staff rather than by a third-party risk provider.
B. Investors should be certain the fund has adopted an independent risk management function, regardless of whether it is carried out by dedicated risk managers at the fund or by a risk service provider.
C. Investors should make sure that all portfolio managers use equivalent levels of liquidity and leverage and maintain these levels over time.
D. Investors should accept that the terms of the fund are unique to the hedge fund and the fund should be the only source of information regarding the terms applied.

Answer: B

Step by step

1. Evaluate each statement against the criteria for assessing a fund's risk management process.
2. Independence of risk measurement from risk taking is what matters; it can be achieved internally or via an external provider, so B is correct.
3. A is wrong (no preference between internal and third-party risk reporting); C is wrong (different strategies need different leverage and liquidity); D is wrong (terms should be benchmarked to similar funds).

Why B is correct: What investors need is independence of the risk function, however it is staffed (dedicated risk managers or a service provider).

Why the others are wrong

- **A:** Risk reporting is relatively objective, so there is little or no preference for internal versus external preparation; what matters is regularity and independence.
- **C:** Different strategies legitimately use different leverage and liquidity; the requirement is consistency with the fund's risk plan and strategy, not equality across managers.
- **D:** Fund terms (fees, high-water marks, redemption limits) should be compared with those of similar funds, not accepted on the fund's word alone.

Q519 Schweser Module Quiz 94.1 Q3

REBUILT — Option B ('written policies are useful') was arguably defensible; changed to the clearly incorrect claim that they give strong assurance.

Which of the following statements regarding the assessment of a fund's risk management process is correct?

A. The periodic valuation of a fund's securities is best performed by the fund manager.

B. The existence of written policies and procedures for risk measurement and monitoring provides strong assurance that risks are properly measured and monitored.

C. The risk reports received by investors are preferably prepared by a third-party risk provider instead of by the fund itself.

D. The key requirement for the information technology resources used to quantify risks is that they measure items consistently.

Answer: D

Step by step

1. IT resources must be reliable and measure items consistently between traders and portfolio managers: D is correct.
2. Valuation should be independent (fund administrator more independent than manager), so A is wrong.
3. Written policies show only intent, so B is wrong; risk reporting is objective so there is no internal/external preference, so C is wrong.

Why D is correct: Consistent measurement across traders and portfolio managers is the key requirement for risk-quantification IT systems.

Why the others are wrong

- **A:** A fund manager is generally less independent than the fund administrator; valuation needs independence and objectivity.
- **B:** Policies and procedures tend to be general and demonstrate only intention; their existence alone gives little assurance.
- **C:** Risk reporting is a relatively objective task, so there is little or no preference between internal and third-party preparation.

LO 94.e

Explain how due diligence can be performed on a hedge fund's operational environment.

Q520 BT answers P2.T8.414.1 · BT notes P2.T9.413.3

In regard to the hedge fund's operational environment, Mirabile advises each of the following as true for investors during due diligence, EXCEPT which is not true?

A. It is important to ensure that the fund offering memo, subscription agreement, limited partnership agreement, investment management agreement, Form ADV, and website are all saying the same thing at the same point in time.

B. Insufficient risk factors are more of a red flag than too many. In addition, overly broad and irrelevant risk factors are also a red flag. In the latter case, the law firm may not have adequately looked at the manager's program or is drafting the OM so broadly that it is looking primarily to protect itself, not the investor.

C. Investors should expect that a hedge fund will make all of the key contacts at their service providers available to them so they can verify the scope of any services being provided. Investors can also obtain internal control letters and audited financial statements from the fund's service providers to make sure there is an independent check on the service providers themselves.

D. The limited partnership agreement should include indemnification provisions such that the fund itself indemnifies the manager and all directors against gross negligence, bad faith, fraud, or willful misconduct of the

manager.

Answer: D

Key correction: Source P2.T9.413.3 had no key (and a mislabelled reference 414.1); the key D is taken from the matching BT answers item.

Step by step

1. Check each statement against the documents-and-disclosure and service-provider guidance.
2. A: consistency across the offering memo, Form ADV, subscription and management agreements is required, so true.
3. B: unclear or overly general risk factors are a concern, so true.
4. C: service-provider contacts, internal control letters and audited statements are the standard checks, so true.
5. D: indemnification should not extend to fraud, gross negligence or malicious intent, so the statement is false.

Why D is correct: Investors should be wary if the manager can be indemnified for fraud, gross negligence, bad faith or willful misconduct; proper indemnification excludes them.

Why the others are wrong

- **A:** True statement; consistency of terms across documents is a core check.
- **B:** True statement; vague or irrelevant risk disclosure warrants further investigation.
- **C:** True statement; verifying with service-provider contacts and reviewing their control letters and audited statements is the recommended evaluation.

Q521 GARP PE2 Q1

A group of newly hired investment analysts at a large wealth management firm is undergoing training on the company's investment practices. Part of the training focuses on hedge fund investments and addresses how the firm evaluates the risk management processes and procedures of a hedge fund being considered for investment. The analysts learn that the firm performs comprehensive due diligence on the hedge fund's investment environment, as well as on its operational environment and business practices. Which of the following is correct about a hedge fund investor's due diligence on the operational environment of a hedge fund?

A. An investor should assess the qualifications of the fund's investment team members but does not need to do the same for other fund personnel.

B. An investor should investigate the fund's compliance practices but does not need to ensure that the fund has an in-house compliance function.

C. An investor should review the fund documents but does not need to verify with the fund's law firm whether they are responsible for the content of these documents.

D. An investor should ask if the fund uses third-party service providers but does not need to inquire about the fund's relationships with these service providers.

Answer: B

Step by step

1. Test each 'does not need to' clause against the operational due diligence guidance.
2. A: internal control, back- and middle-office staff also need adequate experience, training and background checks, so A is wrong.
3. B: the compliance system may be in-house or outsourced; the investor examines its adequacy, not its location, so B is correct.
4. C: the investor must confirm with legal counsel its role in preparing and revising the documents, so C is wrong.
5. D: service providers must be evaluated, including their independence and control environment, so D is wrong.

Why B is correct: Schweser requires examination of the in-house or outsourced compliance system, so an in-house function is not mandatory.

Why the others are wrong

- **A:** Ignores that internal control and back/middle-office staff qualifications must also be assessed.
- **C:** Skips the required confirmation with legal counsel of its involvement in the original documents and revisions.
- **D:** Skips evaluating the relationships, independence and control letters of service providers.

LO 94.f

Explain how a hedge fund's business model risk and its fraud risk can be assessed.

Q522 AnalystPrep Q.2602

REBUILT — Original never stated the broker-dealer was affiliated, and its options were vague; rewrote the stem to state the related-party and manager-valuation facts and options to test the fraud-risk indicators (94.f).

An investor is performing due diligence on a hedge fund. The fund's fact sheet promises returns well above the average of funds in its category. All of the fund's trades are executed through a broker-dealer that is a related party of the fund manager. Illiquid investments make up 20% of total assets, and they are valued only by the manager using a proprietary model built by its quant researchers. Which of the following is the most appropriate conclusion?

A. Trading through the related-party broker is beneficial because it generally produces better quotes, so only the above-average return deserves scrutiny.

B. The 20% illiquid allocation is small, and a model built by the manager's own quants is independent by construction, so no further work is needed.

C. Related-party trading, manager-only valuation of illiquid assets and unusually high stated returns are each fraud-risk indicators, so the investor should seek independent verification of valuations and trade execution.

D. Extensive due diligence before investing eliminates fraud risk, so these features can be accepted if the offering documents disclose them.

Answer: C

Key correction: Source key said C, but its wording relied on an affiliate that the stem did not state; the stem has been rebuilt so C is clearly supported.

Step by step

1. Schweser's fraud-risk factors include trading through a broker or valuator who is a related party.
2. Another factor is a significant concentration of illiquid investments valued only by the manager.
3. A third is unreasonably high stated returns.
4. The fund shows all three, so fraud risk is elevated.
5. Mitigation: independent verification, inquiry of service providers about independence from the manager, background checks.

Why C is correct: Each feature is a listed fraud-risk indicator, and together they call for independent verification rather than reliance on the manager.

Why the others are wrong

- **A:** Wrongly assumes related-party execution is benign; it is a listed fraud-risk factor, and the high return is only one of three flags.
- **B:** Treats a manager-built, manager-run model as independent; independence requires valuation by a party such as the administrator, and 20% illiquid is a meaningful concentration.
- **D:** Fraud risk can exist even after extensive due diligence, and disclosure does not remove the red flags.

Q523 AnalystPrep Q.2603

REBUILT — Original statements (e.g., models 'must be independently audited', illiquidity premium) were not supported by the reading and the all-of-the-above key was arguable; rewrote as fraud-risk indicator statements with one clearly false statement.

Which of the following sets contains only correct statements about assessing a hedge fund's fraud risk?

I. Trading through a broker or using a valuator who is a related party is a fraud-risk indicator.

II. A significant concentration of illiquid investments valued only by the manager is a fraud-risk indicator.

III. Unreasonably high stated investment returns are a fraud-risk indicator.

IV. Valuation by the fund manager is generally more independent than valuation by the fund administrator.

A. I only

B. I and II only

C. I, II and III only

D. I, II, III and IV

Answer: C

Key correction: Source key said D (all accurate); IV is false because administrator valuation is generally more independent, so correct is C.

Step by step

1. I is a listed fraud-risk factor: related-party broker or valuator.
2. II is listed: significant illiquid concentrations, especially valued by the manager only.
3. III is listed: unreasonably high stated returns.
4. IV is reversed: administrator valuation is generally more independent than manager valuation.
5. So I, II and III are correct.

Why C is correct: The first three are listed fraud-risk factors, while IV reverses the independence relationship between administrator and manager valuation.

Why the others are wrong

- **A:** Omits II and III, which are also listed indicators.
- **B:** Omits III; unreasonably high stated returns are a listed indicator.
- **D:** Accepts IV, which is false: the administrator is generally more independent than the manager.

Q524

AnalystPrep Q.2598

An investment management firm hires managers largely through an employee referral programme; employees hired this way are not subject to background checks or rigorous pre-employment interviews. The firm hires a risk manager through the programme and places him on a global equity team. Which of the following is the most accurate statement?

A. Hiring managers through an internal referral programme reduces the risk of employee fraud.

B. Hiring managers through an internal referral programme reduces the cost of conducting background checks, which is a benefit that justifies skipping them.

C. Independent background checks must be carried out for each manager irrespective of the mode of hiring.

D. Managers must never be hired through an internal referral programme because it reduces the number of potential candidates.

Answer: C

Step by step

1. Fraud risk is mitigated by performing extensive background checks on the manager, regardless of how the manager was sourced.
2. A referral does not substitute for independent verification of education, employment history, litigation and regulatory records.
3. So every manager, including referrals, needs an independent background check; C is the only statement consistent with this.

Why C is correct: Fraud risk can persist even after other due diligence, and extensive independent background checks are a core mitigant, so the hiring channel cannot waive them.

Why the others are wrong

- **A:** Wrong: trust in a referral can increase fraud risk because screening is skipped; the error is treating familiarity as verification.
- **B:** Wrong: skipping checks saves cost but shifts the cost to fraud risk; the error is treating cost saving as justification.
- **D:** Wrong: too absolute; referrals are acceptable if the usual checks are applied. The error is overgeneralising from the missing-checks problem.

Q525

AnalystPrep Q.2601

REBUILT — Original keyed option C overstated that high returns are a 'probable case of fraud' and option D was defensible; rebuilt so the correct answer is the reading's fraud-risk indicators requiring further diligence.

An investor is performing due diligence on a hedge fund. The fund's fact sheet claims returns far above the average of its peer category. The fund executes trades through an unaffiliated broker-dealer. The fund values its illiquid investments, which are 20% of total assets, using a model built by its own quant researchers. Which of the following is the most appropriate conclusion?

A. Because the claimed returns exceed the peer average, the investor should invest without further review.

B. The unaffiliated broker-dealer is a related-party transaction and is by itself conclusive evidence of fraud.

C. The unusually high stated returns and the significant share of illiquid assets valued by the manager are fraud-risk indicators that call for more extensive diligence, such as independent valuation and background checks.

D. The fund is certainly fraudulent, so the investor should report it to regulators without further analysis.

Answer: C

Key correction: Source key (AnalystPrep) said C: 'claims of higher returns are generally an indication of misrepresentation and probable fraud'; that wording overstates the evidence, so the item was rebuilt.

Step by step

1. Fraud-risk indicators in the reading include unreasonably high stated returns and significant concentrations of illiquid investments valued only by the manager.

2. The fund shows both: far above-peer returns and 20% illiquid assets valued by an in-house model.
3. Trading through an unaffiliated broker is not a related-party red flag.
4. Indicators raise risk, they do not prove fraud, so the right response is deeper diligence (C).

Why C is correct: Indicators of fraud risk justify additional verification; they are neither reassurance nor proof of fraud.

Why the others are wrong

- **A:** Wrong: treats high returns as a positive; the reading treats unreasonably high returns as a red flag.
- **B:** Wrong: the broker is unaffiliated, so it is not a related-party transaction, and even a related party would be an indicator, not conclusive proof.
- **D:** Wrong: jumps from indicators to certainty of fraud.

Q526

AnalystPrep Q.4855 · Mock 4 Q12

REBUILT — Originals relied on external academic research (Form ADV fraud-prediction statistics and fraud-risk compensation) not in the reading; rebuilt on the reading's fraud-risk mitigation actions.

Which of the following is NOT a recommended action for mitigating a hedge fund manager's fraud risk during due diligence?

A. Checking the SEC website for any prior regulatory infractions by the manager.

B. Checking court records for prior litigation and bankruptcy records for evidence of financial irresponsibility.

C. Inquiring with service providers about their competence and independence from the manager.

D. Accepting the manager's own written account of past infractions and litigation as sufficient, because independent searches are duplicative.

Answer: D

Step by step

1. The reading lists four mitigants: check SEC records for infractions, check court and bankruptcy records, inquire with service providers about competence and independence, and perform extensive background checks.
2. A, B and C are each on that list.
3. D substitutes manager self-reporting for independent verification, so it is not a mitigant.

Why D is correct: Fraud mitigation relies on independent verification from third-party sources, not the manager's own statements.

Why the others are wrong

- **A:** Recommended action, so not the exception; Form ADV and SEC records reveal prior infractions.
- **B:** Recommended action, so not the exception.
- **C:** Recommended action, so not the exception; service providers should be competent and independent of the manager.

Q527

BT answers P2.T8.414.2 · BT notes P2.T9.413.3

Each of the following is a potential indicator of hedge fund FRAUD risk, except which is not (or is the least likely to indicate fraud)?

A. A high percentage of illiquid investments or those marked to market by the manager

B. Personal trading by managers in the same securities or similar securities as the fund

C. Fixed operating costs are above the industry average while manager compensation has a prominent variable component

D. Unusually strong performance claims; e.g., hedge fund performance claims are better than market average over a long period of time

Answer: C

Step by step

1. Fraud indicators in the reading: related-party transactions, illiquid assets valued by the manager, litigation, unreasonably high stated returns, personal trading in the same securities, frequent shorting.
2. A, B and D match listed indicators.
3. C describes the cost structure and variable revenue/compensation, which relates to business model risk (fee-expense gap, reliance on performance fees), not fraud.

Why C is correct: High fixed costs against variable pay is a viability (business model) issue, whereas A, B and D are named fraud indicators.

Why the others are wrong
- **A:** A listed fraud indicator (manager-valued illiquid assets), so not the exception.
- **B:** A listed fraud indicator (personal trading in same securities), so not the exception.
- **D:** A listed fraud indicator (unreasonably high stated returns), so not the exception.

Q528

Mock 4 Q12

REBUILT — Original depended on external academic research; this separate item tests the reading's business model risk assessment (breakeven AUM) with self-contained data.

A hedge fund earns a 1.5% annual management fee on assets under management (AUM) and has fixed operating expenses of $3.0 million per year. Current AUM is $160 million. Ignoring incentive fees, what AUM is required for the fund to break even?

A. $20 million
B. $160 million
C. $200 million
D. $450 million

Answer: C

Step by step
1. Breakeven AUM satisfies fee rate x AUM = expenses.
2. AUM = 3.0 / 0.015 = $200 million.
3. Current management fee revenue is 160 x 0.015 = $2.4 million, a $0.6 million shortfall, so the fund is below breakeven.

Why C is correct: Business model risk is assessed by comparing breakeven AUM with actual AUM; here AUM is $40 million short.

Why the others are wrong
- **A:** Wrong: divides by 15% instead of 1.5% (decimal slip).
- **B:** Wrong: gives current AUM; the fund's revenue of $2.4 million does not cover $3.0 million of costs.
- **D:** Wrong: multiplies expenses by 150 (3.0 x 150) instead of dividing by the fee rate.

Q529

BT Mock Q35

REBUILT — Options B-D were obvious red-flag giveaways and 'most important ingredient' was vague; rebuilt with plausible alternatives.

An investor wants due diligence on a hedge fund to screen specifically for fraud risk. Which approach is most appropriate?

A. Rely on the manager's references and the strength of personal relationships among the manager and related parties.
B. Rely on the fund's strong long-term track record as evidence that fraud is unlikely.
C. Perform independent background checks, search SEC and court records, and inquire with service providers about their competence and independence.
D. Rely on the manager's own valuation of illiquid holdings, since the manager knows them best.

Answer: C

Step by step
1. The reading's mitigants: check SEC infractions, court and bankruptcy records, service provider inquiries, and extensive background checks.
2. C is exactly this set of independent verifications.
3. A, B and D rely on the manager or on unverified signals.

Why C is correct: Fraud screening requires independent, third-party verification rather than reliance on the manager's representations.

Why the others are wrong
- **A:** Wrong: close related-party relationships are themselves a fraud risk, not a safeguard.
- **B:** Wrong: unreasonably strong stated returns are a red flag, not assurance.
- **D:** Wrong: manager-only valuation of illiquid assets is a fraud indicator.

LO 94.g

Describe elements that can be included as part of a due diligence questionnaire.

Q530 Schweser Module Quiz 94.2 Q1

Which of the following items is least likely to be included as requested information on a due diligence questionnaire?

A. Insurance coverage.
B. Returns attribution analysis.
C. Disaster recovery procedures.
D. Anti-money laundering policy.

Answer: B

Step by step

1. Recall the standard DDQ elements: general manager and fund information, execution and trading, service providers, third-party research policy, compliance (in-house counsel, anti-money laundering policy), disaster recovery/business continuity, insurance coverage, key person provisions, investment process, risk controls, and financial statements.
2. Insurance coverage (A), disaster recovery procedures (C) and anti-money laundering policy (D) all appear in that list.
3. Returns attribution analysis is not a questionnaire item; it is an analysis the investor can perform afterward using the information the questionnaire returns. So B is least likely to be requested.

Why B is correct: A DDQ collects information from the manager (operational, compliance, process and risk-control details); return attribution is an analysis the investor performs afterward on that information, not an item requested.

Why the others are wrong

- **A:** Insurance coverage is a standard DDQ item that speaks to the firm's stability; picking it confuses a requested item with an analysis.
- **C:** Disaster recovery and business continuity plans are standard DDQ items; picking it overlooks the operational-resilience section.
- **D:** The existence of an anti-money laundering policy is a standard compliance item on the DDQ; picking it overlooks the compliance section.

Q531 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

An investment consultant is designing a due diligence questionnaire for a pension plan that is considering an allocation to a hedge fund. The plan's investment committee asks why the questionnaire should include a section on the fund's third-party research policy. Which of the following is the most appropriate rationale for including this section?

A. It provides assurance about the stability of the firm, and therefore the safety of invested funds, through business continuity and key person provisions.
B. It indicates the speed and accuracy of transaction processing and whether related-party service providers are used.
C. It provides comfort that the fund and its managers intend to operate in an ethical manner and within the boundaries of the law.
D. It reveals the fund's sources of research information, which allows the investor to assess the extent and quality of the due diligence the fund performs in its investment process.

Answer: D

Step by step

1. The reading lists eight inquiry areas for a questionnaire; each serves a distinct purpose.
2. Third-party research policy is used to determine the fund's sources of research information.
3. This lets the investor judge how much and how good the fund's own due diligence is when it makes investments.
4. Options B, C and A describe the purposes of other inquiry areas.

Why D is correct: Inquiry into third-party research policy identifies the fund's research sources, allowing the investor to assess the extent and quality of the due diligence the fund performs in its investment process.

Why the others are wrong

- **A:** This is the purpose of the disaster recovery, business continuity, insurance and key person inquiry, confused with research policy.
- **B:** This is the purpose of the execution and trading and service provider inquiry (transaction speed/accuracy, related-party providers), not research policy.
- **C:** This is the purpose of the compliance inquiry (in-house legal counsel, anti-money laundering policy), confused with research policy.

Q532 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager at a fund of funds is reviewing the financial information section of a due diligence questionnaire completed by a prospective hedge fund. The fund supplied audited annual statements with an unqualified opinion, along with unaudited quarterly statements. Which of the following statements about this information is most accurate?

A. The audited statements provide objective, historical financial information for assessing performance, while the unaudited interim statements allow a more current assessment of the fund.

B. The unaudited quarterly statements are of no value to the investor, because only audited statements can be used to assess the fund's current condition.

C. The audited statements make information on the composition of invested assets unnecessary, because an unqualified opinion already confirms the fund's risk profile.

D. The audited statements make the fund's reported historical performance redundant, because the auditor independently verifies the strategy's expected future returns.

Answer: A

Step by step

1. The reading states audited statements with an unqualified opinion give objective, historical financial information used to assess performance.
2. Information on the composition of invested assets is also helpful to a potential investor.
3. Interim statements, not necessarily audited, give more timely information for a more current assessment.
4. Only A reflects these points; the others overstate the role of the audit or dismiss interim statements.

Why A is correct: Audited statements with an unqualified opinion supply objective historical information, and interim statements, though not necessarily audited, are more timely and support a more current assessment.

Why the others are wrong

- **B:** Interim statements are explicitly useful because they are more timely, even if unaudited; the error is demanding audit as a precondition for usefulness.
- **C:** Asset composition remains helpful to the investor; an unqualified opinion concerns the statements' fairness, not the fund's risk profile.
- **D:** An audit does not verify future returns, and statements complement rather than replace performance and drawdown information.

Reading 95. Distress Symptoms and Remedies

Roberto Ippolito, Private Capital Investing: The Handbook of Private Debt and Private Equity (Wiley, 2020). Chapter 20.

LO	Learning objective	Qs
95.a	explain different reasons why firms may experience financial distress and identify potential indicators of distress.	2
95.b	describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method.	3
95.c	summarize potential legal challenges that can arise when a firm is in financial distress or is going through a restructuring.	2

LO 95.a

Explain different reasons why firms may experience financial distress and identify potential indicators of distress.

Q533 Schweser Module Quiz 95.1 Q1

Which of the following is a clear financial indicator of credit deterioration?

A. Change in auditors.
B. Material changes in risk tolerance.
C. Use of a sale-leaseback arrangement.
D. Board or senior management resignations.

Answer: C

Step by step

1. Financial indicators: covenant breaches, late financial statements, creative accounting (off-balance-sheet financing, contingent liabilities) and earnings enhancements (sale-leasebacks, write-downs followed by gains on sale, accelerated revenue recognition, cost deferral).
2. Nonfinancial indicators: complex structure, resignations, fiscal year-end change, auditor change, missed commitments, changes in risk tolerance.
3. A sale-leaseback is an earnings enhancement, so it is a financial indicator: C.

Why C is correct: A sale-leaseback artificially boosts reported earnings or cash and is classified as a financial (accounting-based) warning sign.

Why the others are wrong

- **A:** A change in auditors is a nonfinancial indicator; it is a governance/behavioral signal, not a financial-statement item.
- **B:** Material changes in risk tolerance are a nonfinancial indicator reflecting management behavior.
- **D:** Board or senior management resignations are a nonfinancial indicator; candidates confuse a serious warning sign with a financial one.

Q534 Schweser Module Quiz 95.1 Q2

Which of the following scenarios is most likely to result in a rapid, deep cash flow decline and a prolonged period before revenue recovery?

A. Average leverage with weak business operations.
B. Minimal leverage and strong business performance.
C. Excessive leverage combined with a solid business model.
D. Excessive leverage combined with weak business operations.

Answer: D

Step by step

1. Excessive leverage with a solid model: cash flow stabilizes about 10% below default point; revenue recovers within about 3 years.
2. Average leverage with weak operations: cash flow may fall more than 50% below default point; recovery 5-10 years.
3. Excessive leverage with weak operations: cash flow falls 70% or more below default point, with an extended recovery period.

4. The worst combination is D.

Why D is correct: High leverage leaves no cushion while weak operations erode revenues and margins, producing the deepest cash flow fall and longest recovery.

Why the others are wrong

- **A:** Declines are significant (over 50%, 5-10 year recovery) but less severe than when weak operations are combined with excessive leverage.
- **B:** Minimal leverage and strong performance describes a healthy firm, not a distress scenario.
- **C:** A solid business model limits the damage: cash flows stabilize near 10% below default and revenue recovers in about 3 years.

LO 95.b

Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method.

Q535 Schweser Module Quiz 95.2 Q1

Which of the following methods most effectively provides cash to a distressed firm while allowing it to continue using a key operational asset?

A. Asset sale.
B. Equity sponsor.
C. Sale-leaseback.
D. Secured financing.

Answer: C

Step by step

1. The firm needs cash now but must keep using the key operational asset.
2. Sale-leaseback: sell the asset to an investor for cash and lease it back, so operations continue.
3. An outright asset sale gives cash but the firm loses the asset; secured financing needs unencumbered collateral; an equity sponsor does not target a specific asset.

Why C is correct: A sale-leaseback converts the asset into immediate cash while the lease preserves the firm's right to use it.

Why the others are wrong

- **A:** An outright asset sale raises cash but the firm no longer owns or uses the asset; it ignores the requirement to keep using it.
- **B:** An equity sponsor injects capital but is not tied to preserving a specific asset, and distressed firms often cannot attract one; it does not address the asset-use requirement.
- **D:** Secured financing needs unencumbered collateral, which distressed firms usually lack, so it is often infeasible; it is not the most effective choice.

Q536 Schweser Module Quiz 95.2 Q2

Which of the following scenarios is the most effective situation for using open market repurchases to reduce a company's debt?

A. The firm wants to offer alternative securities instead of cash.
B. The firm wants to repurchase only a small percentage of its outstanding bonds.
C. The firm needs to reduce a large proportion of its total debt under strict offer conditions.
D. Individual bondholders approach the firm directly, seeking smaller discounts than the market price.

Answer: B

Step by step

1. Open market repurchases are fast, low cost and discreet, but work best for small amounts (less than about 20% of bonds outstanding).
2. Option B (small percentage) fits this.
3. Large reductions with strict conditions need a cash tender offer; alternative securities need an exchange offer; direct approaches by holders are direct purchases.

Why B is correct: Open market purchases are best when only a small portion of the bonds is to be retired, since larger purchases would push up the market price.

Why the others are wrong

- **A:** Offering alternative securities instead of cash describes an exchange offer, not an open market purchase.
- **C:** Reducing a large proportion of debt under strict offer conditions describes a cash tender offer.
- **D:** Individual holders approaching the firm directly describes direct purchases from bondholders, not purchases in the open market.

Q537 Schweser Module Quiz 95.2 Q3

Which of the following statements best describes a key limitation that can prevent a distressed firm from executing a theoretical restructuring strategy?

A. A firm can always sell any business unit immediately to raise cash.
B. A firm with limited cash reserves may be unable to repurchase debt.
C. The presence of noncore business units always reduces restructuring risk.
D. Firms with simple capital structures typically face significant restructuring challenges.

Answer: B

Step by step

1. Practical limitations include limited liquidity, time to sell units, size of the liquidity shortfall, complex multilayered capital structures, and inability to divest core operations.
2. Limited cash prevents repurchasing debt at a discount, so B is a genuine limitation.
3. The other statements are absolute or reversed.

Why B is correct: Without cash reserves a firm cannot fund a debt repurchase, even when buying debt at a discount would be value-enhancing.

Why the others are wrong

- **A:** Selling business units takes time and is not always possible immediately; 'always' is wrong.
- **C:** Noncore units generally give options to raise cash and reduce risk, but 'always' is too absolute and it is not a limitation.
- **D:** Complex, multilayered capital structures, not simple ones, create restructuring challenges; this reverses the relationship.

LO 95.c

Summarize potential legal challenges that can arise when a firm is in financial distress or is going through a restructuring.

Q538 AnalystPrep Q.5780

In a corporate context, 'bankruptcy' is a term that carries specific legal implications and procedures. Choose the option that most accurately defines bankruptcy as it applies to corporations:

A. A financial situation where a corporation's total debts exceed its assets.
B. A legally declared inability or impairment of a corporation to pay its creditors.
C. The condition of a corporation failing to make timely payments on its debts.
D. A mutually agreed-upon arrangement between a debtor and its creditors to restructure outstanding debt.

Answer: B

Step by step

1. Bankruptcy is a legal status: the firm (or its creditors) files a petition and the court oversees an in-court restructuring (ICR), either reorganization or liquidation.
2. Match each option to a concept: A = balance-sheet insolvency; B = legally declared inability to pay; C = default; D = voluntary out-of-court restructuring.
3. Only B contains the legal declaration that defines bankruptcy.

Why B is correct: Bankruptcy is a court-recognized, legally declared inability to pay creditors, which triggers court-supervised proceedings and protections such as the stay on claims.

Why the others are wrong

- **A:** Describes balance-sheet insolvency (liabilities exceed assets); a firm can be insolvent without any legal declaration.
- **C:** Describes default (a missed payment); default may precede but is not itself a bankruptcy filing.
- **D:** Describes a voluntary out-of-court restructuring (OCR), which is negotiated privately and not court-supervised.

Q539 Schweser Module Quiz 95.2 Q4

Which of the following statements best captures a key difference between out-of-court restructurings (OCRs) and in-court restructurings (ICRs)?

A. OCRs are typically faster, while ICRs can be slower.
B. OCRs eliminate the risk of holdouts among creditors.
C. ICRs avoid reputational damage for the distressed firm.
D. ICRs are always preferred by creditors because they offer more flexibility.

Answer: A

Step by step

1. OCRs are voluntary negotiations with creditors, so they are generally faster and cheaper to execute.
2. ICRs involve formal court supervision, which tends to be slower, costlier and can damage the firm's reputation.
3. Check each option against these facts: only A is consistent.

Why A is correct: OCRs rely on voluntary agreement and avoid court process, making them typically faster and less costly, whereas ICRs are slower and more formal.

Why the others are wrong

- **B:** OCRs actually introduce holdout risk, since individual creditors may refuse to agree hoping for better terms; too many holdouts can force bankruptcy.
- **C:** Reverses the effect: court proceedings are public and can harm the firm's reputation, while OCRs are more discreet.
- **D:** 'Always' is wrong; ICRs are rigid and court-driven, and creditors do not universally prefer them (flexibility is greater in OCRs).

Reading 96. Madoff: A Riot of Red Flags

Greg Gregoriou and Francois Lhabitant, Madoff: A Riot of Red Flags (EDHEC Publication, 2009).

LO	Learning objective	Qs
96.a	explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance.	4
96.b	identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk.	4

LO 96.a

Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance.

Q540 Schweser Module Quiz 96.1 Q1

Bernard L. Madoff Investment Securities, LLC (BMIS) began operations in 1960:

A. as a brokerage business.
B. as a derivatives clearinghouse.
C. as an investment advisory business.
D. conducting electronic trading on the Cincinnati Stock Exchange (CSE).

Answer: A

Step by step

1. BMIS was founded in 1960 as a brokerage that quoted bid and ask prices and executed OTC trades.
2. The investment advisory arm was only established in the late 1980s, as brokerage margins shrank.
3. Involvement with the CSE and its electronic trading came later, after Madoff identified NYSE Rule 390.

Why A is correct: BMIS started as a market-making/brokerage business; the advisory business and CSE electronic trading came decades later.

Why the others are wrong

- **B:** BMIS never operated a derivatives clearinghouse; this confuses the firm with the options it later claimed to trade.
- **C:** The advisory business was set up in the late 1980s, not at founding; it is the business that turned out to be the fraud, which makes it a tempting but anachronistic answer.
- **D:** Madoff joined the CSE and modernized it after discovering Rule 390, well after 1960.

Q541 Schweser Module Quiz 96.1 Q2

Which of the following statements about Madoff's split-strike conversion strategy is incorrect?

A. Part of the strategy involved selling an out-of-the-money call option.
B. A floor value was created by purchasing an out-of-the-money put option.
C. A long position in a portfolio of stocks highly correlated with the S&P 100 was required.
D. Gains on the portfolio were supplemented by gains on the long, out-of-the-money call option position.

Answer: D

Step by step

1. The strategy is a collar: long equity portfolio correlated with the S&P 100, short OTM calls, long OTM puts.
2. A, B and C each match these steps.
3. D is incorrect: the calls were sold (written), not bought, so there was no long call to add gains; the short call caps upside.

Why D is correct: The strategy wrote OTM calls, which cap gains; it did not hold long calls that would supplement gains.

Why the others are wrong

- **A:** Correct description (step 2: write OTM calls), so it is not the incorrect statement.
- **B:** Correct description (step 3: buy OTM puts creates a floor), so not the incorrect statement.
- **C:** Correct description (step 1: long equity portfolio highly correlated with the S&P 100), so not the incorrect

statement.

Q542 Schweser Module Quiz 96.1 Q3

When comparing the reported returns of Madoff's feeder funds with the S&P 100, those returns were:

A. significantly higher than the S&P 100, and more volatile.
B. significantly higher than the S&P 100, and much less volatile.
C. approximately the same as the S&P 100, but with much lower volatility.
D. significantly lower than the S&P 100, but as stable as short-term Treasury bills.

Answer: C

Step by step

1. Reported returns were a steady 8%-12% per year, comparable to the S&P 100.
2. Over 95% of months were positive and there was never a down year, so reported volatility was dramatically lower than the index.

Why C is correct: Returns were similar to the index but implausibly smooth, a hallmark of the fraud.

Why the others are wrong

- **A:** Returns were not significantly higher, and volatility was far lower, not higher.
- **B:** Volatility was much lower, but returns were comparable to, not significantly above, the S&P 100.
- **D:** Returns of 8%-12% were not far below the index and well above T-bill levels; the funds were not T-bill-like.

Q543 Schweser Module Quiz 96.2 Q2

The securities fraud at BMIS came to light when:

A. Harry Markopolos sent a 17-page letter to the SEC detailing the red flags at BMIS.
B. the SEC, after years of investigation, finally gathered enough evidence to indict Madoff.
C. Madoff confessed to his family that his investment advisory business was a giant Ponzi scheme.
D. Madoff's sons discovered evidence of the fraud on their own and immediately contacted authorities.

Answer: C

Step by step

1. Madoff was not uncovered by investigation.
2. On December 10, 2008, amid heavy redemption requests he could not meet, he told his sons, wife and brother the business was a giant Ponzi scheme.
3. His sons then reported him to authorities, and he was arrested the next day.

Why C is correct: The fraud surfaced through Madoff's own confession to family when redemptions overwhelmed him.

Why the others are wrong

- **A:** Warnings to the SEC were sent but did not expose the fraud; they did not trigger action.
- **B:** The SEC did not uncover the fraud through investigation.
- **D:** The sons reported Madoff only after he confessed to them; they did not discover the fraud independently.

LO 96.b

Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk.

Q544 Mock AP Mock 3 Q31

REBUILT — Stem said the red flag led to 'discovery' of the scheme (it was a confession); reworded to ask which feature was a warning sign and made distractors plausible non-features.

Which of the following was a warning sign that should have alerted investors to potential fraud at Bernard L. Madoff Investment Securities (BMIS)?

A. BMIS refused to answer questions about its investment strategy and restricted on-site due diligence.
B. BMIS reported occasional large negative monthly returns that were not explained by market moves.
C. BMIS used a large, well-known audit firm that was independent of its feeder funds' auditors.

D. BMIS charged high management and performance fees that exceeded those of comparable hedge funds.

Answer: A

Step by step

1. Recall the lack of transparency red flag: Madoff would not explain the strategy, did not disclose AUM, and restricted or denied on-site due diligence.
2. Check B: BMIS reported almost no negative months (fewer than 5%), so the red flag was too-smooth returns, not large losses.
3. Check C: the auditor was a three-person firm (Friehling & Horowitz), not a large firm.
4. Check D: BMIS charged no management or performance fees, only brokerage commissions.

Why A is correct: Opacity about strategy and restricted due diligence prevented independent verification of the claimed split-strike conversion strategy, a classic fraud-risk indicator.

Why the others are wrong

- **B:** Reverses the facts: the suspicious feature was the near absence of negative months, not large losses.
- **C:** Reverses the facts: the auditor was a tiny, questionable firm; large firms audited the feeder funds and relied on it.
- **D:** Reverses the fee red flag: BMIS charged no management or performance fees, which was itself odd.

Q545

AnalystPrep Q.4849

REBUILT — Original distractors were implausible and option A was ungrammatical; rewrote stem and options so each wrong option is a realistic misconception.

Bernard Madoff was found guilty of running a Ponzi scheme. Why is a Ponzi scheme not sustainable indefinitely?

A. Payouts to existing investors are funded by new investors' capital rather than real profits, so the scheme fails once inflows fall short of redemptions.

B. Regulators automatically detect the scheme once reported returns exceed those of the S&P 100.

C. The scheme's investment strategy eventually becomes unprofitable as option premiums rise.

D. Feeder funds eventually raise their fees, which erodes returns to the underlying investors.

Answer: A

Step by step

1. A Ponzi scheme pays 'returns' to earlier investors out of money raised from later investors; no real profits are generated.
2. Liabilities (claimed balances plus promised returns) grow faster than the real assets.
3. When new inflows slow or redemptions spike (as in late 2008), the scheme cannot meet withdrawals and collapses.

Why A is correct: The scheme depends on a continuing net inflow of new money; it cannot pay everyone from actual investment profits.

Why the others are wrong

- **B:** Regulators did not detect Madoff despite warnings; collapse came from redemptions, not automatic detection.
- **C:** There was no real strategy being executed, so strategy profitability is irrelevant to the failure.
- **D:** Feeder fee changes do not cause a Ponzi scheme to fail; the structural cash shortfall does.

Q546

AnalystPrep Q.4854

Which of the following best describes a Ponzi scheme?

A. An investment firm forwards funds gathered from new investors to existing investors disguised as returns instead of investing the funds.

B. An investment firm gets into positions that are riskier than investors can tolerate.

C. An investment firm that reports losses even though it has made substantial profits.

D. An investment firm buys securities other than those the investors have been promised as detailed in the prospectus.

Answer: A

Step by step

1. Definition: a Ponzi scheme pays earlier investors with capital from later investors, presenting it as investment returns.
2. Match to the options: only A describes funds being passed from new to existing investors rather than invested.

Why A is correct: A Ponzi scheme's defining feature is paying 'returns' from new investors' money rather than from genuine investment profits.

Why the others are wrong

- **B:** Describes excessive risk-taking, not fraud through recycled capital.
- **C:** Describes misreporting of profits as losses, the opposite of a Ponzi scheme, which overstates performance.
- **D:** Describes style drift or mandate violation; assets are still invested, so it is not a Ponzi scheme.

Q547 Schweser Module Quiz 96.2 Q1

The fee structure at BMIS was a warning sign to investors because Madoff's firm:

A. charged unusually high performance fees.
B. charged unusually high asset management fees.
C. charged only a brokerage commission on trades.
D. did not charge any fees, relying solely on investment returns.

Answer: C

Step by step

1. BMIS charged no management or performance fees; it collected only brokerage commissions on trades.
2. It made little sense that a manager with apparent success would forgo millions in fees, which should have raised suspicion.
3. Option D overstates: commissions were charged, so BMIS did not charge zero fees. Only C is accurate.

Why C is correct: Charging only trade commissions was illogical for a supposedly highly successful manager and also let feeder funds charge high fees, discouraging them from probing.

Why the others are wrong

- **A:** Reverses the fact: BMIS charged no performance fee.
- **B:** Reverses the fact: BMIS charged no asset management fee.
- **D:** Too extreme: BMIS did earn brokerage commissions, so it was not fee-free.

Reading 97. Market-Driven Scenarios: An Approach for Plausible Scenario Construction

Bennett W. Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management (Wiley: 2024). Chapter 5.

LO	Learning objective	Qs
97.a	describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios.	3
97.b	describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios.	3
97.c	explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided.	3
97.d	describe challenges and considerations that may arise when implementing the market-driven scenario approach.	3

LO 97.a

Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios.

Q548

Schweser Module Quiz 97.1 Q1

Which of the following statements best describes the key advantage of the market-driven scenario (MDS) approach compared with traditional stress testing methods used before the Global Financial Crisis (GFC) of 2007-2009? The MDS approach:

A. removes subjectivity entirely by relying on fully automated statistical models.

B. relies solely on historical relationships between market variables to estimate P&L impacts.

C. eliminates the need for covariance matrices by focusing only on individual asset-level shocks.

D. provides a structured framework for aggregating diverse expert opinions into forward-looking scenarios.

Answer: D

Step by step

1. Pre-GFC stress tests were add-ons to VaR or tracking-error models and relied on historical relationships, which broke down during the GFC.
2. MDS gathers views from diverse investment professionals (asset classes, regions, expertise) and aggregates them through a centralized risk-management process, meeting Surowiecki's wisdom-of-crowds criteria.
3. Policy-variable shocks are constrained with statistical tools (Mahalanobis distance, covariance matrix), refined iteratively, then run through a valuation engine to get P&L.
4. The key advantage is therefore a structured, forward-looking framework that aggregates diverse views: D.

Why D is correct: MDS replaces purely historical, model-consistent stress tests with a disciplined process that aggregates diverse expert views into forward-looking scenarios, constrained by statistical tools.

Why the others are wrong

- **A:** MDS reduces but does not eliminate subjectivity; it channels subjective views into a transparent, challengeable process rather than automating them away.
- **B:** This describes the pre-GFC approach (historical relationships), which MDS was designed to overcome.
- **C:** MDS passes policy-variable shocks through a covariance matrix to derive other risk-factor moves, so covariances are central, not eliminated.

Q549

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A chief risk officer at an asset manager is presenting the firm's new market-driven scenario (MDS) process to the investment committee. A portfolio manager objects that because the scenarios rely on views from investors and economists, the output is just opinion and has no value. Which of the following responses is most consistent with the MDS approach?

A. The objection is valid, so MDS output should be used only after replacing expert views with a purely historical covariance model.

B. The process still contains subjectivity, but it forces risk takers to quantify their views, which makes them transparent and open to challenge and refinement.

C. The objection is invalid because aggregating views from many experts through a statistical framework removes subjectivity entirely.

D. The objection is valid for the policy variables but not for the nonpolicy factors, because only the nonpolicy factor shocks are determined subjectively.

Answer: B

Step by step

1. The reading states that MDS retains an element of subjectivity.
2. Its value lies in a structured process that quantifies subjective views.
3. Once quantified, views are transparent and can be challenged and refined through iteration.

Why B is correct: The reading says the approach contains subjectivity yet remains highly valuable because the structured process makes views transparent so they can be challenged and refined iteratively.

Why the others are wrong

- **A:** Reverses the logic: relying only on historical data is precisely the pre-GFC weakness MDS addresses, since historical relationships broke down during the crisis.
- **C:** MDS reduces but does not eliminate subjectivity; the reading explicitly says an element of subjectivity remains.
- **D:** Confuses the roles of the variable sets: policy shocks come from expert judgment, but nonpolicy shocks are implied from them through the covariance matrix, and subjectivity is not confined to one set.

Q550 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk analyst at a multi-asset fund applies the MDS approach as a conditional stress testing framework. The analyst specifies shocks to a small number of policy variables, such as an equity index and 10-year bond yields, and the portfolio also holds a book of equity options. Which of the following statements about the framework is correct?

A. Shocks to the nonpolicy factors are set independently by the experts, and the covariance matrix is used only to check them afterward.

B. The option positions can be handled with the same linear factor-exposure calculation as the cash positions because the policy shocks are small in number.

C. Shocks to the nonpolicy factors are implied from the policy variable shocks using multivariate regression, and the option positions are fully revalued using the full perturbation vector.

D. Keeping the number of policy variables small increases the arbitrariness of the perturbation vector because fewer factors are directly shocked.

Answer: C

Step by step

1. Policy variable shocks (K variables) are specified by experts.
2. The remaining $M - K$ nonpolicy shocks are implied via multivariate regression, equivalent to using the covariance matrix.
3. Linear exposures work for linear assets, but nonlinear payoffs such as options require full revaluation with the full perturbation vector.

Why C is correct: The full perturbation vector comes from regressing nonpolicy factors on policy variables, and positions with embedded optionality require full revaluation rather than a linear approximation.

Why the others are wrong

- **A:** Reverses the mechanics: nonpolicy shocks are implied from the policy shocks through the covariance matrix, not set independently by experts.
- **B:** Linear exposure-based P&L assumes returns are linear in factor shocks; options have nonlinear payoffs, and the number of policy variables is irrelevant to this.
- **D:** The reading says a small number of policy variables helps minimize arbitrariness in the perturbation vector, not increase it.

LO 97.b

Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios.

Q551 Schweser Module Quiz 97.1 Q2

REBUILT — Original stem was ambiguous because the scenario z-score is also used to gauge plausibility; reworded to ask for the correlation-adjusted multivariate distance that is the base statistic.

In the market-driven scenario (MDS) approach, experts propose a vector of shocks to several policy variables. A risk manager wants a single multivariate statistic that accounts for the correlations among the policy variables and measures how many standard deviations the shock vector lies from the historical center, so that implausible outliers can be flagged. Before any adjustment for the number of policy variables, which statistic should the manager use?

A. Scenario z-score.
B. Volatility z-score.
C. Correlation z-score.
D. Mahalanobis distance.

Answer: D

Step by step

1. The required statistic is a multivariate outlier measure that uses the inverse covariance matrix: $\text{MD} = \sqrt{(r-\mu)'\Sigma^{-1}(r-\mu)}$.
2. Because it uses the full covariance matrix, it accounts for correlations among the policy variables and is a multivariate generalization of a z-score.
3. It is the unadjusted base measure: the scenario z-score is derived from it by scaling for the number of policy variables.
4. The volatility z-score (zero-correlation magnitude) and correlation z-score (consistency with covariance structure) are components of the scenario z-score, not outlier-distance measures.
5. Answer: Mahalanobis distance (D).

Why D is correct: The Mahalanobis distance measures, using the inverse covariance matrix, how far a shock vector is from the historical mean in standard-deviation terms while accounting for correlations, which is what identifies multivariate outliers.

Why the others are wrong

- **A:** The scenario z-score is the MD rescaled so it does not grow with the square root of the number of policy variables; the question asks for the measure before that adjustment.
- **B:** The volatility z-score computes the scenario z-score assuming zero correlations, so it ignores correlations and isolates only the magnitude of the shocks.
- **C:** The correlation z-score normalizes the scenario z-score for shock magnitude and shows whether shocks are consistent with the covariance structure; it is not the distance from the center.

Q552 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager reviews a market-driven scenario in which experts have shocked three policy variables. The scenario z-score is 3.4. After the experts double the size of every shock while preserving all signs and relative magnitudes, the risk manager recomputes the statistics. Which of the following is correct?

A. The scenario z-score is unchanged because it is normalized for the number of policy variables.
B. The correlation z-score doubles because it is the scenario z-score divided by the volatility z-score.
C. The volatility z-score is unchanged because it isolates the consistency of the shocks with the covariance structure.
D. The correlation z-score is unchanged because it depends on the signs and relative magnitudes of the factor z-scores rather than their absolute size.

Answer: D

Step by step

1. The volatility z-score captures the magnitude of the shocks (computed with zero correlations).
2. The correlation z-score normalizes the scenario z-score for that magnitude, so it reflects only signs and relative sizes.
3. Doubling every shock scales magnitude measures but leaves the correlation z-score unchanged.

Why D is correct: The reading notes the correlation z-score does not depend on the absolute magnitude of the factor z-scores, so doubling all shocks leaves it unchanged, while the scenario and volatility z-scores would rise.

Why the others are wrong

- **A:** Normalizing for the number of policy variables does not make the score independent of shock size; the scenario z-score measures severity in standard deviations and would increase.

- **B:** The correlation z-score normalizes for magnitude, so the doubling in the numerator is offset by the same scaling in the magnitude component; it does not double.
- **C:** Reverses the definition: the volatility z-score isolates the magnitude of shocks (assuming zero correlations); consistency with the covariance structure is what the correlation z-score reflects. It would double with the shocks.

Q553 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk committee compares two hypothetical scenarios using a risk ruler built from historical events. Scenario X has a scenario z-score of 2.6 and Scenario Y has a scenario z-score of 4.1. A committee member asks the risk manager how to interpret and use these results. Which of the following statements is most accurate?

A. Scenario X is more plausible than Scenario Y, and the z-scores measure relative probability under an elliptical distribution; absolute probabilities require a parametric or nonparametric distributional assumption.

B. Scenario Y is more plausible than Scenario X, and the z-scores directly equal the probabilities of occurrence of the scenarios.

C. Scenario X is more plausible than Scenario Y, and the z-scores can be converted into absolute probabilities without any additional assumptions.

D. Scenario Y is more plausible than Scenario X because a higher scenario z-score indicates a larger shock that the market has already priced in.

Answer: A

Step by step

1. Lower scenario z-score means more plausible: X (2.6) is more plausible than Y (4.1).
2. Scenario z-scores measure relative probability assuming an elliptical distribution of returns.
3. Converting to an absolute probability needs more assumptions: a parametric distribution of z-scores or an empirical (nonparametric) distribution.

Why A is correct: Lower z-scores are more plausible, and absolute probabilities need either a parametric or nonparametric approach.

Why the others are wrong

- **B:** Reverses the ordering: the lower score is more plausible. A z-score is also a number of standard deviations, not a probability.
- **C:** Z-scores only measure relative probabilities; extra assumptions are required for absolute probabilities.
- **D:** Reverses the ordering. A higher z-score indicates a more severe, less plausible scenario, not one already priced in.

LO 97.c

Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided.

Q554 Schweser Module Quiz 97.2 Q1

Which of the following statements is not recommended when developing a market-driven scenario (MDS) analysis?

A. Specifying hypothetical scenarios precisely so they can be translated into market shocks.

B. Using scenario z-scores to assess the plausibility, magnitude, and relative sizing of shocks.

C. Selecting a covariance matrix sample that reflects market conditions consistent with the scenario.

D. Including a large number of highly correlated key risk factors to maximize coverage of different risks.

Answer: D

Step by step

1. Step 1 (define the event): the scenario must be precise, with magnitude and direction not open to interpretation, so A is recommended.
2. Step 2 (select policy variables): use a small number of simple, representative variables to minimize multicollinearity; do not add a variable whose adjusted R-squared against existing ones is 90% or higher. D does the opposite, so it is not recommended.

3. Step 3 (calibrate shocks): scenario z-scores are used to judge plausibility and severity of shock sizes, so B is recommended.
4. Step 4 (generate nonpolicy shocks): the covariance matrix sample period should reflect conditions similar to the scenario, so C is recommended.

Why D is correct: Highly correlated policy variables give redundant information and cause multicollinearity and an unstable perturbation vector, so a small, well-targeted set of policy variables is recommended instead of a large correlated set.

Why the others are wrong

- **A:** Recommended practice (Step 1): precise definition of magnitude and direction. Chosen by those who think vague scenarios are acceptable.
- **B:** Recommended practice (Step 3): scenario z-scores gauge how plausible and severe the proposed shocks are.
- **C:** Recommended practice (Step 4): using a calibration period unlike the scenario would give misleading implied nonpolicy shocks.

Q555 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager is selecting policy variables (Step 2) for a market-driven scenario on the European equity market. The existing policy variables are a European equity index and a 10-year Bund yield. An analyst proposes adding a Europe financials sector index; regressing it on the existing policy variables gives an adjusted R^2 of 93%. The team also wants several additional currency and credit factors included. Which of the following is the most appropriate action?

A. Add the financials index, because a high adjusted R^2 shows it is well explained by the existing variables and so improves the stability of the perturbation vector.

B. Do not add the financials index, because its adjusted R^2 is at or above 90%; if more variables are needed, group implied factors into asset blocks regressed on different subsets of policy variables.

C. Add the financials index, because higher multicollinearity improves coverage of the key market dynamics.

D. Do not add the financials index, and replace the Bund yield with a larger number of highly correlated rate factors to compensate.

Answer: B

Step by step

1. Step 2 recommends a small set of policy variables to minimize multicollinearity.
2. A new variable with adjusted R^2 of 90% or higher versus existing policy variables should be avoided.
3. If a larger set is required, asset blocks reduce multicollinearity.

Why B is correct: With an adjusted R^2 of 93%, the proposed variable is redundant; asset blocks are the recommended way to handle a larger set of factors.

Why the others are wrong

- **A:** A high R^2 means the variable is redundant and creates multicollinearity and instability, not stability.
- **C:** Multicollinearity provides overlapping, redundant information and reduces reliability rather than improving coverage.
- **D:** The conclusion on the index is right, but adding many highly correlated factors worsens the multicollinearity the rule is designed to avoid.

Q556 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

Following the Brexit example, a risk manager applies a Soft Brexit scenario to a portfolio invested 60% in a UK equity index and 40% in a UK sterling bond index. The implied shocks produce a loss of 12% on the equity index and a gain of 4% on the bond index, consistent with the assumption that gilts act as a safe haven. Assuming returns are linear in the shocks, what is the estimated stress-test P&L on the portfolio?

A. -8.8%

B. -7.2%

C. -5.6%

D. -4.0%

Answer: C

Step by step

1. Step 5: portfolio return is the weighted sum of position returns.
2. Equity contribution: $0.60 \times (-12\%) = -7.2\%$.
3. Bond contribution: $0.40 \times 4\% = +1.6\%$.
4. Total: $-7.2\% + 1.6\% = -5.6\%$.

Why C is correct: The sizable equity loss is partly offset by bond gains: $-7.2\% + 1.6\% = -5.6\%$.

Why the others are wrong

- **A:** Subtracts the bond gain as if it were a loss: $-7.2\% - 1.6\% = -8.8\%$.
- **B:** Counts only the equity contribution ($0.60 \times -12\%$) and ignores the safe-haven gain on bonds.
- **D:** Uses a simple equal-weighted average of the two shocks, $(-12\% + 4\%)/2$, instead of the 60/40 weights.

LO 97.d

Describe challenges and considerations that may arise when implementing the market-driven scenario approach.

Q557

Schweser Module Quiz 97.2 Q2

Which of the following statements best describes a key challenge in implementing an MDS approach?

A. Scenario development often depends on expert judgment.
B. Using only historical data is recommended when constructing scenarios.
C. Using a large number of policy variables guarantees more accurate and reliable scenario outputs.
D. Translating economic events into quantitative policy shocks relies solely on objective statistical models.

Answer: A

Step by step

1. The MDS approach defines hypothetical but plausible market shocks to a small set of policy variables, which are passed through a covariance matrix to imply shocks to all other risk factors.
2. Choosing the event, translating it into policy-variable shocks and judging plausibility all require expert judgment, so the approach is inherently subjective.
3. Check each option against this: only A describes a genuine challenge; B, C and D each contradict the reading's guidance.

Why A is correct: MDS scenarios rely on experts to define the event and size the policy-variable shocks, so subjectivity is the core implementation challenge; the structure only makes those views transparent and open to challenge.

Why the others are wrong

- **B:** MDS is forward-looking and exists because purely historical scenarios cannot capture events that have not occurred; relying only on history defeats its purpose (the covariance data is historical, but the shocks are not).
- **C:** A large set of policy variables adds redundancy and arbitrariness to the perturbation and does not guarantee better results; the reading recommends keeping the number of policy variables small.
- **D:** Converting macroeconomic events into policy-variable shocks is inherently subjective and needs judgment, not solely statistical models.

Q558

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager has a market-driven scenario that models an equity market crash. She builds the covariance matrix for implying nonpolicy factor shocks from two years of calm, low-volatility market data, and the experts accept the resulting P&L. Which of the following is the most important concern about this choice?

A. The only concern is that the data window is too short; any window length would be acceptable if it covered a longer history.
B. Calm-period data will overstate correlations among risky assets, so the stress P&L will be too large to be useful.
C. The choice of data window has little effect on the results, since the policy variable shocks alone determine the P&L.
D. Calm-period data will likely understate correlations among risky assets, which spike in crises, so the implied shocks and the severity of the stress will be understated.

Answer: D

Step by step

1. The covariance matrix determines the implied nonpolicy shocks and the z-scores.
2. Correlations among risky assets typically spike during crises.
3. Normal-period data therefore understates stress severity; the sample should reflect similar stressed periods.

Why D is correct: The reading warns that relying only on normal data likely understates the severity of the stress because crisis correlations are higher.

Why the others are wrong

- **A:** Length alone is not the issue; the sample must reflect conditions consistent with the scenario, and there is no universally accepted rule for the window.
- **B:** Reverses the relationship: normal periods produce lower, not higher, correlations among risky assets than crises.
- **C:** Nonpolicy shocks are implied from the covariance matrix, and normal versus stressed data can produce very different outcomes for the same shocks.

Q559

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A firm's risk team uses the Mahalanobis distance and scenario z-scores to screen proposed policy shocks for a market-driven scenario. A newly hired analyst argues that once a shock vector passes the statistical screen, no further judgment is needed. Which of the following best describes why this view is incorrect?

A. Plausibility testing is statistical in nature, but it still requires judgment about what is reasonable versus too extreme, and implied nonpolicy shocks may be unrealistic if the covariance matrix is poorly chosen.

B. Mahalanobis distance can be computed only after the portfolio P&L has been estimated, so it cannot screen shocks beforehand.

C. The statistical screen is unnecessary because the scenario's plausibility is determined entirely by the catalyst chosen in Step 1.

D. Scenario z-scores are valid only for historical events, so they cannot be used to assess hypothetical shocks.

Answer: A

Step by step

1. Plausibility is assessed with Mahalanobis distance or scenario z-scores.
2. There is no objective threshold separating reasonable from too extreme, so judgment is required.
3. Implied nonpolicy shocks depend on the covariance matrix and may not reflect current conditions if it is poorly chosen.

Why A is correct: The reading says plausibility assessment is statistical but still requires judgment, and that poorly chosen covariance matrices can distort implied shocks.

Why the others are wrong

- **B:** The distance is applied to the proposed policy shock vector before P&L estimation to flag outliers.
- **C:** A specific catalyst helps define the event but does not replace statistical plausibility checks.
- **D:** Scenario z-scores are applied to hypothetical future events too, and the risk ruler is used precisely to scale them against past events.

Reading 98. Liquidity Risk Management

Bennett W. Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management (Wiley: 2024). Chapter 7.

LO	Learning objective	Qs
98.a	explain best practices for managing liquidity risk in an investment fund.	2
98.b	explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling.	3
98.c	describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds.	10
98.d	identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed.	3
98.e	explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests.	3

LO 98.a

Explain best practices for managing liquidity risk in an investment fund.

Q560

AnalystPrep Q.3228

REBUILT — Original distractors (17.56%, 2.25%) did not map to realistic errors and the limit wording was loose; clarified that the cap applies to the position value and replaced distractors with error-based options.

XinWin pension fund has a liquidity risk policy that limits each equity holding to a maximum of 45% of the stock's 30-day average value traded. The fund's size is $7 billion. What is the maximum allocation (as a percentage of the fund) that the fund can hold in a stock whose 30-day average value traded is $390 million?

A. 1.25%
B. 2.51%
C. 5.57%
D. 25.07%

Answer: B

Step by step

1. Maximum position value = 0.45 × $390m = $175.5m.
2. Fund size = \$7,000m.
3. Maximum allocation = 175.5/7000 = 0.02507 = 2.51%.

Why B is correct: A liquidity limit tied to average value traded caps the position in dollars (45% of the 30-day average value traded); dividing by fund size converts it to a portfolio weight.

Why the others are wrong

- **A:** 1.25%: halves the limit (uses about 22.5% of average value traded instead of 45%).
- **C:** 5.57%: ignores the 45% factor and divides the full $390m by $7bn.
- **D:** 25.07%: unit slip, dividing $175.5m by $700m instead of $7,000m.

Q561

Schweser Module Quiz 98.1 Q1

Within a robust liquidity risk management framework, which of the following situations most clearly indicates a governance or communication breakdown rather than a measurement or toolkit weakness?

A. The fund's redemption toolkit includes swing pricing mechanisms for stressed markets.
B. Stress testing results trigger predefined limits and prompt partial portfolio rebalancing.
C. The risk function evaluates asset liquidity profiles under normal and stressed conditions.
D. Daily liquidity reports show tightening market depth, but no follow-up discussion occurs between risk management and portfolio management.

Answer: D

Step by step

1. The framework has four elements: liquidity risk measurement, liquidity risk management (limits), portfolio manager awareness (regular discussion and escalation), and a redemption toolkit.

2. A, B and C describe a toolkit tool, limits that work, and measurement under stress: all functioning.
3. D shows a known deterioration in market depth that is not discussed or escalated, a failure of the portfolio manager awareness element.

Why D is correct: Failing to discuss and escalate known liquidity warnings between risk and portfolio management is a governance/communication failure, not a measurement or toolkit gap.

Why the others are wrong

- **A:** Having swing pricing is a toolkit strength, not a breakdown.
- **B:** Stress results triggering limits and rebalancing is the framework working as intended.
- **C:** Assessing liquidity under normal and stressed conditions is sound measurement.

LO 98.b

Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling.

Q562

Schweser Module Quiz 98.1 Q2

Which of the following statements best explains why managing liquidity risk in fixed-income securities is often more difficult than in equity markets?

A. Fixed-income markets have fewer securities but trade more frequently with smaller bid-ask spreads.
B. Equity markets rely primarily on OTC transactions, reducing price transparency relative to fixed income.
C. Asset managers in fixed-income markets typically act as price takers, improving overall market liquidity.
D. Fixed-income markets contain a very large number of securities, many of which trade infrequently, making price discovery and valuation more difficult.

Answer: D

Step by step

1. Equity markets have a manageable number of securities that trade frequently on public exchanges, giving strong price discovery.
2. Fixed-income markets are fragmented: an enormous number of securities (municipal bonds alone about one million CUSIPs), and only a small fraction trade on any given day.
3. Sparse, uneven trading limits observable prices, so valuation, market depth and transaction-cost estimation are harder.
4. Only option D describes this mechanism.

Why D is correct: A very large universe of mostly infrequently traded bonds gives limited price discovery, which makes valuation and liquidity assessment harder than in exchange-traded equities.

Why the others are wrong

- **A:** Reverses the facts: fixed income has far more securities, trades less often, and has wider (not smaller) bid-ask spreads.
- **B:** Reverses the markets: fixed income is predominantly OTC; equities trade on public exchanges with greater transparency.
- **C:** Since the GFC managers have shifted from price takers to price makers, and this has thinned secondary markets rather than improving liquidity.

Q563

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk analyst at a bond fund builds a days-to-liquidate model using FINRA TRACE data. The analyst notes that investment-grade trades above $5 million and high-yield trades above $1 million are reported only as exceeding the cap. Which of the following is the most accurate description of the resulting data-modeling challenge?

A. Capped volumes cause TRACE to understate the number of trades but not their size, so the average trade size can be used without adjustment.
B. The true size of the largest trades is unknown, so models must infer volume from historical patterns, which may be inaccurate if future conditions differ from past trends.
C. Because the cap applies only to high-yield bonds, investment-grade volumes can be taken directly from TRACE without inference.
D. The cap removes the need to model price impact because large trades are excluded from the reported data

entirely.

Answer: B

Step by step

1. TRACE reports exact volume only below the cap; larger trades are flagged only as above the cap.
2. Whether a capped trade was $6 million or $60 million matters greatly for liquidity, so the true volume is ambiguous.
3. Models therefore infer volume from historical trading patterns, which can fail if future conditions differ.

Why B is correct: The cap hides the actual size of large trades, forcing models to infer volume from history, which is unreliable when conditions change.

Why the others are wrong

- **A:** Capped trades are still reported (as above the cap); it is their size that is hidden, so average trade size is distorted. Confuses missing size with missing count.
- **C:** Caps apply to both segments ($5 million investment grade, $1 million high yield). Overlooks the investment-grade cap.
- **D:** Large trades are reported as being above the cap, not excluded, and market impact still must be modeled. Misreads capping as exclusion.

Q564 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A chief investment officer notes that since the Global Financial Crisis her bond fund's traders have increasingly set the price at which they are willing to transact and then seek counterparties at that level, rather than accepting the best of several dealer quotes. Which of the following is the most likely consequence for liquidity risk management in fixed-income portfolios?

A. Price discovery has improved because managers now observe quotes from several dealers before every trade.

B. Secondary markets have become deeper, with higher turnover and narrower bid-ask spreads, so large orders can be executed at once.

C. Secondary markets have become thinner, with lower turnover and wider bid-ask spreads, so managers often break large trades into smaller orders to limit market impact.

D. The cost of liquidity has become independent of the need for immediacy because managers control the transaction price.

Answer: C

Step by step

1. The shift from price taker to price maker is described in the reading.
2. It contributed to thinner secondary markets: lower turnover, wider spreads, greater price impact of large trades.
3. Managers respond by splitting large trades; the cost of liquidity rises with the need for immediacy.

Why C is correct: Price-maker behavior is associated with thinner markets, wider spreads and larger price impact, prompting trade splitting.

Why the others are wrong

- **A:** Describes the earlier price-taker approach of polling several dealers, which is the behavior that has declined.
- **B:** Reverses the effect; the reading links the shift to lower turnover and wider spreads.
- **D:** The cost of liquidity rises as the need for immediacy increases; setting one's own price does not remove this.

LO 98.c

Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds.

Q565 Mock 3 Q56

A hedge fund manager is concerned about a potential increase in investor redemptions and wants to assess the effects of such an event on the fund's liquidity. The manager asks a junior analyst to estimate the average number of days required to liquidate certain securities in the fund. The analyst uses the information presented below to make this estimation for four securities:

Security	Market value in fund (CNY million)	Shares in fund	Average daily trading volume	Maximum daily volume allowed for liqui
A	93.00	500,000	522,000	22%
B	173.04	420,000	1,328,000	12%
C	58.88	256,000	710,000	18%
D	28.80	640,000	848,000	20%

Which of the four securities listed above is expected to take the longest to liquidate?

A. Security A.
B. Security B.
C. Security C.
D. Security D.

Answer: A

Step by step

1. Days to liquidate $= \dfrac{\text{shares held}}{\text{participation rate} \times \text{ADV}}$; market value is not needed.
2. A: $500{,}000/(0.22 \times 522{,}000) = 500{,}000/114{,}840 = 4.35$ days.
3. B: $420{,}000/(0.12 \times 1{,}328{,}000) = 420{,}000/159{,}360 = 2.64$ days.
4. C: $256{,}000/(0.18 \times 710{,}000) = 256{,}000/127{,}800 = 2.00$ days.
5. D: $640{,}000/(0.20 \times 848{,}000) = 640{,}000/169{,}600 = 3.77$ days.
6. Security A has the largest value, 4.35 days.

Why A is correct: Liquidation time is position size divided by the volume the fund may trade each day (participation rate times ADV). A combines a large share count with a high-volume cap that is still small relative to its position.

Why the others are wrong

- **B:** B has the largest market value (CNY 173.04 million) but a large ADV, so it takes only 2.64 days; ranking by market value instead of shares/(rate x ADV) leads here.
- **C:** C is the quickest (2.00 days); it would only be chosen by ranking in the wrong direction.
- **D:** D is the runner-up at 3.77 days; it is chosen if the large share count (640,000) is looked at without dividing by the allowed daily volume.

Q566

Schweser Module Quiz 98.2 Q1

Which of the following statements best describes a key challenge in modeling the liquidity risk of infrequently traded fixed-income securities using the days-to-liquidate approach?

A. Random forest regression cannot handle large feature sets or missing data.
B. Trade frequency and volume follow a linear pattern that basic regression can capture.
C. Bid prices perfectly represent true liquidity conditions, making volume variance irrelevant.
D. Relying on recent trade volumes can bias estimates because latent liquidity requires additional inference.

Answer: D

Step by step

1. For sporadically traded bonds, observed volume understates the volume the market could support (latent liquidity).
2. Observed ADV is therefore a conservative, biased input and the unobserved capacity must be inferred, e.g., from comparable bonds, dealer inventory postings or a 90th-percentile ADV.
3. Daily volume also varies widely, so estimates are noisy.
4. Only D describes this challenge.

Why D is correct: Actual volume is lower than potential volume, so recent trades alone understate liquidity and latent liquidity must be inferred from other data.

Why the others are wrong

- **A:** Reverse of the reading: random forest regression is chosen because it handles missing data, nonlinearity and many explanatory variables.
- **B:** The reading says the relationships are nonlinear, which is why basic linear regression is inadequate.
- **C:** Posted bids are imperfect guides to tradable volume, and high volume variance matters because ADV can be a conservative measure; 'perfectly' and 'irrelevant' are false.

Q567 Schweser Module Quiz 98.2 Q2

When estimating transaction costs using t-cost modeling, why is an intraday, pre-trade benchmark price typically preferred over the previous end-of-day price? The intraday, pre-trade benchmark price:

A. reduces benchmark-related noise.
B. ensures that market impact is zero.
C. allows bid-ask spreads to be ignored.
D. eliminates the need to measure implementation shortfall.

Answer: A

Step by step

1. Implementation shortfall = traded price minus a pre-trade benchmark; it is used to measure market impact.
2. With the previous close as benchmark, any market move between yesterday's close and execution is counted as a cost, adding a return component (noise).
3. An intraday, pre-trade (order-arrival) price reflects prevailing conditions, reducing this noise and being harder to manipulate.
4. Answer A.

Why A is correct: The intraday benchmark strips out price drift unrelated to the trade, so measured implementation shortfall better isolates market impact.

Why the others are wrong

- **B:** No benchmark choice makes market impact zero; impact is what shortfall measures.
- **C:** Bid-ask spreads are a fixed-cost component of t-costs and remain in the model.
- **D:** The benchmark is an input to implementation shortfall, which is still the main measure of market impact.

Q568 AnalystPrep Q.2516 · AnalystPrep Q.2518 · AnalystPrep Q.2534

REBUILT — Three near-identical conceptual items used the non-Schweser term 'Liquidity Duration Statistic'; merged and reworded to the days-to-liquidate approach with tightened distractors.

A risk management unit at an investment fund wants to gauge how long it would take to unwind a large position during a period of heavy redemption requests. It decides to use the days-to-liquidate approach to model asset liquidity. Which of the following best describes what this approach estimates?

A. The expected change in the asset's price over the liquidation horizon, based on recent volatility.
B. The exact amount of cash the fund must hold to meet expected client redemptions.
C. The number of days needed to unwind a position, given the size of the trade, an assumed market participation rate, and the asset's average daily volume.
D. The credit-quality deterioration of the asset's issuer over the liquidation horizon.

Answer: C

Step by step

1. Days-to-liquidate = size of the intended trade / (market participation rate × average daily volume).
2. The participation rate is the share of ADV the fund can sell without significant market impact.
3. The output is therefore a time (in days) to unwind the position in an orderly way, which is an asset-liquidity measure.

Why C is correct: The approach converts trade size, participation rate and ADV into the time needed to unwind a position without moving the market.

Why the others are wrong

- **A:** Confuses an asset-liquidity (time-to-exit) measure with a market-risk forecast; the approach does not predict prices.
- **B:** Confuses asset liquidity with funding/redemption sizing; the approach outputs days, not a cash amount.
- **D:** Confuses liquidity with credit risk; issuer creditworthiness is not an input.

Q569 AnalystPrep Q.2531 · AnalystPrep Q.2522

REBUILT — Q.2522 stated a days-to-liquidate value of 1 that contradicts its own inputs (the correct value is 0.67) and had the same inputs as Q.2531; merged into one calculation item and replaced giveaway distractors with error-based ones.

A fund manager is concerned about the liquidity profile of a security holding. The fund holds 10,000 shares, the average daily trading volume is 100,000 shares, and the fund will not trade more than 15% of the daily volume. Using the days-to-liquidate approach, the number of days needed to liquidate the position is closest to:

A. 0.67
B. 6.67
C. 0.10
D. 1.50

Answer: A
Key correction: Source key (Q.2522) implied a value of 1 day; with the stated inputs the correct value is 10,000 / (0.15 × 100,000) = 0.67 days.
Step by step
1. Days = shares / (participation rate × ADV).
2. Maximum daily sale = 0.15 × 100,000 = 15,000 shares.
3. Days = 10,000 / 15,000 = 0.67.

Why A is correct: The position is smaller than one day's permitted volume (15,000 shares), so it can be unwound in about two-thirds of a day.
Why the others are wrong
- **B:** Entered the participation rate as 1.5% (0.015) instead of 15%: 10,000 / (0.015 × 100,000) = 6.67.
- **C:** Ignored the participation limit: 10,000 / 100,000 = 0.10.
- **D:** Inverted the ratio (15,000 / 10,000 = 1.5) so it shows daily capacity relative to position, not days.

Q570

AnalystPrep Q.2523

A portfolio manager is looking to reduce the holdings of four different securities because of liquidity risks. The following table provides information on each security:

Security	Market Value in Fund (CNY million)	Shares in Fund	Average Daily Trading Volume	Max Daily Volume for Liquidation (%)
P	100.00	400,000	600,000	25
Q	180.04	450,000	1,200,000	15
R	75.88	300,000	650,000	20
S	35.80	550,000	900,000	22

Using the days-to-liquidate approach, which security will require the most days to liquidate fully, given the restrictions?

A. Security P
B. Security Q
C. Security R
D. Security S

Answer: D
Step by step
1. Days = shares / (max daily % × ADV); market value is not needed.
2. P: 400,000 / (0.25 × 600,000) = 400,000 / 150,000 = 2.67.
3. Q: 450,000 / (0.15 × 1,200,000) = 450,000 / 180,000 = 2.50.
4. R: 300,000 / (0.20 × 650,000) = 300,000 / 130,000 = 2.31.
5. S: 550,000 / (0.22 × 900,000) = 550,000 / 198,000 = 2.78, the longest.

Why D is correct: S has the largest holding relative to what it is permitted to sell each day (198,000 shares).
Why the others are wrong
- **A:** P gives 2.67 days, the second longest; it is chosen by ranking on shares / ADV alone (P has the highest ratio, 0.67) and ignoring the different participation limits.
- **B:** Q has the largest market value (CNY 180.04m), but value is irrelevant; Q needs only 2.50 days.
- **C:** R needs 2.31 days, the shortest; a smaller share count does not imply a shorter exit once volume and participation caps are applied.

Q571

AnalystPrep Q.2533

REBUILT — Source key referred to discussions with portfolio managers; rebuilt around the reading's decomposition of the infrequent-trading problem for bonds (probability of a trade and volume given a trade).

A risk analyst adapts the days-to-liquidate approach for an infrequently traded corporate bond by using historical data to fill in missing volume observations, assuming expected volume on no-trade days is zero. Which of the following correctly describes how the infrequent-trading problem is broken down?

A. Estimating the bid-ask spread and then the market impact cost of the trade.
B. Estimating the probability that a trade will occur and then predicting the trade volume if a trade does occur.
C. Estimating the coupon rate and then the residual maturity of the bond.
D. Estimating the participation rate and then the portfolio weight of the bond.

Answer: B
Step by step
1. Because the bond trades only sporadically, observed ADV is unreliable.
2. The problem is split into two components: (1) the probability that a trade occurs and (2) the trade volume if it does occur.
3. With zero expected volume on no-trade days, expected daily volume = P(trade) × E[volume | trade]; a random forest regression can model both pieces and handles nonlinearity and missing data.

Why B is correct: Sporadic trading is modeled as a trade-occurrence probability combined with a conditional volume forecast.
Why the others are wrong
- **A:** Describes the t-cost model (fixed costs and market impact), not the volume-modeling problem.
- **C:** Bond characteristics such as coupon and tenure may be explanatory variables but are not the two components of the problem.
- **D:** Participation rate and portfolio weight are inputs elsewhere in the days-to-liquidate formula, not the decomposition of infrequent trading.

Q572
AnalystPrep Q.2549

REBUILT — Numbers were fine; replaced giveaway distractors (0.1, 0.01, 0.001) with options that reflect realistic calculation errors.
A fund manager does not wish to exceed 10% of the daily volume in any security. The fund holds 10,000 shares of a security whose average daily volume is 100,000 shares. Using the days-to-liquidate approach, the number of days needed to liquidate the position is closest to:
A. 0.10
B. 10.0
C. 0.01
D. 1.0

Answer: D
Step by step
1. Days = shares / (participation rate × ADV).
2. Maximum daily sale = 0.10 × 100,000 = 10,000 shares.
3. Days = 10,000 / 10,000 = 1.0.

Why D is correct: The permitted daily volume exactly equals the holding, so one day suffices.
Why the others are wrong
- **A:** Ignored the participation limit: 10,000 / 100,000 = 0.10.
- **B:** Used 1% instead of 10%: 10,000 / (0.01 × 100,000) = 10.
- **C:** Multiplied the participation rate by the share/ADV ratio (0.10 × 0.10) rather than dividing.

Q573
AnalystPrep Q.2751

REBUILT — Numbers were correct; replaced unexplained distractors with error-based ones and moved the answer position.
Using the days-to-liquidate approach, calculate the number of days needed to liquidate a position, assuming no more than 15% of the daily volume in the security will be sold.

Number of shares held	2,000,000
Average daily volume of the security	750,000

A. 2.7
B. 177.8
C. 17.8
D. 0.056

Answer: C

Step by step

1. Maximum daily sale = 0.15 × 750,000 = 112,500 shares.
2. Days = 2,000,000 / 112,500 = 17.78, about 17.8 days.

Why C is correct: Only 112,500 shares can be sold per day, so the 2,000,000-share position takes about 17.8 days.

Why the others are wrong

- **A:** Ignored the participation limit: 2,000,000 / 750,000 = 2.67.
- **B:** Used 1.5% instead of 15%: 2,000,000 / (0.015 × 750,000) = 177.8.
- **D:** Inverted the formula: (0.15 × 750,000) / 2,000,000 = 0.056.

Q574 AnalystPrep Q.3168

REBUILT — Numbers were correct; replaced weak distractors (32.0, 1.6) with error-based options and moved the answer position.

A fund holds 200,000 shares of a stock. To limit market impact in a low-volume market, the portfolio manager will sell no more than 10% of the daily volume, and current average daily volume is 32,000 shares. Using the days-to-liquidate approach, the number of days needed to liquidate the position is closest to:

A. 6.25
B. 62.5
C. 625
D. 0.016

Answer: B

Step by step

1. Maximum daily sale = 0.10 × 32,000 = 3,200 shares.
2. Days = 200,000 / 3,200 = 62.5.

Why B is correct: The low volume and 10% cap limit sales to 3,200 shares a day, giving 62.5 days.

Why the others are wrong

- **A:** Ignored the participation cap: 200,000 / 32,000 = 6.25.
- **C:** Used 1% instead of 10%: 200,000 / (0.01 × 32,000) = 625.
- **D:** Inverted the ratio: 3,200 / 200,000 = 0.016.

LO 98.d

Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed.

Q575 Schweser Module Quiz 98.3 Q1

REBUILT — Stem said 'hedge fund' but the reading's regulatory permissions are stated for U.S.-based retail funds; reworded to retail fund and clarified the context; answer unchanged.

A U.S.-based retail fund needs an extraordinary liquidity measure to meet unexpected redemption requests when its ordinary liquidity tools are insufficient. Which of the following measures is generally available to it?

A. Gates.
B. Suspension.
C. Swing pricing.
D. Temporary borrowing.

Answer: D

Step by step

1. Temporary borrowing (overdrafts, repos, bank credit lines, interfund lending) is permitted for U.S.-based retail funds, though not for European UCITS funds.
2. Gates (partial redemptions up to a threshold) are permitted for UCITS funds but not for U.S.-based funds.
3. Swing pricing is permitted for UCITS funds; the SEC does not currently allow it for U.S. mutual funds (PRVs exist only for U.S. collective trust funds).
4. Suspension of redemptions is not permitted for U.S. mutual funds without explicit SEC approval.
5. Only temporary borrowing is generally available, so the answer is D.

Why D is correct: U.S. regulation allows funds to bridge redemptions with short-term borrowing (overdrafts, repos, credit lines, interfund loans), whereas gates, swing pricing and suspensions are European (UCITS) tools or require special approval in the U.S.

Why the others are wrong

- **A:** Gates are allowed for UCITS funds in Europe, not for U.S.-based funds; confuses the European regime with the U.S. one.
- **B:** Suspension is the most extreme measure and is not permitted for U.S. mutual funds without explicit SEC approval, so it is not generally available.
- **C:** Swing pricing is widely used for UCITS funds; the SEC does not currently allow it for U.S. mutual funds (only PRVs for collective trust funds).

Q576

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A risk manager at a European UCITS fund is reviewing the fund's extraordinary liquidity measures ahead of a break-glass test. Which of the following statements is correct?

A. Gates are not permitted for UCITS funds, whereas they are available to U.S. mutual funds.
B. Temporary borrowing and gates are both permitted for UCITS funds, but swing pricing is not.
C. Swing pricing and temporary borrowing are both permitted for UCITS funds, but gates are not.
D. Gates and swing pricing are both permitted for UCITS funds, but temporary borrowing is not.

Answer: D

Step by step

1. Temporary borrowing: permitted for U.S. retail funds, not UCITS.
2. Gates: permitted for UCITS, not U.S. mutual funds.
3. Swing pricing: permitted and widely used for UCITS.

Why D is correct: UCITS funds may use gates and swing pricing but not temporary borrowing.

Why the others are wrong

- **A:** Reverses the regimes; gates are allowed for UCITS and not for U.S. funds.
- **B:** Wrongly allows borrowing and denies swing pricing; swing pricing is widely used for UCITS and borrowing is barred.
- **C:** Wrongly allows borrowing and denies gates; applies the U.S. regime to UCITS.

Q577

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A European UCITS fund with EUR 2.0 billion in assets has a gate that limits redemptions to 10% of fund assets. Redemption requests total EUR 350 million. Assuming the gate is activated, how much can be paid out and how much is deferred?

A. Pay EUR 200 million; defer EUR 150 million.
B. Pay EUR 350 million; defer nothing.
C. Pay EUR 150 million; defer EUR 200 million.
D. Pay EUR 175 million; defer EUR 175 million.

Answer: A

Step by step

1. Gate threshold = 10% x EUR 2,000 million = EUR 200 million.
2. Requests of EUR 350 million exceed the threshold.
3. Deferred = 350 - 200 = EUR 150 million.

Why A is correct: A gate permits partial redemptions up to the threshold, so EUR 200 million is paid and EUR 150 million is deferred.

Why the others are wrong

- **B:** Ignores the gate; a gate limits the pace of outflows.
- **C:** Swaps the paid and deferred amounts.
- **D:** Splits requests in half with no basis in the 10% threshold.

LO 98.e

Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests.

Q578 Schweser Module Quiz 98.3 Q2

An analyst is applying the historical redemption-at-risk (HRaR) model to estimate potential redemption demand. Which of the following statements is least likely to represent a key challenge of this approach?

A. Some funds may lack sufficient redemption history because they are new or have been growing rapidly.
B. Limited transparency between asset managers and end investors can reduce the quality of available data.
C. Investor withdrawal behavior may be driven by idiosyncratic factors that are difficult to model statistically.
D. Historical data provides a stable and reliable indicator of future redemption pressures across all market environments.

Answer: D

Step by step

1. HRaR estimates worst-case outflows from historical fund flows (e.g., the 99th-percentile five-day redemption over the fund's life).
2. The reading lists its challenges: short or benign histories (new or fast-growing funds), idiosyncratic investor behavior that cannot be modeled reliably, and poor visibility into end investors when a custodian or other third party sits between fund and investor.
3. Options A, B and C each restate one of these challenges, so each is a genuine challenge.
4. Option D claims history is reliable in all environments, which is the opposite of a challenge and is false since redemptions in stress differ from history. D is the least likely challenge.

Why D is correct: HRaR relies on history, which is limited and may not reflect stress or idiosyncratic behavior; asserting that history is reliable in all market environments is not a challenge but an incorrect claim.

Why the others are wrong

- **A:** This is a stated challenge (new or rapidly growing funds lack meaningful outflow history); picking it misreads 'least likely'.
- **B:** This is a stated challenge (custodians obscure who is redeeming); picking it misreads 'least likely'.
- **C:** This is a stated challenge (idiosyncratic behavior is hard to model); picking it misreads 'least likely'.

Q579 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A U.S. '40 Act fund with total assets of $450 million receives unplanned redemption requests of $120 million. Cash and highly liquid assets of $30 million are available, and pro rata sales that preserve the portfolio's risk profile can raise $50 million. The fund follows a redemption waterfall. How much must be raised from the borrowing layer, and is this within the '40 Act borrowing limit of 33.3% of total fund assets?

A. $70 million, which is within the limit of about $150 million.
B. $40 million, which is within the limit of about $150 million.
C. $90 million, which is within the limit of about $150 million.
D. $40 million, which exceeds a limit of about $15 million.

Answer: B

Step by step

1. Layer 1 cash: $30 million. Layer 2 pro rata sales: $50 million.
2. Shortfall = 120 - 30 - 50 = $40 million from Layer 3.
3. Limit = 33.3% x $450 million = about $150 million; $40 million is within it.

Why B is correct: After Layers 1 and 2 cover $80 million, $40 million of short-term borrowing remains, well under the limit of about $150 million.

Why the others are wrong

- **A:** Omits Layer 1 cash (120 - 50).
- **C:** Omits Layer 2 sales (120 - 30).
- **D:** Correct borrowing amount but misapplies the limit as roughly 3.3% of assets rather than 33.3%.

Q580 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the SchweserNotes 2026 reading in GARP practice-exam style.

A fund manager builds a liquidity optimization model to meet redemptions. The fund holds a position subject to an IPO lockup, trades several bonds in a sector where simultaneous sales could exhaust market demand, and must keep tracking error below a specified level. Which of the following correctly matches each concern with the model constraint that addresses it?

A. Lockup: ADV constraint; sector demand: risk constraint; tracking error: market sector capacity constraint.

B. Lockup: risk constraint; sector demand: time-lag constraint; tracking error: ADV constraint.

C. Lockup: time-lag constraint; sector demand: market sector capacity constraint; tracking error: risk constraint.

D. Lockup: market sector capacity constraint; sector demand: ADV participation constraint only; tracking error: time-lag constraint.

Answer: C

Step by step

1. Fixed time constraints such as IPO lockups are handled by a time-lag constraint in days.
2. Large simultaneous sales that exhaust market absorption are handled by the sector capacity constraint.
3. Tracking error is controlled by a risk constraint with threshold R.

Why C is correct: Each concern maps to its own constraint: time-lag, sector capacity and risk.

Why the others are wrong

- **A:** The ADV constraint limits sales to a participation rate of daily volume, not lockup timing, and the risk constraint does not address sector absorption.
- **B:** Assigns the lockup to the risk constraint and tracking error to ADV; the risk threshold is what controls tracking error.
- **D:** The ADV constraint alone ignores sector-level absorption capacity, and time lags do not control tracking error.

Reading 99. Illiquid Assets

Andrew Ang, Asset Management: A Systematic Approach to Factor Investing (New York, NY: Oxford University Press, 2014). Chapter 13.

LO	Learning objective	Qs
99.a	evaluate the characteristics of illiquid markets.	5
99.b	discuss the relationship between market imperfections and illiquidity.	3
99.c	assess the impact of biases on reported returns for illiquid assets.	18
99.d	explain the process of unsmoothing returns and the effects of unsmoothing.	4
99.e	compare illiquidity risk premiums across and within asset categories.	5
99.f	evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets.	2

LO 99.a

Evaluate the characteristics of illiquid markets.

Q581

AnalystPrep Q.2430

REBUILT — Original options A and B both partly defensible (family holding versus turnover as the reason); rewrote so the disqualifying feature is clearly low trading activity.

Sam Kesler, a portfolio manager, is selecting stocks. The fund's investment policy statement requires that the portfolio consist only of liquid stocks. His candidates are:

Category	Stock	Information
Mid Cap	X	Latest quotes: can be sold at $12.30 and bought at $12.32.
Mid Cap	Y	The majority of the shares is held by a single family. Daily turnover is only 100 shares.
Large Cap	Z	The stock last traded 2 days ago.
Small Cap	A	Latest quotes: can be sold at $4.65 and bought at $8.25.

Which of the following statements about Stock Y is correct?

A. Y should be excluded because its very low daily trading volume shows it trades only in small amounts, a characteristic of illiquid assets.

B. Y should be included because family control indicates stability, which is equivalent to liquidity.

C. Y should be excluded because all mid-cap stocks are illiquid.

D. Y should be included because it has the same market-cap category as X, which is liquid.

Answer: A

Step by step

1. Illiquid assets trade infrequently, in small amounts and with low turnover.
2. Y trades only 100 shares a day, so it fits this profile, regardless of its category.
3. Compare X: bid-ask spread of $0.02 on about $12.31 (0.16%) is tight, consistent with liquidity; A has a spread of $3.60 on a $6.45 mid (about 56%), clearly illiquid; Z's stale last trade also signals illiquidity.
4. Since the policy allows only liquid stocks, Y is excluded because of its low trading activity (answer A).

Why A is correct: Liquidity is revealed by trading activity and spreads; a turnover of only 100 shares a day is the hallmark of an illiquid asset, and the majority family holding is the reason the free float is small.

Why the others are wrong

- **B:** Stability of ownership is not liquidity; concentrated control actually reduces the tradable float.
- **C:** Overgeneralises: liquidity depends on actual trading activity and spreads, not on cap category (X is mid-cap and liquid).
- **D:** Sharing a category with a liquid stock does not make Y liquid; the decision rests on Y's own turnover.

Q582

AnalystPrep Q.2432

REBUILT — Original option D (shares available and transaction costs) was defensible because thin free float raises costs; replaced with unambiguous options so only the free-float effect is correct.

Sam Kesler is screening stocks for a fund that may hold only liquid stocks. Stock Y is a mid-cap stock whose majority shareholding is held by a single family; its daily turnover is only 100 shares.

The family's majority holding most directly decreases:

A. the profitability of the company.
B. the number of shares available for trading (the free float).
C. the company's total market capitalization.
D. the par value of each share.

Answer: B
Step by step
1. Shares held by a controlling family are rarely traded.
2. They are therefore excluded from the effective free float, leaving only a small number of shares to trade.
3. A small tradable float produces low turnover (100 shares a day for Y), which is the characteristic of illiquid assets.
4. Ownership structure does not alter earnings, market capitalization (all shares times price) or par value, so only B is affected.

Why B is correct: A concentrated, locked-up holding removes shares from the tradable supply, which lowers trading volume and thus liquidity.
Why the others are wrong
- **A:** Who owns the shares does not change the firm's operating profitability.
- **C:** Market capitalization counts all outstanding shares at the market price, including the family's shares, so it is unchanged by who holds them.
- **D:** Par value is a fixed legal feature of each share and is unrelated to ownership concentration.

Q583 AnalystPrep Q.3140

The following statements compare a highly liquid asset with an otherwise similar illiquid asset. Which statement is most likely to be false?
A. It is possible to trade a larger quantity of the liquid asset without significantly affecting its price.
B. The liquid asset has a smaller bid-ask spread.
C. The liquid asset has higher price volatility since it trades more often.
D. The liquid asset has higher trading volume.

Answer: C
Step by step
1. Liquid assets trade frequently, in large amounts and with high turnover (D true).
2. Deep markets absorb large orders with little price impact (A true) and market makers quote tight bid-ask spreads (B true).
3. Nothing implies that frequent trading makes prices more volatile; frequent trading generally reflects a deeper market.
4. Thus C is the false statement.

Why C is correct: Liquidity is characterised by high volume, small spreads and low price impact; higher volatility is not a consequence of trading often.
Why the others are wrong
- **A:** True: low price impact for large trades is the definition of a deep, liquid market.
- **B:** True: tighter bid-ask spreads are a standard sign of liquidity.
- **D:** True: higher turnover and volume are characteristic of liquid assets.

Q584 BT answers P2.T8.707.1 · BT notes P2.T8.707.1

According to Andrew Ang, illiquidity can arise from market imperfections such as clientele effects and participation costs, transaction costs, search frictions, asymmetric information, price impact, or funding constraints. He characterizes the effects of these imperfections as "illiquidity." In regard to the characteristics of illiquid markets, based on Ang's research, which of the following statements is TRUE (such that the other statements are generally false)?
A. Normally liquid markets periodically become illiquid.
B. Most individuals hold the majority of their wealth in liquid or highly liquid assets.
C. Most asset classes are liquid, such that genuinely illiquid markets tend to be small and temporary.
D. Technology has virtually eliminated transaction costs, search frictions, asymmetric information, price impact, and funding constraints.

Answer: A

Step by step

1. Ang's characteristics: most asset classes are illiquid; illiquid markets are large; illiquid assets are the bulk of investors' portfolios (about 90% of wealth); liquidity dries up even in liquid markets.
2. A matches the fourth characteristic (e.g., money and commercial paper markets froze in 2007-2009), so it is true.
3. B contradicts the third (the home alone makes illiquid assets most of individual wealth).
4. C contradicts the first two (illiquid markets are large and persistent, e.g., US residential mortgage market $16 trillion).
5. D is false: the existence of vast illiquid markets shows frictions persist despite technology.

Why A is correct: Even liquid markets can freeze in stressed periods, while illiquidity is pervasive in normal times in most asset classes.

Why the others are wrong

- **B:** Reverses the facts: illiquid assets (especially housing) make up roughly 90% of individual wealth.
- **C:** Reverses the facts: most asset classes are illiquid and illiquid markets are large, not small and temporary.
- **D:** Technology has reduced but not eliminated frictions; persistent large illiquid markets are evidence of this.

Q585

Schweser Module Quiz 99.1 Q1

Global liquidity crises generally occur because:

A. governments choose not to engage in monetary policy actions to stimulate economies.

B. financial distress causes markets to freeze.

C. markets for illiquid assets shrink, causing liquidity issues to infect traditional asset classes.

D. transaction costs increase as developing economies get stronger.

Answer: B

Step by step

1. Per the reading, liquidity can dry up even in liquid markets during stressed periods, as in 2007-2009 when repo and commercial paper markets froze.
2. Major liquidity crises have occurred at least once every 10 years, in conjunction with downturns and financial distress.
3. Hence the cause is financial distress leading markets to freeze (B).
4. Illiquid markets are large, not shrinking, and the other options are not given causes.

Why B is correct: Liquidity crises coincide with downturns and financial distress, when investors become unwilling to trade at any price and markets freeze.

Why the others are wrong

- **A:** Crises are not attributed to governments declining to use monetary policy; they stem from financial distress.
- **C:** Reverses the facts: illiquid markets are large, and the crisis starts in normally liquid markets, not from shrinking illiquid ones.
- **D:** Stronger developing economies are unrelated; rising transaction costs are not the cause of freezes.

LO 99.b

Discuss the relationship between market imperfections and illiquidity.

Q586

AnalystPrep Q.2427

REBUILT — Original stem was long and its options were ambiguous (A/B/D arguable); rewrote as a self-contained question on how market imperfections of illiquid assets raise transaction costs.

An endowment manager must fund a withdrawal of a substantial amount in two years, and the board wants transaction costs kept as low as possible. A colleague proposes adding private equity and real estate because of their low correlation with stocks and bonds. Which statement best describes the relationship between market imperfections and these assets' transaction costs?

A. They would lower transaction costs because low correlation with stocks and bonds reduces the need to rebalance.

B. They would raise transaction costs because due diligence fees, search frictions, and asymmetric information make trading them costly and slow, conflicting with the board's objective.

C. They would not affect transaction costs because transaction costs depend only on the size of the endowment.
D. They would lower transaction costs because illiquid assets trade infrequently, so fewer trades are needed.

Answer: B
Step by step
1. Illiquidity arises from market imperfections: participation costs, transaction costs (due diligence, legal, accounting, banking fees), search frictions, asymmetric information, price impact, and funding constraints.
2. Private equity and real estate are exposed to all of these: finding a counterparty can take months or years, and information asymmetry makes buyers wary.
3. Hence trading them (especially to raise cash within two years) is costly and slow, which conflicts with the board's low-cost objective.
4. Diversification benefit is real but does not lower transaction costs.

Why B is correct: Illiquid assets carry high transaction costs from due diligence, search frictions, and asymmetric information, so they conflict with a low-cost objective and a near-term liquidity need.
Why the others are wrong
- **A:** Confuses diversification benefit with trading cost; low correlation does not remove the imperfections that make illiquid assets costly to trade.
- **C:** Ignores that transaction costs are asset-specific and driven by market imperfections, not fund size.
- **D:** Treats infrequent trading as a cost saving; infrequent trading is a symptom of illiquidity, and a forced sale in two years would be costly.

Q587

AnalystPrep Q.3024

REBUILT — Original options were vague ('effect of prices') and used wordplay; reworded options to the Schweser imperfections list with exactly one non-imperfection.

Which of the following is NOT a market imperfection that gives rise to market illiquidity, according to the reading?
A. Asymmetric information between buyers and sellers
B. Price impact of large trades
C. Search frictions in finding a counterparty
D. All participants receiving information simultaneously

Answer: D
Step by step
1. Imperfections encouraging illiquidity: participation costs, transaction costs, search frictions, asymmetric information, price impact, funding constraints.
2. A, B and C are all on this list.
3. Simultaneous information for all participants is an assumption of a perfect market, so it is not a source of illiquidity.

Why D is correct: Equal, simultaneous information is a feature of perfect markets; the other three are recognised sources of illiquidity.
Why the others are wrong
- **A:** Asymmetric information makes investors fear trading with better-informed counterparties (lemons), increasing illiquidity.
- **B:** Large trades move prices, creating liquidity problems.
- **C:** Difficulty finding a counterparty can make transacting take weeks to years.

Q588

Schweser Module Quiz 99.1 Q2

When an investor has difficulty finding a counterparty for a complicated credit product such as a structured debt instrument, this is known as:
A. market participation costs.
B. agency costs.
C. search frictions.
D. selection bias.

Answer: C
Step by step

1. The problem described is locating a willing, capable counterparty, however high the transaction cost.
2. The reading calls this a search friction.
3. Participation costs concern expertise and capital to enter a market; selection bias is a return-reporting bias.

Why C is correct: Difficulty finding a counterparty for a complex or large asset is the definition of a search friction.

Why the others are wrong

- **A:** Participation costs are the time, money, and expertise needed to enter a market, not the difficulty of finding a counterparty.
- **B:** Agency costs arise from conflicts between principals and agents and are not among the imperfections listed for illiquidity.
- **D:** Selection bias is a bias in reported returns of illiquid assets (LO 99.c), not a market imperfection.

LO 99.c

Assess the impact of biases on reported returns for illiquid assets.

Q589

AnalystPrep Q.2428

The bias associated with returns being observed only when the underlying asset values are high is best known as:

A. Selection bias
B. Survivorship bias
C. Infrequent trading
D. None of the above

Answer: A

Step by step

1. Returns on illiquid assets (houses, buyout-fund companies, venture investments) are recorded when a sale or IPO occurs, and owners tend to sell when values are high.
2. Observing only high-value outcomes is sample selection bias.
3. Survivorship bias concerns funds that stop reporting or fail; infrequent trading concerns smoothed returns.

Why A is correct: Selection bias arises because transactions (and therefore reported returns) occur mainly when values are high, so the sample is not representative.

Why the others are wrong

- **B:** Survivorship bias is about poor performers dropping out of the database, not about returns being observed only when values are high.
- **C:** Infrequent trading smooths returns and understates risk; it does not filter observations by the level of value.
- **D:** Incorrect because A describes the bias exactly.

Q590

AnalystPrep Q.2433

Avery Mason, a financial analyst, is investigating the performance of a group of funds based on the following returns:

Fund	Historical Return	Fund Status
1	6%	Active
2	-12%	Closed due to acquisition
3	8%	Active
4	-8%	Closed due to poor performance
5	10%	Active

Mason only considers the funds that are still active for the analysis. The results of his analysis are exposed to:

A. Selection bias
B. Survivorship bias
C. Infrequent sampling
D. None of the above

Answer: B

Step by step

1. Active funds average (6+8+10)/3 = 8.0%; all five funds average (6-12+8-8+10)/5 = 0.8%.
2. Excluding closed funds, which had below-average returns, inflates the average: this is survivorship bias.

3. Selection bias would involve returns recorded only when values are high; infrequent sampling involves smoothed returns. Neither is described.

Why B is correct: Using only funds that survived leaves out the poor performers that closed, so the reported average return is too high.

Why the others are wrong

- **A:** Selection bias concerns returns being recorded when asset values are high (e.g., at sale), not the exclusion of closed funds.
- **C:** Infrequent sampling/trading concerns smoothed, stale prices; the data here are about which funds are included.
- **D:** Incorrect because survivorship bias clearly applies.

Q591

AnalystPrep Q.2439

REBUILT — Original stem never stated how the return data were obtained and mixed in irrelevant transaction-count commentary; rewrote it so the data source (sold buildings only) is explicit and options isolate the alpha/beta direction.

A consultant estimates the alpha and beta of the real estate sector using only the returns on buildings that were sold during the year. Owners in this market typically hold properties until values have recovered before selling. The consultant reports beta = 0.65 and alpha = 3.45%. Which of the following is the most likely implication of the way the data were collected?

A. Alpha is overestimated and beta is underestimated.
B. Alpha and beta are both overestimated.
C. Alpha is underestimated and beta is overestimated.
D. Alpha is correctly estimated but beta is underestimated.

Answer: A

Step by step

1. Returns are observed only when values are high (sample selection bias).
2. Recording only high values inflates reported alpha relative to the true alpha.
3. Fewer low prices are recorded, so reported returns co-move less with the market: beta is too low and the security market line is flattened.
4. So alpha is overstated and beta (risk) is understated.

Why A is correct: Selection bias overstates expected return/alpha and understates systematic risk (beta) and volatility.

Why the others are wrong

- **B:** Beta is understated, not overstated, because low-value observations are missing and measured co-movement with the market falls.
- **C:** Reverses both effects; selection bias raises alpha and lowers beta.
- **D:** Alpha is also biased upward because only high prices are recorded; it is not correctly estimated.

Q592

AnalystPrep Q.2481

A fund manager manages a fund which mainly invests in illiquid assets such as convertible bonds, which are traded infrequently. The manager computes the fund volatility and finds it to be on the lower side. He concludes that the fund has low total risk due to low volatility.

Is the manager's conclusion correct?

A. Yes, because volatility indicates the riskiness of an asset or portfolio. Therefore, low volatility leads to low risk.
B. No, because correlations with other asset classes will be artificially lowered, giving the appearance of low systematic risk.
C. No, because volatility will be artificially lowered, giving the appearance of low total risk.
D. Both B and C.

Answer: D

Step by step

1. Infrequent trading produces smoothed (stale) reported returns.
2. Smoothed returns have artificially low volatility, so total risk appears lower (statement C is true).
3. Smoothed returns also show artificially low correlations (and betas) with liquid asset classes, so systematic risk appears lower (statement B is true).
4. Both B and C are true, so the answer is D.

Why D is correct: Infrequent trading mutes measured volatility, correlations and betas, so measured risk understates true risk.

Why the others are wrong

- **A:** Ignores that measured volatility of infrequently traded assets is biased downward.
- **B:** True but incomplete; volatility is also understated.
- **C:** True but incomplete; correlations are also understated.

Q593

AnalystPrep Q.2545

Which of the following correctly defines survivorship bias?

A. A bias towards instances which made it past some selection process and overlooking others due to a lack of visibility.

B. A bias towards instances with negative outcomes.

C. A bias towards instances which have recently occurred.

D. A bias towards instances which have positive outcomes.

Answer: A

Step by step

1. Survivorship bias arises when only entities that survive (keep reporting, do not fail) enter the sample.
2. Failed or non-reporting funds are not visible, so reported returns are too high.
3. A captures this; B, C and D describe other bias types.

Why A is correct: Survivorship bias is the focus on those that passed a selection (survival) process while ignoring those that dropped out.

Why the others are wrong

- **B:** Survivorship bias skews results toward good outcomes, not negative ones.
- **C:** That describes recency bias, which is a behavioral bias unrelated to the sample composition.
- **D:** Too general; positive outcomes are a consequence of survival, not the definition (selection bias also favors high values).

Q594

AnalystPrep Q.2571

REBUILT — Original asked about unexplained correlation between a database and FoF returns with a debatable key; replaced with a numeric survivorship bias question that tests the quantitative impact of dropped funds.

A database of hedge funds contains five funds with the following annual returns over the sample period:

Fund	Return	Status
A	12%	Reporting
B	9%	Reporting
C	6%	Reporting
D	-4%	Stopped reporting after poor performance
E	-9%	Stopped reporting after poor performance

The database provider only includes funds that reported throughout the period. By how much does the database's average return overstate the average return of all five funds?

A. 2.8%

B. 6.2%

C. 9.0%

D. 15.5%

Answer: B

Step by step

1. Average of all five funds = (12 + 9 + 6 - 4 - 9) / 5 = 14 / 5 = 2.8%.
2. Average of reporting (surviving) funds = (12 + 9 + 6) / 3 = 9.0%.
3. Survivorship bias = 9.0% - 2.8% = 6.2%.

Why B is correct: Dropping poorly performing funds that stop reporting raises the reported average; the bias is the survivor average minus the full-universe average.

Why the others are wrong

- **A:** 2.8% is the true average of all five funds, not the overstatement.
- **C:** 9.0% is the reported (survivor) average, not the difference from the true average.

- **D:** 15.5% is the survivor average minus the average of the dead funds (9.0% - (-6.5%)), not the bias against the full universe.

Q595

AnalystPrep Q.2581

Samuel Ray, a risk analyst, is evaluating the annual performance data of hedge funds and real estate funds for investment purposes. He notes two concerns: (1) for the hedge fund dataset, any fund that discontinues its performance reporting is excluded from the database; (2) for real estate funds, assets are appraised only annually due to their illiquid nature and infrequent trading. What are the likely implications of these data issues for his performance analysis?

A. The mean Sharpe ratio of hedge funds is underestimated, and the mean Sharpe ratio of real estate funds is overestimated.
B. The mean Sharpe ratio of both hedge funds and real estate funds is overestimated.
C. Both hedge funds and real estate funds show an overestimated average volatility.
D. Hedge funds present an overestimated average volatility, while real estate funds show an underestimated average volatility.

Answer: B

Step by step

1. Issue 1 is survivorship bias: poor performers drop out, so average return is overstated and the Sharpe ratio is overstated.
2. Issue 2 is infrequent trading/appraisal smoothing: volatility is understated, so the Sharpe ratio (excess return / volatility) is overstated.
3. Both Sharpe ratios are overestimated.

Why B is correct: Survivorship bias inflates return; infrequent sampling deflates volatility; both inflate the Sharpe ratio.

Why the others are wrong

- **A:** Survivorship bias overstates, not understates, the hedge fund Sharpe ratio.
- **C:** Volatility is understated for illiquid, smoothed assets, not overstated.
- **D:** Hedge fund volatility is not overstated by survivorship, and the answer mixes up the direction of the effect.

Q596

AnalystPrep Q.2582

REBUILT — Source key included look-ahead bias, which is not among the three biases in the reading, and marked the textbook answer (survivorship in active, selection in passive) wrong; rewrote options so exactly one matches the reading.

Alexa Thompson is analyzing a dataset of active and passive funds. She notes: (1) the data of some active funds that performed poorly and subsequently closed are not included in the database; (2) for the passive funds, the dataset includes only funds that have outperformed their benchmark in the recent past. Which biases is Alexa most likely facing?

A. Survivorship bias in the active funds and selection bias in the passive funds.
B. Selection bias in both the active and the passive funds.
C. Survivorship bias in both the active and the passive funds.
D. Infrequent trading in the active funds and survivorship bias in the passive funds.

Answer: A

Key correction: Source key (B) included look-ahead bias, which is outside the reading's three biases; correct per the reading is survivorship (active) and selection (passive), i.e. option D of the original wording.

Step by step

1. Missing closed, poorly performing active funds is survivorship bias.
2. Admitting passive funds only because they recently did well builds a sample of high-return observations: sample selection bias.
3. So: survivorship (active), selection (passive).

Why A is correct: Dropped failed funds are survivorship bias; a sample chosen on good outcomes is selection bias.

Why the others are wrong

- **B:** Labels the closed-fund exclusion as selection bias; the textbook term for it is survivorship bias.
- **C:** Passive funds are chosen for recent outperformance, not by who survives/keeps reporting.
- **D:** Infrequent trading concerns smoothed prices, not missing funds; and the passive issue is selection, not survivorship.

Q597 AnalystPrep Q.2758

People tend to overstate expected returns and understate the risk of illiquid assets. This is due to certain biases that affect reported data. These biases include all of the following, except the:

A. Survivorship bias
B. Infrequent sampling bias
C. Selection bias
D. Availability bias

Answer: D

Step by step

1. The three reporting biases for illiquid assets are survivorship, sample selection, and infrequent trading/sampling.
2. A, B and C are in this list.
3. Availability bias is a behavioral bias, not a reporting bias for illiquid asset returns, so D is the exception.

Why D is correct: Survivorship, selection and infrequent sampling bias reported illiquid returns up and risk down; availability bias is not one of them.

Why the others are wrong

- **A:** Is one of the three biases (overstates returns).
- **B:** Is one of the three biases (understates volatility and correlations).
- **C:** Is one of the three biases (overstates alpha, understates beta and volatility).

Q598 AnalystPrep Q.3025

Expected returns can be overstated and the risk of illiquid assets understated due to some key biases. Survivorship bias is one of these biases. How does survivorship bias arise?

A. The survivorship bias is the result of infrequent trading where risk estimates like volatility, correlations and betas get too low when calculated using past data.
B. The survivorship bias results from the tendency of returns to be only observed as the underlying values of the asset get high as compared to other assets.
C. The survivorship bias results from the tendency of poorly performing funds to stop their activities.
D. All the above.

Answer: C

Step by step

1. Survivorship bias: poorly performing funds stop reporting or fail, and studies include only funds that survived the whole period.
2. Non-surviving funds have below-average returns, so the surviving sample's reported returns are too high (as much as 4% for illiquid assets).
3. Match each option to a bias: A is infrequent trading, B is sample selection bias, C is survivorship bias.

Why C is correct: Failed and non-reporting funds are dropped from the data, so only above-average survivors remain and reported returns are overstated.

Why the others are wrong

- **A:** Describes infrequent trading (return smoothing), not survivorship; confuses the source of the understated volatility, correlation and beta.
- **B:** Describes sample selection bias (returns recorded mainly when values are high), a different bias.
- **D:** Not correct because A and B each describe different biases.

Q599 AnalystPrep Q.5417

REBUILT — Original 'reporting bias' option set blurred survivorship (non-reporting of poor funds) and selection bias and included backfill bias outside the Schweser text; rebuilt around sample selection bias as defined in the reading.

According to Ang, which of the following best describes sample selection bias in the reported returns of illiquid assets?

A. Funds that fail or stop reporting are dropped, so only surviving funds appear in the performance data.
B. Asset values and returns tend to be recorded only when they are high (for example, properties sold after prices recover), so reported alpha is too high and beta and volatility are too low.
C. Infrequent trading smooths appraisal-based returns, so volatility, correlation and beta are understated even if the values are unbiased.

D. Reported returns are too low because only distressed sales are observed, so reported beta is too high.

Answer: B

Step by step

1. Sample selection bias arises because prices/returns are observed when they are high (houses and office buildings sold after recovery; buyout and VC exits when valuations are high).
2. Effect: higher reported alpha than true alpha, lower reported beta and lower variance, so expected return is overstated and risk understated.
3. Only B states both the cause and the direction of these effects.

Why B is correct: Only high values are recorded, so alpha is overstated and beta and volatility are understated.

Why the others are wrong

- **A:** This is survivorship bias (failed or non-reporting funds drop out), not selection bias.
- **C:** This is the infrequent trading (smoothing) bias, a distinct third bias.
- **D:** Reverses the direction: selection bias overstates returns and understates beta.

Q600

BT answers P2.T8.707.2 · BT notes P2.T8.707.2 · BT Mock Q58

Andrew Ang makes an important, provocative statement when he writes "Reported illiquid asset returns are not returns." He claims that people overstate the expected returns and understate the risk of illiquid assets, and he attributes this to three key biases. According to Ang, each of the following is a bias that overstates the expected returns (and/or understates the risk) of illiquid assets EXCEPT which is not accurate?

A. Survivorship bias can inflate returns by 4.0% or more

B. Infrequent sampling (aka, infrequent trading) artificially reduces risk and risk-related metrics such as volatility, correlation and beta

C. Turnover bias decreases the typical time between transactions and tends to artificially increase the expected return by 5.0% or more

D. Selection bias (aka, reporting bias) is a distortion of the sample that artificially increases (i.e., overestimates) alpha and artificially decreases (i.e., underestimates) beta

Answer: C

Step by step

1. The three biases are survivorship, sample selection and infrequent trading.
2. A: survivorship overstates returns (up to about 4% in illiquid markets): accurate.
3. B: smoothed returns lower volatility, correlation and beta: accurate.
4. D: selection bias raises alpha and lowers beta: accurate.
5. C: 'turnover bias' is not one of the three biases, and a shorter time between transactions would reduce, not create, bias: the inaccurate statement.

Why C is correct: There is no 'turnover bias' in Ang's framework; the other three statements correctly describe actual biases.

Why the others are wrong

- **A:** Accurate: survivorship bias overstates returns by up to about 4% in illiquid markets.
- **B:** Accurate: infrequent trading smooths returns and understates volatility, correlation and beta.
- **D:** Accurate: sample selection bias overstates alpha and understates beta.

Q601

BT answers P2.T8.410.2 · BT notes P2.T9.410.2

REBUILT — Original relied on Fung-Hsieh hedge fund database definitions (backfill, measurement bias) outside the Schweser reading and had a selection-bias definition that conflicts with Schweser; rebuilt as an EXCEPT item on Schweser's three biases.

Which of the following statements about biases in the reported returns of illiquid assets is INCORRECT?

A. Survivorship bias arises because poorly performing funds stop reporting or fail, so reported returns are too high.

B. Sample selection bias refers to a fund re-classifying itself ex post into a more favorable strategy; it affects only the reported average return, not risk estimates.

C. Infrequent trading smooths reported returns, so volatility, correlation and beta computed from them are too low.

D. Sample selection bias is especially prevalent in private equity because exits tend to occur when valuations are high.

Answer: B

Step by step

1. A: Failed/non-reporting funds are excluded, so surviving-fund returns overstate true returns: correct.
2. C: Smoothed returns understate volatility, correlation and beta: correct.
3. D: Buyout and venture exits occur when values are high, so recorded returns are high: correct.
4. B: Sample selection bias concerns returns being recorded when values are high, not strategy re-classification, and it lowers reported beta and variance, so it also understates risk: incorrect.

Why B is correct: Selection bias is about observing only high values and it distorts risk measures as well as returns, so B misdefines it.

Why the others are wrong

- **A:** A correct statement of survivorship bias.
- **C:** A correct statement of the infrequent trading bias.
- **D:** A correct statement; PE exits at high valuations drive selection bias.

Q602

GARP PE2 Q11 · Mock 4 Q26

The chief investment officer (CIO) of a large university endowment fund is considering adding some illiquid assets to improve the performance of the university's investment portfolio. The CIO asks an investment manager to prepare a report discussing the characteristics of illiquid asset returns. The manager reviews the dynamics of illiquid assets and the determinants of their returns. Which of the following is a correct statement for the manager to include in the report?

A. Survivorship bias overstates reported returns of illiquid assets by reporting only the returns of those funds that have achieved a return above a required threshold.

B. Reporting bias overstates reported returns of illiquid assets because only the returns of those funds that continue to remain in business over a given time period are considered.

C. Survivorship bias and reporting bias can theoretically be eliminated by including the returns of the entire population of funds.

D. Infrequent trading, although considered a bias, can still generate sufficient data for accurate beta and correlation estimates.

Answer: C

Step by step

1. Survivorship/reporting bias come from missing funds (failed funds or those that stop or never report), so observing the whole fund universe would remove them in theory (in practice impossible for illiquid markets).
2. A and B swap or distort the mechanisms; D contradicts the fact that smoothing understates beta and correlation.
3. Only C is correct.

Why C is correct: Both biases are missing-data problems that disappear in principle if the entire population of funds were observed.

Why the others are wrong

- **A:** Wrong mechanism: survivorship bias drops failed funds; it is not about a return threshold.
- **B:** Describes survivorship (staying in business) rather than reporting bias, which is about funds that do not report.
- **D:** Infrequent trading smooths returns and biases beta and correlation estimates downward, so they are not accurate.

Q603

GARP PE2 Q51

An individual investor wants to invest USD 8 million in exchange-traded funds (ETFs) or private equity funds (PEFs). The investor obtains the previous year's returns for several ETFs and calculates summary statistics such as volatility and correlation based on these returns. The investor also reviews a database of reported returns and volatilities for several PEFs and then selects two potential investments in each asset class. Using the data from the sources described above, the investor generates the following information for the four potential investments:

Asset	Annual 1-year return	Volatility of returns
Broad equity market index ETF (ETF1)	6.5%	11.4%
Growth stock ETF (ETF2)	8.3%	13.6%
Private equity fund 1 (PEF1)	7.4%	12.3%
Private equity fund 2 (PEF2)	10.2%	11.1%
Correlation of returns between ETF1 and ETF2	0.67	
Correlation of returns between ETF1 and PEF1	0.25	
Correlation of returns between PEF1 and PEF2	0.41	

The manager evaluates this information while also considering the potential biases and uncertainties in the reported data. Which of the following conclusions is most appropriate for the investor to make?

A. The correlation of returns between PEF1 and ETF1 is more accurate than the correlation of returns between ETF1 and ETF2.

B. The entire USD 8 million should be allocated to PEF2 because it is clearly the superior investment from a return to risk perspective.

C. The number of assets in PEFs are typically higher than the number of assets in ETFs, which makes PEFs much less risky than ETFs over longer time horizons.

D. Summary statistics computed using reported returns of PEFs can be biased downward, which compromises the reliability of these risk measures.

Answer: D

Step by step

1. Return/volatility: ETF1 0.57, ETF2 0.61, PEF1 0.60, PEF2 0.92 (computed from the table), so PEF2 looks best only on reported figures.
2. PEF figures come from reported (appraisal-based, infrequently traded, survivorship- and selection-biased) returns, so their volatility and correlations are understated.
3. ETF statistics come from market prices and are reliable; hence the PEF risk measures are the unreliable ones (D).

Why D is correct: Reported PEF returns are smoothed and biased, so their volatility, correlation and beta are understated and not reliable.

Why the others are wrong

- **A:** Reversed: ETF-ETF correlation uses market prices and is more accurate than the PEF-based correlation.
- **B:** PEF2's high return/risk ratio (about 0.92) rests on understated volatility, and putting everything in one asset ignores diversification.
- **C:** The number of holdings does not remove the smoothing bias; PEF true risk is higher than reported.

Q604

Mock 1 Q60

REBUILT — Original options (infrequent trading vs smoothed data) overlapped and two were defensible; rewrote the stem to target understated volatility/correlation and replaced options with distinct biases.

An analyst at a pension fund is reviewing an allocation to illiquid alternative assets such as real estate and private equity. The analyst finds that the reported volatilities, betas, and correlations with public equities of these assets are all very low, and is concerned that the fund will understate the true risk of the allocation. Which source of bias in the reported data best explains the analyst's concern?

A. Survivorship bias.

B. Sample selection bias.

C. Infrequent trading.

D. Unsmoothing of reported returns.

Answer: C

Step by step

1. Identify the symptom: volatility, beta and correlation are too LOW, i.e., risk is understated (not returns overstated).
2. Per Schweser, infrequent trading means reported values are stale/smoothed, which dampens measured return volatility, betas and correlations.
3. Survivorship and sample selection biases inflate reported average returns, not risk estimates.
4. Unsmoothing is the remedy that adds noise back, not a bias in the data.
5. Answer: C.

Why C is correct: Illiquid assets trade infrequently, so reported (appraisal-based) returns are smoothed and understate volatility, beta and correlation, which understates risk.

Why the others are wrong

- **A:** Survivorship bias overstates reported returns because failed funds drop out; it does not directly explain low measured volatility, beta and correlation.
- **B:** Sample selection bias (values reported when high) overstates returns; the concern here is understated risk.
- **D:** Unsmoothing is the correction applied to remove the smoothing effect; it is not a source of bias in reported data.

Q605

Mock 3 Q30

A risk analyst at an investment bank is conducting performance analyses of hedge funds and real estate funds. The analyst notes the following two issues regarding the funds' annual performance data:
- Whenever a hedge fund stops reporting its performance, it is removed from the database of hedge funds.
- Assets owned by the real estate funds are valued only once a year due to infrequent trading.
Which of the following best describes the impacts of using the data with the aforementioned issues on the results of the performance analyses?

A. The average Sharpe ratio of hedge funds is understated and the average Sharpe ratio of real estate funds is overstated.

B. The average Sharpe ratio of hedge funds is overstated and the average Sharpe ratio of real estate funds is also overstated.

C. The average volatility of hedge funds is overstated and the average volatility of real estate funds is also overstated.

D. The average volatility of hedge funds is overstated and the average volatility of real estate funds is understated.

Answer: B

Step by step

1. Hedge funds: dropping non-reporting (poor) funds is survivorship bias, so average reported return is too high and the Sharpe ratio (numerator = excess return) is overstated.
2. Real estate: infrequent valuation smooths returns, so measured volatility is too low; with a too-small denominator the Sharpe ratio is overstated.
3. Therefore both average Sharpe ratios are overstated: B.
4. Options C and D claim hedge fund volatility is overstated, which neither issue implies.

Why B is correct: Survivorship bias inflates hedge fund returns and smoothing deflates real estate volatility; both push the Sharpe ratio up.

Why the others are wrong

- **A:** Wrongly treats survivorship bias as lowering hedge fund Sharpe; removing failed funds raises average returns.
- **C:** Volatility of hedge funds is not overstated by survivorship, and real estate volatility is understated, not overstated, by infrequent valuation.
- **D:** Real estate volatility is correctly understated, but hedge fund volatility is not overstated by survivorship bias (the main effect is on returns).

Q606

Schweser Module Quiz 99.1 Q3

REBUILT — Replaced the near-synonym option 'attrition bias' with 'smoothing bias' so that only survivorship bias is defensible.

Blue Sky Funds, a private equity fund, has suffered low returns for the last five years. As a result, the fund has decided to quit reporting returns. The fund did report returns each year for the last 10 years when performance was strong. This problem of reporting leads to:

A. survivorship bias.

B. sample selection bias.

C. infrequent trading bias.

D. smoothing bias.

Answer: A

Step by step

1. Funds have no obligation to report to databases; poor performers stop reporting or fail.
2. The database then contains only funds (or periods) that survived/continued reporting, whose returns are above average.

3. This is survivorship bias, which overstates reported returns.
4. Answer: A.

Why A is correct: When poorly performing funds stop reporting, the database retains only successful funds, so reported returns are too high: survivorship bias.

Why the others are wrong

- **B:** Sample selection bias arises when values are reported only when high (e.g., assets sold at high prices), not when a fund stops reporting altogether.
- **C:** Infrequent trading concerns stale, smoothed valuations that understate risk, not a fund exiting the database.
- **D:** Smoothing is a consequence of infrequent valuation and affects risk estimates; it is not caused by a fund ceasing to report.

LO 99.d

Explain the process of unsmoothing returns and the effects of unsmoothing.

Q607

AnalystPrep Q.2435

REBUILT — Original asked which 'process' accounts for infrequent trading, but both 'Unsmoothing' and 'Filtering' are defensible (unsmoothing is a filtering procedure); rewritten to test what unsmoothing does.

A fund holds illiquid assets that trade infrequently, so its reported returns are smoothed. An analyst applies an unsmoothing (de-smoothing) procedure to the reported return series. Which of the following best describes the effect of unsmoothing?

A. It adds noise back to the reported returns, so estimated volatility rises and correlations with liquid asset classes typically rise

B. It removes noise from the reported returns, so estimated volatility falls and the returns look more stable

C. It leaves volatility unchanged but raises the mean return to offset survivorship bias

D. It eliminates sample selection bias by including the returns of funds that stopped reporting

Answer: A

Step by step

1. Infrequent trading smooths reported returns, so reported volatility, beta and correlations are too low.
2. Unsmoothing is the reverse of the usual filtering: it adds noise back to uncover the true, noisier returns.
3. Schweser example: standard deviation of raw real estate returns 2.25% versus 6.26% unsmoothed; correlation with the S&P 500 rose from 9.2% to 15.8%.
4. Therefore volatility and correlation estimates increase.

Why A is correct: Smoothing mutes volatility and correlation; unsmoothing reverses this by adding noise back, raising risk estimates.

Why the others are wrong

- **B:** Describes standard filtering, which removes noise; unsmoothing does the opposite.
- **C:** Unsmoothing targets risk (volatility, beta, correlation) distortions from infrequent trading, not survivorship bias, and volatility does change.
- **D:** Survivorship/selection biases come from which funds or prices are reported; unsmoothing cannot recover missing funds.

Q608

AnalystPrep Q.2436

REBUILT — Original options mixed a property (autocorrelation) with processes (unsmoothing, filtering) and a vague 'auto regression'; rewritten so each option is a coherent property-plus-consequence pair.

Alco Bank builds an index of artwork values from data reported by auction firms and art advisors. Because art is not actively traded, index values rely on past appraisals and the most recent comparable sales. Sam Park intends to use the index returns to judge the risk of art. Which of the following best describes the index returns?

A. They exhibit positive autocorrelation, so the index volatility is understated relative to true volatility

B. They exhibit negative autocorrelation, so the index volatility is understated relative to true volatility

C. They exhibit positive autocorrelation, so the index volatility is overstated relative to true volatility

D. They exhibit zero autocorrelation, so the index is unbiased but its correlation with equities is understated

Answer: A

Step by step

1. Appraisal-based values lag true values, so each period's reported return contains part of the prior period's return.
2. This smoothing creates positive serial correlation in reported returns.
3. Smoothed returns have lower dispersion, so measured volatility, beta and correlations are understated; unsmoothing is needed.

Why A is correct: Stale, appraisal-based values smooth returns, producing positive autocorrelation and understated risk.

Why the others are wrong

- **B:** Smoothing induces positive, not negative, autocorrelation.
- **C:** Right sign of autocorrelation but smoothing understates, not overstates, volatility.
- **D:** Smoothed returns are autocorrelated; assuming zero autocorrelation ignores the bias.

Q609

BT answers P2.T8.707.3 · BT notes P2.T8.707.3

REBUILT — Original relied on missing equations 13.1/13.2/13.4 and option A ('affects only risk, not expected returns') conflicts with Schweser, which says unsmoothing affects both risk and return estimates; the stem now states the model and the options avoid that ambiguity.

Reported returns of an illiquid asset are smoothed according to $r_t^* = (1-\varphi)r_t + \varphi r_{t-1}^*$, where r_t is the true unobserved return, r_t^* is the reported return and $0 < \varphi < 1$. Unsmoothing recovers the true returns. Each of the following statements about unsmoothing is correct EXCEPT:

A. The unsmoothed returns have a higher variance than the reported returns

B. If $\varphi = 0$, unsmoothing leaves the reported returns unchanged

C. The unsmoothed true returns should show approximately zero autocorrelation

D. Unsmoothing lowers the estimated volatility of the asset and its correlation with liquid asset classes

Answer: D

Key correction: Source key (D) was right for its version (variance of true returns is higher, not lower, than reported), but its option A conflicts with Schweser, which says unsmoothing affects risk and return estimates; option A was removed.

Step by step

1. Rearranging the model gives $r_t = [r_t^* - \varphi r_{t-1}^*]/(1-\varphi)$.
2. If $\varphi = 0$, $r_t = r_t^*$: no change (B true).
3. Dividing by $1-\varphi < 1$ and removing the lagged term gives a noisier series with higher variance (A true) and removes the serial correlation (C true).
4. Higher variance means unsmoothing raises, not lowers, estimated volatility and correlations (D false, the answer).

Why D is correct: Unsmoothing adds noise back, so volatility and correlation estimates increase, which contradicts D.

Why the others are wrong

- **A:** True statement: the true series is noisier than the reported one (true variance exceeds reported variance).
- **B:** True statement: with no smoothing parameter there is nothing to undo.
- **C:** True statement: removing the φr_{t-1}^* carryover eliminates the autocorrelation.

Q610

GARP PE1 Q60 · Mock 3 Q21

REBUILT — The two sources differed only in option C (negative vs positive autocorrelation) and with the lagged-regression option D both were arguably correct; options rewritten so exactly one is correct.

An investor is performing due diligence on a hedge fund that invests in illiquid assets. The investor obtains monthly return data on the fund but is concerned about possible bias in the data, which may provide a misleading impression of the fund's risk profile. Which of the following is the correct assessment?

A. Volatility will be artificially high, giving the appearance of high total risk, which can be corrected by taking into account the resulting positive autocorrelation of returns

B. Correlations with other asset classes will be artificially high, giving the appearance of high systematic risk, which can be corrected by unsmoothing the returns

C. Volatility will be artificially low, giving the appearance of low total risk, which can be corrected by taking into account the resulting positive autocorrelation of returns (unsmoothing)

D. Volatility will be artificially low, giving the appearance of low total risk, which can be corrected by taking into account the resulting negative autocorrelation of returns

Answer: C

Key correction: GARP PE1 key said D (correlations too low, fixed with lagged regressions summing coefficients), which is also a valid correction in its original wording; Mock 3 key said C. Rebuilt so only C is correct.

Step by step

1. Illiquid assets are priced infrequently or by appraisal, so reported returns are smoothed.
2. Smoothing induces positive autocorrelation and lowers measured volatility, beta and correlations.
3. The fix is to unsmooth: use the positive autocorrelation to add noise back and raise the risk estimates.
4. Only C pairs low volatility with positive autocorrelation.

Why C is correct: Smoothed returns are positively autocorrelated and understate risk; unsmoothing using that autocorrelation corrects it.

Why the others are wrong

- **A:** Smoothing makes volatility artificially low, not high.
- **B:** Smoothing makes correlations artificially low, not high.
- **D:** Right direction of bias but smoothing creates positive, not negative, autocorrelation.

LO 99.e

Compare illiquidity risk premiums across and within asset categories.

Q611

BT Mock Q64 · BT answers P2.T8.708.1 · BT notes P2.T8.708.1

REBUILT — Source relied on a scatterplot exhibit that is not reproduced; restated the exhibit's key data in words so the item is self-contained.

Antti Ilmanen's scatterplot (as cited by Andrew Ang) plots long-run average return against an index of illiquidity across asset classes, 1990-2009. Venture capital is the least liquid and has the highest average return (about 16%-17%), while cash and public equities are the most liquid with returns of roughly 4%. In regard to the illiquidity risk premium, which of the following statements is TRUE?

A. In general, illiquid asset classes offer high risk-adjusted returns.

B. The scatterplot demonstrates that large illiquidity risk premiums exist ACROSS asset classes.

C. Large illiquidity risk premiums exist WITHIN many asset classes, but are not demonstrated ACROSS asset classes.

D. Illiquid equities earn the same returns as liquid equities, and illiquid bonds earn the same returns as liquid bonds.

Answer: C

Step by step

1. Across asset classes, the raw scatterplot shows a positive link between illiquidity and return, but it is confounded: reported returns of illiquid assets are biased upward (survivorship, sample selection, infrequent sampling), illiquid classes carry other risks (business risk, leverage), indices are not investable, and manager skill cannot be separated from factor returns.
2. After adjusting for these issues, a substantial premium cannot be demonstrated across asset classes.
3. Within asset classes, less liquid assets (off-the-run vs on-the-run Treasuries, wide-spread corporate bonds, low-turnover stocks) earn clearly higher returns, with other factors roughly held constant.
4. So the statement 'large premiums within, but not demonstrated across, asset classes' (C) is correct.

Why C is correct: Within-class comparisons hold business risk and other characteristics roughly fixed, so they reveal genuine illiquidity premiums; the cross-class chart is contaminated by biases and other risks.

Why the others are wrong

- **A:** Wrong: after adjusting for biases and extra risks, illiquid classes look much less attractive on a risk-adjusted basis. Mistake: reading raw high returns as risk-adjusted returns.
- **B:** Wrong: the raw positive slope is not evidence of a premium once biases and other risks are removed. Mistake: taking the unadjusted chart at face value.
- **D:** Wrong: less liquid stocks and bonds earn higher returns than liquid ones within the same class. Mistake: denying any within-class premium.

Q612 BT answers P2.T8.708.2 · BT notes P2.T8.708.2

REBUILT — Option A used loose wording ('dynamic rebalancing') and the BT notes copy had no key; reworded to match Schweser's 'dynamic factor strategies at the aggregate portfolio level' and sharpened the other options.

Andrew Ang describes four ways an investor can harvest the illiquidity premium. Which of the four is the easiest to implement and can have the greatest effect on portfolio returns?

A. Dynamic factor strategies (countercyclical rebalancing, long illiquid / short liquid) at the aggregate portfolio level

B. Acting as a market maker for individual securities

C. Selecting less liquid securities within each asset class

D. Passive allocation to illiquid asset classes such as real estate

Answer: A

Step by step

1. The four methods are: (1) passive allocation to illiquid asset classes, (2) liquidity security selection within asset classes, (3) market making in individual securities, (4) dynamic factor strategies at the aggregate portfolio level.
2. Method 4 works through countercyclical rebalancing: buy when others want to sell, sell when others want to buy, tilting long illiquid and short liquid.
3. Rebalancing is the simplest way to provide liquidity, so Schweser/Ang rank method 4 as the easiest to implement with the largest return impact.
4. Answer: A.

Why A is correct: Rebalancing at the aggregate level requires no special access or deal sourcing and systematically supplies liquidity when it is scarce, giving the largest impact.

Why the others are wrong

- **B:** Wrong: market making needs scale, infrastructure and counterparty screening to avoid adverse selection; it is harder to implement. Mistake: equating the most visible method with the easiest.
- **C:** Wrong: security-level selection is a valid method but has a smaller impact than aggregate rebalancing. Mistake: confusing the within-class premium with the greatest-impact strategy.
- **D:** Wrong: passive allocation is simple, but it is a static tilt that does not exploit liquidity differences over time and is not ranked as having the greatest effect. Mistake: assuming 'simplest' means 'passive'.

Q613 GARP PE1 Q24 · Mock 4 Q28

The chief investment officer (CIO) of a large university endowment fund is considering whether to add illiquid assets to the university's investment portfolio. The CIO asks an investment analyst to review illiquidity risk premiums across and within asset categories and to prepare a report. Which of the following statements is correct for the analyst to include in the report?

A. Corporate bonds that trade less frequently or have larger bid-ask spreads have lower returns than more liquid corporate bonds.

B. Expected returns of illiquid assets can be overstated due to measurement biases.

C. US Treasury instruments are the only assets that do not exhibit an illiquidity risk premium.

D. Hedge funds that do not place restrictions on withdrawals exhibit higher returns.

Answer: B

Step by step

1. Reported returns of illiquid assets suffer survivorship bias, sample selection bias and infrequent sampling (smoothing), so returns are overstated and risk/correlation understated.
2. Check A: infrequent trading and wide bid-ask spreads raise corporate bond yields, hence returns, so A is reversed.
3. Check C: Treasuries show an on-the-run/off-the-run spread and large liquidity effects in 2007-09, so they do exhibit a premium.
4. Check D: hedge funds WITH withdrawal restrictions earn higher returns, not those without. Only B is correct.

Why B is correct: Biases in reported data inflate observed returns of illiquid assets, so measured premiums overstate true ones.

Why the others are wrong

- **A:** Wrong: less liquid bonds have higher yields and returns. Mistake: reversing the sign of the premium.
- **C:** Wrong: Treasuries do exhibit a premium (off-the-run bills yield more than on-the-run). Mistake: assuming the most liquid market has no illiquidity effects.

- **D:** Wrong: it is funds with lockups/withdrawal restrictions that earn higher returns. Mistake: reversing the relationship.

Q614 Schweser Module Quiz 99.2 Q1

Which of the following variables is NOT an illiquidity factor that affects equity returns?

A. Measures of adverse selection (informed trading).
B. The number of recorded positive returns.
C. Turnover.
D. Volume.

Answer: B

Step by step

1. Illiquidity factors for equities include bid-ask spreads, volume, turnover, signed volume, the Amihud ratio, price impact, informed trading (adverse selection), quote size/depth, trade frequency, number of zero returns, and return autocorrelation.
2. A (adverse selection), C (turnover) and D (volume) are on the list.
3. The relevant count is of zero returns (stale/no trade), not positive returns, so B is not an illiquidity factor.

Why B is correct: Illiquid stocks often show no price change on many days, so the number of zero returns signals illiquidity; positive returns carry no such information.

Why the others are wrong

- **A:** Wrong: informed trading measures are a recognised factor. Mistake: overlooking adverse selection as a liquidity cost.
- **C:** Wrong: low turnover signals illiquidity. Mistake: treating turnover as only a size measure.
- **D:** Wrong: trading volume is a core liquidity measure. Mistake: assuming volume is unrelated to expected returns.

Q615 Schweser Module Quiz 99.2 Q2

Rick Fairview, a general partner and portfolio manager with Fairview Funds, is considering ways to profit from illiquidity risk premiums. Which of the following strategies is most likely to produce the greatest impact on portfolio performance?

A. Acting as a market maker for individual securities.
B. Choosing the most illiquid assets within an asset class, even if the asset class is generally considered liquid.
C. Allocating a portion of the portfolio to illiquid asset classes.
D. Using dynamic factor strategies at the aggregate portfolio level.

Answer: D

Step by step

1. List the four harvesting methods: passive allocation to illiquid classes (C), liquidity security selection (B), market making (A), dynamic factor strategies at the aggregate level (D).
2. Ang/Schweser: dynamic factor strategies (long illiquid, short liquid, countercyclical rebalancing) are the easiest to implement and can have the greatest effect on returns.
3. Answer: D.

Why D is correct: Aggregate-level countercyclical rebalancing systematically harvests the premium and is ranked highest in impact.

Why the others are wrong

- **A:** Wrong: market making is one valid method but is harder to implement and not ranked as highest impact. Mistake: picking the strategy with the most direct liquidity provision.
- **B:** Wrong: security selection captures the within-class premium but has less overall impact. Mistake: overweighting the within-class evidence.
- **C:** Wrong: passive allocation is static and subject to biased return data. Mistake: assuming strategic illiquid allocation is the primary driver.

LO 99.f

Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets.

Q616

AnalystPrep Q.2426

Jude Wang, a portfolio manager, has been asked to manage a university endowment fund. The head of the university informs her that the university plans to withdraw a substantial amount of the fund within the next two years to establish a financial laboratory, and that the board wants transaction costs kept as low as possible. A colleague, Jane Jones, suggests including private equity and real estate in the fund because these asset classes have low correlation with stocks and bonds, offer better diversification, and would make it easier to generate excess return. Which of the following does NOT support the idea of including private equity and real estate in the portfolio?

A. It would provide a better diversification effect to the portfolio.
B. It would help Wang to generate an excess return.
C. The university needs a substantial amount of funds within the next two years for the laboratory.
D. It would improve the portfolio because of the assets' low correlation with stocks and bonds.

Answer: C

Step by step

1. Identify the arguments in favor of illiquid assets: low correlation with stocks and bonds (diversification) and the potential for alpha because illiquid markets are less efficient.
2. Identify the constraints: illiquid assets trade infrequently and carry large transaction costs; only liquid assets can be 'eaten' (consumed).
3. The university will withdraw a large amount within two years, so a liquidity event is needed soon and the board wants low transaction costs.
4. Forced sales of private equity or real estate in that window would mean long search times, due diligence delays and steep discounts, so this fact argues against the allocation. This is option C.

Why C is correct: A known large cash need in the near term and a low-transaction-cost mandate are exactly the conditions under which illiquid assets are unsuitable, since the investor cannot easily sell them when cash is needed.

Why the others are wrong

- **A:** This is a genuine benefit of illiquid assets, so it supports inclusion; picking it misreads the 'does NOT support' wording.
- **B:** The reading notes that less efficient illiquid markets and wide dispersion of manager skill create alpha opportunities, so this supports inclusion.
- **D:** Low correlation with stocks and bonds is a standard rationale for the allocation, so it supports inclusion.

Q617

BT answers P2.T8.708.3 · BT notes P2.T8.708.3

You are consulting to a large endowment fund that is determining its asset allocation budget, including the target allocation to, and hurdle rates for, illiquid assets. Four staff members make the assertions below. According to Ang, each of these is true EXCEPT which is probably false?

A. Phillip says: The longer the time between liquidity events for an asset (i.e., the less liquid the asset), the LOWER its optimal allocation in the portfolio.
B. Debra says: The longer we need to wait to exit an investment (i.e., the later the arrival of liquidity events), the HIGHER should be our illiquidity hurdle rate.
C. Peter says: Because of the nature of factor risk and idiosyncratic risk in illiquid markets, it is really important to assign (or delegate to) genuinely skilled investors for the illiquid allocation.
D. Mary says: It is not very important to identify skill in the illiquid asset class because we can easily benchmark against tradeable indexes, which lets us separate factor risk from manager skill (alpha).

Answer: D

Step by step

1. A: Infrequent liquidity events reduce optimal holdings, so the allocation falls as the wait lengthens. True.
2. B: Investors must demand a higher illiquidity premium (hurdle rate) when trading is rarer. True.
3. C: There is no market portfolio of illiquid assets, idiosyncratic risk is high and alpha is widely dispersed, so manager skill is crucial. True.
4. D: There are no tradeable indices for illiquid assets (e.g., NCREIF is not investable), so factor risk cannot easily

be separated from alpha and skill identification is very important. False, so D is the answer.

Why D is correct: Illiquid asset classes lack investable benchmarks, so manager skill cannot be cleanly separated from factor exposure and is therefore critically important, which contradicts Mary's claim.

Why the others are wrong

- **A:** True statement: less frequent liquidity events lower the optimal allocation; chosen only by misreading 'EXCEPT'.
- **B:** True statement: rarer trading requires a larger illiquidity premium, so the hurdle rate rises with the wait.
- **C:** True statement: high idiosyncratic risk, agency problems and dispersed alpha make skilled managers essential.